

SURREAL ALGEBRA

## Set-Sized Quotients of the Omnific Integers

The universal constant-term quotient, exact cardinal thresholds,  
support thresholds, and what survives in large quotients

Merged research report built from twenty-four manuscripts (03, 04, 06–11 of one batch,  
12–16 of the next, 17–19 of a third, 20 of a fourth, 21 of a fifth, 22 of a sixth, 23 of a seventh,  
24–27 of an eighth)

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### Abstract

Write an omnific integer uniquely as  $n + s$  with  $n \in \mathbb{Z}$  and  $s$  purely infinite, and let  $\text{ct}(n + s) = n$ . Every additive and multiplicative map from  $\mathbf{Oz}$  to a set-sized ring, even a noncommutative, nonreduced or nonunital one, has the form  $x \mapsto \text{ct}(x)e$  with  $e$  idempotent; every set-sized  $\mathbf{Oz}$ -module is an abelian group with scalar action through  $\text{ct}$ . The same holds for  $D + \Pi_K$  over any set field  $K$  and unital  $D \subseteq K$ , for the Gaussian ring  $\mathbf{Oz}[i]$  (with  $\mathbb{Z}[i]$ ), and, on the finite side, for standard part on the finite surreal and surcomplex rings. Countable supports already force this collapse while finite supports do not.

Over set-sized exponent groups the theorem becomes a detection threshold. For each of five explicit set-sized rings (four integer parts inside  $\mathbf{Oz}$ , of sizes  $\kappa^{\aleph_0}$ ,  $\kappa^{<\kappa}$ ,  $\kappa$  and  $\lambda$ , and a lexicographic Hahn ring of size  $\kappa^{<\kappa}$ ), the least size of a ring or module detecting any nonzero purely infinite element equals the size of the ring; two of these values, and the exact thresholds of two examples in fixed exponent groups, are new to this merge. The report also keeps the large-quotient side (a copy of  $\mathbf{No}$  and a Cantor algebra inside  $\mathbf{Oz}/(1 + \omega^a)$ ), the integral and complete integral closures of  $\mathbf{Oz}$ , the arithmetic of integer-valued polynomials on  $\mathbf{Oz}$ , and a negative answer to a posted question on natural ordinal arithmetic. Five later manuscripts add the ordinary homological and tensor algebra of subrings of  $\mathbf{Oz}$ : for set-sized finite-support cores the arithmetic quotient has flat dimension  $d$  when the exponents form a coordinatewise cone of rank  $d$  and 1 when they form a totally ordered cone, and coefficient-lattice ideals have explicit tensor, Rees and duality algebra; and they add a fresh-scale image-gap theorem for polynomial and rational maps. Three manuscripts of a third batch add exact relation counts and Tor groups of these ideals, a sharp bound on the ordinary outputs of integer-valued polynomials, and the integral closure of  $\mathbf{Oz}$  modulo the ideal generated by  $\Pi$  as a Boolean power of the algebraic integers. A fourth adds that the Boolean algebra of that power is atomless and a proper class, split uniformly over every set-sized subring, with integrality local to the exponent group of an element. A fifth shows that for the coordinatewise cone of rank  $d$  the projective dimension of the arithmetic quotient is  $d + 1$ . A sixth certifies rational functions of several variables by finitely many values at one fresh scale, removes set-sized exceptional sets in every number of variables, and transfers ordinary constant test sets exactly. A seventh shows that over one cyclic exponent group the extended ideal of the integral closure is not radical, while the Boolean algebra of the reduced fibre is still atomless. Two manuscripts of an eighth batch show, with global choice, that  $1 + \omega^a$  lies in characteristic-zero maximal class ideals of  $\mathbf{Oz}$ , modulo which exactly the monomials  $\omega^b$  with  $b > na$  for every  $n$  vanish and whose residue fields embed in  $\mathbf{No}[i]$ ; and that a globally smooth semialgebraic map preserving  $\mathbf{Oz}^r$  is an integer-valued polynomial, while finitely smooth ones need not be even piecewise polynomial. Two more compute the flat dimensions of radical monomial quotients of cone cores by the topology of their faces, and build set-sized omnific integer parts whose henselization at the constant-term ideal has the maximal number of minimal primes although its completion is  $\mathbb{Q}$ . It is an AI-assisted merge of unrefereed manuscripts; nothing in it is verified in Lean.

**Status.** Proof-bearing research report assembled from twenty-four AI-assisted manuscripts written on 22 and 23 September 2026. The sources call their main results candidate original; priority is not certified, no named conjecture is claimed solved, the proofs have not been refereed, and no statement is formalized. Results added in the merge are marked [merge]; the source of every other statement is tagged [NN].

# Contents

|           |                                                                                |           |
|-----------|--------------------------------------------------------------------------------|-----------|
| <b>1</b>  | <b>Introduction</b>                                                            | <b>4</b>  |
| 1.1       | The question and the answer . . . . .                                          | 4         |
| 1.2       | Main theorems . . . . .                                                        | 5         |
| 1.3       | Sources, provenance and where the merge chose . . . . .                        | 7         |
| 1.4       | Relation to the repository . . . . .                                           | 16        |
| <b>2</b>  | <b>Conventions, notation, and the constant-term retraction</b>                 | <b>19</b> |
| 2.1       | Foundations . . . . .                                                          | 19        |
| 2.2       | Normal forms and the orientation of exponents . . . . .                        | 20        |
| 2.3       | The rings . . . . .                                                            | 20        |
| 2.4       | Notation and renamings . . . . .                                               | 21        |
| <b>3</b>  | <b>Support gaps and the purely infinite ideal</b>                              | <b>22</b> |
| <b>4</b>  | <b>The collision mechanism</b>                                                 | <b>24</b> |
| 4.1       | Scaled-field collision . . . . .                                               | 25        |
| 4.2       | The geometric telescope . . . . .                                              | 26        |
| 4.3       | Division of an arbitrary purely infinite element . . . . .                     | 27        |
| 4.4       | Separation principles . . . . .                                                | 27        |
| <b>5</b>  | <b>The universal set-sized quotient</b>                                        | <b>28</b> |
| 5.1       | The theorem . . . . .                                                          | 28        |
| 5.2       | Consequences for rings and representations . . . . .                           | 30        |
| <b>6</b>  | <b>Quotients, closures, completions and localizations</b>                      | <b>34</b> |
| 6.1       | Every quotient and its small image . . . . .                                   | 34        |
| 6.2       | Finite congruences, closures and invisible primes . . . . .                    | 36        |
| 6.3       | Localizations and presentations . . . . .                                      | 38        |
| 6.4       | Diophantine observation . . . . .                                              | 40        |
| <b>7</b>  | <b>Modules and derivations over the full class ring</b>                        | <b>41</b> |
| <b>8</b>  | <b>The finite side: standard part is universal</b>                             | <b>44</b> |
| <b>9</b>  | <b>The support threshold: finite supports fail, countable supports suffice</b> | <b>45</b> |
| <b>10</b> | <b>Fixed exponent groups: cardinal lower bounds and exact thresholds</b>       | <b>48</b> |
| 10.1      | The bounds of the sources . . . . .                                            | 48        |
| 10.2      | The field bound . . . . .                                                      | 49        |
| 10.3      | Criteria that kill the whole ideal below a cardinal . . . . .                  | 51        |
| 10.4      | The organizing theorem: the threshold is the size of the ring . . . . .        | 52        |
| <b>11</b> | <b>Set-sized rings with exact thresholds</b>                                   | <b>54</b> |
| 11.1      | Countable supports over a two-armed group [03] . . . . .                       | 54        |
| 11.2      | Supports of size $< \kappa$ over a two-sided group [06] . . . . .              | 58        |
| 11.3      | A controlled field of size exactly $\kappa$ [08] . . . . .                     | 62        |
| 11.4      | Countable supports over a long one-armed group [09] . . . . .                  | 65        |
| 11.5      | Supports of size $< \kappa$ over a one-armed group [11] . . . . .              | 68        |
| <b>12</b> | <b>Flat ideals and homological reflection</b>                                  | <b>70</b> |
| <b>13</b> | <b>Homological and tensor algebra of omnific cores</b>                         | <b>74</b> |
| 13.1      | Coordinate-cone and ordered-cone cores . . . . .                               | 75        |
| 13.2      | Flat face ideals and a Boolean resolution . . . . .                            | 77        |

|           |                                                                                   |            |
|-----------|-----------------------------------------------------------------------------------|------------|
| 13.3      | The multigraded Tor and exact flat dimensions . . . . .                           | 79         |
| 13.4      | Coefficient lattices and their syzygies . . . . .                                 | 81         |
| 13.5      | Flat and projective dimensions; the ordered cone . . . . .                        | 84         |
| 13.6      | The tensor normal form . . . . .                                                  | 89         |
| 13.7      | Rees equations . . . . .                                                          | 93         |
| 13.8      | Homomorphisms, ideal classes and duals . . . . .                                  | 95         |
| 13.9      | Relations, Tor and moduli of lattice ideals . . . . .                             | 100        |
| 13.10     | Exact projective dimensions and derived arithmetic . . . . .                      | 111        |
| 13.11     | Polyhedral cores: face topology and flat dimension . . . . .                      | 126        |
| <b>14</b> | <b>Large internal fields and the binomial quotient</b>                            | <b>149</b> |
| 14.1      | Compressed exponent fields . . . . .                                              | 150        |
| 14.2      | An ordered additive compression of the surreals . . . . .                         | 151        |
| 14.3      | Transfer by inverting a purely infinite element . . . . .                         | 152        |
| 14.4      | The binomial quotient and its localization . . . . .                              | 154        |
| 14.5      | Exact binomial kernels . . . . .                                                  | 155        |
| 14.6      | The dyadic tower and its Cantor algebra . . . . .                                 | 157        |
| 14.7      | Gaussian binomial quotients . . . . .                                             | 160        |
| 14.8      | Boundaries of the large-field results . . . . .                                   | 161        |
| 14.9      | Class residue fields beyond the constant term . . . . .                           | 162        |
| <b>15</b> | <b>Integral closure, complete integral closure, and invisible equations</b>       | <b>175</b> |
| 15.1      | Uniform denominators . . . . .                                                    | 176        |
| 15.2      | Almost integrality and complete integral closure . . . . .                        | 177        |
| 15.3      | Integral closure: reciprocals, small units and density . . . . .                  | 177        |
| 15.4      | Constant slices, truncation, and a unit equation . . . . .                        | 180        |
| 15.5      | The surcomplex and Gaussian versions . . . . .                                    | 182        |
| 15.6      | Set-valued valuations cannot contain the omnific ring . . . . .                   | 183        |
| 15.7      | An equation invisible to every set-sized image . . . . .                          | 184        |
| 15.8      | The arithmetic fibre of the normalization . . . . .                               | 186        |
| 15.9      | Proper-class Boolean branching of the fibre . . . . .                             | 193        |
| 15.10     | Ramification in a cyclic workspace . . . . .                                      | 202        |
| 15.11     | Henselian branching at an omnific augmentation . . . . .                          | 215        |
| <b>16</b> | <b>Arithmetic of polynomial maps on the omnific lattices</b>                      | <b>223</b> |
| 16.1      | Lattice separation and an omitted congruence type . . . . .                       | 224        |
| 16.2      | Polynomials preserving the omnific lattices . . . . .                             | 225        |
| 16.3      | Congruences modulo every omnific integer . . . . .                                | 227        |
| 16.4      | $p$ -adic characters and completions . . . . .                                    | 229        |
| 16.5      | Rational self-maps and gaps at arbitrary surreal scales . . . . .                 | 232        |
| 16.6      | Fresh scales: image gaps, fibers and surjective self-maps . . . . .               | 236        |
| 16.7      | Finite tests, fresh probes and sampling . . . . .                                 | 240        |
| 16.8      | Ordinary outputs, canonical translations and composition . . . . .                | 243        |
| 16.9      | Scope of the polynomial arithmetic . . . . .                                      | 248        |
| 16.10     | Finite tests at a single new scale . . . . .                                      | 249        |
| 16.11     | Semialgebraic preservers: smooth rigidity and bounded-scale flexibility . . . . . | 257        |
| <b>17</b> | <b>Natural ordinal arithmetic is not a quotient of the omnific integers</b>       | <b>270</b> |
| <b>18</b> | <b>Boundaries, non-claims and open questions</b>                                  | <b>271</b> |
| 18.1      | What the results do not say . . . . .                                             | 271        |
| 18.2      | Non-claims of the sources . . . . .                                               | 272        |
| 18.3      | Open questions and their status . . . . .                                         | 286        |
| <b>19</b> | <b>Verification record and formalization routes</b>                               | <b>302</b> |

|                                     |     |
|-------------------------------------|-----|
| 19.1 Finite checks . . . . .        | 302 |
| 19.2 Formalization routes . . . . . | 305 |

# 1 Introduction

## 1.1 The question and the answer

The omnific integers  $\mathbf{Oz} = \mathbb{Z} \oplus \Pi$  form a discretely ordered subring of  $\mathbf{No}$  whose fraction field is all of  $\mathbf{No}$ . Their purely infinite ideal  $\Pi$  is a proper class; each of its elements has a set-sized normal form, but below any positive exponent there is a proper class of smaller exponent scales. The eight manuscripts of the first merge (03, 04, 06–11) ask the same question: *what can an ordinary, set-sized ring or module see of an omnific integer?* All eight give the same answer, by different proofs, and two later manuscripts (13, 15) reprove it independently: exactly its ordinary integer constant term. This is much stronger than the observation that a proper-class field cannot embed in a set: the source is a ring, its homomorphisms have large kernels, and the targets may have zero divisors, nilpotents, noncommutative multiplication and any characteristic (06, 11). Nor is it a consequence of divisibility of  $\Pi$  by every integer, which only disposes of targets of positive characteristic (04, 07).

The sources then go in different directions: exact cardinal thresholds for set-sized integer parts (03, 06, 08, 09, 11), the finite side and the support threshold (04, 11), the large quotients in which a purely infinite element becomes a unit (07), integral closure (03), polynomial maps (10), and natural ordinal arithmetic (11). Three later manuscripts add the ordinary homological and tensor algebra of subrings of  $\mathbf{Oz}$  with finite supports or with coefficient lattices (12, 14; [Section 13](#)), and fresh-scale arguments for polynomial and rational maps (16; [Sections 16.6](#) and [16.7](#)). Three manuscripts of a third batch extend three of these directions: relation modules, Tor groups and isomorphism classes of coefficient-lattice ideals (17; [Section 13.9](#)), the ordinary-output loci of integer-valued polynomials (18; [Section 16.8](#)), and the arithmetic fibre of the integral closure (19; [Section 15.8](#)). A manuscript of a fourth batch determines how that fibre branches: its Boolean algebra of idempotents is atomless and a proper class, and integrality is local to the exponent group of an element (20; [Section 15.9](#)). A manuscript of a fifth batch determines the projective dimensions of the coordinate-cone cores, which 12 had bounded, and computes the Tor algebra of lattice quotients, with arithmetic applications (21; [Section 13.10](#)). A manuscript of a sixth batch certifies rational functions of several variables by finitely many values at a single fresh scale, removes set-sized exceptional sets in every number of variables and lifts ordinary constant test sets exactly; much of its one-variable content re-derives 16's (22; [Section 16.10](#)). A manuscript of a seventh batch decides the fixed-workspace questions on that fibre for cyclic exponent groups, with opposite answers: the extended ideal is not radical, while the Boolean algebra of the reduced fibre is atomless, both by ramification over the place  $\omega^a = 0$  of the rational functions in  $\omega^a$  (23; [Section 15.10](#)). Two manuscripts of an eighth batch go beyond set-sized targets and beyond polynomial maps. With global choice, every proper class ideal lies in a maximal class ideal, so  $1 + \omega^a$  lies in characteristic-zero maximal ideals of  $\mathbf{Oz}$  whose residue fields are proper-class fields embedded in  $\mathbf{No}[i]$ , and the monomials vanishing there are exactly the  $\omega^b$  with  $b > na$  for every ordinary  $n$  (24; [Section 14.9](#)). Globally smooth semialgebraic maps preserving  $\mathbf{Oz}^r$  are the integer-valued polynomials, while for every finite order of smoothness an explicit bump, built at the same scale threshold, agrees with no finite family of polynomials on  $\mathbf{Oz}$  (25; [Section 16.11](#)). Two further manuscripts of that batch extend the cores and the fibre. Over an arbitrary rational polyhedral cone, the flat dimension of the quotient keeping a face is the largest dimension of a face disjoint from it, so that the cone is simplicial exactly when every facet quotient has flat dimension one (26; [Section 13.11](#)). And a set-sized omnific integer part  $\mathbb{E} \cap \mathbf{Oz}$  of any infinite cardinality  $\kappa$  can have an idempotent constant-term ideal whose local ring has completion  $\mathbb{Q}$  but a henselization with  $2^\kappa$  minimal primes, all with complex fraction field (27; [Section 15.11](#)). This report prints the common spine once and keeps every result of every source.

## 1.2 Main theorems

**Theorem A** (Universal set-sized quotient). *Let  $K$  be a set field and  $D \subseteq K$  a unital subring. Every additive multiplicative map from  $D + \Pi_K$  to a set-sized ring kills  $\Pi_K$  and factors uniquely through  $\text{ct}$ ; unital maps correspond bijectively to unital maps from  $D$ ; every set-sized module is annihilated by  $\Pi_K$ . For  $\mathbf{Oz}$  the unique unital map to a set-sized ring  $S$  is  $x \mapsto \text{ct}(x)1_S$ , and unital maps  $\mathbf{Oz}[i] \rightarrow S$  correspond to  $j \in S$  with  $j^2 = -1$ . The kernels admitting set-sized quotients are exactly the  $\text{ct}^{-1}(\mathbf{a})$ ; for  $\mathbf{Oz}$  they are  $\Pi$  and  $n\mathbf{Oz}$  ( $n \geq 1$ ). (Theorems 5.1 and 6.2 and Corollaries 5.2 and 5.3; the eight sources of the first merge, manuscripts 13 and 15, and manuscript 05 of the sibling report.)*

**Theorem B** (The finite side and the support threshold). *Standard part is the universal set-sized quotient of the finite surreal and surcomplex rings, and more generally of  $D + \mathfrak{m}_k$  (04). For every infinite  $\lambda$ , the rings of normal forms with supports of size at most  $\lambda$  (full surreal exponents) already have universal set-sized quotient  $D$  on both sides, while the finite-support rings admit the augmentation  $\sum c_g \omega^g \mapsto \sum c_g$ ; a map on the finite-support ring extends to the countable-support ring if and only if it factors through  $\text{ct}$  or  $\text{st}$  on the respective side (the failure for the augmentation is 04's and 11's, the criterion is 11's). (Theorems 8.2 and 9.1.)*

**Theorem C** (Detection threshold equals size). *For each of five explicit set-sized rings  $A$ , namely four integer parts inside  $\mathbf{Oz}$ , of cardinality  $\kappa^{\aleph_0}$  (cf  $\kappa > \aleph_0$ ; 03),  $\kappa^{<\kappa}$  ( $\kappa$  regular uncountable; 06),  $\kappa$  (every infinite  $\kappa$ ; 08) and  $\lambda$  ( $\lambda$  infinite,  $\lambda^{\aleph_0} = \lambda$ ; 09), and 11's lexicographic Hahn ring of cardinality  $\kappa^{<\kappa}$  (coefficient field with at most  $\max(\kappa, 2^{\aleph_0})$  elements), the least cardinality of a ring target or module detecting any nonzero purely infinite element is exactly  $|A|$ , in ZFC. (Theorem 10.10.) The values for 06's and 11's rings are new here: 06 and 11 prove exactness only under  $\kappa^{<\kappa} = \kappa$ , and the unconditional value  $\kappa^{<\kappa}$  follows from 06's own collision lemma (Theorem 11.11 and Proposition 11.23). The same thresholds hold for the first and second Ext groups of the homological package of 06, which extends to all five rings (Theorems 12.2 and 12.3).*

**Theorem D** (The field bound in a fixed exponent group). *In a set-sized Hahn integer part, a ring or module detecting an element divisible by  $X^{2^c}$  has at least as many elements as the field  $L_c(F)$  of elements of the ambient field  $F$  supported in the scale subgroup  $H_c$  (03, generalized through 06's lemma). This bound is exact in 04's example  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ , where it equals  $2^{\aleph_0}$ , and in an example with rational coefficients it exceeds by far all the monomial and weighted counts of 04, 07 and 10, showing that 10's bound need not be attained. (Proposition 10.3 and Examples 10.4 and 10.5.)*

**Theorem E** (Large internal fields). *If a nonzero purely infinite element becomes a unit in a nonzero ring, the ring contains an explicitly embedded copy of  $\mathbf{No}$ . The nonzero quotient  $\mathbf{Oz}/(1 + \omega^a)$ , which has no nonzero set-sized image, contains an algebraically closed field containing  $\mathbf{No}[i]$ , exact cyclotomic algebras over it, and the Cantor algebra of locally constant functions on  $\mathbb{Z}_2$  (07; one proposition of 06). With global choice, every proper class ideal of a commutative class ring lies in a maximal class ideal, a principle equivalent to global choice; so  $1 + \omega^a$  lies in at least  $2^{\aleph_0}$  maximal class ideals of  $\mathbf{Oz}$ , all of characteristic 0, modulo each of which  $\omega^b$  ( $b \geq 0$ ) vanishes if  $b > na$  for every ordinary  $n$  and is a unit otherwise; these residue fields embed in  $\mathbf{No}[i]$ , and the maps  $\mathbf{Oz} \rightarrow \mathbf{No}[i]$  so obtained separate points, those vanishing on some  $1 + \omega^a$  without preserving strong summability (24). (Theorems 14.7, 14.15, 14.19, 14.24, 14.30, 14.34, 14.37, 14.46 and 14.53.)*

**Theorem F** (Closures and invisible equations). *The complete integral closure of  $\mathbf{Oz}$  is  $\mathbf{No}$ , while its integral closure is a proper, fine-dense subring with zero conductor that no set generates; its constant slice is the real algebraic integers, it has no finite quotients, and no nontrivial set-valued valuation of  $\mathbf{No}$  contains  $\mathbf{Oz}$  (03). For  $\gamma > 0$  and  $m \geq 2$  the equation  $X^m = \omega^\gamma + 1$  has no root in*

$\mathbf{Oz}[i]$ , yet its image in every set-sized ring has the root 1 (08). The extended ideal  $\Pi\mathcal{N}$  is radical,  $\mathcal{N}/\Pi\mathcal{N}$  is the finite-partition Boolean power  $\overline{\mathbb{Z}}[\text{Idem}]$  of the ring of all complex algebraic integers, complexification doubles it integrally, and every set-sized domain image of  $\mathcal{N}$  or  $\mathcal{N}_{\mathbb{C}}$  is  $\overline{\mathbb{Z}}$  or  $\overline{\mathbb{F}_p}$  (19). Its Boolean algebra  $\text{Idem}(\mathcal{N}/\Pi\mathcal{N})$  is atomless and a proper class, one idempotent splits every nonzero element of any set-sized subring of  $\mathcal{N}/\Pi\mathcal{N}$ , and no set-indexed family of maps to set-sized rings is jointly faithful on it; an element of  $\mathcal{N}$  supported in a set-sized group  $G$  is integral over the omnific integers supported in  $G$  (20). In a cyclic workspace  $\mathbb{Z} + \omega^a \mathbb{R}[\omega^a]$  the extended ideal of the integral closure is not radical, its fibre having nilpotents of every index, while the Boolean algebra of the reduced fibre is atomless of cardinality and order density  $2^{\aleph_0}$ , and the bounded-denominator Puiseux limit is reduced (23). For every infinite cardinal  $\kappa$  a real closed  $\mathbb{E} \subseteq \mathbf{No}$  of cardinality  $\kappa$  has an omnific integer part  $\mathbb{E} \cap \mathbf{Oz}$  whose local ring at the constant-term ideal has completion  $\mathbb{Q}$ , while its henselization has exactly  $2^\kappa$  minimal primes, all with fraction field  $\mathbb{E}[i]$  (27). (Theorems 15.5, 15.12, 15.22, 15.25, 15.29, 15.38, 15.42, 15.46, 15.57, 15.66, 15.73, 15.79, 15.86, 15.100, 15.111 and 15.120 and Corollary 15.110.)

**Theorem G** (Polynomial maps). The ring of finite-degree surreal polynomials preserving  $\mathbf{Oz}^r$  is  $\Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$  with universal set-sized image  $\text{Int}(\mathbb{Z}^r)$ ; a Newton least-common-multiple condition decides congruence preservation modulo every omnific integer; the  $p$ -adic completion is  $C(\mathbb{Z}_p^r, \mathbb{Z}_p)$  and most of its characters are not evaluations; rational self-maps of  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  are polynomial and the bijections are affine (10). For every unital  $D \subseteq k \in \{\mathbb{R}, \mathbb{C}\}$  and every polynomial  $f$  of degree at least two with coefficients in a set-sized Hahn field  $K_G$ , a fresh monomial  $t$  has  $f(D + \Pi_k) \cap (f(t) + K_G) = \{f(t)\}$ , every root of  $f(Y) = f(t) + c$  ( $c \neq 0$ ) having an explicit forbidden block of negative exponents; rational self-maps of  $D + \Pi_k$  are polynomials and the surjective ones are the affine maps  $uX + a$ ,  $u \in D^\times$ ;  $d+1$  points test degree  $d$  and no fewer do, while no set-sized sample tests all rational self-maps (16). A polynomial of degree  $d \geq 1$  preserving  $\mathbf{Oz}$  (or  $\mathbf{Oz}[i]$ ) sends at most  $d$  inputs to  $\mathbb{Z}$  (or  $\mathbb{Z}[i]$ ) unless  $f(X) = \text{ct}_*(f)(X - \eta)$  for a unique purely infinite  $\eta$ , which the two highest coefficients determine; in degree at least two at most one coset  $b + \mathbb{Z}$  (or  $b + \mathbb{Z}[i]$ ) has more than  $d$  preimages (18). For rational functions of  $m$  variables with coordinate degrees at most  $d$ , the values on an ordinary box and on  $m$  grids with one coordinate at a single fresh monomial, at most  $(d+1)^m + m(2d^2 + 2d + 1)^{m-1}$  of them, certify preservation of  $\mathbf{Oz}$  (similarly of  $\mathbf{Oz}[i]$ ); set-sized exceptional sets are removable in every number of variables;  $\mathcal{N}_m(\mathbf{Oz}) \cong \mathbf{Oz} \otimes_{\mathbb{Z}} \text{Int}(\mathbb{Z}^m)$  and  $\mathcal{N}_m(\mathbf{Oz}[i]) \cong \mathbf{Oz}[i] \otimes_{\mathbb{Z}[i]} \text{Int}(\mathbb{Z}[i]^m)$ ; and a finite subset of  $\mathbb{Z}^m$  (or  $\mathbb{Z}[i]^m$ ) tests preservation of  $\mathbf{Oz}$  (or  $\mathbf{Oz}[i]$ ) exactly when it tests  $\text{Int}(\mathbb{Z}^m)$  (or  $\text{Int}(\mathbb{Z}[i]^m)$ ) (22). Over a real closed field with a cofinal discretely ordered subring  $B$ , a globally  $C^\infty$  semialgebraic map preserving  $B^r$  has coordinates in  $\mathcal{N}_r(B)$  ( $C^{\mathbb{D}^2}$  suffices at graph degree  $D$ ), and every semialgebraic preserver of  $B$  is polynomial on its two tails; for every ordinary  $k$  a semialgebraic  $C^k$  map sends  $\mathbf{Oz}$  into  $\Pi$  and agrees with no finite family of polynomials on  $\mathbf{Oz}$ , and no set of sample points tests preservation of  $\mathbf{Oz}$  even for globally smooth quadratic algebraic branches (25). (Theorems 16.7, 16.14, 16.36, 16.43, 16.46, 16.51, 16.59, 16.61, 16.70, 16.75, 16.80, 16.82, 16.95, 16.102, 16.112 and 16.117 and Corollaries 16.20 and 16.29.)

**Theorem H** (Natural ordinal arithmetic). The Grothendieck ring of the ordinals under natural operations is not a quotient of  $\mathbf{Oz}$ ; every unital ring map from  $\mathbf{Oz}$  to it has image  $\mathbb{Z}$ . This answers negatively the quotient question in MathOverflow question 188430 of Jesse Elliott (11; independently 13). (Theorem 17.1.)

**Theorem I** (Homological and tensor algebra of omnific cores). Let  $D \subseteq K$  be  $\mathbb{Z} \subset \mathbb{R}$  or  $\mathbb{Z}[i] \subset \mathbb{C}$ . The set-sized subring  $A_d = D + \bigoplus_{0 \neq \alpha \in \mathbb{Q}_{\geq 0}^d} KX^\alpha$  of  $\mathbf{Oz}$  (or  $\mathbf{Oz}[i]$ ), whose exponents form a coordinatewise cone of rank  $d$ , has  $\text{Tor}_{>0}^{A_d}(D, D) = 0$  but  $\text{fd}_{A_d} D = d$ , with a complete multigraded computation



of  $\mathrm{Tor}^{A_d}(D, A_d/(X_i : i \in F))$ , and its finite quotients are those of  $D$  (12). Enlarging the cone to all positive exponents of the same ordered group is a nonflat extension over which  $\mathrm{fd} D = 1$  and  $\mathrm{pd} D = 2$ , as for the totally ordered cones of [Section 12](#) (12, with the merge). Over  $A_d$  itself  $\mathrm{pd}_{A_d} D = d + 1$ , one more than the flat dimension, with an explicit free telescope resolution and an embedding  $D^{\mathbb{N}}/D^{(\mathbb{N})} \hookrightarrow \mathrm{Ext}_{A_d}^{d+1}(D, A_d)$  (21). For a coefficient lattice  $M \subset K$  of rank  $r \geq 2$ , the module  $L(M) = M + \Pi$  over  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  (or over the set-sized model  $A_1$ ) has first syzygy  $\Pi^{r-1}$ , tensor square  $\Pi \oplus (M \otimes M)$  whose torsion is  $\ker(M \otimes M \rightarrow M^2)$ , nonlinear Rees equations exactly the ordinary relations among the generators (the first one in degree  $d$  for  $(\omega, 2^{1/d}\omega)$ ), isomorphism type determined by  $M$  up to a scalar, ideal classes realizing  $\mathrm{Pic}$  of an order, and dual  $\Pi$  (14). Over  $\mathbf{Oz}$  such a module has no presentation by a set of generators and a set of relations; over the lexicographic model  $R_{\Gamma_{\kappa, < \kappa}}^{11} \subset \mathbf{Oz}$  every finite free presentation of it needs exactly  $\kappa$  relations, for every uncountable regular  $\kappa$  (17) and for  $\kappa = \aleph_0$  (merge); there, for  $J = X^g L(M)$  and  $J' = X^h L(N)$ ,  $\mathrm{Tor}_1(J, A/J') \cong \mathrm{Tor}_2(A/J, A/J') \cong \mathcal{T}(M, N) = \ker(M \otimes_D N \rightarrow K)$ , and the ideals  $(\omega, \alpha\omega)$ ,  $\alpha \notin \mathrm{Frac}(D)$ , form  $2^{\aleph_0}$  isomorphism classes, the orbits of  $\mathrm{GL}_2(D)$  acting by fractional linear maps (17). Over  $A_d$  the projective dimensions of  $L(M)$  and  $A_d/J_g(M)$  are  $d + 1$  and  $d + 2$ ; over  $T_{\mathbb{Q}}$  and over the class ring read in two universes, the second  $\mathrm{Tor}$  of the quotients by  $J_g(\mathfrak{o}_1)$  and  $J_h(\mathfrak{o}_2)$  for two orders vanishes exactly when their fields are linearly disjoint, the self- $\mathrm{Tor}$  product is the antisymmetrized boundary tensor, and the framed second  $\mathrm{Tor}$  of an order recovers its multiplication (21). Over the core of any pointed rational polyhedral cone the quotient keeping a face  $F$  has flat dimension the largest dimension of a face disjoint from  $F$ ; hence the cone is simplicial exactly when every facet quotient has flat dimension one (26). ([Theorems 13.7, 13.9, 13.11, 13.14, 13.18, 13.21, 13.24, 13.25, 13.27, 13.29, 13.31, 13.34, 13.38, 13.48, 13.53, 13.54, 13.56, 13.75](#) and [13.79](#) and [Corollaries 13.8](#) and [13.47](#).)

### 1.3 Sources, provenance and where the merge chose

The report is built from twenty-four manuscripts. Eight (03, 04, 06–11), dated 22 September 2026 and placed in commit `be06fc8`, were merged first; their numbers are arrival numbers in their batch. Five (12–16), dated 23 September 2026 and placed in commit `cf350b1`, are items 02–06 of the following batch; they are numbered here by the file prefixes under which their programs and data are stored in this directory. Three (17–19), also dated 23 September 2026 and placed in commit `f4c9504`, are items 01, 03 and 05 of a third batch, numbered in the same way; the other seven items of that batch are merged into the reports on omnific-preserving automorphisms and on omnific groups and lattices [\[48, 49\]](#). One (20), also dated 23 September 2026 and placed in commit `c6359e4`, is item 01 of a fourth batch, numbered in the same way; the other eight items of that batch were placed in other reports. One (21), also dated 23 September 2026 and placed in commit `66d7e55`, is item 01 of a fifth batch, numbered in the same way; the other eight items of that batch were placed in other reports. One (22), also dated 23 September 2026 and placed in commit `9d28e28`, is item 09 of a sixth batch, numbered in the same way; the other eight items of that batch were placed in other reports. One (23), also dated 23 September 2026 and placed in commit `7d04483`, is item 01 of a seventh batch, numbered in the same way; the other eight items of that batch were placed in other reports. Four (24–27), also dated 23 September 2026 and placed in commit `a7a435f`, are items 02, 04, 06 and 07 of an eighth batch, numbered by their file prefixes, which continue this directory’s local sequence; the other five items were placed in other reports. None has a reviewer, and none is shipped with this report.

|    | Manuscript (pages)                                                                     | Pin     | Contribution and placement                                                                                                                                                                                                                                               |
|----|----------------------------------------------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 03 | <i>Cardinal Visibility and Normalization of Omnific Integers</i> (25)                  | cd80e5e | field bound ( <a href="#">Proposition 10.3</a> ); model $A_\kappa^{03}$ with threshold $\kappa^{\aleph_0}$ ( <a href="#">Section 11.1</a> ); flatness, Tor; integral and complete integral closure, valuations ( <a href="#">Section 15</a> )                            |
| 04 | <i>What Set-Sized Algebra Can See of Surreal Arithmetic</i> (24)                       | cd80e5e | $D + \Pi_k$ theorem via separation and telescope; finite side ( <a href="#">Section 8</a> ); support threshold with negative side ( <a href="#">Section 9</a> ); fixed-group bounds; localizations; modules, derivations, algebras                                       |
| 06 | <i>The Universal Set-Sized Quotient of the Omnific Integers</i> (25)                   | a5c2a97 | <b>base</b> : coefficient-ring theorem for any set field, scaled-field collision lemma, direct module argument; model $A_\kappa^{06}$ ; homological package ( <a href="#">Section 12</a> ); matrix representations; polynomial tests; prime-reservoir proposition        |
| 07 | <i>Small Quotients and Large Internal Fields of the Omnific Integers</i> (27)          | 37eefca | theorem for any set field via weighted certificates; completions; compression $\mathbf{No} \hookrightarrow \mathcal{F}_a$ , transfer, binomial kernel, Cantor algebra ( <a href="#">Section 14</a> )                                                                     |
| 08 | <i>What Small Rings Can See of Omnific Integers</i> (26)                               | 37eefca | Hartogs formulation, nonunital maps; closure formula; invisible equation; class-indexed derivations; criterion; model $A_\kappa^{08}$ for every infinite $\kappa$ ; cyclic-module lemma                                                                                  |
| 09 | <i>Set-Sized Shadows of Omnific Integers</i> (20)                                      | 37eefca | direct division of $D + \Pi_k$ elements; model $A_\lambda^{09}$ , residue fields, non-arithmetic maximal ideals, $\aleph_1$ generators; Gaussian ramification                                                                                                            |
| 10 | <i>Set-Sized Shadows of Omnific Arithmetic</i> (26)                                    | a5c2a97 | theorem for any set field incl. maps on $\Pi_K$ ; quantitative bound; endomorphisms; numerical polynomials, $p$ -adic completion and characters, lattice rigidity ( <a href="#">Section 16</a> )                                                                         |
| 11 | <i>What Set-Sized Algebra Can See of the Omnific Integers</i> (20)                     | 37eefca | pairwise-difference families; localization and presentation calculus; extension criterion; coinitality theorem; residual targets; ordinal ring ( <a href="#">Section 17</a> )                                                                                            |
|    | Later manuscript (pages)                                                               | Pin     | Contribution and placement                                                                                                                                                                                                                                               |
| 12 | <i>Arithmetic Invisibility and Homological Dimension in Omnific Integer Cores</i> (21) | none    | coordinate-cone cores $A_d$ : Boolean flat resolution, multigraded Tor, $\text{fd } D = d$ ; noncoherence certificate; ordered-cone collapse to $\text{fd } D = 1$ ; projective bounds ( <a href="#">Section 13</a> ; attained by 21's <a href="#">Corollary 13.47</a> ) |
| 13 | <i>Small-Target Rigidity and Arithmetic Specialization</i> (21)                        | 71e9606 | independent proofs of the universal theorem and of the ordinal answer (credited in <a href="#">Section 5.1</a> and <a href="#">Section 17</a> ); its fiber and norm theorems belong to the Diophantine report [47]                                                       |
| 14 | <i>Arithmetic Tensors in the Omnific Integers</i> (19)                                 | 71e9606 | coefficient-lattice modules: syzygy, tensor normal form, symmetric and exterior powers, Rees equations, Hom, ideal classes, duals, matrix kernels, $\text{fd}$ over $A_0$ ( <a href="#">Section 13</a> )                                                                 |
| 15 | <i>Omnific Integers Have Only Ordinary Set-Sized Representations</i> (21)              | 71e9606 | an independent proof of the universal theorem and its consequences (credited throughout); two one-line additions ( <a href="#">Corollary 5.5</a> and <a href="#">Proposition 7.3</a> )                                                                                   |
| 16 | <i>Fresh-Scale Image Gaps and Rational Rigidity for Omnific Integers</i> (23)          | 71e9606 | image gap and forbidden layers, fresh fibers, rational collapse and surjective maps over any constant ring, optimal and Gaussian finite tests, rational probes, sampling obstructions ( <a href="#">Sections 16.6</a> and <a href="#">16.7</a> )                         |

|    | Third-batch manuscript<br>(pages)                                                              | Pin     | Contribution and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----|------------------------------------------------------------------------------------------------|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 17 | <i>Proper-Class Relations and Arithmetic Endomorphisms of Omnific Ideals</i> (19)              | 71e9606 | independent proofs of 14's matrix, syzygy, Hom, tensor and power-growth theorems and of 12's intersection (credited in <a href="#">Section 13</a> ); no set-presentation, exactly $\kappa$ relations, $\text{Tor}_1 \cong \text{Tor}_2 \cong \mathcal{T}(M, N)$ with the self-Tor range and field test, $\text{GL}_2(D)$ moduli, orders, an embedding of $R_{\Gamma, < \kappa}^{11}$ in <b>Oz</b> ( <a href="#">Section 13.9</a> )                                                                                          |
| 18 | <i>Polynomial Rigidity of Omnific Integer Lattices</i> (21)                                    | befe739 | ordinary-output dichotomy over any discretely ordered ring, canonical translation, rich targets, composition alignment, a smaller Gaussian test, matrix and jet reductions ( <a href="#">Section 16.8</a> ); independent proofs of 10's and 16's classification, tests, rational collapse and surjections (credited in <a href="#">Section 16</a> )                                                                                                                                                                         |
| 19 | <i>Normalization and Arithmetic Fibres of the Omnific Integers</i> (23)                        | befe739 | radical extended ideal, saturation, real-rooted lifting, the residue with $v^2 = -1$ , the Boolean power $\overline{\mathbb{Z}}[\text{Idem}]$ , integral doubling, domain images and kernel detection, a finite-axiom criterion ( <a href="#">Section 15.8</a> ); independent proofs of four results of 03 and of the universal theorem (credited)                                                                                                                                                                          |
|    | Fourth-batch manuscript<br>(pages)                                                             | Pin     | Contribution and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 20 | <i>Proper-Class Boolean Branching in the Normalization of the Omnific Integers</i> (24)        | 4cdeaec | monic radical certificates, support projection and exact descent, one-scale rational functions, independent quadratic branches, uniform splitting, rank extension of set-sized Hahn fibres, relative Boolean freeness, an explicit ordinal family, atomless proper-class Idem, reduced fibres of fixed Hahn fields, no jointly faithful set of small targets, $2^\kappa$ branches through prescribed data ( <a href="#">Section 15.9</a> ); independent proofs of 19's fibre theory and of the universal theorem (credited) |
|    | Fifth-batch manuscript<br>(pages)                                                              | Pin     | Contribution and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 21 | <i>Exact Homological Dimensions and Arithmetic Intersections in Omnific Integer Rings</i> (25) | 9693b28 | $\text{pd}_{A_d} D = d + 1$ and the face quotients, by a free Koszul telescope and an embedded $D^{\mathbb{N}}/D^{(\mathbb{N})}$ in the top Ext; exact projective dimensions of lattice modules over $A_d$ ; a flat DGA resolution, the first-Tor window, the self-Tor product, a linear-disjointness test, framed reconstruction of orders, the square-zero constant-term shadow ( <a href="#">Section 13.10</a> ); independent proofs of 17's second Tor, self-Tor rank and Tor against tail-killed modules (credited)    |

|    | Sixth-batch manuscript<br>(pages)                  | Pin     | Contribution and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----|----------------------------------------------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22 | <i>Finite Tests at New Surreal Scales</i> (24)     | bcac55a | single-scale certificates for rational functions of $m$ variables over $\mathbf{Oz}$ and $\mathbf{Oz}[i]$ by resultant specialization, common tests for a set of functions, removal of set-sized exceptions in $m$ variables, a coordinate-degree Gaussian grid, tensor base change, exact lifting of constant test sets, image-ideal generators, birational rigidity, fresh scales in a fixed Hahn workspace ( <a href="#">Section 16.10</a> ); independent proofs of 16's Laurent blocks, detector, finite tests, one-variable certificate, no-set-test theorem and one-variable removal of exceptions, and of 10's grid test and decomposition (credited)                                                                                                                               |
|    | Seventh-batch manuscript<br>(pages)                | Pin     | Contribution and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 23 | <i>Ramification in Omnific Normalizations</i> (25) | 343dc2c | in the cyclic workspace $\mathcal{A}(\mathbb{Z}a)$ : a nonradical extended ideal with nilpotents of every index and exact refinement thresholds, a residue criterion for $\mathcal{N}(\mathbb{Z}a)$ with $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$ , the ramification filtration, a reduced bounded-denominator Puiseux limit, Prüfer local structure with value group $\mathbb{Q}$ , an idempotent nilradical, the associated graded ring and completion, splitting by fresh ramification with atomless Idem of size and density $2^{N_0}$ and $2^{2^{N_0}}$ branches, a periodic resolution over the coefficient fibre, a Cantor fragment ( <a href="#">Section 15.10</a> ); independent proofs of the first equality of 20's rational slice and of 20's multiquadratic lemma (credited) |

|    | Eighth-batch manuscript<br>(pages)                                                           | Pin     | Contribution and placement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|----------------------------------------------------------------------------------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24 | <i>Beyond the Constant Term:<br/>Class Residue Fields of the<br/>Omnific Integers</i> (26)   | 51c1cc7 | in GBC: maximal class extensions of proper class ideals, equivalent to global choice for all class rings; characteristic-zero maximal ideals above $1 + \omega^a$ ; the exact zero-or-unit monomial cutoff, truncation to $\text{cvx}(\mathbb{Z}a)$ , Bézout identities between separated scales; $2^{\aleph_0}$ maximal ideals per scale and an $\mathbf{On} \times \mathbb{Z}_2$ family; separation by characteristic-zero maximal ideals and embeddings of their residue fields in $\mathbf{No}[i]$ , the binomial specializations not strongly summable; term-closed and strongly sum-closed primes (Section 14.9); its field core re-derives 07's (credited) |
| 25 | <i>Smooth Rigidity and<br/>Bounded-Scale Flexibility of<br/>Omnific-Preserving Maps</i> (23) | 738a390 | over a real closed field with a cofinal discrete subring: polynomial tails of semialgebraic preservers, the two-end quotient, globally $C^\infty$ semialgebraic preservers are integer-valued polynomials in every dimension, $C^{D^2}$ at graph degree $D$ ; support-field absorption, the exact square-root threshold, $C^k$ bumps and shears, real-semialgebraic surcomplex forms, a smooth quadratic defeating every set of tests, removal of set-sized exceptions, set-sized models (Section 16.11); its numerical-polynomial description is 10's (credited)                                                                                                 |
| 26 | <i>Face Topology and<br/>Arithmetic Homology in<br/>Omnific Integer Rings</i> (30)           | 7b256e0 | cores $A_{\mathcal{C}}$ over rational polyhedral cones: flat face ideals, radical monomial ideals as face antichains, flat dimension of their quotients by join homology with cap-module witnesses, the disjoint-face formula, simpliciality by facet quotients, cube cones, homological epimorphisms, arithmetic modules over a principal ideal domain, nonflatness of the passage to $\mathbf{Oz}$ (Section 13.11); its orthant cases re-derive 12's and 21's (credited)                                                                                                                                                                                        |
| 27 | <i>Maximal Henselian<br/>Branching at Omnific<br/>Augmentations</i> (24)                     | fc27d87 | a seeded real closed $\mathbb{E}$ of cardinality $\kappa$ with $\mathbb{E} \cap \mathbf{Oz}$ an integer part and idempotent constant-term ideal; henselization with $2^\kappa$ minimal primes, generic fibre $\text{LC}(X, \mathbb{E}[i])$ , forced complexification, completion $\mathbb{Q}$ , strict and Gaussian factors, projectives, finite étale algebras, $2^\kappa$ centred valuation components (Section 15.11); its splitter re-derives 20's (credited)                                                                                                                                                                                                 |

The pins are the repository commits the manuscripts inspected (cd80e5e0adc2, a5c2a97df4b5, 37eefca1a309, all of 22 September 2026, Pacific time; and 71e9606d297c, 23 September 2026, for 13–16). The commit 71e9606 precedes the placement of the first eight manuscripts, so 13–16 saw neither this report nor its sibling; 12 records no commit, only access to the repository through a GitHub connector on 23 September 2026. Of the third batch, 17 also pins 71e9606 and saw neither report, and 18 and 19 pin befe739b56f4 (23 September 2026), which follows the placement be06fc8 but precedes the writing of this report: at that commit this directory held manuscript 06 as its text and the other first-batch manuscripts only as programs and audit files, so 18 and 19 saw 06's text but not 03's normalization, 10's polynomials or the second batch. 20 pins 4cdeaec43ff1 (23 September 2026), which also precedes the writing of this report: at that commit this directory again held 06's text, so 20 saw 06's small-image theorem but neither 16's Laurent blocks nor 19; it read 19 as a saved draft (“the preceding normalization manuscript”) and does not assert that 19 was in the repository. 21 pins 9693b28c24e6 (23 September 2026),

which follows the writing of the second merge and precedes that of the third (aae58bb): at that commit this report was built from thirteen manuscripts and held 12's and 14's section, including [Proposition 13.12](#) and the paragraph after it, but not 17's [Section 13.9](#) or the material of 18–20. 22 pins bcac55ae6fd3 (23 September 2026), a commit that contains the writing of the third merge (aae58bb) but not those of the fourth and fifth: at that commit this report held 16's and 18's subsections, including [Theorems 16.51](#) and [16.52](#) and [Corollary 16.50](#) and [Question 18.14](#), but not 20's or 21's material. 22 read the repository and catalogue READMEs, the openings of the Diophantine and holonomic-rigidity reports and the fractions audit of the Diophantine report, not this report. 23 pins 343dc2c47121 (23 September 2026), which contains the writing of the fifth merge (a8f35b5) but not that of the sixth: at that commit this report held 20's and 21's material but not 22's; 23 read its normalization, fibre and branching sections. 24 pins 51c1cc702402 and 25 pins 738a39017285 (both 23 September 2026); both commits contain the writing of the sixth merge (9b463f7) but not that of the seventh (b9cae81), and their text of this report is the same (Git blob ef7aef1), so neither saw 23's [Section 15.10](#). 24 read the large-quotient section, [Corollary 14.10](#) and the README sentence on nonarithmetic class primes; 25 read selected ranges, among them the numerical polynomials, the rational-map results and 22's non-claim on semialgebraic maps, through a connector whose long responses were sometimes truncated. Manuscript 05 of the same batch, *Omnific Integers and Diophantine Rigidity*, is merged into the sibling report [47]. It also proves the unital universal theorem for **Oz** (its “set-target rigidity”), the clearing of set supports and common monomial divisors, and it asks the “exact size boundary” question that 03, 06, 08, 09 and 11 answer; 03 credits the first three to 05 and 04 credits its finite-quotient background to manuscript 02 of the same sibling report. Those credits are kept.

### Where the merge chose.

- *Base.* 06 is the base: its coefficient-ring theorem over any set field (also proved by 07 and 10) is the most general statement, and its scaled-field collision lemma (with 03's field bound, a special case) is the mechanism that yields the sharp cardinal bounds where counting monomials does not suffice (03, 06, 11 and the fixed-group examples; 08 and 09 reach their thresholds by counting monomials). The proof is written in 08's choice-free Hartogs form, with 08's and 10's nonunital forms and 06's direct module argument. 04's explicit telescope, 09's and 10's direct division, 11's pairwise-difference families and 07's weighted certificates are kept as second routes ([Section 5.1](#)).
- *Support threshold.* Printed once, in 11's formulation (every infinite support bound, with the extension criterion) together with 04's negative side ([Theorem 9.1](#)).
- *Thresholds.* Organized by one theorem ([Theorem 10.10](#)) with the five constructions of [Section 11](#); generator counts are listed per construction and no general law is stated.
- *Homological algebra.* 06's package is proved under three hypotheses that all five constructions satisfy ([Theorem 12.2](#)).
- *Foundations.* NBG with choice for sets; global choice, stated by 04, 06, 09 (and 05) and by 17, 19 and 20, is used only for the class ultrafilters of 19 and 20, and the results depending on them say so ([Section 2.1](#) and [Remark 15.75](#)). (Status (batch 34): 24 adds an essential use, maximal class ideals ([Theorem 14.24](#)), whose unrestricted form is equivalent to global choice ([Theorem 14.30](#)); every result of [Section 14.9](#) built on it states GBC. 25 reads its class-sized support fields through set-sized ones and uses no global choice.)
- *Later manuscripts.* 13 and 15 reprove the universal theorem and its consequences; they



are credited where each result is printed, and only two one-line additions of 15 are printed ([Corollary 5.5](#) and [Proposition 7.3](#)); 13's answer to Elliott's question is credited in [Section 17](#). 12 and 14 form one new section ([Section 13](#)), placed after the homological package it is to be compared with; the ring  $A_0 = A_1$  and the two-generator syzygy are printed once, the syzygy in 14's rank generality over 12's rings. 16's results sit with 10's polynomial maps; where they overlap (rational collapse, affine surjections, transfer, grids) the general- $D$  statement is printed once with 16's fresh-scale proof and 10's statements keep their finite-difference and lattice proofs as a second route. 14's criterion for unimodular projective directions and 13's fiber and norm theorems are printed in the Diophantine report [\[47\]](#), not here.

- *Third batch.* Each of 17–19 joins the section it extends, with its new results in one new subsection. About half of 17 re-derives 14 (matrix kernels, lattice syzygy, Hom and isomorphism types, orders, tensor torsion, power growth, flat dimensions) and 12 (the intersection  $X^g A \cap cX^g A$ ); these are printed once, under 14's and 12's labels with 17 added as a proving source, and 17's new statements are in [Section 13.9](#). 18's description of  $\mathcal{N}_r$ , its tests, its rational collapse and its classification of surjections repeat 10 and 16 and are credited there; its dichotomy package is [Section 16.8](#). 19 re-derives 03's constant slices, the absence of finite quotients, the complexification defect (with a different witness) and the universal theorem (with 04's telescope); these are credited, and its fibre theory is [Section 15.8](#). Genuinely different proofs are kept as short second routes.
- *Fourth batch.* 20 answers 19's question on the Boolean algebra and joins 19's section in one new subsection ([Section 15.9](#)), placed after 19's. About a third of 20 re-derives 19 (contraction, radicality, saturation, the residue  $\mathbf{v}$ , the coefficient copy, spectral idempotents and the Boolean power, doubling and  $w_{\pm}$ , factorization of set-sized images, domain images, class ultrafilters and detection) with the same constructions; these are printed once, under 19's labels with 20 added as a proving source, and 20's proof of the universal theorem for  $\mathbf{Oz}$  is credited at [Theorem 5.1](#). 20's dominant-scale lemma is 16's Laurent-block theorem, which 20 could not see; it is printed as 16's with 20's surjectivity step added ([Lemma 15.62](#)). 20's class-ultrafilter lemma, which starts from a set with the finite intersection property, is printed beside 19's ([Lemma 15.78](#)).
- *Fifth batch.* 21 answers the first clause of 12's and 14's question [Question 18.12](#) and joins their section in one new subsection ([Section 13.10](#)), placed at the end of [Section 13](#) so that no earlier statement is renumbered; the bounds it sharpens ([Proposition 13.12](#)) carry pointers to it. Its second part re-derives 17's second-Tor identification with its rank, the self-Tor rank, the pair  $\mathbb{Z}[\sqrt{2}]$ ,  $\mathbb{Z}[\sqrt{3}]$  and the Tor groups against modules killed by the tail, which 21 could not see; these are printed once, under 17's labels with 21 added as a proving source ([Theorems 13.34](#) and [13.42](#) and [Corollary 13.36](#)), and 21's extension of them to the class ring in a two-universe reading is stated in [Section 13.10](#). 21's flat face resolution, flat dimensions and coefficient syzygy repeat 12's and 14's ([Theorems 13.4](#), [13.9](#) and [13.11](#) and [Corollary 13.8](#)), as 21 says, and are cited, not reprinted. Its three questions are merged into [Questions 18.12](#), [18.13](#) and [18.15](#).
- *Sixth batch.* 22 joins [Section 16](#) in one new subsection ([Section 16.10](#)), placed after [Section 16.9](#) at the end of the section so that no earlier statement is renumbered. About half of 22 re-derives 10's and 16's results, which were in this report at 22's pin but which 22 did not read; these are printed once, under their labels with 22 added as a proving source ([Lemmas 16.5](#) and [16.30](#) and [Theorems 16.6](#), [16.7](#), [16.9](#), [16.31](#), [16.46](#) and [16.51](#)), and its one-variable certificate, detector, multivariate collapse and one-variable removal of exceptions are credited at [Corollary 16.50](#) and [Theorems 16.33](#), [16.49](#) and [16.52](#). 22's witness for the no-set-test theorem is kept as a second witness ([Remark 16.73](#)), and its multivariate certificate is kept beside 16's smaller probe

certificate and compared with it (Remark 16.71). Its question on Gaussian constant tests is merged into Question 18.14, and its other questions form Questions 18.19 and 18.20.

- *Seventh batch.* 23 decides for cyclic exponent groups the fixed-workspace clause of Question 18.17(b), which 20 left open, and joins Section 15 in one new subsection (Section 15.10), placed after 20's at the end of the section so that no earlier statement is renumbered. 20's multi-quadratic lemma, which 23 reproves, is printed once under 20's label with 23 added as a proving source (Lemma 15.69); the first equality of 20's rational slice (Theorem 15.60) is contained in 23's Gaussian and fixed-ideal corollary (Corollary 15.94), and 23 credits it. The independent family of size  $2^{\aleph_0}$  in the reduced cyclic fibre is 20's (Theorem 15.70 with  $G = \{0\}$ ), which 23 did not cite; 23's own family is kept. Its Puiseux radicality is kept with 19's finite-axiom theorem as a second route (Remark 15.102). Two of its questions are merged into Question 18.17(b), (c), and its other seven form Questions 18.21 to 18.27.
- *Eighth batch.* 24 answers, with global choice, the existence clause of Question 18.4 and joins Section 14 in one new subsection (Section 14.9), placed after Section 14.8 at the end of the section so that no earlier statement is renumbered; 25 answers the first clause of Question 18.20(a) for globally smooth semialgebraic maps and joins Section 16 in one new subsection (Section 16.11), placed after 22's at the end of that section. 24's field core (its Section 7) and its dyadic lemma re-derive 07's results in a weaker form and are printed once, under 07's labels (Proposition 14.1, with 24 added as a proving source, and Lemma 14.14, Corollaries 14.10 and 14.17, and Theorem 14.15, credited), 24's degree proof being kept as a second route (Remark 14.45); its class back-and-forth is credited to the surcomplex-automorphism and first- $\kappa$  reports. 25's numerical-polynomial descriptions are 10's and are credited, not reprinted (Theorems 16.6 and 16.7, and Theorem 16.9 for  $r = 1$ ). The two sources state the same scale threshold independently, 24 as  $b \gg a$  and 25 as  $b > na$  for all  $n$ ; it is written  $a \ll b$  throughout, as in (2.4), and 24's  $C_a$  and 25's  $H_a$ , one and the same group, are written  $\text{cvx}(\mathbb{Z}a)$  (Remark 16.111). 24's Question 12.6 is merged into Question 18.5, 25's Questions 12.7 and 12.9 into Question 18.20(a), (b); their other questions form Questions 18.28 to 18.34 and Questions 18.35 to 18.44. 26 answers, for flat dimensions of radical monomial quotients, the non-simplicial-cone clause of Question 18.12 and joins Section 13 in one new subsection (Section 13.11), placed after 21's at the end of the section. Its orthant cases re-derive 12's face ideals, resolution and arithmetic Tor vanishing and 21's face quotients (a rational simplicial core is isomorphic to  $A_d$ , a merge observation); they are credited, and its nonflatness certificate is kept as a second route to 12's. Its questions on projective and global dimensions are merged into Question 18.12, and its other ten form Questions 18.45 to 18.54. 27 answers no numbered question; it joins Section 15 in one new subsection (Section 15.11), after 23's at the end of the section. About a third of it reconstructs 20's splitting machinery, printed once under 20's labels (Lemmas 15.53, 15.55 and 15.62 and Theorems 15.65 and 15.72), with 27's quadratic proof of the splitter kept as a second route (Remark 15.122); its twelve questions form Questions 18.55 to 18.66.

**Results added in the merge.** Each carries the tag [merge] and a complete proof: the field bound for an arbitrary subfield of a fixed-group Hahn field (Proposition 10.3); the exact thresholds  $2^{\aleph_0}$  in 04's example and in the rational-coefficient example (Examples 10.4 and 10.5), which answer 04's question for its own example and show that 10's bound need not be attained; the criterion (b') in Theorem 10.7; the exact value  $\kappa^{<\kappa}$  of the sizes and thresholds of  $A_\kappa^{06}$  and of 11's example, in ZFC (Lemma 11.10, Theorem 11.11, and Proposition 11.23), which answers the first half of 06's question on cardinal arithmetic (08 answers the second half); the organizing theorem itself (Theorem 10.10)



and the reconciliation of the realized cardinals (Remark 10.11); the homological package and the Ext thresholds for all five constructions (Theorems 12.2 and 12.3); the negative-side extension criterion (Theorem 9.1(iv)); the embedding  $\mathbf{No} \hookrightarrow \text{Frac}(\mathbf{Oz}/P)$  for primes not containing  $\Pi$  (Remark 14.13); and the explicit elements of the normalization provided by 08's equation (Remark 15.27). From the later manuscripts: the syzygy, noncoherence and nonflatness theorems in the common generality of 12's and 14's rings (Theorem 13.9 and Proposition 13.10); the flat dimensions of lattice modules and cyclic quotients over every core, which removes a superfluous hypothesis of 12 (Theorem 13.11); exact projective dimensions where 12 gives bounds, and  $\text{pd } D = 2$  for every ring of Section 12 with countably generated tail (Proposition 13.12); and the comparison of the two kinds of flat-dimension statements (Remark 13.16). From the third batch: 17's relation count, flat dimensions and Tor formula for  $\kappa = \aleph_0$ , with  $\text{pd } \Pi = 1$  and  $\text{pd } L(M) = 2$  there (Remark 13.32); 17's Tor formula over 12's cores and 14's  $A_0$ , which gives  $\text{rank Tor}_2(A/H, A/H) = 4 - \text{rank}(M^2)$  for 12's cyclic modules without the hypothesis  $c^2 \in D$  (Theorem 13.34 and Corollary 13.35); the comparison of 18's and 16's Gaussian tests (Proposition 16.64); and the explicit candidates for idempotents of  $\mathcal{N}/\Pi\mathcal{N}$  and the domain-image form of 08's invisible roots (Remarks 15.47 and 15.52). From the fourth batch: the descent  $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$  asked by 03, with its radical and Gaussian forms, from 20's support projection (Corollary 15.59); the splitting by 03's unit, which decides both candidates of Remark 15.52 (Proposition 15.68); the observation that 20's splitting results need no global choice (Remark 15.75); and the remark that  $\mathbf{B}_\mathbb{R}$  is residually set-sized although no set of small targets is jointly faithful (after Theorem 15.79). From the fifth batch: 21's projective dimensions of lattice modules over  $A_d$  for every unital  $D \subseteq K$  (21 assumes a principal ideal domain only to have a basis; Theorem 13.48); and the observation that 21's cores  $A_d$  satisfy every hypothesis of Proposition 13.12(v) except (H2) and have  $\text{pd } D = d + 1$ , so (H2) cannot be dropped there (Remark 13.49). From the sixth batch: the comparison of 22's single-scale certificate with 16's probe certificate (Remark 16.71); the conclusion of 16's no-set-test theorem for  $\mathcal{A}_{D,k}$  with every unital  $D \neq k$  (Remark 16.88); and the observation that where 22's criterion gives no fresh exponent, 18's discrete-gap collapse still holds (after Proposition 16.86). From the seventh batch: the comparison with the full class, that  $\Pi\mathcal{N} \cap K_{\mathbb{Z}a}$  is the radical of 23's extended ideal, so that the map of the cyclic arithmetic fibre to  $\mathcal{N}/\Pi\mathcal{N}$  kills exactly its nilradical and the radical in Theorem 15.57 cannot be removed (Remark 15.119), with the explicit  $\sqrt{\omega^{2a} + \omega^a} = \omega^{a/2}\sqrt{1 + \omega^a} \in \Pi\mathcal{N}$  (Example 15.87); the second route to 23's Puiseux radicality through 19's finite-axiom theorem, with the Boolean-power form, doubling and absolute integral closedness of the Puiseux fibre and the atomlessness of its Boolean algebra (Remark 15.102); the Boolean-power form of the cyclic reduced fibre, with an atomless parameter of size and density  $2^{\aleph_0}$ , and for the real algebraic fragment the ring of locally constant  $\mathbb{Z}$ -valued functions on Cantor space (Remark 15.114); and 19's  $v_T$  as a witness inside the cyclic workspace (Remark 15.104). From the eighth batch: 19's and 20's class-ultrafilter lemmas as the Boolean-ring case of 24's maximal class extensions, for class families as well as sets (Remark 14.28); a second route to 24's monomial cutoff through 07's localization and the support description of  $\mathbf{Oz}[\omega^{-a}]$  (Remark 14.38); a copy of  $\mathbf{No}[i]$ , not only an algebraically closed Hahn field, in every residue field above  $1 + \omega^a$  (Corollary 14.44); the identification of 24's first idempotents with 07's (Remark 14.45); the extension of 07's point evaluations on the Cantor subalgebra to homomorphisms of all of  $\mathcal{Q}_a$  onto fields containing  $\mathcal{E}_a$  (Remark 14.47); the observation that the embedded  $\mathcal{F}_a$  alone does not distinguish the maximal ideals above one binomial (status of Question 18.4); the containment of Lemma 4.2(ii), Lemma 14.6 and 16's freshness in 25's absorption theorem (after Theorem 16.105); the equivalence of 24's and 25's thresholds with absorption of  $K_{\text{cvx}(\mathbb{Z}a)}$  and with  $\omega^b/(1 + \omega^a) \in \mathbf{Oz}$  (Remark 16.111);  $C^k$  examples showing that 25's removal of exceptions needs global smoothness (Remark 16.120); the isomorphism of a rational simplicial cone core with  $A_d$ , through which 26's simplicial face dimensions are 21's and the projective dimension

of every simplicial face quotient is the upper endpoint of 26's interval (Section 13.11); the flat dimensions  $d$  and  $d + 1$  of rank-at-least-two lattice modules and their quotients over every cone core, hence weak global dimension at least  $d + 1$  when  $K \neq \text{Frac } D$  (Remark 13.86); the identification of 27's  $\sqrt{\Pi\mathcal{N}}$  with  $\Pi\mathcal{N}$  (Section 15.11); and the comparison of the branch counts of 20, 23 and 27 (Remark 15.143). Each follows from lemmas of the sources; none is a new method.

## 1.4 Relation to the repository

The foundations report defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  (found:eq:omnific in [43]), and omnific integers appear as kernels and integer parts in the trigonometry, gamma-zeta, analysis and euclidean-three-space reports; the sibling report [47] treats their Diophantine geometry. The euclidean-three-space report proves the group analogue of the finite-side theorem for  $\text{SO}(3, \mathbf{No})$  (Section 8). The first-kappa report proves closedness of bounded-support Hahn fields, which contains the closure step of 06's construction (06 proves that step itself; Section 11.2). Its Part II (batch 32) treats the class fields of surreals with fewer than  $\kappa$  terms, whose omnific parts are the bounded rings of Theorem 9.1 (i) for  $D = \mathbb{Z}$ ,  $k = \mathbb{R}$  and  $\lambda = \kappa$ ; see the notes in and after the proof of that theorem. The  $\kappa$ -bounded coefficient rings  $\mathbb{Z} + K((G^{<0}))_\kappa$  of L'Innocente-Mantova [23] are the published precedent for the bounded-support rings of 06 and 11. The Hahn-evaluation report [53], cited by 12, separates finite polynomial evaluation at  $\omega$  from evaluation of unrestricted series; 12's embeddings use finite normal forms only. The repository now has actual omnific and Gaussian omnific rings and shared support, degree, unit and ideal results; their exact scope is mapped in docs/FORMALIZATION.md. The full-class universal quotient theorem and the later cardinal classifications still need their own formal proofs.

**Stale statements corrected.** Three sources report a repository code search for “omnific” with no hits (06, 08, 11; 09 reports an explicitly incomplete result). At their pins the repository already defined  $\mathbf{Oz}$  at found:eq:omnific and used it in the reports listed above; what it lacked was a report devoted to omnific arithmetic, and the collection now has two (this one and [47]). 03's companion draft and 04's “prior manuscript” were not in the repository at their pins; they are now manuscripts 05 and 02 of [47]. 07's description of the staging directory (“Gamma-zeta ZIP archives”) was true at its pin and is a snapshot statement. Of the later manuscripts, 13 and 15 (and 15's audit file) report that a repository search for “omnific” returned no matches; at their pin 71e9606 the foundations report already defined  $\mathbf{Oz}$  at found:eq:omnific. 13 and 15 call the universal theorem a proposed, respectively candidate-original, contribution, and 13 calls its answer to Elliott's question one; the same theorem and the same answer are in the eight manuscripts of the first merge, dated a day earlier (Theorems 5.1 and 17.1), which 13 and 15 could not see, so both are independent re-derivations. 16 found no integer-valued-polynomial treatment in the repository; at its pin that was accurate, and the collection now has 10's (Section 16). 14's “companion manuscripts” are manuscripts 02 and 05 of the Diophantine report [47] and 03 of this one. Of the third batch, 17 proposes as additions the constant-matrix descent criterion, the coefficient-lattice Hom and isomorphism classification and the multiplication-kernel computation of Tor, with their consequences for orders and ideal powers; at its pin 71e9606 none of these was in the repository, but the collection now prints the descent criterion, the Hom classification, the multiplication kernel of the tensor square and the power growth as 14's, and the intersection  $(\omega^\gamma) \cap (\alpha\omega^\gamma) = \omega^\gamma\Pi$  as 12's (Theorems 13.9, 13.18, 13.24 and 13.28 and Corollary 13.23), so 17 is credited there as an independent second source; its catalogue count of 49 main texts is a snapshot. 18 reports that targeted repository searches for integer-valued polynomial and rational terminology returned no matches; at its pin bfe739 manuscript 10 was present only

as its check program `code/10-omnific-arithmetic-checks.py`, and the collection now prints 10's and 16's integer-valued polynomials, finite tests and rational self-maps (Section 16). 19 reports that the catalogue lists nine omnific manuscripts awaiting integration; that was true at its pin, and the first batch has since been written into this report and its sibling; 19's "repository quotient manuscript" was 06's text, now this report. 20 (pin 4cdeaec) reports that the catalogue separates the main reports from companions awaiting integration and cites this report under 06's title and date; both were true at its pin, and 19, which 20 read only as a saved draft, has since been placed and written into this report (Section 15.8). 20 proposes support-local descent, the dominant-scale branch construction, relative freeness, the answers on atoms and size and the representation bounds as new; the identification of a fresh scale with Laurent blocks over the old Hahn field is 16's (Theorem 16.31 and Corollary 16.32), which was not in the repository at 20's pin, and the reproved antecedents are 19's, as 20 says. 21 (pin 9693b28) quotes Proposition 13.12(iii) as  $d \leq \text{pd}_{A_d} D \leq d + 1$  with the next paragraph leaving the value undetermined for  $d \geq 2$ , and (iv) as bounds for lattice modules; both were accurate at its pin, and 21 itself now settles them (Corollary 13.47 and Theorem 13.48). 21 lists as its contributions, beside the dimensions, a uniform arbitrary-scale calculation of all Tor groups of two lattice quotients, the additive second-Tor identification (which it calls a dimension-shift consequence of the report's tensor calculation), the self-Tor ranks, second Tor as a linear-disjointness test and the Tor groups against constant-term modules. At its pin the report did not yet contain 17's Section 13.9, written in aae58bb after 21's pin; the collection now prints the second-Tor identification with its rank, the self-Tor range, the pair  $\mathbb{Z}[\sqrt{2}]$ ,  $\mathbb{Z}[\sqrt{3}]$  and the Tor groups against modules killed by the tail as 17's (Theorems 13.34 and 13.42 and Corollary 13.36), and 21 is credited there as a further source. The "quadratic self-Tor example" that 21 finds in the report is 12's  $\text{Tor}_2(A/H, A/H) \cong D \oplus D$  (Theorem 13.11). 22 (pin bcac55a) states, in its introduction and in its audit (22-finite-tests-SOURCE\_AUDIT.md, its Section 4), that the precise statements of its proposed package were not located in the selected repository material it inspected. That material did not include this report, which at 22's pin already contained the Laurent-block detector at one fresh scale (Theorems 16.31 and 16.33), workspace-uniform rational certificates with  $d + 2$  values in one variable and fresh probes in several (Corollary 16.50), a rational function of numerator degree 0 and denominator degree 1 with no pole on  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  that deceives any set (Theorem 16.51; 16's  $\omega^\beta/(X - \frac{1}{2})$  is  $2\omega^\beta/(2X - 1)$ , with coefficients in  $\mathbf{Oz}$ ), the empty-or-proper-class alternative in one variable (Theorem 16.52) and Gaussian tests with a one-variable lifting of ordinary test sets (Propositions 16.47 and 16.64); so its novelty list is incomplete, and what is new in it is listed in Section 16.10. 23 (pin 343dc2c) read this report's normalization, fibre and branching sections and states accurately that Question 18.17(b) leaves fixed-workspace radicality and atomlessness open and that Theorem 15.57 keeps the radical; it credits Theorem 15.60. It presents its independent family of continuum size and the cardinality of the fibre as new. At its pin this report already stated, after Theorem 15.70, that one Laurent scale carries continuum many independent idempotents over  $\overline{\mathbb{Z}}_{\mathbb{R}}$ , which with  $G = \{0\}$  is an independent family of size  $2^{\aleph_0}$  in  $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$ ; so the lower-bound half of 23's cardinality statement for the reduced fibre is 20's. Its Puiseux radicality also follows from 19's Theorem 15.51, which was in the report at its pin. What is new in 23 is listed in Section 15.10. 24 (pin 51c1cc7) states accurately that this report refrains from deriving maximal class ideals by an unqualified Zorn argument and that the README does not claim nonarithmetic class primes; it credits the field core and the Cantor-type subalgebra to this report and presents their promotion to maximal residue fields as its point. Its Section 7 is weaker than the report at its pin, which already embedded  $\mathbf{No}$  in  $\mathcal{F}_a$  and  $\mathbf{No}[i]$  in  $\mathcal{E}_a$  (Theorem 14.4 and Corollary 14.17); its class back-and-forth is in the surcomplex-automorphism and first- $\kappa$  reports, which it does not cite; and its recursion is the mechanism of 19's and 20's class ultrafilters. 25 (pin 738a390) states accurately that 22's non-claim leaves algebraic and semialgebraic maps unclassified; its tail theorem generalizes 18's

collapse (Theorem 16.66), which it credits only as context, and its no-set-test and exception theorems refine 16's and 22's, which it credits in general terms (Section 16.11). 26 (pin 7b256e0) read this report's opening and 12's orthant results and credits them; it presents its arbitrary-cone package as its contribution and credits 21's exact projective dimension without using it. Its orthant cases re-derive 12's and 21's results, and it does not notice that a rational simplicial core is isomorphic to  $A_d$ , so that its simplicial face dimensions are 21's (Section 13.11). 27 (pin fc27d87) reconstructs 20's splitter from 20's shipped proof audit and credits it; it does not cite 08's  $A_\kappa^{08} = F_\kappa^{08} \cap \mathbf{Oz}$  or the Diophantine report's local ring  $\mathbf{Oz}_\Pi$  and its example  $T^2 - T - \omega$ , and its source file reports that a repository search for "hensel" found nothing, whereas the word occurs in sixteen reports at its pin, including this one and the Diophantine report (Section 15.11). The shipped audit files keep these statements verbatim; the README records the corrections.

**Corrections to the sources.** 03's source lost a relation symbol and prints " $C \in B[i]$ " for  $C \neq B[i]$  (Proposition 15.18). 09 attributes the separated-scale division to Proposition 8.2.1 of [23]; the displayed separated-scale factorization is Fact 8.0.1 (Gonshor's Theorem 8.6) with its example on p. 46, as 11 cites, while Proposition 8.2.1 is the truncation criterion for divisibility cited by 08 (Section 4.2). For infinite  $\lambda$ , 09's hypothesis  $\lambda \geq 2^{\aleph_0}$  is implied by  $\lambda^{\aleph_0} = \lambda$  (Section 11.4). 04's remark on its  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  example is true but misleading (Remark 10.6). 06's remark that  $|A_\kappa^{06}/xA_\kappa^{06}| \geq \kappa$  "under the cardinal-arithmetic hypothesis" needs no hypothesis and in fact gives  $\kappa^{<\kappa}$  (Section 12). 06, 04 and 09 state global choice, which is never used (Section 2.1). 12 assumes  $c^2 \in D$  for its flat dimensions  $d+1$  and 2 of cyclic modules, but needs it only for its  $\text{Tor}_2$  formula (Theorem 13.11), and gives only bounds for projective dimensions that are exact in rank one (Proposition 13.12; 21 has since determined them for every  $d$ , Corollary 13.47 and Theorem 13.48). 14 credits the Diophantine report's example  $[1 : \sqrt{2}]$  but not its rationality theorem for primitive directions, of which 14's criterion is the Gaussian and principal-ideal extension (Section 13.8). 16's rational collapse has the D-ring precedent that 16 itself credits (Theorem 16.42). 17 assumes  $\kappa$  uncountable for its exact relation count, but the proof uses only regularity, so  $\kappa = \aleph_0$  is allowed (Remark 13.32). 19 did not see 03's text and reproves four of its results without credit; they are credited here (Section 15.8). The real analogue of 18's question on minimal tests with omnific nodes is answered by 16's optimality theorem; the Gaussian question itself stays open (Question 18.16). 20's dominant-scale lemma repeats 16's Laurent blocks without credit (it could not see them); they are credited (Lemma 15.62). 21 repeats 17's Tor results without credit (it could not see them); they are credited (Section 13.10); its proofs contain no error found in this merge. 22 repeats 10's and 16's results without credit (it did not read this report); they are credited (Section 16.10), and its proofs contain no error found in this merge. 23 does not cite 20's continuum family or 19's finite-axiom theorem (see above; credited in Section 15.10); its proofs contain no error found in this merge. 24 and 25 repeat 07's field core and 10's numerical polynomials with credit, and 18's collapse, 16's and 22's sampling and exception theorems without precise citation (credited in Sections 14.9 and 16.11); no error was found in their proofs. 26 and 27 repeat 12's, 21's and 20's results with general credit (Sections 13.11 and 15.11); 27's statement that a repository search for "hensel" returned no match is inaccurate at its pin, and no error was found in their proofs. No source contains a false theorem.

## 2 Conventions, notation, and the constant-term retraction

### 2.1 Foundations

We work in von Neumann–Bernays–Gödel class theory (NBG) with the axiom of choice for sets. Every surreal number, every normal-form support and every coefficient field or target ring called *set-sized* is a set; the carriers  $\mathbf{No}$ ,  $\mathbf{Oz}$ ,  $\mathbf{No}[i]$  and most rings of Sections 3 and 5 are proper classes. A homomorphism with a class source is a class function obeying the usual identities, and its restriction to a set is a set.

The sources state different foundations. Manuscript 06 works in Gödel–Bernays set theory with global choice, 09 in GBC, and 04 in “GBC, with ordinary choice for sets” (GBC already contains global choice); 07, 08 and 11 work in NBG with choice for sets, 10 in NBG with choice, and 03 in NBG for the classes and ZFC for its set-sized constructions; 07 remarks that global choice is unnecessary. *No argument of the first two batches uses global choice.* Of the third batch, 17 and 19 state Gödel–Bernays class theory with global choice; 17’s uses of it (choosing preimages, a Schanuel splitting) are avoided below, and 19’s one essential use, a class ultrafilter on a Boolean algebra that may be a proper class, is stated with its hypothesis (Lemma 15.48); only the results built on it (Theorem 15.49 and Corollary 15.50) need global choice. 20 states the same foundation; its splitting results need only choice for sets (Remark 15.75), and its extension of branch maps (Theorem 15.80) uses class ultrafilters as 19’s detection does. 23 uses only set-sized rings and choice for sets. 24 works in Gödel–Bernays class theory with global choice and uses it essentially: its maximal class ideals (Theorem 14.24) come from a recursion with set-valued stages along a global well-order, and for all class rings the principle is equivalent to global choice (Theorem 14.30); the results of Section 14.9 that need it say so, while its monomial cutoff, truncation and support-prime classification do not. 25 reads proper-class support fields pointwise, through set-sized divisible hulls, and uses no global choice. 26 uses only set-sized rings and modules, the full class ring appearing only as the target of an embedding and of a finite relation; 27 uses set-sized fields, rings and spectra, set-length recursion and choice for sets, with no class Zorn lemma. The collision step of the universal theorem (Theorem 5.1) is choice-free in the formulation of 08, which we adopt: for a set  $S$ , its Hartogs ordinal is an ordinal that does not inject into  $S$ ; it exists in ZF. The same device is used in the rotation-group theorem `e3:cut:thm:smallquotient` of [44]. Choice for sets enters only where it is visibly needed: maximal ideals of set-sized rings (Section 11), cardinal arithmetic such as  $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |\{[\kappa]^{\leq \aleph_0}\}|)$ , and the existence of residue fields.

**Definition 2.1** (Set-sized quotients and detection). A class ideal  $J$  of a class ring  $R$  *admits a set-sized quotient* if there is a surjective class ring homomorphism  $R \rightarrow S$  onto a set-sized ring with kernel  $J$ . A ring homomorphism  $\varphi$  *detects*  $x$  if  $\varphi(x) \neq 0$ ; a module  $M$  *detects*  $x$  if  $xM \neq 0$ .

This is the convention of 04 (“set-realizable quotient”), 08, 09 (“set-sized realization”), 10 (“set-sized quotient presentation”) and 11. Manuscripts 06 and 07 alternatively represent a class quotient by Scott codes (the members of a coset of least set-theoretic rank form a nonempty set); every quotient statement below may be read in either way, and no set or class whose members are arbitrary proper classes is formed. Notation such as  $\text{Hom}(\mathbf{Oz}, S)$  abbreviates a statement about individual class maps and their factorizations; after the universal theorem these maps are coded by their restrictions to a set-sized coefficient ring.

**The universe-relative reading.** Manuscripts 04, 10 and 11 add an equivalent reading: fix a lower universe containing the coefficient fields, targets and supports, and place the surreal carrier in



a higher universe; “set-sized” then means small in the lower universe. The size restriction must be carried in the statement. A set-sized fragment of surreal arithmetic viewed in a larger universe has its own identity map as a set-sized target, so it does not satisfy the full-class theorem (11 insists that its main theorem is relative to one fixed universe, not to fragments viewed in a larger one).

## 2.2 Normal forms and the orientation of exponents

A nonzero surreal has a unique Conway normal form  $x = \sum_{\gamma \in S} r_\gamma \omega^\gamma$  with nonzero real coefficients and a *reverse well-ordered set*  $S$  of exponents (every nonempty subset has a greatest element) [12, 20, 5]. We use *large monomials*:  $\omega^\gamma$  is infinitely large when  $\gamma > 0$ , and  $\omega^{\gamma+\delta} = \omega^\gamma \omega^\delta$  is Conway’s monomial law, not an exponential. For an abstract ordered abelian group  $G$  we write formal monomials  $X^g$  with the same orientation, and  $k((X^G))$  for the Hahn series over a set field  $k$  with reverse well-ordered set supports in  $G$  (for  $G = \mathbf{No}$  a proper class of series). The usual small-monomial notation is recovered by  $t^g = X^{-g}$ ; sources that write  $t$  with well-ordered supports reverse all signs. A family is strongly summable if the union of its supports is reverse well ordered and each exponent receives finitely many contributions; strong sums are coefficientwise and are *not* limits of partial sums in the fine order topology.

*Purely infinite* refers to support, not to size or sign:  $-\omega$  is purely infinite, while  $\omega - 1$  is infinite but not purely infinite. This is the convention of the foundations report, which defines  $\mathbf{Oz} = \Pi \oplus \mathbb{Z}$  with  $\Pi$  the purely infinite normal forms (`found:eq:omnific` in [43]); the gamma-zeta report uses the same symbol [52]. The trigonometry report writes the same class  $\Pi$  as the normal forms supported at *negative  $t$ -exponents* (`trigonometry:eq:split` in [45]), because it uses  $t = \omega^{-1}$ ; the two descriptions agree after the sign change  $t^g = \omega^{-g}$ .

## 2.3 The rings

Let  $k$  be a set field,  $G$  an ordered abelian group (a set, or the class  $\mathbf{No}$ ), and  $D \subseteq k$  a unital subring. Put

$$\Pi_k(G) = \{f \in k((X^G)) : \text{supp } f \subseteq G_{>0}\}, \quad \mathfrak{m}_k(G) = \{f \in k((X^G)) : \text{supp } f \subseteq G_{<0}\}, \quad (2.1)$$

$$\mathcal{A}_{D,k}(G) = D + \Pi_k(G), \quad \mathcal{B}_{D,k}(G) = D + \mathfrak{m}_k(G). \quad (2.2)$$

Both sums are direct as additive groups, and the zero series belongs to both ideals. We drop  $(G)$  when  $G = \mathbf{No}$  and write  $\Pi = \Pi_{\mathbb{R}}$ . For  $k = \mathbb{R}$  and  $k = \mathbb{C}$  the identification  $X^g \mapsto \omega^g$  gives  $\mathbb{R}((X^{\mathbf{No}})) = \mathbf{No}$  and  $\mathbb{C}((X^{\mathbf{No}})) = \mathbf{No}[i]$  (a union of two reverse well-ordered supports is reverse well ordered, so complexification is coefficientwise), and

$$\mathbf{Oz} = \mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbb{Z} \oplus \Pi, \quad \mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz} = \mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}, \quad \Pi_{\mathbb{C}} = \Pi + i\Pi. \quad (2.3)$$

We call  $\mathbf{Oz}[i]$  the *Gaussian omnific ring*; it is a subring of  $\mathbf{No}[i]$ , not the surcomplex field, and not the integral closure of  $\mathbb{Z}$  in  $\mathbf{No}[i]$  (08, 09). The finite surreal and surcomplex rings are  $\mathcal{O}_{\mathbb{R}} = \mathcal{B}_{\mathbb{R},\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}} = \mathcal{B}_{\mathbb{C},\mathbb{C}}$ , and  $\mathcal{A}_{\mathbb{R},\mathbb{R}} = \mathbb{R} + \Pi$  is the nonnegative-support ring (03’s  $P_{\mathbb{R}}$ ). For a general set field  $k$  (06, 07, 10) the class field  $k((X^{\mathbf{No}}))$  need not be realized inside the surreals; nothing below requires such a realization.

**Lemma 2.2** (Constant term [03, 04, 06, 07, 08, 09, 10, 11, 13, 14, 15, 16]). *The map  $\text{ct} : \mathcal{A}_{D,k}(G) \rightarrow D$  extracting the coefficient of  $X^0$  is a surjective unital ring homomorphism with kernel  $\Pi_k(G)$ , split by the inclusion of constants. On  $\mathcal{B}_{D,k}(G)$  the same coefficient is a split surjective ring homomorphism with kernel  $\mathfrak{m}_k(G)$ ; on  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$  it is the standard part, written  $\text{st}$ .*



*Proof.* The nonnegative support cone is closed under addition of exponents. Thus sums and Hahn products of such series still have nonnegative support; cancellation only removes exponents. A sum of two nonnegative exponents is zero only when both are zero, so the constant coefficient of a product is the product of the constant coefficients and lies in  $D$ . This also verifies that  $\mathcal{A}_{D,k}(G)$  is a subring. Additivity, the kernel and the section follow from the unique decomposition (2.2); surjectivity follows because every  $d \in D$  is a constant series. The same argument with nonpositive exponents proves all assertions for  $\mathcal{B}_{D,k}(G)$ .  $\square$

*Remark 2.3* (What `ct` is not). The map `ct` is defined on a ring of generally infinite elements; it is not the standard part, and coefficient extraction at 0 is not multiplicative on **No**:  $\omega \cdot \omega^{-1} = 1$  although both factors have zero constant coefficient. This is the warning recorded in [50] and `diff:warn:ct` of [54]. Nor is `ct` order preserving:  $\omega - N > 0$  but `ct`( $\omega - N$ ) =  $-N$  for every ordinary  $N \geq 1$  (04, 08, 09, 11). On  $\mathcal{A}_{D,k}$  the two maps `ct` and `st` receive different names because they have different domains (04), and (8.1) below shows that they cannot be combined.

**Lemma 2.4** (Degree and units [03, 04, 07, 08, 09, 10, 11, 13, 14, 16]). *For  $x \neq 0$  in  $k((X^G))$  put  $\deg_\omega(x) = \max \text{supp } x$ . Then  $\deg_\omega(xy) = \deg_\omega(x) + \deg_\omega(y)$  for  $x, y \neq 0$ , so  $\mathcal{A}_{D,k}(G)$  and  $\mathcal{B}_{D,k}(G)$  are domains, and*

$$\mathcal{A}_{D,k}(G)^\times = D^\times, \quad \mathbf{Oz}^\times = \{\pm 1\}, \quad \mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}.$$

*In particular every nonconstant element of  $\mathcal{A}_{D,k}(G)$ , such as  $1 + \omega^a$  or  $\omega^{\sqrt{2}} + \omega + 1$ , is a nonunit.*

*Proof.* The leading coefficient of a product is the product of the leading coefficients, which is nonzero because  $k$  is a field; the degree identity follows. A nonzero element of  $\mathcal{A}_{D,k}(G)$  has nonnegative degree, so  $xy = 1$  forces both degrees to be 0, i.e. both factors are constants in  $D$  with  $xy = 1$  in  $D$ . The two unit lists are the units of  $\mathbb{Z}$  and  $\mathbb{Z}[i]$ .  $\square$

Here  $\deg_\omega$  is the largest growth exponent (04); it is not the ordinal-valued degree of the order type of a support used in parts of the factorization literature [23].

## 2.4 Notation and renamings

The manuscripts use many symbols for the purely infinite ideal and several for the constant term. We use the repository's  $\Pi$  and the common `ct`, and resolve every collision by renaming in the new material. The renamings, and the tags distinguishing the different set-sized constructions, are listed below; Section 13 and Section 16.6 list the further renamings of 12, 14 and 16, Sections 13.9, 15.8 and 16.8 those of 17, 18 and 19, Section 15.9 those of 20, Section 13.10 those of 21, Section 16.10 those of 22, Section 15.10 those of 23, and Sections 13.11, 14.9, 15.11 and 16.11 those of 24 to 27.

| Object                                         | In the sources                                                                                                                                                                                                    | Here                                                                                                                                               |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| purely infinite ideal                          | $\mathcal{I}$ (03, 06, 09, 14, 21), $\mathcal{J}_k$ (04), $\mathfrak{J}$ (07, 08, 11), $\mathfrak{P}$ (10), $J_k$ (13), $\mathcal{I}_K$ (15), $\mathcal{P}_k$ (16), $\mathbf{m}$ (17), $\mathcal{J}$ (18, 19, 20) | $\Pi$ , $\Pi_k$ , $\Pi_{\mathbb{C}}$                                                                                                               |
| constant term                                  | $c_0$ (06), $\pi$ , $\pi_D$ (10), $\varepsilon_D$ (09), $\varepsilon$ (12), $\text{ct}$ (others)                                                                                                                  | $\text{ct}$                                                                                                                                        |
| coefficientwise constant term on polynomials   | $\Pi$ (10), $c_0$ (06)                                                                                                                                                                                            | $\text{ct}_*$                                                                                                                                      |
| $D + \Pi_k$                                    | $\mathcal{A}_{D,k}$ (04), $A_{D,k}$ (06), $A(D, K)$ (07), $R_D$ (08), $\mathcal{A}_D$ (09), $R(D, K)$ (10), $\mathcal{Z}_{K,D}$ (11), $A(D, k, G)$ (03), $A_D$ (13), $R_A$ (15), $\mathcal{A}_D$ (16)             | $\mathcal{A}_{D,k}$ , $\mathcal{A}_{D,k}(G)$                                                                                                       |
| Gaussian omnific ring                          | $\text{GOz}$ (06), $\mathbf{Og}$ (09)                                                                                                                                                                             | $\mathbf{Oz}[i]$                                                                                                                                   |
| surcomplex field                               | $\mathbf{No}(i)$ (04, 07, 08, 09, 12, 15, 16, 19)                                                                                                                                                                 | $\mathbf{No}[i] = \mathbf{No}[i]$ [50]                                                                                                             |
| Hahn field over $H_a$                          | $L_a$ (06), $\mathcal{F}_a$ (07), $L_h$ (03, set version)                                                                                                                                                         | $\mathcal{F}_a(k)$ ; $L_h$ for 03's set field                                                                                                      |
| localization $\mathbf{Oz}[\omega^{-a}]$        | $L_a$ (07)                                                                                                                                                                                                        | $\mathbf{Oz}[\omega^{-a}]$ (never $L_a$ )                                                                                                          |
| binomial quotient $\mathbf{Oz}/(1 + \omega^a)$ | $B_a$ (07)                                                                                                                                                                                                        | $\mathcal{Q}_a$                                                                                                                                    |
| 07's compression map $a\Psi$                   | $E_a$                                                                                                                                                                                                             | $\Psi_a$                                                                                                                                           |
| integral closure                               | $B, C, \text{Int}_F(A)$ (03), $N_{\mathbb{R}}, N_{\mathbb{C}}, \text{Int}_A(F)$ (19, 20)                                                                                                                          | $\mathcal{N}, \mathcal{N}_{\mathbb{C}}, \text{IC}_F(A)$                                                                                            |
| integer-valued polynomials                     | $\text{Int}(\mathbb{Z}^r)$ , $I_r$ (10)                                                                                                                                                                           | $\text{Int}(\mathbb{Z}^r)$                                                                                                                         |
| set-sized integer parts                        | $A_{\kappa}$ (03, 08, 17), $R_{\kappa}$ (06), $A_{\lambda,D}$ (09), $R_{\Gamma, < \kappa}$ (11)                                                                                                                   | $A_{\kappa}^{03}, A_{\kappa}^{06}, A_{\kappa}^{08}, A_{\lambda,D}^{09}, R_{\Gamma, < \kappa}^{11}$ (17's is $R_{\Gamma_{\kappa}, < \kappa}^{11}$ ) |
| their ideals                                   | $I_{\kappa}, \mathcal{J}_{\kappa}, I_{\lambda,k}$                                                                                                                                                                 | $\Pi_{\kappa}^{03}, \Pi_{\kappa}^{06}, \Pi_{\kappa}^{08}, \Pi_{\lambda}^{09}$                                                                      |
| small exponents below a support                | $\text{Tiny}(S)$ (09), $U(S)$ (10), $G_{\ll h}$ (03)                                                                                                                                                              | $U_G(S), H_h(G)$                                                                                                                                   |
| binomial series                                | $B_m(t)$ (08)                                                                                                                                                                                                     | $\mathbf{b}_m(t)$                                                                                                                                  |

Throughout, for  $a > 0$  in an ordered group  $G$ ,

$$H_a(G) = \{h \in G : n|h| < a \text{ for every ordinary } n \geq 1\}, \quad H_a = H_a(\mathbf{No}), \quad (2.4)$$

and  $h \ll a$  means  $h > 0$  and  $h \in H_a(G)$ . This is a comparison of *exponents*:  $h \ll a$  does not say that  $\omega^h$  is small (it is infinite whenever  $h > 0$ ). For a set  $S \subseteq G_{>0}$  we write  $U_G(S) = \{a > 0 : na < s \text{ for all } s \in S, n \geq 1\}$  (09's  $\text{Tiny}(S)$ , 10's  $U(S)$ ). Letters  $\kappa, \lambda, \mu, \theta$  denote cardinals and are restated locally;  $\mu = \text{cf } \kappa$  and  $\theta = \kappa^{\aleph_0}$  in the construction of 03.

### 3 Support gaps and the purely infinite ideal

This section collects the facts that use the full surreal exponent class. All eight manuscripts of the first merge prove them, as do 13 and 15 and manuscript 05 of the sibling report [47], and 14 and 16 use them; of the third batch, 17 and 19 reprove the gaps, the units and the common divisors, and 18 uses the constant term. They are background for the new results, not claimed as new. Throughout,  $k$  is a set field and  $D \subseteq k$  a unital subring.

**Lemma 3.1** (Gaps below and above a set [03, 04, 06, 07, 08, 09, 10, 11, 13, 15, 16]).

(i) Every set of surreal numbers has a strict upper and a strict lower bound in  $\mathbf{No}$ .

(ii) For every set  $S \subseteq \mathbf{No}_{>0}$  there is  $h > 0$  with  $nh < s$  for all  $s \in S$  and all ordinary  $n \geq 1$ ; in particular  $0 < h < S$ .

(iii) For such  $S$  the class  $U_{\mathbf{No}}(S)$  contains the interval  $(0, h)$ , which is not a set, and for every ordinal  $\Lambda$  the exponents

$$a_\alpha = \frac{h}{\omega^{\alpha+1}} \quad (\alpha < \Lambda) \quad (3.1)$$

are distinct elements of  $U_{\mathbf{No}}(S)$ , each  $\ll h$ .

*Proof.* For a set  $T$ , the cuts  $\{T \mid \emptyset\}$  and  $\{\emptyset \mid T\}$  are set cuts and give (i). For (ii), apply the set cut  $\{0 \mid T\}$  to the set  $T = \{s/n : s \in S, n \geq 1\}$  of positive surreals; for  $S = \emptyset$  take  $h = 1$ . For (iii), every  $0 < u < h$  satisfies  $nu < nh < s$ , so  $(0, h) \subseteq U_{\mathbf{No}}(S)$ . If this interval were a set, (ii) would give a positive strict lower bound  $\ell$ ; since  $h/2$  is in the interval,  $0 < \ell/2 < h$ , contradicting that lower bound. The denominators  $\omega^{\alpha+1}$  are the Conway monomials of the ordinals  $\alpha + 1$  in the *exponent* field; they strictly increase with  $\alpha$  and exceed every ordinary integer, so the  $a_\alpha$  are distinct, positive and satisfy  $na_\alpha < h$ .  $\square$

Only the existence of a separator is used, not its simplicity or birthday (08). The sources use several equivalent families in place of (3.1):  $a/\omega^{\alpha+1}$  (04, 05, 06, 08, 09, 03),  $a/(\omega + \alpha + 1)$  with the field sum of embedded ordinals (07),  $\delta\omega^{-(\alpha+1)}$  (11),  $\epsilon\xi/\kappa$  (10),  $\epsilon/(\bar{\alpha} + 2)$  with  $\bar{\alpha}$  the surreal ordinal (13), and  $g\omega^{-(\xi+1)}$  (15). Each is a set whenever  $\alpha$  ranges over a set.

**Example 3.2** (Surreal gaps where real gaps fail [03, 06, 08, 10, 11]). The series  $f = \sum_{n \geq 1} \omega^{1/n}$  lies in  $\Pi$ ; its support has no positive *real* lower bound but has positive surreal lower bounds, for instance  $1/\omega$ . With  $a = 2/\omega$  and  $b = 1/\omega$  one has  $na < 1/m$  for all ordinary  $m, n$  (10). In the fixed exponent group  $\mathbb{R}$  the same series is purely infinite but is not divisible by any  $X^h$ ,  $h > 0$ , inside the nonnegative-support ring (03); this is where the passage from monomials to the whole ideal fails in a fixed group (Section 10).

**Proposition 3.3** (Common monomial divisors [03, 04, 06, 07, 08, 09, 10, 11, 13, 14, 15]). *Let  $A = \mathcal{A}_{D,k}$  and  $\Pi_k$  its purely infinite ideal.*

- (i) *Every set-sized family  $\mathcal{F} \subseteq \Pi_k$  satisfies  $\mathcal{F} \subseteq X^a \Pi_k$  for some  $a > 0$ .*
- (ii) *Each  $f \in \Pi_k$  is a single product of two elements of  $\Pi_k$ ; hence  $\Pi_k^2 = \Pi_k$ , and  $\Pi_k = \bigcup_{a>0} X^a A$  is a directed union of principal ideals.*
- (iii)  *$\Pi_k$  is not generated, as an ideal of  $A$ , by any set of its elements.*
- (iv)  *$d\Pi_k = \Pi_k$  for every  $0 \neq d \in D$ ; in particular  $n\Pi_k = \Pi_k$  for every ordinary integer  $n$  whose image in  $k$  is nonzero. In characteristic zero this includes every nonzero ordinary integer.*

*Proof.* (i) The union of the supports of the members of  $\mathcal{F}$  is a set of positive surreals; choose  $0 < a <$  that union by Lemma 3.1; for an empty union take any  $a > 0$ . Translating each support by  $-a$  leaves it positive and reverse well ordered, so  $X^{-a}f \in \Pi_k$ . (ii) Apply (i) to  $\{f\}$ :  $f = X^a(X^{-a}f)$  with both factors in  $\Pi_k$ . If  $0 < b < a$  then  $X^a A \subseteq X^b A$ . Each  $X^a A$  is contained in  $\Pi_k$ , and (i) puts every member of  $\Pi_k$  in one of these ideals; choosing  $b$  below any two positive indices proves directedness. The ideal product  $\Pi_k^2$  consists of finite sums of products; the single-product factorization proves one inclusion, and the ideal property the other. (iii) If a set generated  $\Pi_k$ , (i) would place the generated ideal inside  $X^a A$ ; but  $X^{a/2} \in \Pi_k$  is not in  $X^a A$ , since its unique quotient by  $X^a$  in the ambient field is  $X^{-a/2}$ , which has a negative exponent. (iv) Divide every coefficient by  $d$ , which is invertible in  $k$ ; this preserves the support. Multiplication by  $d$  gives the reverse inclusion.  $\square$

The qualification on  $n$  is essential when  $k$  is not assumed to have characteristic zero. If  $k$  has characteristic  $p > 0$ , then  $p\Pi_k = 0$ , whereas  $X^a \in \Pi_k$  is nonzero for  $a > 0$ . The unrestricted integer-divisibility clause in the earlier assembly was therefore false in that generality; the coefficient statement  $d\Pi_k = \Pi_k$  for  $d \neq 0$  is unchanged.

Idempotence and divisibility do not by themselves prove the universal theorem: an idempotent, rationally divisible ideal can act nontrivially on infinite set-sized modules, and a homomorphism to an infinite characteristic-zero ring need not kill an arbitrarily divisible additive subgroup (04, 06, 07, 08). When  $k$  has characteristic zero and the target has positive characteristic, the last item already suffices (Remark 5.10).

$$n\mathbf{Oz} = n\mathbb{Z} + \Pi = \text{ct}^{-1}(n\mathbb{Z}) \quad (n \geq 1), \quad d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}} \quad (0 \neq d \in \mathbb{Z}[i]). \quad (3.2)$$

Indeed  $n\Pi = \Pi$  and  $d\Pi_{\mathbb{C}} = \Pi_{\mathbb{C}}$  by Proposition 3.3(iv). Consequently

$$\Pi = \bigcap_{n \geq 1} n\mathbf{Oz} = \bigcap_{p \text{ prime}} p\mathbf{Oz}, \quad \Pi_{\mathbb{C}} = \bigcap_{n \geq 1} n\mathbf{Oz}[i], \quad (3.3)$$

since a nonzero ordinary integer (or Gaussian integer) is not divisible by every  $n$  or every prime (08, 10, 11). For a general  $D$  the intersection formula can fail: for  $D = \mathbb{Q}$  every element of  $\mathcal{A}_{\mathbb{Q}, \mathbb{R}}$  is divisible by every ordinary positive integer (11).

**Proposition 3.4** (Clearing and fraction fields [03, 04, 06, 07, 08, 09, 11, 14, 15, 16]).

- (i) For every set  $E \subseteq k((X^{\mathbf{No}}))$  there is  $h > 0$  with  $X^h E \subseteq \Pi_k$ . For  $E \subseteq \mathbf{No}$  or  $E \subseteq \mathbf{No}[i]$  this reads  $\omega^h E \subseteq \Pi$ , respectively  $\omega^h E \subseteq \Pi_{\mathbb{C}}$ .
- (ii)  $\text{Frac}(\mathcal{A}_{D,k}) = k((X^{\mathbf{No}}))$ ; in particular  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  and  $\text{Frac}(\mathbf{Oz}[i]) = \mathbf{No}[i]$ .
- (iii) Inverting the positive monomials gives the whole field: if  $\mathcal{U} = \{X^a : a > 0\} \cup \{1\}$  then  $\mathcal{U}^{-1}\mathcal{A}_{D,k} = k((X^{\mathbf{No}}))$ .
- (iv) Neither  $\mathbf{No}$  nor  $\mathbf{No}[i]$ , nor any localization of  $\mathcal{A}_{D,k}$  in which some nonzero element of  $\Pi_k$  becomes a unit, admits a unital homomorphism to a nonzero set-sized ring or a nonzero set-sized unital module.

*Proof.* (i) The union of the supports of the members of  $E$  is a set; by Lemma 3.1(i) choose  $h > 0$  larger than  $-g$  for each exponent  $g$  in it. Then every exponent of  $X^h x$ ,  $x \in E$ , is positive. (ii) and (iii) For  $x \in k((X^{\mathbf{No}}))$  apply (i) to  $\{x\}$ :  $x = (X^h x)/X^h$  with numerator and denominator in  $\mathcal{A}_{D,k}$ ; the reverse inclusions hold because  $\mathcal{A}_{D,k}$  is a subring of the field. (iv) is proved after the universal theorem (Corollary 5.9).  $\square$

Part (i) is manuscript 05's ‘‘Clearing supports’’ theorem, now in the sibling report [47]; 03 credits it to 05 and adds the algebra-wide formulation of Section 15. Clearing one element is weaker than clearing all powers of it with one denominator, which Section 15 also supplies (03).

## 4 The collision mechanism

Every proof of the universal theorem combines two finite multiplicative identities with a cardinal comparison. This section states the identities once, in the forms used by the different sources. The first (Lemma 4.1, from 06, and in a special form from 03) gives a uniform proof of the sharp cardinal bounds of Sections 10 and 11, including those that counting monomials cannot reach; the others are genuinely different routes and are kept as such.

## 4.1 Scaled-field collision

**Lemma 4.1** (Scaled-field collision [06; 03]). *Let  $A$  be a commutative subring of a field  $F$ , let  $L \subseteq F$  be a subfield and let  $0 \neq m \in F$  satisfy  $mL \subseteq A$ .*

- (i) *If  $\varphi : A \rightarrow S$  is a ring homomorphism,  $S$  any ring (not necessarily commutative, unital or reduced), and  $\varphi(m^2) \neq 0$ , then  $x \mapsto \varphi(mx)$  is injective on  $L$ .*
- (ii) *If  $M$  is a unital  $A$ -module,  $v \in M$  and  $m^2v \neq 0$ , then  $x \mapsto (mx)v$  is injective on  $L$ .*

*For set-sized  $L$  and targets, consequently, if  $|L| > |S|$  then  $\varphi(m^2) = 0$ , and if  $|L| > |M|$  then  $m^2M = 0$ .*

*Proof.* Since  $1 \in L$ ,  $m \in A$ . For distinct  $x, y \in L$  the element  $m/(x - y)$  lies in  $mL \subseteq A$ , and in  $A$

$$(mx - my) \cdot \frac{m}{x - y} = m^2. \quad (4.1)$$

If  $\varphi(mx) = \varphi(my)$ , applying  $\varphi$  gives  $\varphi(m^2) = 0$ ; no cancellation in  $S$  is used. For the module, apply  $m/(x - y)$  to  $(mx)v = (my)v$ ; commutativity of  $A$  turns the result into  $m^2v = 0$ .  $\square$

The proof of (i) uses only that  $A$  is closed under addition and multiplication and contains  $mL$ ; so it applies verbatim when  $A$  is a nonunital subring such as  $\Pi_K$ , which is how it is used in [Theorem 5.1\(iii\)](#). The module version bounds  $|M|$  itself. Deriving it through the ring  $\text{End}_{\mathbb{Z}}(M)$  would only bound  $|\text{End}_{\mathbb{Z}}(M)|$ , which can be much larger than  $|M|$  (06, 03, 09); this matters for every set-sized threshold below. In 03 the lemma appears as the identity  $(X^{h/2}(f - f'))(X^{h/2}/(f - f')) = X^h$  for  $f \neq f'$  in a smaller-scale field (03's “smaller-scale field obstruction”), i.e. the case  $m = X^{h/2}$  of (4.1).

**Lemma 4.2** (The scale subgroup and its Hahn field [03, 06, 07]). *Let  $G$  be an ordered abelian group (a set or **No**) and  $a \in G_{>0}$ .*

- (i)  *$H_a(G)$  is a convex subgroup of  $G$ , divisible if  $G$  is, and  $H_c(G) = H_{2c}(G)$  for every  $c \in G_{>0}$ .*
- (ii)  *$k((X^{H_a(G)}))$  is a subfield of  $k((X^G))$ , and  $X^c k((X^{H_a(G)})) \subseteq \Pi_k(G)$  for every  $c \in G$  with  $2c \geq a$ ; for instance  $c = a$ , or  $c = a/2$  when  $a/2 \in G$ .*
- (iii) *For  $G = \mathbf{No}$  the field  $\mathcal{F}_a(k) = k((X^{H_a}))$  is a proper class: it contains the distinct monomials  $X^{a/\omega^{\alpha+1}}$  for every ordinal  $\alpha$ .*

*Proof.* (i) If  $h, h' \in H_a(G)$  then  $2n|h| < a$  and  $2n|h'| < a$  give  $n|h + h'| < a$ ; negation and convexity are immediate, and if  $G$  is divisible then  $h/m$  satisfies the same inequalities. For every  $c > 0$ ,  $n|h| < c$  for all  $n$  is equivalent to  $n|h| < 2c$  for all  $n$ . (ii) Supports of sums and products of series supported in a subgroup stay in it, and the Hahn inverse of  $f = cX^g(1 + u)$ ,  $\text{supp } u < 0$ , is  $c^{-1}X^{-g} \sum_{n \geq 0} (-u)^n$ , whose support lies in the subgroup generated by  $\text{supp } f$ . The geometric family is Hahn summable by the Neumann support lemma; for a class exponent group this argument takes place in the set-sized subgroup generated by the support of  $f$ . If  $2c \geq a$  and  $h \in H_a(G)$  then  $2|h| < a \leq 2c$ , so  $c + h > 0$ ; hence  $X^c f$  has positive support for  $f \in k((X^{H_a(G)}))$ . (iii) Put  $a_\alpha = a/\omega^{\alpha+1}$ . Each  $a_\alpha \ll a$  lies in  $H_a$ , and the exponents are distinct. If  $\mathcal{F}_a(k)$  were a set, restricting this family to its Hartogs ordinal would give the forbidden injection into that set.  $\square$

The field  $\mathcal{F}_a = \mathcal{F}_a(\mathbb{R})$  is 07's  $\mathcal{F}_a$  and 06's  $L_a$  with  $k = \mathbb{R}$ ; it is real closed and contains a copy of all of **No** ([Proposition 14.1](#) and [Theorem 14.4](#)).

## 4.2 The geometric telescope

**Lemma 4.3** (Bounded-exponent telescope [03, 04, 05, 06, 07, 08, 09, 10, 11, 15]). *Let  $G$  be an ordered abelian group, and let  $0 < b < a$  and  $a \ll c$  in  $G$ . Put  $d = a - b$ . Then*

$$q_{c;a,b} = \sum_{n \geq 0} X^{c-a-nd} \in \Pi_k(G), \quad (X^a - X^b) q_{c;a,b} = X^c. \quad (4.2)$$

*The support of  $q_{c;a,b}$  is countable and strictly decreasing. For every ordinary  $N \geq 0$ ,*

$$(X^a - X^b) \sum_{n=0}^N X^{c-a-nd} = X^c - X^{c-(N+1)d}, \quad (4.3)$$

*and the remainder monomial  $X^{c-(N+1)d}$  is still infinitely large.*

*Proof.* For each ordinary  $n \geq 0$ ,  $a + nd \leq (n+1)a < c$ , so every exponent is positive; the exponents strictly decrease with  $n$ , so every nonempty set of them has a largest element. Multiplication by  $X^a$  gives the exponents  $c - nd$  and multiplication by  $X^b$  gives  $c - (n+1)d$ ; at  $c$  the coefficient is 1 and at every  $c - md$ ,  $m \geq 1$ , it is  $1 - 1 = 0$ . Each coefficient receives at most two contributions, so this is a Hahn identity. The finite identity is the same computation truncated; its remainder exponent  $c - (N+1)d \geq c - (N+1)a > 0$ .  $\square$

*Remark 4.4* (No limit is taken and no sum is transported). The remainders in (4.3) are positive monomials; they do not tend to 0 in the order, valuation or fine topology, not even at the ordinary scale 1. The infinite identity is a coefficient identity, not a limit of (4.3) (04, 06, 07, 08, 10, 03). A target homomorphism is never applied term by term to  $q_{c;a,b}$ : the series is first a single element of the source ring, and only the finite product identity is mapped (06, 08, 09, 11). This is why no continuity, valuation or strong-additivity hypothesis on homomorphisms appears anywhere. It is also why a ring of finite-support series behaves differently (Section 9): there  $q_{c;a,b}$  is absent and the nonzero remainder of (4.3) cannot be removed.

**Example 4.5** (A witness for  $\omega$  [04, 10, 15]). With  $c = 1$ ,  $b = 1/\omega$  and  $a = 2/\omega$ ,  $(\omega^{2/\omega} - \omega^{1/\omega}) \sum_{n \geq 0} \omega^{1-(n+2)/\omega} = \omega$ , while the finite partial products leave the nonzero remainders  $\omega^{1-(N+1)/\omega}$  (04). In the lexicographic group  $\mathbb{Z} \times \mathbb{Z}$  with  $c = (1, 0)$ ,  $a = (0, 2)$ ,  $b = (0, 1)$  the quotient is  $\sum_{n \geq 0} X^{(1, -n-2)}$  and the  $N$ -th remainder is  $X^{(1, -N-1)}$  (10); under a compatible monomial realization each remainder is still infinite. 15 uses the same series  $q = \sum_{n \geq 0} \omega^{1-(n+2)/\omega}$  to block the extension of the finite-support augmentation (Section 9).

**Precedents.** The countable geometric-series product with a monomial result is already in L’Innocente–Mantova [23, Remark 3.4.4], used there to show that  $\deg \circ \text{ot}$  is not multiplicative when the value group does not embed in  $\mathbb{R}$ ; 04 names it a direct precursor. The factorization of a series whose support meets two distinct nonzero Archimedean classes is [23, Fact 8.0.1] (Gonshor’s [20, Thm. 8.6]), with the geometric factorization displayed on p. 46 of the arXiv version; 11 cites it as the precedent of the division below. Manuscript 09 credited “Proposition 8.2.1 on separated-scale division”; that proposition is the truncation criterion ( $b$  divides  $c$  if and only if  $b$  divides a truncation  $T_\Delta(c)$ ), whose proof divides the higher-scale part of  $c$  by  $b$ , and 08 cites it as the precedent for higher-scale divisibility. We credit all three places; none of the sources claims the geometric mechanism as new.



**Lemma 4.6** (Weighted certificate [07]). *Let  $K$  be a set field,  $r, s \in K^\times$ ,  $a > 0$  and  $\delta > \eta > 0$  with  $\delta \ll a$ . Then*

$$\frac{X^a}{rX^\delta - sX^\eta} = \frac{1}{r} \sum_{k \geq 0} \left(\frac{s}{r}\right)^k X^{a-\delta-k(\delta-\eta)} \in \Pi_K, \quad (4.4)$$

*If instead  $\delta = \eta > 0$ ,  $\delta \ll a$  and  $r \neq s$ , the quotient is  $(r-s)^{-1}X^{a-\delta} \in \Pi_K$ .*

*Proof.* The exponents are those of Lemma 4.3; multiplying by  $rX^\delta - sX^\eta$  gives, at exponent  $a - k(\delta - \eta)$  with  $k \geq 1$ , the two contributions  $r \cdot r^{-1}(s/r)^k$  and  $-s \cdot r^{-1}(s/r)^{k-1}$ , which cancel. At exponent  $a$  only the initial term contributes, with coefficient 1. The equal-exponent case follows by factoring  $(r-s)X^\delta$ ; its quotient exponent is positive because  $\delta < a$ .  $\square$

### 4.3 Division of an arbitrary purely infinite element

The telescope divides a monomial. Manuscripts 09, 10 and 11 divide an arbitrary element directly, which removes the separate reduction to monomials.

**Lemma 4.7** (Subordinate division [09, 10, 11, 13, 15]). *Let  $G$  be an ordered abelian group,  $0 \neq s = \sum_{g \in S} c_g X^g \in \Pi_k(G)$  and  $a > b > 0$  in  $U_G(S)$ . Put  $d = a - b$ . Then*

$$\frac{s}{X^a - X^b} = \sum_{g \in S} \sum_{n \geq 0} c_g X^{g-a-nd} \in \Pi_k(G), \quad (4.5)$$

*so  $s = (X^a - X^b)q$  with  $q \in \Pi_k(G)$ . The support of  $q$  lies in  $\{g - a - nd : g \in S, n \in \mathbb{N}\}$ , of cardinality at most  $\max(|S|, \aleph_0)$ ; it is countable when  $S$  is.*

*Proof.* In the ambient Hahn field  $(X^a - X^b)^{-1} = X^{-a}(1 - X^{-d})^{-1} = X^{-a} \sum_{n \geq 0} X^{-nd}$ . The product with  $s$  is a Hahn product: the sum of two reverse well-ordered sets is reverse well ordered and each exponent has finitely many representations (11 proves this two-support lemma in full; it is the standard Neumann support lemma). Every exponent that can occur satisfies  $g - a - nd \geq g - (n+1)a > 0$  because  $(n+1)a < g$ ; cancellation cannot create new exponents. Hence the quotient lies in  $\Pi_k(G)$ .  $\square$

The inverse of  $X^a - X^b$  is not in the ring; only its product with the larger-scale  $s$  is (11).

### 4.4 Separation principles

**Lemma 4.8** (Separation [04, 09, 11; 03]). *Let  $R$  be a ring (possibly a proper class),  $e \in R$ , and  $(x_t)_{t \in T}$  a set-indexed family with  $e \in (x_s - x_t)R$  for all  $s \neq t$ .*

- (i) *If  $\varphi : R \rightarrow S$  is a ring homomorphism with  $\varphi(e) \neq 0$ , then  $t \mapsto \varphi(x_t)$  is injective; hence  $|T| \leq |S|$  if  $S$  is a set.*
- (ii) *If  $R$  is commutative,  $M$  an  $R$ -module,  $v \in M$  and  $ev \neq 0$ , then  $t \mapsto x_tv$  is injective.*
- (iii) *If for every cardinal  $\lambda$  such a family with  $|T| = \lambda$  exists, then every homomorphism from  $R$  to a set-sized ring kills  $e$ , and  $e$  annihilates every set-sized  $R$ -module.*

*Proof.* If  $\varphi(x_s) = \varphi(x_t)$  with  $e = (x_s - x_t)q$ , then  $\varphi(e) = (\varphi(x_s) - \varphi(x_t))\varphi(q) = 0$ ; neither commutativity of  $S$  nor preservation of 1 is used. For (ii), commutativity lets us write  $ev = q(x_s - x_t)v = 0$ . For the ring assertion in (iii), choose a family whose index set cannot inject into the

target, and apply (i). For a module over a commutative  $R$ , apply (ii) in the same way. For a possibly noncommutative  $R$ , use instead the action homomorphism  $R \rightarrow \text{End}_{\mathbb{Z}}(M)$ : the endomorphisms of the underlying set-sized additive group form a set-sized ring, so the ring assertion makes the action of  $e$  zero. This last route proves annihilation but uses the size of  $\text{End}_{\mathbb{Z}}(M)$ , not the sharper bound by  $|M|$  in (ii).  $\square$

Item (i) is 04's separation lemma, (ii) is 09's orbit separation and 03's exponent-family obstruction, and (iii) is 11's collision-divisibility principle.

**Theorem 4.9** (Divisibility-separated families [11, 15]). *Let  $\mathcal{S} \subseteq \Pi_k$  be a set and  $\Lambda$  an ordinal. There are distinct monomials  $(u_\alpha)_{\alpha < \Lambda}$  in  $\mathcal{A}_{D,k}$  with*

$$s \in (u_\alpha - u_\beta)\Pi_k \quad (s \in \mathcal{S}, \alpha \neq \beta).$$

*Proof.* Choose  $\delta > 0$  strictly below the union of the supports of the members of  $\mathcal{S}$  (Lemma 3.1), put  $a_\alpha = \delta\omega^{-(\alpha+1)}$  and  $u_\alpha = X^{a_\alpha}$ . For  $\alpha < \beta$ ,  $0 < a_\beta < a_\alpha \leq \delta/\omega$ , so  $na_\alpha < \delta$  for every  $n$ , and Lemma 4.7 applies to each nonzero  $s$ ; for  $s = 0$  take quotient zero. Reversing the indices only changes the sign of the quotient. The empty family  $\mathcal{S}$  causes no restriction on the choice of  $\delta$ .  $\square$

The coefficients of the  $u_\alpha$  are 1; no independence of the coefficients of  $s$  is used, and cancellations inside  $s$  or the quotient are harmless (11). 15 proves the same with the family  $u_\xi = \omega^{g\omega^{-(\xi+1)}}$ ,  $\xi < \theta$ , for every ordinal  $\theta$ , and simultaneously for a set of elements  $s$ .

## 5 The universal set-sized quotient

### 5.1 The theorem

**Theorem 5.1** (Universal set-sized quotient [04, 06, 07, 08, 09, 10, 11, 13, 15; 03, 05]). *Let  $K$  be a set field,  $D \subseteq K$  a unital subring, and  $A = \mathcal{A}_{D,K} = D + \Pi_K$  with the full surreal exponent class.*

- (i) *Let  $S$  be a set-sized associative ring, not necessarily unital or commutative, and let  $\varphi : A \rightarrow S$  be an additive and multiplicative class map, not necessarily preserving 1. Then  $\varphi(\Pi_K) = 0$ , and  $\varphi = \psi \circ \text{ct}$  for a unique ring homomorphism  $\psi : D \rightarrow S$ , namely  $\psi = \varphi|_D$ .*
- (ii) *For every set-sized unital ring  $S$ , restriction to  $D$  and composition with  $\text{ct}$  are mutually inverse bijections, natural in  $S$ ,*

$$\text{Hom}_1(A, S) \longleftrightarrow \text{Hom}_1(D, S).$$

- (iii) *Every additive and multiplicative map from the nonunital ring  $\Pi_K$  to a set-sized ring is zero.*
- (iv) *Every set-sized unital  $A$ -module  $M$  satisfies  $\Pi_K M = 0$ , and its action is  $f \cdot v = \text{ct}(f)v$ .*

*Proof.* We follow 06's scaled-field argument in 08's choice-free Hartogs form.

*Positive monomials are killed.* Let  $\varphi : A \rightarrow S$  be as in (i), or an additive multiplicative map defined only on  $\Pi_K$  as in (iii). Fix  $g > 0$ , put  $m = X^{g/2}$  and  $L = \mathcal{F}_g(K) = K((X^{H_g}))$ . By Lemma 4.2(ii) with  $c = g/2$ ,  $mL \subseteq \Pi_K$ ; also  $m/(x - y) \in mL$  for distinct  $x, y \in L$ , so the identity (4.1) takes place inside  $\Pi_K$ . Let  $\lambda$  be the Hartogs ordinal of the set  $S$ , and put  $b_\alpha = g/\omega^{\alpha+1}$  for  $\alpha < \lambda$ ; these are distinct elements of  $H_g$  (Lemma 3.1(iii)), so  $\alpha \mapsto X^{b_\alpha}$  is an injection  $\lambda \rightarrow L$ . If  $\varphi(X^g) = \varphi(m^2) \neq 0$ , Lemma 4.1(i) makes  $x \mapsto \varphi(mx)$  injective on  $L$ , and composing gives an injection  $\lambda \rightarrow S$ , which is impossible. Only the restriction of the explicitly defined class map to the

set  $\lambda$  is used; there is no choice of representatives from a proper class. Hence  $\varphi(X^g) = 0$  for every  $g > 0$ .

*The whole ideal is killed.* For  $f \in \Pi_K$ , [Proposition 3.3](#) gives  $f = X^a h$  with  $a > 0$  and  $h \in \Pi_K$ , so  $\varphi(f) = \varphi(X^a)\varphi(h) = 0$ . This proves (iii), and for (i) it gives  $\varphi(d + f) = \varphi(d)$  for  $d \in D$ ,  $f \in \Pi_K$ . Thus  $\varphi = \varphi|_D \circ \text{ct}$ , and  $\psi = \varphi|_D$  is forced because  $\text{ct}$  is surjective.

*The correspondence.* If  $\varphi$  is unital, so is  $\psi$ . Conversely, for a unital  $\psi : D \rightarrow S$  the composite  $\psi \circ \text{ct}$  is a unital homomorphism by [Lemma 2.2](#), and its restriction to  $D$  is  $\psi$ . Post-composition with a map of targets preserves both constructions, which proves (ii).

*Modules.* Let  $M$  be a set-sized unital  $A$ -module and  $v \in M$ . If  $X^g v \neq 0$ , [Lemma 4.1\(ii\)](#) injects  $L$ , hence the Hartogs ordinal of  $M$ , into  $M$ ; so  $X^g M = 0$  for every  $g > 0$ . For  $f = X^a h$  and any  $v \in M$ ,  $fv = X^a(hv) = 0$ , because  $hv$  is another vector of  $M$ . Thus  $\Pi_K M = 0$ . Writing  $f = \text{ct}(f) + (f - \text{ct}(f))$  gives  $f \cdot v = \text{ct}(f)v$ .  $\square$

(Status (batch 33): multiplicativity in (i) cannot be weakened to additivity. The coefficient map  $x \mapsto [\omega^\omega]x$  is additive on  $\mathbf{Oz}$  but not multiplicative: it kills  $\omega^{\omega/2}$  and sends its square to 1. The holonomic-rigidity report [\[56\]](#) uses it to show that one fixed Conway coefficient of the orbit  $y_{n+1} = \omega y_n$  of a purely infinite omnific integer can encode any real sequence (`hol:rc:thm:coding`); this does not conflict with (i), since every  $y_n$  has constant term 0 (`hol:rc:rem:quotients`).)

**Sources and routes.** The statement is proved in the following generality by the sources: any set field  $K$  and unital  $D \subseteq K$  (06, 07, 10);  $k \in \{\mathbb{R}, \mathbb{C}\}$  and any unital  $D \subseteq k$  (04, 09, 11);  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and complex coefficients with any unital  $D \subseteq \mathbb{C}$  (08). Nonunital maps are treated by 08 (as a theorem), 06 (as a remark) and 10 (including maps defined on  $\Pi_K$  alone). Noncommutative targets are allowed in all eight sources. Manuscript 05 of the sibling report [\[47\]](#) proves the unital case for  $\mathbf{Oz}$  with the telescope and a cardinal  $\kappa > |A|$ , and 03 reproves that case (credited to 05) through its exponent-family obstruction. Two later manuscripts prove the theorem again, independently and without seeing the others: 13 for any set field  $k$  and unital  $D \subseteq k$  (unital maps, noncommutative targets), by direct division of an arbitrary purely infinite element with the Hartogs family  $\epsilon/(\bar{\alpha} + 2)$  (the route of 09, 10 and 11 below), and 15 for  $k \in \{\mathbb{R}, \mathbb{C}\}$  and unital  $D \subseteq k$ , with nonunital maps in a remark, by the separating families of [Theorem 4.9](#) and a Hartogs ordinal, which is 08's formulation; 15 also states the universe-relative reading of [Section 2.1](#). Neither adds a hypothesis-free generalization. Of the third batch, 17 proves the unital case for  $D_0 + \Pi_k$  with  $k \in \{\mathbb{R}, \mathbb{C}\}$  and any set-sized unital  $D_0 \subseteq k$  by 06's scaled-field collision, choosing preimages by global choice, and declares it background; 19 proves the unital case for  $\mathbf{Oz}$  by the explicit telescope below with a cardinal  $\lambda > |S|$  and credits it to 06's text. Both are credited here as further proving sources; neither adds generality. So is 20, of the fourth batch, which proves the unital case for  $\mathbf{Oz}$  by the same telescope with the exponents  $a/\omega^{\alpha+1}$  ( $\alpha < \lambda$ ,  $\lambda > |S|$ ), crediting this report. The proofs differ, and the following are genuinely different routes, kept here in brief.

- *Explicit telescope* (04, 05, 06 appendix, 08). If  $\varphi(X^c) \neq 0$ , then for distinct  $a_\alpha, a_\beta \ll c$  the telescope [\(4.2\)](#) puts  $X^c$  in  $(X^{a_\alpha} - X^{a_\beta})A$ , so [Lemma 4.8\(i\)](#) makes  $\alpha \mapsto \varphi(X^{a_\alpha})$  injective, which is impossible for  $\alpha$  ranging over a set larger than  $S$ ; [Proposition 3.3](#) then kills all of  $\Pi_K$ . This is manuscript 05's proof, which the sibling report [\[47\]](#) quotes and refers here for; 05 chooses a cardinal  $\kappa > |S|$  with global choice, where the Hartogs ordinal suffices. The route uses only countable supports, which is why it survives in the support-bounded rings of [Section 9](#).
- *Direct division of an arbitrary element* (09, 10, 11). If  $\varphi(s) \neq 0$  for  $s \in \Pi_K$ , [Lemma 4.7](#) gives  $s \in (X^a - X^b)\Pi_K$  for all  $a > b$  in  $U_{\mathbf{No}}(\text{supp } s)$ , and [Theorem 4.9](#) supplies such families of every

set size, so [Lemma 4.8\(iii\)](#) applies to  $s$  itself; no separate reduction to monomials is needed. 10 phrases this as an injection  $U(\text{supp } s) \rightarrow S$ ,  $a \mapsto \varphi(X^a)$ , valid also for nonunital maps on  $\Pi_K$ ; 11 phrases it as divisibility-separated families.

- *Weighted certificates* (07). With  $\delta_\alpha = a/(\omega + \alpha + 1)$ ,  $\alpha < \kappa^+$  for a cardinal  $\kappa \geq |S|$ , pairwise differences of the monomials  $X^{\delta_\alpha}$  divide  $X^a$ , and (4.4) even separates all  $rX^\delta$ ,  $r \in K^\times$ .
- *Modules through the endomorphism ring* (04, 07, 08, 10, 11). A set-sized module gives a unital ring map  $A \rightarrow \text{End}_{\mathbb{Z}}(M)$  into a set-sized, generally noncommutative ring, and (ii) applies. This is valid for the full class but bounds only  $|\text{End}_{\mathbb{Z}}(M)|$ ; the direct argument above (06, 09, 03) and 08's cyclic-module argument ([Lemma 10.9](#)) are the ones that give set-sized thresholds.

**A group-level strengthening.** The report on omnific groups and lattices [49], written concurrently, proves that for  $n \geq 3$  the purely infinite congruence kernel of the elementary group  $E_n(\mathbf{Oz})$  (and of  $E_n(\mathbf{Oz}[i])$ ) has no nontrivial homomorphism to a set-sized group; applied to the elementary matrices  $e_{12}(\psi(x))$  this implies (i) for unital maps. (Status (batch 32): that report's Part IV extends this to every root system  $\Phi$  without  $A_1$  component and every unital  $D \subseteq k$ : the congruence kernels of  $E_\Phi(\mathcal{A}_{D,k})$  and of  $\text{St}_\Phi(\mathcal{A}_{D,k})$  have no nontrivial homomorphism to a set-sized group, and  $\text{ct}$  is a split universal set-sized quotient of the Steinberg group (`ogl:ch:thm:intrinsic`, `ogl:ch:thm:steinberg`); for a commutative ring  $I$  without unit and with  $2I = I$ , set-invisibility of  $I$  is equivalent to that of the congruence kernel of  $E_\Phi$  over its Dorroh unitization (`ogl:ch:thm:equivalence`); for a Lie lattice  $\mathfrak{l}$  the constant term is a universal set-sized quotient of  $\mathfrak{l}(\mathcal{A}_{D,k})$  if  $\mathfrak{l}_{\mathbb{Q}}$  is perfect, and no universal set-sized quotient exists otherwise (`ogl:ch:thm:liepositive`, `ogl:ch:thm:lienegative`); and these statements transfer to the bounded rings of [Theorem 9.1](#) and to exact regular-cardinal thresholds (`ogl:ch:cor:support`, `ogl:ch:cor:thresholds`).)

## 5.2 Consequences for rings and representations

**Corollary 5.2** (Omnific integers [03, 04, 05, 06, 07, 08, 09, 10, 11, 13, 15]). *For every set-sized unital ring  $S$  the only unital homomorphism  $\mathbf{Oz} \rightarrow S$  is  $x \mapsto \text{ct}(x)1_S$ . Additive multiplicative maps  $\mathbf{Oz} \rightarrow S$  into a set-sized ring correspond bijectively to idempotents  $e \in S$ , via  $x \mapsto \text{ct}(x)e$ .*

*Proof.* There is one unital map  $\mathbb{Z} \rightarrow S$ . A ring map  $\psi : \mathbb{Z} \rightarrow S$  is determined by  $e = \psi(1)$ , which satisfies  $e^2 = e$ . Conversely  $a \mapsto ae$  is additive and  $(ae)(be) = abe^2 = abe$ , so every idempotent defines one, even if  $S$  is noncommutative or lacks an identity. Here  $ae$  means an integer multiple in the additive group of  $S$ . Apply [Theorem 5.1\(i\),\(ii\)](#).  $\square$

**Corollary 5.3** (Gaussian omnific integers [04, 06, 07, 08, 09, 10, 11, 13, 15]). *For a set-sized unital ring  $S$ , unital maps  $\mathbf{Oz}[i] \rightarrow S$  correspond bijectively to elements  $j \in S$  with  $j^2 = -1_S$ , via  $x + iy \mapsto \text{ct}(x)1_S + \text{ct}(y)j$  ( $x, y \in \mathbf{Oz}$ ). Every such map discards all positive-exponent complex coefficients, not only their real parts.*

*Proof.* Apply [Theorem 5.1](#) with  $(D, K) = (\mathbb{Z}[i], \mathbb{C})$ . A unital map from  $\mathbb{Z}[i] = \mathbb{Z}[T]/(T^2 + 1)$  is a choice of a root of  $T^2 + 1$ ; in a noncommutative  $S$  the subring generated by  $1_S$  and  $j$  is commutative, so evaluation is valid. The element  $j$  need not be central in all of  $S$ ; centrality of the image would be an additional algebra-structure condition.  $\square$

For completeness, nonunital additive multiplicative maps to a set-sized ring  $S$  are similarly described by pairs  $(e, j)$  with

$$e^2 = e, \quad ej = je = j, \quad j^2 = -e,$$

through  $x + iy \mapsto \text{ct}(x)e + \text{ct}(y)j$ . Necessity follows by applying the map to 1,  $i$  and their products. Conversely these relations give  $(ae + bj)(ce + dj) = (ac - bd)e + (ad + bc)j$  for integers  $a, b, c, d$ ; combine this map on  $\mathbb{Z}[i]$  with the constant term. For a unital map,  $e = 1_S$ . This is a consequence of [Theorem 5.1\(i\)](#), added in the review.

The image of  $i$  is part of the data: these are ring maps, not maps of complex algebras. For  $(D, K) = (k, k)$  with  $k \in \{\mathbb{R}, \mathbb{C}\}$ , every unital map from  $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$  or  $\mathcal{A}_{\mathbb{C}, \mathbb{C}}$  to a set-sized ring factors through the constant coefficient in  $k$ , and if the maps are required to fix  $k$ , the constant term is the unique such map to  $k$  (04). For any unital  $D \subseteq \mathbb{C}$ , the ring of complex-coefficient normal forms with positive support apart from a constant in  $D$  has universal set-sized quotient  $D$  (08); this is the case  $K = \mathbb{C}$  of the theorem.

**Corollary 5.4** (What all small observations identify [04]). *For  $x, y \in \mathcal{A}_{D, k}$ ,  $\text{ct}(x) = \text{ct}(y)$  if and only if  $\varphi(x) = \varphi(y)$  for every unital homomorphism  $\varphi$  from  $\mathcal{A}_{D, k}$  to a set-sized ring. Thus  $\Pi_k$  is the common kernel of all set-sized ring observations.*

*Proof.* The forward direction is the theorem; for the converse use the target  $D$  and the map  $\text{ct}$ .  $\square$

This does not say that the elements of  $\Pi_k$  are zero:  $\omega \neq 0$  in  $\mathbf{Oz}$ , yet every set-target ring homomorphism sends it to 0 (04).

**Corollary 5.5** (A common monomial in a set of kernels [15]). *Let  $(\varphi_j : \mathcal{A}_{D, k} \rightarrow B_j)_{j \in J}$  be a set-indexed family of ring homomorphisms, the targets possibly proper classes, together with specified nonzero  $b_j \in \ker \varphi_j$ . Then some nonzero monomial  $X^\delta$  lies in every  $\ker \varphi_j$ ; in particular the family is not jointly injective.*

*Proof.* The specified witnesses have a set of inverses in the ambient Hahn field, by class replacement. Apply [Proposition 3.4\(i\)](#) to that set: some  $\delta > 0$  has  $X^\delta/b_j \in \Pi_k$  for all  $j$ , so  $X^\delta \in b_j \mathcal{A}_{D, k} \subseteq \ker \varphi_j$ . If  $J$  is empty any positive  $\delta$  works. In every case  $X^\delta \neq 0$ , so it and 0 have the same images under the entire family.  $\square$

The witnesses are specified so that no choice over a proper class of families is needed; nothing is claimed about an intersection indexed by the class of all nonzero elements (15).

**Corollary 5.6** (No faithful set-sized representation [04, 06, 07, 08, 11]). *No nonzero element of  $\Pi_k$  acts nontrivially on a set-sized unital  $\mathcal{A}_{D, k}$ -module; in particular  $\mathcal{A}_{D, k}$  has no faithful set-sized module. For every set-sized field  $F$  and set-sized  $F$ -vector space  $V$  (of any dimension), every unital ring map  $\mathbf{Oz} \rightarrow \text{End}_F(V)$  is  $x \mapsto \text{ct}(x) \text{id}_V$ , and every unital ring map  $\mathbf{Oz}[i] \rightarrow \text{End}_F(V)$  is  $x + iy \mapsto \text{ct}(x) \text{id}_V + \text{ct}(y)J$  with  $J^2 = -\text{id}_V$ .*

*Proof.* The module assertion is [Theorem 5.1\(iv\)](#), and  $\Pi_k$  contains a nonzero monomial, so no such module is faithful. The ring  $\text{End}_F(V)$  is a set, since its elements are functions on the set  $V$ . Apply [Corollaries 5.2](#) and [5.3](#) to this ring. The zero vector space is allowed; its endomorphism ring is the zero ring and the same formulas still apply.  $\square$

**Proposition 5.7** (Finite matrix representations [06, 11]). *Let  $n \geq 1$ .*

- (i) The only unital map  $\mathbf{Oz} \rightarrow M_n(F)$ ,  $F$  a set-sized field, is  $x \mapsto \text{ct}(x)I_n$ .
- (ii) Up to conjugacy in  $\text{GL}_n(\mathbb{C})$ , the unital maps  $\mathbf{Oz}[i] \rightarrow M_n(\mathbb{C})$  are indexed by  $p \in \{0, \dots, n\}$ , with  $i \mapsto \text{diag}(iI_p, -iI_{n-p})$ .
- (iii) A unital map  $\mathbf{Oz}[i] \rightarrow M_n(\mathbb{R})$  exists exactly when  $n$  is even, and then all such maps are conjugate to the one sending  $i$  to  $n/2$  diagonal blocks  $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ .

*Proof.* (i) is [Corollary 5.2](#). By [Corollary 5.3](#) a map in (ii) or (iii) is a matrix  $J$  with  $J^2 = -I_n$ . Over  $\mathbb{C}$  its minimal polynomial divides  $(T - i)(T + i)$ , which has distinct roots, so  $J$  is diagonalizable with eigenvalues  $\pm i$ , and the multiplicity  $p$  of  $i$  determines the conjugacy class. Over  $\mathbb{R}$ ,  $J$  makes  $\mathbb{R}^n$  a complex vector space by  $(u + iv)w = uw + vJw$ . It is finite-dimensional over  $\mathbb{C}$  because any real basis also spans over  $\mathbb{C}$ . If  $v_1, \dots, v_r$  is a complex basis, then  $v_1, Jv_1, \dots, v_r, Jv_r$  is a real basis: spanning and independence are precisely the real and imaginary parts of a complex linear combination. Hence  $n = 2r$ , and  $J$  acts on each ordered pair  $(v_j, Jv_j)$  by the displayed block. Conversely those blocks square to  $-I_n$ . In (ii), every  $p$  gives a map and distinct  $p$  give different eigenspace dimensions, so the list is both exhaustive and without repetitions.  $\square$

No norm bound, analytic hypothesis or finite-dimensionality is used to kill the purely infinite part; finite dimension enters only the classification of the possible images of  $i$  (06). For an infinite-dimensional set-sized space the reduction to one operator  $J$  with  $J^2 = -1$  remains valid.

**Algebras.** A set-sized associative  $\mathcal{A}_{D,k}$ -algebra is a set-sized ring with a unital map from  $\mathcal{A}_{D,k}$  to its center; by the theorem this map factors through  $D$ , so set-sized  $\mathcal{A}_{D,k}$ -algebras and  $D$ -algebras coincide, with the same underlying rings and morphisms; noncentral structure maps are covered by the noncommutative form (04). Enlarging the set-sized algebra in which a calculation takes place therefore never yields a faithful algebraic model of the full ring, although such calculations can be faithful on a chosen set-sized subring (04). Nor does the theorem forbid set-sized rings from embedding into  $\mathbf{Oz}$ : the inclusion of  $\mathbb{Z}[\omega]$  detects  $\omega$ ; it concerns maps defined on all of  $\mathbf{Oz}$ , and it makes no claim that  $\mathbf{Oz}$  and  $\mathbb{Z}$  have the same first-order theory (15).

**Corollary 5.8** (No ordered specialization, no small retract, rigid endomorphisms [08, 10, 11; 05]).

- (i) There is no order-preserving unital map from  $\mathbf{Oz}$  to a nonzero ordered set-sized ring (11; 09 records the relevant value  $\text{ct}(\omega - 1) = -1$ ).
- (ii) A set-sized unital subring  $A \subseteq \mathbf{Oz}$  is the image of a ring retraction  $\mathbf{Oz} \rightarrow A$  if and only if  $A = \mathbb{Z}$  (08).
- (iii) Every unital class ring endomorphism  $\sigma$  of  $\mathbf{Oz}$  satisfies  $\text{ct} \circ \sigma = \text{ct}$ ; hence every automorphism preserves  $\Pi$ . Every automorphism of  $\mathbf{Oz}[i]$  induces the identity or complex conjugation on  $\mathbb{Z}[i]$  (10; the first assertion is also 05's, see [47]).

*Proof.* (i) The unique map sends the positive element  $\omega - 1$  to  $-1 < 0$ . (ii) A retraction fixes 1, because the subring has the same identity and the retraction fixes that subring pointwise. It is therefore  $f \mapsto \text{ct}(f)1_A$  by [Corollary 5.2](#), whose image is  $\mathbb{Z}$ ; and  $\text{ct}$  is a retraction onto  $\mathbb{Z}$ . (iii)  $\text{ct} \circ \sigma$  is a unital map to the set  $\mathbb{Z}$ , hence equals  $\text{ct}$ . For an automorphism apply this to  $\sigma$  and  $\sigma^{-1}$ . In the Gaussian case,  $\text{ct} \circ \sigma : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$  factors through  $\text{ct}$  by [Theorem 5.1](#); hence  $\sigma(\Pi_{\mathbb{C}}) \subseteq \Pi_{\mathbb{C}}$ . The same argument for  $\sigma^{-1}$  gives equality. Thus  $\sigma$  and  $\sigma^{-1}$  induce mutually inverse ring automorphisms of  $\mathbb{Z}[i]$  fixing  $\mathbb{Z}$ , and  $i$  goes to a root of  $T^2 + 1$  in  $\mathbb{Z}[i]$ , i.e. to  $\pm i$ .  $\square$



Passing to a small workspace is therefore not a homomorphic truncation of the full ring; the workspace may embed without being a retract (08). The endomorphism statement does not say that every endomorphism is the identity:  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  composed with  $\text{ct}$  is a noninjective endomorphism.

**Corollary 5.9** (Fields and unit-forcing localizations [04, 06, 09, 11]). *Let  $0 \neq u \in \Pi_k$ . The class rings  $k((X^{\mathbf{No}}))$  (in particular  $\mathbf{No}$  and  $\mathbf{No}[i]$ ) and  $\mathcal{A}_{D,k}[u^{-1}]$  admit no unital homomorphism to a nonzero set-sized ring and no nonzero set-sized unital module.*

*Proof.* A unital map from either ring to a set-sized  $S$  restricts to  $\mathcal{A}_{D,k}$  and so kills  $u$ ; but  $u$  is invertible in the source, so  $1_S = \varphi(u)\varphi(u^{-1}) = 0$ . A nonzero set-sized module would give such a map to the nonzero set-sized ring  $\text{End}_{\mathbb{Z}}(M)$ .  $\square$

This is not merely the observation that a proper-class field cannot inject into a set: the localizations need not be fields (04). For example  $\mathbf{Oz}[\omega^{-1}]$  is a nonzero class domain, not a field (Proposition 6.11), with no nonzero set-sized module. This completes the proof of Proposition 3.4(iv).

*Remark 5.10* (Positive characteristic and additive observations). Suppose the target additive group is killed by a positive integer  $n$  and  $n$  is nonzero in the coefficient field  $k$ . Then no cardinal argument is needed:  $f = n(f/n)$  with  $f/n \in \Pi_k$  gives  $\varphi(f) = n\varphi(f/n) = 0$  (04, 07). This applies to positive-characteristic targets when  $k$  has characteristic zero, in particular for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ . It does not apply when  $k$  and the target both have characteristic  $p$ , since division by  $p$  is then unavailable; the full theorem still follows from the collision argument. More generally, an ideal of  $\mathcal{A}_{D,k}$  containing a nonzero element of  $D$  contains  $\Pi_k$  (07, 15), by Proposition 3.3(iv). For characteristic-zero targets such as  $\mathbb{Q}$ ,  $\mathbb{R}$  or  $\text{End}_{\mathbb{Z}}(M)$  for infinite torsion-free  $M$ , this bounded-torsion shortcut is unavailable (11). The theorem concerns multiplicative maps: for a fixed  $\gamma > 0$  the coefficient extraction  $x \mapsto [\omega^\gamma]x$  is a nonzero additive map  $\mathbf{Oz} \rightarrow \mathbb{R}$  detecting  $\omega^\gamma$ , but it is not multiplicative, since  $\omega^{\gamma/2}$  has coefficient 0 at  $\gamma$  while its square has coefficient 1 (04, 07, 09, 10, 11, 15). Valuations also detect nonconstant parts, but not by unital ring maps into set-sized rings (04).

*Remark 5.11* (What the theorem does not say). The theorem is a universal property relative to set-sized targets, not an isomorphism  $\mathbf{Oz} \cong \mathbb{Z}$ . The ideal  $\Pi$  is nonzero, idempotent, not nilpotent and not generated by any set, and  $\mathbf{Oz}$  is a domain (09, 08). The regular module  $\mathbf{Oz}$  and the ideal  $\Pi$  are proper-class modules on which  $\Pi$  acts nontrivially, so the module correspondence is not a Morita equivalence of unrestricted module theories (06). The identity of  $\mathbf{Oz}$  and the inclusion  $\mathbf{Oz} \hookrightarrow \mathbf{No}$  are faithful class-valued maps outside the theorem (09). Individual monomials can be represented faithfully in small rings, for instance in polynomial rings or Hahn subfields, and every finite calculation lies in a set-sized subring with a faithful regular representation; the obstruction concerns one global homomorphism defined on *all* of the class ring, including the collision witnesses at all lower scales (04, 07, 08). In the language of 11, the set-sized reflection is  $\text{Ref}_{\text{set}}(\mathbf{Oz}) = \mathbb{Z}$  and  $\text{Ref}_{\text{set}}(\mathbf{Oz}[i]) = \mathbb{Z}[i]$ : a surjection  $q : R \rightarrow D$  onto a set-sized ring is a universal set-sized quotient if every map from  $R$  to a set-sized ring factors uniquely through  $q$ ; this refers to the explicit universal property, not to a category of all class rings with set-sized hom-sets.

**Theorem 5.12** (Residually set-sized targets [11, 13]). *Call a class ring  $B$  residually set-sized if for each nonzero  $b \in B$  there are a set-sized ring  $A_b$  and a unital map  $\pi_b : B \rightarrow A_b$  with  $\pi_b(b) \neq 0$  (the quantifier is elementwise; no set of detecting maps is formed). Every unital map  $h : \mathcal{A}_{D,k} \rightarrow B$  into a residually set-sized ring kills  $\Pi_k$  and factors uniquely through  $D$ . For a set-sized commutative ring  $E$  and any class  $X$  of indeterminates, the ring  $E[X]$  of finite polynomials is residually set-sized.*

*Proof.* If  $h(s) \neq 0$  for some  $s \in \Pi_k$ , then  $\pi \circ h$  detects  $s$  for a detecting map  $\pi$  with set-sized target, contrary to Theorem 5.1. A nonzero polynomial  $f \in E[X]$  uses a finite set  $Y \subseteq X$  of variables;

sending the other variables to 0 gives a unital map  $E[X] \rightarrow E[Y]$  fixing  $f$ , with set-sized target. This substitution is defined on each finite polynomial separately, so it needs no choice of a set containing all of  $X$ . Once  $h(\Pi_k) = 0$ , the factorization is  $h(d + u) = h(d)$ , with unique restriction  $h|_D$ , just as in the original theorem.  $\square$

## 6 Quotients, closures, completions and localizations

Throughout,  $A = \mathcal{A}_{D,k}$  with  $k$  a set field and  $D \subseteq k$  a unital subring, and  $\Pi_k = \ker \text{ct}$ .

### 6.1 Every quotient and its small image

**Theorem 6.1** (Small reflection of an arbitrary quotient [06, 07, 09, 11, 15]). *Let  $J$  be a class ideal of  $A$ . Then  $\text{ct}(J)$  is an ideal of  $D$ , and for every set-sized unital ring  $S$*

$$\text{Hom}_1(A/J, S) \cong \text{Hom}_1(D/\text{ct}(J), S),$$

*the universal map being  $[x] \mapsto \text{ct}(x) \bmod \text{ct}(J)$ . Its kernel is  $(\Pi_k + J)/J$ , and  $\text{ct}^{-1}(\text{ct}(J)) = \Pi_k + J$ .*

*Proof.*  $\text{ct}(J)$  is a set by separation inside  $D$ . For  $c = \text{ct}(j)$  with  $j \in J$  and  $d \in D$ , one has  $dc = \text{ct}(dj) \in \text{ct}(J)$ ; addition and negation work similarly, so it is an ideal. A unital map from  $A/J$  is a unital map  $\varphi : A \rightarrow S$  vanishing on  $J$ ; by [Theorem 5.1](#),  $\varphi = \psi \circ \text{ct}$  for a unique  $\psi$ , and  $\varphi(J) = 0$  if and only if  $\psi(\text{ct}(J)) = 0$ . If  $\text{ct}(a) = \text{ct}(b)$  with  $b \in J$ , then  $a - b \in \Pi_k$ , which gives the kernel and the preimage; conversely every element of  $\Pi_k + J$  has its constant term in  $\text{ct}(J)$ . The map  $[x] \mapsto [\text{ct}(x)]$  is well defined because changing  $x$  by an element of  $J$  changes its constant term by an element of  $\text{ct}(J)$ . It is surjective on constant representatives, and the constructions commute with maps of target rings.  $\square$

Thus a quotient can be a nonzero ring while all of its set-sized images are zero; a quotient presentation of a proper-class ring is not automatically a set-sized ring (09). “Reflection” is only a name for the displayed mapping property; no category of all proper classes is used (07).

**Theorem 6.2** (Ideals with set-sized quotient [04, 06, 07, 08, 09, 10, 11, 13, 15]). *For a class ideal  $J$  of  $A$  the following are equivalent:*

- (i)  $J$  admits a set-sized quotient ([Definition 2.1](#));
- (ii)  $\Pi_k \subseteq J$ ;
- (iii)  $J = \text{ct}^{-1}(\mathfrak{a}) = \Pi_k + \mathfrak{a}$  for an ideal  $\mathfrak{a}$  of  $D$ .

*Then  $\mathfrak{a} = J \cap D = \text{ct}(J)$  is unique and  $A/J \cong D/\mathfrak{a}$ . For  $\mathbf{Oz}$  the kernels are exactly*

$$\Pi \text{ (quotient } \mathbb{Z}) \quad \text{and} \quad n\mathbf{Oz} = n\mathbb{Z} + \Pi \text{ (quotient } \mathbb{Z}/n\mathbb{Z}, n \geq 1),$$

*the case  $n = 1$  giving the zero ring. For  $\mathbf{Oz}[i]$  they are  $\Pi_{\mathbb{C}}$  (quotient  $\mathbb{Z}[i]$ ) and  $d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}}$  for  $0 \neq d \in \mathbb{Z}[i]$ , with finite quotient  $\mathbb{Z}[i]/(d)$  of  $|d|^2$  elements; associates give the same ideal.*

*Proof.* (i) $\Rightarrow$ (ii) by [Theorem 5.1](#) applied to the quotient map. (ii) $\Rightarrow$ (iii): if  $\Pi_k \subseteq J$  then  $d + f \in J$  ( $f \in \Pi_k$ ) if and only if  $d \in J \cap D$ . (iii) $\Rightarrow$ (i):  $A \rightarrow D \rightarrow D/\mathfrak{a}$  realizes the quotient. In this situation  $\text{ct}(J) = J \cap D$ : subtracting the purely infinite part of an element of  $J$  leaves its constant term in  $J$ . This proves uniqueness of  $\mathfrak{a}$ , not merely existence of some set-sized target. For  $\mathbf{Oz}$  the ideals of

$\mathbb{Z}$  are  $n\mathbb{Z}$ ,  $n \geq 0$ ;  $n = 0$  gives  $\Pi$  and  $n \geq 1$  gives  $n\mathbf{Oz}$  by (3.2). For  $\mathbf{Oz}[i]$ ,  $\mathbb{Z}[i]$  is a principal ideal domain (division with remainder by rounding the real and imaginary parts of a complex quotient), and (3.2) gives  $d\mathbf{Oz}[i] = d\mathbb{Z}[i] + \Pi_{\mathbb{C}}$ . Multiplication by  $d = b + ci$  on  $\mathbb{Z}[i] \cong \mathbb{Z}^2$  has matrix  $\begin{pmatrix} b & -c \\ c & b \end{pmatrix}$  of determinant  $b^2 + c^2 = |d|^2$ , the index of its image; more generally a nonzero ideal containing  $d$  contains the positive integer  $d\bar{d}$ , so it has finite index without any appeal to unique factorization (07, 06, 08).  $\square$

The distinction between  $n = 0$  and  $n \geq 1$  matters: the characteristic-zero kernel is  $\Pi$ , not  $0\mathbf{Oz}$ ; one must not set  $n = 0$  in  $n\mathbf{Oz} = n\mathbb{Z} + \Pi$  (04, 06, 07, 08, 09, 10, 11). For  $D = k$ , the only proper kernel with a set-sized quotient is  $\Pi_k$ , with quotient  $k$  (04).

**Corollary 6.3** (Small primes, maximal ideals and fields [06, 07, 08, 10, 11, 13, 15]). *The prime class ideals of  $\mathbf{Oz}$  with set-sized quotient are  $\Pi$  and  $p\mathbf{Oz}$  ( $p$  an ordinary prime), with quotient domains  $\mathbb{Z}$  and  $\mathbb{F}_p$  and fraction fields  $\mathbb{Q}$  and  $\mathbb{F}_p$ ; the maximal ones are the  $p\mathbf{Oz}$ . A unital map from  $\mathbf{Oz}$  to a set-sized field has kernel  $\Pi$  in characteristic 0 (its image is  $\mathbb{Z}$ , not a field) and  $p\mathbf{Oz}$  in characteristic  $p$ ; the only nonzero set-sized field quotients are the  $\mathbb{F}_p$ . For  $\mathbf{Oz}[i]$  replace these by  $\Pi_{\mathbb{C}}$  and the ideals generated by Gaussian primes, with quotient domains  $\mathbb{Z}[i]$  and the corresponding finite fields.*

*Proof.* Under Theorem 6.2, the quotient is exactly  $D/\mathfrak{a}$ , so prime and maximal ideals in the stated class correspond to prime and maximal ideals of  $D$ . For  $D = \mathbb{Z}$  these are  $(0)$  and  $(p)$ , with  $(0)$  not maximal. A unital map to a field restricts on  $\mathbb{Z}$  to its prime-subring map, giving kernel  $(0)$  or  $(p)$  according to its characteristic. In characteristic zero that image is  $\mathbb{Z}$ , even though its fraction field is  $\mathbb{Q}$ . For  $D = \mathbb{Z}[i]$ , the same argument uses its zero ideal and Gaussian prime ideals, whose quotient rings are respectively  $\mathbb{Z}[i]$  and finite fields.  $\square$

In particular  $\Pi$  is prime but not maximal (it lies in  $p\mathbf{Oz}$ ). These results classify only ideals with set-sized quotients:  $(0)$  is prime in  $\mathbf{Oz}$  with a proper-class quotient, and no Zorn argument for a proper class of ideals is used to assert that proper class ideals lie in maximal class ideals; a maximal ideal outside this list, if one is given, has a proper-class residue field (07, 08, 11). (Status (batch 34): with global choice, 24 proves that every proper class ideal lies in a maximal class ideal (Theorem 14.24), by a recursion with set-valued stages rather than a Zorn argument, and that  $1 + \omega^a$  lies in characteristic-zero maximal class ideals of  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  outside this list (Theorem 14.34); the classification above, of the ideals with set-sized quotients, is unaffected.)

**Proposition 6.4** (Principal quotients [04]). *Let  $f \in A$  be nonconstant and  $\beta = \deg_{\omega}(f) > 0$ . For distinct  $\gamma, \delta \in (0, \beta)$ ,  $X^{\gamma} - X^{\delta} \notin fA$ ; hence  $A/(f)$  has sets of pairwise distinct residues of every set cardinality and cannot be realized as a set-sized ring. More generally, for arbitrary  $f \in A$ , the ideal  $(f)$  admits a set-sized quotient exactly when  $f \in D \setminus \{0\}$ , and then  $fA = fD + \Pi_k$  and  $A/(f) \cong D/fD$ . Among the ideals with set-sized quotient,  $\Pi_k$  is the only one not generated by a set; every other one is generated by a subset of  $D$ , and for  $\mathbf{Oz}$  these are the principal ideals  $n\mathbf{Oz}$ ,  $n \geq 1$ .*

*Proof.* The difference has degree  $\max(\gamma, \delta) < \beta$ , while a nonzero multiple  $fq$ ,  $q \in A$ , has degree  $\beta + \deg_{\omega}(q) \geq \beta$  (Lemma 2.4); the interval  $(0, \beta)$  contains sets of every cardinality (Lemma 3.1). For  $0 \neq f \in D$ ,  $f\Pi_k = \Pi_k$  (Proposition 3.3); the zero ideal would inject the class  $A$  into a set. For the last assertion write the ideal as  $\Pi_k + \mathfrak{a}$ ; if  $\mathfrak{a} = 0$  use Proposition 3.3(iii); otherwise some  $0 \neq d \in \mathfrak{a}$  gives  $d\Pi_k = \Pi_k$ , so the ideal generated by the set  $\mathfrak{a}$  already contains  $\Pi_k$ .  $\square$

The size of  $A/(f)$  must be distinguished from the size of a further image of that quotient. By Theorem 6.1, its universal set-sized image is  $D/(\text{ct}(f))$ , because  $\text{ct}(fA) = \text{ct}(f)D$ . Thus  $\mathbf{Oz}/(\omega)$  is

not a set-sized ring but surjects onto  $\mathbb{Z}$ ;  $\mathbf{Oz}/(2 + \omega)$  surjects onto  $\mathbb{Z}/2\mathbb{Z}$ ; and  $\mathbf{Oz}/(1 + \omega)$  has only the zero set-sized image. All three source quotients are nonzero, since their generators are nonunits. This corrects the earlier wording that the nonconstant principal quotient itself “admits no set-sized quotient”.

**Theorem 6.5** (Quotient-size alternative [08, 13]). *Let  $J$  be a class ideal of  $\mathbf{Oz}$  with  $\Pi \not\subseteq J$ . For every set cardinal  $\mu$  there is a set of at least  $\mu$  elements of  $\mathbf{Oz}$  pairwise incongruent modulo  $J$ .*

*Proof.* Choose  $f \in \Pi \setminus J$  and write  $f = \omega^c h$  with  $c > 0$ ,  $h \in \Pi$ ; then  $\omega^c \notin J$ . For an ordinal  $\lambda$  of cardinality at least  $\mu$  use the exponents  $c/\omega^{\beta+1}$ ,  $\beta < \lambda$ : if two of the monomials were congruent modulo  $J$ , their difference would lie in  $J$ ; multiplying by the telescope quotient of (4.2) would put  $\omega^c$  in  $J$ , a contradiction. Every multiplier belongs to the source ring. The family is indexed by the set  $\lambda$ , so no cardinality of a proper class is used.  $\square$

A statement that a quotient “is not small” always has this concrete meaning: for every set cardinal it contains a larger set of distinct elements; no “cardinality of  $\mathbf{No}$ ” is used (07).

## 6.2 Finite congruences, closures and invisible primes

For a class ideal  $J$  of  $\mathbf{Oz}$  define its finite-congruence closure  $\bar{J}^{\text{fin}} = \bigcap_{n \geq 1} (J + n\mathbf{Oz})$ ; the witness in  $J$  may depend on  $n$  (08).

**Theorem 6.6** (Closure of every ideal [08, 09]). *If  $\text{ct}(J) = d\mathbb{Z}$  with  $d \geq 0$ , then  $\bar{J}^{\text{fin}} = J + \Pi = \text{ct}^{-1}(d\mathbb{Z})$ . In particular  $J$  is dense for finite congruences exactly when  $\text{ct}(J) = \mathbb{Z}$ , and the closure of 0 is  $\Pi$ .*

*Proof.*  $\text{ct}(J)$  is an ideal of  $\mathbb{Z}$ . Since  $\Pi \subseteq n\mathbf{Oz}$ , the ideal  $J + n\mathbf{Oz}$  contains  $\Pi$  and is the full preimage of its constant ideal  $\text{ct}(J) + n\mathbb{Z} = d\mathbb{Z} + n\mathbb{Z} = \text{gcd}(d, n)\mathbb{Z}$ . Hence  $J + n\mathbf{Oz} = \text{ct}^{-1}(\text{gcd}(d, n)\mathbb{Z})$  by (3.2). Preimages commute with intersections. If  $d > 0$ , the term  $n = d$  is  $d\mathbb{Z}$  and all terms contain it; if  $d = 0$ , the intersection is  $\bigcap_{n \geq 1} n\mathbb{Z} = 0$ , since a nonzero integer cannot be divisible by an integer larger than its absolute value. Finally Theorem 6.1 identifies the preimage with  $J + \Pi$ .  $\square$

**Corollary 6.7** (Dense proper principal ideals [08, 09]). *If  $u \in \mathbf{Oz}$  is nonconstant with  $\text{ct}(u) = \pm 1$ , then  $(u)$  is proper and dense for finite congruences, and every unital map from  $\mathbf{Oz}$  to a nonzero set-sized ring sends  $u$  to  $\pm 1$ . Every nonconstant irreducible  $p \in \mathbf{Oz}$  has  $\text{ct}(p) = \pm 1$ ; hence  $(p)$  is proper, profinitely dense, and  $\mathbf{Oz}/(p)$  has no nonzero set-sized ring image.*

*Proof.*  $u$  is a nonunit by Lemma 2.4; apply Theorem 6.6 and Corollary 5.2. For an irreducible  $p = c + f$ ,  $0 \neq f \in \Pi$ : if  $c = 0$  then  $p = 2(p/2)$ , and if  $|c| > 1$  then  $p = c(p/c)$  with  $p/c = 1 + f/c$  nonconstant, both factorizations into nonunits; so  $c = \pm 1$ . In fact even a nonunital additive multiplicative map  $\varphi$  to a set-sized ring that kills  $p$  must be zero: Corollary 5.2 gives  $0 = \varphi(p) = \pm \varphi(1)$ , and then  $\varphi(x) = \varphi(1)\varphi(x) = 0$  for every  $x$ .  $\square$

The converse of the second statement is false:  $\omega^2 - 1 = (\omega - 1)(\omega + 1)$  has constant term  $-1$  and is reducible. The constant term is a complete invariant for set-sized ring observations, not for factorization inside  $\mathbf{Oz}$  (09).

**Corollary 6.8** (A prime quotient invisible to all set-sized rings [06, 08, 09, 11, 15]). *Let  $q = \omega^{\sqrt{2}} + \omega + 1$ . Then  $\mathbf{Oz}/(q)$  is a nonzero proper-class integral domain of characteristic 0; it admits no unital map to a nonzero set-sized ring and no nonzero set-sized unital module.*

*Proof.* L’Innocente and Mantova proved that  $q$  is prime in  $\mathbf{Oz}$  [23, Theorem B], answering Gonshor’s question [20, p. 117]; this primality is imported, not reproved. So  $(q)$  is a proper prime ideal and  $\mathbf{Oz}/(q)$  is a nonzero domain. Since  $\text{ct}(q) = 1$ , Theorem 6.1 gives the zero ring as its universal set-sized image. On a set-sized module, inflated to  $\mathbf{Oz}$ ,  $q$  acts both as  $\text{ct}(q) = 1$  and as 0, so the module is zero. More concretely, Proposition 6.4 supplies sets of pairwise distinct residues of every set cardinality, so the quotient has no set-sized realization. No nonzero integer  $n$  lies in  $(q)$ : if  $n = qa$ , then  $a \neq 0$  and  $0 = \deg_\omega(n) = \sqrt{2} + \deg_\omega(a) > 0$ , since a nonzero omnific integer has nonnegative degree. Thus the map  $\mathbb{Z} \rightarrow \mathbf{Oz}/(q)$  is injective and the domain has characteristic 0.  $\square$

This is no contradiction with the maximal-ideal theorem for set-sized rings: the source quotient is not a set, and the corollary does not say that a particular maximal class ideal fails to exist (06, 09). The same obstruction needs no primality. (Related (batch 35): the proof of Corollary 6.8 uses only that  $q$  is prime and  $\text{ct}(q) = 1$ ; such a  $q$  is nonconstant, hence infinite, which is what keeps the nonzero integers out of  $(q)$ . It therefore applies to each of the continuum many pairwise nonassociate primes  $P_\eta = 1 + \sum_{n \in \mathfrak{B}_\eta} \omega^{\rho_n}$  of  $\mathbf{Oz}$ , with real  $\rho_n$  decreasing to 0, of `isc:hj:main:prime` and `isc:hj:cor:nonassociate` in [80], whose primality is imported there from the one-row theorem of Pitteloud and Biljaković–Kochetov–Kuhlmann: the prime ideals  $(P_\eta)$  are distinct, and each  $\mathbf{Oz}/(P_\eta)$  is a nonzero proper-class domain of characteristic 0 with no unital map to a nonzero set-sized ring and no nonzero set-sized unital module.)

**Proposition 6.9** (Unit-constant ideals [06, 07]). *If a class ideal  $J$  of  $\mathbf{Oz}$  contains an element with constant term 1, then  $\mathbf{Oz}/J$  has no nonzero set-sized unital module and no unital map to a nonzero set-sized ring; if  $J$  is proper,  $\mathbf{Oz}/J$  is a nonzero proper-class ring. More generally, if  $\text{ct}(f) = c$ , the small reflection of  $\mathbf{Oz}/(f)$  is  $\mathbb{Z}/c\mathbb{Z}$ .*

*Proof.* Write  $u \in J$  for the element with  $\text{ct}(u) = 1$ . On a set-sized quotient module, inflated to  $\mathbf{Oz}$ , it acts as the identity by Theorem 5.1 and as zero because  $u \in J$ ; hence the module is zero. Similarly a unital map to a set-sized ring sends  $u$  to both 1 and 0, so that target is the zero ring. If  $J$  is proper, the quotient is nonzero, and its identity map rules out a set-sized realization. For a principal ideal,  $\text{ct}(fa) = c \text{ct}(a)$  and constant representatives realize every integer multiple of  $c$ , so  $\text{ct}((f)) = c\mathbb{Z}$ ; apply Theorem 6.1.  $\square$

For example  $\omega + 1$  and  $1 + \omega^a$  ( $a > 0$ ) are nonunits, so  $\mathbf{Oz}/(1 + \omega^a)$  is a nonzero class ring with no nonzero set-sized image; Section 14 shows how much it contains. The benefit of  $q$  in Corollary 6.8 is that the quotient is a domain (06). Under any unital map to a set-sized ring,  $\omega^a$  becomes 0 and  $1 + \omega^a$  becomes 1, which is the arithmetic reason (07).

**Theorem 6.10** (Arithmetic completions [04, 06, 07, 08, 09, 10, 11, 13, 15]). *All finite quotient rings of  $\mathbf{Oz}$  are the  $\mathbb{Z}/n\mathbb{Z}$ , and*

$$\widehat{\mathbf{Oz}} := \varprojlim_{n \geq 1} \mathbf{Oz}/n\mathbf{Oz} \cong \widehat{\mathbb{Z}}, \quad \varprojlim_{r \geq 1} \mathbf{Oz}/p^r \mathbf{Oz} \cong \mathbb{Z}_p \quad (p \text{ prime}).$$

*The canonical maps are  $\text{ct}$  followed by the usual injections; their kernel is  $\Pi$  and their image is the diagonal copy of  $\mathbb{Z}$ . For the Gaussian ring,  $\widehat{\mathbf{Oz}[i]} \cong \widehat{\mathbb{Z}[i]} = \widehat{\mathbb{Z}}[T]/(T^2 + 1)$  and  $\varprojlim_{r \geq 1} \mathbf{Oz}[i]/\ell^r \mathbf{Oz}[i] \cong \mathbb{Z}_\ell[i] = \mathbb{Z}_\ell[T]/(T^2 + 1)$  for an ordinary prime  $\ell$  (not necessarily a domain), with kernel  $\Pi_{\mathbb{C}}$ .*

*Proof.* Index the first limit by positive integers under divisibility: for  $n \mid m$  the transition map is reduction from modulus  $m$  to modulus  $n$ . The maps induced by  $\text{ct}$  commute with these reductions and identify each  $\mathbf{Oz}/n\mathbf{Oz}$  with  $\mathbb{Z}/n\mathbb{Z}$ . Conversely, Theorem 6.2 lists every finite quotient, including



the zero ring at  $n = 1$ . Applying the compatible coordinate isomorphisms and their inverses gives the isomorphism of limits. Restricting to  $p^r$  gives the second limit. The canonical coordinates are  $\text{ct}(x) \bmod n$ , respectively  $\text{ct}(x) \bmod p^r$ . Their common kernel is  $\Pi$ : a nonzero integer is not divisible by every positive integer, or by every power of a fixed prime. Surjectivity of  $\text{ct}$  gives exactly the diagonal integer image, not all of the completion.

For  $\mathbf{Oz}[i]$ , every finite quotient has positive characteristic  $n$  (take  $n = 1$  for the zero ring), so its kernel contains  $n\mathbf{Oz}[i] = n\mathbb{Z}[i] + \Pi_{\mathbb{C}}$ . Thus the ideals  $n\mathbb{Z}[i]$  are cofinal among finite-index ideals. Each quotient  $\mathbb{Z}[i]/n\mathbb{Z}[i]$  has a unique pair of coordinates  $a + bi$  with  $a, b \in \mathbb{Z}/n\mathbb{Z}$ , and reduction acts on the coordinates separately. Compatible pairs therefore give  $\widehat{\mathbb{Z}} \oplus \widehat{\mathbb{Z}}i$ , with multiplication  $(a, b)(c, d) = (ac - bd, ad + bc)$ ; this is precisely  $\widehat{\mathbb{Z}}[T]/(T^2 + 1)$ . The same coordinate argument along  $\ell^r$  gives  $\mathbb{Z}_{\ell}[T]/(T^2 + 1)$  without any irreducibility assumption on  $T^2 + 1$ . The Gaussian canonical map has coordinates  $\text{ct}(x)$  in these residue rings; its kernel is  $\Pi_{\mathbb{C}}$  and its image is the diagonal copy of  $\mathbb{Z}[i]$ .  $\square$

These isomorphisms also preserve the inverse-limit topologies, since each residue isomorphism is a homeomorphism between finite discrete rings. The diagonal images are dense: finitely many profinite coordinates are simultaneously represented by one residue at their least common multiple, and finitely many prime-power coordinates by one residue at their largest power. Lift that residue to an ordinary integer, or to an ordinary Gaussian integer in the rank-two case. This density statement uses the inverse-limit topology.

The finite quotient system is an ordinary set-indexed system; no inverse limit over a proper class of targets is formed, and  $\mathbf{Oz}$  is not claimed to be an object of the category of small topological rings (08, 09). The finite-congruence topology on  $\mathbf{Oz}$  is not Hausdorff:  $\Pi$  is the closure of 0. (Status (batch 32): this is forced. By `opa:tc:thm:oztopology` in [48], every ring topology on  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  in which every Hahn-summable family of its elements with sum in the ring converges to its sum along finite partial sums is pulled back along  $\text{ct}$  from a ring topology on  $\mathbb{Z}$  or  $\mathbb{Z}[i]$ , and none is Hausdorff; the finite-congruence and  $p$ -adic topologies are such pullbacks (`opa:tc:rem:congruence`.) It is unrelated to the fine order topology and to Hahn valuation topologies (07, 10, 11). For the asserted compatibility with strong sums in these pullback topologies, only finitely many summands have a nonzero constant coefficient. Once those summands are included, every further finite partial sum has the same constant term as the Hahn sum and is therefore in every pulled-back neighborhood of it. This proves compatibility directly; it does not claim convergence in a Hahn valuation topology. Every  $g \in \Pi$  is divisible in  $\mathbf{Oz}$  by every power of  $p$ , but this does not represent  $g$  as a nonzero  $p$ -adic integer; for example  $\omega^{\omega} + \pi\omega + 17$  maps to 17 in every finite quotient and in every  $\mathbb{Z}_p$  (07). The finite-quotient and completion statements were already in the Diophantine manuscripts of the sibling report [47] (04 cites them as prior); the universal theorem explains why passing from finite targets to arbitrary set-sized targets recovers nothing more.

### 6.3 Localizations and presentations

**Proposition 6.11** (Adjoining one monomial inverse [04, 07]). *For  $a > 0$ ,*

$$\mathcal{A}_{D,k}[X^{-a}] = \{x \in k((X^{\mathbf{No}})) : \text{supp } x \subseteq [-Na, \infty) \text{ for some } N \in \mathbb{N}\},$$

*independently of  $D$ . This is a proper subring of  $k((X^{\mathbf{No}}))$  and not a field; it admits no unital map to a nonzero set-sized ring (Corollary 5.9). In particular  $\mathbf{Oz}[\omega^{-a}]$  is a nonzero class domain, not a field, with no nonzero set-sized unital module.*



*Proof.* The powers of  $X^a$  are nonzero, so the localization embeds in the ambient field. Every fraction is  $fX^{-Na}$  with  $f \in \mathcal{A}_{D,k}$  and  $N \in \mathbb{N}$ ; its support is bounded below by  $-Na$ . Conversely, if  $\text{supp } x \subseteq [-Na, \infty)$ , then  $X^{(N+1)a}x$  has strictly positive support, so  $x = X^{(N+1)a}x/X^{(N+1)a}$  lies in the localization; the extra factor  $X^a$  removes the restriction on the coefficient at the lower endpoint, which explains the independence from  $D$ . The exponent  $a\omega$  exceeds every  $Na$ , so  $X^{-a\omega}$  is not in the ring although  $X^{a\omega}$  is. The missing inverse proves both properness and failure to be a field. The ring is a nonzero domain because it is a subring of the ambient field. Since  $X^a \in \Pi_k$  is now invertible, [Corollary 5.9](#) gives the target and module obstructions.  $\square$

The same argument shows  $\omega^{-\omega} \notin \mathbf{Oz}[1/\omega]$ :  $\omega^{-\omega} = a/\omega^n$  would force  $a = \omega^{n-\omega} \notin \mathbf{Oz}$  (11).

**Theorem 6.12** (Quotient-localization calculus [08, 09, 11, 15]). *Let  $R = \mathcal{A}_{D,k}$ ,  $J$  a class ideal and  $S \subseteq R$  a multiplicative class containing 1. Put  $D_0 = D/\text{ct}(J)$  and  $T = \{\text{ct}(s) + \text{ct}(J) : s \in S\} \subseteq D_0$ . For every set-sized commutative ring  $B$ ,*

$$\text{Hom}_1((R/J)[S^{-1}], B) \cong \text{Hom}_1(D_0[T^{-1}], B);$$

*if  $0 \in T$  the right side is empty for  $B \neq 0$ . For  $R = \mathbf{Oz}$ ,  $J = 0$  and  $S$  a set-sized multiplicative subset, the same holds for noncommutative  $B$ , with universal formula  $f/s \mapsto \text{ct}(f)1_B(\text{ct}(s)1_B)^{-1}$ .*

*Proof.* Here all displayed maps preserve the identity, and  $S$  in the left localization means its image in  $R/J$ . A unital map from the left ring is a unital map  $h : R \rightarrow B$  killing  $J$  and sending each  $s \in S$  to a unit. By [Theorem 5.1](#),  $h = g \circ \text{ct}$  uniquely, and the conditions become  $g(\text{ct}(J)) = 0$  and  $g(\text{ct}(s)) \in B^\times$ , which are exactly the conditions for a map  $D_0[T^{-1}] \rightarrow B$ ; the sets  $\text{ct}(J)$  and  $T$  are sets even when  $J$  and  $S$  are classes. Explicitly the natural map sends  $f/s$  to  $[\text{ct}(f)]/[\text{ct}(s)]$ . Every denominator of the target is represented by some  $s \in S$ , and every numerator by a constant  $d \in D$ , so the map is surjective. If  $0 \in T$ , inverting it gives  $0 = 1$ ; the localized target is then the zero ring, whose only unital maps are to the zero ring. This explains the stated empty Hom-set for nonzero  $B$ . For the noncommutative assertion the denominators  $\text{ct}(s)1_B$  are central. If they are units, their inverses are central too, and the fraction formula respects the localization relation; if any is not a unit there is no such map. Uniqueness follows because a map must send  $s^{-1}$  to the inverse of its prescribed image (08).  $\square$

**Example 6.13** (Three effects [08, 11, 15]).  $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/(\omega + 2)]) = \mathbb{Z}[1/2]$ ;  $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/\omega]) = 0$ ;  $\text{Ref}_{\text{set}}(\mathbf{Oz}/(\omega + 1)) = 0$ ; and  $\text{Ref}_{\text{set}}(\mathbf{Oz}[1/n]) = \mathbb{Z}[1/n]$  for a nonzero ordinary integer  $n$ . The second ring is a nonzero domain inside  $\mathbf{No}$  and the third is nonzero because  $\omega + 1$  is not a unit; both have no nonzero set-sized representation. A nonzero class ring can therefore have no nonzero set-sized unital module, so set-sized existence arguments for modules cannot be applied to a proper class of class ideals (11). If  $\text{ct}(s) = 0$  for some  $s \in S$ , the displayed localization of  $\mathbb{Z}$  is the zero ring (08). In the language of 09: no unital map to a nonzero set-sized ring can make a nonzero  $f \in \Pi_k$  invertible or make an element with unit constant term zero.

**Theorem 6.14** (Algebraic points forget infinite coefficients [06, 09, 11, 15]). *Let  $X$  be a set of commuting variables,  $(f_j)_{j \in E}$  a set-indexed family in  $\mathcal{A}_{D,k}[X]$ ,  $B = \mathcal{A}_{D,k}[X]/(f_j)$  and  $B_0 = D[X]/(\text{ct}_* f_j)$ , where  $\text{ct}_*$  applies  $\text{ct}$  to coefficients. For every set-sized commutative ring  $S$  the natural map  $B \rightarrow B_0$  induces  $\text{Hom}_1(B_0, S) \cong \text{Hom}_1(B, S)$ ; thus  $\text{Ref}_{\text{set}}(B) = B_0$ .*

*Proof.* All maps here are unital. A map  $\mathcal{A}_{D,k}[X] \rightarrow S$  is specified by a map on  $\mathcal{A}_{D,k}$  and images of the commuting variables. Its coefficient map factors uniquely through  $\text{ct}$ . Polynomial evaluation is a finite sum of finite products, so the values of  $f_j$  and  $\text{ct}_* f_j$  agree for every  $j$ . Vanishing of

these relations is exactly the condition to factor through the corresponding quotient; this gives both directions of the claimed correspondence and their uniqueness. The ring  $D[X]$  is a set of finite polynomial expressions, and its ideal generated by the relations is a set, so  $B_0$  is a set. The coefficient map fixes constants and variables, hence is surjective before and after passing to the quotients.  $\square$

06 states this for finitely many variables and 11 for a set of variables; 09 states the corresponding reduction of localization and relation problems. The relation  $\omega T = 1$  reduces to  $0 = 1$ , so it has no solution in a nonzero set-sized ring under any unital representation of its coefficients, although  $T = \omega^{-1}$  solves it in **No** (06). For  $B = \mathbf{Oz}[T]/(T^2 - \omega)$  one gets  $\text{Ref}_{\text{set}}(B) = \mathbb{Z}[T]/(T^2)$ : a set-sized field sends  $T$  to 0 and a set-sized ring may send it to any square-zero element, while the actual root  $\omega^{1/2} \in \mathbf{Oz}$  gives a map to the class ring **Oz** (11). These are universal-property statements, not quantifier elimination, not a reconstruction of class-valued solutions, and not an assertion that a scheme with a proper-class coordinate ring is an ordinary scheme (06, 11).

## 6.4 Diophantine observation

If  $P \in \mathbb{Z}[X_1, \dots, X_m]$ ,  $x \in \mathbf{Oz}^m$  and  $\varphi : \mathbf{Oz} \rightarrow S$  is a unital set-target map, then  $\varphi(P(x)) = P(\text{ct}(x_1)1_S, \dots, \text{ct}(x_m)1_S)$ , even for noncommutative  $S$ , since the coordinates are integer scalars (04). This does *not* say that  $P(x) = 0$  whenever  $P(\text{ct}(x)) = 0$  (take  $P(X) = X$  and  $x = \omega$ ), and it does not classify infinite points of curves; inequalities, supports, valuations and nonmultiplicative coefficient observations remain available (04).

**Proposition 6.15** (Conservativity for integer coefficients [08, 13]). *A finite system of polynomial equations with coefficients in  $\mathbb{Z}$  has a solution in **Oz** if and only if it has one in  $\mathbb{Z}$ ; more generally, positive existential ring formulas with parameters in  $\mathbb{Z}$  have the same truth value in **Oz** and  $\mathbb{Z}$ . With an inequation this fails:  $\exists x \exists y (x^2 = 2y^2 \wedge y \neq 0)$  holds in **Oz** (take  $x = \sqrt{2}\omega$ ,  $y = \omega$ ) but not in  $\mathbb{Z}$ .*

*Proof.* Both homomorphisms  $\text{ct}$  and  $\mathbb{Z} \hookrightarrow \mathbf{Oz}$  fix every integer parameter. They commute with evaluation of ring terms, hence preserve atomic equations. Conjunction and disjunction preserve this implication, and an existential witness is sent to its image under the homomorphism. Induction on the finite formula therefore proves preservation of positive existential formulas in each direction. For a system of equations, the forward direction simply takes the constant term of each coordinate and the reverse direction includes the integer solution. A nonzero integer solution of  $x^2 = 2y^2$  would give a rational square root of 2;  $\text{ct}$  maps the omnific witnesses to  $x = y = 0$ , preserving the equation but not the inequation.  $\square$

The same retraction argument applies to  $D \hookrightarrow \mathcal{A}_{D,k} \rightarrow D$  for every coefficient pair in the report: positive existential formulas with parameters in  $D$  have the same truth value in the two rings. In particular it applies to  $\mathbb{Z}[i] \hookrightarrow \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$ . This is a direct consequence of the proof, not a statement about formulas with negations or unrestricted parameters.

The parameter restriction is essential (the coefficient  $\omega^\gamma + 1$  of [Theorem 15.25](#) is not an integer), and so is positivity (08). The sibling report [47] proves the same constant-term transfer for equations in its Diophantine setting.

## 7 Modules and derivations over the full class ring

All modules in this section and on the finite side below are unital. A derivation is taken relative to the specified module action, not merely to its underlying set.

**Theorem 7.1** (Set-sized modules [04, 06, 07, 08, 09, 10, 11, 13, 15]). *Let  $A = \mathcal{A}_{D,k}$ . Restriction along  $D \hookrightarrow A$  and inflation along  $\text{ct}$  are mutually inverse equivalences between set-sized unital  $A$ -modules and set-sized unital  $D$ -modules; the action of  $A$  on a set-sized module is  $(d + f)m = dm$  ( $d \in D, f \in \Pi_k$ ). The equivalence is the identity on underlying abelian groups and on morphisms, and it is exact. In particular set-sized  $\mathbf{Oz}$ -modules are exactly abelian groups with  $xm = \text{ct}(x)m$ , and set-sized  $\mathbf{Oz}[i]$ -modules are exactly  $\mathbb{Z}[i]$ -modules inflated along  $\text{ct}$ .*

*Proof.* Theorem 5.1(iv) gives  $\Pi_k M = 0$  and the action formula. Conversely, a  $D$ -module inflates along  $\text{ct}$ , and restriction recovers the  $D$ -action because  $\text{ct}|_D = \text{id}$ . An additive map between two such modules is  $A$ -linear exactly when it is  $D$ -linear: if  $a = d + f$ , then  $h(am) = h(dm) = dh(m) = ah(m)$ . Restricting an inflated action recovers the original action, and inflating a restricted action recovers it by the displayed formula. Kernels and cokernels remain set-sized and carry the induced actions. Exactness is equality of the same underlying image and kernel subgroups, so the correspondence preserves and reflects exact sequences.  $\square$

A  $\mathbf{Oz}[i]$ -module structure on an abelian group is the choice of an endomorphism  $J$  with  $J^2 = -\text{id}$ , with inverse  $-J$  (08). Category language abbreviates the correspondence of individual objects and morphisms: each action is coded by its restriction to the set  $D \times M$ , and no class of arbitrary class actions is formed (04, 08, 11). The regular module  $A$  is a proper class on which  $\Pi_k$  acts nontrivially, so there is no conflict with recovering a ring from its regular module (04). The equivalence can transport extension problems between the two small-module categories, but it should not be rephrased as a statement about homological dimensions computed with arbitrary class-sized modules: the rank-one free module  $A$  is not an object of the small-module category (07). Section 12 treats the homological algebra over set-sized integer parts.

**Proposition 7.2** (An additive splitting that is not a module splitting [08, 15]). *The exact sequence of class  $\mathbf{Oz}$ -modules  $0 \rightarrow \Pi \rightarrow \mathbf{Oz} \xrightarrow{\text{ct}} \mathbb{Z} \rightarrow 0$ , with  $\mathbb{Z}$  carrying the action  $x \cdot n = \text{ct}(x)n$ , does not split as a sequence of  $\mathbf{Oz}$ -modules, although it splits as a sequence of additive groups and of rings.*

*Proof.* The inclusion  $n \mapsto n$  is an additive and unital ring section of  $\text{ct}$ . It need not be a module section: a module section  $s$  would give  $e = s(1)$  with  $\text{ct}(e) = 1$  and  $ue = s(u \cdot 1) = s(0) = 0$  for every  $u \in \Pi$ . With  $u = \omega$  and  $\mathbf{Oz}$  a domain,  $e = 0$ , a contradiction.  $\square$

**Proposition 7.3** (Projectivity depends on the size convention [15]). *With the constant-term action,  $\mathbb{Z}$  is projective in the category of set-sized  $\mathbf{Oz}$ -modules, although  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  has no  $\mathbf{Oz}$ -linear section (Proposition 7.2). For every set-sized  $\mathbf{Oz}$ -module  $M$ ,  $\text{Hom}_{\mathbf{Oz}}(M, \mathbf{Oz}) = 0$ .*

*Proof.* By Theorem 7.1 the category in question is the category of abelian groups. Explicitly, given a surjection  $g : M \rightarrow N$  in that category and a map  $h : \mathbb{Z} \rightarrow N$ , choose  $m_1 \in M$  with  $g(m_1) = h(1)$ . Then  $n \mapsto nm_1$  is a lift of  $h$ ; it is  $\mathbf{Oz}$ -linear because both actions factor through  $\text{ct}$ . This proves projectivity within the stated category. The nonsplit surjection  $\text{ct} : \mathbf{Oz} \rightarrow \mathbb{Z}$  lies outside it, since its domain is a proper class. If  $f : M \rightarrow \mathbf{Oz}$  is linear and  $m \in M$ , then  $\omega f(m) = f(\omega m) = f(0) = 0$ , so  $f(m) = 0$  in the domain  $\mathbf{Oz}$ .  $\square$

A nonzero free  $\mathbf{Oz}$ -module contains an injective copy of  $\mathbf{Oz}$  on any one basis vector, so it cannot be set-sized. The set-sized module category nevertheless has many projectives; this is a warning against importing the size convention of ordinary module theory, not a Morita statement (15).

**Theorem 7.4** (Derivations into set-sized modules [04, 06, 08, 09, 10, 11, 13]). *Let  $A = \mathcal{A}_{D,k}$  and  $M$  a set-sized  $A$ -module. Every derivation  $\partial : A \rightarrow M$  (an additive class map with  $\partial(xy) = x\partial(y) + y\partial(x)$ ) vanishes on  $\Pi_k$ , and restriction gives a natural bijection  $\text{Der}_{\mathbb{Z}}(A, M) \cong \text{Der}_{\mathbb{Z}}(D, M)$  with inverse  $\delta \mapsto \delta \circ \text{ct}$ . Consequently:*

- (i) every derivation  $\mathbf{Oz} \rightarrow M$  is zero;
- (ii)  $\text{Der}_{\mathbb{Z}}(\mathbf{Oz}[i], M) \cong M[2] = \{m \in M : 2m = 0\}$  via  $\partial \mapsto \partial(i)$ , and  $\text{Der}_{\mathbb{Z}[i]}(\mathbf{Oz}[i], M) = 0$ ;
- (iii) every  $k$ -linear derivation of  $\mathcal{A}_{k,k}$  or of  $\mathcal{O}_k$  ( $k = \mathbb{R}, \mathbb{C}$ ) into a set-sized module is zero, and  $\text{Der}_D(A, M) = 0$ .

*Proof.*  $\Pi_k M = 0$  by Theorem 7.1, and each  $x \in \Pi_k$  is a product  $uv$  with  $u, v \in \Pi_k$  (Proposition 3.3), so  $\partial(x) = u\partial(v) + v\partial(u) = 0$ . Thus  $\partial(d + x) = \partial(d)$ , proving that restriction is injective. Also  $\partial(1) = \partial(1 \cdot 1) = 2\partial(1)$ , so  $\partial(1) = 0$  and additivity forces  $\partial(n) = 0$  for every integer  $n$ . This explains the subscript  $\mathbb{Z}$ . Conversely, for a derivation  $\delta$  of  $D$ ,  $\delta \circ \text{ct}$  satisfies the Leibniz rule because  $A$  acts on  $M$  through  $\text{ct}$ . A derivation of  $\mathbb{Z}$  therefore vanishes. For  $\mathbb{Z}[i]$ , write  $m = \partial(i)$ ; additivity forces  $\partial(a + bi) = bm$ . Differentiating  $i^2 + 1 = 0$  gives  $2im = 0$ , equivalently  $2m = 0$ , since  $i$  acts invertibly. Conversely, if  $2m = 0$ , then

$$(a + bi)(dm) + (c + di)(bm) = (ad + bc)m + 2bd im = (ad + bc)m,$$

which is the derivative of  $(a + bi)(c + di)$  under this formula. It kills  $\mathbb{Z}[i]$  exactly when  $m = 0$ . All these identifications commute with module homomorphisms, which gives naturality. A  $D$ -linear derivation kills  $D$ , hence kills  $A$ ; a  $k$ -linear derivation of  $k$  vanishes for the same reason. On the finite side, Theorem 8.2 applied to the endomorphism ring of the underlying group makes  $\mathfrak{m}_k$  act by zero. Its idempotence (Lemma 8.4) then kills every infinitesimal under a derivation, so the same argument applies.  $\square$

**A second route (09).** Since  $M$  is a  $D$ -module, the square-zero extension  $D \ltimes M$  with  $(d, m)(e, n) = (de, dn + em)$  is a set-sized ring, and  $a \mapsto (\text{ct}(a), \partial(a))$  is a unital ring map  $A \rightarrow D \ltimes M$ : additivity is componentwise, the Leibniz rule is exactly the second coordinate of multiplicativity, and  $\partial(1) = 0$  gives the identity  $(1, 0)$ . The universal theorem forces this map to factor through  $D$ , so its second coordinate vanishes on  $\Pi_k$ .

**Example 7.5** (The ramification at 2 survives [09, 11]). Give  $\mathbb{F}_2$  the  $\mathbb{Z}[i]$ -module structure  $i \mapsto 1$ . Then  $\partial(a + bi + f) = b \bmod 2$  is a nonzero  $\mathbb{Z}$ -derivation  $\mathbf{Oz}[i] \rightarrow \mathbb{F}_2$ . Indeed the coefficient action is  $a + bi \mapsto a + b \bmod 2$ , which respects  $i^2 + 1 = 0$  in characteristic two;  $\partial(i) = 1$  and the preceding product calculation proves Leibniz. The statement “all set-valued derivations of the Gaussian omnific ring vanish” is false; only the positive-support directions vanish, and the surviving derivation reflects ordinary ramification of  $\mathbb{Z}[i]$  at 2.

The coefficient-linearity qualification in (iii) cannot be dropped: the theorem reduces arbitrary derivations of  $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$  or  $\mathcal{O}_{\mathbb{R}}$  to derivations of  $\mathbb{R}$  over  $\mathbb{Q}$ , and such derivations can be nonzero (04). To see this algebraically, choose a transcendence basis  $T$  of  $\mathbb{R}/\mathbb{Q}$  and  $t_0 \in T$ . The basis is nonempty, since the algebraic numbers are countable and  $\mathbb{R}$  is not. On  $\mathbb{Q}(T)$  prescribe  $\delta(t_0) = 1$  and  $\delta(t) = 0$  for  $t \neq t_0$ , and use formal differentiation and the quotient rule. This derivation extends uniquely to  $\mathbb{R}$ :

on adjoining a separable algebraic element  $\alpha$  with minimal polynomial  $p$ , set  $\delta(\alpha) = -(\delta_* p)(\alpha)/p'(\alpha)$ . The denominator is nonzero in characteristic zero. Choose a polynomial representative modulo  $p$  for this value of  $\delta(\alpha)$  and use it as the derivative of the indeterminate. The resulting derivation on the polynomial ring preserves  $(p)$  and therefore descends to the simple extension. Iteration handles each finite subextension, and uniqueness makes these extensions agree on intersections. Their union gives a derivation on  $\mathbb{R}$  still satisfying  $\delta(t_0) = 1$ . No continuity or canonical choice is asserted. For a set-sized commutative  $D$ -algebra  $B$ , every  $D$ -algebra map  $A \rightarrow B[\varepsilon]/(\varepsilon^2)$  kills  $\Pi_k$ , so relative first-order deformations with set-sized coefficients cannot detect  $\Pi_k$ ; no assertion about an absolute proper-class module of Kähler differentials is made (11).

**Proposition 7.6** (Class-valued derivations [06, 08, 10]). *For  $\eta \in \mathbf{No}$  let  $\ell_\eta : \mathbf{No} \rightarrow \mathbb{R}$  extract the coefficient of  $\omega^\eta$  from the normal form of its argument; it is additive. The formula*

$$D_\eta\left(\sum_x r_x \omega^x\right) = \sum_x \ell_\eta(x) r_x \omega^x$$

*defines a derivation  $\mathbf{No} \rightarrow \mathbf{No}$  which restricts to a derivation  $\mathbf{Oz} \rightarrow \Pi$  into the class module  $\Pi$ , preserves strongly summable families, and satisfies  $D_\eta(\omega^{\omega^\theta}) = \omega^{\omega^\theta}$  if  $\eta = \theta$  and 0 otherwise. Every finite family of distinct  $D_\eta$  is linearly independent over  $\mathbb{R}$ , and  $D_0(\omega) = \omega$ . Coefficientwise complexification gives the analogous derivations of  $\mathbf{No}[i]$  and  $\mathbf{Oz}[i]$ .*

*Proof.* Coefficient extraction on the normal form of an exponent is additive, so  $\ell_\eta(x + y) = \ell_\eta(x) + \ell_\eta(y)$ . The output support of  $D_\eta$  is a subset of its input support, hence remains a reverse well-ordered set. The exponent 0 is killed because  $\ell_\eta(0) = 0$ , and no negative exponent appears for an omnific input; this proves that the restricted image lies in  $\Pi$ . The map is  $\mathbb{R}$ -linear, since it multiplies each coefficient by a fixed real number. For a strongly summable family, the output support union is contained in the original union, and each coefficient still receives only finitely many contributions. Distributing  $\ell_\eta(z)$  over that finite sum proves both summability of the differentiated family and commutation with its strong sum.

For the product of two normal forms, the Hahn support condition makes the coefficient sum over pairs  $x + y = z$  finite. At each such pair,  $\ell_\eta(z)r_x s_y = \ell_\eta(x)r_x s_y + r_x \ell_\eta(y)s_y$ ; summing proves Leibniz without exchanging non-summable families. The exponent  $\omega^\theta$  has coefficient 1 at  $\theta$  and 0 elsewhere, giving the displayed values. If  $\sum_{j=1}^n c_j D_{\eta_j} = 0$  with distinct  $\eta_j$ , evaluation at  $\omega^{\omega^{\eta_j}}$  gives  $c_j \omega^{\omega^{\eta_j}} = 0$ ; the monomial is nonzero, so every  $c_j = 0$ . Finally  $\ell_0(1) = 1$ . For complexification,  $D_\eta(x + iy) = D_\eta(x) + iD_\eta(y)$ ; expanding a product proves the complex Leibniz identity and gives image in  $\Pi_{\mathbb{C}}$  on  $\mathbf{Oz}[i]$ .  $\square$

The proposition combines three sources, none of which states all of it: 08 introduces the family  $D_\eta$  for every  $\eta$ , as derivations  $\mathbf{Oz} \rightarrow \Pi$ , with the diagonal values and linear independence; the case  $\eta = 0$  is 06's Euler-type derivation  $\mathcal{D}$  (06 defines it on  $\mathbf{No}$ , with the complex extension) and 10's exponent-weighted derivation  $D_h$  ( $h = \ell_0$ ; 10 records preservation of strongly summable families). The general- $\eta$  statements on  $\mathbf{No}$  and on strong sums follow by the same coefficientwise proof. There is no conflict with Theorem 7.4: the target  $\Pi$  is a proper-class module with its internal multiplication. Berarducci and Mantova's surreal derivation with  $\partial\omega = 1$  [6, Theorem A and the following normalization, p. 2], restricted to  $\mathbf{Oz}$  with values in  $\mathbf{No}$ , is another nonzero class-valued boundary example (09). None of the  $D_\eta$  is asserted to be the Berarducci–Mantova derivation, a fine derivative, or compatible with surreal exponentiation (06, 08, 10), and no uniqueness is claimed for the established derivation (09). In differential-field terminology a “small derivation” preserves infinitesimals [6, Theorem B]; that condition does not mean its target is a set. The word “small” in Theorem 7.4 cannot be dropped.



## 8 The finite side: standard part is universal

This section is manuscript 04's; no other source treats the negative-support side. Its proof is shorter because the ambient valuation produces the needed quotients directly, and it is a different route rather than a sign reversal of the positive side (04). Let  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $D \subseteq k$  a unital subring, and for  $x \neq 0$  let  $v(x) = -\deg_\omega(x)$ , so that  $v(x/y) = v(x) - v(y)$ ; a nonzero normal form lies in  $\mathfrak{m}_k$  exactly when  $v(x) > 0$ .

**Lemma 8.1** (Division of an infinitesimal by small differences [04]). *Let  $0 \neq \varepsilon \in \mathfrak{m}_k$  and  $a = v(\varepsilon) > 0$ . For  $0 < \gamma < \delta < a$ ,  $\varepsilon/(\omega^{-\gamma} - \omega^{-\delta}) \in \mathfrak{m}_k$ .*

*Proof.* The two exponents are distinct, and  $-\gamma > -\delta$ , so their difference is nonzero with largest exponent  $-\gamma$  and valuation  $\gamma$ . Division takes place in the ambient field. The quotient has valuation  $a - \gamma > 0$ , i.e. its largest exponent is negative and therefore every supported exponent is negative. It consequently lies in  $\mathfrak{m}_k$ , not merely in the ambient field.  $\square$

**Theorem 8.2** (Universal set-sized residue on the finite side [04]). *For every set-sized unital ring  $S$ , restriction gives a bijection  $\text{Hom}_1(\mathcal{B}_{D,k}, S) \rightarrow \text{Hom}_1(D, S)$  with inverse  $\psi \mapsto \psi \circ \text{st}$ . Every such homomorphism annihilates  $\mathfrak{m}_k$ . In particular every unital map from  $\mathcal{O}_{\mathbb{R}}$  or  $\mathcal{O}_{\mathbb{C}}$  to a set-sized ring factors through  $\text{st} : \mathcal{O}_{\mathbb{R}} \rightarrow \mathbb{R}$ , respectively  $\text{st} : \mathcal{O}_{\mathbb{C}} \rightarrow \mathbb{C}$ .*

*Proof.* Suppose  $0 \neq \varepsilon \in \mathfrak{m}_k$  and  $\varphi(\varepsilon) \neq 0$ , put  $a = v(\varepsilon)$  and let  $H$  be the Hartogs ordinal of the underlying set of  $S$ . The explicit exponents  $\gamma_\beta = a/\omega^{\beta+1}$ ,  $\beta < H$ , are distinct and lie in  $(0, a)$ ; class replacement makes this a set-indexed family (Lemma 3.1). If two images  $\varphi(\omega^{-\gamma_\beta})$  agreed, order the two exponents and apply Lemma 8.1. It writes  $\varepsilon$  as their monomial difference times an element of  $\mathfrak{m}_k \subseteq \mathcal{B}_{D,k}$ . Its image would then be zero, a contradiction. Thus  $\beta \mapsto \varphi(\omega^{-\gamma_\beta})$  injects  $H$  into  $S$ , contradicting the definition of  $H$ . Hence  $\varphi(\mathfrak{m}_k) = 0$  and  $\varphi(d + \varepsilon) = \varphi(d)$ . The unique factor is the restriction to  $D$ , and conversely  $\psi \circ \text{st}$  extends every  $\psi$  because  $\text{st}$  fixes constants. This argument neither sums images of an infinite family nor imposes any topology on the target.  $\square$

The collision argument also works for a nonunital additive multiplicative map: it only uses additivity and one finite product identity. Such maps likewise factor uniquely through  $\text{st}$ , with a possibly nonunital map on  $D$ .

For a target without a specified coefficient-field structure, the induced map on  $\mathbb{R}$  or  $\mathbb{C}$  is not presumed to be the identity: the assertion is factorization through the residue field, not uniqueness of maps from  $\mathcal{O}_{\mathbb{C}}$  to an arbitrary ring (04). The statement involves no topology on  $S$  and so is stronger than a statement about continuous maps into topological rings; it must also be distinguished from the existence of standard part, which is classical and part of the repository's formal infrastructure [51].

**Corollary 8.3** (Quotients, modules and derivations on the finite side [04]). *The kernels of  $\mathcal{B}_{D,k}$  admitting a set-sized quotient are exactly  $\mathfrak{m}_k + \mathfrak{a}$  for ideals  $\mathfrak{a}$  of  $D$ ; the only nonzero set-sized quotients of  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$  are  $\mathbb{R}$  and  $\mathbb{C}$ , up to the residue identification. Set-sized unital  $\mathcal{B}_{D,k}$ -modules are exactly set-sized unital  $D$ -modules inflated along  $\text{st}$  (set-sized  $\mathcal{O}_k$ -modules are  $k$ -vector spaces), no faithful set-sized module exists, and  $\text{Der}_{\mathbb{Z}}(\mathcal{B}_{D,k}, M) \cong \text{Der}_{\mathbb{Z}}(D, M)$  for set-sized  $M$ . For  $0 \neq \varepsilon \in \mathfrak{m}_k$ , the localization  $\mathcal{B}_{D,k}[\varepsilon^{-1}]$  admits no unital map to a nonzero set-sized ring and no nonzero set-sized unital module.*



*Proof.* A set-sized quotient map kills  $\mathfrak{m}_k$  by [Theorem 8.2](#). Its kernel is therefore the full preimage of its intersection  $\mathfrak{a}$  with  $D$ , namely  $\mathfrak{m}_k + \mathfrak{a}$ ; conversely  $\text{st}$  followed by  $D \rightarrow D/\mathfrak{a}$  realizes each such quotient. If  $D = k$ , the field has only the ideals 0 and  $k$ , so its only nonzero quotient is  $k$  itself.

For a set-sized unital module  $M$ , the action is a unital map to the set-sized ring  $\text{End}_{\mathbb{Z}}(M)$ . The finite-side theorem makes  $\mathfrak{m}_k$  act by zero, and restriction and inflation recover each other exactly as on the positive side. The element  $\omega^{-1} \neq 0$  lies in  $\mathfrak{m}_k$ , so no such module is faithful. By [Lemma 8.4](#), every infinitesimal is a product of two infinitesimals; the Leibniz rule therefore kills it. Restriction of a derivation to  $D$  and composition with  $\text{st}$  are inverse, since the action factors through  $\text{st}$ . Finally, inverting a nonzero infinitesimal makes a killed element a unit; a unital map to a nonzero set-sized ring is impossible. A nonzero set-sized unital module would supply such a map to its nonzero endomorphism ring.  $\square$

**Lemma 8.4** (Idempotence on the finite side [04]).  $\mathfrak{m}_k^2 = \mathfrak{m}_k$  as an ideal of  $\mathcal{B}_{D,k}$ .

*Proof.* For  $0 \neq \varepsilon \in \mathfrak{m}_k$  with  $a = v(\varepsilon)$ ,  $\varepsilon = \omega^{-a/2}(\omega^{a/2}\varepsilon)$  with both factors of valuation  $a/2 > 0$ . The zero element is  $0 \cdot 0$ , so every element lies in  $\mathfrak{m}_k^2$ . Conversely  $\mathfrak{m}_k^2 \subseteq \mathfrak{m}_k$  because  $\mathfrak{m}_k$  is an ideal. In particular, the proof gives a single product for each element, which is stronger than a finite sum of products.  $\square$

**Two residues that cannot be combined (04).** Every element of  $k((X^{\mathbf{No}}))$  has a unique decomposition along  $\Pi_k \oplus k \oplus \mathfrak{m}_k$ , obtained by splitting its support at 0; so  $\mathcal{A}_{k,k} \cap \mathcal{O}_k = k$  and  $\mathcal{A}_{k,k} + \mathcal{O}_k$  is the whole field additively. The two constant-coefficient maps agree on the intersection, yet no unital ring map on the field extends both:

$$\text{ct}(\omega) = 0, \quad \text{st}(\omega^{-1}) = 0, \quad \omega \cdot \omega^{-1} = 1. \quad (8.1)$$

The obstruction is multiplicative, not an ambiguity of the additive decomposition, and [Corollary 5.9](#) sharpens it: adjoining one inverse to the positive-support ring already destroys every nonzero set-sized target.

**Relation to the rotation groups.** The euclidean-three-space report proves that every class homomorphism from  $\text{SO}(3, \mathbf{No})$  to a set-sized group factors through standard part (`e3:cut:thm:smallquotient`), that every nontrivial set-sized image is  $\text{SO}(3, \mathbb{R})$  (`e3:cut:cor:onlyimage`), and the analogues for  $\text{SO}(n, \mathbf{No})$ ,  $n \geq 3$ , and  $\text{SU}(n, \mathbf{No}[i])$ ,  $n \geq 2$  [44]. [Theorem 8.2](#) is the ring analogue on the finite side; neither result implies the other. The group proof uses exact commutators and normal generation before a choice-free Hartogs contradiction; the ring proof uses divisibility by differences of monomials. Manuscripts 06, 07, 08, 09, 10 and 11 acknowledge the rotation theorem as the repository’s conceptual predecessor for universal set-sized quotients, 04 refers to related size obstructions in other algebraic settings, and none of them uses the rotation theorem as a lemma.

## 9 The support threshold: finite supports fail, countable supports suffice

The universal theorem does not need normal forms of every set cardinality. Keep the *full* exponent class  $\mathbf{No}$  and bound only the size of each individual support. For  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $D \subseteq k$  unital, and an infinite cardinal  $\lambda$ , put

$$\mathcal{A}_{D,k}^{<\lambda} = D + \{f \in \Pi_k : |\text{supp } f| < \lambda\}, \quad \mathcal{B}_{D,k}^{<\lambda} = D + \{f \in \mathfrak{m}_k : |\text{supp } f| < \lambda\},$$

and, in 11's notation,  $R_{\leq \lambda} = \mathcal{A}_{D,k}^{\leq \lambda +}$  and  $R_{\text{fin}} = \mathcal{A}_{D,k}^{\leq \aleph_0}$ . These are rings: the support of a sum lies in a finite union and that of a product in the image of the product of the two supports under addition, and for an infinite  $\lambda$  finite unions and products of sets of size  $< \lambda$  (or  $\leq \lambda$ ) keep that bound; regularity of  $\lambda$  is not needed. They remain proper classes: the positive-side ring contains every  $\omega^a$  with  $a > 0$ , and the finite-side ring every  $\omega^{-a}$  with  $a > 0$  (04, 11). In each case the bound is on one element's support, not on the number of elements of the ring.

**Theorem 9.1** (Support threshold [04, 11; 03, 07, 09]).

(i) For every uncountable cardinal  $\lambda$  and every set-sized unital ring  $S$ , restriction induces bijections

$$\text{Hom}_1(\mathcal{A}_{D,k}^{\leq \lambda}, S) \cong \text{Hom}_1(D, S), \quad \text{Hom}_1(\mathcal{B}_{D,k}^{\leq \lambda}, S) \cong \text{Hom}_1(D, S),$$

with inverses given by *ct* and *st*. In particular countably supported normal forms ( $\lambda = \aleph_1$ , i.e.  $R_{\leq \aleph_0}$ ) already have universal set-sized quotient  $D$  on both sides, and so does  $R_{\leq \mu}$  for every infinite  $\mu$ .

(ii) For finite supports both assertions fail: the augmentation

$$E\left(d + \sum_{g \in F} c_g \omega^g\right) = d + \sum_{g \in F} c_g \quad (F \text{ finite})$$

is a unital ring homomorphism from each finite-support ring to  $k$ , sending every monomial in its domain to 1: in particular  $\omega$  on the positive side and  $\omega^{-1}$  on the finite side; it does not factor through *ct* or *st*.

(iii) (Extension criterion.) For a set-sized unital ring  $S$  and a unital map  $h : R_{\text{fin}} \rightarrow S$  the following are equivalent: (a)  $h$  extends to  $R_{\leq \aleph_0}$ ; (b)  $h$  extends to  $R_{\leq \mu}$  for some infinite  $\mu$ ; (c)  $h = g \circ \text{ct}$  for a map  $g : D \rightarrow S$ ; (d)  $h$  extends to  $\mathcal{A}_{D,k}$ . Whenever an extension exists it is unique at each of these support levels. In particular  $E$  does not extend to  $R_{\leq \aleph_0}$ .

(iv) [merge] The same equivalence holds on the finite side, with  $\mathcal{B}_{D,k}^{\leq \aleph_0}$ ,  $\mathcal{B}_{D,k}^{\leq \aleph_1}$ ,  $\mathcal{B}_{D,k}^{\leq \mu^+}$ ,  $\mathcal{B}_{D,k}$  and *st* in place of  $R_{\text{fin}}$ ,  $R_{\leq \aleph_0}$ ,  $R_{\leq \mu}$ ,  $\mathcal{A}_{D,k}$  and *ct*, where  $\mu$  is infinite as in (iii).

*Proof.* (i), positive side. The zero element is already killed, so fix  $0 \neq s \in \Pi_k$  with  $|\text{supp } s| < \lambda$ . For  $a > b$  in  $U_{\mathbf{No}}(\text{supp } s)$ , the quotient of Lemma 4.7 has support of cardinality at most  $\max(|\text{supp } s|, \aleph_0) < \lambda$ , and the monomials  $\omega^a$  are in the ring; so the separating families of Theorem 4.9, applied to the singleton  $\{s\}$ , and every individual quotient lie in  $\mathcal{A}_{D,k}^{\leq \lambda}$ . Take a family indexed by the Hartogs ordinal of  $S$ . A collision of two monomial images and the finite identity  $s = (\omega^a - \omega^b)q$  force  $\varphi(s) = 0$ , as in Lemma 4.8(iii) (11). No sum of the whole separating family is formed; its index set may be much larger than  $\lambda$ . Alternatively (04), every monomial and every telescope  $q_{c;a,b}$  has at most countable support, so the explicit-telescope route (Lemma 4.3 with Lemma 4.8(i); see the second routes after Theorem 5.1) kills all positive monomials inside the bounded ring (the field-collision proof of Theorem 5.1 cannot use the whole Hahn field over  $H_g$  here, since shifting a support does not reduce its cardinality), and the factorization  $s = \omega^a(\omega^{-a}s)$  of Proposition 3.3 only translates the support, keeping its cardinality. A third route (batch 32) is in the first- $\kappa$  report [46]: its reservoir lemma (`fkcsb:lem:reservoir`) supplies, below every positive scale, a set field of arbitrarily large cardinality whose members have at most countable support, so the scaled-field collision of Lemma 4.1 can be run with that field inside the bounded ring; for  $D = \mathbb{Z}$ ,  $k = \mathbb{R}$  and  $\lambda = \kappa$  this is its `fkcsb:thm:smallquotient`, which it identifies with the present statement. The field used in that route is  $k(X^h : h \in H)$  for a sufficiently large set

group  $H$  of exponents infinitesimal relative to the chosen scale. An individual rational expression uses finitely many monomials. Inverting its denominator by a Hahn geometric series introduces at most countably many exponents; finite sums and products retain that bound. Scaling shifts each support into the positive side without changing its size. Thus a large field of elements with individually countable supports supplies the collision witnesses.

(i), finite side (04). For  $0 \neq \varepsilon$  with  $|\text{supp } \varepsilon| < \lambda$ ,  $a = v(\varepsilon)$ ,  $0 < \gamma < \delta < a$  and  $d = \delta - \gamma$ ,

$$\frac{\varepsilon}{\omega^{-\gamma} - \omega^{-\delta}} = \varepsilon \omega^\gamma \sum_{n \geq 0} \omega^{-nd},$$

the geometric series being the inverse of  $1 - \omega^{-d}$ . Its support lies in  $\{\beta + \gamma - nd : \beta \in \text{supp } \varepsilon, n \in \mathbb{N}\}$ , of cardinality at most  $|\text{supp } \varepsilon| \cdot \aleph_0 < \lambda$ , and its largest exponent is  $-a + \gamma < 0$ . So the quotient of [Lemma 8.1](#) stays in the bounded ring. The geometric family is Hahn summable, and its product with  $\varepsilon$  is an ordinary Hahn product; a homomorphism is applied only to the resulting finite identity. The Hartogs family in the proof of [Theorem 8.2](#) consequently still works. The cardinal estimate uses one support times a countable set; since  $\lambda$  is uncountable, their maximum is below  $\lambda$  even when  $\lambda$  is singular. A valuation inequality alone would not control the support size; this check is needed.

(ii) Additivity and  $E(1) = 1$  are clear; for finite supports  $E(xy) = \sum_{\alpha, \beta} c_\alpha d_\beta = E(x)E(y)$  by expanding the finite double sum. Coincident exponents in a product merely regroup this finite sum, so there is no ambiguity from cancellation. The ring  $k$  is set-sized, and  $E(\omega) = 1 \neq 0 = \text{ct}(\omega)$  on the positive side, while  $E(\omega^{-1}) = 1 \neq 0 = \text{st}(\omega^{-1})$  on the finite side.

(iii) An extension to some  $R_{\leq \mu}$  factors through  $\text{ct}$  by (i), giving (a) $\Rightarrow$ (c) and (b) $\Rightarrow$ (c); an extension to  $\mathcal{A}_{D,k}$  factors by [Theorem 5.1](#), giving (d) $\Rightarrow$ (c). Conversely  $g \circ \text{ct}$  is defined on every ring in the list and extends  $h$ , and two extensions to one ring both factor through  $\text{ct}$  and agree on  $D$ . Since  $E(\omega) = 1 \neq \text{ct}(\omega)$ ,  $E$  violates (c).

(iv) Use (i) on the finite side, [Theorem 8.2](#) and  $\text{st}$ . Here  $|\text{supp } f| < \mu^+$  means  $|\text{supp } f| \leq \mu$ ; since  $\mu$  is infinite, this extension ring contains every countably supported element. Thus (i) applies, and the extension  $g \circ \text{st}$  exists on every ring of the list with the same uniqueness argument. Replacing this ring by  $\mathcal{B}_{D,k}^{< \aleph_0}$  would make extension automatic and the assertion false for augmentation; the infinite support allowance is essential.  $\square$

*Remark 9.2* (Why the threshold is about closure, not continuity). No axiom that target maps preserve countable sums is added: certain countable sums lie *inside* the source ring, where they satisfy finite identities. An extension of  $E$  to the countable-support ring would send both monomials in [Example 4.5](#) to 1 and the product to 1, forcing  $(1 - 1)E(q) = 1$  (04, 11). A map that assigns different values to some monomials is handled by the full theorem: a large family forces a collision somewhere (11). The separating families may have cardinality far above  $\lambda$ ; their members are distinct elements, not terms of one series, and each witness separately has countable support (04, 07, 11).

The finite-support augmentation appears independently in 03 (“two necessary warnings”), 04, 09, 11 and 15 (which blocks its extension with the single series of [Example 4.5](#)), and 12’s cores carry it too ([Proposition 13.2](#)); 07 describes the same boundary in words (on a finite-support fragment the certificate (4.2) need not exist, and a faithful interpretation in a set-sized target can exist), and manuscript 05 of the sibling report [47] observes that  $\mathbb{Z} + \omega\mathbb{R}[\omega]$  has many real evaluation maps that do not extend to **Oz**. Proper-class size alone therefore does not prove the universal theorem: the finite-support rings have a proper class of monomials and arbitrarily small positive exponents, yet  $E$  sees every monomial as 1 (04). Neither size nor additive divisibility alone suffices; the proof

needs many subordinate scales together with enough support closure for geometric division (11). For  $D = \mathbb{Z}$ , finite targets still kill the positive-support ideal of  $R_{\text{fin}}$  ( $f = n(f/n)$  in characteristic  $n$ ), so its finite quotient theory is ordinary congruence theory: for a finite unital target of characteristic  $n \geq 1$ , the equation  $f = n(f/n)$  kills each positive finite-support element, and a surjective map is determined by its ordinary integer scalars. Every kernel of a finite quotient is therefore  $n\mathbb{Z} + \Pi_{\text{fin}}$ , where  $\Pi_{\text{fin}} = \Pi_k \cap R_{\text{fin}}$ , and every such kernel occurs. Having profinite completion  $\widehat{\mathbb{Z}}$  is thus weaker than having  $\mathbb{Z}$  universal for all set-sized targets (09).

The quotient and module classifications extend to the bounded rings of (i), and so does the derivation statement, because the monomial factorizations proving idempotence preserve support cardinality (04). On the positive side the cofactor only translates the support; on the finite side  $\varepsilon = \omega^{-a/2}(\omega^{a/2}\varepsilon)$  does the same, with the other factor a single monomial. For modules apply the bounded-ring mapping property to the set-sized ring  $\text{End}_{\mathbb{Z}}(M)$ ; the resulting annihilation and these factorizations then give the derivation correspondence. 04 does *not* infer that these smaller rings have fraction field  $k((X^{\mathbf{No}}))$ : arbitrary normal forms need not have the required support size. The result is also different from completion or omitted-type questions for bounded-support series over a *fixed* set-sized exponent group (compare [46]): here the exponent class is still large enough to supply a new separating family for every proposed target (04). The fixed-group question is the subject of the next section.

(Status (batch 32): equal set-sized quotients do not make the bounded rings isomorphic. With  $D = \mathbb{Z}$ ,  $k = \mathbb{R}$ , the first- $\kappa$  report shows that the rings of (i) for  $\lambda = \kappa$  and  $\lambda = \kappa'$  are not isomorphic as abstract rings when  $\text{cf } \kappa \neq \text{cf } \kappa'$ , since their fraction fields are not (**fkcsbcor:rings** in [46]); its Part II also classifies the set-presented gaps and the saturation of those fraction fields.)

## 10 Fixed exponent groups: cardinal lower bounds and exact thresholds

Over a fixed set-sized exponent group the full-class theorem cannot hold verbatim: the ring is itself a set, and its identity map is a set-sized target that retains every element (03, 04, 07, 08, 10, 11). What survives is a lower bound on the size of a target that detects a given element. This section compares the bounds proved by the sources, proves a field bound that dominates them, and states the organizing theorem of the report: for each of the set-sized constructions of Section 11, the exact detection threshold is the size of the ring.

Throughout,  $\Gamma$  is an ordered abelian set group,  $k$  a set field,  $D \subseteq k$  a unital subring,  $F \subseteq k((X^\Gamma))$  a subfield containing  $k$  and every monomial  $X^g$ ,  $\Pi(F) = \{f \in F : \text{supp } f \subseteq \Gamma_{>0}\}$  and  $A = D + \Pi(F)$ . The case  $F = k((X^\Gamma))$  is the ring  $\mathcal{A}_{D,k}(\Gamma)$  of (2.2); the constructions of Section 11 take  $F$  to be a countable-support, a  $< \kappa$ -support, or a controlled subfield.

### 10.1 The bounds of the sources

**Proposition 10.1** (Monomial-count bounds [03, 04, 07, 10, 13, 15]). *Let  $A = \mathcal{A}_{D,k}(\Gamma)$  and  $\varphi : A \rightarrow S$  a ring homomorphism.*

- (i) *If  $\varphi(X^a) \neq 0$  for some  $a > 0$ , then  $\gamma \mapsto \varphi(X^\gamma)$  is injective on  $H_a(\Gamma)_{>0}$  (04); more generally  $rX^\delta \mapsto \varphi(rX^\delta)$  is injective on  $\{rX^\delta : r \in k^\times, 0 < \delta \in H_a(\Gamma)\}$ , and the same holds if  $X^a$  divides some  $g$  with  $\varphi(g) \neq 0$  (07).*
- (ii) *If  $0 \neq x \in \Pi_k(\Gamma)$  and  $\varphi(x) \neq 0$ , then  $a \mapsto \varphi(X^a)$  is injective on  $U_\Gamma(\text{supp } x)$ ; this also holds*

for a nonunital map defined on  $\Pi_k(\Gamma)$  alone (10; 15's local criterion is this bound, and 13's criterion is the same injection on any set  $T \subseteq \Gamma_{>0}$  with  $s > a + n(a - b)$  for  $a > b$  in  $T$ ,  $s \in \text{supp } x$ ).

- (iii) On the negative side, if  $0 \neq \varepsilon \in \mathfrak{m}_k(\Gamma)$ ,  $a = v(\varepsilon)$  and  $\varphi(\varepsilon) \neq 0$  for a map  $\varphi$  on  $\mathcal{B}_{D,k}(\Gamma)$ , then  $\gamma \mapsto \varphi(X^{-\gamma})$  is injective on  $(0, a)_\Gamma$  (04).
- (iv) For  $h > 0$ , an  $A$ -module  $M$  and  $v \in M$  with  $X^h v \neq 0$ ,  $a \mapsto X^a v$  is injective on  $\{a \in \Gamma : a \ll h\}$  (03).

No divisibility of  $\Gamma$  is needed.

*Proof.* For (i), subtract two distinct weighted monomials  $rX^\delta$  and  $sX^\eta$ . If  $\delta \neq \eta$ , order them and apply the weighted telescope; if they are equal, then  $r \neq s$  and use  $(r - s)^{-1}X^{a-\delta}$ . In both cases the quotient multiplying the difference to  $X^a$  has positive support. All exponents are integer combinations of  $a, \delta, \eta$ , hence lie in  $\Gamma$ ; the coefficients lie in  $k$ . This proves the claimed injections by Lemma 4.8(i). If  $g = X^a q$  is detected, then  $X^a$  must be detected as well.

For (ii), the quotient in Lemma 4.7 belongs to  $\Pi_k(\Gamma)$ , so the same argument works even when the map is defined only on that nonunital ring. For (iii), the valuation of  $\varepsilon/(X^{-\gamma} - X^{-\delta})$  is  $a - \min(\gamma, \delta) > 0$ ; this is the proof of Lemma 8.1, unchanged for any coefficient field and ordered abelian group. The quotient belongs to  $\mathfrak{m}_k(\Gamma)$ . For (iv), if  $X^a v = X^b v$  with distinct  $a, b \ll h$ , the telescope gives  $X^h = (X^a - X^b)q$  with  $q \in A$ . Acting by  $q$  and using commutativity of  $A$  contradicts  $X^h v \neq 0$ . No step uses halves or division of an exponent by an integer; this explains why divisibility of  $\Gamma$  is unnecessary.  $\square$

**Example 10.2** (Where these bounds are vacuous [04, 08, 10, 13]). If  $\Gamma$  is Archimedean,  $H_a(\Gamma) = \{0\}$  and the telescopes do not exist. For  $\Gamma = \mathbb{Z}$  a reverse well-ordered positive support is finite,  $\mathcal{A}_{\mathbb{Z}, \mathbb{R}}(\mathbb{Z}) = \mathbb{Z} + X\mathbb{R}[X]$ , and evaluation at  $X = 1$  is a map to the set field  $\mathbb{R}$  detecting  $X$  (10). An arbitrary fixed Hahn workspace cannot inherit the full theorem merely because its notation resembles **No** (08).

## 10.2 The field bound

The monomial counts are far from optimal. The following bound is 03's “smaller-scale field obstruction”, obtained there for countable-support fields over divisible groups; the general form below is a direct application of 06's scaled-field collision lemma and is recorded as a merge statement.

**Proposition 10.3** (Field bound [03, 06] [merge]). *Let  $c \in \Gamma_{>0}$  and  $L_c(F) = F \cap k((X^{H_c(\Gamma)}))$ .*

- (i)  $L_c(F)$  is a subfield of  $F$  and  $X^c L_c(F) \subseteq \Pi(F) \subseteq A$ .
- (ii) If  $\varphi : A \rightarrow S$  is a ring homomorphism with  $\varphi(X^{2c}) \neq 0$ , then  $f \mapsto \varphi(X^c f)$  is injective on  $L_c(F)$ ; if  $M$  is an  $A$ -module and  $X^{2c} v \neq 0$ , then  $f \mapsto (X^c f)v$  is injective on  $L_c(F)$ .
- (iii) Consequently, if  $x \in X^{2c} A$  (for instance  $x = X^{2c}$ , or  $0 \neq x \in \Pi(F)$  with  $2c$  a strict lower bound of  $\text{supp } x$ ), every ring homomorphism and every module detecting  $x$  has cardinality at least  $|L_c(F)|$ .

*Proof.* (i)  $k((X^{H_c(\Gamma)}))$  is a subfield by Lemma 4.2(ii), so  $L_c(F)$  is an intersection of two subfields. For  $h \in H_c(\Gamma)$ ,  $|h| < c$ , so  $c + h > 0$  and  $X^c f \in F$  has positive support. (ii) is Lemma 4.1 with  $m = X^c$

and  $L = L_c(F)$ , inside the field  $k((X^\Gamma))$ . More explicitly, for distinct  $f, g \in L_c(F)$  the multiplier  $X^c/(f - g)$  belongs to  $X^c L_c(F) \subseteq A$ , and

$$(X^c f - X^c g) \frac{X^c}{f - g} = X^{2c}.$$

This gives both the ring injection and the module injection by acting on the specified vector; it bounds the underlying set of  $M$ , not its endomorphism ring. (iii) If  $x = X^{2c}y$  with  $y \in A$  and  $\varphi(x) \neq 0$ , then  $\varphi(X^{2c}) \neq 0$ ; if  $xv \neq 0$  then  $X^{2c}(yv) \neq 0$ . If  $2c < \text{supp } x$  then  $X^{-2c}x \in F$  has positive support, so it lies in  $\Pi(F) \subseteq A$ . This lower-bound condition is required for that example of (iii); arbitrary positive support in a fixed group need not have a positive strict lower bound.  $\square$

For  $F = H_\omega(k, \Gamma)$  and  $c = h/2$  this is exactly 03's theorem ( $H_{h/2} = H_h$ ). The bound contains 04's and 10's monomial counts whenever those monomials lie in  $H_c(\Gamma)$ , and 07's weighted count, because  $L_c(F)$  contains every  $rX^\delta$  with  $r \in k$  and  $\delta \in H_c(\Gamma)$ ; it also contains  $k$  itself, so with real coefficients every target detecting an element as in (iii), in particular a monomial  $X^{2c}$ , has at least  $2^{\aleph_0}$  elements (03). The following examples show that it can be strictly and decisively larger.

**Example 10.4** (The exact threshold in 04's example [merge]). Let  $\Gamma = \mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  (first coordinate dominant),  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $A = \mathcal{A}_{D,k}(\Gamma)$  with full Hahn supports, and  $a = (1, 0)$ . 04 computes the positive part  $H_a(\Gamma)_{>0} = \{(0, q) : q \in \mathbb{Q}_{>0}\}$ , so  $H_a(\Gamma) = \{0\} \oplus \mathbb{Q}$ , and notes that retaining  $X^a$  requires only a countable family of distinct images. With  $c = (1/2, 0)$ ,  $H_c(\Gamma) = \{0\} \oplus \mathbb{Q}$  and  $L_c = k((X^{\{0\} \oplus \mathbb{Q}})) \cong k((t^{\mathbb{Q}}))$  has cardinality  $2^{\aleph_0}$ ; and  $|A| \leq |k((X^\Gamma))| \leq 2^{\aleph_0} \cdot (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ , since  $\Gamma$  is countable. Hence the least cardinality of a ring target, and of a module, detecting any nonzero  $x \in X^{(1,0)}A$  is exactly  $2^{\aleph_0}$ , attained by the identity map and the regular module. The same holds on the negative side: if  $\varepsilon \in \mathfrak{m}_k(\Gamma)$  has  $v(\varepsilon) = (1, 0)$ , then with  $c = (1/4, 0)$  the element  $X^{-2c}$  divides  $\varepsilon$  in  $\mathcal{B}_{D,k}(\Gamma)$  (the cofactor  $X^{2c}\varepsilon$  has largest exponent  $(-1/2, 0) < 0$ ),  $X^{-c}L_c \subseteq \mathfrak{m}_k(\Gamma)$ , and Lemma 4.1 gives the bound  $2^{\aleph_0}$ , which is again attained by the identity of the negative-support ring and its regular module: that ring also has at most continuum many elements.

In this example 07's weighted bound ( $|k^\times| \cdot \aleph_0 = 2^{\aleph_0}$ ) is also attained, while 04's and 10's monomial counts give only  $\aleph_0$ . With a countable coefficient field all three source bounds fail:

**Example 10.5** (None of the monomial or weighted counts is attained [merge]). Let  $k = D = \mathbb{Q}$ ,  $\Gamma = \bigoplus_{n < \omega} \mathbb{Q}e_n$  with the least nonzero index dominant, and  $A = \mathcal{A}_{\mathbb{Q},\mathbb{Q}}(\Gamma)$  with full Hahn supports. For  $x = X^{e_0}$  and  $c = e_0/2$ ,  $H_c(\Gamma) = \bigoplus_{n \geq 1} \mathbb{Q}e_n$ , and  $L_c$  contains the  $2^{\aleph_0}$  distinct series  $\sum_{n \in B} X^{e_n}$ ,  $B \subseteq \omega \setminus \{0\}$  (the exponents  $e_n$  decrease with  $n$ , so each support is reverse well ordered). Since  $\Gamma$  is countable,  $|A| \leq 2^{\aleph_0} \cdot \aleph_0^{\aleph_0} = 2^{\aleph_0}$ . So the exact detection threshold of  $X^{e_0}$  is  $2^{\aleph_0}$ , attained by the identity map and regular module of  $A$ , while 04's bound  $|H_{e_0}(\Gamma)_{>0}|$ , 07's bound  $|\mathbb{Q}^\times| \cdot |H_{e_0}(\Gamma)_{>0}|$  and 10's bound  $|U_\Gamma(\{e_0\})|$  are all  $\aleph_0$ . Indeed the group and coefficient field are countable, the positive tail contains infinitely many  $e_n$ , and membership of  $n$  in  $B$  is recovered from the coefficient of  $X^{e_n}$ . Thus the lower field bound really counts continuum many distinct series; it is not just a count of monomials.

These examples settle the two example-level questions of the sources and leave the general ones open: 04's question on optimal small targets in a fixed exponent group is answered for its own example  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  (the answer is  $2^{\aleph_0}$ ), and for 10's question (when is the bound of Proposition 10.1(ii) attained?) Example 10.5 shows, in ZFC, that the bound need not be attained. The same phenomenon appears in 03's ring  $A_{\aleph_1}^{03}$  of Section 11.1: there  $|U_\Gamma(S)| \leq \aleph_1$ , while the exact threshold is  $2^{\aleph_0}$ , so if the continuum hypothesis fails, 10's bound is not attained there either. Remark 10.6 records the corresponding correction to a sentence of 04.



*Remark 10.6* (A correction to 04’s commentary). 04 remarks that in  $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$  “this is only a countable family” which “cannot contradict the existence of a sufficiently large infinite set target, such as the ring itself”. The statement is true, but the ring itself has  $2^{\aleph_0}$  elements and, by [Example 10.4](#), no smaller target detects  $X^{(1,0)}$ ; the countable family is not the obstruction that decides the answer.

### 10.3 Criteria that kill the whole ideal below a cardinal

To pass from monomials to the whole ideal one also needs a gap: every element must be divisible by a positive monomial inside the ring. 03’s example  $\sum_{n \geq 1} X^{1/n}$  over  $\Gamma = \mathbb{R}$  shows that this can fail ([Example 3.2](#)), and 09’s uncountable-support series  $f_*$  ([Remark 11.21](#)) shows the same for full supports over a long group.

**Theorem 10.7** (Quantitative criterion [08] [merge]). *Let  $\Gamma$  be divisible,  $\kappa$  an infinite cardinal, and suppose*

- (a) *every nonzero  $f \in \Pi(F)$  has a positive exponent  $\epsilon \in \Gamma$  strictly below its support;*
- (b) *for every  $c > 0$  in  $\Gamma$ , the set  $\{a \in \Gamma : a \ll c\}$  has at least  $\kappa$  elements.*

*Then every ring homomorphism from  $A$  to a ring of cardinality  $< \kappa$  factors uniquely through  $\text{ct} : A \rightarrow D$ . [merge] The same conclusion holds, with the same proof, if (b) is replaced by (b’):  $|L_c(F)| \geq \kappa$  for every  $c > 0$ .*

*Proof.* Fix  $c > 0$ . Under (b), for two distinct  $a > b \ll c$  the series  $q_{c;a,b}$  equals the rational expression  $X^c/(X^a - X^b)$  in  $k((X^\Gamma))$ , whose numerator and denominator lie in the field  $F$ ; so  $q_{c;a,b} \in \Pi(F) \subseteq A$ , and a nonzero image of  $X^c$  would make the images of  $\kappa$  monomials distinct. Under (b’) use [Proposition 10.3\(ii\)](#) with  $c/2$  in place of  $c$ . So every positive monomial is killed. For  $f \in \Pi(F)$ , (a) gives  $f = X^\epsilon g$  with  $g = X^{-\epsilon} f \in F$  of positive support, so  $f$  is killed. Writing an arbitrary element as  $d + f$  now gives the unique factor  $\varphi|_D$  through  $\text{ct}$ ; no unital hypothesis on  $\varphi$  was used.  $\square$

Condition (a) is a gap property of the chosen field and (b) a supply of lower Archimedean scales; neither says that homomorphisms preserve strongly summable families (08). 08 states the theorem for a subfield  $K \subseteq \mathbb{R}$  and constants  $\mathbb{Z}$ ; the proof uses neither restriction, and it is stated here in the setting of this section.

**Theorem 10.8** (Coinitiality criterion [11]). *Let  $\Gamma$  be divisible,  $\kappa$  uncountable, and  $\text{coin}(\Gamma_{>0}) \geq \kappa$  (every set of fewer than  $\kappa$  positive elements has a positive strict lower bound). Put  $R_{\Gamma, < \kappa} = D + \{s \in \Pi_k(\Gamma) : |\text{supp } s| < \kappa\}$ , a ring for every infinite  $\kappa$ . Then every unital map  $R_{\Gamma, < \kappa} \rightarrow S$  with  $|S| < \kappa$  factors uniquely through  $\text{ct}$ .*

*Proof.* Let  $s \neq 0$  have  $|\text{supp } s| < \kappa$ . Choose  $\delta > 0$  strictly below  $\text{supp } s$ , and then a positive strict lower bound  $\eta$  of the countable set  $\{\delta/n : n \geq 1\}$  (possible since  $\kappa > \aleph_0$ ); every  $a \in E = (0, \eta)$  satisfies  $na < \delta$ . The set  $E$  has at least  $\kappa$  elements: otherwise it would have a positive strict lower bound  $b$ , but then  $b \in E$  (as  $b < \eta/2 \in E$ ), which is absurd. For distinct  $a, b \in E$ , [Lemma 4.7](#) gives  $s \in (X^a - X^b)R_{\Gamma, < \kappa}$  with a quotient of support size at most  $\max(|\text{supp } s|, \aleph_0) < \kappa$ . The inequality is valid without regularity, since it is the maximum of two fixed cardinals below the uncountable  $\kappa$ . All  $X^a$  for  $a \in E$  lie in the source. Since  $|S| < \kappa \leq |E|$ , two images collide, and applying the map to this finite product identity kills  $s$ . The zero element needs no argument. Restriction to  $D$  then supplies the unique factorization of the map.  $\square$

This theorem concerns an ordinary set-sized source and cannot exclude its identity representation; the target bound is indispensable. Regularity of  $\kappa$  is not needed for the theorem itself, only for the examples of [Section 11.5](#) (11).

**Lemma 10.9** (Cyclic modules [08]). *Let  $A$  be a commutative ring,  $I$  an ideal and  $\kappa$  a cardinal such that every ring homomorphism from  $A$  onto a ring of cardinality  $< \kappa$  kills  $I$ . Then  $IM = 0$  for every  $A$ -module  $M$  with  $|M| < \kappa$ .*

*Proof.* For  $m \in M$ , commutativity makes  $\text{Ann}(m) = \{a \in A : am = 0\}$  a two-sided ideal. The map  $A/\text{Ann}(m) \rightarrow Am$ ,  $[a] \mapsto am$ , is well defined and bijective: equality of the two images is exactly  $(a - b)m = 0$ . Thus its source has cardinality  $|Am| \leq |M| < \kappa$ . Apply the hypothesis to the surjective quotient ring map  $A \rightarrow A/\text{Ann}(m)$  to obtain  $I \subseteq \text{Ann}(m)$ . This holds for every  $m$ , so  $IM = 0$ . The set  $Am$  is used only to count elements; it is not asserted to be a subring of  $M$ .  $\square$

Applying the ring statement to  $\text{End}_{\mathbb{Z}}(M)$  would need an estimate of  $|\text{End}_{\mathbb{Z}}(M)|$ , which may exceed  $\kappa$  even when  $|M| < \kappa$ ; the lemma avoids that hidden cardinal-arithmetic assumption (08). [Proposition 10.3\(ii\)](#) gives the module bound directly (06, 03, 09).

## 10.4 The organizing theorem: the threshold is the size of the ring

The five constructions of [Section 11](#) are summarized here; their notation is tagged by source because they are different rings. The first three rows use ordinary integer constants, with the stated Gaussian variants for 03 and 06. In the last two rows  $D$  is a unital subring of the displayed coefficient field. The last column records the least size of an ideal generating set, not a support bound or the size of the ring.

| ring                        | exponent group                                                | supports              | coefficients                            | hypotheses                                               | size                | gen. of $\Pi_A$ |
|-----------------------------|---------------------------------------------------------------|-----------------------|-----------------------------------------|----------------------------------------------------------|---------------------|-----------------|
| $A_{\kappa}^{03}$           | $\bigoplus_{\kappa^{\text{op}} + \kappa} \mathbb{Q}$ , 2 arms | countable             | $\mathbb{R}$                            | cf $\kappa > \aleph_0$                                   | $\kappa^{\aleph_0}$ | cf $\kappa$     |
| $A_{\kappa}^{06}$           | $\text{span}_{\mathbb{Q}}\{\omega^{\pm(\alpha+1)}\}$ , 2 arms | $< \kappa$            | $\mathbb{R}$                            | $\kappa$ reg. unctbl.                                    | $\kappa^{<\kappa}$  | $\kappa$        |
| $A_{\kappa}^{08}$           | $\bigoplus_{\xi < \kappa} \mathbb{Q}$ , 1 arm                 | countable, controlled | $\mathbb{R} \cap \overline{\mathbb{Q}}$ | $\kappa$ infinite                                        | $\kappa$            | cf $\kappa$     |
| $A_{\lambda, D}^{09}$       | $\bigoplus_{\omega_1 \times \lambda} \mathbb{Q}$ , 1 arm      | countable             | $k \in \{\mathbb{R}, \mathbb{C}\}$      | $\lambda \geq \aleph_0$ , $\lambda^{\aleph_0} = \lambda$ | $\lambda$           | $\aleph_1$      |
| $R_{\Gamma, < \kappa}^{11}$ | $\bigoplus_{\alpha < \kappa} \mathbb{Q}$ , 1 arm              | $< \kappa$            | $ K  \leq \max(\kappa, 2^{\aleph_0})$   | $\kappa$ reg. unctbl.                                    | $\kappa^{<\kappa}$  | $\kappa$        |

All are ZFC constructions. The sizes are proved in [Section 11](#); for  $A_{\kappa}^{06}$  and  $R_{\Gamma, < \kappa}^{11}$  the size  $\kappa^{<\kappa}$  is a merge result (06 and 11 compute the size only under  $\kappa^{<\kappa} = \kappa$ ).

**Theorem 10.10** (Detection threshold equals size [03, 08, 09] [merge]). *Let  $A$  be any of the five rings of the table, with purely infinite ideal  $\Pi_A$ , and let  $0 \neq x \in \Pi_A$ . Then*

$$\min\{|S| : \varphi : A \rightarrow S, \varphi(x) \neq 0\} = \min\{|M| : M \text{ an } A\text{-module}, xM \neq 0\} = |A|.$$

*Every homomorphism from  $A$  to a ring of cardinality  $< |A|$  factors uniquely through  $\text{ct}$ , every  $A$ -module of cardinality  $< |A|$  is annihilated by  $\Pi_A$ , and every quotient  $A/J$  either is  $D/\text{ct}(J)$  with  $\Pi_A \subseteq J$  or has cardinality exactly  $|A|$ . The same holds for the Gaussian versions of  $A_{\kappa}^{03}$ ,  $A_{\kappa}^{06}$  and  $A_{\lambda, \mathbb{Z}[i]}^{09}$ .*

*Proof.* Use the ambient field  $F$  and coefficient field of the chosen construction: the countable-support field for 03 and 09, the  $< \kappa$ -support field for 06 and 11, and the controlled field for 08. In each case

$A = D + \Pi(F)$ ,  $F$  contains all its exponent-group monomials, and  $|F| = |A|$  by the size calculations in [Section 11](#). We first verify that every smaller-scale field has this same size:

$$|L_c(F)| = |A| \quad (c > 0).$$

For 03 this is [Lemma 11.4](#); for 06 it is [Lemma 11.10](#); and for 11 it is [Proposition 11.23](#) together with its ambient-field upper bound. For 08, the field  $L_c(F)$  contains the  $\kappa$  monomials  $X^{e_\xi}$  with  $\xi$  beyond the least nonzero coordinate of  $c$ , by [Lemma 11.13](#). They lie in the controlled field and have exponents in  $H_c$ ; the upper bound is  $|F| = \kappa$ . For 09, apply [Lemma 11.17](#) to  $\{c\}$ . A single later row gives  $\lambda$  distinct monomials in  $L_c(F)$ , and  $|F| = \lambda$  gives the upper bound. The real and Gaussian coefficient choices use the same monomial families and their respective field-size bounds.

Now fix  $0 \neq x \in \Pi(F)$ . Each construction supplies a positive strict lower bound  $g$  of  $\text{supp } x$ : use [Lemmas 11.3, 11.7, 11.15](#) and [11.17](#) for 03, 06, 08 and 09, respectively, and [Proposition 11.22](#) with  $|\text{supp } x| < \kappa$  for 11. Divisibility of the group gives  $c = g/2 > 0$ . Since  $F$  is a field,  $b = X^{-2c}x \in F$ ; all its exponents are positive, so  $b \in \Pi(F)$  and  $x = X^{2c}b$ . The field bound in [Proposition 10.3\(iii\)](#) now shows that every ring target or module detecting  $x$  has at least  $|L_c(F)| = |A|$  elements. This proof does not require a homomorphism to preserve 1 or the target to be commutative. For modules it counts the orbit of a vector directly. The identity map and the regular module, with vector 1, attain both bounds. Thus both minima exist and equal  $|A|$  for each such  $x$ .

A homomorphism into a smaller ring therefore kills all of  $\Pi(F)$ . Writing  $a = d + f$  shows that its factor through  $\text{ct}$  is exactly its restriction to  $D$ , and the section  $D \rightarrow A$  makes that factor unique. The same decomposition supplies the unique induced  $D$ -action on every smaller module. If  $\Pi(F) \subseteq J$ , the assignment  $a + J \mapsto \text{ct}(a) + \text{ct}(J)$  is a ring isomorphism  $A/J \rightarrow D/\text{ct}(J)$ . If this inclusion fails, the quotient map detects some nonzero purely infinite element; hence  $|A| \leq |A/J| \leq |A|$ . These alternatives are cases determined by ideal containment, not necessarily disjoint lists of cardinalities. The zero ring and, when  $J = \Pi(F)$ , the quotient  $D$  are included.  $\square$

For  $A_\kappa^{03}$ ,  $A_\kappa^{08}$  and  $A_{\lambda,D}^{09}$  the theorem is proved in the sources (03 with the field bound; 08 and 09 by counting monomials, which suffices because there the number of small monomials already equals  $|A|$ ). For  $A_\kappa^{06}$  06 proves the lower bound  $\kappa$  and exactness under  $\kappa^{<\kappa} = \kappa$ , and for  $R_{\Gamma,<\kappa}^{11}$  11 proves sharpness under  $\kappa^{<\kappa} = \kappa$  and  $|K| \leq \kappa$ ; the unconditional exact value  $\kappa^{<\kappa}$  is new here ([Theorem 11.11](#) and [Proposition 11.23](#)). The quotient alternative is 08's and 09's for their rings and a one-line consequence for the others.

*Remark 10.11* (Reconciling the thresholds [merge]).

- *Realized cardinals.* Since  $(\kappa^{\aleph_0})^{\aleph_0} = \kappa^{\aleph_0}$ , and conversely an infinite cardinal  $\theta$  with  $\theta^{\aleph_0} = \theta$  has  $\text{cf } \theta > \aleph_0$  by König's theorem (so  $\kappa = \theta$  is allowed in 03), the thresholds realized by 03 are exactly the infinite cardinals  $\theta$  with  $\theta^{\aleph_0} = \theta$ ; these are 09's cardinals  $\lambda$ . The countable-cofinality obstruction is the diagonal argument in [Lemma 11.6](#): for infinite  $\theta$  of countable cofinality it gives  $\theta^{\aleph_0} > \theta$ .

06 and 11 realize  $\theta = \kappa^{<\kappa}$  for regular uncountable  $\kappa$ . These sizes also satisfy  $\theta^{\aleph_0} = \theta$ . Indeed, realize  $\theta$  as the cardinality of  $\bigcup_{\nu < \kappa} (\{\nu\} \times {}^\nu \kappa)$ , where  $\nu$  ranges over cardinals. A countable sequence of these functions has its lengths bounded by some infinite  $\rho < \kappa$ , by regularity. For a fixed  $\rho$ , there are at most  $\kappa^{\rho \cdot \aleph_0} = \kappa^\rho$  such sequences, including their length tags. Summing over fewer than or equal to  $\kappa$  choices of  $\rho$  gives the upper bound  $\theta$ ; constant sequences give the lower bound.

Thus 08 is the only one of these five families reaching an infinite size  $\theta$  below  $2^{\aleph_0}$  or with  $\theta^{\aleph_0} > \theta$ ; it reaches every infinite cardinal. The continuum obstruction has an algebraic explanation in

the real-coefficient models: if  $rX^c$  lies in the ring for every real  $r$ , every target detecting  $X^{2^c}$  has at least  $2^{\aleph_0}$  elements by [Lemma 4.1](#) with  $L = \mathbb{R}$ . 08 instead uses real algebraic coefficients and a countable-stage closure. The cardinal calculation above also covers 11's smaller coefficient fields, where this particular real-coefficient argument is unavailable.

- *The same ring, different bounds.*  $A_{\aleph_1}^{06}$  and  $A_{\aleph_1}^{03}$  are isomorphic: both are the countable-support real Hahn integer parts over the lexicographic sum  $\bigoplus_J \mathbb{Q}$  with  $J$  of order type  $\omega_1^{\text{op}} + \omega_1$  (06's generators satisfy  $\epsilon_\beta \ll \epsilon_\alpha \ll h_\alpha \ll h_\beta$  for  $\alpha < \beta$ ). 06 proves the threshold is at least  $\aleph_1$  and exactly  $\aleph_1$  under CH; 03 proves it is exactly  $2^{\aleph_0}$  in ZFC; 06's own example already notes that without CH its ring need not have size  $\aleph_1$ . These agree.
- *Generators.* The generator counts are cf  $\kappa$  (03, 08),  $\kappa$  (06, and for 11's ring by the argument of (H3) in [Section 12](#)) and  $\aleph_1$  (09). In each case the proof exhibits a coinital chain of monomials in the positive cone of the exponent group of that length; the counts differ because the constructions differ, and no general formula is asserted for other exponent groups or support restrictions. For example, 08 at  $\kappa = \aleph_0$  has threshold and generator count  $\aleph_0$ , while 09 at  $\lambda = 2^{\aleph_0}$  has threshold  $2^{\aleph_0}$  and generator count  $\aleph_1$ ; equality of the latter two cardinals is precisely CH. The generator proof for 11 is given directly after [Proposition 11.23](#).
- *Fraction fields.*  $\text{Frac } A = F$  is proved for 03 and 06, which use a second (reversed) arm to clear negative exponents; 08 explicitly does not claim it for its one-armed controlled field, and 09 and 11 do not address it.
- *Hypotheses.* Regularity in 06 and 11 bounds the union of fewer than  $\kappa$  supports, each of size less than  $\kappa$ . For 03,  $\mu = \text{cf } \kappa > \aleph_0$  bounds the union of fewer than  $\mu$  countable supports. In particular a single support has a gap in each case. In 09 a countable union of finite coordinate supports has its first coordinates bounded in  $\omega_1$ , without regularity of  $\lambda$ . 08 replaces these full-support bounds with its finite-coordinate invariant, which confines each element to finitely many coordinates and works even when  $\text{cf } \kappa = \aleph_0$ . These are different reasons for the gap property; a cardinal estimate alone does not supply it.

## 11 Set-sized rings with exact thresholds

Five constructions replace the proper-class supply of scales by a cardinal. They are different rings and are tagged by source; all are ZFC constructions of sets. The displayed real/integer models and the stated Gaussian versions have embeddings into  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , respectively. The general coefficient choices in 09 and 11 are kept as abstract Hahn rings; no such embedding is asserted for every coefficient field. For example, 11 permits  $D = K = \mathbb{F}_p$ , giving a ring of characteristic  $p$ , which cannot embed unitally in either characteristic-zero omnific ring. For every one of them [Theorem 10.10](#) gives the exact detection threshold; this section supplies the constructions and the size computations it uses. Manuscript 05 (sibling report [\[47\]](#)) asked which set-sized exponent groups admit analogues of the common-divisor and set-target theorems (its “exact size boundary” question); 03, 06, 08, 09 and 11 each answer it with a sharp family, and none claims a classification of all exponent groups or support restrictions.

### 11.1 Countable supports over a two-armed group [03]

For an ordered abelian group  $G$  and  $k \in \{\mathbb{R}, \mathbb{C}\}$  let  $H_\omega(k, G)$  be the Hahn series over  $k$  with *at most countable* reverse well-ordered support (any countable reverse order type is allowed; the subscript

means countable support, not order type  $\omega$ ). It is the  $\aleph_1$ -bounded Hahn field in the terminology of [23].

**Lemma 11.1** (Countable-support Hahn fields [03]).  *$H_\omega(k, G)$  is a field. If  $G$  is divisible,  $H_\omega(\mathbb{R}, G)$  is real closed,  $H_\omega(\mathbb{C}, G)$  is algebraically closed, and  $H_\omega(\mathbb{C}, G) = H_\omega(\mathbb{R}, G)[i]$ .*

*Proof.* Given finitely many series, the subgroup  $H \leq G$  generated by the union of their supports is countable: its elements are finite integer combinations of a countable set. The series therefore lie in the full Hahn field  $k((X^H))$ , and every series in that field still has countable support in  $G$ . Sums, products, additive inverses and the inverse of each nonzero series consequently remain in  $H_\omega(k, G)$ .

If  $G$  is divisible, replace  $H$  by its divisible hull inside  $G$ , namely its rational span; this is still countable. The full Hahn field over this divisible group is real closed for  $k = \mathbb{R}$ , and algebraically closed for  $k = \mathbb{C}$  [23, Fact 2.1.1]. Thus every complex-coefficient nonconstant polynomial has a root in a countable-support subfield. On the real side, the same argument supplies a square root of each positive element and a root of every odd-degree polynomial, proving real closedness in the leading-term order. Finally, real and imaginary parts have supports contained in the original countable support, and the union of two countable supports is countable. This proves both inclusions in  $H_\omega(\mathbb{C}, G) = H_\omega(\mathbb{R}, G)[i]$ .  $\square$

With  $A(D, k, G) = D + \{f \in H_\omega(k, G) : \text{supp } f \subseteq G_{>0}\}$ , the constant term is a split surjection, the units are  $D^\times$  (Lemma 2.4), and for  $(D, k) = (\mathbb{Z}, \mathbb{R})$  the ring is a discretely ordered integer part of  $H_\omega(\mathbb{R}, G)$ : write  $x = P + r + \epsilon$  by positive, zero and negative exponents and take  $P + \lfloor r \rfloor$ , except  $P + r - 1$  when  $r \in \mathbb{Z}$  and  $\epsilon < 0$ . Indeed,  $\epsilon$  is infinitesimal, so when  $r \notin \mathbb{Z}$  it cannot cross either endpoint of the interval  $[\lfloor r \rfloor, \lfloor r \rfloor + 1)$ . When  $r \in \mathbb{Z}$ , its sign decides between  $P + r$  and  $P + r - 1$ . Truncating preserves countable support, so the chosen floor lies in the ring. Every positive element of this ring is either a positive integer or has positive leading exponent and hence exceeds every real constant; in particular there is no element strictly between 0 and 1. Discarding all negative exponents without the endpoint correction would send  $-X^{-g}$  to 0, which is not its floor. The same rule gives the omnific integer part of **No** (03, 06, 08; compare [45]).

**The construction.** Let  $\kappa$  be an infinite cardinal with  $\mu = \text{cf } \kappa > \aleph_0$ ,  $\theta = \kappa^{\aleph_0}$ ,  $J_\kappa = \kappa^{\text{op}} + \kappa$  with indices  $p_\alpha$  (reversed arm) and  $q_\alpha$  (forward arm), and

$$G_\kappa = \bigoplus_{j \in J_\kappa} \mathbb{Q}e_j,$$

ordered lexicographically with the *least* index of a nonzero vector deciding its sign; earlier indices are larger scales ( $ne_{j'} < e_j$  for  $j < j'$ ).  $\iota(e_{p_\alpha}) = \omega^\alpha$ ,  $\iota(e_{q_\alpha}) = \omega^{-(\alpha+1)}$  extends to an ordered additive embedding  $G_\kappa \hookrightarrow \mathbf{No}$ , and  $\sum a_g X^g \mapsto \sum a_g \omega^{\iota(g)}$  is an ordered field embedding. Put

$$F_\kappa^{03} = H_\omega(\mathbb{R}, G_\kappa), \quad A_\kappa^{03} = A(\mathbb{Z}, \mathbb{R}, G_\kappa) = F_\kappa^{03} \cap \mathbf{Oz}, \quad \Pi_\kappa^{03} = \ker(\text{ct}|_{A_\kappa^{03}}),$$

and use the superscript  $\mathbb{C}$  for the parallel construction over  $\mathbb{C}$  and  $\mathbb{Z}[i]$ :  $F_\kappa^{03, \mathbb{C}} = F_\kappa^{03}[i]$ ,  $A_\kappa^{03, \mathbb{C}} = A_\kappa^{03}[i]$ ,  $\Pi_\kappa^{03, \mathbb{C}} = \Pi_\kappa^{03} + i\Pi_\kappa^{03}$ .

**Lemma 11.2** (Room on both sides [03]).

- (i) Every family of fewer than  $\mu$  elements of  $G_\kappa$  is bounded above and below in  $G_\kappa$ .
- (ii) Every family of fewer than  $\mu$  positive elements has a positive strict lower bound.

(iii) For every  $h > 0$ , a final segment of the forward arm, of cardinality  $\kappa$ , consists of basis vectors  $e_{q_\alpha} \ll h$ .

*Proof.* Let  $E$  be a family of fewer than  $\mu$  vectors. Each vector has finite coordinate support, so fewer than  $\mu$  indices occur altogether. Since  $\mu = \text{cf } \kappa$ , their ordinal labels are bounded below  $\kappa$  on each arm. Choose  $\beta < \kappa$  above all reversed-arm labels that occur. Then  $p_\beta$  precedes every occurring index, including every index of the forward arm. Lexicographic comparison gives  $-e_{p_\beta} < v < e_{p_\beta}$  for every  $v \in E$ , proving (i).

For a family of positive vectors choose  $\gamma < \kappa$  above all its forward-arm labels. The index  $q_\gamma$  follows every occurring index. The first nonzero coordinate of each vector is positive and occurs before  $q_\gamma$ , so each vector exceeds  $ne_{q_\gamma}$  for every positive integer  $n$ . In particular  $e_{q_\gamma}$  is a positive strict lower bound, proving (ii). Empty families impose no restriction on these choices. For (iii), apply the same comparison to the single vector  $h$ : every  $\alpha$  beyond its finitely many forward-arm labels satisfies  $e_{q_\alpha} \ll h$ . Such a final segment has cardinality  $\kappa$ , because removing a bounded initial segment of the infinite cardinal  $\kappa$  leaves  $\kappa$  ordinals.  $\square$

The reversed arm supplies a denominator large enough to shift a countable support upward; the forward arm supplies a positive divisor below a countable positive support and many indistinguishable smaller scales (03).

**Lemma 11.3** (Small families have common divisors [03]). *Every family of fewer than  $\mu$  elements of  $\Pi_\kappa^{03}$  (or  $\Pi_\kappa^{03, \mathbb{C}}$ ) has a common divisor  $X^g$ ,  $g > 0$ , with all quotients in the ideal.*

*Proof.* For a family  $E$  with  $|E| < \mu$ , the union  $T = \bigcup_{f \in E} \text{supp } f$  has size at most  $|E| \cdot \aleph_0 < \mu$  when  $E$  is nonempty, because  $\mu$  is regular and uncountable. All members of  $T$  are positive. Choose  $0 < g < T$  by Lemma 11.2(ii). Then for every  $f \in E$ , the series  $X^{-g}f$  has positive, countable, reverse well-ordered support, so it lies in the same purely infinite ideal. Translation neither changes support cardinality nor the coefficient field. If  $T$  is empty, every member is zero and any positive  $g$  works. The argument applies equally to real and complex coefficients.  $\square$

**Lemma 11.4** (Every smaller-scale field has size  $\theta$  [03]). *For either coefficient pair, the field, the ring, its purely infinite ideal and every  $L_h = H_\omega(k, H_h(G_\kappa))$  ( $h > 0$ ) have cardinality  $\theta = \kappa^{\aleph_0}$ .*

*Proof.* Write  $F$ ,  $A$  and  $\Pi_A$  for the field, ring and ideal of the chosen coefficient pair. The finite-coordinate group  $G_\kappa$  has cardinality  $\kappa$ . There are at most  $\kappa^{\aleph_0}$  countable supports in it, and each support admits at most  $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0} \leq \theta$  coefficient functions, for either  $k = \mathbb{R}$  or  $k = \mathbb{C}$ . This gives the common upper bound  $\theta$  for the field and all the indicated subsets.

For the lower bound choose a forward tail  $T \subseteq \kappa$  of size  $\kappa$  with  $e_{q_\alpha} \ll h$  for every  $\alpha \in T$ , using Lemma 11.2(iii). Each countable  $S \subseteq T$  determines

$$f_S = \sum_{\alpha \in S} X^{e_{q_\alpha}} \in L_h.$$

Its support is reverse well ordered: a nonempty subset has a largest exponent at its least ordinal index. Its coefficients recover  $S$ , so these series are distinct. Also  $L_h$  contains all constant series  $k$ . Hence Lemma 11.6 gives  $|L_h| \geq \max(|[\kappa]^{\leq \aleph_0}|, 2^{\aleph_0}) = \theta$ . Finally  $H_h(G_\kappa) = H_{h/2}(G_\kappa)$ , and every exponent  $u$  in this subgroup satisfies  $|u| < h/2$ . Thus multiplication by  $X^{h/2}$  is an injection  $L_h \rightarrow \Pi_A$ . Together with  $\Pi_A \subseteq A \subseteq F$ , this proves all the claimed equalities.  $\square$



**Theorem 11.5** (Cardinal realization [03]). *Take  $(A, F, D) = (A_\kappa^{03}, F_\kappa^{03}, \mathbb{Z})$  or  $(A_\kappa^{03, \mathbb{C}}, F_\kappa^{03, \mathbb{C}}, \mathbb{Z}[i])$ , with ideal  $\Pi_A$ :*

- (i)  $|F| = |A| = \theta$  and  $\text{Frac}(A) = F$ ;
- (ii) *every unital map  $A \rightarrow R$  with  $|R| < \theta$  factors uniquely through  $\text{ct}$  (for the Gaussian ring the factorization through the Gaussian constant term is unique for a given map; different square roots of  $-1$  give different maps);*
- (iii) *every  $A$ -module of cardinality  $< \theta$  is annihilated by  $\Pi_A$ , and for every  $0 \neq f \in \Pi_A$  the least cardinality of a ring target detecting  $f$ , and of a module on which  $f$  acts nontrivially, is exactly  $\theta$ ;*
- (iv)  $\Pi_A$  needs exactly  $\mu$  generators; it is flat with  $\Pi_A^2 = \Pi_A$ , the module  $D = A/\Pi_A$  is not flat, and  $\text{Tor}_n^A(D, N) = 0$  for every  $D$ -module  $N$  and  $n > 0$ .

*No inaccessible cardinal, CH or other additional axiom is used.*

*Proof.* (i) The sizes are Lemma 11.4. For  $x \in F$ , apply Lemma 11.2(i) to the countable set  $\{0\} \cup \{-g : g \in \text{supp } x\}$ . Its strict upper bound  $h$  is positive and satisfies  $h + g > 0$  for every exponent of  $x$ . Therefore  $X^h x \in \Pi_A$ ,  $X^h \in \Pi_A \setminus \{0\}$  and  $x = (X^h x)/X^h$ . Conversely  $A \subseteq F$  and  $F$  is a field, proving the fraction-field assertion with the matching real or complex coefficients.

(ii)–(iii) We prove the lower bound directly, supplying this case of Theorem 10.10. For  $0 \neq f \in \Pi_A$ , Lemma 11.3 gives  $f = X^g b$  with  $g > 0$  and  $b \in \Pi_A$ . Set  $c = g/2$ . By Proposition 10.3(iii), a ring map detecting  $f$  has a target of size at least  $|L_c| = \theta$ ; a module with  $fv \neq 0$  has the same bound, using the vector  $bv$ . The identity map and the regular module  $A$ , with vector 1, attain it. Thus a map to a smaller ring kills every element of  $\Pi_A$ , and its unique factor is its restriction to  $D$ : for  $d \in D$  and  $p \in \Pi_A$ ,  $\varphi(d + p) = \varphi(d)$ . If  $|M| < \theta$ , no product  $fv$  can be nonzero, so  $\Pi_A M = 0$ . The lower-bound argument also works for nonunital ring maps. In the unital Gaussian case the restriction to  $D = \mathbb{Z}[i]$  is fixed by the chosen map; the theorem does not prescribe one square root of  $-1$  for all maps.

(iv) Choose a strictly increasing cofinal sequence  $(\alpha_\xi)_{\xi < \mu}$  in  $\kappa$  and put  $m_\xi = X^{e_{q_{\alpha_\xi}}}$ . A countable positive support involves only countably many coordinate indices; these are bounded on the forward arm because  $\text{cf } \kappa > \aleph_0$ . Consequently, for sufficiently large  $\xi$ ,  $e_{q_{\alpha_\xi}}$  is strictly below every exponent of  $f$ , and  $f/m_\xi \in \Pi_A$ . Moreover  $m_\xi A \subsetneq m_\eta A$  when  $\xi < \eta$ : the forward quotient is a positive monomial and the reverse quotient has negative exponent. Thus

$$\Pi_A = \bigcup_{\xi < \mu} m_\xi A.$$

This exhibits  $\mu$  generators. Any family of fewer than  $\mu$  elements of  $\Pi_A$  is contained in  $X^h A$  for some  $h > 0$  by Lemma 11.3. Its generated ideal is therefore contained there too, whereas  $X^{h/2} \in \Pi_A$  is missing: its unique quotient by  $X^h$  in  $F$  is  $X^{-h/2} \notin A$ . Hence fewer generators cannot suffice.

Each  $m_\xi A$  is free of rank one since  $A$  is a domain, and the displayed union is a directed colimit indexed by the set  $\mu$ . It is flat [29, Lemma 10.39.3]. The factorization  $f = X^g b$  above has both factors in  $\Pi_A$ , so  $\Pi_A^2 = \Pi_A$ . If  $N$  is a  $D$ -module inflated along  $\text{ct}$ , then  $\Pi_A N = 0$  and every elementary tensor  $f \otimes n = (X^g b) \otimes n = b \otimes X^g n$  vanishes. The flat resolution  $0 \rightarrow \Pi_A \rightarrow A \rightarrow D \rightarrow 0$  therefore gives  $\text{Tor}_n^A(D, N) = 0$  for every  $n > 0$ . Nevertheless  $D$  is not flat: for  $0 \neq x \in \Pi_A$  the injection  $x A \hookrightarrow A$ , after tensoring with  $D$ , becomes the zero map between two copies of the nonzero module  $D$ . These are the parts of Theorem 12.2 needed here; no assertion about the other four models is used.  $\square$

For  $\kappa = \aleph_1$ :  $2^{\aleph_0} \leq \aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}$ , so  $A_{\aleph_1}^{03}$  has the size of the continuum, every target of size below the continuum sees only the constant term, whether or not CH holds, and the ideal needs  $\aleph_1$  generators. For every infinite  $\rho$ ,  $\kappa = 2^\rho$  gives  $\theta = 2^\rho$  with  $\text{cf}(2^\rho) > \rho$ ; so arbitrarily large thresholds are realized in ZFC. This is a realization theorem for this family, not a classification of ordered exponent groups. The case  $\text{cf } \kappa = \aleph_0$  is excluded because countable supports can then run cofinally through an arm (03).

**Lemma 11.6** (Cardinal facts [03]). *For every infinite  $\kappa$ ,  $\kappa^{\aleph_0} = \max(2^{\aleph_0}, |[\kappa]^{\leq \aleph_0}|)$ ; for every infinite  $\rho$ ,  $(2^\rho)^{\aleph_0} = 2^\rho$  and  $\text{cf}(2^\rho) > \rho$ .*

*Proof.* A sequence in  $\kappa$  is determined by its nonempty countable range and one of at most  $\aleph_0^{\aleph_0} = 2^{\aleph_0}$  sequences in that range. This gives the upper bound by the stated maximum. Conversely, binary sequences embed in  $\kappa^{\aleph_0}$ ; and, by choice, select an enumeration of each nonempty at-most-countable subset of  $\kappa$ , repeating entries for finite subsets. Different ranges give different sequences. Adjoining the empty subset does not change this infinite cardinality. These two lower bounds prove the first formula.

For infinite  $\rho$ , cardinal exponentiation gives  $(2^\rho)^{\aleph_0} = 2^{\rho \cdot \aleph_0} = 2^\rho$ . For the cofinality claim, suppose the binary functions on  $\rho$  were covered by families  $(E_\xi)_{\xi < \tau}$  with  $0 < \tau \leq \rho$  and  $|E_\xi| < 2^\rho$  for every  $\xi$ . Partition  $\rho$  into  $\tau$  disjoint blocks  $B_\xi$ , each of size  $\rho$ . There are  $2^\rho$  binary patterns on  $B_\xi$ , but fewer than  $2^\rho$  restrictions of members of  $E_\xi$ , so choose a missing pattern on each block. The assembled function differs from every member of  $E_\xi$  on its corresponding block, contradicting the cover. A cofinal sequence of length at most  $\rho$  in  $2^\rho$  would give just such a cover by smaller initial segments. Thus  $\text{cf}(2^\rho) > \rho$ , the required instance of König's theorem.  $\square$

## 11.2 Supports of size $< \kappa$ over a two-sided group [06]

Fix an uncountable regular cardinal  $\kappa$ . In the additive group of **No** put

$$h_\alpha = \omega^{\alpha+1}, \quad \epsilon_\alpha = \omega^{-(\alpha+1)} \quad (\alpha < \kappa), \quad \Gamma_\kappa^{06} = \text{span}_{\mathbb{Q}}\{h_\alpha, \epsilon_\alpha : \alpha < \kappa\}$$

(finite rational combinations of distinct Conway monomials, hence linearly independent). This is a divisible ordered group of cardinality  $\kappa$ , and  $\epsilon_\beta \ll \epsilon_\alpha \ll h_\alpha \ll h_\beta$  for  $\alpha < \beta$ ; here  $\ll$  refers to the outer exponents.

**Lemma 11.7** (Simultaneous upper and lower scales [06]). *For every  $T \subseteq \Gamma_\kappa^{06}$  with  $|T| < \kappa$  there is  $\beta < \kappa$  with  $\epsilon_\beta \ll |t| \ll h_\beta$  for all  $t \in T \setminus \{0\}$ .*

*Proof.* Each member of  $T$  uses finitely many of the generators. The union of their index sets has cardinality less than  $\kappa$ , so regularity gives  $\beta < \kappa$  strictly above every index occurring. The empty union allows any  $\beta$ . For a nonzero  $t$ , the largest Conway monomial in its finite rational expression determines the Archimedean class of  $|t|$ ; its coefficient cannot cancel because the expression uses distinct monomials. Both  $h_\alpha$  and  $\epsilon_\alpha$ , for  $\alpha < \beta$ , have classes strictly between those of  $\epsilon_\beta$  and  $h_\beta$ . Thus for every positive integer  $n$ ,  $n\epsilon_\beta < |t|$  and  $n|t| < h_\beta$ , as required.  $\square$

For  $k \in \{\mathbb{R}, \mathbb{C}\}$  put  $K_{\kappa, k}^{06} = \{f \in k((X^{\Gamma_\kappa^{06}})) : |\text{supp } f| < \kappa\}$ ,  $K_\kappa^{06} = K_{\kappa, \mathbb{R}}^{06}$ ,  $\Pi_\kappa^{06} = \{f \in K_\kappa^{06} : \text{supp } f > 0\}$  and  $A_\kappa^{06} = \mathbb{Z} + \Pi_\kappa^{06}$  (06's  $R_\kappa$ );  $X^g \mapsto \omega^g$  makes them subsets of **No** and **No**[ $i$ ].

**Proposition 11.8** (Field closure and integer part [06]).  *$K_{\kappa, \mathbb{R}}^{06}$  is real closed,  $K_{\kappa, \mathbb{C}}^{06} = K_{\kappa, \mathbb{R}}^{06}[i]$  is algebraically closed,  $A_\kappa^{06} = K_\kappa^{06} \cap \mathbf{Oz}$ ,  $\text{Frac}(A_\kappa^{06}) = K_\kappa^{06}$ , and  $A_\kappa^{06}$  is a discretely ordered integer part of  $K_\kappa^{06}$ .*

*Proof.* For finitely many series, let  $S$  be the union of their supports and  $\Delta = \text{span}_{\mathbb{Q}} S \subseteq \Gamma_{\kappa}^{06}$ . Then

$$|\Delta| \leq \max(|S|, \aleph_0) < \kappa.$$

All the chosen series lie in the full Hahn field  $k((X^{\Delta}))$ , and every support in that field has size less than  $\kappa$ . This proves closure under sums, products, negatives and nonzero inverses. Applied to the finitely many coefficients of a polynomial over  $K_{\kappa, \mathbb{C}}^{06}$ , the same argument places them in  $\mathbb{C}((X^{\Delta}))$ . This field is algebraically closed because  $\Delta$  is divisible [27, Corollary 4]; hence the polynomial splits within the support bound. Splitting coefficients into real and imaginary parts preserves that bound, and the union of two supports of size less than  $\kappa$  still has size less than  $\kappa$ . Therefore  $K_{\kappa, \mathbb{C}}^{06} = K_{\kappa, \mathbb{R}}^{06}[i]$ . The real field is ordered by its leading coefficient and has this algebraically closed quadratic extension, so it is real closed. Only uncountability of  $\kappa$ , not regularity, was used in this paragraph.

The embedding  $X^g \mapsto \omega^g$  identifies  $A_{\kappa}^{06}$  with  $K_{\kappa}^{06} \cap \mathbf{Oz}$ : the normal-form characterization of  $\mathbf{Oz}$  requires exactly nonnegative exponents and an integer coefficient at zero. For  $f \in K_{\kappa}^{06}$ , apply Lemma 11.7 to  $\text{supp } f$ . Every  $g$  in that support satisfies  $h_{\beta} + g > 0$ , including  $g = 0$ , since  $h_{\beta} > 0$ . Consequently  $X^{h_{\beta}} f \in \Pi_{\kappa}^{06}$  and  $X^{h_{\beta}} \in A_{\kappa}^{06} \setminus \{0\}$ , giving  $f = (X^{h_{\beta}} f)/X^{h_{\beta}}$  and the fraction-field equality.

For the integer part, use the floor rule in Section 11.1. Its positive truncation and corrected integer constant retain support of size less than  $\kappa$ , so the floor belongs to  $A_{\kappa}^{06}$ . A positive element of this ring is either a positive integer or has positive leading exponent and positive leading coefficient, in which case it exceeds every real constant. Thus there is no ring element strictly between 0 and 1. The sign condition is essential:  $-X^g$  for  $g > 0$  also has positive leading exponent but is negative.  $\square$

Field closure uses only that  $\kappa$  is uncountable; regularity is used for simultaneous gaps and generation (06). The first-kappa report proves the closedness of  $\{f \in k((t^{\Gamma})) : |\text{supp } f| < \kappa\}$  for every uncountable  $\kappa$ , possibly singular (`fkcthm:main` in [46]); 06's closure argument is not a novelty claim, and singular support bounds may preserve field closure while failing the simultaneous gap property used here. The two families  $h_{\alpha}$  and  $\epsilon_{\alpha}$  are deliberate: the  $\epsilon_{\alpha}$  provide smaller positive gaps and the  $h_{\alpha}$  shift supports upward, and a one-sided family would not automatically give the fraction-field assertion (06).

The failure of the gap property at a singular bound can be seen directly. If the same formulas are used with an uncountable singular  $\kappa$ , choose a strictly increasing cofinal sequence  $(\alpha_{\xi})_{\xi < \text{cf } \kappa}$  in  $\kappa$ . The series

$$f = \sum_{\xi < \text{cf } \kappa} X^{\epsilon_{\alpha_{\xi}}}$$

has positive reverse well-ordered support of size  $\text{cf } \kappa < \kappa$ . Every  $g > 0$  in  $\Gamma_{\kappa}^{06}$  has a finite coordinate expression, so eventually  $\epsilon_{\alpha_{\xi}} \ll g$ . Thus  $f/X^g$  has a negative exponent and is not in the integer-part ring. This gives an element with no positive monomial divisor there, although the support-bounded field is still closed under field operations and algebraic roots. The regular cardinal hypothesis is retained in the results below.

**Proposition 11.9** (A chain of principal ideals [06]).  $\Pi_{\kappa}^{06} = \bigcup_{\alpha < \kappa} X^{\epsilon_{\alpha}} A_{\kappa}^{06}$  is a strictly increasing union; the ideal is idempotent and its least number of generators is  $\kappa$ .

*Proof.* Write  $A = A_{\kappa}^{06}$  and  $I = \Pi_{\kappa}^{06}$ . For  $\alpha < \beta$ ,

$$X^{\epsilon_{\alpha}} = X^{\epsilon_{\beta}} X^{\epsilon_{\alpha} - \epsilon_{\beta}} \quad \text{with} \quad X^{\epsilon_{\alpha} - \epsilon_{\beta}} \in I.$$

Hence  $X^{\epsilon_\alpha} A \subseteq X^{\epsilon_\beta} A$ . The inclusion is strict: the only possible quotient of  $X^{\epsilon_\beta}$  by  $X^{\epsilon_\alpha}$  in the ambient field has negative exponent, so is not in  $A$ . For  $f \in I$ , [Lemma 11.7](#) gives  $\epsilon_\alpha < g$  for every  $g \in \text{supp } f$ , and therefore  $X^{-\epsilon_\alpha} f \in I$ . This proves the union; it also factors every  $f$  as a product of two members of  $I$ , proving  $I^2 = I$ . Zero causes no exception, since its quotient is zero.

The displayed  $\kappa$  monomials generate  $I$ . Conversely, if  $E \subseteq I$  has cardinality less than  $\kappa$ , regularity gives  $|\bigcup_{f \in E} \text{supp } f| < \kappa$ . Apply the gap lemma to this union to find a single  $a = \epsilon_\beta > 0$  with  $E \subseteq X^a A$ . Every finite  $A$ -linear combination of members of  $E$  still lies in  $X^a A$ . But  $X^{a/2} \in I$  does not: its unique quotient by  $X^a$  is  $X^{-a/2} \notin A$ . Thus  $E$  cannot generate  $I$ .  $\square$

**Lemma 11.10** (Size of the ring and of every smaller-scale field [merge]). *For every  $c > 0$  in  $\Gamma_\kappa^{06}$ , the field  $L_c = \{f \in K_\kappa^{06} : \text{supp } f \subseteq H_c(\Gamma_\kappa^{06})\}$  has cardinality at least  $\kappa^{<\kappa}$ , and*

$$|K_{\kappa, \mathbb{R}}^{06}| = |K_{\kappa, \mathbb{C}}^{06}| = |A_\kappa^{06}| = |\Pi_\kappa^{06}| = \kappa^{<\kappa}.$$

*Proof.* First count the series from above. The exponent group has size  $\kappa$ . A support of size  $\nu < \kappa$ , together with its real or complex coefficient function, has at most  $\kappa^\nu (2^{\aleph_0})^\nu$  possible descriptions. For infinite  $\nu$ ,  $(2^{\aleph_0})^\nu = 2^\nu \leq \kappa^\nu$ , so this is at most  $\kappa^\nu$ . Finite supports together contribute at most  $\kappa \cdot 2^{\aleph_0} \leq \kappa^{\aleph_0} \leq \kappa^{<\kappa}$ . There are at most  $\kappa$  possible cardinal sizes  $\nu$ , and  $\kappa^{<\kappa} \geq \kappa$ , so summing these bounds gives  $|K_{\kappa, k}^{06}| \leq \kappa^{<\kappa}$  for either coefficient field.

For the lower bound, all coordinates occurring in the finite expression of  $c$  have indices below some  $\alpha_0 < \kappa$ . The final segment  $T = (\alpha_0, \kappa)$  has cardinality  $\kappa$ , and  $\epsilon_\alpha \ll c$  for every  $\alpha \in T$ . Each subset  $B \subseteq T$  of size less than  $\kappa$  gives

$$f_B = \sum_{\alpha \in B} X^{\epsilon_\alpha} \in L_c.$$

The support is reverse well ordered, since a nonempty subset has largest exponent at its least ordinal index, and its cardinality is  $|B| < \kappa$ . Its exponents lie in  $H_c(\Gamma_\kappa^{06})$ . The coefficients recover  $B$ , so the map  $B \mapsto f_B$  is injective.

To justify the subset count, for any set  $T$  of size  $\kappa$  one has  $|[T]^{<\kappa}| = \kappa^{<\kappa}$ . The upper bound follows by enumerating each subset, giving at most  $\sum_{\nu < \kappa} \kappa^\nu = \kappa^{<\kappa}$  possibilities. For the reverse bound fix  $0 < \nu < \kappa$  and a bijection  $\nu \times T \rightarrow T$ . The graph of each function  $\nu \rightarrow T$  becomes a distinct subset of  $T$  of size  $\nu < \kappa$ . Hence  $|[T]^{<\kappa}| \geq \kappa^\nu$  for every such  $\nu$ , and taking the supremum proves the claim.

Thus  $|L_c| \geq \kappa^{<\kappa}$ , and the upper bound already proved gives equality. By [Proposition 10.3\(i\)](#), multiplication by  $X^c$  embeds  $L_c$  into  $\Pi_\kappa^{06}$ . The inclusions of the ideal in the ring, the ring in the real field and the real field in its complexification give all remaining equalities. The same lower-bound series have real coefficients, so they also work in the complex smaller-scale field; its upper bound is the one proved above.  $\square$

**Theorem 11.11** (The exact threshold of  $A_\kappa^{06}$  [06] [merge]). *Let  $\kappa$  be uncountable and regular. Every homomorphism  $A_\kappa^{06} \rightarrow S$  with  $|S| < \kappa^{<\kappa}$  factors through  $\text{ct}$ , every  $A_\kappa^{06}$ -module  $M$  with  $|M| < \kappa^{<\kappa}$  satisfies  $\Pi_\kappa^{06} M = 0$ , and for each  $0 \neq x \in \Pi_\kappa^{06}$  the least cardinality of a ring target detecting  $x$ , and of a module on which  $x$  acts nontrivially, is exactly  $\kappa^{<\kappa}$ . The same holds for  $A_\kappa^{06}[i]$  with  $\mathbb{Z}[i]$ .*

*Proof.* Set  $A = A_\kappa^{06}$ ,  $I = \Pi_\kappa^{06}$  and  $\theta = \kappa^{<\kappa} = |A|$ . For  $0 \neq x \in I$ , [Lemma 11.7](#) supplies  $\epsilon_\beta < \text{supp } x$ . Put  $c = \epsilon_\beta/2$  and  $b = X^{-2c}x \in I$ . Then  $x = X^{2c}b$ , and [Proposition 10.3\(iii\)](#), together with [Lemma 11.10](#), bounds every ring or module detecting  $x$  from below by  $|L_c| = \theta$ . In the module case,  $xv \neq 0$  implies  $X^{2c}(bv) \neq 0$ , so the injection is into the module itself. The identity map of  $A$  and

the regular module with vector 1 attain this bound. This proves the exact threshold directly, and supplies the present model's case of [Theorem 10.10](#).

Every map  $\varphi$  into a smaller ring therefore kills  $I$ . Since  $A = \mathbb{Z} \oplus I$ , it satisfies  $\varphi = \varphi|_{\mathbb{Z}} \circ \text{ct}$ , and this factor is unique by restricting to  $\mathbb{Z}$ . Every module of size less than  $\theta$  has  $xv = 0$  for all  $x \in I$  and  $v \in M$ , which is exactly  $IM = 0$ . The ring argument uses only additivity and multiplicativity, so includes nonunital maps.

For the Gaussian version, real and imaginary parts give

$$A[i] = \mathbb{Z}[i] + I_{\mathbb{C}}, \quad I_{\mathbb{C}} = I + iI = \{f \in K_{\kappa, \mathbb{C}}^{06} : \text{supp } f > 0\}.$$

Finite unions of supports preserve the bound. Thus  $|A[i]| = \theta$ : it contains  $A$  and is contained in  $K_{\kappa, \mathbb{C}}^{06}$ , both of that size. The gap lemma depends only on exponents, and the complex smaller-scale fields also have size  $\theta$  by the preceding proof. The same collision and attainment arguments apply, with factorization through  $\mathbb{Z}[i]$  and annihilation by  $I_{\mathbb{C}}$ .  $\square$

06 proves the lower bound  $\kappa$  (with the explicit telescope family of the  $\kappa$  monomials  $X^{a+\epsilon_\alpha}$ , each witness having countable support) and, under the extra hypothesis  $\kappa^{<\kappa} = \kappa$ , that  $|A_\kappa^{06}| = \kappa$  and the threshold is exactly  $\kappa$ ; that is the special case of the theorem where  $\kappa^{<\kappa} = \kappa$ . The lower bounds of 06 need no cardinal arithmetic, and neither does the exact value  $\kappa^{<\kappa}$ , which comes from 06's own collision lemma (compare 03's field bound). This answers the first half of 06's question "Cardinal arithmetic without the matching hypothesis": the least size of a target detecting a fixed nonzero  $x \in \Pi_\kappa^{06}$  is  $\kappa^{<\kappa}$  in ZFC. The second half ("a different integer-part construction of size exactly  $\kappa$  with the same lower-bound property under weaker hypotheses") is answered by 08 in ZFC for every infinite  $\kappa$  ([Theorem 11.16](#)), for a ring of size exactly  $\kappa$  whose fraction field, unlike  $\text{Frac } A_\kappa^{06} = K_\kappa^{06}$ , is not claimed to be its ambient field.

**Example 11.12** (The first uncountable cardinal [06]). Under CH,  $\aleph_1^{<\aleph_1} = \aleph_1^{\aleph_0} = 2^{\aleph_0} = \aleph_1$ , so  $A_{\aleph_1}^{06}$  has size  $\aleph_1$ : every action on a countable abelian group is an ordinary integer action, while the regular module of size  $\aleph_1$  sees all purely infinite elements. Without CH the lower bound at  $\aleph_1$  holds, the ring has size  $\aleph_1^{\aleph_0} = 2^{\aleph_0}$ , and by [Theorem 11.11](#) the exact threshold is  $2^{\aleph_0}$ ; this ring is isomorphic to  $A_{\aleph_1}^{03}$  ([Remark 10.11](#)).

*Proof.* Every cardinal below  $\aleph_1$  is at most countable, so  $\aleph_1^{<\aleph_1} = \aleph_1^{\aleph_0}$ . In ZFC,

$$2^{\aleph_0} \leq \aleph_1^{\aleph_0} \leq (2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0}.$$

Apply [Theorem 11.11](#); CH only identifies this already computed threshold with  $\aleph_1$ . A countable module is below the threshold in either case and hence is acted on through  $\text{ct}$ .

For the stated isomorphism, the assignments  $e_{p_\alpha} \mapsto h_\alpha$  and  $e_{q_\alpha} \mapsto \epsilon_\alpha$  extend to an ordered group isomorphism  $G_{\aleph_1} \rightarrow \Gamma_{\aleph_1}^{06}$ : both groups have these rational bases in the same two-arm order, with the first nonzero coordinate determining the sign. Reindexing Hahn exponents by this map preserves sums, products, support order and support cardinality. At  $\kappa = \aleph_1$  both fields allow exactly the at-most-countable supports, so this gives a coefficient-preserving field isomorphism. It restricts to  $A_{\aleph_1}^{03} \cong A_{\aleph_1}^{06}$  and commutes with  $\text{ct}$ , because it preserves positivity and the zero exponent. The same construction applies with complex coefficients and Gaussian integer constants. This is a reindexing isomorphism; the two embedded subrings of  $\mathbf{No}$  need not be identified element by element.  $\square$

### 11.3 A controlled field of size exactly $\kappa$ [08]

Let  $\kappa$  be any infinite cardinal and  $\Gamma_\kappa^{08} = \bigoplus_{\xi < \kappa} \mathbb{Q}e_\xi$ , ordered lexicographically with the least nonzero index deciding the sign, so  $0 < ne_\eta < e_\xi$  for  $\xi < \eta$ ; it is divisible of cardinality  $\kappa$ , and each nonzero element has a finite coordinate support.

**Lemma 11.13** (Scale and support bounds [08]). *For every  $c > 0$  there are  $\kappa$  positive  $a \ll c$  (namely  $e_\xi$  for  $\xi$  beyond the least nonzero index of  $c$ ). If  $S \subseteq (\Gamma_\kappa^{08})_{>0}$  has fewer than  $\text{cf } \kappa$  elements, some  $\beta < \kappa$  has  $e_\beta \ll s$  for all  $s \in S$ .*

*Proof.* Let  $\xi_0$  be the least nonzero coordinate of  $c$ . For every  $\xi > \xi_0$  and every positive integer  $n$ , the leading coordinate of  $c - ne_\xi$  is still the positive coordinate of  $c$  at  $\xi_0$ . Hence  $e_\xi \ll c$ . The final segment  $(\xi_0, \kappa)$  has cardinality  $\kappa$ , proving the first assertion.

For the second, each  $s \in S$  has finite coordinate support. Their union has size less than  $\text{cf } \kappa$  and is bounded below  $\kappa$ ; this also holds when  $\text{cf } \kappa = \aleph_0$ , since then  $S$  is finite. Choose  $\beta < \kappa$  strictly above every coordinate occurring. Each positive  $s$  has its first nonzero coordinate before  $\beta$ , so the same comparison gives  $e_\beta \ll s$ . If  $S$  is empty, any  $\beta$  works.  $\square$

Let  $K = \mathbb{R} \cap \overline{\mathbb{Q}}$  be the countable real closed field of real algebraic numbers,  $H = K((X^{\Gamma_\kappa^{08}}))$ , and  $H_{\text{ctbl}} \subseteq H$  the series with at most countable support. This is a real closed subfield: finitely many series lie in  $K((X^\Delta))$  for the countable divisible subgroup  $\Delta$  generated by their supports. The full Hahn field is real closed [23, Fact 2.1.1] and all its supports are countable. This supplies field operations, positive square roots and odd-degree polynomial roots within  $H_{\text{ctbl}}$ .

A *truncation* restricts a series to an initial segment of its support in decreasing order, including the empty and whole support. If the support has countable reverse order type  $\alpha$ , there are  $\alpha + 1$  such initial segments, hence at most countably many truncations. Arbitrary subseries are not being adjoined.

For a subfield  $E \subseteq H_{\text{ctbl}}$ , its real closure here means its relative algebraic closure: all elements of  $H_{\text{ctbl}}$  algebraic over  $E$ . This is a real closed subfield, since positive square roots and odd-degree roots exist in  $H_{\text{ctbl}}$  and remain algebraic over  $E$ . Let  $F_0$  be this closure of the field generated by  $K$  and all  $X^g$ . Given  $F_n$ , adjoin all truncations of its elements and take the same relative algebraic closure to obtain  $F_{n+1}$ . These are increasing subfields; put  $F = \bigcup_{n < \omega} F_n$ .

**Lemma 11.14** (The controlled field [08]).  *$F$  is real closed and truncation-closed, has cardinality  $\kappa$ , consists of series with countable support, and contains  $K$  and every monomial.*

*Proof.* The field generated by  $K$  and all monomials has size  $\kappa$ : there are  $\kappa$  generators, every rational expression uses only finitely many, and the distinct monomials already give the lower bound. An algebraic extension of an infinite field of size  $\kappa$  has size at most  $\kappa$ , because there are  $\kappa$  nonzero polynomials and each has finitely many roots. Thus  $|F_0| = \kappa$ .

If  $|F_n| = \kappa$ , its elements together have at most  $\kappa \cdot \aleph_0 = \kappa$  truncations. Adjoining these elements and then taking an algebraic extension again gives size at most  $\kappa$ ; the monomials give equality. Induction proves  $|F_n| = \kappa$  for all  $n$ , and the countable union has size  $\kappa$  as well. This calculation uses neither regularity nor  $\kappa^{\aleph_0} = \kappa$ .

The union is a field because the stages increase, and it is real closed: a positive element, or the finitely many coefficients of an odd-degree polynomial, belongs to some real closed stage, which supplies the required root. Every truncation of  $f \in F_n$  belongs to  $F_{n+1}$ , proving truncation closure



of  $F$ . Finally every stage lies in  $H_{\text{ctbl}}$  and contains  $K$  and the monomials, proving the remaining assertions.  $\square$

**Lemma 11.15** (Finite-coordinate invariant [08]). *Every  $f \in F$  has  $\text{supp } f \subseteq \Gamma_E = \bigoplus_{\xi \in E} \mathbb{Q}e_\xi$  for some finite  $E \subseteq \kappa$  (its support need not be finite).*

*Proof.* For finite  $E \subseteq \kappa$  put  $H_E = K((X^{\Gamma_E}))$ . The group  $\Gamma_E$  is countable and divisible, so  $H_E$  is real closed and lies in  $H_{\text{ctbl}}$ . Also  $H_E, H_{E'} \subseteq H_{E \cup E'}$ , giving a directed family. Its coefficients all belong to the same field  $K$ ; allowing infinitely many series terms does not introduce additional basis coordinates.

Every element of the field generated by  $K$  and the monomials is a rational expression in finitely many monomials. The union of their finite coordinate sets is finite, so the expression belongs to some  $H_E$ . If  $f \in F_0$ , choose a nonzero polynomial witnessing its algebraicity over that generated field. Its finitely many coefficients lie in one  $H_E$ , so  $f$  is algebraic over  $H_E$ . A real closed field has no proper ordered algebraic extension; since  $H_E(f)$  lies in the ordered field  $H_{\text{ctbl}}$ , it follows that  $f \in H_E$ .

Suppose the assertion holds for  $F_n$ . A truncation of an element of  $H_E$  still has all its exponents in  $\Gamma_E$ , and hence belongs to  $H_E$ . Each rational expression in  $F_n$  and the adjoined truncations therefore belongs to some  $H_E$ , by taking the finite union of the coordinate sets used by its inputs. For an element of  $F_{n+1}$ , apply the preceding algebraicity argument to the finitely many coefficients of a defining polynomial. This proves the invariant at every stage, and thus throughout  $F$ .  $\square$

This invariant removes regularity: even at a singular  $\kappa$  of countable cofinality, the coordinates in one element are confined to a finite set. The proof uses countable supports to control the number of truncations and finite coordinate sets to find smaller positive scales (08). For example,  $\sum_{n < \omega} X^{e_n}$  has a countable reverse well-ordered support and lies in  $H_{\text{ctbl}}$ , but it is not in  $F$ : its exponents use infinitely many basis coordinates. Thus countable support is a necessary condition on members of  $F$ , not a description of all its members. This distinction allows the construction to have size  $\kappa$  even when the full countable-support field is larger.

The map  $\iota(\sum q_\xi e_\xi) = \sum q_\xi \omega^{-\xi}$  is an increasing additive embedding of  $\Gamma_\kappa^{08}$  into  $\mathbf{No}$  (the least nonzero coordinate decides the sign on both sides), and  $\Phi(\sum a_g X^g) = \sum a_g \omega^{\iota(g)}$  is an ordered field embedding  $H \hookrightarrow \mathbf{No}$  preserving truncations and constant terms. Put

$$F_\kappa^{08} = \Phi(F), \quad A_\kappa^{08} = F_\kappa^{08} \cap \mathbf{Oz} = \Phi(\{f \in F : \text{supp } f \geq 0, \text{ ct}(f) \in \mathbb{Z}\}),$$

and  $\Pi_\kappa^{08} = \ker(\text{ct}|_{A_\kappa^{08}})$ . In calculations below we identify a Hahn series with its image under  $\Phi$ ; in particular  $X^{e_0}$  denotes the actual surreal  $\omega$ , since  $\iota(e_0) = 1$ . The ring has size  $\kappa$ : it is contained in  $\Phi(F)$  of that size and contains the  $\kappa$  distinct positive monomials  $X^{e_\xi}$ . It is a discretely ordered integer part of  $F_\kappa^{08}$ . Indeed, for  $x \in F$  its positive part  $P$  is a truncation, hence lies in  $F$ , and its constant coefficient  $r$  belongs to  $K \subseteq F$ . Apply the floor rule of Section 11.1 to  $x = P + r + \epsilon$ ; the result belongs to  $F$  and has precisely the support and constant-term conditions defining  $A_\kappa^{08}$ . Discreteness is inherited from  $\mathbf{Oz}$ .

**Theorem 11.16** (Sharp models at every infinite cardinal [08]). *For every infinite cardinal  $\kappa$ :*

- (i)  $F_\kappa^{08}$  is a truncation-closed real closed subfield of  $\mathbf{No}$  with countable supports,  $|F_\kappa^{08}| = |A_\kappa^{08}| = \kappa$ , and  $A_\kappa^{08}$  is an integer part of it;
- (ii) every homomorphism from  $A_\kappa^{08}$  to a ring of cardinality  $< \kappa$  factors through  $\text{ct}$ ;

- (iii) every quotient ring of  $A_\kappa^{08}$  is a quotient  $\mathbb{Z}/n\mathbb{Z}$  ( $n \geq 0$ ) under  $\text{ct}$  or has cardinality exactly  $\kappa$ ;
- (iv)  $\Pi_\kappa^{08}$  needs exactly  $\text{cf } \kappa$  generators;
- (v) every unital  $A_\kappa^{08}$ -module of cardinality  $< \kappa$  has action  $f \cdot m = \text{ct}(f)m$ ;
- (vi) the positive-characteristic maximal residue fields are the  $\mathbb{F}_p$ , with maximal ideals  $pA_\kappa^{08}$ ; characteristic-zero maximal residue fields exist and all have cardinality exactly  $\kappa$ .

No hypothesis such as  $CH$ ,  $GCH$ ,  $\kappa^{\aleph_0} = \kappa$  or an inaccessible cardinal is used.

*Proof.* Write  $A = A_\kappa^{08}$  and  $I = \Pi_\kappa^{08}$ . (i) follows from Lemma 11.14 and the embedding and floor argument above. In particular all ring quotients used below are sets.

(ii) Each  $f \in I$  has support in some  $\Gamma_E$  with  $E$  finite, by Lemma 11.15. Choose  $\beta < \kappa$  strictly above every member of  $E$ . Each nonzero exponent in this positive support has its first nonzero coordinate in  $E$ , so exceeds  $e_\beta$ . Also Lemma 11.13 gives  $\kappa$  scales below every  $c > 0$ . Apply Theorem 10.7 to obtain annihilation of  $I$ . The factor is the restriction to  $\mathbb{Z}$ , since  $\varphi(n + f) = \varphi(n)$  for  $n \in \mathbb{Z}$ ,  $f \in I$ ; it is unique. The argument includes nonunital ring homomorphisms.

(iii) If  $I \subseteq J$ , the constant term identifies  $A/J \cong \mathbb{Z}/\text{ct}(J)$ , and every ideal of  $\mathbb{Z}$  is  $n\mathbb{Z}$  for a unique  $n \geq 0$ . If  $I \not\subseteq J$ , the quotient map detects some element of  $I$ , so (ii) gives  $|A/J| \geq \kappa$ ; the opposite inequality follows from  $|A| = \kappa$ . The alternatives need not be disjoint: for  $\kappa = \aleph_0$ , the quotient  $A/I \cong \mathbb{Z}$  also has size  $\kappa$ . The cases  $n = 0$  and  $n = 1$  include  $\mathbb{Z}$  and the zero ring, respectively.

(iv) Choose a cofinal  $C \subseteq \kappa$  of size  $\text{cf } \kappa$ . For  $f \in I$ , choose  $\beta \in C$  beyond a finite coordinate set for  $f$ . Then  $X^{-e_\beta} f \in F$  has positive support, so lies in  $I$ . Thus the monomials  $X^{e_\beta}$ ,  $\beta \in C$ , generate  $I$ . Conversely, a family of fewer than  $\text{cf } \kappa$  elements has its finite coordinate sets bounded by some  $\beta < \kappa$ , even when  $\text{cf } \kappa = \aleph_0$ . All its members, and hence their generated ideal, lie in  $X^{e_\beta} A$ . This ideal misses  $X^{e_\beta/2} \in I$ , whose unique quotient by  $X^{e_\beta}$  in the field is  $X^{-e_\beta/2} \notin A$ . No smaller family can generate  $I$ .

(v) Apply Lemma 10.9 to (ii): for each  $m \in M$ , the quotient ring  $A/\text{Ann}(m)$  is in bijection with  $Am \subseteq M$ , so (ii) kills  $I$  in that quotient. Thus  $IM = 0$ . Since the action is unital, the remaining action of  $\mathbb{Z}$  is its ordinary additive action, giving  $f \cdot m = \text{ct}(f)m$ .

(vi) For every nonzero integer  $n$ , the constant  $1/n$  belongs to  $K \subseteq F$ , and multiplication by it preserves positive supports. Thus  $nI = I$  and  $nA = n\mathbb{Z} + I$ . This assertion is not used at  $n = 0$ , where  $0I = 0 \neq I$ . For a prime  $p$ , it gives  $A/pA \cong \mathbb{F}_p$ . Any maximal ideal with residue characteristic  $p$  contains  $pA$ , and therefore equals it, since  $pA$  is already maximal.

For characteristic zero,  $u = 1 + X^{e_0}$  is a nonunit by Lemma 2.4. Hence  $uA$  is proper, and the ordinary maximal-ideal theorem for the set-sized ring  $A$  gives a maximal ideal  $M \supseteq uA$ . If  $0 \neq n \in \mathbb{Z}$  belonged to  $M$ , the identity  $X^{e_0} = n(X^{e_0}/n)$ , with  $X^{e_0}/n \in I$ , would put  $X^{e_0}$  and then  $1 = u - X^{e_0}$  in  $M$ , a contradiction. So  $A/M$  has characteristic zero. Finally, any characteristic-zero maximal residue field of size less than  $\kappa$  would, by (ii), be a surjective image of  $\mathbb{Z}$ . Characteristic zero makes that map injective, forcing the residue ring to be isomorphic to  $\mathbb{Z}$ , which is not a field. Every such residue field therefore has size at least  $\kappa$ , and exactly  $\kappa$  because it is a quotient of  $A$ .  $\square$

For every  $0 \neq f \in \Pi_\kappa^{08}$ , (ii) and (v) give the lower bound  $\kappa$  for a ring target detecting  $f$  or a unital module on which  $f$  acts nontrivially. The identity map and the regular module  $A_\kappa^{08}$  with vector 1 attain it, so the stated cardinal boundary is sharp for each such element.

At  $\kappa = \aleph_0$  the theorem restricts only finite targets, and the characteristic-zero residue fields are countable and detect infinite monomials. For regular  $\kappa$  the generator count is  $\kappa$ ; at a singular  $\kappa$  it

is smaller, so a countably generated ideal can be invisible to all ring targets below a much larger singular  $\kappa$  of countable cofinality. Two resources are separated: monomial generators coinital in the support scales (cf  $\kappa$ ) and pairwise collision witnesses below a fixed scale ( $\kappa$ ) (08).

In the residue field of a maximal ideal containing  $1 + \omega$  the image of  $\omega$  is  $-1$ , not  $0$ , and that field has size  $\kappa$ ; so for every infinite  $\kappa$  the subring  $A_\kappa^{08} \subseteq \mathbf{Oz}$  has a characteristic-zero field quotient of cardinality  $\kappa$  in which  $\omega \mapsto -1$ . This quotient map cannot extend to  $\mathbf{Oz}$ , where every set-sized image sends  $\omega$  to  $0$ : an explicit extension obstruction between a set-sized integer part and the class ring (08). 08 does not identify  $\text{Frac}(A_\kappa^{08})$  with  $F_\kappa^{08}$ , imposes no initial-segment condition or birthday cutoff on the embedding, and designs the construction for truncation, support control and cardinality, not for closure under a derivation or an exponential; the isomorphism types of the residue fields are not classified.

## 11.4 Countable supports over a long one-armed group [09]

Fix an infinite cardinal  $\lambda$  with  $\lambda^{\aleph_0} = \lambda$ . Then  $2^{\aleph_0} \leq \lambda^{\aleph_0} = \lambda$ , since  $2 \leq \lambda$ . Thus 09's separate continuum lower bound is redundant under the stated infinitude hypothesis. The qualification matters:  $0$  and  $1$  also satisfy  $\lambda^{\aleph_0} = \lambda$  and would not give this construction. For example,  $\lambda = 2^\nu$  works for every infinite  $\nu$ , because  $(2^\nu)^{\aleph_0} = 2^{\nu \cdot \aleph_0} = 2^\nu$ . Neither regularity of  $\lambda$  nor CH is assumed. Let  $L_\lambda = \omega_1 \times \lambda$  with the lexicographic order and  $\Gamma_\lambda^{09} = \bigoplus_{\ell \in L_\lambda} \mathbb{Q}e_\ell$ , the least nonzero index deciding the sign.

**Lemma 11.17** (A large small-exponent tail [09]). *For every countable  $S \subseteq (\Gamma_\lambda^{09})_{>0}$  there is  $\alpha_0 < \omega_1$  with  $e_{(\alpha,\beta)} \in U_\Gamma(S)$  for all  $\alpha_0 \leq \alpha < \omega_1$  and  $\beta < \lambda$ ; in particular  $U_\Gamma(S)$  contains  $\lambda$  elements, and every countable positive set has a positive strict lower bound.*

*Proof.* Let  $C$  be the set of first coordinates appearing in the finite index supports of elements of  $S$ . It is countable. Set  $\alpha_0 = \sup\{\alpha + 1 : \alpha \in C\} < \omega_1$ , with supremum  $0$  if  $C$  is empty. For  $s \in S$ , its least nonzero coordinate precedes every  $(\alpha, \beta)$  with  $\alpha \geq \alpha_0$ , and its coefficient there is positive. Subtracting  $ne_{(\alpha,\beta)}$  leaves that leading coordinate unchanged. Hence  $0 < ne_{(\alpha,\beta)} < s$  for every positive integer  $n$ , which is exactly the required small-exponent condition. The single row  $\{e_{(\alpha_0,\beta)} : \beta < \lambda\}$  already gives  $\lambda$  distinct elements, and  $e_{(\alpha_0,0)}$  is a strict positive lower bound for  $S$ . Only the regularity of  $\omega_1$ , not of  $\lambda$ , is used here.  $\square$

For  $k \in \{\mathbb{R}, \mathbb{C}\}$  and a unital subring  $D \subseteq k$ , let  $H_{\lambda,k} = H_\omega(k, \Gamma_\lambda^{09})$ , let  $\Pi_{\lambda,k}^{09}$  be its positive-support part, and put  $A_{\lambda,D}^{09} = D + \Pi_{\lambda,k}^{09}$ . Write  $A = A_{\lambda,D}^{09}$  and  $\Pi_A = \Pi_{\lambda,k}^{09}$  throughout this subsection. The ambient field is closed under its field operations by Lemma 11.1; its real version is real closed and its complex version is its algebraically closed complexification. The ring  $A$  is a domain,  $\text{ct} : A \rightarrow D$  is a split surjection with kernel  $\Pi_A$ , and  $A^\times = D^\times$  by Lemma 2.4. The subordinate quotients of Lemma 4.7 remain in  $\Pi_A$ : their support bound is still countable. This is a closure assertion for those particular quotients, not for division by arbitrary nonzero elements of  $A$ .

Here  $|\Gamma_\lambda^{09}| = |L_\lambda| = \lambda$ : finite rational combinations give the upper bound and the basis gives the lower bound. There are at most  $\lambda^{\aleph_0} = \lambda$  countable supports and, for each one, at most  $|k|^{\aleph_0} = 2^{\aleph_0} \leq \lambda$  coefficient assignments. Consequently  $|H_{\lambda,k}| \leq \lambda$ . The  $\lambda$  distinct positive basis monomials belong to  $\Pi_A$ , so  $|\Pi_A| = |A| = |H_{\lambda,k}| = \lambda$ , for every allowed  $D$ .

Regard  $\lambda$  as its initial ordinal and set  $j(\alpha, \beta) = \lambda \cdot \alpha + \beta$ , using ordinal operations. This is an order embedding of  $L_\lambda$  into the ordinals. Thus

$$\iota\left(\sum_{\ell} q_\ell e_\ell\right) = \sum_{\ell} q_\ell \omega^{-j(\ell)}$$

is an ordered additive embedding: the finite sum's leading coefficient is the coefficient at its least index. Reindexing normal forms by  $\iota$  gives a field embedding

$$\Phi_{\mathbb{R}} : H_{\lambda, \mathbb{R}} \longrightarrow \mathbf{No}, \quad \sum_{\gamma} c_{\gamma} X^{\gamma} \longmapsto \sum_{\gamma} c_{\gamma} \omega^{\iota(\gamma)}.$$

The support remains reverse well ordered, and injectivity follows from uniqueness of normal forms. Real and imaginary parts extend this to  $\Phi_{\mathbb{C}} : H_{\lambda, \mathbb{C}} \rightarrow \mathbf{No}[i]$ . These embeddings send  $A_{\lambda, \mathbb{Z}}^{09}$  into  $\mathbf{Oz}$  and  $A_{\lambda, \mathbb{Z}[i]}^{09}$  into  $\mathbf{Oz}[i]$ ; the former is the integer part of  $\Phi_{\mathbb{R}}(H_{\lambda, \mathbb{R}})$  by the truncation and endpoint rule following [Lemma 11.1](#). We identify these rings with their images only after applying  $\Phi$ . In particular,  $X^{e_{\ell}} \mapsto \omega^{\omega^{-j(\ell)}}$ , and  $X^{e(0,0)}$  maps to the actual  $\omega$ ; the group coordinates describe the exponents, not the monomials themselves.

**Theorem 11.18** (Cardinal barrier [09]). *Let  $A = A_{\lambda, D}^{09}$  and  $\Pi_A = \Pi_{\lambda, k}^{09}$ .*

- (i) *If  $\varphi : A \rightarrow B$  is a unital ring map with  $\varphi(\Pi_A) \neq 0$ , then  $|\varphi(A)| \geq \lambda$ .*
- (ii) *If  $M$  is an  $A$ -module with  $\Pi_A M \neq 0$ , then  $|M| \geq \lambda$ .*
- (iii) *All maps to rings of size  $< \lambda$ , and all modules of size  $< \lambda$ , factor through ct.*
- (iv) *For every ideal  $J$ , either  $\Pi_A \subseteq J$  and  $A/J \cong D/\text{ct}(J)$ , or  $|A/J| = \lambda$ .*

*In particular, in ZFC there is an explicit  $A \subseteq \mathbf{Oz}$  of cardinality  $2^{\aleph_0}$  (take  $\lambda = 2^{\aleph_0}$ ,  $D = \mathbb{Z}$ ) whose maps to countable rings factor through ct, whose quotients are finite, countably infinite, or of cardinality  $2^{\aleph_0}$  (the last for every quotient not induced from  $\mathbb{Z}$ ), and whose countable modules are abelian groups with constant-term action.*

*Proof.* Choose  $f \in \Pi_A$  with  $\varphi(f) \neq 0$  for (i), or  $f \in \Pi_A$  and  $v \in M$  with  $fv \neq 0$  for (ii). Its support is countable, so [Lemma 11.17](#) supplies a set  $E$  of  $\lambda$  positive small exponents. For distinct  $a, b \in E$ , [Lemma 4.7](#), with the sign changed if necessary, gives  $f = (X^a - X^b)q_{a,b}$  with  $q_{a,b} \in \Pi_A$ . Therefore

$$a \longmapsto \varphi(X^a), \quad a \longmapsto X^a v$$

are injective into  $\varphi(A)$  and  $M$ , respectively, by [Lemma 4.8](#). Only a finite product identity is transported by  $\varphi$ ; no infinite sum is transported. The ring argument also works for nonunital maps and noncommutative targets.

If the target or module has size less than  $\lambda$ , this forces annihilation of all of  $\Pi_A$ . Since  $A = D \oplus \Pi_A$  additively, restriction to  $D$  gives the unique factorization in (iii). In (iv), if  $\Pi_A \subseteq J$ , the map  $a + J \mapsto \text{ct}(a) + \text{ct}(J)$  is a well-defined ring isomorphism  $A/J \rightarrow D/\text{ct}(J)$ . Otherwise the quotient map detects some  $f \in \Pi_A$ , giving  $\lambda \leq |A/J| \leq |A| = \lambda$ .

For the final assertion, Cantor's theorem gives  $\aleph_0 < 2^{\aleph_0}$ , so countable targets and modules are below the threshold without CH. Every ideal of  $\mathbb{Z}$  is  $n\mathbb{Z}$  with  $n \geq 0$ :  $n = 0$  gives the countably infinite quotient  $\mathbb{Z}$ ,  $n = 1$  the zero ring, and  $n \geq 2$  the finite cyclic rings. Conversely these quotients all occur via ct. A countable module has exactly its underlying abelian group action pulled back along ct.  $\square$

When the continuum exceeds  $\aleph_1$  the last statement also excludes all intermediate quotient cardinalities; this is conditional only on that cardinal inequality (09).

**Lemma 11.19** (Recovering the coefficient field [09]). *Let  $\varphi : A \rightarrow F$  be a unital map to a field with  $\varphi(\Pi_A) \neq 0$ . There is a unique embedding  $\sigma : k \hookrightarrow F$  with  $\varphi(rg) = \sigma(r)\varphi(g)$  for  $r \in k$ ,  $g \in \Pi_A$ , and  $\sigma|_D = \varphi|_D$ ; in particular  $F$  has characteristic 0.*

*Proof.* Choose  $f \in \Pi_A$  with  $\varphi(f) \neq 0$  and define  $\sigma(r) = \varphi(rf)/\varphi(f)$ . Multiplication by a coefficient preserves positive support, so  $rf \in \Pi_A$  even when  $r \notin A$ . Additivity follows from  $(r+s)f = rf + sf$ , and  $\sigma(1) = 1$ . The identity  $(rf)(sf) = (rsf)f$  gives

$$\sigma(r)\sigma(s) = \frac{\varphi((rf)(sf))}{\varphi(f)^2} = \frac{\varphi(rsf)}{\varphi(f)} = \sigma(rs).$$

Thus  $\sigma : k \rightarrow F$  is a unital field homomorphism, hence injective. For  $d \in D$ ,  $\varphi(df) = \varphi(d)\varphi(f)$  gives  $\sigma(d) = \varphi(d)$ . For any  $g \in \Pi_A$ , apply  $\varphi$  to  $(rg)f = (rf)g$  and divide by  $\varphi(f)$  to obtain  $\varphi(rg) = \sigma(r)\varphi(g)$ , including when  $\varphi(g) = 0$ . Any embedding with this property must give the same ratio at the chosen  $f$ , proving uniqueness and independence of its choice. The canonical copy of  $k$  is therefore relative to  $\varphi$ ; it is not obtained by assuming that every coefficient already belongs to  $A$ .  $\square$

**Theorem 11.20** (Residue fields at the threshold [09]). *Let  $A = A_{\lambda,D}^{09}$ .*

- (i) *For every  $0 \neq f \in \Pi_A$  there is a maximal ideal  $\mathfrak{m}$  with  $f \equiv 1 \pmod{\mathfrak{m}}$  and  $|A/\mathfrak{m}| = \lambda$ .*
- (ii) *Every maximal ideal not containing  $\Pi_A$  has a residue field of cardinality  $\lambda$  with a canonical copy of  $k$ .*
- (iii) *For  $D = \mathbb{Z}$ , the maximal ideals containing  $\Pi_A$  are the  $pA$ , with residue fields  $\mathbb{F}_p$ , and every other maximal ideal has a characteristic-zero residue field of cardinality  $\lambda$ .*
- (iv) *There are at least  $\lambda$  distinct maximal ideals not containing  $\Pi_A$ .*

*Consequently the least cardinality of a field (and of a module) detecting a given  $0 \neq f \in \Pi_A$  is exactly  $\lambda$ ; and for  $D = \mathbb{Z}$  the intersection of all finite-index ideals is  $\Pi_A \neq 0$  while  $\text{Jac}(A) = 0$ .*

*Proof.* (i) Since  $0 \neq f \in \Pi_A$ , the element  $1 - f$  is nonconstant, hence a nonunit by Lemma 2.4. The ideal  $(1 - f)A$  is proper. The ordinary maximal-ideal theorem for the set-sized ring  $A$  gives a maximal ideal  $\mathfrak{m}$  containing it. In the nonzero residue field,  $f$  then has image 1. The barrier gives  $|A/\mathfrak{m}| \geq \lambda$ , and the quotient upper bound gives equality.

(ii) The quotient map detects an element of  $\Pi_A$ , so the same cardinality argument applies. Its canonical coefficient-field embedding is exactly the one constructed in Lemma 11.19. For (iii), every nonzero integer  $n$  satisfies  $n\Pi_A = \Pi_A$ , since division of its coefficients by  $n$  stays in  $\Pi_A$ . Hence  $pA = p\mathbb{Z} + \Pi_A$  and  $A/pA \cong \mathbb{F}_p$ . The ideals containing  $\Pi_A$  correspond under ct to ideals of  $\mathbb{Z}$ ; their maximal members are precisely these  $pA$ . All remaining maximal ideals have the fields of (ii), whose embedded  $\mathbb{R}$  forces characteristic zero.

(iv) For  $\ell < \ell'$  put  $\gamma = e_\ell$  and  $a = e_{\ell'}$ . Then  $na < \gamma$  for every positive integer  $n$ . The exponents  $\gamma - na$  are positive and strictly decreasing, so they form a countable reverse well-ordered support. Thus

$$q = - \sum_{n \geq 1} X^{\gamma - na} \in \Pi_A, \quad (1 - X^a)q = X^\gamma, \quad 1 = (1 - X^\gamma) + (1 - X^a)q.$$

The product is a Hahn product: the term at  $\gamma$  has coefficient 1, and at each  $\gamma - na$  the two contributions cancel. No passage of an infinite sum through a quotient map is used. The principal ideals  $(1 - X^{e_\ell})A$  are proper by the units criterion and pairwise comaximal by this identity. Choose one maximal ideal above each of them. No maximal ideal can contain two, and none can contain  $\Pi_A$ , because it would contain both  $X^{e_\ell}$  and  $1 - X^{e_\ell}$ . Since  $|L_\lambda| = \lambda$ , this gives at least  $\lambda$  distinct maximal ideals of the required kind, using choice only for a set-indexed family.

For any fixed  $0 \neq f \in \Pi_A$ , (i) gives a field target of size  $\lambda$  and, by restriction of scalars, an  $A$ -module of that size on which  $f$  acts as the identity. Together with the barrier, this proves both exact minima. When  $D = \mathbb{Z}$ , every finite-index ideal contains  $\Pi_A$  by the barrier; conversely, an element  $n + g$  with  $n \neq 0$  is excluded from  $pA$  by choosing a prime  $p$  not dividing  $n$ . Thus the intersection of all finite-index ideals is exactly  $\Pi_A$ , which is nonzero since it contains the basis monomials. Finally, every nonzero  $x = n + g$  is avoided by a maximal ideal: use such a  $pA$  if  $n \neq 0$ , and apply (i) to  $g = x$  if  $n = 0$ . This proves  $\text{Jac}(A) = 0$  despite the nonzero finite residual.  $\square$

All of this concerns set-sized rings and is not an application of the maximal-ideal theorem to **Oz**. The distinction between finite ring probes and field probes is sharp: the extra information is carried by residue fields of the full workspace cardinality (09). The number of maximal ideals and the isomorphism types of the residue fields are not determined; distinct maximal ideals need not give nonisomorphic fields.

The generator count is independent of  $\lambda$ . If  $\alpha < \alpha'$ , then  $e_{(\alpha,0)} - e_{(\alpha',0)} > 0$ , giving inclusion of the corresponding principal ideals. For every  $f \in \Pi_A$ , Lemma 11.17 supplies  $\alpha$  such that shifting its countable support by  $-e_{(\alpha,0)}$  still leaves positive support. Consequently

$$\Pi_{\lambda,k}^{09} = \bigcup_{\alpha < \omega_1} X^{e_{(\alpha,0)}} A$$

is an increasing union and gives  $\aleph_1$  generators. Conversely, the union of the supports of a countable family in  $\Pi_A$  is countable. Choose  $h$  small relative to that union by Lemma 11.17. Every member then lies in  $X^{2h}A$ , whereas  $X^h$  does not: its only possible quotient is  $X^{-h}$ , which has negative support. No countable family generates  $\Pi_A$ , so its least number of generators is exactly  $\aleph_1$ . Flatness is treated in Theorem 12.2.

*Remark 11.21* (Why the support restriction is real [09]). In the unrestricted Hahn field over  $\Gamma_\lambda^{09}$  the series  $f_* = \sum_{\alpha < \omega_1} X^{e_{(\alpha,0)}}$  has reverse well-ordered but uncountable support with no positive lower bound in  $\Gamma_\lambda^{09}$  (a late enough  $e_{(\alpha,0)}$  lies below any given  $h > 0$ ); it is not divisible by any positive monomial within the nonnegative-support ring and is not in  $A_{\lambda,D}^{09}$ . No assertion is made about the quotient classification of that larger unrestricted ring.

Indeed, the support is strictly decreasing in  $\alpha$ , so each nonempty subset has a greatest element, indexed by its least ordinal. It has cardinality  $\aleph_1$ , beyond the allowed support bound. For any  $h > 0$ , choose  $\alpha$  greater than the first coordinate of the leading index of  $h$ . Then  $0 < e_{(\alpha,0)} < h$ , and the coefficient of  $X^{e_{(\alpha,0)}-h}$  in  $X^{-h}f_*$  is nonzero. Hence division by  $X^h$  produces a negative exponent; uniqueness of the quotient in the ambient Hahn field rules out any other nonnegative-support quotient.

## 11.5 Supports of size $< \kappa$ over a one-armed group [11]

For uncountable  $\kappa$ , Theorem 10.8 (11) applies to any divisible ordered abelian set group  $\Gamma$  with  $\text{coin}(\Gamma_{>0}) \geq \kappa$ . The coinitality condition says that every set of fewer than  $\kappa$  positive elements has a positive strict lower bound. In the one-arm example below, set

$$F_{\kappa,K} = \{f \in K((X^\Gamma)) : |\text{supp } f| < \kappa\}, \quad I = \{f \in F_{\kappa,K} : \text{supp } f \subseteq \Gamma_{>0}\}, \quad A = R_{\Gamma,<\kappa}^{11} = D + I,$$

where  $K$  is a set field and  $D \subseteq K$  a unital subring. The subscript on  $F$  is a support-cardinality bound; arbitrary reverse well-ordered supports of that size are allowed.



**Proposition 11.22** (Long lexicographic groups [11]). *For  $\kappa$  uncountable and regular and  $\Gamma = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$  with the least nonzero index dominant,  $|\Gamma| = \kappa$  and  $\text{coin}(\Gamma_{>0}) = \kappa$ . The group  $\Gamma_\kappa^{06}$  also has coinitality  $\kappa$ , so  $A_\kappa^{06} = R_{\Gamma_\kappa^{06}, < \kappa}^{11}$  with  $(D, K) = (\mathbb{Z}, \mathbb{R})$  is an instance of [Theorem 10.8](#).*

*Proof.* The group has at least  $\kappa$  elements because its basis vectors are distinct, and at most  $\kappa$  because every element is a finite rational linear combination of them. For  $g > 0$  with least nonzero index  $\alpha$ , every  $\beta > \alpha$  satisfies  $ne_\beta < g$  for every positive integer  $n$ : subtraction leaves the leading coefficient of  $g$  unchanged. Hence the basis vectors form a coinital family of size  $\kappa$ .

If  $S \subseteq \Gamma_{>0}$  has size less than  $\kappa$ , regularity of  $\kappa$  bounds its set of leading indices below some  $\beta < \kappa$ . Then  $e_\beta < s$  for every  $s \in S$ ; in particular  $S$  is not coinital. This proves the reverse inequality and hence the exact coinitality. For  $\Gamma_\kappa^{06}$ , [Lemma 11.7](#) gives the lower bound, and the coinital family  $(e_\alpha)_{\alpha < \kappa}$  gives the upper bound. Substituting  $(D, K) = (\mathbb{Z}, \mathbb{R})$  in the definition of  $R^{11}$  gives the asserted identity of that support-restricted ring with  $A_\kappa^{06}$ .  $\square$

11 proves sharpness under the displayed extra hypotheses  $|K| \leq \kappa$  and  $\kappa^{<\kappa} = \kappa$ : then  $|R_{\Gamma, < \kappa}^{11}| = \kappa$  and its identity map is a target of size  $\kappa$  not factoring through  $\text{ct}$ . 11 states explicitly that these hypotheses are assumptions for the sharpness example, not assertions about all uncountable regular cardinals, and that regularity enters only the lexicographic example.

**Proposition 11.23** (Exact size and threshold [merge]). *In [Proposition 11.22](#), for every set field  $K$  and every  $c > 0$  the field  $L_c = \{f \in K((X^\Gamma)) : |\text{supp } f| < \kappa, \text{supp } f \subseteq H_c(\Gamma)\}$  has at least  $\kappa^{<\kappa}$  elements. If  $|K| \leq \max(\kappa, 2^{\aleph_0})$  (which includes 11's hypothesis  $|K| \leq \kappa$ ), then  $|R_{\Gamma, < \kappa}^{11}| = \kappa^{<\kappa}$ , and the least cardinality of a ring target or module detecting any nonzero element of the positive-support ideal is exactly  $\kappa^{<\kappa}$ .*

*Proof.* First  $F_{\kappa, K}$  is a field. Finitely many of its series have a union  $S$  of supports of size less than  $\kappa$ . The subgroup generated by  $S$  has size at most  $\max(|S|, \aleph_0) < \kappa$ , so the full Hahn field over that subgroup contains the series and still lies in  $F_{\kappa, K}$ . It supplies their field operations and nonzero inverses. This argument only uses uncountability of  $\kappa$ , not regularity. Since  $H_c(\Gamma)$  is a subgroup,  $L_c$  is the intersection of two subfields and is therefore a field.

Let  $\alpha_0$  be the least nonzero index of  $c$ . The final segment  $T = (\alpha_0, \kappa)$  has cardinality  $\kappa$ , and  $e_\alpha \ll c$  for every  $\alpha \in T$ . Every subset  $B \subseteq T$  of size less than  $\kappa$  gives

$$f_B = \sum_{\alpha \in B} X^{e_\alpha} \in L_c.$$

The support is reverse well ordered, since its greatest element in a nonempty subset has the least ordinal index, and its size is  $|B| < \kappa$ . The coefficient  $1_K \neq 0$  recovers  $B$ , even in positive characteristic. Thus  $B \mapsto f_B$  is injective. The subset count proved in [Lemma 11.10](#),  $|[T]^{<\kappa}| = \kappa^{<\kappa}$ , gives the lower bound without any restriction on  $|K|$ .

Now assume  $|K| \leq \max(\kappa, 2^{\aleph_0})$ . Supports of size  $\nu < \kappa$  have at most  $\kappa^\nu$  descriptions. For infinite  $\nu$ , their coefficient assignments number at most

$$|K|^\nu \leq \max(\kappa^\nu, 2^\nu) = \kappa^\nu.$$

All finite supports together contribute at most  $\kappa \cdot \max(\kappa, 2^{\aleph_0}) \leq \kappa^{\aleph_0} \leq \kappa^{<\kappa}$  possibilities, including constant series. There are at most  $\kappa$  possible support cardinalities, so  $|F_{\kappa, K}| \leq \kappa^{<\kappa}$ . Multiplication by  $X^c$  injects  $L_c$  into  $I$ , by [Proposition 10.3\(i\)](#). Together with  $I \subseteq A \subseteq F_{\kappa, K}$ , this proves

$$|L_c| = |I| = |A| = |F_{\kappa, K}| = \kappa^{<\kappa}.$$

For  $0 \neq x \in I$ , coinitality gives  $g > 0$  strictly below its support. Put  $c = g/2$ . Then  $x = X^{2c}b$  for  $b = X^{-2c}x \in I$ , since translation preserves support cardinality. [Proposition 10.3\(iii\)](#) gives the lower bound  $|L_c| = \kappa^{<\kappa}$  for every ring target or module detecting  $x$ . The identity map and the regular module with vector 1 attain it. This establishes the threshold directly, independently of the organizing theorem that summarizes it.  $\square$

Thus the exact boundary in this example is  $\kappa^{<\kappa}$  under the stated coefficient-field bound. It specializes to 11's boundary  $\kappa$  when  $\kappa^{<\kappa} = \kappa$ , and sharpens the original coinitality estimate when  $\kappa^{<\kappa} > \kappa$ .

For completeness, the ideal requires exactly  $\kappa$  generators, regardless of  $D$ . The positive basis monomials give a strictly increasing chain with

$$I = \bigcup_{\alpha < \kappa} X^{e_\alpha} A.$$

For  $\alpha < \beta$ ,  $e_\alpha - e_\beta > 0$  gives the inclusion; its strictness follows because  $X^{e_\beta - e_\alpha}$  has negative support. For any  $f \in I$ , regularity bounds the leading indices of its fewer than  $\kappa$  support exponents, so some  $e_\alpha$  lies strictly below all of them. The shifted series  $X^{-e_\alpha} f \in I$  proves the union and also  $I^2 = I$ . If  $E \subseteq I$  has size less than  $\kappa$ , regularity gives  $|\bigcup_{f \in E} \text{supp } f| < \kappa$ . A common strict lower bound  $h > 0$  then puts every member of  $E$ , and every finite  $A$ -linear combination of them, in  $X^h A$ . But  $X^{h/2} \in I$  is outside that ideal because its unique quotient is  $X^{-h/2} \notin A$ . So no such  $E$  generates  $I$ . The regularity used for this union of supports is distinct from the field-closure argument above.

With  $(D, K) = (\mathbb{Z}, \mathbb{R})$  this ring is a Hahn integer part of the kind treated in this section; 11 does not embed it in **No**, and no embedding is used in this section. Manuscript 17 supplies one:  $e_\alpha \mapsto \omega^{-\alpha}$  embeds  $\Gamma$  in  $(\mathbf{No}, +)$  as an ordered group, and hence  $R_{\Gamma, < \kappa}^{11}$  in **Oz** for  $(D, K) = (\mathbb{Z}, \mathbb{R})$  and in **Oz**[ $i$ ] for  $(\mathbb{Z}[i], \mathbb{C})$  ([Lemma 13.30](#)).

## 12 Flat ideals and homological reflection

All rings, modules, resolutions, tensor products and Tor, Ext groups in this section are ordinary set-sized objects over the set-sized rings of [Section 11](#). No class-sized resolution is formed, and no directed-colimit theorem is applied to the proper-class union  $\Pi = \bigcup_{a > 0} \omega^a \mathbf{Oz}$  (03, 06, 09). The package is 06's, proved there for  $A_\kappa^{06}$ ; 03 and 09 prove its flatness and Tor parts for their rings. The proofs use only the following properties, which all five constructions have:

- (H1)  $A$  is a commutative domain,  $D \subseteq A$  is a unital subring,  $A = D \oplus I$  additively, and  $\text{ct} : A \rightarrow D$  is the ring retraction with kernel  $I$ , identifying  $A/I \cong D$ ;
- (H2)  $A$  is built over a divisible exponent group  $\Gamma$ , and every  $f \in I$  has the form  $f = X^g b$  with  $g \in \Gamma_{>0}$  and  $b \in I$ ; hence  $I = \bigcup_{h \in \Gamma_{>0}} X^h A$ , a directed set-indexed union of principal ideals;
- (H3)  $I$  is not finitely generated.

(H2) is the gap property used in [Theorem 10.10](#) (a positive strict lower bound  $g$  of  $\text{supp } f$  gives  $b = X^{-g} f \in I$ ). (H3) holds because the least number of generators is  $\text{cf } \kappa$ ,  $\kappa$ ,  $\text{cf } \kappa$ ,  $\aleph_1$ , respectively  $\kappa$  ([Theorem 11.5\(iv\)](#), [Proposition 11.9](#), [Theorem 11.16\(iv\)](#), the chain after [Theorem 11.20](#), and for  $R_{\Gamma, < \kappa}^{11}$  the coinital chain  $(X^{e_\alpha})_{\alpha < \kappa}$  of [Proposition 11.22](#): fewer than  $\kappa$  elements have a common monomial divisor  $X^h$ , and  $X^h A$  misses  $X^{h/2}$ ); these counts are infinite. In particular  $I \neq 0$ , and  $I \neq A$  since  $A$  is generated by 1 as an ideal. We use  $D$  as an  $A$ -module via  $\text{ct}$ . The constant

inclusion  $D \hookrightarrow A$  is a ring section and a  $D$ -linear additive section, but is not asserted to be  $A$ -linear. All splitting claims below specify this latter module structure.

**Lemma 12.1** (Projective ideals of a domain [06]). *A nonzero projective ideal of a commutative domain is finitely generated.*

*Proof.* Choose  $0 \neq a \in J$ . For every  $A$ -linear  $f : J \rightarrow A$  and  $x \in J$ ,

$$af(x) = f(ax) = xf(a).$$

If  $f(a) = 0$ , cancellation of  $a$  in the domain gives  $f(x) = 0$  for every  $x$ , so  $f = 0$ .

Projectivity gives maps  $j : J \rightarrow \bigoplus_{s \in S} Ae_s$  and  $p : \bigoplus_{s \in S} Ae_s \rightarrow J$  with  $pj = \text{id}_J$  [30, Tag 05CF]. Write  $f_s$  for the  $s$ th coordinate of  $j$ , and  $u_s = p(e_s)$ . Each  $j(x)$  has finite support, and applying  $p$  yields  $x = \sum_s f_s(x)u_s$ . The set  $S_0 = \{s : f_s(a) \neq 0\}$  is finite. For  $s \notin S_0$  the first paragraph gives  $f_s = 0$  identically, so every  $x \in J$  is a linear combination of the fixed finite family  $(u_s)_{s \in S_0}$ . No finite generation assumption on the projective module was used.  $\square$

**Theorem 12.2** (Homological package [03, 06, 09] [merge]). *Let  $(A, I, D)$  satisfy (H1)–(H3). Then:*

- (i)  *$I$  is flat,  $I^2 = I$  (each element is a product of two elements of  $I$ ), and  $I$  is not projective;*
- (ii) *for every  $D$ -module  $N$  (inflated along  $\text{ct}$ ),  $I \otimes_A N = 0$  and  $\text{Tor}_n^A(D, N) = 0$  for  $n > 0$ ; in particular  $\text{Tor}_n^A(D, D) = 0$ ;*
- (iii)  *$D$  is not flat:  $\text{fd}_A D = 1$ , and  $\text{Tor}_1^A(D, A/xA) \cong D$  for every  $0 \neq x \in I$ ;*
- (iv) *for all  $D$ -modules  $N, M$  and all  $n \geq 0$ , inflation induces natural isomorphisms  $\text{Ext}_A^n(N, M) \cong \text{Ext}_D^n(N, M)$ ;*
- (v)  *$\text{pd}_A D \geq 2$ ;*
- (vi) *the sequence  $0 \rightarrow I \rightarrow A \xrightarrow{\text{ct}} D \rightarrow 0$  does not split as a sequence of  $A$ -modules, so  $\text{Ext}_A^1(D, I) \neq 0$ ;*
- (vii) *if  $(x_\alpha)_{\alpha \in \Lambda}$  generates  $I$ ,  $P = \bigoplus_\alpha Ae_\alpha \rightarrow I$ ,  $e_\alpha \mapsto x_\alpha$ , and  $J$  is its kernel, then  $\text{Ext}_A^2(D, J) \neq 0$ .*

*Proof.* (i) For every  $h > 0$ , multiplication by  $X^h$  identifies  $A$  with  $X^h A$ , since  $A$  is a domain. These are free rank-one  $A$ -modules. If  $0 < h' < h$ , then  $X^h = X^{h'} X^{h-h'}$  with multiplier in  $A$ , so  $X^h A \subseteq X^{h'} A$ . For two such ideals, any positive exponent below both indices gives a common larger member. Thus (H2) expresses  $I$  as their set-indexed filtered colimit. It is flat by [29, Lemma 10.39.3]. The factorization in (H2) puts every  $f \in I$  in  $I^2$ , while the ideal property gives  $I^2 \subseteq I$ . Finally  $I$  is nonzero; if it were projective, Lemma 12.1 would contradict (H3).

(ii) Let  $N$  be a  $D$ -module with its inflated  $A$ -action, so  $IN = 0$ . For a pure tensor  $f \otimes n$ , factor  $f = ab$  with  $a, b \in I$  by (H2). The balancing relation gives

$$f \otimes n = b \otimes an = 0.$$

Pure tensors generate, hence  $I \otimes_A N = 0$ . Equivalently this is the standard identification  $I \otimes_A N \cong (I/I^2) \otimes_D N$  with  $I/I^2 = 0$ . By (i),  $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$  is a flat resolution of  $D$  of length one. Tensoring it with  $N$  gives the complex  $0 \rightarrow 0 \rightarrow N$  in positive and zero homological degrees; thus every positive  $\text{Tor}_n^A(D, N)$  vanishes. Flat resolutions compute these groups [33, Section 15.60].

(iii) That resolution gives  $\text{fd}_A D \leq 1$ . For  $0 \neq x \in I$ , the domain hypothesis makes  $0 \rightarrow A \xrightarrow{x} A \rightarrow A/xA \rightarrow 0$  a free resolution. Tensor it with  $D$ . The differential  $D \xrightarrow{x} D$  is zero because  $x$

acts through  $\text{ct}(x) = 0$ , and therefore  $\text{Tor}_1^A(D, A/xA) \cong D \neq 0$ . This proves  $\text{fd}_A D = 1$ . The same obstruction can be seen without derived notation: the injection  $xA \hookrightarrow A$  becomes the zero map  $D \rightarrow D$  after tensoring with  $D$ , so tensoring by  $D$  is not left exact.

(iv) Choose a projective  $A$ -resolution  $P_\bullet \rightarrow N$  in nonnegative homological degrees. By (ii), the augmented complex  $D \otimes_A P_\bullet \rightarrow D \otimes_A N \cong N$  is exact. Each term is projective over  $D$ : tensoring a split realization of  $P_j$  as a summand of a free  $A$ -module makes it a summand of a free  $D$ -module. Consequently this is a projective  $D$ -resolution of  $N$ . For every degree the tensor–Hom adjunction is

$$\text{Hom}_A(P_j, M) \cong \text{Hom}_D(D \otimes_A P_j, M), \quad f \mapsto (d \otimes p \mapsto df(p)),$$

with inverse  $g \mapsto (p \mapsto g(1 \otimes p))$ . The maps commute with the differentials, so taking cohomology gives the stated isomorphisms, including  $n = 0$ .

To identify the map with inflation, take a projective  $D$ -resolution  $Q_\bullet \rightarrow N$ . Viewed as an exact complex of  $A$ -modules, it admits a comparison map  $P_\bullet \rightarrow Q_\bullet$  lifting the identity of  $N$ . Its adjoint  $D \otimes_A P_\bullet \rightarrow Q_\bullet$  is a comparison between projective  $D$ -resolutions, hence a homotopy equivalence. The induced cochain map realizes the canonical inflation map from  $\text{Ext}_D$  to  $\text{Ext}_A$ ; the displayed isomorphism in the other direction is its inverse. Comparison maps for morphisms of  $N$  and  $M$  give naturality. This does not require  $D$  to be flat over  $A$ .

(v) If  $\text{pd}_A D \leq 1$ , choose a projective presentation  $0 \rightarrow Q_1 \rightarrow Q_0 \rightarrow D \rightarrow 0$  with  $Q_0, Q_1$  projective. Schanuel’s lemma, applied alongside  $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$ , gives  $I \oplus Q_0 \cong Q_1 \oplus A$  [30, Tag 00O3]. The right side is projective, so its direct summand  $I$  is projective, contradicting (i).

(vi) An  $A$ -linear section  $s : D \rightarrow A$  would give  $r = s(1)$  with  $\text{ct}(r) = 1$ . Choose  $0 \neq x \in I$ . Since  $x$  acts by zero on  $D$ ,

$$xr = s(x \cdot 1) = s(0) = 0.$$

The domain hypothesis forces  $r = 0$ , contradicting  $\text{ct}(r) = 1$ . The nonsplit short exact sequence thus represents a nonzero class in  $\text{Ext}_A^1(D, I)$ . The existing ring section is not  $A$ -linear: it sends  $x \cdot 1 = 0$  to 0, whereas multiplying its value at 1 by  $x$  gives  $x$ .

(vii) The prescribed generating family makes  $P \rightarrow I$  surjective. If  $0 \rightarrow J \rightarrow P \rightarrow I \rightarrow 0$  split,  $I$  would be a direct summand of the free module  $P$ , again contradicting (i). Let  $\eta$  be its nonzero class in  $\text{Ext}_A^1(I, J)$ . The long exact sequence from  $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$  contains

$$0 = \text{Ext}_A^1(A, J) \longrightarrow \text{Ext}_A^1(I, J) \longrightarrow \text{Ext}_A^2(D, J) \longrightarrow \text{Ext}_A^2(A, J) = 0.$$

Thus its connecting map is an isomorphism and carries  $\eta$  to a nonzero class. In Yoneda form this is the splice  $0 \rightarrow J \rightarrow P \rightarrow A \rightarrow D \rightarrow 0$ , where  $P \rightarrow A$  is the composite  $P \rightarrow I \hookrightarrow A$ .  $\square$

In standard terms,  $A \rightarrow D$  is a homological ring epimorphism ( $D \otimes_A D = D$ , all higher self-Tor vanish) but not a flat one (03, 06); this terminology is optional. The exact value of  $\text{pd}_A D$  is not determined in general: only  $\text{pd}_A D \geq 2$  and  $\text{fd}_A D = 1$  are proved without a generator restriction (06). When  $I$  is countably generated,  $\text{pd}_A D = 2$  by the telescope described below (Proposition 13.12(v), from 12). Hypothesis (H2) is where the totally ordered exponent cone enters: for 12’s coordinatewise cones the flat dimension of  $D$  is any  $d \geq 1$ , and the two kinds of statement are compared in Remark 13.16. The nonzero  $\text{Tor}_1^A(D, A/xA)$  does not contradict (ii), because  $A/xA$  is not a  $D$ -module with its given  $A$ -action. Indeed,  $I \subseteq xA$  would imply  $I = xA$ , since  $x \in I$ , contradicting (H3). Hence  $I$  acts nontrivially on  $A/xA$ . Likewise the nonzero Ext witnesses in (vi) and (vii) are not modules annihilated by  $I$ : otherwise (iv), with first variable  $D$ , would make their Ext groups zero. Finally,  $I/I^2 = 0$  is not a finite-presentation or smoothness statement (09).

**The countably generated case.** Choose a decreasing sequence  $h_n > 0$  so that  $I = \bigcup_{n \geq 0} X^{h_n} A$ : (H2) and directedness allow each new principal ideal to contain the preceding one and the next member of a countable generating family. Divisibility lets the indices decrease strictly. Put  $a_n = X^{h_n - h_{n+1}}$ . The transition between the free rank-one modules is multiplication by  $a_n$ , and the telescope is

$$0 \longrightarrow \bigoplus_{n \geq 0} A e_n \xrightarrow{\delta} \bigoplus_{n \geq 0} A e_n \xrightarrow{\pi} I \longrightarrow 0,$$

where  $\delta(e_n) = e_n - a_n e_{n+1}$  and  $\pi(e_n) = X^{h_n}$ . The composite is zero. If  $\delta(\sum r_n e_n) = 0$ , its zeroth coordinate gives  $r_0 = 0$ , and consecutive coordinates give every  $r_n = 0$ , so  $\delta$  is injective. Modulo its image, a finite sum supported through  $N$  reduces to a single coefficient times  $e_N$ . If its image under  $\pi$  vanishes, that coefficient times  $X^{h_N}$  is zero; the domain hypothesis makes the coefficient zero. This proves exactness at the middle term, and the union proves surjectivity of  $\pi$ . The telescope is a length-one free resolution of  $I$ . Splicing with  $0 \rightarrow I \rightarrow A \rightarrow D \rightarrow 0$  gives  $\text{pd}_A D \leq 2$ ; (v) gives equality. Among the five families the countably generated case occurs precisely for 08 with  $\text{cf } \kappa = \aleph_0$ , by their exact generator counts. No conclusion  $\text{pd}_A D = 2$  is drawn for the other cases.

**Theorem 12.3** (Exact size of cohomological witnesses [06] [merge]). *Let  $A$  be one of the five rings of Section 10.4, with  $D = \mathbb{Z}$  (or any admissible  $D$ ), and  $\tau = |A|$ . For every  $A$ -module  $M$  with  $|M| < \tau$  and every  $n > 0$ ,  $\text{Ext}_A^n(D, M) = 0$ , and for any two  $A$ -modules of size  $< \tau$  the Ext groups over  $A$  are those of the underlying  $D$ -modules. For  $n \in \{1, 2\}$ ,*

$$\min\{|M| : \text{Ext}_A^n(D, M) \neq 0\} = \tau.$$

For  $A_\kappa^{06}$  this minimum is  $\kappa^{<\kappa}$  in ZFC; under  $\kappa^{<\kappa} = \kappa$  it is  $\kappa$ , which is 06's statement. For  $A_\kappa^{03}$ ,  $A_\kappa^{08}$  and  $A_{\lambda,D}^{09}$  it is  $\kappa^{\aleph_0}$ ,  $\kappa$  and  $\lambda$ .

*Proof.* If  $|M| < \tau$ , the detection theorem Theorem 10.10 gives  $IM = 0$ . Thus  $M$  has the unique induced  $D$ -module structure, and Theorem 12.2(iv) gives

$$\text{Ext}_A^n(D, M) \cong \text{Ext}_D^n(D, M) = 0 \quad (n > 0),$$

since  $D$  is a free rank-one module over itself. Applying (iv) to two small modules gives their claimed Ext comparison in every degree. In particular every coefficient module detecting a nonzero class in degree 1 or 2 must have at least  $\tau$  elements.

For degree 1, take  $M = I$ . The model-size calculations give  $|I| = |A| = \tau$ , and the nonsplit sequence in Theorem 12.2(vi) is a nonzero class. For degree 2, choose a generating set  $(x_\alpha)_{\alpha \in \Lambda}$  for  $I$  with  $|\Lambda| \leq \tau$ , for example all of  $I$ . Since  $\tau$  is infinite, the free module  $P = \bigoplus_{\alpha \in \Lambda} A e_\alpha$  has size at most  $\tau$ : its elements are finite lists of indices and coefficients, and  $\sum_{m < \omega} (|\Lambda| \cdot \tau)^m \leq \tau$ . Let  $J = \ker(P \rightarrow I)$ . Then  $|J| \leq \tau$ , and (vii) supplies  $\text{Ext}_A^2(D, J) \neq 0$ . The lower bound already proved forces  $|J| = \tau$ . Both minima are therefore attained. Substituting each model's proved size gives the displayed values; no additional cardinal arithmetic assumption or minimal generating presentation is needed.  $\square$

The Ext groups are computed in the full category of  $A$ -modules, not only among small modules; this is why Theorem 12.2(iv) is an ambient derived-functor statement and not merely an equivalence of small module categories (06). The two witnesses show that a quotient can look homologically projective against every small coefficient module while failing projectivity in the ambient category. The whole argument works for the Gaussian versions with  $D = \mathbb{Z}[i]$ , where  $\text{Ext}_D$  means extensions of  $\mathbb{Z}[i]$ -modules (06). Finally, for any of the five models and any  $0 \neq x \in I$ , the module  $A/xA$  is

detected by  $I$ , as shown above. The detection theorem and quotient upper bound give  $|A/xA| = |A|$ . In particular  $|A_\kappa^{06}/xA_\kappa^{06}| = \kappa^{<\kappa}$ ; 06 stated a lower bound with the qualifier “under the cardinal-arithmetic hypothesis”, which is not needed. The minima in [Theorem 12.3](#) concern the size of the coefficient module  $M$ , not the cardinality of the Ext group itself.

### 13 Homological and tensor algebra of omnific cores

This section is the part contributed by manuscripts 12 and 14 ([Section 1.3](#)). Both study ordinary commutative algebra of rings of the form “constants plus a tail” inside  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , and they share one set-sized model: 14’s finite-support ring  $A_0$  is 12’s rank-one coordinate-cone core  $A_1$ , and 12’s two-generator relation module is the case  $r = 2$  of 14’s lattice syzygy. Each is printed once ([Lemma 13.1](#) and [Theorem 13.9](#)). Manuscript 12 computes flat and projective dimensions of set-sized finite-support cores ([Sections 13.1 to 13.3](#) and [13.5](#)); 14 computes tensor products, symmetric and exterior powers, Rees algebras, homomorphisms and duals of coefficient-lattice modules, over the full class ring and over  $A_0$  ([Sections 13.4](#) and [13.6 to 13.8](#)). Manuscript 17 of the third batch re-derives, independently, 14’s matrix, syzygy, Hom, tensor-torsion and power-growth theorems and 12’s intersection; these are printed once, with 17 added to their sources, and 17’s new material (presentations and relation counts, the Tor identification, moduli, orders, the lexicographic models inside  $\mathbf{Oz}$ ) is [Section 13.9](#). Manuscript 21 of a fifth batch determines the projective dimensions that 12 bounds and computes the Tor algebra of lattice quotients ([Section 13.10](#)). Manuscript 26 of an eighth batch replaces the coordinate cone by an arbitrary rational polyhedral cone ([Section 13.11](#)). Labels carry the sub-prefixes **osq:hd**: (12), **osq:tn**: (14), **osq:rel**: (17), **osq:da**: (21) and **osq:pc**: (26).

**Scope.** The rings of [Sections 13.1 to 13.3](#) and [13.5](#) and 14’s model  $A_0$  have *finite* supports. They are therefore outside the universal theorem, like the finite-support rings of [Section 9](#): the augmentation that sends every monomial to 1 is a unital map to the coefficient field that detects the tail ([Proposition 13.2](#)). What 12 calls arithmetic invisibility is invisibility to *finite* rings and to modules on which the tail acts by zero; these rings are not invisible to set-sized rings in the sense of [Theorem 5.1](#).

**Notation.** 12’s augmentation  $\varepsilon$  is  $\text{ct}$ ; its tail  $M$  is  $\Pi_\Lambda$  (the letter  $M$  is 14’s coefficient lattice); its ordered-cone ring  $T = D \oplus P$  with monomials  $t^\gamma$  is  $T_\Gamma = D \oplus \Pi_\Gamma^{\text{fin}}$  with monomials  $X^\gamma$  (in this report  $t^g$  means  $X^{-g}$ , [Section 2.2](#)); its test modules  $N_F$  are  $V_F$ ; its cyclic module  $Q = A/(X_1, cX_1)$  is written  $A/H$ . 14’s ideal  $\mathcal{I}$ , its rings  $B = k + \mathcal{I}$ ,  $A = D + \mathcal{I}$  and its field  $F$  are  $\Pi_k$ ,  $\mathcal{A}_{k,k}$ ,  $\mathcal{A}_{D,k}$  and  $k((X^{\mathbf{No}}))$ ; its set-sized  $P$ ,  $A_0$ ,  $B_0$  are  $\Pi_\mathbb{Q}^{\text{fin}}$ ,  $T_\mathbb{Q}$  and  $k + \Pi_\mathbb{Q}^{\text{fin}}$ ; its multiplication map  $\mu_{M,N}$ , kernel  $K(M, N)$  and carrier  $E(M, N)$  are  $\mathfrak{m}_{M,N}$ ,  $\mathcal{T}(M, N)$  and  $\mathbf{E}(M, N)$ ; its multiplier ring  $\mathcal{O}(M)$  and orders  $\mathcal{O}$  in number fields  $K$  are  $\mathfrak{o}(M)$  and  $\mathfrak{o} \subset K_0$  (the letter  $\mathcal{O}$  denotes the finite surreal rings); the saturated lattice  $L$  of its matrix theorem is  $W_D$ . 17’s positive-support ideal  $\mathfrak{m}$  (which it warns is not the infinitesimal ideal; here  $\mathfrak{m}$  is the negative-support ideal) is  $\Pi$ , its  $F = \text{Frac}(D)$  is written  $\text{Frac}(D)$  (14’s  $F$  was the fraction field  $k((X^{\mathbf{No}}))$ ), its lattices  $L, M$  are  $M, N$ , its ideals  $J_L^\gamma = \omega^\gamma(L + \mathfrak{m})$  are  $J_\gamma(M) = X^\gamma L(M)$ , its multiplier ring  $E(L)$  and conductor  $(M : L)$  are  $\mathfrak{o}(M)$  and  $(N : M)$ , its multiplication map  $\mu_{L,M}$  and kernel  $T(L, M)$  are  $\mathfrak{m}_{M,N}$  and  $\mathcal{T}(M, N)$  (the letter  $\mu$  was 14’s generator count), its kernel dimensions  $d, r$  are  $q, \ell$ , and its set-sized models  $A_\kappa$ ,  $\mathfrak{m}_\kappa$  are  $R_{\Gamma_\kappa, <\kappa}^{11}$  and  $\Pi_\kappa^{11}$ .



### 13.1 Coordinate-cone and ordered-cone cores

Throughout Sections 13.1 to 13.3,  $K$  is a field and  $D \subseteq K$  a unital subring, so  $D$  is a domain. Let  $\Lambda$  be  $[d] = \{1, \dots, d\}$  ( $d \geq 1$ ) or  $\mathbb{N}_{>0}$ , and let  $P_\Lambda = \mathbb{Q}_{>0}^{(\Lambda)}$  be the monoid of finitely supported vectors  $\alpha = (\alpha_i)_{i \in \Lambda}$  with nonnegative rational coordinates. In the monoid algebra  $K[P_\Lambda]$  (finite sums,  $X^\alpha X^\beta = X^{\alpha+\beta}$ ) put

$$\Pi_\Lambda = \bigoplus_{0 \neq \alpha \in P_\Lambda} KX^\alpha, \quad A_\Lambda = D \oplus \Pi_\Lambda, \quad (13.1)$$

and write  $A_d, \Pi_d$  for  $\Lambda = [d]$  and  $X_i = X^{e_i}$ . The coefficient at  $X^0$  is a split surjective ring homomorphism  $\text{ct} : A_\Lambda \rightarrow D$  with kernel  $\Pi_\Lambda$ , because a sum of two nonnegative vectors is zero only when both are. For a nonzero divisible subgroup  $\Gamma \subseteq \mathbb{R}$  put

$$\Pi_\Gamma^{\text{fn}} = \bigoplus_{\gamma \in \Gamma_{>0}} KX^\gamma, \quad T_\Gamma = D \oplus \Pi_\Gamma^{\text{fn}}, \quad (13.2)$$

the finite-support ring of Section 9 over the fixed ordered group  $\Gamma$ , again with  $\text{ct} : T_\Gamma \rightarrow D$ . The difference between the two families is the exponent cone: coordinatewise nonnegative vectors in  $A_\Lambda$ , the positive cone of a totally ordered group in  $T_\Gamma$ .

**Lemma 13.1** (Domains and embeddings [12; 14]).

- (i)  $K[P_\Lambda]$ ,  $A_\Lambda$  and  $T_\Gamma$  are domains, and  $\Pi_\Lambda$  is a nonzero prime ideal.
- (ii) Let  $(\gamma_i)_{i \in \Lambda}$  be positive surreals linearly independent over  $\mathbb{Q}$ . Then  $X^\alpha \mapsto \omega^{\sum_i \alpha_i \gamma_i}$  defines an injective ring homomorphism  $\mathbb{R}[P_\Lambda] \rightarrow \mathbf{No}$ , and, applied to real and imaginary parts,  $\mathbb{C}[P_\Lambda] \rightarrow \mathbf{No}[i]$ ; it maps  $\mathbb{Z} + \Pi_\Lambda$  into  $\mathbf{Oz}$  and  $\mathbb{Z}[i] + \Pi_\Lambda$  into  $\mathbf{Oz}[i]$ . The  $\gamma_i$  may be taken real:  $\gamma_i = \theta^{i-1}$  for a transcendental real  $\theta > 1$ . Likewise  $X^\gamma \mapsto \omega^\gamma$  embeds  $T_\Gamma$  in  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  for  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ .
- (iii) If the  $\gamma_i$  are positive reals linearly independent over  $\mathbb{Q}$  and  $\Gamma = \sum_i \mathbb{Q}\gamma_i$ , then  $X_i \mapsto X^{\gamma_i}$  is an injective ring homomorphism  $A_\Lambda \hookrightarrow T_\Gamma$  compatible with  $\text{ct}$ . For  $d = 1$  and  $\gamma_1 = 1$  it is an isomorphism  $A_1 \cong T_\mathbb{Q}$ ; this ring, with  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , is 14's set-sized model  $A_0$ .

*Proof.* (i) Given two nonzero elements of  $K[P_\Lambda]$ , their finite supports involve only finitely many coordinates  $i_1, \dots, i_s$ . Choose a common positive denominator  $n$  for all their rational exponents. Both elements lie in the subalgebra generated by  $X_{i_1}^{1/n}, \dots, X_{i_s}^{1/n}$ . Distinct products of these generators have distinct exponent vectors, so this subalgebra is a polynomial ring over  $K$ , hence a domain. Their product is therefore nonzero in  $K[P_\Lambda]$ . Its subring  $A_\Lambda$  is a domain as well. In  $T_\Gamma$ , the largest exponents of two nonzero finite sums add, and the coefficient at that sum is the product of their nonzero leading coefficients; no other pair reaches that exponent. This proves that  $T_\Gamma$  is a domain. Finally,  $A_\Lambda/\Pi_\Lambda \cong D$  is a nonzero domain, so  $\Pi_\Lambda$  is proper and prime, and  $X_1 \in \Pi_\Lambda$  makes it nonzero.

(ii) The map  $\ell(\alpha) = \sum_i \alpha_i \gamma_i$  is defined by a finite sum, is additive, and is injective by rational linear independence. It is strictly positive on nonzero vectors of  $P_\Lambda$ . Send  $\sum c_\alpha X^\alpha$  to  $\sum c_\alpha \omega^{\ell(\alpha)}$ . The law  $\omega^u \omega^v = \omega^{u+v}$  and finite distributivity show that this preserves products and the unit. Distinct exponents and uniqueness of Conway normal forms show that it is injective. For complex coefficients, the same construction on real and imaginary parts gives the homomorphism into  $\mathbf{No}[i] = \mathbf{No}[i]$ . Positive nonconstant exponents and an integer, respectively Gaussian-integer, constant coefficient give the stated omnific memberships.

If  $\gamma_i = \theta^{i-1}$ , any finite rational relation among the  $\gamma_i$  would give a nonzero rational polynomial vanishing at the transcendental number  $\theta$ ; hence these positive real choices work also for countably many indices. For  $T_\Gamma$  use the inclusion  $\Gamma \subseteq \mathbb{R} \subseteq \mathbf{No}$  in place of  $\ell$ , with the same normal-form argument.

(iii) With real  $\gamma_i$ , the same  $\ell$  is an injective monoid map  $P_\Lambda \rightarrow \Gamma_{\geq 0}$  and has zero fiber  $\{\mathbf{0}\}$ . Reindexing the finite sum by  $\ell$  gives the required embedding over  $D$ , and preserves the coefficient at zero. In rank one with  $\gamma_1 = 1$ , every nonnegative rational exponent is reached and every allowed coefficient is unchanged. The map is therefore onto  $T_\mathbb{Q}$ , giving the stated isomorphism. In higher rank only an embedding is asserted: the coordinate cone need not fill the positive cone of the ordered group.  $\square$

Thus  $d$  counts independent monomial directions, not Archimedean scales: with  $\gamma_i = \theta^{i-1}$  all exponents lie in the single Archimedean class of 1 (12). Every element of every ring in this subsection is a finite sum; no topology, convergence or infinite summation is involved.

**Proposition 13.2** (Finite targets only [12]). *Let  $A$  be  $A_\Lambda$  or  $T_\Gamma$  with tail  $\Pi$ , and  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ .*

- (i) *For  $0 \neq a \in D$ ,  $a\Pi = \Pi$  and  $A/aA \cong D/aD$ . Every unital map from  $A$  to a finite ring, and every finite  $A$ -module, factors through  $\text{ct}$ ;  $\hat{A} \cong \hat{D}$  for the profinite completions,  $\varprojlim_k A/p^k A \cong \varprojlim_k D/p^k D$  for every ordinary prime  $p$ ,  $\bigcap_k p^k A = \Pi$ , and  $\text{Tor}_{>0}^A(D, E) = 0$  for every finite  $A$ -module  $E$ .*
- (ii) *The augmentation  $\sum_\alpha c_\alpha X^\alpha \mapsto \sum_\alpha c_\alpha$  is a unital ring homomorphism  $A \rightarrow K$  sending every monomial to 1; it does not factor through  $\text{ct}$ .*

*Proof.* (i) Dividing each of the finitely many tail coefficients by  $0 \neq a \in D$  stays in  $K$  and preserves its positive exponent. Thus  $a\Pi = \Pi$ , and the additive decomposition gives  $aA = aD \oplus \Pi$ . The map  $A \rightarrow D/aD$ ,  $f \mapsto \text{ct}(f) \bmod aD$ , is surjective with this kernel, proving  $A/aA \cong D/aD$ .

Let  $\varphi : A \rightarrow R$  be unital with  $R$  finite. Some positive integer  $n$  annihilates  $R$  (including  $n = 1$  for the zero ring). Since these coefficient pairs have characteristic zero,  $n\Pi = \Pi$ . For  $x \in \Pi$  write  $x = ny$  with  $y \in \Pi$ ; then  $\varphi(x) = n\varphi(y) = 0$ . The quotient property gives unique factorization through  $\text{ct}$ . Likewise the additive group of a finite  $A$ -module  $E$  is annihilated by some  $n \geq 1$ . For  $e \in E$ ,  $xe = (ny)e = y(ne) = 0$ , so  $\Pi E = 0$  and its action factors uniquely through  $D$ . This argument does not assume a commutative target ring.

Every finite-index ideal  $J$  contains  $\Pi$ , by applying the preceding argument to  $A/J$ , and it contains  $nA$  for a positive integer annihilating that quotient. Conversely  $A/nA \cong D/nD$  is finite. The ideals  $nA$  are therefore cofinal among finite-index ideals. The displayed isomorphisms commute with reduction whenever one integer divides another, identifying the inverse limits  $\hat{A} \cong \hat{D}$ . Restricting to  $n = p^k$  gives the stated  $p$ -adic limits. Moreover

$$\bigcap_{k \geq 1} p^k A = \left( \bigcap_{k \geq 1} p^k D \right) \oplus \Pi = \Pi,$$

because an ordinary integer divisible by every  $p^k$  is zero; apply this to both integer coordinates for  $D = \mathbb{Z}[i]$ . Thus  $A$  itself is not separated for the  $p$ -adic topology: its map to that completion has kernel  $\Pi$ .

Finally  $E$  is an inflated  $D$ -module. The Tor assertion follows from the face resolution and [Corollary 13.5](#) below for  $A_\Lambda$ . For  $T_\Gamma$ , use the flat tail in [Theorem 13.13](#) and [Theorem 12.2\(ii\)](#). These are algebraic resolution arguments; the completion isomorphisms alone do not compute Tor.

(ii) Every sum is finite, so regrouping the finitely many terms of a product proves that its coefficient sum is the product of the two coefficient sums. This gives a unital map to  $K$ . It sends  $X_1$  to 1 for  $A_\Lambda$ , and  $X^\gamma$  to 1 for any  $\gamma > 0$  in  $T_\Gamma$ . Since these monomials have constant term zero, the map cannot factor through  $\text{ct}$ . This second augmentation is distinct from the constant-coefficient retraction.  $\square$

Hence all the cores, and 14's  $A_0$ , have the same finite quotients and profinite completion as  $D$ , while [Corollary 13.8](#) below shows that their homological dimensions differ; this limits any reconstruction of an omnific subring from its finite arithmetic shadows (12). Item (ii) is the reason the word “invisible” must be read with (i)’s restriction.

### 13.2 Flat face ideals and a Boolean resolution

For a finite nonempty  $S \subseteq \Lambda$  put  $p_S = \prod_{i \in S} X_i$  and let

$$I_S = \bigoplus_{\alpha \in P_\Lambda, \alpha_i > 0 \ (i \in S)} KX^\alpha, \quad (13.3)$$

an ideal of  $A_\Lambda$  with  $I_S = \bigcap_{i \in S} I_{\{i\}}$  and  $\Pi_\Lambda = \sum_i I_{\{i\}}$ .

**Lemma 13.3** (Face ideals [12]). *Let  $A = A_\Lambda$  and  $\Pi = \Pi_\Lambda$ .*

- (i)  $I_S = \bigcup_{n \geq 0} p_S^{2^{-n}} A$ , an increasing chain of free rank-one  $A$ -modules; hence  $I_S$  is flat.
- (ii)  $\Pi^2 = \Pi$ ,  $I_S^2 = I_S$  and  $\Pi I_S = I_S$ ; for every  $D$ -module  $E$ , regarded as an  $A$ -module through  $\text{ct}$ ,  $I_S \otimes_A E = 0$ .
- (iii) Neither  $\Pi$  nor any  $I_S$  is finitely generated.

*Proof.* (i) Put  $u_n = p_S^{2^{-n}}$ . Multiplication by  $u_n$  identifies  $A$  with  $u_n A$ , because  $A$  is a domain. The relation  $u_n = u_{n+1} u_{n+1}$  makes these principal ideals an increasing chain, each contained in  $I_S$ . If  $f \neq 0$  lies in  $I_S$ , there are only finitely many positive coordinates  $\alpha_i$ , with  $i \in S$ , among its finitely many exponents. Choose  $n$  so that  $2^{-n}$  is strictly below all of them. Subtracting  $2^{-n}$  in each coordinate of  $S$  leaves nonnegative exponent vectors still positive on  $S$ . Consequently  $f/u_n$  is a tail in  $A$ , even if its coefficients are arbitrary in  $K$  rather than in  $D$ . The zero element lies in every member of the chain. This proves the exhaustion. Each member is free of rank one, so the filtered-colimit flatness theorem gives flatness of  $I_S$  [29, Lemma 10.39.3].

(ii) Every nonzero tail monomial admits the factorization

$$kX^\alpha = (kX^{\alpha/2})X^{\alpha/2}.$$

Both factors belong to  $\Pi$ , and both belong to  $I_S$  when  $\alpha_i > 0$  for every  $i \in S$ . Summing finitely many monomials gives  $\Pi \subseteq \Pi^2$  and  $I_S \subseteq I_S^2$ ; the opposite inclusions follow from the ideal property. Since  $I_S \subseteq \Pi$ , we also obtain  $I_S = I_S^2 \subseteq \Pi I_S \subseteq I_S$ . For an inflated  $D$ -module  $E$ , write each  $x \in I_S = \Pi I_S$  as a finite sum  $\sum_j a_j b_j$  with  $a_j \in \Pi$ ,  $b_j \in I_S$ . Tensor balancing gives  $x \otimes e = \sum_j b_j \otimes a_j e = 0$ . Pure tensors generate, so  $I_S \otimes_A E = 0$ .

(iii) Both ideals are nonzero. If a finite family generated either one, discard its zero members and let  $\delta > 0$  be the smallest total degree  $|\alpha| = \sum_i \alpha_i$  among all monomials occurring in that family.

Multiplication by any element of  $A$  adds a nonnegative total degree; addition and cancellation create no new exponent. Every element of the generated ideal therefore has all its monomials in total degree at least  $\delta$ . Choose a positive rational  $q < \delta$  for  $\Pi$ , or  $0 < q < \delta/|S|$  for  $I_S$ . The monomial  $X_1^q \in \Pi$ , respectively  $p_S^q \in I_S$ , has smaller total degree, a contradiction. The same argument works when  $\Lambda$  is countably infinite, since the proposed generating family and each of its supports are finite.  $\square$

Flatness of every  $I_S$  does not imply flatness of their sum  $\Pi$ ; the failure begins at  $d = 2$  ([Corollary 13.8](#)). For  $p \geq 1$  put  $C_p = \bigoplus_{S \subseteq \Lambda, |S|=p} I_S$  and  $C_0 = A$ ; for  $S = \{s_1 < \dots < s_p\}$  let  $d_p$  send  $f$  in the summand  $I_S$  to  $\sum_{r=1}^p (-1)^{r-1} f$  in the summands  $I_{S \setminus \{s_r\}}$  (for  $p = 1$  the target is  $A$  and the map is the inclusion). Deleting two indices in either order gives opposite signs, so  $d_{p-1}d_p = 0$ .

**Theorem 13.4** (Coordinate-intersection resolution [12]). *The augmented complex*

$$\dots \longrightarrow C_2 \xrightarrow{d_2} C_1 \xrightarrow{d_1} A_\Lambda \xrightarrow{\text{ct}} D \longrightarrow 0$$

*is exact and every  $C_p$  is flat. For  $\Lambda = [d]$ ,  $C_p = 0$  for  $p > d$ , and this is a flat resolution of  $D$  of length  $d$ ; also for  $\Lambda = \mathbb{N}_{>0}$  the complex is exact.*

*Proof.* The inclusions defining the differentials are  $A_\Lambda$ -linear and preserve exponent degree. In degree  $0$  the augmented complex is just  $D \xrightarrow{\text{id}} D$ . Fix a nonzero exponent  $\alpha$  and write  $T = \{i : \alpha_i > 0\}$ , a finite nonempty set even in infinite rank. The degree- $\alpha$  part of  $A$  is  $K$ , and that of  $I_S$  is  $K$  exactly for  $\emptyset \neq S \subseteq T$ . Thus the component is the augmented simplex complex on  $T$ , with a  $p$ -element subset in homological degree  $p$  and the empty subset in degree zero.

Here is an explicit contraction of this component. Let  $t = \min T$ , and write  $[S]$  for its basis vector, including  $[\emptyset]$ . Define

$$h[S] = \begin{cases} [\{t\} \cup S], & t \notin S, \\ 0, & t \in S. \end{cases}$$

The element  $t$  is first in the ordered list. If  $t \notin S$ , the term deleting  $t$  in  $dh[S]$  is  $[S]$ , and each other deletion has the opposite sign to its counterpart in  $hd[S]$ . If  $t \in S$ , only deleting  $t$  survives after applying  $h$ , and it gives  $[S]$ . For the empty subset,  $dh[\emptyset] = [\emptyset]$ . Hence  $dh + hd = 1$  in every nonzero exponent degree.

These are contractions of coefficient complexes. They may depend on  $\alpha$  and are not claimed to be  $A_\Lambda$ -linear. They nevertheless prove exactness of the  $A_\Lambda$ -module complex: a cycle has only finitely many exponent components, and its component preimages under  $h$  form a finite sum in the required direct sum. Each  $C_p$  for  $p > 0$  is a direct sum of the flat ideals  $I_S$ , hence flat;  $C_0 = A$  is free. If  $\Lambda = [d]$  there are no subsets of size greater than  $d$ , giving the stated length. The same component proof works for  $\Lambda = \mathbb{N}_{>0}$  without forming an infinite-support chain.  $\square$

**Corollary 13.5** (Arithmetic modules are Tor-acyclic [12]). *For every  $D$ -module  $E$ ,  $D \otimes_{A_\Lambda} E \cong E$  and  $\text{Tor}_j^{A_\Lambda}(D, E) = 0$  for  $j > 0$ ; in particular  $A_\Lambda \rightarrow D$  is a homological ring epimorphism.*

*Proof.* Give  $E$  the action  $ae = \text{ct}(a)e$ . The map  $D \otimes_A E \rightarrow E$ ,  $d \otimes e \mapsto de$ , has inverse  $e \mapsto 1 \otimes e$ ; tensor balancing through the surjection  $A \rightarrow D$  verifies both identities. Now tensor the flat resolution of [Theorem 13.4](#) with  $E$ . Tensor product commutes with direct sums, and each  $I_S \otimes_A E$  vanishes by [Lemma 13.3\(ii\)](#). The resulting complex has only  $E$  in degree zero, so it computes zero Tor in every positive degree. This also holds for the unbounded-length resolution in countable rank: it is still a flat resolution in nonnegative homological degrees [33, Section 15.60]. Taking  $E = D$  gives the

self-Tor vanishing; together with the multiplication isomorphism  $D \otimes_A D \cong D$ , this is the asserted homological epimorphism.  $\square$

This concerns every module on which the tail acts by zero; it is why such modules cannot see the flat dimension computed next (12).

**Proposition 13.6** (Formal lifting and differentials [12; 13]). *Let  $A$  be  $A_\Lambda$  or  $T_\Gamma$  with tail  $\Pi$ . Then  $A \rightarrow D$  is formally étale for nilpotent lifting problems, but neither flat nor finitely presented as a ring map, and  $\Omega_{A/D} \otimes_A D = 0$ : every  $D$ -derivation of  $A$  into a module killed by  $\Pi$  vanishes.*

*Proof.* Halving each monomial exponent gives  $\Pi^2 = \Pi$  in either family, so  $\Pi^n = \Pi$  for every  $n \geq 1$ . Consider a commutative  $A$ -algebra  $R$  with structure map  $f : A \rightarrow R$ , a nilpotent ideal  $J \subseteq R$  with  $J^n = 0$ , and a compatible map  $g : D \rightarrow R/J$ . Compatibility says that  $f(\Pi) \subseteq J$ . Thus  $f(\Pi) = f(\Pi^n) \subseteq J^n = 0$ . The map  $f$  factors uniquely through a ring map  $\tilde{g} : D \rightarrow R$ . Its reduction is  $g$  because both agree after precomposition with the surjection  $A \rightarrow D$ . This proves existence and uniqueness of the required lift. It is the idempotent-ideal mechanism of [31, Tag 060H].

For nonflatness choose a nonzero monomial  $u \in \Pi$ . Multiplication by  $u$  is an injective endomorphism of the domain  $A$ , whereas after tensoring with  $D$  it becomes the zero map  $D \rightarrow D$ . Since  $D \neq 0$ , tensoring fails to preserve this injection.

The ideal  $\Pi_\Lambda$  is not finitely generated by Lemma 13.3(iii). In  $T_\Gamma$ , a finite nonzero generating family would have a smallest positive exponent  $\delta$  among its monomials. Products with  $T_\Gamma$  cannot introduce an exponent below  $\delta$ , but  $X^{\delta/2} \in \Pi$  has such an exponent, a contradiction. For completeness, a finitely presented quotient  $A/I$  must have  $I$  finitely generated as an ideal. Indeed, write  $A/I = A[Y_1, \dots, Y_m]/(r_1, \dots, r_s)$  as an  $A$ -algebra and lift the images of  $Y_j$  to elements  $a_j \in A$ . Substitution  $Y_j \mapsto a_j$  sends the relation ideal onto  $I$ : it maps into  $I$ , and every constant polynomial from  $I$  lies in that relation ideal. Hence the finitely many  $r_i(a_1, \dots, a_m)$  generate  $I$ . Apply this with  $I = \Pi$  to exclude finite presentation of  $A \rightarrow D$ .

Finally, if  $E$  is killed by  $\Pi$ , a  $D$ -derivation  $\partial : A \rightarrow E$  is zero on  $D$  and satisfies  $\partial(uv) = u\partial v + v\partial u = 0$  for  $u, v \in \Pi$ . Since  $\Pi = \Pi^2$ , it vanishes on  $\Pi$  and therefore on  $A$ . Apply this to the universal derivation followed by  $\Omega_{A/D} \rightarrow \Omega_{A/D} \otimes_A D$ . This target is killed by  $\Pi$  and is generated over  $D$  by the images of differentials, all now zero, so  $\Omega_{A/D} \otimes_A D = 0$ . The lifting assertion concerns  $A \rightarrow D$ ; the displayed differential module concerns the section  $D \rightarrow A$ , followed by reduction along  $A \rightarrow D$ . These are different directions of ring maps.  $\square$

The mechanism is classical and not a novelty claim (12). Manuscript 13 proves the same statements for the full omnific ring in a two-universe reading (Remark 13.16), and Theorem 7.4 is the class-ring analogue for derivations. “Nilpotent” is a test condition on  $R$ : the cores are domains.

### 13.3 The multigraded Tor and exact flat dimensions

For a finite nonempty  $F \subseteq \Lambda$  let  $J_F = (X_i : i \in F)A_\Lambda$  and  $V_F = A_\Lambda/J_F$ . For finite  $S$  put  $\mathbf{e}_S = \sum_{i \in S} \mathbf{e}_i$ ; the notation  $K\langle \mathbf{e}_S \rangle$  means a copy of the  $D$ -module  $K$  placed in multidegree  $\mathbf{e}_S$  (not a subring of  $A_\Lambda$ ).

**Theorem 13.7** (Boolean-face Tor formula [12]). *For  $\Lambda$  finite or countably infinite and  $F$  as above,*

$$\begin{aligned} \mathrm{Tor}_0^{A\Lambda}(D, V_F) &\cong D\langle \mathbf{0} \rangle, & \mathrm{Tor}_1^{A\Lambda}(D, V_F) &\cong \bigoplus_{i \in F} D\langle \mathbf{e}_i \rangle, \\ \mathrm{Tor}_p^{A\Lambda}(D, V_F) &\cong \bigoplus_{S \subseteq F, |S|=p} K\langle \mathbf{e}_S \rangle & (p \geq 2). \end{aligned}$$

In particular  $\mathrm{Tor}_p = 0$  for  $p > |F|$ .

*Proof.* Tensor the flat resolution with  $V_F = A/J_F$ . The degree-zero term is  $V_F$  and each positive summand becomes  $I_S \otimes_A A/J_F \cong I_S/J_F I_S$ . We compute this quotient complex directly; the preceding degreewise contraction was not an  $A$ -linear splitting to which tensoring could be applied.

Fix  $\alpha \neq \mathbf{0}$  and put  $T = \{i : \alpha_i > 0\}$ . Only subsets  $S \subseteq T$  can contribute in this degree. If some  $j \in F$  has  $\alpha_j > 1$ , subtraction of  $\mathbf{e}_j$  leaves a nonnegative vector with the same support  $T$ . Every coefficient in each contributing  $I_S$ , and also in  $A$ , is therefore a multiple of  $X_j$  with cofactor in that same module. The entire quotient complex is zero in this degree.

Assume now  $\alpha_j \leq 1$  for all  $j \in F$ , and set  $U = \{j \in F : \alpha_j = 1\} \subseteq T$ . For nonempty  $S \subseteq T$ , the coefficient space  $(I_S)_\alpha$  is  $K$ . The product  $X_j I_S$  reaches it exactly when  $j \in U \setminus S$ : subtracting  $\mathbf{e}_j$  is nonnegative and retains positivity at every index of  $S$ . In that event it reaches all of  $K$ . Thus

$$(I_S/J_F I_S)_\alpha = \begin{cases} K, & U \subseteq S \subseteq T, S \neq \emptyset, \\ 0, & \text{otherwise.} \end{cases}$$

The differentials delete one index, with their original signs, whenever the target survives. The term  $(V_F)_\alpha$  must be computed separately because  $A_0 = D$ .

*Case  $U = \emptyset$ .* No generator of  $J_F$  reaches this degree, so  $(V_F)_\alpha = K$ . All faces survive and the augmented simplex contraction proves zero homology.

*Case  $U \neq \emptyset$  and  $T \setminus U \neq \emptyset$ .* Choose  $j \in U$ . Subtracting  $\mathbf{e}_j$  leaves a nonzero vector, so the cofactor lies in a full  $K$ -coefficient tail and  $(V_F)_\alpha = 0$ . Choose  $t \in T \setminus U$ . On surviving faces define  $h[S] = 0$  if  $t \in S$  and otherwise  $h[S] = (-1)^{|\{s \in S : s < t\}|} [S \cup \{t\}]$ . The target still contains  $U$ . Deleting  $t$  contributes  $[S]$ ; all other deletions cancel in  $dh + hd$ , or both corresponding terms are zero because they delete an index of  $U$ . Thus  $dh + hd = 1$  and this component is again acyclic.

*Case  $T = U$  and  $|U| \geq 2$ .* The only surviving positive summand is  $S = U$ , a copy of  $K$  in degree  $|U|$ . There is no larger face, and deletion of any index kills the target. Also  $(V_F)_{\mathbf{e}_U} = 0$ , since removing one of its at least two unit coordinates leaves a nonzero tail. This gives precisely  $K\langle \mathbf{e}_U \rangle$  in homology.

*Case  $T = U = \{j\}$ .* Here  $\alpha = \mathbf{e}_j$ . The only way  $J_F$  reaches this degree is by multiplying  $X_j$  by a constant in  $A$ , whose coefficient must lie in  $D$ . Consequently  $(V_F)_{\mathbf{e}_j} = K/D$ , an additive  $D$ -module quotient. The component complex is the quotient map  $K \rightarrow K/D$ , with  $K$  in homological degree one. Its kernel is  $D$  and its cokernel zero, giving  $D\langle \mathbf{e}_j \rangle$  in degree one. This includes  $D = K$ , when the target is zero and the same kernel formula holds.

In exponent degree  $\mathbf{0}$  only  $D$  in homological degree zero remains. All elements and preimages used above have finite support, so the homology of the full graded complex is the direct sum of these component homologies, also in countably infinite rank. The listed components give exactly the asserted formula and the vanishing above  $|F|$ .  $\square$

For  $|S| = p \geq 2$  the class of  $k \in K$  is represented by  $kX^{\mathbf{e}_S}$  in  $I_S/J_F I_S$ ; in degree one only coefficients  $d \in D$  give cycles  $dX_i$ . Treating the constant coefficient as if it lay in  $K$  gives a wrong



answer (12). Scaling coordinates by positive rationals  $b_i$  ( $i \in F$ ) is an automorphism of  $A_\Lambda$  fixing the coefficients, so the same formula holds for  $A_\Lambda/(X_i^{b_i} : i \in F)$  with  $\mathbf{e}_S$  replaced by  $\sum_{i \in S} b_i \mathbf{e}_i$  (12).

**Corollary 13.8** (Exact flat dimensions [12]). *For  $\Lambda = [d]$ ,  $\text{fd}_{A_d} D = d$  and  $\text{fd}_{A_d} \Pi_d = d - 1$ ; for  $\Lambda = \mathbb{N}_{>0}$  both are infinite. All positive self-Tor groups  $\text{Tor}_{>0}^{A_d}(D, D)$  vanish.*

*Proof.* For finite rank, the flat resolution has length  $d$ , so  $\text{fd}_{A_d} D \leq d$ . Take  $F = [d]$  in Theorem 13.7. The degree- $d$  group is  $D$  if  $d = 1$  and  $K$  if  $d \geq 2$ , with its displayed multidegree; either is nonzero. A flat resolution of length less than  $d$  would make this group zero, so  $\text{fd}_{A_d} D = d$ .

Truncating the resolution at its image  $\Pi_d \subset A_d$  gives a flat resolution of  $\Pi_d$  of length  $d - 1$ . For  $d \geq 2$ , the long exact Tor sequence of  $0 \rightarrow \Pi_d \rightarrow A_d \rightarrow D \rightarrow 0$ , with  $A_d$  free, gives

$$\text{Tor}_{d-1}^{A_d}(\Pi_d, V_{[d]}) \cong \text{Tor}_d^{A_d}(D, V_{[d]}) \neq 0.$$

This proves the matching lower bound. For  $d = 1$ , the ideal  $\Pi_1 = I_{\{1\}}$  is nonzero and flat, so its flat dimension is zero.

In countably infinite rank, choose a finite  $F$  of size  $n$  for each  $n \geq 1$ . The same formula gives nonzero  $\text{Tor}_n(D, V_F)$  for arbitrarily large  $n$ . Taking  $n \geq 2$  and dimension shifting gives nonzero  $\text{Tor}_{n-1}(\Pi_\Lambda, V_F)$  as well. Neither module can have a finite-length flat resolution. Finally the vanishing of all positive self-Tor groups is Corollary 13.5; it is compatible with these lower bounds because the modules  $V_F$  are different test modules.  $\square$

For  $d = 2$  the groups are  $D^2$  and  $K$  (the latter represented by  $kX_1X_2$ ), and for  $d = 3$  they are  $D^3, K^3, K$ : vertices, edges and the full face (12). The finite audit of 12 checks this formula for  $D = \mathbb{Q} \subset K = \mathbb{Q}(\sqrt{2})$  (Section 19.1).

### 13.4 Coefficient lattices and their syzygies

The next results need only a ring of the following shape, which covers 12's cores, 12's ordered-cone rings and 14's rings.

**Setting (L).**  $K$  is a field,  $D \subseteq K$  a unital subring, and  $A = D \oplus \Pi$  is one of: (a)  $A_\Lambda$  with  $\Pi = \Pi_\Lambda$ ; (b)  $T_\Gamma$  with  $\Pi = \Pi_\Gamma^{\text{fin}}$ ; (c) the class ring  $\mathcal{A}_{D,K}$  with  $\Pi = \Pi_K$  (full surreal exponents,  $K$  a set field). In each case  $B = K \oplus \Pi$  is a ring containing  $A$ ,  $\Pi$  is an ideal of  $B$  and a  $K$ -vector space,  $K \cap \Pi = 0$ , all three lie in a field, and  $\Pi$  is not finitely generated in (a), (b) and not generated by a set in (c) (Lemma 13.3(iii), Proposition 13.6, Proposition 3.3(iii)). A *coefficient lattice* is a nonzero module  $M = Dm_1 + \cdots + Dm_r \subseteq K$ , with  $r \geq 1$  and  $m_1, \dots, m_r$  linearly independent over  $D$  (equivalently over  $\text{Frac } D$ );  $r = \text{rank } M$ . Put

$$L(M) = M + \Pi \subseteq B, \quad J_g(M) = X^g L(M) \subseteq A$$

for a nonzero monomial  $X^g \in \Pi$  (a vector  $g \neq \mathbf{0}$  in (a),  $g \in \Gamma_{>0}$  in (b),  $g > 0$  in (c)). For  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $D$  is a principal ideal domain and every nonzero finitely generated  $D$ -submodule of  $K$  is free, so these are exactly 14's coefficient lattices.

The direct sums in Setting (L) are additive decompositions. A lattice's  $D$ -rank  $r$  counts independent constants; it is different from the fraction-field rank of  $L(M)$  as an  $A$ -module. In the set-sized cases all modules and tensor products below have their usual meanings. For the class case, finite free modules, maps and kernels are explicit class constructions; tensor statements use the balanced carriers specified in Section 13.6. No set of generators for the tail is assumed.

**Theorem 13.9** (Generators and the first syzygy [14; 12; 17] [merge]). *In Setting (L), let  $M$  have basis  $m_1, \dots, m_r$ .*

(i)  $L(M) = \sum_j Am_j$ ,  $\Pi L(M) = \Pi$  and  $L(M)/\Pi L(M) \cong M$ ; the least number of generators of  $L(M)$  is  $r$ , although  $L(M)$  has rank one over the fraction field.

(ii) The sequence

$$0 \longrightarrow \Pi^{r-1} \xrightarrow{\sigma} A^r \xrightarrow{\pi} L(M) \longrightarrow 0, \quad \pi(a) = \sum_j a_j m_j,$$

$$\sigma(u_2, \dots, u_r) = \left( - \sum_{j \geq 2} \frac{m_j}{m_1} u_j, u_2, \dots, u_r \right),$$

is exact. For  $r = 1$ ,  $L(M) \cong A$ ; for  $r \geq 2$ ,  $L(M)$  is not finitely presented in (a), (b), and in (c) the kernel is not generated by any set.

(iii) (12's certificate.) Let  $c \in K \setminus \text{Frac}(D)$  and  $H = (X^g, cX^g)A$ . Then  $X^g A \cap cX^g A = X^g \Pi$  and  $0 \rightarrow \Pi \xrightarrow{m \mapsto (-cm, m)} A^2 \xrightarrow{(a, b) \mapsto X^g a + cX^g b} H \rightarrow 0$  is exact; in (a) and (b),  $A$  is not coherent.

*Proof.* (i) Every basis vector is nonzero. Since  $\Pi$  is a  $K$ -vector space,  $m_1 \Pi = \Pi$ , and therefore  $\sum_j Am_j$  contains both  $M$  and  $\Pi$ . Conversely each  $a_j m_j$  has its constant part in  $Dm_j$  and its tail in  $\Pi$ , so this sum is exactly  $L(M)$ . The ideal property in  $B$  gives  $\Pi L(M) \subseteq \Pi$ , while  $\Pi m_1 = \Pi$  gives the reverse inclusion. Taking the constant part identifies  $L(M)/\Pi L(M)$  with  $M \cong D^r$ , with  $A$  acting through  $D$  on this quotient.

Any  $q$  generators of  $L(M)$  give  $q$  generators of  $D^r$ . Tensoring the resulting surjection  $D^q \rightarrow D^r$  with  $\text{Frac}(D)$  shows  $q \geq r$  by vector-space dimension; the displayed  $r$  generators attain this bound. To compare this with fraction-field rank, choose  $0 \neq u \in \Pi$ . Every  $k \in K$  equals  $(ku)/u$ , with numerator and denominator in  $A$ , so  $K \subseteq \text{Frac}(A)$ . Thus  $L(M)$  is a nonzero submodule of  $\text{Frac}(A)$  and spans that one-dimensional vector space after localization. This is rank one over  $\text{Frac}(A)$ , independently of  $r$ .

(ii) Write  $a_j = d_j + u_j$  uniquely with  $d_j \in D$ ,  $u_j \in \Pi$ . If  $\sum_j a_j m_j = 0$ , its constant part is  $\sum_j d_j m_j = 0$ . Independence of the basis over  $D$  forces every  $d_j = 0$ . The remaining relation is  $u_1 = -\sum_{j \geq 2} (m_j/m_1) u_j$ . Its right side lies in  $\Pi$  because  $\Pi$  is a  $K$ -space. Conversely every tuple of this form is a relation. The last  $r - 1$  coordinates recover the input to  $\sigma$ , proving injectivity and the asserted description of the kernel. Surjectivity of  $\pi$  is (i).

For  $r = 1$ , multiplication by  $m_1$  gives the isomorphism  $A \cong L(M)$ . For  $r \geq 2$ , projection from  $\Pi^{r-1}$  onto any coordinate is surjective. A finite generating family for this kernel would therefore generate  $\Pi$ , contradicting Setting (L). In (a) and (b), if  $L(M)$  had any finite presentation, the kernel of its displayed finite free surjection would be finitely generated. One can see the independence of the chosen presentation from Schanuel's lemma: comparing it with a finite presentation makes this kernel a direct summand of a finitely generated module [30, Tag 00O3]. In (c), the same coordinate projection shows directly that the displayed kernel cannot be generated by a set; the stronger statement about arbitrary presentations is treated in Theorem 13.29.

(iii) The assumption  $c \notin \text{Frac}(D)$  gives  $c \neq 0$  and makes  $1, c$  independent over  $D$ . Apply (ii) to  $M = D + Dc$ , and multiply the presentation by  $X^g$ . Multiplication is injective in the ambient field and takes  $L(M)$  onto  $H$ , so the stated sequence is exact. For the intersection, cancel  $X^g$  in  $X^g(d + u) = cX^g(e + v)$ . Comparing constant parts gives  $d = ce$ . If  $e \neq 0$ , this would put  $c = d/e$  in  $\text{Frac}(D)$ ; hence  $e = d = 0$ . Thus the intersection is contained in  $X^g \Pi$ . Conversely  $c\Pi = \Pi$  puts

every element of  $X^g\Pi$  in both principal ideals. The intersection is therefore exactly  $X^g\Pi$ , which is isomorphic to the non-finitely generated module  $\Pi$ .

For (a) and (b), this rules out coherence: in a coherent ring the intersection of two finitely generated ideals is finitely generated [32, Section 10.90]. Equivalently, the finitely generated ideal  $H \cong L(D + Dc)$  is not finitely presented by (ii), contradicting the definition of a coherent ring. The existence of  $c \in K \setminus \text{Frac}(D)$  is part of this certificate.  $\square$

**Theorem 13.9(iii)** is 12's noncoherence certificate for  $A_\Lambda$  and  $T_\Gamma$  whenever  $K \neq \text{Frac}(D)$ ; no principal-ideal hypothesis on the unital subring  $D$  is needed. (i) and (ii) are 14's theorem for  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and  $A_0$  with  $D$  a principal ideal domain. Stating both at once in Setting (L) is the merge's; the proofs are the sources'. All compatible relations between two generators with the same monomial and incompatible constant lines are forced into the infinitely divisible tail (12). No novelty is attached to noncoherence as such: rings of the form  $D + M$  and their coherence have a long history, including Dobbs–Papick [14]; 14 uses no valuation-ring hypothesis on  $B$ , and the useful information is the exact syzygy (14). Manuscript 17 proves (i), (ii) and the intersection formula of (iii) again, independently, for  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and the models  $R_{\Gamma_\kappa, < \kappa}^{11}$ , and adds that in case (c) the module  $L(M)$  with  $r \geq 2$  has no presentation by sets (**Theorem 13.29**); it credits the kernel shape  $\Pi^{r-1}$  and the flat dimension one of the  $D + M$  setting to Dobbs [13] and the intersection mechanism to Dobbs–Papick.

**Proposition 13.10** (Antisymmetric torsion and nonflatness [14] [merge]). *In Setting (L), let  $M, N$  be coefficient lattices and  $\mathbf{m}_{M,N} : M \otimes_D N \rightarrow K$  the multiplication map. On  $\mathbf{E}(M, N) = \Pi \oplus (M \otimes_D N)$  the rule*

$$(d + h) \cdot (u, t) = (du + hu + h \mathbf{m}_{M,N}(t), dt) \quad (d \in D, h \in \Pi) \quad (13.4)$$

*is an  $A$ -module structure, and  $(m + u) \otimes (n + v) \mapsto (mv + un + uv, m \otimes n)$  is a well-defined  $A$ -linear map  $\beta : L(M) \otimes_A L(N) \rightarrow \mathbf{E}(M, N)$ . If  $\text{rank } M \geq 2$  and  $m_1, m_2$  are basis vectors, then  $\tau = m_1 \otimes m_2 - m_2 \otimes m_1$  is a nonzero element of  $L(M) \otimes_A L(M)$  with product 0. Consequently, in (a) and (b),  $L(M)$  and every ideal  $J_g(M)$  are not flat, hence neither projective nor invertible.*

*Proof.* Write  $\mu = \mathbf{m}_{M,N}$ . The first coordinate in (13.4) lies in  $\Pi$ , since that is an ideal of  $B$  stable under all  $K$ -scalars. For  $a = d + h$  and  $b = e + v$  in  $A$ , their product has constant part  $de$  and tail  $dv + eh + hv$ . Expanding the action twice gives

$$a \cdot (b \cdot (u, t)) = (abu + (dv + eh + hv)\mu(t), det) = (ab) \cdot (u, t).$$

Here  $\mu(et) = e\mu(t)$  accounts for the cross term. Additivity in both arguments and the identity action follow directly, proving that this is an  $A$ -module.

The decompositions  $x = m + u$  and  $y = n + v$  in  $L(M)$  and  $L(N)$  are unique. Define the biadditive map  $b(x, y) = (xy - mn, m \otimes n)$ . For  $a = d + h \in A$  the constant part of  $ax$  is  $dm$ , so

$$b(ax, y) = (axy - dm n, dm \otimes n) = a \cdot b(x, y) = b(x, ay),$$

where  $dm \otimes n = m \otimes dn$  in  $M \otimes_D N$ . Thus  $b$  is balanced and induces the stated  $A$ -linear map  $\beta$ . The action is generally not the direct-sum action: the term  $h\mu(t)$  couples the two coordinates.

For  $N = M$ , the image of the antisymmetric tensor is

$$\beta(\tau) = (0, m_1 \otimes_D m_2 - m_2 \otimes_D m_1).$$

In the free  $D$ -module with basis  $m_i \otimes_D m_j$ , the coefficient of  $m_1 \otimes_D m_2$  is  $1 \neq 0$ . Hence this image, and therefore  $\tau$ , is nonzero, including in characteristic two. Multiplication sends  $\tau$  to zero because the ambient field is commutative.

In (a) and (b), multiplication by  $X^g$  identifies  $L(M)$  with the ideal  $J = J_g(M) \subset A$ ; both  $X^g M$  and  $X^g \Pi$  lie in  $\Pi$ . The induced tensor isomorphism carries  $\tau$  to a nonzero element of  $J \otimes_A J$  whose product is zero. If  $J$  were flat, tensoring the injection  $J \hookrightarrow A$  with  $J$  would give an injection  $J \otimes_A J \rightarrow A \otimes_A J \cong J$ . This map is exactly multiplication, a contradiction. Isomorphism transfers nonflatness to  $L(M)$ . Projective modules and invertible ideals are flat, giving the remaining conclusions. For class carriers the same balanced-map calculation is interpreted as described immediately below.  $\square$

In (c) the tensor product is 14's explicit carrier (Section 13.6); there the same element shows that tensoring  $J_g(M) \hookrightarrow A$  with  $J_g(M)$  is not injective, which is 14's statement that  $L(M)$  is not flat over  $\mathbf{Oz}$ . Over  $A_0$  this is 14's corollary; allowing any unital  $D \subseteq K$  and 12's rings is the merge's (the proof is 14's, and no principal-ideal hypothesis is used in it).

### 13.5 Flat and projective dimensions; the ordered cone

**Theorem 13.11** (Lattice modules and cyclic quotients [12, 14] [merge]). *Let  $A$  be  $A_d$  ( $d \geq 1$ ) or  $T_\Gamma$ , and put  $\delta = d$ , respectively  $\delta = 1$ . Let  $M$  be a coefficient lattice of rank  $r \geq 2$  and  $J = J_g(M)$ . Then*

$$\mathrm{fd}_A L(M) = \delta, \quad \mathrm{fd}_A(A/J) = \delta + 1,$$

*and for every  $A$ -module  $N$  and  $j \geq 2$ ,  $\mathrm{Tor}_j^A(L(M), N) \cong \mathrm{Tor}_j^A(D, N)^{r-1} \cong \mathrm{Tor}_{j+1}^A(A/J, N)$ . Over  $A_{\mathbb{N}_{>0}}$  both dimensions are infinite. In particular, for  $c \in K \setminus \mathrm{Frac}(D)$  and  $H = (X^g, cX^g)A$ ,  $\mathrm{fd}_{A_d}(A_d/H) = d + 1$  and  $\mathrm{fd}_{T_\Gamma}(T_\Gamma/H) = 2$  (with  $g = \mathbf{e}_1$  giving  $H = (X_1, cX_1)$  in  $A_d$ ), and over  $A_0 = T_\mathbb{Q}$ ,  $\mathrm{fd} L(M) = 1$  and  $\mathrm{fd}(A_0/J_e(M)) = 2$ . If moreover  $c^2 \in D$ , then  $\mathrm{Tor}_2^A(A/H, A/H) \cong D + cD \cong D \oplus D$ .*

*Proof.* Use the two exact sequences

$$0 \rightarrow \Pi^{r-1} \rightarrow A^r \rightarrow L(M) \rightarrow 0, \quad 0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0,$$

with  $J \cong L(M)$  by multiplication by  $X^g$ . For  $A_d$ , Corollary 13.8 gives  $\mathrm{fd} \Pi = d - 1$ . For  $T_\Gamma$ , the principal-ideal exhaustion in Theorem 13.13 makes  $\Pi$  flat. Splicing flat resolutions into the displayed sequences gives  $\mathrm{fd} L(M) \leq \delta$  and  $\mathrm{fd}(A/J) \leq \delta + 1$ .

For  $j \geq 2$ , the free middle terms make the long exact sequences give

$$\mathrm{Tor}_j(L(M), N) \cong \mathrm{Tor}_{j-1}(\Pi, N)^{r-1} \cong \mathrm{Tor}_j(D, N)^{r-1},$$

where the second comparison uses  $0 \rightarrow \Pi \rightarrow A \rightarrow D \rightarrow 0$ . Similarly  $\mathrm{Tor}_{j+1}(A/J, N) \cong \mathrm{Tor}_j(J, N)$ , and the isomorphism  $J \cong L(M)$  gives the claimed comparison. These maps are natural in  $N$  after the lattice basis and  $X^g$  are fixed.

If  $\delta = d \geq 2$ , take  $N = V_{[d]}$ :  $\mathrm{Tor}_d(D, N) \cong K \neq 0$  by Theorem 13.7, and  $r - 1 \geq 1$  ensures that its indicated direct power is nonzero. This attains both upper bounds. For  $\delta = 1$ , the preceding antisymmetric tensor gives a concrete degree-one obstruction. With  $N = A/J$ , the long exact sequence of  $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$  tensored with  $J$  identifies  $\mathrm{Tor}_1(A/J, J)$  with the kernel of  $J \otimes_A J \rightarrow J$ . It is nonzero. Symmetry of  $\mathrm{Tor}$  over the commutative ring then gives  $\mathrm{Tor}_1(J, A/J) \neq 0$ , and dimension shifting gives  $\mathrm{Tor}_2(A/J, A/J) \neq 0$ . Thus  $\mathrm{fd} L(M) = 1$  and  $\mathrm{fd}(A/J) = 2$ . In infinite

rank, use  $V_F$  with arbitrarily large finite  $|F| = n \geq 2$  in the same comparison. Both dimensions are infinite.

The two-generator instances use the lattice  $U = D + Dc$ , of rank two because  $c \notin \text{Frac}(D)$ . Assume additionally  $c^2 \in D$  for the last assertion. Then

$$H = X^g U \oplus X^g \Pi, \quad H\Pi = X^g \Pi, \quad cH \subseteq H.$$

For the middle identity use  $U\Pi = \Pi$  and  $\Pi^2 = \Pi$ ; for the last,  $cU \subseteq U$  follows from  $c^2 \in D$  and  $c\Pi = \Pi$ . Tensor the presentation  $0 \rightarrow \Pi \xrightarrow{m \mapsto (-cm, m)} A^2 \rightarrow H \rightarrow 0$  with  $A/H$ . Its long exact sequence, together with dimension shifting, identifies

$$\text{Tor}_2(A/H, A/H) \cong \text{Tor}_1(H, A/H) \cong \ker(\Pi/X^g \Pi \rightarrow (A/H)^2).$$

The map sends  $m$  to  $(-cm, m)$  modulo  $H$ . Its kernel consists exactly of the classes with  $m \in H$ : that condition is necessary from the second coordinate and sufficient from  $cH \subseteq H$ . The kernel is therefore  $H/X^g \Pi \cong U$ , by taking the coefficient of  $X^g$ . The action factors through  $D$  since  $H\Pi = X^g \Pi$ , and  $U \cong D \oplus D$  via its basis  $1, c$ . No condition  $c^2 \in D$  was used in the flat-dimension bounds.  $\square$

**Correction to 12 and the relation to 14.** 12 states  $\text{fd}(A_d/H) = d+1$  and  $\text{fd}(T_\Gamma/H) = 2$  under the extra hypothesis  $c^2 \in D$ , which it uses only to compute  $\text{Tor}_2(A/H, A/H)$ ; for  $d \geq 2$  its own shift argument never uses it, and for  $d = 1$  and  $T_\Gamma$  the lower bound 2 follows from 14's antisymmetric torsion tensor. The hypothesis is needed only for the formula  $\text{Tor}_2(A/H, A/H) \cong D \oplus D$ . 14's flat-dimension proposition for  $A_0$  is the case  $A = T_\mathbb{Q} = A_1$  of the theorem. Since a nonzero Tor in degree  $n$  forces weak global dimension at least  $n$ : if  $K \neq \text{Frac } D$  (for instance  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$  with  $c = \sqrt{2}$ ), then  $\text{w.gl.dim } A_d \geq d+1$ ,  $\text{w.gl.dim } T_\Gamma \geq 2$ , and the countable-rank core has infinite weak global dimension (12, 14). Neither source claims the exact weak global dimension of any of these rings.

**Proposition 13.12** (Projective dimensions [12] [merge]).

- (i) (Telescope, 12.) Let  $A$  be a domain and  $I = \bigcup_{n \geq 0} x_n A$  for nonzero  $x_n$  with  $x_n = x_{n+1} u_n$ ,  $u_n \in A$ . With  $F = \bigoplus_{n \geq 0} A e_n$ , the sequence  $0 \rightarrow F \xrightarrow{\Delta} F \xrightarrow{\pi} I \rightarrow 0$ ,  $\Delta(e_n a) = e_n a - e_{n+1} u_n a$ ,  $\pi(e_n a) = x_n a$ , is exact; so  $\text{pd}_A I \leq 1$ . This applies to every  $I_S$  ( $x_n = p_S^{2^{-n}}$ ) and to  $\Pi_\Gamma^{\text{fin}}$  ( $x_n = X^{2^{-n}g}$ ,  $g \in \Gamma_{>0}$ ).
- (ii) [merge]  $\text{pd}_{A_\Lambda} I_S = 1$  and  $\text{pd}_{T_\Gamma} \Pi_\Gamma^{\text{fin}} = 1$ .
- (iii)  $d \leq \text{pd}_{A_d} D \leq d+1$  (12), and [merge]  $\text{pd}_{A_1} D = \text{pd}_{T_\Gamma} D = 2$ . (The upper bound is attained for every  $d$ : Corollary 13.47, 21.)
- (iv) [merge] Over  $T_\Gamma$  (in particular over  $A_0$ ), for a lattice of rank at least two,  $\text{pd } L(M) = 2$  and  $\text{pd}(T_\Gamma/J_g(M)) = 3$ . Over  $A_d$ ,  $d \leq \text{pd } L(M) \leq d+1$  and  $d+1 \leq \text{pd}(A_d/J_g(M)) \leq d+2$  (12, for  $M = D + Dc$ ). (The upper bounds are attained: Theorem 13.48, 21.)
- (v) [merge] Let  $(A, I, D)$  satisfy (H1)–(H3) of Section 12, with  $A$  consisting of series of nonnegative support (as do all rings of Section 11 and  $T_\Gamma$ ). If  $I$  is countably generated, equivalently if  $\Gamma_{>0}$  has a countable coinital subset, then  $\text{pd}_A I = 1$  and  $\text{pd}_A D = 2$ .

*Proof.* (i) The relation  $x_n = x_{n+1} u_n$  gives  $\pi \Delta = 0$ , and the union gives surjectivity of  $\pi$ . If a finite vector is nonzero, its least nonzero coordinate is unchanged by  $\Delta$ , proving injectivity. Modulo  $\text{im } \Delta$

the relation  $e_na = e_{n+1}u_na$  moves every term of a finite vector to one index  $N$ . Thus  $v \equiv e_Na$ , with  $x_Na = \pi(v)$ . If  $\pi(v) = 0$ , cancellation in the domain gives  $a = 0$ , proving exactness at the middle term. Both copies of  $F$  are free. The face ideals have the stated exhaustion by [Lemma 13.3\(i\)](#). For  $T_\Gamma$  fix  $g > 0$ . Since  $\Gamma \subseteq \mathbb{R}$ , for any finite nonzero tail choose  $n$  so large that  $2^{-n}g$  is strictly below every exponent in its support. Dividing by  $X^{2^{-n}g}$  then leaves a tail in  $T_\Gamma$ . Divisibility ensures these exponents lie in  $\Gamma$ , and the Archimedean property gives their cointiality.

(ii) The ideals in question are nonzero and not finitely generated. By [Lemma 12.1](#) they cannot be projective, while (i) gives projective dimension at most one. Hence their dimension is one.

(iii) Each  $C_p$  for  $p > 0$  in the finite face resolution has a length-one free resolution: take direct sums of the telescopes in (i). Here  $C_0 = A_d$  is already free. Dimension shifting along the finite resolution gives  $\text{pd } D \leq d + 1$ . Explicitly, in  $0 \rightarrow U \rightarrow V \rightarrow W \rightarrow 0$  one has  $\text{pd } W \leq \max(\text{pd } V, \text{pd } U + 1)$ ; iterate this from the leftmost  $C_d$  to  $D$ . The lower bound is  $\text{pd } D \geq \text{fd } D = d$ . For  $T_\Gamma$ , the sequence  $0 \rightarrow \Pi \rightarrow T_\Gamma \rightarrow D \rightarrow 0$  gives  $\text{pd } D \leq 2$ . A projective dimension at most one for  $D$  would make this kernel  $\Pi$  projective by Schanuel's lemma, contrary to (ii). Thus the dimension is two, also for  $A_1 \cong T_\mathbb{Q}$ .

(iv) Over  $T_\Gamma$  the presentation of  $L(M)$  has kernel  $\Pi^{r-1}$  of projective dimension one. It gives  $\text{pd } L(M) \leq 2$ . If this dimension were at most one, Schanuel's lemma would make  $\Pi^{r-1}$  projective, hence also its summand  $\Pi$ , a contradiction. Thus  $\text{pd } L(M) = \text{pd } J = 2$ . The sequence  $0 \rightarrow J \rightarrow T_\Gamma \rightarrow T_\Gamma/J \rightarrow 0$  gives the upper bound three for the quotient. If the quotient had projective dimension at most two, its first syzygy  $J$  would have dimension at most one, again a contradiction. Over  $A_d$ , truncation of the face resolution and the same dimension inequalities give  $\text{pd } \Pi_d \leq d$ , then  $\text{pd } L(M) \leq d + 1$  and  $\text{pd}(A_d/J) \leq d + 2$ . Their flat dimensions give the displayed lower bounds.

(v) Suppose a countable family generates  $I$ , and write each generator as  $f_n = X^{g_n}b_n$  with  $g_n > 0$ ,  $b_n \in I$ , by (H2). For any  $h > 0$ , the monomial  $X^h \in I$  is a finite linear combination of these generators. It therefore lies in  $X^gA$  for  $g$  the minimum of the finitely many  $g_n$  used. Cancelling the monomial in the ambient series field shows  $X^{h-g} \in A$ . The nonnegative-support hypothesis implies  $g \leq h$ , so the  $g_n$  form a cointial family.

Conversely, enumerate a countable cointial family as  $q_n > 0$ . Divisibility lets us choose recursively  $h_n > 0$  with  $h_n \leq q_n$  and  $h_{n+1} < h_n$ ; this sequence is still cointial. If  $f = X^gb$  as in (H2), choose  $n$  with  $h_n \leq g$ . Then  $f = X^{h_n}(X^{g-h_n}b) \in X^{h_n}A$ , proving  $I = \bigcup_n X^{h_n}A$ . The transition factors are  $X^{h_n-h_{n+1}} \in A$ , so (i) applies. Nonfinite generation and [Lemma 12.1](#) exclude projectivity, giving  $\text{pd } I = 1$ . The same quotient argument as in (iii) gives  $\text{pd } D = 2$ .  $\square$

Items (ii)–(v) sharpen 12's bounds  $1 \leq \text{pd}_{A_1} D \leq 2$  and are consistent with 06's  $\text{pd} \geq 2$  ([Theorem 12.2\(v\)](#)). For  $d \geq 2$  these items left the value  $\text{pd}_{A_d} D \in \{d, d + 1\}$  undetermined; 21 shows that it is  $d + 1$  for every  $d$ , and that the lattice bounds of (iv) are attained ([Corollary 13.47](#) and [Theorem 13.48](#)). Hypothesis (H2) in (v) cannot be dropped:  $A_d$  has a countably generated tail and satisfies (H1) and (H3), but not (H2), and  $\text{pd}_{A_d} D = d + 1$  ([Remark 13.49](#), a merge remark). Among the five constructions of [Section 11](#), (v) applies exactly to  $A_\kappa^{08}$  with  $\text{cf } \kappa = \aleph_0$ : the tail of each construction needs as many generators as the length of its cointial chain ([Remark 10.11](#)), namely  $\text{cf } \kappa$  for 03 and 08,  $\kappa$  for 06 and 11,  $\aleph_1$  for 09, and only 08 allows countable cofinality. [Question 18.3](#) is re-scoped accordingly.



**Theorem 13.13** (Ordered-cone cores [12]).  $(T_\Gamma, \Pi_\Gamma^{\text{fin}}, D)$  satisfies (H1)–(H3) of Section 12. Consequently  $\Pi_\Gamma^{\text{fin}} = \bigcup_{\delta \in \Gamma_{>0}} X^\delta T_\Gamma$  is flat and idempotent but not projective,  $\text{Tor}_{>0}^{T_\Gamma}(D, N) = 0$  for every  $D$ -module  $N$ ,  $\text{fd}_{T_\Gamma} D = 1$  with  $\text{Tor}_1(D, T_\Gamma/X^\gamma T_\Gamma) \cong D$  for  $\gamma > 0$ , and the other items of Theorem 12.2 hold;  $T_\Gamma \rightarrow D$  is formally étale and not flat,  $\text{pd}_{T_\Gamma} D = 2$ , and  $T_\Gamma$  is not coherent if  $K \neq \text{Frac}(D)$ .

*Proof.* The domain, additive decomposition and constant-term retraction give (H1). For a nonzero tail  $f$ , let  $g$  be half the least of its finitely many positive exponents. Then  $f = X^g(X^{-g}f)$ , and the quotient has strictly positive exponents and arbitrary coefficients in  $K$ , hence is again a tail. The zero tail has the same factorization with zero quotient and any  $g > 0$ . This proves (H2), including the directed principal-ideal exhaustion. For (H3), a finite nonzero family of tails has a smallest exponent  $\delta > 0$  among its monomials. Multiplication by  $T_\Gamma$  cannot introduce smaller exponents, so it cannot generate  $X^{\delta/2} \in \Pi_\Gamma^{\text{fin}}$ .

The homological package now gives flatness, idempotence, nonprojectivity, all the asserted Tor conclusions and flat dimension one. For the displayed nonzero  $\text{Tor}_1$ , use the free resolution given by multiplication by  $X^\gamma$  for  $\gamma > 0$ . Formal lifting is Proposition 13.6, and the countable exhaustion in Proposition 13.12(i)–(iii) gives projective dimension two. Finally, if  $K \neq \text{Frac}(D)$ , choose  $c \in K \setminus \text{Frac}(D)$  and apply the two-generator noncoherence certificate of Theorem 13.9(iii).  $\square$

**Coefficient hypothesis for noncoherence.** The final clause above needs the displayed restriction. Without it the earlier unconditional claim is false: take  $D = K$  and  $\Gamma = \mathbb{Q}$ . Then  $T_\mathbb{Q} = K[\mathbb{Q}_{\geq 0}]$ . Any finite family lies in a common polynomial subring  $K[X^{1/n}]$ . Its ideal there is generated by a greatest common divisor, with a polynomial Bézout identity expressing that divisor in terms of the family. The same divisibilities and identity hold after extending to  $T_\mathbb{Q}$ , so every finitely generated ideal of  $T_\mathbb{Q}$  is principal. A nonzero principal ideal in this domain is free of rank one; the zero ideal is finitely presented as well. Thus  $T_\mathbb{Q}$  is coherent, despite its non-finitely generated tail and the projective dimension two of its constant quotient. This counterexample does not affect the integer/real or Gaussian/complex cases, where  $K \neq \text{Frac}(D)$ . No complete coherence classification is asserted.

**Theorem 13.14** (Same exponent group, different dimension [12]). Let  $(\gamma_i)_{i \in \Lambda}$  be positive reals linearly independent over  $\mathbb{Q}$ ,  $\Gamma = \sum_i \mathbb{Q}\gamma_i$ , and  $A_\Lambda \hookrightarrow T_\Gamma$  the inclusion of Lemma 13.1(iii). Then  $\Pi_\Lambda T_\Gamma = \Pi_\Gamma^{\text{fin}}$  and  $T_\Gamma \otimes_{A_\Lambda} D \cong D$ . If  $|\Lambda| \geq 2$  the extension is not flat:  $(X_1 A_\Lambda \cap X_2 A_\Lambda) T_\Gamma \neq X_1 T_\Gamma \cap X_2 T_\Gamma$ . The flat dimension of  $D$  drops from  $d$  to 1 in rank  $d$ , and from  $\infty$  to 1 in countable rank.

*Proof.* The image of  $\Pi_\Lambda$  lies in the ideal  $\Pi_\Gamma^{\text{fin}}$ , giving one inclusion. Conversely, for  $\gamma > 0$  choose a positive rational  $q$  with  $q\gamma_1 < \gamma$ , using that these are real numbers. Then

$$kX^\gamma = X_1^q kX^{\gamma - q\gamma_1} \in \Pi_\Lambda T_\Gamma.$$

The second factor is a tail even for  $k \notin D$ , so the identity covers every coefficient. Finite sums give equality of the ideals. The usual base-change identity for a quotient therefore gives

$$T_\Gamma \otimes_{A_\Lambda} D \cong T_\Gamma / \Pi_\Lambda T_\Gamma = T_\Gamma / \Pi_\Gamma^{\text{fin}} \cong D.$$

This identity does not require flatness.

If  $|\Lambda| \geq 2$ , choose and relabel two coordinates so that  $\gamma_1 < \gamma_2$ ; independence excludes equality. Every monomial in  $X_1 A_\Lambda \cap X_2 A_\Lambda$  has both first and second coordinates at least one. Its real

exponent is therefore at least  $\gamma_1 + \gamma_2$ . Multiplication by  $T_\Gamma$  adds only nonnegative real exponents, and cancellation cannot create a new smaller exponent. Thus this lower bound holds throughout the extended intersection ideal. But

$$X^{\gamma_2} = X_1 X^{\gamma_2 - \gamma_1} \in X_1 T_\Gamma \cap X_2 T_\Gamma$$

has smaller exponent and does not lie in that ideal.

A flat extension would preserve this intersection: tensor the exact sequence  $0 \rightarrow I \cap I' \rightarrow A \rightarrow A/I \oplus A/I'$  and use flatness also to identify the tensors of ideals with their extended ideals inside the target ring. Its exactness would give equality of the two intersections, contrary to the exhibited monomial. Finally [Corollary 13.8](#) and [Theorem 13.13](#) compute the two quotient flat dimensions, in finite and countably infinite rank. The nonflat extension, rather than any change in the constant ring or ordered exponent group, accounts for their difference.  $\square$

**Example 13.15** (Two directions inside  $\mathbf{Oz}$  [12]). With  $\gamma_1 = 1$ ,  $\gamma_2 = \sqrt{2}$  and  $(D, K) = (\mathbb{Z}, \mathbb{R})$ , the core  $A_2 = \mathbb{Z} + \bigoplus_{a, b \in \mathbb{Q}_{\geq 0}, a+b>0} \mathbb{R} \omega^{a+b\sqrt{2}}$  lies in the ring  $T_{\mathbb{Q}+\mathbb{Q}\sqrt{2}} \subseteq \mathbf{Oz}$ , which also contains  $\omega^{\sqrt{2}-1} \notin A_2$ ; the identity  $\omega^{\sqrt{2}} = \omega \cdot \omega^{\sqrt{2}-1}$  is the extra intersection element, and the two arithmetic quotients have flat dimensions 2 and 1. Replacing  $(\mathbb{Z}, \mathbb{R})$  by  $(\mathbb{Z}[i], \mathbb{C})$  gives the same example in  $\mathbf{Oz}[i]$ .

*Remark 13.16* (Flat dimension one is a property of the cone [merge]). The report now contains two kinds of statements about the flat dimension of the arithmetic quotient, and they concern different rings.

- $\text{fd}_A D = 1$  holds whenever (H1)–(H3) hold: the tail is a directed union of principal monomial ideals, because the positive exponents form the positive cone of a totally ordered group and every family of tails of the relevant size (finite for finite supports, of fewer than  $\kappa$  or countably many elements in the constructions, a set for the class ring) has a common monomial divisor in the ring. This covers the five constructions of [Section 11](#) ([Theorem 12.2](#)), including 06’s ring  $A_\kappa^{06}$ ; the full omnific ring in the two-universe reading, where 13 proves that the tail is flat, idempotent and not generated by a family from the smaller universe, that  $\text{Tor}_{>0}(D, D) = 0$  and that  $D$  is not flat, so  $\text{fd } D = 1$ ; 14’s  $A_0$ ; and 12’s  $T_\Gamma$  ([Theorem 13.13](#)).
- For the coordinatewise cone  $\mathbb{Q}_{\geq 0}^d$  the flat dimension is  $d$  ([Corollary 13.8](#)): under the embedding, with  $\gamma_1 < \gamma_2$ , the quotient  $\omega^{\epsilon(\gamma_2 - \gamma_1)}$  of  $X_2^\epsilon$  by  $X_1^\epsilon$  lies in  $\mathbf{Oz}$  but not in  $A_d$ , so the tail of  $A_d$  is not a directed union of principal ideals; it is a sum of  $d$  flat ideals that is not flat for  $d \geq 2$ . (Status (batch 34): 26 proves  $\text{fd } D = d$  for the core of every pointed full-dimensional rational polyhedral cone of dimension  $d$ , and that its passage to a total-order core, or to  $\mathbf{Oz}$ , is not flat for  $d \geq 2$ ; [Theorems 13.75](#) and [13.88](#).)

So “flat dimension one” must never be asserted for an arbitrary subring of  $\mathbf{Oz}$ :  $A_d \subseteq T_\Gamma \subseteq \mathbf{Oz}$ , with all exponents in one Archimedean class, have flat dimensions  $d$  and 1, and  $A_d \rightarrow T_\Gamma$  is not flat. Being a subring of a ring with small quotient flat dimension imposes no bound on the subring, and conversely these core calculations do not transfer homological dimensions to the full class ring  $\mathbf{Oz}$ . The separate flat-dimension-one assertion for that class ring uses the two-universe interpretation described above; it does not follow from the embeddings of the cores. The projective dimension of  $D$  behaves differently again: under (H1)–(H3) it is 2 when the tail is countably generated, in particular for  $T_\Gamma$ , and is otherwise known only to be at least 2; for  $A_d$  it is  $d + 1$ , one more than the flat dimension ([Proposition 13.12](#), with 21’s [Corollary 13.47](#); before 21 it was known only to be  $d$  or  $d + 1$ ).

### 13.6 The tensor normal form

For the rest of this section (14) fix  $(D, k) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , and let  $(A, \Pi, B)$  be either the class rings  $(\mathcal{A}_{D,k}, \Pi_k, \mathcal{A}_{k,k})$ , that is  $A = \mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , or the set-sized model  $(T_{\mathbb{Q}}, \Pi_{\mathbb{Q}}^{\text{fin}}, k + \Pi_{\mathbb{Q}}^{\text{fin}})$  with  $T_{\mathbb{Q}} = A_0$ . Coefficient lattices are those of Setting (L). The tensor and power results in this subsection hold in both cases (14). Their proofs use Setting (L), the principal-ideal property of  $D$ , and the fact that every  $u \in \Pi$  is a product  $u = ca$  with  $c, a \in \Pi$  (Proposition 3.3(ii); in  $T_{\mathbb{Q}}$  split off a smaller monomial). This interface is 14's proposal for a formalization (Section 19.2). For the class rings a tensor product means an explicit carrier with the balanced universal property proved by finite identities (for two lattices,  $\Pi \times (M \otimes_D N)$ ); no quotient of a proper class by a class relation and no class colimit is used (14).

Here  $MN$  denotes the  $D$ -span of the products  $mn$ , not just the set of individual products. It is nonzero and finitely generated inside  $k$ , hence free over the principal ideal domain  $D$ ; thus  $L(MN)$  is again defined. In a class universal property the target is any fixed class  $A$ -module  $W$  equipped with a specified map. The proof gives one explicit factorization into  $W$  and proves uniqueness for that map; it does not form a class of all class modules. Maps called balanced below are biadditive and satisfy  $b(ax, y) = a b(x, y) = b(x, ay)$ . The finite formulas also apply to the ordinary tensor products in the set-sized model.

**Lemma 13.17** (Absorption [14]). *For every lattice  $N$  and  $0 \neq n_0 \in N$ , multiplication is an isomorphism  $\Pi \otimes_A L(N) \cong \Pi$  with inverse  $u \mapsto (u/n_0) \otimes n_0$ ; indeed  $u \otimes v = (uv/n_0) \otimes n_0$  for  $u \in \Pi$ ,  $v \in L(N)$ .*

*Proof.* Multiplication maps  $\Pi \times L(N)$  to  $\Pi$ , since  $L(N) \subset B$  and  $\Pi$  is an ideal of  $B$ , and is  $A$ -bilinear. Its proposed inverse is defined because  $u/n_0 \in \Pi$  for every  $u \in \Pi$ . It is  $A$ -linear in  $u$ , and multiplication after this inverse is the identity.

To prove the other composition, write  $u = ca$  with  $c, a \in \Pi$ . This includes  $u = 0$ , for which the desired identity is already zero. For  $v \in L(N)$ , scalar stability gives  $v/n_0 \in B$ , so  $b = a(v/n_0) \in \Pi \subseteq A$ . Only elements of  $A$  are moved between the tensor factors in the calculation

$$u \otimes v = c \otimes av = c \otimes bn_0 = cb \otimes n_0 = (uv/n_0) \otimes n_0.$$

In particular this does not move the possibly nonintegral scalar  $v/n_0$  across a tensor sign. Pure tensors generate by finite sums, so the identity proves that the two maps are inverse. It also shows that changing the auxiliary choice  $n_0$  does not change the inverse of multiplication.

For the class interpretation, let  $b' : \Pi \times L(N) \rightarrow W$  be any specified balanced map. Define  $H(u) = b'(u/n_0, n_0)$ . This is  $A$ -linear, and the same sequence of balancing identities gives  $b'(u, v) = H(uv)$ . Multiplication is onto, since  $u = (u/n_0)n_0$ , so this factorization is unique. Thus the explicit carrier  $\Pi$  has the required universal property without first constructing a quotient of a class of formal tensors.  $\square$

**Theorem 13.18** (Exact tensor normal form [14; 17]). *The map  $\beta$  of Proposition 13.10 is an  $A$ -module isomorphism  $L(M) \otimes_A L(N) \cong E(M, N)$ . Multiplication corresponds to  $(u, t) \mapsto u + \mathfrak{m}_{M,N}(t)$ , and there is an exact sequence*

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow L(M) \otimes_A L(N) \longrightarrow L(MN) \longrightarrow 0, \quad \mathcal{T}(M, N) = \ker(M \otimes_D N \rightarrow MN),$$

*with  $\mathcal{T}(M, N)$  embedded as  $(0, \mathcal{T}(M, N))$ . This is exactly the  $A$ -torsion submodule; if it is nonzero its annihilator is exactly  $\Pi$ , and  $\text{rank}_D \mathcal{T}(M, N) = \text{rank } M \cdot \text{rank } N - \text{rank}(MN)$ .*

*Proof.* Write  $\mu = \mathfrak{m}_{M,N}$  and choose  $0 \neq n_0 \in N$ . In the ordinary tensor product define  $j(w) = (w/n_0) \otimes n_0$  for  $w \in \Pi$ , and let  $s : M \otimes_D N \rightarrow L(M) \otimes_A L(N)$  be the  $D$ -linear map induced by the constant inclusions. Then

$$\beta(j(w)) = (w, 0), \quad \beta(s(t)) = (0, t).$$

Consequently  $\beta\psi = 1$  for  $\psi(u, t) = j(u) + s(t)$ . For the other composition expand  $(m + u) \otimes (n + v)$ . The terms  $u \otimes n$  and  $u \otimes v$  become  $j(un)$  and  $j(uv)$  by absorption. For the remaining mixed term, factor  $v = ca$  with  $c, a \in \Pi$ . Balancing gives  $m \otimes v = cm \otimes a$ , whose first factor is in  $\Pi$ , so absorption gives  $j(mv)$ . The constant term is  $s(m \otimes_D n)$ . Hence  $\psi\beta$  is the identity on pure tensors and thus on every finite tensor sum. Since  $\beta$  is  $A$ -linear, its inverse  $\psi$  is  $A$ -linear too; the constant map  $s$  by itself is only asserted to be  $D$ -linear.

We verify the class universal property explicitly as well. For a fixed balanced map  $b : L(M) \times L(N) \rightarrow W$ , put  $j_b(w) = b(w/n_0, n_0)$  and let  $s_b : M \otimes_D N \rightarrow W$  be induced by the restriction of  $b$  to the two constant lattices. This latter map uses the finite free  $D$ -tensor basis, so it is defined by finite sums even when  $W$  is a class module. Define

$$H(u, t) = j_b(u) + s_b(t).$$

Absorption proves  $j_b$  is  $A$ -linear. For  $h \in \Pi$ , first on  $t = m \otimes_D n$  and then by addition, it also gives

$$h s_b(t) = j_b(h\mu(t)).$$

Together with  $D$ -linearity of  $s_b$ , this is exactly the cross term needed for  $H((d + h) \cdot (u, t)) = (d + h)H(u, t)$  under (13.4). The four-term expansion just used gives  $H\beta(x, y) = b(x, y)$ . Finally the elements  $(w, 0)$  are the images of  $(w/n_0, n_0)$ , and  $(0, m \otimes_D n)$  are the images of  $(m, n)$ . They generate every pair by finite sums, so a factorization must have the displayed values and is unique. This proves the claimed universal property directly on  $\mathbf{E}(M, N)$ .

Multiplication is  $(u, t) \mapsto u + \mu(t)$  under the normal form. Its image is  $\Pi + MN = L(MN)$ . If  $u + \mu(t) = 0$ , the first term lies in  $\Pi$  and the second in  $k$ ; their intersection is zero, so  $u = 0$  and  $\mu(t) = 0$ . This proves exactness and identifies the kernel with  $(0, \mathcal{T}(M, N))$ . Its  $A$ -action is  $(d + h)(0, t) = (0, dt)$ , so  $\Pi$  annihilates it. Since  $\Pi$  contains a nonzero monomial, every element of this kernel is  $A$ -torsion. Conversely, any torsion element maps to zero in the submodule  $L(MN)$  of a field, which is torsion free over the domain  $A$ . The kernel is therefore exactly the torsion submodule.

The  $D$ -module  $M \otimes_D N$  is finite free, so its submodule  $\mathcal{T}(M, N)$  is finite free as well. If it is nonzero, no nonzero  $d \in D$  annihilates it. The formula for its action then shows that its annihilator in  $A$  is precisely  $\Pi$ . If the kernel is zero its annihilator is all of  $A$ , which explains the qualification in the statement. Finally the surjection of finite free  $D$ -modules  $M \otimes_D N \rightarrow MN$  gives  $\text{rank}_D \mathcal{T}(M, N) = \text{rank } M \text{ rank } N - \text{rank}(MN)$  by tensoring with  $\text{Frac}(D)$  and taking dimensions.  $\square$

For  $M = D + D\alpha$  with  $\alpha \notin \text{Frac } D$ , the tensor  $\tau = 1 \otimes \alpha - \alpha \otimes 1$  is nonzero and is killed by every  $h \in \Pi$ . If  $\alpha^2 = c \in D$ , write a constant tensor in the ordered basis  $1 \otimes 1, 1 \otimes \alpha, \alpha \otimes 1, \alpha \otimes \alpha$  with coefficients  $(a, b, e, f) \in D^4$ . Its product is  $(a + cf) + (b + e)\alpha$ , which vanishes exactly when  $a = -cf$  and  $e = -b$ . Thus it equals  $b\tau + f(\alpha \otimes \alpha - c(1 \otimes 1))$ , and these two generators are independent by their displayed coordinates. This proves the stated quadratic basis, not just its rank (14). Manuscript 17 proves the exact sequence and the torsion statement independently over its set-sized models, as a consequence of its Tor identification (Theorem 13.34). Here “torsion” means  $A$ -torsion. The kernel is free over  $D$  and hence is also free as an abelian group in both the integer and Gaussian cases (14). Its annihilation by tails is compatible with the absence of ordinary-integer torsion.

**Proposition 13.19** (Several factors [14]). *For lattices  $M_1, \dots, M_n$ ,  $\bigotimes_{j,A} L(M_j) \cong \Pi \oplus \bigotimes_{j,D} M_j$  with the action (13.4) for the  $n$ -fold product map; the torsion is its kernel, killed by  $\Pi$ . For equal factors, permutations act trivially on the first coordinate and by permuting tensor factors on the second.*

*Proof.* First let  $n \geq 1$ , put  $T_n = \bigotimes_{j,D} M_j$ , and let  $\mu_n : T_n \rightarrow k$  be the product map. On  $\Pi \oplus T_n$  use

$$(d + h) \cdot (u, t) = ((d + h)u + h\mu_n(t), dt).$$

The associativity computation of Proposition 13.10 applies to any  $D$ -linear  $\mu_n$ . Define on a tuple  $x_j = m_j + u_j$  the map

$$\beta_n(x_1, \dots, x_n) = \left( \prod_j x_j - \prod_j m_j, \bigotimes_{j,D} m_j \right).$$

The first coordinate is in  $\Pi$ , since expansion removes the all-constant term. This map is additive in each slot. Multiplying one slot by  $d + h$  multiplies its constant part by  $d$ ; the first coordinate becomes  $(d + h) \prod_j x_j - d \prod_j m_j$ . This is exactly the first coordinate of  $(d + h) \cdot \beta_n(x_1, \dots, x_n)$ , proving  $A$ -multilinearity.

Choose  $0 \neq c_j \in M_j$  and define

$$j_n(w) = \frac{w}{c_2 \cdots c_n} \otimes c_2 \otimes \cdots \otimes c_n, \quad \psi_n(u, t) = j_n(u) + s_n(t),$$

where  $s_n$  is induced by the constant inclusions. For  $n = 1$  the empty denominator is 1,  $j_1(w) = w$ , and the formula is simply  $L(M_1) = \Pi + M_1$ . The map  $j_n$  is  $A$ -linear;  $s_n$  is initially only  $D$ -linear.

For  $n \geq 2$ , every term in the expansion of a pure tensor either has all factors constant or contains a tail. If a tail lies in a slot other than the first, write it as  $ab$  with  $a, b \in \Pi$  and move the scalar  $a \in A$  to the first slot. That first factor is now in  $\Pi$ , since  $aL(M_1) \subseteq \Pi$ . With a tail in the first slot, apply absorption to that slot and slot  $j$ , for  $j = 2, \dots, n$ , replacing the latter by  $c_j$  and multiplying the first by the old factor divided by  $c_j$ . The first slot stays in  $\Pi$ . The result is  $j_n$  of the product of the original factors. Thus every term with a tail reduces to its product, and the all-constant term is  $s_n$  of its  $D$ -tensor. This proves  $\psi_n \beta_n = 1$ , while the definitions give  $\beta_n j_n(w) = (w, 0)$  and  $\beta_n s_n(t) = (0, t)$ .

For a specified class-valued  $A$ -multilinear map, the same reductions define its factorization: replace the displayed tensors in  $j_n$  and  $s_n$  by values of that map, extended on the finite free  $D$ -basis of  $T_n$ . The identity  $hs_n(t) = j_n(h\mu_n(t))$  verifies  $A$ -linearity of the factorization, exactly as in the two-factor proof. The values on  $(w, 0)$  and constant pure tensors force uniqueness. Hence no iteration of a class quotient is needed.

Multiplication is  $(u, t) \mapsto u + \mu_n(t)$ , with image  $L(M_1 \cdots M_n)$ , where the product lattice is the  $D$ -span of all products. Its kernel is  $(0, \ker \mu_n)$  because  $\Pi \cap k = 0$ . The same annihilation and torsion-free-image argument as before identifies it with the full  $A$ -torsion. For equal factors, the two products in the first coordinate of  $\beta_n$  are unchanged by permutation; only the order in the  $D$ -tensor coordinate changes. This proves the permutation assertion. With the empty-family convention, the case  $n = 0$  reads  $A \cong \Pi \oplus D$  with its usual multiplication action and zero torsion, separately from the positive-factor construction.  $\square$

**Theorem 13.20** (Exterior and symmetric powers [14]). *Let  $r = \text{rank } M$ .*

- (i) *For  $j \geq 2$ ,  $\bigwedge_A^j L(M) \cong \bigwedge_D^j M$  with  $A$  acting through  $\text{ct}$ : a free  $D$ -module of rank  $\binom{r}{j}$  with annihilator  $\Pi$  when nonzero.*

- (ii) For  $n \geq 1$ ,  $\text{Sym}_A^n L(M) \cong \Pi \oplus \text{Sym}_D^n M$  with the action (13.4) for the product map  $\mathfrak{m}_n : \text{Sym}_D^n M \rightarrow k$ ; the product map has image  $L(M^n)$  ( $M^n$  the  $D$ -span of  $n$ -fold products) and torsion kernel  $\ker \mathfrak{m}_n$ , of rank  $\binom{r+n-1}{n} - \text{rank}(M^n)$ , annihilated by  $\Pi$ .

*Proof.* (i) We first show that every wedge of at least two factors containing a tail vanishes. If the other factor is a nonzero constant  $m \in M$ , then  $u/m \in \Pi \subset A$  and  $u \wedge m = (u/m)(m \wedge m) = 0$ . If both factors  $u, v$  are tails, choose any  $0 \neq m_0 \in M$  and write them as  $(u/m_0)m_0$  and  $(v/m_0)m_0$ , with both scalars in  $A$ . Their wedge is again zero. A zero factor gives zero directly. Expanding the remaining factors into constants and tails proves the claim in every degree  $j \geq 2$ .

It follows that every wedge is represented by constants and that  $\Pi$  acts by zero: multiplying a constant wedge by a tail puts a tail in its first slot. The constant inclusion therefore induces an  $A$ -linear surjection from  $\bigwedge_D^j M$ , inflated through  $\text{ct}$ , onto  $\bigwedge_A^j L(M)$ . In the other direction use

$$(x_1, \dots, x_j) \mapsto \text{ct}(x_1) \wedge_D \cdots \wedge_D \text{ct}(x_j),$$

where  $\text{ct}(m + u) = m$  on  $L(M)$ . This map is alternating and  $A$ -multilinear into the inflated target, because  $\text{ct}(ax) = \text{ct}(a)\text{ct}(x)$ . It is inverse on constant wedges; the preceding expansion makes the other composition the identity. For a class target, every alternating multilinear map has the same tail-vanishing identities. It consequently factors uniquely through this finite free  $D$ -exterior carrier, which proves the class universal property directly.

If  $m_1, \dots, m_r$  is a basis of  $M$ , the wedges with strictly increasing indices form a basis of  $\bigwedge_D^j M$ . Their number is  $\binom{r}{j}$  [34, Section 10.13]; for  $j > r$  the module is zero. When it is nonzero, its action through  $D$  and its  $D$ -freeness make the annihilator exactly  $\Pi$ . The exclusion of  $j = 1$  matters: the first exterior power is  $L(M)$  itself and retains its tail.

(ii) Put  $T_n = M^{\otimes_D n}$  and let  $R_n$  be the  $D$ -submodule generated by all  $t - \sigma t$  for permutations  $\sigma$ . Then  $T_n/R_n = \text{Sym}_D^n M$ . By the preceding proposition, permutation differences in the  $A$ -tensor power have the form  $(0, t - \sigma t)$ . Their product in  $k$  is zero, so their  $A$ -multiples are

$$(d + h) \cdot (0, t - \sigma t) = (0, d(t - \sigma t)).$$

Thus the entire relation submodule is exactly  $0 \oplus R_n$ . In particular it imposes no relation on the tail coordinate, and the quotient is  $\Pi \oplus \text{Sym}_D^n M$  with the stated coupled action. In the class case this is the explicit carrier: the quotient by  $R_n$  is taken only in the set-sized finite free  $D$ -module  $T_n$ . The multilinear factorization just proved descends to this carrier precisely when the map is symmetric, which verifies its universal property without a quotient of a proper class.

Multiplication has image  $\Pi + M^n = L(M^n)$  and kernel  $(0, \ker \mathfrak{m}_n)$ , again by  $\Pi \cap k = 0$ . This kernel is killed by  $\Pi$  and is exactly the  $A$ -torsion, because the image lies in a field and  $\Pi$  contains a nonzero element. The monomials  $m_1^{a_1} \cdots m_r^{a_r}$  with  $a_i \geq 0$  and  $\sum_i a_i = n$  form a basis of the ordinary symmetric power; there are  $\binom{r+n-1}{n}$  of them. The product lattice  $M^n$  is nonzero and finite free over  $D$ . Tensoring its surjective product map with  $\text{Frac}(D)$  therefore gives the asserted kernel rank. For  $n = 1$  the kernel is zero and the formula reduces to  $L(M)$ ; for  $n = 0$  the usual symmetric power is  $A$ , outside the displayed range.  $\square$

For each  $r \geq 2$ , choose an integer coefficient lattice of rank  $r$ , for example the span of  $1, \theta, \dots, \theta^{r-1}$  for a transcendental real  $\theta$ , and scale its module by a positive monomial to obtain a rank-one ideal of  $\mathbf{Oz}$ . Its  $r$ th exterior power is nonzero by (i), while all its exterior powers above degree  $r$  vanish. Thus the assertion is a family of examples with arbitrarily large nonvanishing degree, not one finitely generated ideal with nonzero exterior powers in every degree. After passing to the fraction field, every exterior power of degree at least two vanishes, in agreement with fraction-field rank one: each element of the exterior carrier was already killed by a nonzero tail (14).



### 13.7 Rees equations

Write  $F = \text{Frac}(A)$ . The Rees algebra of the fractional module  $L(M)$  is

$$\mathcal{R}(L(M)) = A \oplus \bigoplus_{n \geq 1} L(M)^n T^n \subseteq F[T].$$

Here  $T$  records the grading; it is independent of the monomial  $X^e$  used to place  $L(M)$  inside  $A$ . The powers are finite sums of products. In fact  $L(M)L(N) = L(MN)$ : expanding products gives one inclusion, products of constants span  $MN$ , and any  $u \in \Pi$  is  $m_0(u/m_0)$  for a fixed nonzero  $m_0 \in M$ , with  $u/m_0 \in \Pi$ . Thus  $L(M)^n = M^n + \Pi$  for  $n \geq 1$ .

For  $e > 0$ , the degree- $n$  map  $uT^n \mapsto X^{ne}uT^n$ , with the identity in degree zero, identifies this algebra with the Rees algebra of  $J_e(M)$  [35, Tag 052P]. The exponents add under multiplication, so this is a graded  $A$ -algebra isomorphism. The module isomorphism  $L(M) \rightarrow J_e(M)$  likewise induces a graded isomorphism of symmetric algebras, compatible with their maps to the Rees algebras. An ideal is *of linear type* if  $\text{Sym} \rightarrow \mathcal{R}$  is an isomorphism. Its *first nonlinear degree* is the least degree of a nonzero element of this kernel, after the linear relations have been imposed (14). The polynomial variables below are presentation variables, distinct from the monomials in  $A$ .

**Theorem 13.21** (All nonlinear Rees equations are arithmetic [14]). *Let  $m_1, \dots, m_r$  be a basis of  $M$  and  $\mathfrak{q}_M = \ker(D[X_1, \dots, X_r] \rightarrow k[T], X_i \mapsto m_i T)$ . The constant-coefficient map  $D[X_1, \dots, X_r] \rightarrow \text{Sym}_A L(M)$  is injective, and under it  $\ker(\text{Sym}_A L(M) \rightarrow \mathcal{R}(L(M))) = \mathfrak{q}_M$ ; this is all the  $A$ -torsion of the symmetric algebra, it is killed by  $\Pi$ , and it is generated by any set of homogeneous generators of  $\mathfrak{q}_M$ , finitely many of which suffice. With  $\mathfrak{l}_M = (hX_j - \frac{m_j}{m_1} hX_1 : h \in \Pi, 2 \leq j \leq r) \subseteq A[X_1, \dots, X_r]$ ,*

$$\text{Sym}_A L(M) \cong A[\mathbf{X}]/\mathfrak{l}_M, \quad \mathcal{R}(L(M)) \cong A[\mathbf{X}]/(\mathfrak{l}_M + \mathfrak{q}_M A[\mathbf{X}]).$$

*Proof.* The linear relations of the generator map are those of Theorem 13.9(ii). They give the usual symmetric-algebra presentation [34, Tag 00DM]. We verify the presentation by a finite reduction, which also explains its meaning for the class rings in Setting (L).

Put  $S_n = \text{Sym}_D^n M$  and let  $\mathfrak{m}_n : S_n \rightarrow M^n$  be multiplication. By Theorem 13.20(ii), the positive-degree carrier is  $G_n = \Pi \oplus S_n$ ; put  $G_0 = A$ . Elements of the graded carrier are finite sequences of these components. For  $n, m \geq 1$ , its multiplication is

$$(u, t)(v, s) = (uv + u\mathfrak{m}_m(s) + v\mathfrak{m}_n(t), ts) \in G_{n+m}.$$

Degree zero acts by the coupled  $A$ -action already established. The identity  $\mathfrak{m}_{n+m}(ts) = \mathfrak{m}_n(t)\mathfrak{m}_m(s)$  verifies associativity by expansion, and the formula agrees with products of symmetric tensors. Thus it realizes the symmetric algebra on an explicit carrier, also when  $\Pi$  is a proper class.

Send  $X_i$  to  $(0, m_i) \in G_1$ . For a homogeneous polynomial  $P$  of positive degree  $n$ , split each coefficient into its ordinary constant and its tail, writing  $P = P_0 + P_\Pi$ . Modulo  $\mathfrak{l}_M$ , every tail monomial reduces to

$$hX_{i_1} \cdots X_{i_n} \equiv h \frac{m_{i_1} \cdots m_{i_n}}{m_1^n} X_1^n.$$

Indeed, replace one variable at a time using a defining linear relation; every new coefficient remains in  $\Pi$ , since  $\Pi$  is a  $k$ -vector space. Consequently  $P \equiv P_0 + (u/m_1^n)X_1^n$ , where  $u = P_\Pi(m_1, \dots, m_r) \in \Pi$ . Its image in  $G_n$  is  $(u, t_{P_0})$ , with  $t_{P_0}$  the element of  $S_n$  represented by  $P_0$ . The ordinary symmetric monomials form a  $D$ -basis, so this image vanishes exactly when  $P_0 = 0$  and  $u = 0$ . Every  $(u, t) \in G_n$  has such a representative. In degree zero the map is the identity of  $A$ . This proves the first

presentation and the injection of  $D[\mathbf{X}]$  degree by degree. For class rings, the displayed quotient notation denotes this explicit carrier and kernel description; no quotient of a proper class is needed.

In positive degree the map to the Rees algebra is  $(u, t) \mapsto u + \mathfrak{m}_n(t)$ . Since  $\Pi \cap k = 0$ , its kernel is exactly  $(0, \ker \mathfrak{m}_n)$ . Under the constant-polynomial injection this is the degree- $n$  part of  $\mathfrak{q}_M$ . The latter ideal is homogeneous because  $X_i \mapsto m_i T$  preserves degrees. It has no degree-zero or degree-one part, by injectivity of  $D \rightarrow k$  and independence of the basis of  $M$ .

Every kernel element is killed by  $\Pi$ . Conversely, any  $A$ -torsion element maps to zero because  $F[T]$  is torsion-free over  $A$ . A fixed nonzero element of  $\Pi$  kills the whole kernel, so the kernel is precisely all the  $A$ -torsion. Moreover, the action of  $A[\mathbf{X}]$  on it factors through the coefficientwise map  $\text{ct} : A[\mathbf{X}] \rightarrow D[\mathbf{X}]$ . Therefore any homogeneous ideal generators of  $\mathfrak{q}_M$  generate the Rees kernel as an ideal of the symmetric algebra. The ring  $D[\mathbf{X}]$  is Noetherian, so a finite subfamily suffices. Taking the inverse image of this kernel in  $A[\mathbf{X}]$  gives  $\mathfrak{l}_M + \mathfrak{q}_M A[\mathbf{X}]$ , as asserted.  $\square$

The nonlinear kernel is finitely generated *as an ideal of the symmetric algebra*. This is not finite generation as an  $A$ -module. In fact, if  $0 \neq q \in \mathfrak{q}_M$  is homogeneous, the elements  $qX_1^j$  remain nonzero in the symmetric algebra and have unbounded degrees. Finitely many  $A$ -module generators have bounded degrees, since  $A$  acts in degree zero, and hence cannot generate this kernel. For  $r \geq 2$ , the linear relation module is itself not finitely (set-) generated (Theorem 13.9). These are statements about different scalar actions, and neither gives a finite presentation of  $L(M)$  (14).

**Theorem 13.22** (The algebraic degree is the first nonlinear degree [14]). *Let  $\alpha \in k \setminus \text{Frac}(D)$ ,  $M = D + D\alpha$  and  $J = (X^e, \alpha X^e) \subseteq A$ .*

- (i) *If  $\alpha$  is transcendental over  $\text{Frac}(D)$ , then  $J$  is of linear type, although it is not flat and not finitely presented.*
- (ii) *If  $\alpha$  is algebraic of degree  $d$  (necessarily  $d \geq 2$ ), let  $f \in D[Z]$  be primitive irreducible with  $f(\alpha) = 0$  and  $F_f(X, Y) = X^d f(Y/X)$ . Then  $\mathcal{R}(J) \cong A[X, Y]/(hY - \alpha hX \ (h \in \Pi), F_f(X, Y))$ ; the kernel of  $\text{Sym}_A J \rightarrow \mathcal{R}(J)$  is generated by  $F_f$ , and the first nonlinear degree is exactly  $d$ .*

*Proof.* The ideal  $\mathfrak{q}_M$  is homogeneous. For a homogeneous  $G(X, Y)$  of degree  $n$ , evaluation gives  $G(T, \alpha T) = T^n G(1, \alpha)$ , so membership is equivalent to  $g(\alpha) = 0$ , where  $g(Z) = G(1, Z)$  has degree at most  $n$ . Dehomogenization is injective on this fixed homogeneous component: the coefficient of  $Z^i$  in  $g$  is the coefficient of  $X^{n-i}Y^i$  in  $G$ . Thus transcendence forces  $G = 0$  in every degree, and  $\mathfrak{q}_M = 0$ .

In the algebraic case, clearing denominators in the minimal polynomial and removing its content gives the primitive  $f$  of degree  $d$ . It need not be monic; no integrality assumption on  $\alpha$  is required. For nonzero  $G \in \mathfrak{q}_M$ , the minimal polynomial divides  $g$  over  $\text{Frac}(D)$ . Gauss's lemma over the unique factorization domain  $D$  gives  $g = fh$  with  $h \in D[Z]$ . In particular  $n \geq d$  and  $\deg h \leq n - d$ .

If  $f(Z) = \sum_{i=0}^d f_i Z^i$ , its homogenization means the polynomial  $F_f = \sum_{i=0}^d f_i X^{d-i} Y^i$ . Similarly  $H = X^{n-d} h(Y/X)$  is a polynomial, because  $\deg h \leq n - d$ . Homogenizing the product identity gives  $G = F_f H$  in  $D[X, Y]$ , without inverting  $X$ . Conversely  $F_f(1, \alpha) = 0$ , so  $\mathfrak{q}_M = (F_f)$ . There is no nonzero component below degree  $d$ , and  $F_f$  is nonzero in degree  $d$ .

By Theorem 13.21,  $F_f$  remains nonzero after imposing the linear relations: the constant-polynomial map into the symmetric algebra is injective. This proves the exact first nonlinear degree, as well as the presentation. Monomial scaling transfers the conclusion from  $L(M)$  to  $J$  and preserves grading. Finally  $1, \alpha$  form a  $D$ -basis, so nonflatness and failure of finite presentation follow from Proposition 13.10 and Theorem 13.9 in both cases.  $\square$

**Corollary 13.23** (A family of every degree [14; 17]). *For  $d \geq 2$  let  $J_d = (\omega, 2^{1/d}\omega) \subseteq \mathbf{Oz}$ . Then  $\mathcal{R}_{\mathbf{Oz}}(J_d) \cong \mathbf{Oz}[X, Y]/(hY - 2^{1/d}hX \ (h \in \Pi), \ Y^d - 2X^d)$ , there is no nonlinear equation in degrees  $2, \dots, d-1$ , and for  $n \geq 1$*

$$\begin{aligned} \mu_{\mathbf{Oz}}(J_d^n) &= \min(n+1, d), & \text{rank}_{\mathbb{Z}} \ker(\text{Sym}_{\mathbf{Oz}}^n J_d \rightarrow J_d^n) &= \max(0, n-d+1), \\ & & \text{rank}_{\mathbb{Z}} \ker(J_d^{\otimes n} \rightarrow J_d^n) &= 2^n - \min(n+1, d), \end{aligned}$$

where  $\mu$  is the least number of generators. All these kernels are killed by  $\Pi$ .

*Proof.* Let  $\alpha = 2^{1/d}$  be the positive ordinary real root. The polynomial  $Z^d - 2$  is Eisenstein at 2: its nonleading coefficients are divisible by 2 and its constant coefficient is not divisible by 4. Thus  $\alpha$  has degree exactly  $d$  over  $\mathbb{Q}$ , and [Theorem 13.22](#) gives the Rees presentation and first nonlinear degree.

For  $M = \mathbb{Z} + \mathbb{Z}\alpha$ , products of  $n$  basis elements give

$$M^n = \sum_{i=0}^n \mathbb{Z}\alpha^i = \bigoplus_{i=0}^{\min(n, d-1)} \mathbb{Z}\alpha^i.$$

The second equality holds as integer lattices, not just after tensoring with  $\mathbb{Q}$ : if  $i = qd + r$  with  $0 \leq r < d$ , then  $\alpha^i = 2^q\alpha^r$ , and the first  $d$  powers are independent over  $\mathbb{Q}$ . The product law gives  $J_d^n = \omega^n L(M^n)$ . Its least number of generators is therefore  $\text{rank}_{\mathbb{Z}} M^n = \min(n+1, d)$  by [Theorem 13.9\(i\)](#).

The ordinary symmetric and tensor powers of the rank-two lattice  $M$  have ranks  $n+1$  and  $2^n$ , respectively. Both product maps surject onto  $M^n$ . Their kernel ranks are consequently  $n+1 - \min(n+1, d) = \max(0, n-d+1)$  and  $2^n - \min(n+1, d)$ , by [Theorem 13.20\(ii\)](#) and [Proposition 13.19](#). The same results identify these ordinary kernels with the kernels over  $\mathbf{Oz}$  and show that  $\Pi$  kills them. The  $\mathbf{Oz}$ -linear isomorphism given by multiplication by  $\omega$  transfers these identifications to the tensor and symmetric powers of  $J_d$ . For  $n = 1$  both kernels vanish. When  $d = 2$ , the excluded range  $2, \dots, d-1$  is empty and the first nonlinear equation is already quadratic.  $\square$

For  $d = 5$  the two-generated ideal  $(\omega, 2^{1/5}\omega)$  has powers needing 2, 3, 4, 5, 5,  $\dots$  generators, a symmetric kernel of ranks 0, 0, 0, 0, 1, 2,  $\dots$ , and a nonzero antisymmetric tensor already in its tensor square: symmetric and tensor obstructions occur at different stages (14). Manuscript 17 proves the generator count  $\min(n+1, d)$  independently, from the product law  $J_g(M)J_h(N) = J_{g+h}(MN)$ , for every  $\alpha$  of degree  $d$  over  $\text{Frac}(D)$ , and adds that it is  $n+1$  for every  $n$  when  $\alpha$  is transcendental; the lattices  $M^n$  need not stabilize when their ranks do, and the count is not a reconstruction theorem (17). For instance, for  $\alpha = \sqrt{2}/2$  the lattice  $M = \mathbb{Z} + \mathbb{Z}\alpha$  has  $M^2 = \frac{1}{2}\mathbb{Z} + \mathbb{Z}\alpha$ . Every  $M^n$  has rank two for  $n \geq 1$ , but  $\alpha^{2^q} = 2^{-q}$  supplies unbounded denominators, so this increasing chain of lattices cannot eventually stabilize.

### 13.8 Homomorphisms, ideal classes and duals

For lattices put  $(N : M) = \{c \in k : cM \subseteq N\}$  and  $\mathfrak{o}(M) = (M : M)$ . More generally, for nonzero  $A$ -submodules  $U, V$  of  $F = \text{Frac}(A)$ , write  $(V : U) = \{q \in F : qU \subseteq V\}$ . The lattice colon is restricted to the ordinary coefficient field  $k$ ; the module colon permits a scalar in  $F$ . Below,  $U^* = \text{Hom}_A(U, A)$  is the algebraic  $A$ -dual, with its natural  $A$ -module structure.

**Theorem 13.24** (Scalar homomorphisms and isomorphism types [14; 17]). *Identifying a scalar with multiplication by it,  $\text{Hom}_A(L(M), L(N)) = \Pi + (N : M)$ , composition is multiplication, and  $L(M) \cong L(N)$  if and only if  $N = cM$  for some  $c \in k^\times$ . Hence  $\text{End}_A L(M) = \Pi + \mathfrak{o}(M)$  and  $\text{Aut}_A L(M) = \mathfrak{o}(M)^\times$ . The ring  $\mathfrak{o}(M)$  is finite free over  $D$  and integral over  $D$ ; in the real case  $\mathbb{Q}\mathfrak{o}(M) \subseteq \mathbb{R}$  is a number field of degree at most  $\text{rank } M$  (not necessarily totally real).*

*Proof.* We first justify the scalar description for arbitrary nonzero  $A$ -submodules  $U, V \subseteq F$ , including  $\Pi$  and  $B$ . Fix  $0 \neq u_0 \in U$  and an  $A$ -linear map  $\varphi : U \rightarrow V$ . For each  $u \in U$ , choose a nonzero  $a \in A$  clearing the two denominators, so  $au_0, au \in A$ . By  $A$ -linearity,

$$(au_0)\varphi(u) = \varphi((au_0)u) = \varphi((au)u_0) = (au)\varphi(u_0).$$

Cancelling  $au_0$  in  $F$  gives  $\varphi(u) = qu$ , where  $q = \varphi(u_0)/u_0$  is independent of  $u$ . Conversely every  $q \in (V : U)$  defines such a map, uniquely. This also includes the zero map, coded by  $q = 0$ . For class modules, each specified class map therefore has one scalar code; Hom is realized by this explicit class of scalars, not by a collection of arbitrary class functions. Addition and composition become scalar addition and multiplication.

Now suppose  $qL(M) \subseteq L(N)$  and choose  $0 \neq m_0 \in M$ . Since  $qm_0 \in L(N) \subseteq B$  and  $B$  is a  $k$ -algebra,  $q = (qm_0)/m_0 \in B$ . Write uniquely  $q = c + h$  with  $c \in k$ ,  $h \in \Pi$ . For  $m \in M$  and  $v \in \Pi$ ,

$$(c + h)(m + v) = cm + (cv + hm + hv),$$

and the parenthesized term lies in  $\Pi$ . Thus the required inclusion is equivalent to  $cM \subseteq N$ , proving the formula for Hom and, on taking  $N = M$ , for End.

If the map is an isomorphism, its inverse is multiplication by  $q^{-1}$ . Both scalars lie in  $B$ , so  $q \in B^\times = k^\times$ : the leading-degree argument of Lemma 2.4 applies both to the full class ring and to the finite-support model. Hence an isomorphism forces  $N = qM$ . Conversely any nonzero  $c \in k$  with  $N = cM$  gives an isomorphism because  $c\Pi = \Pi$ . The automorphisms are exactly the constants  $c$  with  $cM = M$ , equivalently  $c, c^{-1} \in \mathfrak{o}(M)$ .

The map  $c \mapsto cm_0$  embeds  $\mathfrak{o}(M)$  as a  $D$ -submodule of the finite free module  $M$ . Since  $D$  is a principal ideal domain, this proves finite generation and freeness. For  $c \in \mathfrak{o}(M)$ , multiplication by  $c$  has a matrix over  $D$  on a basis of  $M$ . Its monic characteristic polynomial annihilates this endomorphism by Cayley–Hamilton. Applying it to  $m_0 \neq 0$  shows that the same polynomial vanishes at  $c$  in  $k$ , proving integrality.

In the real case,  $E = \mathbb{Q}\mathfrak{o}(M)$  is a finite-dimensional  $\mathbb{Q}$ -algebra inside  $\mathbb{R}$ , and  $c \mapsto cm_0$  embeds it in  $\mathbb{Q}M$ , of dimension  $\text{rank } M$ . Multiplication by a nonzero element of  $E$  is an injective endomorphism of a finite-dimensional vector space, hence surjective; its inverse image of 1 is the multiplicative inverse. Thus  $E$  is a number field with the stated degree bound.  $\square$

Manuscript 17 proves the Hom, isomorphism and automorphism statements independently for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  (for the ideals  $J_\gamma(M)$ , with  $\text{Hom}_A(J_\gamma(M), J_\eta(N)) = \omega^{\eta-\gamma}((N : M) + \Pi)$ ), and sharpens the degree bound: the field spanned by  $\mathfrak{o}(M)$  has degree dividing  $\text{rank } M$  (Theorem 13.39).

The qualification “not necessarily totally real” is essential: for  $M = \mathbb{Z}[\sqrt[3]{2}] \subset \mathbb{R}$ , the multiplier ring is  $M$  itself, since  $1 \in M$  and  $M$  is a ring. Its rational span  $\mathbb{Q}(\sqrt[3]{2})$  has two nonreal embeddings, despite its chosen embedding in  $\mathbb{R}$ .

For the next theorem and example only, specialize to the real case  $D = \mathbb{Z}$ ,  $k = \mathbb{R}$ . Let  $K_0 \subset \mathbb{R}$  be a number field and  $\mathfrak{o} \subset K_0$  an order, that is, a subring whose additive group is a full integer lattice in  $K_0$ . An invertible fractional  $\mathfrak{o}$ -ideal  $P$  is a coefficient lattice: it is finitely generated over  $\mathfrak{o}$ , hence

over  $\mathbb{Z}$ , and torsion-free inside  $K_0$ . Its rational span is  $K_0$ , since  $p\mathfrak{o} \subseteq P$  for any  $0 \neq p \in P$ ; thus its integer rank is  $[K_0 : \mathbb{Q}]$ .

For these modules put  $U \odot V = (U \otimes_A V)/(A\text{-torsion})$ . By [Theorem 13.18](#), multiplication realizes  $L(M) \odot L(N)$  as  $L(MN)$ : its kernel is exactly the torsion and its image is  $L(MN)$ . In the class case this image is the explicit carrier intended by the quotient notation.

**Theorem 13.25** (Ideal classes survive torsion-free multiplication [14]).  *$P \mapsto L(P)$  distinguishes exactly the ideal classes of invertible fractional  $\mathfrak{o}$ -ideals, and under  $\odot$  these isomorphism types form a group isomorphic to  $\text{Pic}(\mathfrak{o})$  with identity  $[L(\mathfrak{o})]$ . If  $[K_0 : \mathbb{Q}] \geq 2$  this is not a subgroup of  $\text{Pic}(A)$  and its identity is not  $[A]$ .*

*Proof.* By [Theorem 13.24](#),  $L(P) \cong L(Q)$  exactly when  $Q = cP$  for some  $c \in \mathbb{R}^\times$ . Choose  $0 \neq p \in P$ . Then  $cp \in Q \subset K_0$ , so  $c = (cp)/p \in K_0^\times$ . This is precisely equality of the fractional-ideal classes in  $\text{Pic}(\mathfrak{o})$ ; the reverse implication is multiplication by  $c$ .

The product formula gives  $L(P) \odot L(Q) \cong L(PQ)$ . Replacing  $P, Q$  by  $aP, bQ$  replaces  $PQ$  by  $abPQ$ , so the operation is well-defined on their isomorphism types. Associativity and commutativity follow from fractional-ideal multiplication;  $\mathfrak{o}P = P$  gives the identity and  $PP^{-1} = \mathfrak{o}$  gives the inverse. These statements prove the asserted group isomorphism, including its identity.

Put  $d = [K_0 : \mathbb{Q}]$ . Each  $P$  has integer rank  $d$ , so [Theorem 13.9\(i\)](#) says that  $L(P)$  needs exactly  $d$  generators. If  $d \geq 2$ , the nonflatness obstruction of [Proposition 13.10](#) rules out invertibility over  $A$ : an invertible module is finitely generated projective, hence flat [81, Tag 0AFW]. For the class ring, invertibility would likewise give a finite dual basis and hence the equational criterion contradicted there. Moreover  $L(\mathfrak{o})$  needs  $d \geq 2$  generators whereas  $A = L(\mathbb{Z})$  needs one, so these modules are not isomorphic. Thus this realization of the ideal-class group does not take its values in  $\text{Pic}(A)$ .  $\square$

The group has its own idempotent identity inside the multiplicative semigroup of lattice-module types:  $L(\mathfrak{o}) \odot L(\mathfrak{o}) \cong L(\mathfrak{o})$ . When  $[K_0 : \mathbb{Q}] \geq 2$  this identity differs from the semigroup identity  $A$ . In the class setting, choose ordinary fractional-ideal representatives indexed by the set  $\text{Pic}(\mathfrak{o})$  and apply  $P \mapsto P + \Pi$ . The scalar criterion and product formula express the group law on this indexed family without forming a collection of all class modules (14).

**Example 13.26** (A class of order two [14]). In  $K_0 = \mathbb{Q}(\sqrt{10})$  with  $\mathfrak{o} = \mathbb{Z} + \mathbb{Z}\sqrt{10}$  let  $P = 2\mathbb{Z} + \mathbb{Z}\sqrt{10}$ . It is an  $\mathfrak{o}$ -ideal,  $P^2 = 4\mathbb{Z} + 2\sqrt{10}\mathbb{Z} + 10\mathbb{Z} = 2\mathfrak{o}$ , so  $P$  is invertible. It is not principal:  $P = c\mathfrak{o}$  with  $c = a + b\sqrt{10} \in P$  would give  $|a^2 - 10b^2| = [\mathfrak{o} : P] = 2$ , impossible modulo 5. Hence  $L(P) \not\cong L(\mathfrak{o})$  and  $L(P) \odot L(P) \cong L(\mathfrak{o})$ ; both have rank one and need two generators, and the tensor square of  $L(P)$  has torsion of rank  $4 - 2 = 2$ . As ideals,  $\omega L(P) = (2\omega, \sqrt{10}\omega)$  and  $\omega L(\mathfrak{o}) = (\omega, \sqrt{10}\omega)$ .

To spell out the arithmetic in the example, multiplication by  $\sqrt{10}$  sends the generators  $2, \sqrt{10}$  of  $P$  to  $2\sqrt{10}, 10 \in P$ , so  $P$  is an  $\mathfrak{o}$ -ideal. The constants 4 and 10 generate  $2\mathbb{Z}$ , giving  $P^2 = 2\mathfrak{o}$  and the explicit inverse  $P^{-1} = \frac{1}{2}P$ . If  $P = c\mathfrak{o}$ , then  $c \in P \subset \mathfrak{o}$ , since  $1 \in \mathfrak{o}$ . For  $c = a + b\sqrt{10}$ , multiplication on the basis  $1, \sqrt{10}$  has matrix  $\begin{pmatrix} a & 10b \\ b & a \end{pmatrix}$ . Its determinant has absolute value  $[\mathfrak{o} : c\mathfrak{o}] = [\mathfrak{o} : P] = 2$ . But  $a^2 - 10b^2 \equiv a^2 \pmod{5}$  cannot be 2 or  $-2$ , since the squares modulo 5 are 0, 1, 4. This proves nonprincipality and hence exact order two of the ideal class. The rank-one assertion is over  $\text{Frac}(A)$ ; the rank-two torsion assertion is over  $\mathbb{Z}$ .



We now return to both coefficient pairs  $(D, k)$  fixed above.

**Theorem 13.27** (Uniform duals [14]).  *$(A : \Pi) = B$  and  $(A : B) = \Pi$  (in both cases of this subsection). For a lattice  $M$  of rank at least two,  $L(M)^* = \text{Hom}_A(L(M), A) = \Pi$ ,  $L(M)^{**} = B$ , the canonical bidual map is the inclusion  $L(M) \hookrightarrow B$ , and  $B/L(M) \cong k/M$  with  $A$  acting through  $\text{ct}$ . Moreover  $\Pi^* = B$ ,  $B^* = \Pi$ ,  $\Pi^{**} = \Pi$ ,  $B^{**} = B$ , and  $J_e(M)^* = X^{-e}\Pi$ ,  $J_e(M)^{**} = X^e B$ .*

*Proof.* The scalar argument in Theorem 13.24 identifies the dual of any nonzero  $A$ -submodule  $U \subseteq F$  with  $(A : U)$ . First suppose  $q\Pi \subseteq A$ . Since  $X^1 \in \Pi$ ,  $qX^1 \in A$ , and  $q = X^{-1}(qX^1)$  has the corresponding shifted normal form. In the model  $T_{\mathbb{Q}}$  this proves, before inspecting exponents, that  $q$  has finite rational support; an arbitrary rational function need not have that property. If a nonzero coefficient of  $q$  occurred at a negative exponent  $\gamma$ , choose  $0 < \delta < -\gamma$  in the same exponent group (for example  $\delta = -\gamma/2$ ). The coefficient at  $\gamma + \delta < 0$  of  $qX^\delta$  would be unchanged and nonzero, contradicting  $qX^\delta \in A$ . Thus  $q$  has only nonnegative exponents and belongs to  $B$ . Conversely  $B\Pi \subseteq \Pi$  because  $\Pi$  is an ideal of  $B$ . This proves  $(A : \Pi) = B$ .

If  $qB \subseteq A$ , then  $q = q \cdot 1 \in A$ , so write  $q = d + u$  with  $d \in D$  and  $u \in \Pi$ . Testing all constants  $c \in k \subset B$  gives  $dc \in D$ . If  $d \neq 0$ , taking  $c = 1/(2d)$  would give  $1/2 \in D$ , impossible for either  $D = \mathbb{Z}$  or  $D = \mathbb{Z}[i]$ . Hence  $d = 0$  and  $q \in \Pi$ . The reverse inclusion is again  $\Pi B \subseteq \Pi$ , proving  $(A : B) = \Pi$ . In particular  $\Pi^* = B$  and  $B^* = \Pi$ .

For a lattice, Theorem 13.24 gives  $L(M)^* = \Pi + (D : M)$ . A nonzero  $c \in (D : M)$  would give an injective  $D$ -linear map  $M \rightarrow D$ , contradicting  $\text{rank}_D M \geq 2$  after tensoring with  $\text{Frac}(D)$ . Thus  $(D : M) = 0$ , the first dual is  $\Pi$ , and the second is  $B$ . Under these scalar identifications, evaluation at  $x \in L(M)$  sends the functional with scalar  $u \in \Pi$  to  $ux$ . Its scalar code in  $\text{Hom}_A(\Pi, A) = B$  is therefore  $x$  itself. This proves that the canonical bidual map is the stated inclusion. The same evaluation calculation for  $\Pi$  and  $B$  gives their displayed bidual identities and proves both modules reflexive.

The surjective map  $B \rightarrow k/M$  given by  $c + h \mapsto c + M$  has kernel  $L(M)$ . For  $a = d + v \in A$ , its value on  $a(c + h)$  is  $dc + M$ , so the  $A$ -action on  $k/M$  is through  $\text{ct} : A \rightarrow D$ . This is an additive module quotient; no ring structure on  $k/M$  is asserted. In the class case the ordinary set-sized module  $k/M$  explicitly realizes the displayed quotient.

Finally, for  $0 \neq t \in F$  and  $U \subseteq F$ ,  $(tU)^* = t^{-1}U^*$ , since  $q(tU) \subseteq A$  is equivalent to  $qt \in U^*$ . Applying this twice with  $t = X^e$  gives  $J_e(M)^* = X^{-e}\Pi$  and  $J_e(M)^{**} = X^e B$ .  $\square$

The rank restriction matters: if  $M = Dm$  has rank one, then  $L(M) = mA$ , its dual is  $m^{-1}A$ , and its bidual is  $mA$ . For rank at least two, the two nonisomorphic modules of Example 13.26 have the same dual and reflexive hull. Their difference is retained by the coefficient lattice but lost under double dualization (14).

**Theorem 13.28** (Common-scale constant matrices [14; 17]). *Let  $C \in k^{m \times n}$ ,  $e > 0$ ,  $W = \ker_k C$ ,  $q = \dim_k W$ ,  $W_D = W \cap D^n$  and  $\ell = \text{rank}_D W_D$ . Then  $\ker_A(X^e C) \cong A^\ell \oplus \Pi^{q-\ell}$ . The image of  $X^e C : A^n \rightarrow A^m$  is finitely presented if and only if  $\ell = q$ , that is, if and only if  $W$  is defined over  $\text{Frac}(D)$ , and it is then free of rank  $n - q$ .*

*Proof.* All entries of  $X^e C$  belong to  $\Pi \subseteq A$ , and the nonzero common scalar  $X^e$  does not change the kernel. Decompose  $x \in A^n$  uniquely as  $x = d + u$  with  $d \in D^n$  and  $u \in \Pi^n$ . Because  $Cd$  is constant and  $Cu$  is a tail vector,  $Cx = 0$  is equivalent to the two equations  $Cd = 0$  and  $Cu = 0$ .

The submodule  $W_D \subseteq D^n$  is saturated: if  $0 \neq a \in D$  and  $av \in W_D$ , then  $aCv = 0$  in  $k^m$ , so  $Cv = 0$ . Hence  $D^n/W_D$  is finitely generated and torsion-free over the principal ideal domain  $D$ ,



and is free. The quotient map therefore splits. Choose a  $D$ -basis  $v_1, \dots, v_\ell$  of  $W_D$  and extend it to a  $D$ -basis of  $D^n$ . In particular these first vectors are  $k$ -independent; extend them instead to a  $k$ -basis  $v_1, \dots, v_q$  of  $W$ .

Gaussian elimination over  $k$  gives

$$\{u \in \Pi^n : Cu = 0\} = \left\{ \sum_{j=1}^q h_j v_j : h_j \in \Pi \right\}.$$

Indeed, the coordinates in this kernel basis are fixed  $k$ -linear combinations of the coordinates of  $u$ , and hence remain in the  $k$ -vector space  $\Pi$ . Independence remains valid over  $F$ , as a nonzero  $q$ -rowed minor over  $k$  remains nonzero in  $F$ . Combining constants and tails now gives the explicit  $A$ -linear isomorphism

$$A^\ell \oplus \Pi^{q-\ell} \longrightarrow \ker_A(X^e C), \quad (a_1, \dots, a_\ell, h_{\ell+1}, \dots, h_q) \longmapsto \sum_{j \leq \ell} a_j v_j + \sum_{j > \ell} h_j v_j.$$

The decomposition above proves surjectivity and the independence proves injectivity. The construction uses only finite matrices and also defines the class-module isomorphism.

If  $q > \ell$ , projection onto one tail coordinate shows that the kernel is not finitely generated (and not set-generated in the class case). This excludes every finite presentation of the image, not just the displayed presentation [82, Tag 0517]. To see this, a hypothetical finite presentation with finitely generated relation module  $R$  and free module  $A^a$  would, by the finite-free pullback argument of Schanuel's lemma, give

$$\ker_A(X^e C) \oplus A^a \cong R \oplus A^n.$$

The right side is finitely generated, so its direct summand  $\ker_A(X^e C)$  would be too. Only finitely many lifts of free basis vectors are used, so the argument applies to the class presentation as well.

If  $q = \ell$ , use the chosen extension of the  $D$ -basis of  $W_D$  to a basis of  $D^n$ . After this invertible change of columns, the first  $q$  columns of  $C$  are zero and the other  $n - q$  are independent over  $k$ , hence over  $F$ . Their multiples by  $X^e$  form an  $A$ -basis of the image. Thus the image is free of rank  $n - q$  and in particular finitely presented.

Finally put  $F_0 = \text{Frac}(D)$ . Clearing the finitely many denominators of a vector gives  $F_0 W_D = W \cap F_0^n$ . It follows that  $\ell$  is the  $F_0$ -dimension of this intersection. Equality  $\ell = q$  is therefore equivalent to  $W$  having a basis over  $k$  consisting of vectors in  $F_0^n$ , which is the asserted meaning of “defined over  $F_0$ ”. The formulas include  $q = 0$  (zero kernel) and  $C = 0$  (kernel  $A^n$  and zero image).  $\square$

The matrix theorem concerns constant matrices at one common monomial scale; several support levels need more analysis (14, 17). Manuscript 17 proves it independently for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  and adds that  $\ell = \dim_{\text{Frac } D}(W \cap \text{Frac}(D)^n)$  (a nonzero vector of  $W \cap \text{Frac}(D)^n$  has a common denominator in  $D$ ), so that the kernel is set-generated exactly when  $W$  is spanned by vectors over  $\text{Frac}(D)$ ; its computation of  $q$  and  $\ell$  and the relation counts over the models are in Section 13.9. Its companion, 14's criterion that a point  $[r_1 : \dots : r_n] \in \mathbb{P}^{n-1}(k)$  has a unimodular representative in  $A^n$  if and only if it is  $\text{Frac}(D)$ -rational, is in the real case the rationality theorem for primitive constant directions of the Diophantine report [47], which 14 does not credit (it credits only that report's example  $[1 : \sqrt{2}]$ , to a companion manuscript); the Gaussian and principal-ideal form is printed there with that theorem, not here.

**The Gaussian case and the reach of the results (14).** The tensor, power, Rees, scalar-homomorphism, dual and matrix results above treat both  $(D, k) = (\mathbb{Z}, \mathbb{R})$  and  $(\mathbb{Z}[i], \mathbb{C})$ . Their proofs use the stated properties of these rings, including the principal-ideal property of  $D$ , the common-factor property of  $\Pi$ , and, for the scalar and dual descriptions, the support and unit calculations in  $B$ . They are not consequences of the principal-ideal hypothesis alone. Over  $\mathbf{Oz}[i] = \mathbb{Z}[i] + \Pi_{\mathbb{C}}$ , ranks are  $\mathbb{Z}[i]$ -ranks, multiplication kernels are killed by  $\Pi_{\mathbb{C}}$ , and the degree in [Theorem 13.22](#) is taken over  $\mathbb{Q}(i)$ .

For real algebraic  $\alpha$ , put  $K_0 = \mathbb{Q}(\alpha) \subset \mathbb{R}$ . Since  $i \notin K_0$ , the tower formula gives  $[K_0(i) : \mathbb{Q}] = 2[K_0 : \mathbb{Q}] = 2[K_0(i) : \mathbb{Q}(i)]$ . Thus the degree of  $\alpha$  over  $\mathbb{Q}(i)$  equals its degree over  $\mathbb{Q}$ . Consequently [Corollary 13.23](#) holds over  $\mathbf{Oz}[i]$  with the same ranks over  $\mathbb{Z}[i]$ ; in particular  $(\omega, \sqrt{2}\omega)$  has the single nonlinear equation  $Y^2 - 2X^2$ . The ideal-class argument also has a Gaussian version: take an order over  $\mathbb{Z}[i]$  in a finite extension of  $\mathbb{Q}(i)$  inside  $\mathbb{C}$ , and replace its integer rank by its  $\mathbb{Z}[i]$ -rank. The displayed  $\mathbb{Q}(\sqrt{10})$  example above remains a real example; no assertion that extending coefficients preserves its nontrivial ideal class is needed.

Complex conjugation transports the construction for  $M$  to that for  $\overline{M}$ . These results concern the omnific *rings*: after extending scalars to their fraction fields, the modules become vector spaces and the arithmetic torsion vanishes. They do not classify arbitrary finitely generated omnific ideals, whose generators may lie at several support levels.

### 13.9 Relations, Tor and moduli of lattice ideals

This subsection is manuscript 17's new material; its results that repeat 12 and 14 are printed above, with 17 added to their sources. As in [Section 13.6](#),  $(D, k)$  is  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , so that  $\text{Frac } D$  is  $\mathbb{Q}$  or  $\mathbb{Q}(i)$ ; a lattice is a coefficient lattice  $M \subset k$  of Setting (L), of rank  $r$ , and  $J_g(M) = X^g L(M)$ . The class ring is  $A = \mathcal{A}_{D,k}$ , that is  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$ . 17 works in class theory with global choice; over the class ring,  $\text{Hom}$  means the class of scalar codes of [Theorem 13.24](#), and ordinary Tor groups, flat resolutions and weak dimensions are taken over set-sized rings only (17). A module  $P$  has a *set-presentation* if there is an exact sequence  $A^{(T)} \rightarrow A^{(S)} \rightarrow P \rightarrow 0$  of finite-support free modules on sets  $S, T$ ; equivalently, some surjection from a set-indexed free module onto  $P$  has a set-generated kernel. All linear combinations and equational factorizations below are finite. The adjective “set-indexed” restricts the basis, not the coefficients: when  $A$  is a class ring,  $A^{(S)}$  is the class of finitely supported coefficient vectors on the set  $S$ .

**Theorem 13.29** (No presentation by sets [17]). *Let  $A = \mathcal{A}_{D,k}$ , let  $M$  be a lattice of rank  $r \geq 2$  and  $g > 0$ .*

- (i)  *$J_g(M)$  and  $L(M)$  have no set-presentation. In particular, for  $\alpha \in k \setminus \text{Frac } D$  and  $\gamma > 0$  the two-generated ideal  $(\omega^\gamma, \alpha\omega^\gamma)$  has none, and the intersection  $\omega^\gamma A \cap \alpha\omega^\gamma A = \omega^\gamma \Pi_k$  of two principal ideals has no set of generators. Thus  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  are neither coherent nor finite-conductor domains, in the strong sense that an intersection of two principal ideals need not be generated by a set.*
- (ii)  *$J_g(M)$  fails the equational criterion of flatness:  $a_j = X^g m_j$  and  $x = (X^g m_2, -X^g m_1, 0, \dots, 0)$  satisfy  $\sum_j a_j x_j = 0$ , but there is no factorization  $x_j = \sum_t b_{jt} y_t$  with  $y_t \in J_g(M)$ ,  $b_{jt} \in A$  and  $\sum_j a_j b_{jt} = 0$  for every  $t$ .*

*Proof.* (i) Multiplication by  $X^g$  transfers the presentation in [Theorem 13.9\(ii\)](#) to

$$0 \longrightarrow K_M \longrightarrow A^r \xrightarrow{\pi} J \longrightarrow 0, \quad K_M \cong \Pi_k^{r-1}, \quad J = J_g(M).$$

Suppose  $\pi' : A^{(S)} \rightarrow J$  is another surjection, with  $S$  a set and  $H = \ker \pi'$  generated by a set. Form the explicit class pullback

$$P = \{(x, y) \in A^r \oplus A^{(S)} : \pi(x) = \pi'(y)\}.$$

Both projections split, as follows. Every  $z \in J$  has a unique expression  $z = X^g(\sum_j d_j m_j + u)$  with  $d_j \in D$  and  $u \in \Pi_k$ . It has the specified preimage  $\sigma(z) = (d_1 + u/m_1, d_2, \dots, d_r)$  under  $\pi$ . This  $\sigma$  is a section as a map of classes; it is not asserted to be  $A$ -linear. Instead assign to each basis vector  $e_s$  the pair  $(\sigma(\pi'(e_s)), e_s)$  and extend by finite  $A$ -linear combinations. The resulting map is an  $A$ -linear section of  $P \rightarrow A^{(S)}$ , whose kernel is  $K_M \oplus 0$ . For the projection  $P \rightarrow A^r$ , choose preimages under  $\pi'$  of the images of the finitely many standard basis vectors of  $A^r$ . Extending those choices linearly gives a section, with kernel  $0 \oplus H$ .

These splittings are the concrete Schanuel isomorphisms

$$K_M \oplus A^{(S)} \cong P \cong H \oplus A^r.$$

No choice indexed by all elements of a proper class is involved. The right side is generated by a set, and an image of a set-generated module is set-generated. Since  $r - 1 \geq 1$ , project the left side onto a copy of  $\Pi_k$ . This contradicts [Proposition 3.3\(iii\)](#), proving that no set-presentation exists. The same holds for  $L(M) \cong J$ .

For the stated two-generated ideal take  $M = D + D\alpha$ , whose rank is two because  $\alpha \notin \text{Frac}(D)$ . The intersection is  $\omega^\gamma \Pi_k$  by [Theorem 13.9\(iii\)](#), and multiplication by  $\omega^\gamma$  identifies this module with  $\Pi_k$ . Thus it too has no set of generators. In particular the ideal is not finitely presented and the intersection is not finitely generated, excluding coherence and the finite-conductor property, respectively.

(ii) With  $a_j = X^g m_j$ , the vector in the statement is  $(a_2, -a_1, 0, \dots, 0) \in J^r$ , and its scalar product with  $(a_1, \dots, a_r)$  is zero by commutativity. In a hypothetical finite equational factorization, every column  $(b_{jt})_j$  lies in  $K_M$ . All its coordinates therefore lie in  $\Pi_k$ , by [Theorem 13.9\(ii\)](#). Consequently each  $x_j = \sum_t b_{jt} y_t$  belongs to

$$\Pi_k J = X^g \Pi_k L(M) = X^g \Pi_k.$$

But  $x_1/X^g = m_2$  is a nonzero ordinary constant, and  $k \cap \Pi_k = 0$ . This is the required contradiction. The argument uses only finite equations, so its class-ring assertion requires no class category of tensor products.  $\square$

The kernel shape  $\Pi^{r-1}$ , flat dimension one and the intersection mechanism are classical for  $D + M$  rings (Dobbs [13], Dobbs–Papick [14]); 17 presents neither as new and does not infer any factorization property from noncoherence. Over a set-sized ring, (ii) is the equational criterion [29, Lemma 10.39.11]; over the class ring it is stated without a category of class tensor products (17).

**Lemma 13.30** (The lexicographic models inside **Oz** [17]; [merge] for  $\kappa = \aleph_0$ ). *Let  $\kappa$  be an infinite regular cardinal and  $\Gamma_\kappa = \bigoplus_{\alpha < \kappa} \mathbb{Q}e_\alpha$ , ordered by the sign of the coefficient at the least index where it is nonzero (the group of [Proposition 11.22](#)).*

- (i)  $e_\alpha \mapsto \omega^{-\alpha}$  extends to an injective order-preserving homomorphism  $\iota : \Gamma_\kappa \rightarrow (\mathbf{No}, +)$ .
- (ii) Every subset of  $\Gamma_\kappa^{>0}$  with fewer than  $\kappa$  elements has a positive strict lower bound; the  $e_\alpha$  are coinital in  $\Gamma_\kappa^{>0}$ , whose coinitality is exactly  $\kappa$ .
- (iii)  $X^g \mapsto \omega^{\iota(g)}$  embeds  $R_\kappa^{11} := R_{\Gamma_\kappa, < \kappa}^{11}$ , with  $(D, K) = (\mathbb{Z}, \mathbb{R})$ , respectively  $(\mathbb{Z}[i], \mathbb{C})$ , in **Oz**, respectively **Oz**[ $i$ ], as the ring of normal forms with constant term in  $D$  and fewer than  $\kappa$  further exponents, all in  $\iota(\Gamma_\kappa)_{>0}$ . We write  $\Pi_\kappa^{11}$  for its positive-support ideal.

*Proof.* (i) Define  $\iota(\sum_{\alpha} q_{\alpha} e_{\alpha}) = \sum_{\alpha} q_{\alpha} \omega^{-\alpha}$ , a finite sum. Here  $-\alpha$  is the additive negative of the ordinal regarded as a surreal number; differences such as  $\beta - \alpha$  use surreal addition, not ordinal subtraction. The definition is additive and  $\mathbb{Q}$ -linear, and  $\iota(e_0) = 1$ . If  $\alpha$  is the least nonzero index, divide the image by  $\omega^{-\alpha} > 0$ . The result is  $q_{\alpha}$  plus finitely many terms  $q_{\beta} \omega^{\alpha-\beta}$  with  $\beta > \alpha$ . Each latter term is infinitesimal, so the sign is the sign of the nonzero rational  $q_{\alpha}$ . Thus every nonzero vector keeps its lexicographic sign, proving injectivity and preservation of the order.

(ii) Let  $S \subseteq \Gamma_{\kappa}^{>0}$  have size less than  $\kappa$ . The union  $U$  of its finite sets of nonzero coordinate indices has size less than  $\kappa$ . Regularity gives  $\beta = \sup\{\alpha + 1 : \alpha \in U\} < \kappa$ , with  $\beta = 0$  if  $S$  is empty. Every element of  $S$  has its leading positive coordinate before  $\beta$ , so  $0 < e_{\beta} < s$  for every  $s \in S$ . The same argument applied to a singleton shows that the family  $(e_{\alpha})_{\alpha < \kappa}$  is coinital. Conversely a family  $S$  of size less than  $\kappa$  cannot be coinital: none of its members lies below its positive strict lower bound  $e_{\beta}$ . This proves the exact coinitality.

(iii) For a series  $f = \sum_g c_g X^g$ , set  $\Phi(f) = \sum_g c_g \omega^{\iota(g)}$ . An order embedding preserves reverse well-ordering of the support, and injectivity preserves its cardinality. Thus this is an allowed set-supported Conway normal form. Uniqueness of coefficients gives injectivity; coefficientwise addition and the finite convolution formula for multiplication show that  $\Phi$  is a unital ring map, since  $\iota(g + h) = \iota(g) + \iota(h)$  (Sections 2.2 and 2.3). For supports  $S, T$ , the new supports lie in  $S \cup T$  and  $S + T$ , respectively. If  $|S|, |T| < \kappa$ , then both have size less than the infinite cardinal  $\kappa$ ; in the countable case both supports are finite. Regularity was needed in (ii), not in this two-support estimate.

The coefficient at zero stays in  $D$ , and every other exponent stays positive. Conversely a normal form with the stated support pulls back uniquely along  $\iota$  and has the same support bound, giving the asserted image. For complex coefficients this argument is applied to the real and imaginary parts; their finite union of supports still satisfies the bound.  $\square$

There are two levels of monomials here:  $e_{\alpha}$  maps to the surreal exponent  $\omega^{-\alpha}$ , whereas the ring monomial  $X^{e_{\alpha}}$  maps to  $\omega^{\omega^{-\alpha}}$ . In particular  $X^{e_0}$  maps to  $\omega$ , not to 1.

11 does not embed its ring in **No** (Section 11.5); (iii) places the ring of Propositions 11.22 and 11.23 inside **Oz**. For  $\kappa = \aleph_0$  this notation extends the ring definition to finite-support series; it does not assert that the corresponding finite-support ambient collection is a field. Like the finite-support cores of 12 and 14, this ring lies outside Theorem 5.1: the coefficient sum is a unital map to  $K$  detecting its tail, by the proof of Proposition 13.2(ii). In both cases  $R_{\kappa}^{11}$  has the properties listed in Setting (L) of Section 13.4, with  $B = K + \Pi_{\kappa}^{11}$  and  $\Pi_{\kappa}^{11}$  not finitely generated, so Theorem 13.9(i),(ii) and Proposition 13.10 hold for it with the same proofs (17 proves them for its models).

**Theorem 13.31** (Exact relation counts [17]; [merge] for  $\kappa = \aleph_0$ ). *Let  $\kappa$  be an infinite regular cardinal,  $A = R_{\kappa}^{11}$  and  $\Pi = \Pi_{\kappa}^{11}$ .*

- (i)  $\Pi^2 = \Pi = \bigcup_{\delta \in \Gamma_{\kappa}^{>0}} X^{\delta} A$ , a directed union of free modules of rank one;  $\Pi$  is flat, and  $\text{gen}_A(\Pi) = \kappa$ .
- (ii) For  $C \in k^{m \times n}$  and  $e \in \Gamma_{\kappa}^{>0}$ , with  $q$  and  $\ell$  as in Theorem 13.28,  $\ker_A(X^e C) \cong A^{\ell} \oplus \Pi^{q-\ell}$ ; it needs exactly  $\ell$  generators if  $q = \ell$  and exactly  $\kappa$  if  $q > \ell$ .
- (iii) For a lattice  $M$  of rank  $r \geq 2$  and  $g \in \Gamma_{\kappa}^{>0}$ , the kernel of every surjection from a finite free  $A$ -module onto  $J_g(M)$  needs exactly  $\kappa$  generators; for the natural surjection it is  $\Pi^{r-1}$ .

*Proof.* (i) Let  $0 \neq f \in \Pi$ . Its support has fewer than  $\kappa$  positive exponents, so Lemma 13.30(ii) supplies a positive strict lower bound  $\delta$ . Translation by  $-\delta$  preserves reverse well-ordering and

support cardinality, and every translated exponent is still positive. Hence  $X^{-\delta}f \in \Pi$  and  $f = X^\delta(X^{-\delta}f) \in \Pi^2$ . The reverse inclusion follows because  $\Pi$  is an ideal, proving idempotence. This factorization also puts  $f$  in  $X^\delta A$ ; the zero element belongs to every member of the union.

If  $0 < \delta' < \delta$ , then  $X^\delta = X^{\delta'}X^{\delta-\delta'}$ , with the last factor in  $A$ . Thus  $X^\delta A \subseteq X^{\delta'} A$ . For any two indices choose a positive index smaller than both; the union is directed by inclusion. Each principal ideal is free of rank one, since its generator is nonzero in a domain. Its directed union  $\Pi$  is therefore flat [29, Lemma 10.39.3].

Choose  $e_\alpha < \delta$  in the preceding factorization. Then  $f \in X^{e_\alpha} A$ , so the  $\kappa$  monomials  $X^{e_\alpha}$  generate  $\Pi$ . For the reverse bound, suppose a family  $\{f_i : i \in I\}$  with  $|I| < \kappa$  generated it. Regularity implies that  $\bigcup_i \text{supp}(f_i)$  still has size less than  $\kappa$ . Choose a positive strict lower bound  $\delta$  for this union. Each  $f_i$  then lies in  $X^\delta A$ , so its generated ideal is contained there. However  $X^{\delta/2} \in \Pi$  does not belong to  $X^\delta A$ : cancelling  $X^\delta$  would put the negative monomial  $X^{-\delta/2}$  in  $A$ . This proves  $\text{gen}_A \Pi = \kappa$ . For  $\kappa = \aleph_0$ , the regularity step is just the fact that a finite union of finite supports is finite.

(ii) The kernel construction in Theorem 13.28 uses the additive decomposition  $A = D \oplus \Pi$ , stability of  $\Pi$  under  $k$ -scalars, and the principal-ideal property of  $D$ . These hold in this model, and give the same isomorphism  $A^\ell \oplus \Pi^{q-\ell}$ . If  $q > \ell$ , projection onto one copy of  $\Pi$  forces at least  $\kappa$  generators. Conversely, the union of generating families for its finitely many summands has cardinality at most  $\kappa$ , since  $\kappa$  is infinite. If  $q = \ell$ , the standard basis gives  $\ell$  generators and tensoring with  $\text{Frac}(A)$  shows that fewer cannot suffice. This includes  $\ell = 0$ , when the kernel is zero.

(iii) The natural kernel is  $\Pi^{r-1}$  by Theorem 13.9(ii). For any surjection  $A^m \rightarrow J_g(M)$  with kernel  $H$ , the finite-free pullback argument gives

$$H \oplus A^r \cong \Pi^{r-1} \oplus A^m.$$

If  $H$  had  $\lambda < \kappa$  generators, then the left side would have at most  $\lambda + r < \kappa$  generators. Projecting the right side onto a copy of  $\Pi$  contradicts (i), since  $r - 1 \geq 1$ . Conversely the right side has  $\kappa$  generators; transport them to the left side and project onto  $H$ . This gives a generating family of size at most  $\kappa$ , proving equality for every such surjection.  $\square$

For uncountable  $\kappa$ , (i) is the case  $R_{\Gamma, < \kappa}^{11}$  of Theorem 12.2(i) together with the generator count recorded in (H3). Statement (iii) concerns numbers of generators; for uncountable  $\kappa$ , the cardinality thresholds  $\kappa^{< \kappa}$  of Proposition 11.23 are a different invariant of the same ring. That threshold assertion is not extended to  $\aleph_0$ : the coefficient-sum map above already detects its finite-support tail. In the class ring the relation module needs more than a set of generators (Theorem 13.29); a set-sized model measures the same obstruction by the regular cardinal  $\kappa$  (17).

*Remark 13.32* (The countable case [merge]). 17 states Lemma 13.30 and Theorem 13.31 for uncountable regular  $\kappa$ , but the proofs above also apply to  $\aleph_0$ . Put  $A = R_{\aleph_0}^{11}$ . For  $r \geq 2$  every  $L(M)$  is countably presented but not finitely presented. More precisely, every surjection from a finite free  $A$ -module onto  $L(M)$  has a kernel needing exactly  $\aleph_0$  generators. This is a finite generating map with infinitely many relations, not a finite presentation. The natural map has kernel  $\Pi^{r-1}$ , so choosing its countable generating family gives a presentation  $A^{(\mathbb{N})} \rightarrow A^r \rightarrow L(M) \rightarrow 0$ .

The projective dimensions can also be read directly. With  $x_n = X^{e_n}$  one has  $\Pi = \bigcup_n x_n A$  and  $x_n = x_{n+1} X^{e_n - e_{n+1}}$ . The transition factor lies in  $A$ , since  $e_n > e_{n+1}$ . Thus the telescope of Proposition 13.12(i) gives  $\text{pd}_A \Pi \leq 1$ . Nonfinite generation and Lemma 12.1 rule out projectivity, so  $\text{pd}_A \Pi = 1$ . The exact sequence  $0 \rightarrow \Pi \rightarrow A \rightarrow D \rightarrow 0$  then gives  $\text{pd}_A D = 2$ : a value at most one would make its first syzygy  $\Pi$  projective. Likewise the natural presentation with kernel  $\Pi^{r-1}$  gives  $\text{pd}_A L(M) = \text{pd}_A J_g(M) = 2$ , since a smaller value would make the direct summand  $\Pi$  projective.

Finally  $0 \rightarrow J_g(M) \rightarrow A \rightarrow A/J_g(M) \rightarrow 0$  gives  $\text{pd}_A(A/J_g(M)) = 3$  by the same dimension shift. For uncountable  $\kappa$ , the tail needs  $\kappa$  generators and these projective dimensions remain open in this report ([Questions 18.3](#) and [18.15](#)).

**Proposition 13.33** (Flat dimensions over the models [17]). *Let  $\kappa$  be infinite and regular,  $A = R_\kappa^{11}$  and  $M$  a lattice of rank  $r$ . Then  $\text{Tor}_j^A(J_g(M), -) = 0$  for  $j \geq 2$ . If  $r \geq 2$ , then  $\text{fd}_A J_g(M) = 1$  and  $\text{fd}_A(A/J_g(M)) = 2$ ; hence the weak global dimension of  $A$  is at least 2.*

*Proof.* Write  $J = J_g(M)$ . The natural presentation  $0 \rightarrow \Pi^{r-1} \rightarrow A^r \rightarrow J \rightarrow 0$  is a flat resolution by [Theorem 13.31](#)(i). It gives  $\text{fd}_A J \leq 1$  and  $\text{Tor}_j^A(J, N) = 0$  for every  $N$  and  $j \geq 2$ . Splicing with  $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$  gives  $\text{fd}_A(A/J) \leq 2$ .

For  $r \geq 2$  we exhibit a nonzero obstruction, without using the general coefficient-Tor formula below. Put  $a_j = X^g m_j$ . In  $J \otimes_A J$  consider

$$\tau = a_1 \otimes a_2 - a_2 \otimes a_1.$$

Its product is zero. The map  $J \rightarrow M$  given by  $X^g(m + u) \mapsto m$  is  $A$ -linear when  $M$  has action through  $\text{ct}$ . Applying it in both factors sends  $\tau$  to  $m_1 \otimes m_2 - m_2 \otimes m_1$  in  $M \otimes_D M$ . This is nonzero because the two displayed tensors are distinct members of its  $D$ -basis. Thus  $\tau \neq 0$ .

The long exact sequence obtained by tensoring  $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$  with  $J$  identifies  $\text{Tor}_1^A(J, A/J)$  with the kernel of multiplication  $J \otimes_A J \rightarrow J$ , since  $\text{Tor}_1^A(J, A) = 0$ . It is nonzero by  $\tau$ , proving  $\text{fd}_A J = 1$ . The same short exact sequence in the first variable gives  $\text{Tor}_2^A(A/J, A/J) \cong \text{Tor}_1^A(J, A/J) \neq 0$ . Hence  $\text{fd}_A(A/J) = 2$ , and the supremum defining weak global dimension is at least two.  $\square$

This is the analogue over  $R_\kappa^{11}$  of [Theorem 13.11](#) for  $T_\Gamma$  (14's  $A_0$ , where 14 proves it); both rings have a totally ordered exponent cone, as [Remark 13.16](#) requires. No upper bound on the weak global dimension of  $R_\kappa^{11}$  and no global dimension is claimed; Dobbs's global-dimension theorem for valuation pullbacks does not apply, because  $K + \Pi_\kappa^{11}$  is not a valuation ring (17).

**Theorem 13.34** (Tor as coefficient multiplication [17; 21]; [merge] over the cores). *Let  $A$  be  $R_\kappa^{11}$  for an infinite regular  $\kappa$  (with  $K = k$ ), or one of the set-sized rings  $A_\Lambda$ ,  $T_\Gamma$  of Setting (L)(a), (b) with any unital  $D \subseteq K$ . Let  $M, N$  be lattices of ranks  $r, s$ , let  $X^g, X^h \in \Pi$  be nonzero monomials,  $I = J_g(M)$  and  $J = J_h(N)$ . Then, with  $A$  acting on the right through  $\text{ct}$ ,*

$$\text{Tor}_1^A(I, A/J) \cong \text{Tor}_2^A(A/I, A/J) \cong \mathcal{T}(M, N) = \ker(\mathfrak{m}_{M,N} : M \otimes_D N \rightarrow K),$$

*a  $D$ -module of rank  $rs - \text{rank}(MN)$  (ranks are dimensions after tensoring with  $\text{Frac } D$ ). There is an exact sequence*

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow I \otimes_A J \xrightarrow{\text{mult}} X^{g+h}(MN + \Pi) \longrightarrow 0,$$

*whose first term is the torsion submodule of  $I \otimes_A J$  and is killed by  $\Pi$ .*

*Proof.* Put  $F = \text{Frac}(D)$ . A lattice here is finite free by definition; no principal-ideal or Noetherian hypothesis on  $D$  is used. Fix a basis  $m_1, \dots, m_r$  of  $M$ . Scaling the presentation in [Theorem 13.9](#)(ii) by  $X^g$  gives

$$0 \longrightarrow K_M \longrightarrow A^r \xrightarrow{(a_j) \mapsto X^g \sum_j m_j a_j} I \longrightarrow 0, \quad K_M = \{u \in \Pi^r : \sum_j m_j u_j = 0\}.$$



The kernel is a  $B$ -module and a  $K$ -vector space: multiplication preserves both the tail coordinates and their linear relation. Consequently  $JK_M \subseteq X^h K_M$ . For the reverse inclusion, choose  $0 \neq n \in N$ . For every  $u \in K_M$ ,  $X^h u = (X^h n)(u/n) \in JK_M$ , since  $u/n \in K_M$ . Thus  $JK_M = X^h K_M$ .

Tensor the presentation with  $A/J$ . Since  $A^r$  is free, its positive Tor groups vanish, and the long exact sequence gives

$$\mathrm{Tor}_1^A(I, A/J) \cong \ker(K_M/JK_M \longrightarrow (A/J)^r) = (K_M \cap J^r)/X^h K_M.$$

For  $x \in K_M \cap J^r$ , write  $x = X^h y$  with  $y \in (N + \Pi)^r$ . This expression is unique in the ambient field. Cancellation of  $X^h$  gives  $\sum_j m_j y_j = 0$ . Use the coefficient map on  $B = K \oplus \Pi$  coordinatewise to set  $c = \mathrm{ct}(y) \in N^r$ . The constant part of that relation is  $\sum_j m_j c_j = 0$ . Conversely, every such tuple  $c$  lifts to  $x = X^h c \in K_M \cap J^r$ . The kernel of  $x \mapsto c$  is exactly  $X^h K_M$ : a zero constant tuple means  $y \in \Pi^r$ , and its unchanged relation then means  $y \in K_M$ . For  $a \in A$ , one has  $\mathrm{ct}(ay) = \mathrm{ct}(a)c$ , so this quotient map is  $A$ -linear when the coefficient tuples have action through  $\mathrm{ct}$ .

The basis isomorphism

$$N^r \longrightarrow M \otimes_D N, \quad (c_j) \longmapsto \sum_j m_j \otimes c_j$$

identifies the equation  $\sum_j m_j c_j = 0$  with  $\mathcal{T}(M, N)$ . Changes of basis give the same coefficient tensor. Dimension shifting in  $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$  now identifies  $\mathrm{Tor}_2^A(A/I, A/J)$  with  $\mathrm{Tor}_1^A(I, A/J)$ . Finally, localization at  $D \setminus \{0\}$  is exact. Applying it to

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow M \otimes_D N \xrightarrow{m_{M,N}} MN \longrightarrow 0$$

gives the rank  $rs - \mathrm{rank}(MN)$ , because  $M \otimes_D N \cong D^{rs}$ . Here  $MN$  is the  $D$ -span of the products. Over an arbitrary  $D$  this rank computation asserts neither freeness nor finite generation of the kernel.

Tensoring  $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$  with  $I$  gives an injection of  $\mathrm{Tor}_1^A(I, A/J)$  into  $I \otimes_A J$ , whose image is the kernel of multiplication. The coefficient description sends  $t = \sum_j m_j \otimes c_j \in \mathcal{T}(M, N)$  to

$$\lambda(t) = \sum_j (X^g m_j) \otimes (X^h c_j).$$

Indeed, the tuple  $X^h c \in K_M \cap J^r$  represents the connecting class just computed. There is also a direct check that this tensor cannot vanish when  $t \neq 0$ . The  $A$ -balanced map

$$(a, b) \longmapsto \mathrm{ct}(X^{-g}a) \otimes_D \mathrm{ct}(X^{-h}b), \quad a \in I, \ b \in J,$$

induces  $\rho : I \otimes_A J \rightarrow M \otimes_D N$  with  $\rho\lambda(t) = t$ . Balancing follows from the coefficient action through  $D$  on both factors.

The image of multiplication is  $IJ = X^{g+h}(MN + \Pi)$ . Products with a tail factor lie in  $\Pi$ ; conversely every tail  $u$  is  $m(u/m)$  with  $0 \neq m \in M$  and  $u/m \in \Pi \subseteq N + \Pi$ . This proves the stated exact sequence. Its kernel is killed by  $\Pi$ , hence by the nonzero element  $X^h$  of the domain  $A$ , so every kernel element is  $A$ -torsion. Conversely any torsion tensor maps to zero in  $IJ$ , since  $IJ$  is a submodule of a field. The kernel is therefore exactly the torsion submodule. It is nevertheless torsion free over  $D$ , being a submodule of  $D^{rs}$ ; the two uses of torsion have different base rings.  $\square$

For  $R_\kappa^{11}$  with uncountable  $\kappa$  this is 17's theorem. Its proof uses only the properties of Setting (L) and the presentation of [Theorem 13.9\(ii\)](#), which is why it applies to 12's cores and to 14's  $A_0 = T_{\mathbb{Q}}$ ;

that extension is the merge's. The exact sequence is 14's [Theorem 13.18](#) for the class ring and for  $A_0$ , where the class tensor product is an explicit carrier; the Tor identifications are 17's. The proof shows which coefficient relations survive: a nonzero relation  $\sum_j m_j \otimes c_j$  is represented in  $I \otimes_A J$  by  $\sum_j (X^g m_j) \otimes (X^h c_j)$ , whose product is zero although the tensor is not (17). Manuscript 21, whose pin precedes 17's placement, proves the identification of  $\text{Tor}_2$  with  $\mathcal{T}(M, N)$ , the action through ct and the rank again, independently and by the same kernel computation, over  $T_{\mathbb{Q}}$  and over the class ring in a two-universe reading; it adds that  $\text{Tor}_n = 0$  for  $n \geq 3$  there, and the first Tor ([Theorem 13.51](#)).

**Corollary 13.35** (12's cyclic modules without  $c^2 \in D$  [merge]). *In the setting of [Theorem 13.11](#), let  $c \in K \setminus \text{Frac}(D)$ ,  $M = D + Dc$  and  $H = X^g L(M) = (X^g, cX^g)$ . Then  $\text{Tor}_2^A(A/H, A/H) \cong \mathcal{T}(M, M)$  has rank  $4 - \text{rank}(M^2)$ , which is 2 if  $c$  is quadratic over  $\text{Frac}(D)$  and 1 otherwise. If  $c^2 \in D$ , it is free with basis  $1 \otimes c - c \otimes 1$  and  $c \otimes c - c^2(1 \otimes 1)$ , which is 12's  $\text{Tor}_2(A/H, A/H) \cong D \oplus D$ .*

*Proof.* Put  $F = \text{Frac}(D)$  and apply [Theorem 13.34](#) with  $N = M$ . The elements  $1, c$  are  $F$ -independent, and  $FM^2 = F + Fc + Fc^2$ . Its dimension is two exactly when  $c$  satisfies a quadratic equation over  $F$ ; otherwise it is three, including both algebraic degree at least three and the transcendental case. Subtracting from four gives the asserted ranks.

If  $c^2 \in D$ , write a tensor in the  $D$ -basis  $1 \otimes 1, 1 \otimes c, c \otimes 1, c \otimes c$  as

$$t = a(1 \otimes 1) + b(1 \otimes c) + b'(c \otimes 1) + d(c \otimes c).$$

Its product is  $(a + dc^2) + (b + b')c$ . Independence of  $1, c$  therefore makes the product zero exactly when  $a = -dc^2$  and  $b' = -b$ . Equivalently,

$$t = b(1 \otimes c - c \otimes 1) + d(c \otimes c - c^2(1 \otimes 1)).$$

The coordinates of  $1 \otimes c$  and  $c \otimes c$  recover  $b, d$ , so these two generators are a basis, not just a spanning family.  $\square$

The hypothesis  $c^2 \in D$  supplies the displayed basis and the isomorphism in [Theorem 13.11](#); it is sufficient, not necessary. More generally, if  $c^2 = u + vc$  with  $u, v \in D$ , the same coordinate calculation gives the basis

$$1 \otimes c - c \otimes 1, \quad c \otimes c - u(1 \otimes 1) - v(1 \otimes c).$$

For example, over  $D = \mathbb{Z}$  the golden ratio  $c = (1 + \sqrt{5})/2$  has  $c^2 = 1 + c \notin \mathbb{Z}$ , but the kernel is still free of rank two. Without such an integral relation, the corollary still computes the rank; it does not infer freeness over an arbitrary coefficient ring. Nonvanishing holds for every  $c \notin \text{Frac}(D)$ .

**Corollary 13.36** (The self-Tor range [17; 21]). *In [Theorem 13.34](#) let  $N = M$ ,  $J = I$  and  $\ell = \text{rank}(M^2)$ . Then*

$$\text{rank Tor}_2^A(A/I, A/I) = r^2 - \ell, \quad \frac{r(r-1)}{2} \leq r^2 - \ell \leq r(r-1).$$

*For  $(D, k) = (\mathbb{Z}, \mathbb{R})$  and  $(\mathbb{Z}[i], \mathbb{C})$  both ends occur for every  $r \geq 1$ .*

*Proof.* Choose  $0 \neq m \in M$ . Multiplication by  $m$  embeds  $M$  in  $M^2$ , while the  $r(r+1)/2$  products  $m_i m_j$  with  $i \leq j$  span  $M^2$ . After localization at  $D \setminus \{0\}$  these give  $r \leq \ell \leq r(r+1)/2$ . Subtract from  $r^2$  and use [Theorem 13.34](#).

For the lower endpoint, take  $M = D + D\alpha_2 + \cdots + D\alpha_r$  with the  $\alpha_j \in k$  algebraically independent over  $F = \text{Frac}(D)$ . Such a finite family exists in both cases:  $F$  is countable, the field generated by finitely many chosen elements is countable, and its algebraic elements in  $k$  form a countable set; choose the next element outside it. The products are precisely the distinct monomials  $1, \alpha_j, \alpha_i \alpha_j$  with  $2 \leq i \leq j \leq r$ , so algebraic independence gives  $\ell = r(r+1)/2$ .

For the upper endpoint, use an order  $M$  of rank  $r$ , with  $1 \in M$ . Then  $M^2 = M$ . Over  $\mathbb{Z}$ , take  $M = \mathbb{Z}[\theta]$  with the positive real root  $\theta = 2^{1/r}$ . The polynomial  $X^r - 2$  is Eisenstein at 2, so  $1, \theta, \dots, \theta^{r-1}$  is a  $\mathbb{Z}$ -basis. Over  $\mathbb{Z}[i]$ , choose a complex root of  $X^r - (1+i)$  and use its powers in the same way; Eisenstein at the Gaussian prime  $1+i$  gives degree  $r$  over  $\mathbb{Q}(i)$ . Monicity reduces higher powers to this basis, proving both finite freeness and closure under multiplication. For  $r = 1$  the independent family is empty and both constructions give  $M = D$ ; both endpoints are then zero.  $\square$

21 proves the rank  $r^2 - \ell$  and the value  $r(r-1)$  for orders again, independently, and splits  $\text{Tor}_2$  into alternating and symmetric relations (Corollary 13.55); it does not state the range.

**Proposition 13.37** (The maximal self-obstruction detects a field [17]). *In Corollary 13.36 suppose  $1 \in M$  and put  $V = \text{Frac}(D)M$ . Then  $\text{rank } M^2 = r$  if and only if  $V$  is a field (for the multiplication of  $k$ ), if and only if  $\text{rank } \text{Tor}_2^A(A/I, A/I) = r(r-1)$ . This does not imply that  $M$  itself is a ring.*

*Proof.* Put  $F = \text{Frac}(D)$  and let  $V^2$  denote the  $F$ -span of all products of elements of  $V$ . Thus  $V^2 = FM^2$ , with dimensions  $\dim_F V = r$  and  $\dim_F V^2 = \text{rank } M^2$ . Since  $1 \in V$ , one has  $V \subseteq V^2$ . Equality of dimensions is therefore equivalent to  $V = V^2$ , hence to closure of  $V$  under multiplication. If this holds and  $0 \neq v \in V$ , the  $F$ -linear map  $w \mapsto vw$  is an injective endomorphism of the finite-dimensional space  $V$ , since  $V \subseteq k$  and  $k$  is a field. It is surjective, so  $vw = 1$  for some  $w \in V$ . Thus  $V$  is a field. Conversely, if  $V$  is a field then  $V^2 = V$ . The equivalence with the maximal second-Tor rank follows from Corollary 13.36.

The lattice need not be closed under multiplication. For  $M = \mathbb{Z} + \mathbb{Z} \cdot (1/\sqrt{2})$  one has  $FM = \mathbb{Q}(\sqrt{2})$  and  $\text{rank } M^2 = 2$ . But  $(1/\sqrt{2})^2 = 1/2 \notin M$ : an equality  $1/2 = a + b/\sqrt{2}$  with integers  $a, b$  forces  $b = 0$  by irrationality of  $\sqrt{2}$ , then the impossible equality  $a = 1/2$ .  $\square$

For the following moduli and endomorphism-order results, return to  $(D, k) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , as fixed at the start of this subsection. In particular,  $D$  is a countable principal ideal domain. The arbitrary-coefficient generality of the preceding Tor theorem does not extend these order statements to arbitrary  $D$ .

**Theorem 13.38** (Rank-two moduli [17]). *For  $\alpha \in k \setminus \text{Frac}(D)$  put  $M_\alpha = D + D\alpha$ . For  $\alpha, \beta \in k \setminus \text{Frac}(D)$  and  $\gamma, \eta > 0$ , the ideals  $J_\gamma(M_\alpha) = (\omega^\gamma, \alpha\omega^\gamma)$  and  $J_\eta(M_\beta)$  of  $\mathcal{A}_{D,k}$  are isomorphic if and only if  $\beta = (a\alpha + b)/(c\alpha + d)$  for some  $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{GL}_2(D)$ . There are exactly  $2^{\aleph_0}$  isomorphism classes of these two-generated ideals; all have rank one over the fraction field and relation module isomorphic to  $\Pi_k$ .*

*Proof.* Multiplication by  $\omega^\gamma$  identifies  $L(M_\alpha)$  with  $J_\gamma(M_\alpha)$ , so the monomial scales do not affect the isomorphism type. By Theorem 13.24, an isomorphism exists exactly when  $M_\beta = sM_\alpha$  for some  $s \in k^\times$ . Both  $(1, \beta)$  and  $(s, s\alpha)$  are then  $D$ -bases of this rank-two lattice. Write their change of basis as

$$\begin{pmatrix} 1 \\ \beta \end{pmatrix} = \begin{pmatrix} d & c \\ b & a \end{pmatrix} \begin{pmatrix} s \\ s\alpha \end{pmatrix}.$$

Its determinant is  $ad - bc \in D^\times$ . The first row gives  $1 = s(c\alpha + d)$ , and division in the second gives  $\beta = (a\alpha + b)/(c\alpha + d)$ .

Conversely, suppose the displayed fractional-linear equation holds with a matrix in  $\mathrm{GL}_2(D)$ . Its denominator is nonzero: if  $c \neq 0$ , vanishing would put  $\alpha = -d/c \in \mathrm{Frac}(D)$ ; if  $c = 0$ , invertibility forces  $d \neq 0$ . Set  $s = (c\alpha + d)^{-1}$ . The same invertible change of basis gives  $M_\beta = sM_\alpha$ , hence an isomorphism of the ideals, explicitly multiplication by  $\omega^{\eta-\gamma}s$ .

These transformations preserve  $k \setminus \mathrm{Frac}(D)$ . Indeed, if  $\beta \in \mathrm{Frac}(D)$ , then  $(a - \beta c)\alpha + (b - \beta d) = 0$  forces both coefficients to vanish, contradicting  $ad - bc \neq 0$ . Their orbits therefore classify the ideals. There are at most countably many transforms of each  $\alpha$ , because  $D$  and  $\mathrm{GL}_2(D)$  are countable. Put  $\lambda$  for the number of orbits. The parameter set has size  $2^{\aleph_0}$ , so

$$2^{\aleph_0} \leq \lambda \cdot \aleph_0 = \max(\lambda, \aleph_0), \quad \lambda \leq 2^{\aleph_0}.$$

Since the continuum is uncountable, these inequalities give  $\lambda = 2^{\aleph_0}$ , without a continuum-hypothesis assumption. Finally, [Theorem 13.9](#) applied to the basis  $(1, \alpha)$  gives the relation module  $\Pi_k$  and fraction-field rank one. The coefficient rank is two; in particular the two displayed ideal generators cannot be replaced by one.  $\square$

The count  $2^{\aleph_0}$  and the  $\mathrm{GL}_2(D)$  form are not in 14, whose [Theorem 13.24](#) they refine.

**Theorem 13.39** (Orders from endomorphism rings [17]). *Let  $M$  be a lattice of rank  $r$ ,  $V = \mathrm{Frac}(D)M$  and  $K_V = \{c \in k : cV \subseteq V\}$ . Then  $K_V$  is a field, finite over  $\mathrm{Frac} D$ , and  $[K_V : \mathrm{Frac} D]$  divides  $r$ . The ring  $\mathfrak{o}(M)$  is an order in  $K_V$ : it is finite free and integral over  $D$ , with  $\mathrm{Frac} \mathfrak{o}(M) = K_V$  and  $\mathrm{rank}_D \mathfrak{o}(M) = [K_V : \mathrm{Frac} D]$ . In the class ring  $\mathrm{End}_A(J_\gamma(M)) = \mathfrak{o}(M) + \Pi_k = \mathcal{A}_{\mathfrak{o}(M),k}$ , so  $\mathrm{End}_A(J_\gamma(M))/\Pi_k \cong \mathfrak{o}(M)$  is its universal set-sized quotient.*

*Proof.* Put  $F = \mathrm{Frac}(D)$ . The set  $K_V$  contains  $F$  and is closed under addition, negation and multiplication. If  $0 \neq c \in K_V$ , multiplication by  $c$  is an injective  $F$ -linear endomorphism of  $V$ , since  $V \subseteq k$ . Finite dimension makes it surjective: for every  $v \in V$  one has  $v = cw$  with  $w \in V$ . Thus  $c^{-1}V \subseteq V$ , proving that  $K_V$  is a field. Fix  $0 \neq v_0 \in V$ . The map  $c \mapsto cv_0$  embeds  $K_V$   $F$ -linearly in  $V$ , so  $[K_V : F]$  is finite. Also  $V$  is a  $K_V$ -vector space, spanned over  $K_V$  by any  $F$ -basis. The vector-space dimension formula gives

$$r = \dim_F V = [K_V : F] \dim_{K_V} V,$$

which proves the divisibility assertion.

By definition,  $\mathfrak{o}(M) = \{c \in k : cM \subseteq M\} \subseteq K_V$ . Its action on a  $D$ -basis of  $M$  embeds it as a  $D$ -submodule of  $\mathrm{End}_D(M) \cong D^{r \times r}$ . Since  $D$  is a principal ideal domain, this submodule is finite free. For  $c \in \mathfrak{o}(M)$ , Cayley–Hamilton annihilates the matrix of multiplication by  $c$  by its monic characteristic polynomial. Applying this endomorphism identity to any nonzero member of  $M$  shows that the same polynomial vanishes at  $c$  in  $k$ . Hence  $c$  is integral over  $D$ .

Conversely, let  $c \in K_V$ . In a  $D$ -basis of  $M$ , which is also an  $F$ -basis of  $V$ , the matrix of multiplication by  $c$  has finitely many entries in  $F$ . Clear their denominators with one  $0 \neq a \in D$ . Then  $acM \subseteq M$ , so  $ac \in \mathfrak{o}(M)$  and  $c = (ac)/a \in F\mathfrak{o}(M)$ . Therefore

$$F\mathfrak{o}(M) = K_V = \mathrm{Frac} \mathfrak{o}(M), \quad \mathrm{rank}_D \mathfrak{o}(M) = \dim_F K_V.$$

This proves that  $\mathfrak{o}(M)$  is an order in precisely this field, rather than merely in an unspecified finite extension.

Scaling by  $\omega^\gamma$  leaves endomorphism scalars unchanged. Thus [Theorem 13.24](#) identifies the endomorphism ring with  $\mathfrak{o}(M) + \Pi_k$ . The constant-term map onto  $\mathfrak{o}(M)$  has kernel  $\Pi_k$ . Apply

**Theorem 5.1** with coefficient ring  $\mathfrak{o}(M)$ : every unital map from this class ring into any set-sized ring factors uniquely through that constant-term map. This application requires a unital coefficient subring, not that  $\mathfrak{o}(M)$  itself be a principal ideal domain.  $\square$

**Corollary 13.40** (Every order occurs [17]). *If  $\mathfrak{o} \subset k$  is an order over  $D$  in a finite extension of  $\text{Frac } D$ , then  $\text{End}_A(J_\gamma(\mathfrak{o})) = \mathfrak{o} + \Pi_k$  and  $\text{Aut}_A(J_\gamma(\mathfrak{o})) = \mathfrak{o}^\times$ .*

*Proof.* An order  $\mathfrak{o}$  here contains 1 and is finite free over  $D$ , so it is an allowed coefficient lattice. If  $c\mathfrak{o} \subseteq \mathfrak{o}$ , then  $c = c \cdot 1 \in \mathfrak{o}$ . Conversely every  $c \in \mathfrak{o}$  preserves  $\mathfrak{o}$  by multiplication. Hence  $\mathfrak{o}(\mathfrak{o}) = \mathfrak{o}$ , and Theorem 13.24 gives the endomorphism ring. An automorphism scalar and its inverse both lie in  $k + \Pi_k$ , whose only units are  $k^\times$  (Lemma 2.4). Thus an automorphism has no tail part; it is a constant  $c$  with  $c\mathfrak{o} = \mathfrak{o}$ , equivalently  $c \in \mathfrak{o}^\times$ . This proves the asserted automorphism group, even though the endomorphism ring itself contains the whole tail.  $\square$

14's Theorem 13.25 realizes the ideal classes of such an order as well; the corollary realizes the order itself as the constant quotient of an endomorphism ring.

**Example 13.41** (A unit that is not an omnific integer [17]). For  $J = (\omega, \sqrt{2}\omega) \subseteq \mathbf{Oz}$ , the coefficient lattice is the order  $\mathbb{Z}[\sqrt{2}]$ , so  $\text{End}_{\mathbf{Oz}}(J) = \mathbb{Z}[\sqrt{2}] + \Pi$ . Multiplication by  $1 + \sqrt{2}$  sends the two generators to

$$\omega + \sqrt{2}\omega, \quad 2\omega + \sqrt{2}\omega,$$

respectively. Its matrix on this generating pair, with images in columns, is  $\begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ , of determinant  $-1$ . The inverse scalar is  $\sqrt{2} - 1$ , since  $(1 + \sqrt{2})(\sqrt{2} - 1) = 1$ . These are automorphisms of the ideal;  $1 + \sqrt{2} \notin \mathbf{Oz}$  because it is a noninteger real constant. Its positive integer powers are distinct, as  $1 + \sqrt{2} > 1$ . The generating pair is a  $\mathbb{Z}$ -basis of the scaled coefficient lattice, not an  $\mathbf{Oz}$ -basis of  $J$ ; its relations over  $\mathbf{Oz}$  are the tail relations already described. Over  $\mathbf{Oz}[i]$ , the same generating pair has endomorphism order  $\mathbb{Z}[i, \sqrt{2}]$ .

In contrast, let  $\alpha$  be algebraic of degree at least three over  $F = \text{Frac}(D)$ , or transcendental over  $F$ . A multiplier of  $D + D\alpha$  belongs to that lattice because 1 does, so it has the form  $a + b\alpha$  with  $a, b \in D$ . If its product with  $\alpha$  lies in  $D + D\alpha$ , there are  $u, v \in D$  with  $b\alpha^2 + (a - v)\alpha - u = 0$ . Independence of  $1, \alpha, \alpha^2$  over  $F$  forces  $b = 0$  (and  $u = 0, a = v$ ). Thus  $\mathfrak{o}(D + D\alpha) = D$ . The ideal  $(\omega, \alpha\omega)$  has only the automorphisms  $\pm 1$  in the real case, and  $\pm 1, \pm i$  in the Gaussian case.

**Theorem 13.42** (Modules killed by the tail [17; 21]). *Let  $A = R_\kappa^{11}$ ,  $M$  a lattice of rank  $r$ , and  $P$  a  $D$ -module on which  $A$  acts through  $\text{ct}$ . Then  $J_g(M) \otimes_A P \cong M \otimes_D P \cong P^r$  and  $\text{Tor}_j^A(J_g(M), P) = 0$  for  $j \geq 1$ . Over the class ring  $\mathcal{A}_{D,k}$ , every set-sized unital module  $P$  is of this kind (Theorem 5.1(iv)), and  $M \otimes_D P$  is a set-sized model of  $J_g(M) \otimes P$ : every balanced map on  $J_g(M) \times P$  factors through  $M \times P$ .*

*Proof.* Write  $J = J_g(M)$ . By Theorem 13.9(i),  $\Pi J = X^g \Pi$ , and the map  $X^g(m + u) \mapsto m$  induces  $J/\Pi J \cong M$ . Since  $\Pi P = 0$ , the formula

$$X^g(m + u) \otimes p \mapsto m \otimes_D p$$

defines a map  $J \otimes_A P \rightarrow M \otimes_D P$ : multiplying the first input by  $a \in A$  multiplies its coefficient by  $\text{ct}(a)$ , which acts on  $P$  as  $a$  does. Its inverse is induced by the  $D$ -balanced map  $(m, p) \mapsto X^g m \otimes p$ . Indeed, the difference  $X^g u$  lies in  $\Pi J$ , and every finite sum  $\sum_i a_i j_i$  with  $a_i \in \Pi$ ,  $j_i \in J$  tensors to  $\sum_i j_i \otimes a_i p = 0$ . The two maps are therefore inverse. After a basis of  $M$  is fixed,  $M \otimes_D P \cong P^r$ . No torsion-freeness or flatness of  $P$  over  $D$  is needed.

For the set-sized ring  $R_\kappa^{11}$ , use its natural presentation  $0 \rightarrow K_M \rightarrow A^r \rightarrow J \rightarrow 0$ , with  $K_M \cong \Pi^{r-1}$ . The tail is idempotent and flat by [Theorem 13.31\(i\)](#), so  $\Pi K_M = K_M$  and  $K_M$  is flat, including the zero kernel when  $r = 1$ . Every  $u \in K_M$  is a finite sum  $\sum_i a_i u_i$  with  $a_i \in \Pi$ ; balancing gives  $u \otimes p = \sum_i u_i \otimes a_i p = 0$ . Hence  $K_M \otimes_A P = 0$ . The long exact sequence gives an injection  $\text{Tor}_1^A(J, P) \hookrightarrow K_M \otimes_A P$ , and for  $j \geq 2$  gives  $\text{Tor}_j^A(J, P) \cong \text{Tor}_{j-1}^A(K_M, P) = 0$  by flatness.

For the class ring, the universal quotient theorem makes every set-sized unital module  $P$  satisfy  $\Pi_k P = 0$ . Fix any target module and a specified balanced biadditive map on  $J \times P$ . The same finite-sum calculation shows it kills  $\Pi_k J \times P$ , so its value on  $(X^g(m+u), p)$  depends only on  $(m, p)$ . The resulting map on  $M \times P$  is  $D$ -balanced and therefore factors uniquely through the set-sized tensor module  $M \otimes_D P$ . Conversely its canonical map is balanced for the action of the class ring through  $\text{ct}$ . This proves the stated universal property using one fixed map and target at a time; it constructs no collection of all class maps and invokes no class-ring Tor functor.  $\square$

This extends [Theorem 12.2\(ii\)](#) and [Corollary 13.5](#) from  $D$  to the lattice ideals. For  $r \geq 2$  it does not make  $J_g(M)$  flat. In the set-sized model,  $A/J_g(M)$  detects nonzero second Tor by [Corollary 13.36](#); this test module is not killed by the tail. For example,  $X^{g/2} \in \Pi$  has nonzero image modulo  $J_g(M)$ , since  $X^{-g}X^{g/2} = X^{-g/2} \notin M + \Pi$ . Over the class ring, every *set-sized* test module is killed by the tail, so the tensor functor on these tests agrees, after choosing a lattice basis, with that of a free module of rank  $r$ . The ideal's fraction-field rank is still one, and its equational nonflatness is proved in [Theorem 13.29](#); the set-sized tests do not detect it. No full-class Tor functor is inferred (17). Manuscript 21 obtains the same Tor groups, in the equivalent form  $\text{Tor}_1(A/J_g(M), P) \cong M \otimes_D P$  and  $\text{Tor}_{\geq 2}(A/J_g(M), P) = 0$ , independently over  $T_{\mathbb{Q}}$  and over the class ring in a two-universe reading, together with a square-zero model ([Theorem 13.57](#)).

**Computing  $q$  and  $\ell$  (17).** If the entries of  $C \in k^{m \times n}$  lie in a known  $\text{Frac}(D)$ -span with a basis  $\theta_1, \dots, \theta_t$  that is independent over  $\text{Frac } D$ , write  $C = \sum_j \theta_j C_j$  with  $C_j \in \text{Frac}(D)^{m \times n}$ ; then  $Cx = 0$  for  $x \in \text{Frac}(D)^n$  exactly when every  $C_j x = 0$ , so

$$\ell = n - \text{rank}_{\text{Frac } D} \begin{pmatrix} C_1 \\ \vdots \\ C_t \end{pmatrix}, \quad q = n - \text{rank}_k C,$$

and [Theorems 13.28](#) and [13.31](#) decide the kernel type and its generator count. This assumes an exact description of the coefficient span, as number-field data provide; it is not an algorithm for arbitrary real input or floating-point data (17).

**Examples (17).** All Tor statements here are over  $R_\kappa^{11}$  (or the cores). For  $M = \mathbb{Z} + \mathbb{Z}\sqrt{2}$  the multiplication matrix in the bases  $(1 \otimes 1, 1 \otimes \alpha, \alpha \otimes 1, \alpha \otimes \alpha)$ ,  $\alpha = \sqrt{2}$ , has rows  $(1, 0, 0, 2)$  and  $(0, 1, 1, 0)$ , and  $\mathcal{T}(M, M)$  has basis  $\alpha \otimes 1 - 1 \otimes \alpha$  (commutativity) and  $\alpha \otimes \alpha - 2(1 \otimes 1)$  (the quadratic equation), so  $\text{Tor}_2(A/J, A/J) \cong \mathbb{Z}^2$  for  $J = (\omega, \sqrt{2}\omega)$ ; the ideal has two generators, its relation module needs  $\kappa$  generators, and all its powers have two generators; over  $\mathbb{Z}[i]$  the same matrix gives  $\mathbb{Z}[i]^2$ . For  $\alpha = \sqrt[3]{2}$  and for a transcendental  $\tau$  the matrices coincide (rank three, kernel spanned by the antisymmetric tensor), and so do the orders ( $\mathbb{Z}$ ), but the power sequences  $\min(n+1, 3)$  and  $n+1$  differ: generator counts, first relation modules, orders and self-Tor ranks do not determine one another. For  $M = \mathbb{Z} + \mathbb{Z}\sqrt{2}$  and  $N = \mathbb{Z} + \mathbb{Z}\sqrt{3}$ ,  $\text{rank } MN = 4$ , so  $\mathcal{T}(M, N) = 0$  and  $\text{Tor}_2(A/J_g(M), A/J_h(N)) = 0$ , although each self-Tor has rank two; the same holds over  $\mathbb{Z}[i]$ , since  $1, \sqrt{2}, \sqrt{3}, \sqrt{6}$  stay independent over  $\mathbb{Q}(i)$ . 21 gives the same pair independently, as the first case of its linear-disjointness test ([Theorem 13.53](#)).



### 13.10 Exact projective dimensions and derived arithmetic

This subsection is manuscript 21's material (Section 1.3). 21 pins 9693b28, at which this section held 12's and 14's results but not yet 17's Section 13.9. Its first part answers the first clause of Question 18.12: for every  $d \geq 1$ ,  $\text{pd}_{A_d} D = d + 1$ , with an explicit countably generated free resolution and an explicit copy of  $D^{\mathbb{N}}/D^{(\mathbb{N})}$  in the top Ext group; it also determines the projective dimensions of the coordinate face quotients and of the lattice modules over  $A_d$ , which Proposition 13.12(iii), (iv) bound. Its second part computes the Tor groups of two lattice quotients over  $T_{\mathbb{Q}}$  and over the class ring read in two universes. There its second-Tor formula with the rank count, the pair  $\mathbb{Z}[\sqrt{2}]$ ,  $\mathbb{Z}[\sqrt{3}]$  and the Tor groups against modules killed by the tail repeat 17's Theorems 13.34 and 13.42 and Corollary 13.36, which 21 could not see; they are printed there, with 21 credited as a further source, and are only cited here. New in the second part are the first-Tor window formula, a flat DGA resolution with the complete self-Tor product, the linear-disjointness test and the framed reconstruction of orders. Labels carry the sub-prefix **osq:da**.

**Notation.** 21's field  $k$  and ring  $A_d$  are  $K$  and  $A_d$  of Section 13.1; its tail  $\mathcal{I}_d$  is  $\Pi_d$ , and its embedding  $\gamma_i = \omega^{i-1}$  is an instance of Lemma 13.1(ii) (the report uses  $\theta^{i-1}$ ; both are valid). Its lattices  $L, M$  and its module  $M_L = L + \mathcal{I}_d$  are  $M, N$  and  $L(M)$ : the letters are swapped. Its monomial  $t_a = \omega^a$  ( $a > 0$ ) is  $X^a$  here, so its ideal  $J_{a,L} = t_a(L + \mathcal{I})$  is  $J_a(M)$ ; the letter  $t$  is not used, because in this report  $t^g$  means  $X^{-g}$  (Section 2.2). Its face quotient  $B_U$  is  $F_U$  (the letter  $B$  is the ring  $K \oplus \Pi$ ), its  $c = |U|$  is written  $|U|$  (the letter  $c$  is an element of  $K \setminus \text{Frac } D$ ), its ideals  $Q_n$  are  $J_U^{(n)}$ , its test module  $A/(X_i : i \in U)$  is  $V_U$  of Section 13.3, and its complexes  $K_n, F$ , shift  $\sigma$ , telescope  $\mathcal{T}$  and dual differential  $\delta_N$  are  $\text{Kos}_n, \text{Kos}, \text{sh}, \text{Tel}$  and  $\Delta_P^{\vee}$  (the letter  $K$  is the field,  $\mathcal{T}$  the multiplication kernel). In the second part its rings  $R_0$  and  $R$  are  $T_{\mathbb{Q}}$  and  $\mathcal{A}_{D,k}$ , its ideal  $\mathcal{I}$  is  $\Pi$ , its kernel  $\mathcal{K}(L, M) = \ker(L \otimes_D M \rightarrow LM)$  is  $\mathcal{T}(M, N)$  (the same kernel, since  $MN \subseteq k$ ), its syzygy module  $W_L$  is  $\text{Syz}_M$ , its quotients  $S, T$  are  $A/I, A/J$  ( $T$  is 12's ordered-cone ring), its  $H_j = \text{Tor}_j(S, S)$  are  $T_j$  (the letter  $H$  is 12's cyclic module), its boundary map  $\lambda$  is  $\mathfrak{b}$  ( $\lambda$  is a cardinal in Section 11.4), its quotient  $Q_2$  is  $\text{Ind}_2$ , its boundary lattice  $C_{a,b}$  and window  $\mathcal{H}_k(0, c)$  are  $\mathbb{L}_{g,h}$  and  $\Pi_{(0,e)}$ , its fields  $E, F$  over  $F_0$  are  $K_1, K_2$  over  $\text{Frac } D$ , and its orders  $O$  and kernels  $\mathcal{K}_O$  are  $\mathfrak{o}$  and  $\mathcal{T}(\mathfrak{o}, \mathfrak{o})$ .

**Face quotients and a free telescope.** Fix  $d \geq 1$ ,  $A = A_d$ ,  $\Pi = \Pi_d$  and a nonempty  $U \subseteq [d]$ . With the face ideals  $I_S$  of (13.3) put

$$\mathfrak{f}_U = \sum_{i \in U} I_{\{i\}}, \quad F_U = A/\mathfrak{f}_U.$$

The ideal  $\mathfrak{f}_U$  is spanned by the terms  $KX^{\alpha}$  with  $\alpha_i > 0$  for some  $i \in U$ , so  $F_U$  keeps exactly the terms whose coordinates in  $U$  vanish:  $F_U \cong A_{d-|U|}$ , with the same  $(D, K)$ , if  $|U| < d$ , and  $F_{[d]} = A/\Pi = D$ . For  $1 \leq p \leq |U|$  let  $C_p^U = \bigoplus_{S \subseteq U, |S|=p} I_S$ , with the differentials of Theorem 13.4, and  $C_0^U = A$ .

**Proposition 13.43** (Face quotients: flat resolution and flat dimension [21]). *The complex  $0 \rightarrow C_{|U|}^U \rightarrow \cdots \rightarrow C_1^U \rightarrow A \rightarrow F_U \rightarrow 0$  is exact, every  $C_p^U$  is flat, and  $\text{fd}_A F_U = |U|$ .*

*Proof.* The differentials are the signed face inclusions, so deleting two indices in opposite orders proves that their composites vanish. To prove exactness, fix a multidegree  $\alpha$  and put  $T = \{i \in U : \alpha_i > 0\}$ . If  $T = \emptyset$ , every positive term vanishes in this degree, and  $A \rightarrow F_U$  is an isomorphism there. This

includes the constant degree, whose coefficient module is  $D$ , as well as degrees supported outside  $U$ , whose coefficient module is  $K$ .

If  $T \neq \emptyset$ , the quotient component is zero. The component of  $I_S$  is  $K$  exactly when  $\emptyset \neq S \subseteq T$ , and the degree-zero chain term  $A_\alpha$  is also  $K$ . These form the augmented simplex complex on  $T$ . Insert  $t = \min T$  when it is absent and send a face already containing  $t$  to zero. As in [Theorem 13.4](#), the term deleting  $t$  is the identity and all other deletions cancel in pairs, giving  $dh + hd = 1$ . This is a contraction of the coefficient complex, not an asserted  $A$ -linear contraction. A chain has finitely many monomials, so summing the preimages of its components proves exactness of the  $A$ -module complex. Each  $I_S$  is flat by [Lemma 13.3\(i\)](#), hence so is every finite sum  $C_p^U$ . Thus  $\text{fd}_A \mathbf{F}_U \leq |U|$ .

For the opposite inequality, tensor this flat resolution with  $V_U = A/J_U$ , where  $J_U = (X_i : i \in U)A$ . The standard quotient tensor identification gives  $I_S \otimes_A V_U = I_S/J_U I_S$ . Consider the multidegree  $\mathbf{e}_U = \sum_{i \in U} \mathbf{e}_i$ . The coefficient of  $I_U$  there is  $K$ , and that of  $J_U I_U$  is zero: subtracting any  $\mathbf{e}_i$  leaves zero in coordinate  $i$ , whereas an exponent of  $I_U$  must be strictly positive. The top tensor term therefore contains a copy of  $K$  in that degree.

If  $|U| \geq 2$ , the face  $U \setminus \{i\}$  is nonempty, and for every  $a \in K$ ,

$$aX^{\mathbf{e}_U} = X_i(aX^{\mathbf{e}_{U \setminus \{i\}}}) \in J_U I_{U \setminus \{i\}}.$$

All components of the top differential vanish on this copy of  $K$ . There is no incoming differential from a higher term, so it survives in  $\text{Tor}_{|U|}^A(\mathbf{F}_U, V_U)$ . For  $U = \{i\}$ , the target is instead  $A/J_U$ ; its component in degree  $\mathbf{e}_i$  is  $K/D$ , because that of  $X_i A$  is  $D$ . The differential is the quotient map  $K \rightarrow K/D$ , with nonzero kernel  $D$ . Thus the top Tor is nonzero also in this case, including when  $D = K$  and the target component is zero. In either case, a shorter flat resolution would force this Tor group to vanish, proving the lower bound.  $\square$

For  $n \geq 0$  put  $J_U^{(n)} = (X_i^{2^{-n}} : i \in U)A$  and  $u_n = p_U^{2^{-n-1}}$ ; since  $u_n X_i^{2^{-n-1}} = X_i^{2^{-n}} p_{U \setminus \{i\}}^{2^{-n-1}}$ , one has  $u_n J_U^{(n+1)} \subseteq J_U^{(n)}$ . Let  $\text{Kos}_n$  be the Koszul complex over  $A$  of the  $|U|$  elements  $X_i^{2^{-n}}$  ( $i \in U$ ): it is free on basis vectors  $e_S$  ( $S \subseteq U$ ) in degree  $|S|$ , with  $\partial e_S = \sum_{r=1}^p (-1)^{r-1} X_{s_r}^{2^{-n}} e_{S \setminus \{s_r\}}$  for  $S = \{s_1 < \dots < s_p\}$ . The maps  $f_n(e_S) = p_S^{2^{-n-1}} e_S$  (with  $f_n(e_\emptyset) = e_\emptyset$ ) are chain maps  $\text{Kos}_n \rightarrow \text{Kos}_{n+1}$ , because  $X_i^{2^{-n-1}} p_S^{2^{-n-1}} = X_i^{2^{-n}} p_{S \setminus \{i\}}^{2^{-n-1}}$  for  $i \in S$ .

**Theorem 13.44** (A free telescope resolution [21]).

- (i) In each degree  $p \geq 1$  the direct system  $(\text{Kos}_n, f_n)$  has colimit  $C_p^U$ , through the maps sending  $e_S$  at stage  $n$  to  $p_S^{2^{-n}}$  in the summand  $I_S$ , and the colimit of the Koszul differentials is the differential of [Proposition 13.43](#); in degree 0 the system is constant with value  $A$ .
- (ii) Let  $\text{Kos} = \bigoplus_{n \geq 0} \text{Kos}_n$ , let  $\text{sh} : \text{Kos} \rightarrow \text{Kos}$  send the  $n$ th summand to the  $(n+1)$ st by  $f_n$ , and put  $\Delta = 1 - \text{sh}$ . The mapping cone  $\text{Tel}_j = \text{Kos}_j \oplus \text{Kos}_{j-1}$ ,  $d(x, y) = (\partial x + \Delta y, -\partial y)$ , is a complex of countably generated free  $A$ -modules in degrees  $0, \dots, |U| + 1$ , and the augmentations of the  $\text{Kos}_n$  make it a free resolution of  $\mathbf{F}_U$  of length  $|U| + 1$ .

No regularity of  $(X_i^{2^{-n}})_{i \in U}$  is used, and the individual  $\text{Kos}_n$  need not be acyclic: for  $D \neq K$  the sequence need not be regular (for  $c \in K \setminus D$ ,  $X_2$  kills the nonzero class of  $cX_1$  in  $A_2/X_1 A_2$ ).

*Proof.* (i) Fix a nonempty  $S \subseteq U$ . The stage- $n$  map  $Ae_S \rightarrow I_S$ ,  $ae_S \mapsto ap_S^{2^{-n}}$ , is injective because  $A$  is a domain. It is compatible with the transition:

$$p_S^{2^{-n-1}} p_S^{2^{-n-1}} = p_S^{2^{-n}}.$$

The images form the increasing chain of principal ideals in Lemma 13.3(i), with union  $I_S$ . In particular, the union contains arbitrary  $K$ -coefficients: for a given finite support choose  $2^{-n}$  strictly smaller than all coordinates on  $S$ , so division by  $p_S^{2^{-n}}$  leaves a tail in  $A$ . Taking the finite direct sum over faces of size  $p$  gives  $C_p^U$ . In degree zero the transition is the identity on  $A$ .

For  $S = \{s_1 < \dots < s_p\}$ , the  $r$ th term of the stage Koszul differential maps to

$$(-1)^{r-1} X_{s_r}^{2^{-n}} p_{S \setminus \{s_r\}}^{2^{-n}} = (-1)^{r-1} p_S^{2^{-n}}$$

in the face summand  $S \setminus \{s_r\}$ . This is exactly the signed inclusion in  $C^U$ . When  $p = 1$ , the target is  $A$  and  $p_\emptyset = 1$ , so the same computation gives the inclusion  $I_{\{s_1\}} \rightarrow A$ .

(ii) Regard Kos as a homological complex and set its terms outside degrees  $0, \dots, |U|$  to zero. The chain-map identity for each  $f_n$  gives  $\partial\Delta = \Delta\partial$ . Hence

$$d^2(x, y) = (\partial^2 x + \partial\Delta y - \Delta\partial y, \partial^2 y) = (0, 0).$$

The map  $\Delta$  is injective degreewise. Indeed, a nonzero element of the direct sum has a least nonzero stage, and the component of  $\Delta$  at that stage is unchanged. The cokernel identifies each stage element with its image under  $f_n$  in the next stage, which is precisely the direct-limit relation. Part (i) therefore gives an exact sequence of complexes

$$0 \longrightarrow \text{Kos} \xrightarrow{\Delta} \text{Kos} \longrightarrow C^U \longrightarrow 0.$$

The map  $q : \text{Tel} \rightarrow C^U$ ,  $q(x, y) = [x]$ , is a surjective chain map, since  $[\Delta y] = 0$ . Its kernel consists uniquely of pairs  $(\Delta z, y)$ . In the coordinates  $(z, y)$ , with  $z \in \text{Kos}_j$  and  $y \in \text{Kos}_{j-1}$ , the kernel differential is

$$d(z, y) = (\partial z + y, -\partial y).$$

Define  $h(z, y) = (0, z)$  in degree  $j + 1$ . Then

$$dh(z, y) = (z, -\partial z), \quad hd(z, y) = (0, \partial z + y),$$

so  $dh + hd = 1$ , including the endpoint degrees. The kernel is contractible; the long exact homology sequence now shows that  $q$  induces isomorphisms on homology. By Proposition 13.43, this homology is  $\mathbf{F}_U$  in degree zero and zero in positive degrees.

Explicitly, the telescope augmentation sends a finite stage tuple  $(a_n)_n \in \text{Tel}_0 = \bigoplus_n A$  to  $\sum_n a_n \bmod \mathfrak{f}_U$ . It kills the degree-one boundary: the Koszul terms lie in  $\mathfrak{f}_U$ , and the  $\Delta$  terms cancel under the sum. Each telescope term is a countable direct sum of finite free modules. The last possible term is  $\text{Tel}_{|U|+1} = \text{Kos}_{|U|}$ , giving the claimed free resolution and length bound.

The argument used neither exactness of the individual Koszul complexes nor regularity of their defining sequences. For the stated failure of regularity, choose  $c \in K \setminus D$ . The element  $cX_1$  has nonzero class modulo  $X_1A_2$ , since in degree  $\mathbf{e}_1$  that ideal has only coefficients in  $D$ . But  $X_2(cX_1) = X_1(cX_2)$  and  $cX_2 \in A_2$ , so  $X_2$  annihilates that class. The transition system and its cone, not an assumption on these finite sequences, provide the resolution.  $\square$

For  $|U| = 1$  this is 12's telescope  $0 \rightarrow F \rightarrow F \rightarrow I_S \rightarrow 0$  of Proposition 13.12(i), placed in the cone; 21 assembles the telescopes of all faces into one resolution (21 credits countable telescopes as a standard tool, citing Positselski [28]).

**Theorem 13.45** (The top Ext group [21]). *For an  $A$ -module  $P$  put  $G_n(P) = P/J_U^{(n)}P$ ; multiplication by  $u_n$  maps  $G_{n+1}(P)$  to  $G_n(P)$ . Then, naturally in  $P$ ,*

$$\mathrm{Ext}_A^{|U|+1}(\mathbf{F}_U, P) \cong \mathrm{coker}\left(\Delta_P^\vee : \prod_{n \geq 0} G_n(P) \longrightarrow \prod_{n \geq 0} G_n(P), (y_n)_n \longmapsto (y_n - u_n y_{n+1})_n\right),$$

which is  $\varprojlim^1$  of the inverse system  $(G_n(P))_n$ .

*Proof.* First, the transition on  $G_n(P)$  is well defined. The identity  $u_n J_U^{(n+1)} \subseteq J_U^{(n)}$  gives  $u_n J_U^{(n+1)}P \subseteq J_U^{(n)}P$ , so changing a representative in  $G_{n+1}(P)$  does not change its image in  $G_n(P)$ .

The top term of the free resolution is  $\mathrm{Tel}_{|U|+1} = \bigoplus_{n \geq 0} Ae_{U,n}$ . A homomorphism from this direct sum is specified by arbitrary values on its basis; these values need not have finite support. Consequently its dual with values in  $P$  is  $\prod_{n \geq 0} P$ . There is no higher term, so the desired Ext group is this product modulo the image of the preceding coboundary.

A functional on  $\mathrm{Tel}_{|U|} = \mathrm{Kos}_{|U|} \oplus \mathrm{Kos}_{|U|-1}$  is a pair  $(\varphi, \psi)$ . Its coboundary, evaluated at a top element  $y$ , is

$$y \longmapsto \varphi(\Delta y) - \psi(\partial y).$$

Since  $\Delta e_{U,n} = e_{U,n} - u_n e_{U,n+1}$ , the first contribution ranges over

$$(z_n - u_n z_{n+1})_n, \quad (z_n)_n \in \prod_{n \geq 0} P.$$

Write  $U = \{i_1 < \dots < i_{|U|}\}$ . The second contribution in coordinate  $n$  is

$$-\sum_{r=1}^{|U|} (-1)^{r-1} X_{i_r}^{2^{-n}} \psi(e_{U \setminus \{i_r\}, n}).$$

Its possible values are exactly  $J_U^{(n)}P$ . These choices are independent for all faces and stages, because  $\psi$  is a functional on a direct sum of the indicated free summands. Thus the second contribution ranges over the whole product  $\prod_n J_U^{(n)}P$ , not just its direct sum.

Let  $W = \prod_n J_U^{(n)}P$  and let  $L : \prod_n P \rightarrow \prod_n P$  send  $z$  to  $(z_n - u_n z_{n+1})_n$ . We have proved

$$\mathrm{Ext}_A^{|U|+1}(\mathbf{F}_U, P) \cong \left(\prod_n P\right) / (\mathrm{im} L + W).$$

Coordinatewise quotient identifies  $(\prod_n P)/W$  with  $\prod_n G_n(P)$ . Moreover  $L(W) \subseteq W$  by the transition containment checked above. Its induced endomorphism on that quotient is exactly  $\Delta_P^\vee$ , so the displayed quotient is its cokernel.

Every  $A$ -linear map  $P \rightarrow P'$  respects the ideals, the transition multipliers and the preceding coboundaries. The identification is therefore natural in  $P$ . Finally, for a countable inverse system of modules, the two-product complex with differential  $1 - \text{shift}$  computes the derived inverse limit; its first cohomology is this cokernel [33, Tags 07KV and 0CQC]. No surjectivity of the transitions or Mittag-Leffler condition is assumed here.  $\square$

**Theorem 13.46** (A copy of  $D^{\mathbb{N}}/D^{(\mathbb{N})}$  in the top Ext group [21]). *Send  $(c_n) \in D^{\mathbb{N}}$  to the class of the constant representatives  $(c_n) \in \prod_n G_n(A)$  in  $\mathrm{Ext}_A^{|U|+1}(\mathbf{F}_U, A)$ . This  $D$ -linear map has kernel exactly  $D^{(\mathbb{N})}$ , the finitely supported sequences, so  $D^{\mathbb{N}}/D^{(\mathbb{N})} \hookrightarrow \mathrm{Ext}_A^{|U|+1}(\mathbf{F}_U, A)$ .*

*Proof.* The constant inclusions  $D \rightarrow G_n(A)$ , followed by the cokernel map in [Theorem 13.45](#), define the stated  $D$ -linear map. Suppose a sequence  $c = (c_n)$  maps to zero. Choose representatives  $y_n \in A$  for a preimage under  $\Delta_A^\vee$ . Then

$$y_n - u_n y_{n+1} - c_n \in J_U^{(n)} \quad (n \geq 0).$$

Set  $m_j = p_U^{1-2^{-j}}$ , including  $m_0 = 1$ . The geometric sum of the transition exponents gives  $m_j = u_0 \cdots u_{j-1}$  and  $m_j u_j = m_{j+1}$ . Moreover, for every  $i \in U$ ,

$$m_j X_i^{2^{-j}} = X_i p_{U \setminus \{i\}}^{1-2^{-j}} \in J_U,$$

where an empty product is 1. Multiplying the  $j$ th relation by  $m_j$  and summing for  $0 \leq j \leq N$  therefore telescopes to

$$y_0 - \sum_{j=0}^N c_j m_j - m_{N+1} y_{N+1} \in J_U.$$

Fix  $j \leq N$ . All coordinates on  $U$  of the exponent of  $m_j$  equal  $1 - 2^{-j} < 1$ . Every monomial of  $J_U$  has at least one coordinate on  $U$  at least 1, so it cannot contribute to this coefficient. Nor can  $m_{N+1} y_{N+1}$ : its coordinates on  $U$  are all at least  $1 - 2^{-N-1} > 1 - 2^{-j}$ , since every exponent of  $y_{N+1}$  is coordinatewise nonnegative. Finally the  $m_j$  are pairwise distinct because  $U$  is nonempty. Comparing their coefficients gives

$$[m_j]y_0 = c_j \quad (0 \leq j \leq N).$$

The representative  $y_0$  is fixed as  $N$  varies. Taking  $N \geq j$  for each  $j$  forces it to contain a distinct monomial for every nonzero  $c_j$ . But  $A_d$  consists of finite sums. Thus  $c$  is finitely supported. No assertion about convergence or infinite Hahn sums is used: the obstruction lies in a single element of  $A_d$ .

Conversely, suppose  $c$  has finite support. For each  $n$  define the finite sum

$$y_n = \sum_{j \geq n} c_j p_U^{2^{-n} - 2^{-j}}.$$

The term  $j = n$  is the constant  $c_n \in D$ , and terms with  $j > n$  have positive exponent on  $U$ , so  $y_n \in A$ . Multiplication by  $u_n = p_U^{2^{-n-1}}$  turns the exponent  $2^{-n-1} - 2^{-j}$  in  $y_{n+1}$  into  $2^{-n} - 2^{-j}$ . All terms with  $j > n$  cancel, leaving  $y_n - u_n y_{n+1} = c_n$  exactly. Their quotient classes give the required preimage. The kernel is therefore exactly  $D^{(\mathbb{N})}$ , and passage to this quotient proves the injection.  $\square$

**Corollary 13.47** (Exact projective dimensions of the coordinate cores [21]). *For every  $d \geq 1$  and every nonempty  $U \subseteq [d]$ ,  $\text{pd}_{A_d} F_U = |U| + 1$  and  $\text{fd}_{A_d} F_U = |U|$ . In particular*

$$\text{pd}_{A_d} D = d + 1, \quad \text{pd}_{A_d} \Pi_d = d,$$

*and  $\text{Ext}_{A_d}^{d+1}(D, A_d) \cong \text{Ext}_{A_d}^d(\Pi_d, A_d)$  contains a copy of  $D^{\mathbb{N}}/D^{(\mathbb{N})}$ .*

*Proof.* Since  $D$  is a unital subring of a field,  $1 \neq 0$  in  $D$ . The sequence  $(1, 1, \dots)$  defines a nonzero element of  $D^{\mathbb{N}}/D^{(\mathbb{N})}$ , hence a nonzero element of  $\text{Ext}_{A_d}^{|U|+1}(F_U, A)$  by [Theorem 13.46](#). A projective resolution of length at most  $|U|$  would make this Ext group zero, so  $\text{pd}_A F_U \geq |U| + 1$ . The free telescope has length  $|U| + 1$ , giving the opposite bound. The flat dimension is [Proposition 13.43](#).

For  $U = [d]$  the quotient is  $D$ , so  $\text{pd}_{A_d} D = d+1$ . Apply  $\text{Hom}_{A_d}(-, P)$  to  $0 \rightarrow \Pi_d \rightarrow A_d \rightarrow D \rightarrow 0$ . Because  $A_d$  is free, its long exact sequence gives, naturally for every target  $P$ ,

$$\text{Ext}_{A_d}^j(\Pi_d, P) \cong \text{Ext}_{A_d}^{j+1}(D, P) \quad (j \geq 1).$$

For  $j > d$  the right side vanishes for every  $P$ , giving  $\text{pd } \Pi_d \leq d$ . For  $j = d \geq 1$  and  $P = A_d$ , it contains the nonzero copy of  $D^{\mathbb{N}}/D^{(\mathbb{N})}$  already constructed. Thus  $\text{pd } \Pi_d \geq d$ , and the same isomorphism gives the asserted embedding in its top Ext group. The argument includes  $d = 1$ , where the tail is flat but has projective dimension one.  $\square$

So for every  $d$  the projective dimension of the arithmetic quotient exceeds its flat dimension by exactly one, as in rank one, and the upper bound of [Proposition 13.12\(iii\)](#) is attained. For  $d = 2$  and  $U = \{1, 2\}$ , a preimage of the constant sequence 1 would need its first representative to contain all the monomials  $1, (X_1 X_2)^{1/2}, (X_1 X_2)^{3/4}, (X_1 X_2)^{7/8}, \dots$ , whose coordinates are below 1; so  $\text{Ext}_{A_2}^3(D, A_2) \neq 0$ , although the flat resolution of [Theorem 13.4](#) has length two. When  $D \neq K$ , the finite Koszul complexes need not be resolutions, as the preceding nonregularity example shows (21, its worked appendix). Every finite initial segment of the equations  $y_n - u_n y_{n+1} = 1$  has a solution, obtained by choosing a last value and solving backwards; the obstruction is that a solution of all of them would force infinitely many monomials into  $y_0$ , so no finite computation proves it (21).

**Theorem 13.48** (Lattice modules over the coordinate cores [21] [merge]). *Let  $d \geq 1$ , let  $D \subseteq K$  be any unital subring,  $M$  a coefficient lattice of Setting (L) of rank  $r \geq 2$ ,  $0 \neq g \in P_{[d]}$  and  $J = J_g(M) \subseteq A_d$ . Then*

$$\text{pd}_{A_d} L(M) = d + 1, \quad \text{pd}_{A_d}(A_d/J) = d + 2,$$

and for every  $A_d$ -module  $P$  there are natural isomorphisms  $\text{Ext}_{A_d}^{d+1}(L(M), P) \cong \text{Ext}_{A_d}^{d+2}(A_d/J, P) \cong \text{Ext}_{A_d}^{d+1}(D, P)^{r-1}$ ; for  $P = A_d$  these groups contain  $(D^{\mathbb{N}}/D^{(\mathbb{N})})^{r-1}$ . If  $K \neq \text{Frac } D$ , the global dimension of  $A_d$  is at least  $d+2$ . For  $r = 1$ ,  $L(M) \cong A_d$  and  $A_d/J$  has projective and flat dimension 1.

*Proof.* Fix a  $D$ -basis of  $M$  and use [Theorem 13.9\(ii\)](#):

$$0 \longrightarrow \Pi_d^{r-1} \longrightarrow A_d^r \longrightarrow L(M) \longrightarrow 0.$$

For every target  $P$ , the free middle term and the sequence  $0 \rightarrow \Pi_d \rightarrow A_d \rightarrow D \rightarrow 0$  give

$$\text{Ext}_{A_d}^{j+1}(L(M), P) \cong \text{Ext}_{A_d}^j(\Pi_d, P)^{r-1} \cong \text{Ext}_{A_d}^{j+1}(D, P)^{r-1} \quad (j \geq 1).$$

These are natural in  $P$  after the basis is fixed; the finite power comes from the displayed direct sum of syzygies. Since  $\text{pd } \Pi_d = d$ , splicing its projective resolution with this presentation gives  $\text{pd } L(M) \leq d+1$ . At  $j = d$  and  $P = A_d$ , the last group contains  $(D^{\mathbb{N}}/D^{(\mathbb{N})})^{r-1}$  by [Theorem 13.46](#). It is nonzero since  $r-1 \geq 1$ , so  $\text{pd } L(M) = d+1$ .

Multiplication by the nonzero monomial  $X^g$  is an isomorphism  $L(M) \rightarrow J$ . Apply the long exact sequence to  $0 \rightarrow J \rightarrow A_d \rightarrow A_d/J \rightarrow 0$ . For every  $j \geq 1$  it gives

$$\text{Ext}_{A_d}^{j+1}(A_d/J, P) \cong \text{Ext}_{A_d}^j(J, P).$$

The upper bound now shifts to  $d+2$ , while the preceding nonzero group shifts to degree  $d+2$ . This proves equality, the displayed natural identifications, and their copies of the coefficient-sequence



quotient. If  $K \neq \text{Frac } D$ , choose  $c \in K \setminus \text{Frac } D$ . Then  $D + Dc$  has rank two, so its cyclic ideal quotient supplies a module of projective dimension  $d + 2$ , giving the global lower bound.

For the separate rank-one case, write  $M = Dm_1$ , with  $m_1 \neq 0$ . Then  $L(M) = m_1 A_d \cong A_d$  and  $J = qA_d$ , where  $q = m_1 X^g$  is a nonzero nonunit: it is a tail monomial and the constant coefficient of every multiple of it is zero. Thus  $A_d/J \neq 0$  and

$$0 \longrightarrow A_d \xrightarrow{q} A_d \longrightarrow A_d/J \longrightarrow 0$$

is a free resolution of length one. This quotient is not flat. If it were, tensoring the injection  $q : A_d \rightarrow A_d$  with it would make multiplication by  $q$  injective on  $A_d/J$ . That map is zero on a nonzero module. Hence its flat dimension is one, and its projective dimension is also one.  $\square$

21 states the theorem for a principal ideal domain  $D$ , which it uses only to give  $M$  a basis; in Setting (L) a lattice has a basis by definition, and extending the statement to every unital  $D \subseteq K$  is the merge's (the proof is 21's). The flat dimensions  $d$  and  $d + 1$  of  $L(M)$  and  $A_d/J$  are [Theorem 13.11](#) (12, 14), which 21 proves again over  $A_d$ . When  $K \neq \text{Frac } D$ , the rank-two lattice used above gives the additional lower bound  $\text{w.gl.dim } A_d \geq d + 1$ . Without that coefficient hypothesis the face quotient still gives  $\text{w.gl.dim } A_d \geq d$ . These bounds alone determine neither global dimension in the general coefficient setting.

The coefficient hypothesis cannot be omitted from the stronger weak bound. For  $d = 1$  and  $D = K$ , one has  $A_1 = K[\mathbb{Q}_{\geq 0}]$ . Every finite family lies in a polynomial ring  $K[X^{1/n}]$ ; a greatest common divisor and its Bézout identity show that every finitely generated ideal of  $A_1$  is principal, as in the earlier coherence counterexample. If  $P$  is torsion free, tensoring a nonzero principal ideal into  $A_1$  is multiplication by its generator on  $P$ , hence injective. The finitely generated ideal criterion therefore makes every torsion-free module flat [29, Lemma 10.39.5]. The kernel of a free presentation of any module is torsion free, so every module has flat dimension at most one. The quotient  $A_1/(X)$  is nonzero and not flat, by the multiplication argument just given. Consequently  $\text{w.gl.dim } A_1 = 1$ , while  $\text{pd}_{A_1} K = 2$  still holds. This is a counterexample to an unconditional lower bound of two for weak global dimension.

*Remark 13.49* (Consistency with the earlier bounds; (H2) cannot be dropped from [Proposition 13.12\(v\)](#) [merge]).

- (i) For  $d = 1$ ,  $\text{pd}_{A_1} D = 2$  is the merge value of [Proposition 13.12\(iii\)](#), and  $\text{pd } L(M) = 2$ ,  $\text{pd}(A_1/J_g(M)) = 3$  are its item (iv) for  $T_{\mathbb{Q}} = A_1$ ; 21 adds an explicit free resolution and an explicit nonzero class. For every  $d$  the values of [Corollary 13.47](#) and [Theorem 13.48](#) are the upper ends of the bounds  $d \leq \text{pd } D \leq d + 1$ ,  $d \leq \text{pd } L(M) \leq d + 1$  and  $d + 1 \leq \text{pd}(A_d/J_g(M)) \leq d + 2$  of [Proposition 13.12\(iii\)](#), (iv).
- (ii) The tail  $\Pi_d$  is countably generated, by the monomials  $X_i^{2^{-n}}$  ( $i \in [d]$ ,  $n \geq 0$ ): a term  $kX^\alpha$  with  $\alpha_i > 0$  equals  $X_i^{2^{-n}} \cdot kX^{\alpha - 2^{-n}\mathbf{e}_i}$  for  $2^{-n} < \alpha_i$ , and the second factor lies in  $\Pi_d$ . Moreover  $A_d$  satisfies (H1) ([Lemma 13.1\(i\)](#)) and (H3) ([Lemma 13.3\(iii\)](#)), and under the embeddings of [Lemma 13.1\(ii\)](#) its elements have nonnegative support. For  $d \geq 2$  it fails (H2): if  $X_1 + X_2 = X^g b$  with  $g > 0$  and  $b \in \Pi_d$ , then under an embedding  $X^\beta \mapsto \omega^{\ell(\beta)}$  of [Lemma 13.1\(ii\)](#) the exponents  $\gamma_1 - g$ ,  $\gamma_2 - g$  of  $b$  are  $\ell(\beta_1)$ ,  $\ell(\beta_2)$  with  $\beta_1, \beta_2 \in P_{[d]}$ ; linear independence gives  $\beta_1 - \beta_2 = \mathbf{e}_1 - \mathbf{e}_2$ , so the second coordinate of  $\beta_2$  is at least 1, and  $g = \ell(\mathbf{e}_2 - \beta_2) \leq 0$  because  $\mathbf{e}_2 - \beta_2$  has no positive coordinate. (Under (H2) the tail would also be flat, by [Theorem 12.2\(i\)](#), whereas  $\text{fd } \Pi_d = d - 1 \geq 1$  by [Corollary 13.8](#); compare [Remark 13.16](#).) Since  $\text{pd}_{A_d} D = d + 1 > 2$ , the conclusion  $\text{pd } D = 2$  of [Proposition 13.12\(v\)](#) fails for  $A_d$ : a countably generated tail does not suffice without (H2), and the rings  $A_d$  realize every value  $\text{pd } D \geq 2$ .

- (iii)  $T_\Gamma$  and 17's countable model  $R_{\aleph_0}^{11}$  satisfy (H1)–(H3) and have  $\text{pd } D = 2$  (Theorem 13.13, Remark 13.32). For the same exponent group, the contrast of Theorem 13.14 between the coordinate cone and the totally ordered cone is now  $d+1$  against 2 for the projective dimension, as it is  $d$  against 1 for the flat dimension.
- (iv) Modules killed by the tail do not see the projective dimension:  $\text{Tor}_{>0}^{A_d}(D, P) = 0$  for every  $D$ -module  $P$  (Corollary 13.5, 12), although  $\text{pd}_{A_d} D = d+1$ ; the witness of Theorem 13.46 uses the target  $A_d$  itself (21).

Nothing here concerns the uncountable constructions of Section 11 or  $R_\kappa^{11}$  with  $\kappa$  uncountable, which satisfy (H2) and whose projective dimensions stay open (Question 18.3).

**Lattice quotients over the ordered core and the class ring.** For the rest of this subsection  $(D, k)$  is  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ , and  $A$  is one of two rings:  $T_\mathbb{Q}$  (14's  $A_0$ , 21's  $R_0$ ), with positive rational scales; or the class ring  $\mathcal{A}_{D,k}$ , that is  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , with positive surreal scales, read in two universes as in the universe-relative reading of Section 2.1. In that reading 21 fixes Grothendieck universes  $\mathcal{U} \in \mathcal{V}$  containing  $k$ , uses the surreals with  $\mathcal{U}$ -small sign sequences and the normal forms with  $\mathcal{U}$ -small supports, and forms all modules, resolutions, tensor products and Tor and Ext groups in  $\mathcal{V}$ , where this ring is a set; this is 13's reading of Remark 13.16. It is explicit size bookkeeping, not an assertion that a proper class is an ordinary ring, and it differs from 17's convention in Section 13.9, under which ordinary Tor is formed over set-sized rings only. In both cases every  $\mathcal{U}$ -small (for  $T_\mathbb{Q}$ : finite) family of tails has a common monomial divisor  $X^\epsilon$ ,  $\epsilon > 0$ , so  $\Pi = \bigcup_{\epsilon > 0} X^\epsilon A$  is a directed union of free ideals, hence flat, and  $\Pi^2 = \Pi$  (Theorem 13.13 for  $T_\mathbb{Q}$ ; the proof of Proposition 3.3, with  $\mathcal{U}$ -small in place of set-sized, for the class ring). These are the only rings of this part; for  $A_d$  with  $d \geq 2$  the tail is not flat and nothing below is asserted there. Let  $M, N \subset k$  be lattices of ranks  $r, s$  with bases  $m_1, \dots, m_r$  and  $n_1, \dots, n_s$ , let  $g, h > 0$  be scales,  $I = J_g(M)$  and  $J = J_h(N)$ . For  $u \in A^r$  put  $m(u) = \sum_j m_j u_j \in B$ , and let  $\text{Syz}_M = \{u \in \Pi^r : m(u) = 0\} \cong \Pi^{r-1}$  (Theorem 13.9(ii)).

**Theorem 13.50** (A flat DGA resolution [21]). *The complex  $\mathbf{P}_\bullet$  with  $\mathbf{P}_0 = A$ ,  $\mathbf{P}_1 = A^r$ ,  $\mathbf{P}_2 = \text{Syz}_M$ ,  $\mathbf{P}_j = 0$  for  $j \geq 3$ , differential  $u \mapsto X^g m(u)$  on  $\mathbf{P}_1$  and the inclusion on  $\mathbf{P}_2$ , is a flat resolution of  $A/I$ . With the  $A$ -module structure, the product*

$$u \cdot v = X^g(m(u)v - m(v)u) \in \mathbf{P}_2 \quad (u, v \in \mathbf{P}_1),$$

*and all products of positive degrees summing to at least three equal to zero, it is a unital, associative, graded-commutative differential graded algebra, and the augmentation  $\mathbf{P}_\bullet \rightarrow A/I$  is a quasi-isomorphism of differential graded algebras.*

*Proof.* Write  $d_1(u) = X^g m(u)$  and let  $d_2$  be the inclusion of  $\text{Syz}_M$  into  $A^r$ . The relation  $m(x) = 0$  for  $x \in \text{Syz}_M$  gives  $d_1 d_2 = 0$ . The presentation in Theorem 13.9(ii), scaled by the nonzero monomial  $X^g$ , identifies  $\ker d_1$  with  $\text{Syz}_M$  and  $\text{im } d_1$  with  $I$ . Since  $d_2$  is injective, this proves exactness of the augmented complex. Its terms in degrees zero and one are free; its degree-two term is isomorphic to  $\Pi^{r-1}$  and is flat because  $\Pi$  is flat in the two rings under consideration. This includes  $r = 1$ , when that term is zero.

The proposed product is well defined. Indeed,  $X^g m(u), X^g m(v) \in I \subseteq \Pi$ , so each coordinate of  $X^g(m(u)v - m(v)u)$  lies in the ideal  $\Pi$ . Applying  $m$  gives

$$X^g(m(u)m(v) - m(v)m(u)) = 0,$$

so the vector belongs to  $\text{Syz}_M$ . The formula is  $A$ -bilinear, changes sign on interchanging  $u, v$ , and is zero when  $u = v$ . Products involving degree zero are the prescribed  $A$ -module actions, with

unit  $1 \in A$ . They obey associativity by bilinearity. If three factors all have positive degree, either parenthesization is zero: a first product is either zero or has degree at least two, and multiplying it by the remaining factor has degree at least three. This proves associativity. Graded commutativity follows from the alternating degree-one formula, commutativity of the scalar action, and the zero higher products.

Use homological degrees, so the Leibniz rule is  $d(ab) = d(a)b + (-1)^{|a|}a d(b)$ . In degrees  $(1, 1)$  it reads

$$d_2(u \cdot v) = X^g m(u)v - X^g m(v)u = d_1(u)v - u d_1(v).$$

For  $u \in P_1$  and  $x \in P_2$  the product  $ux$  is zero, while its required differential is

$$d_1(u)x - u \cdot d_2(x) = X^g m(u)x - X^g(m(u)x - m(x)u) = 0.$$

In the opposite order the degree-two sign is positive, and

$$d_2(x) \cdot u + x d_1(u) = X^g(m(x)u - m(u)x) + X^g m(u)x = 0.$$

Here both equalities use  $m(x) = 0$ . For two degree-two factors, each term on the right has degree three and is zero. Cases involving degree zero are exactly the  $A$ -linearity of the differential, since  $d(A) = 0$ . Thus all degree pairs satisfy the Leibniz rule.

Finally the quotient map in degree zero and the zero maps above it form a multiplicative augmentation:  $d_1$  lands in  $I$ , and positive-degree products have positive degree. Exactness already shows that it induces  $H_0(P_\bullet) \cong A/I$  and zero homology above degree zero. It is therefore the claimed quasi-isomorphism of differential graded algebras.  $\square$

The bounded complex of flat modules  $P_\bullet$  is  $K$ -flat, so  $P_\bullet \otimes_A A/J$  computes  $A/I \otimes_A^L A/J$  and its homology computes the Tor groups. Its multiplication also computes their usual product. Indeed, the chain map  $P_\bullet \otimes_A P_\bullet \rightarrow P_\bullet$  lifts multiplication on  $A/I$  through the multiplicative augmentation. Derived base change to  $A/J$  and passage to homology give precisely the canonical Tor product [33, Tags 06XY and 068G]. In particular this applies to the self-Tor algebra when  $J = I$ . This is the ordinary derived product; 21 does not claim that the model is a cofibrant simplicial commutative algebra, determines higher operations, or is  $E_\infty$ -formal.

**Theorem 13.51** (All Tor groups of two lattice quotients [17; 21]).

- (i)  $\text{Tor}_2^A(A/I, A/J) \cong \mathcal{T}(M, N)$ , a free  $D$ -module of rank  $rs - \text{rank}(MN)$  on which  $A$  acts through  $\text{ct}$ , independently of the scales  $g, h$ ; and  $\text{Tor}_n^A(A/I, A/J) = 0$  for  $n \geq 3$ .
- (ii)  $\text{Tor}_0^A(A/I, A/J) = A/(I + J)$  and  $\text{Tor}_1^A(A/I, A/J) \cong (I \cap J)/IJ$ . Put  $e = \min(g, h)$  and let  $\mathbb{L}_{g,h}$  be  $N$  if  $g < h$ ,  $M$  if  $h < g$ , and  $M \cap N$  if  $g = h$ . Then, as  $A$ -modules and as  $D$ -modules respectively,

$$\text{Tor}_1^A(A/I, A/J) \cong \frac{\mathbb{L}_{g,h} + \Pi}{X^e(MN + \Pi)}, \quad \text{Tor}_1^A(A/I, A/J) \cong \mathbb{L}_{g,h} \oplus \Pi_{(0,e)} \oplus (k/MN)X^e,$$

where  $\Pi_{(0,e)}$  is the group of tails supported in the open interval  $(0, e)$ . The second decomposition is coefficientwise; it is not an  $A$ -module splitting.

*Proof.* (i) Put  $S = \text{Syz}_M$ . Tensoring the flat resolution of Theorem 13.50 with  $A/J$  identifies its degree-two homology with

$$\ker(S/JS \longrightarrow (A/J)^r) = \frac{S \cap J^r}{JS}.$$

There are no boundaries from degree three. The module  $S$  is stable under multiplication by  $B = k + \Pi$  and by  $k$ . Thus  $JS \subseteq X^h S$ . Conversely choose  $0 \neq n \in N$ ; then  $X^h x = (X^h n)(x/n) \in JS$  for every  $x \in S$ . Consequently  $JS = X^h S$ .

For  $x \in S \cap J^r$ , divide its coordinates by  $X^h$  to write  $x = X^h y$ ,  $y \in (N + \Pi)^r$ . Cancellation gives  $m(y) = 0$ . Its constant tuple  $c = \text{ct}(y) \in N^r$  satisfies  $\sum_j m_j c_j = 0$ . Conversely such a tuple lifts to  $X^h c$ ; it has tail coordinates and satisfies the same relation, hence belongs to  $S \cap J^r$ . The kernel of this coefficient map is  $X^h S$ : zero constants mean  $y \in \Pi^r$  and  $m(y) = 0$ . The basis map

$$(c_j) \mapsto \sum_j m_j \otimes c_j$$

therefore identifies  $H_2$  with  $\mathcal{T}(M, N)$ , exactly as in [Theorem 13.34](#). For  $a \in A$ , taking constants of  $ay$  multiplies this tuple by  $\text{ct}(a)$ , proving the asserted  $A$ -action. The resulting coefficient module involves neither scale  $g$  nor scale  $h$ .

Both  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  are principal ideal domains. Hence the kernel  $\mathcal{T}(M, N) \subseteq M \otimes_D N \cong D^{rs}$  is finite free. Localizing the exact coefficient sequence

$$0 \longrightarrow \mathcal{T}(M, N) \longrightarrow M \otimes_D N \longrightarrow MN \longrightarrow 0$$

at  $D \setminus \{0\}$  gives its rank  $rs - \text{rank}(MN)$ . All homology in degree at least three vanishes because the resolution has length at most two. Over  $T_{\mathbb{Q}}$  its quotient has flat dimension exactly two when  $r \geq 2$  by [Theorem 13.11](#); for  $r = 1$  the ideal is principal and the dimension is one. The vanishing assertion includes both cases. For the universe-relative class ring the same argument takes place among ordinary modules in the larger universe; no tensor product of literal proper-class modules is needed.

(ii) Right exactness gives  $(A/I) \otimes_A (A/J) \cong A/(I + J)$ . Tensor the sequence  $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$  with  $A/J$ . Since  $A$  is free, the initial part of the long exact sequence is

$$0 \longrightarrow \text{Tor}_1^A(A/I, A/J) \longrightarrow I/IJ \longrightarrow A/J.$$

The last map is induced by inclusion, so its kernel is  $(I \cap J)/IJ$ . This formula does not assume that  $I$  or  $J$  is flat.

If  $g < h$ , every element of  $J$  belongs to  $I$ , because  $X^{h-g}(N + \Pi) \subseteq \Pi \subseteq M + \Pi$ . Thus  $I \cap J = J$  and  $I + J = I$ ; if  $h < g$  the roles reverse. If  $g = h$ , cancellation of  $X^g$  and uniqueness of the constant and tail parts give

$$I \cap J = X^g((M \cap N) + \Pi), \quad I + J = X^g((M + N) + \Pi).$$

In all three cases the intersection is  $X^{\max(g, h)}(\mathbb{L}_{g, h} + \Pi)$ . Also  $(M + \Pi)(N + \Pi) = MN + \Pi$ , where the left side denotes the additive span of products. Products involving a tail stay in  $\Pi$ , and, for a fixed  $0 \neq m \in M$ , every tail  $w$  is  $m(w/m)$  with  $w/m \in \Pi \subseteq N + \Pi$ . This proves  $IJ = X^{g+h}(MN + \Pi)$ . Since  $g + h - \max(g, h) = e$ , division by  $X^{\max(g, h)}$  is an  $A$ -linear identification with the first displayed quotient.

To obtain the second identification, write a representative uniquely as

$$l + w_{<e} + cX^e + w_{>e}, \quad l \in \mathbb{L}_{g, h}, \quad w_{<e} \in \Pi_{(0, e)}, \quad c \in k,$$

where  $w_{>e}$  has support strictly above  $e$ . Truncation to each of these support pieces preserves admissibility; translation by  $-e$  carries  $w_{>e}$  to a tail. The denominator  $X^e(MN + \Pi)$  consists

exactly of arbitrary such  $w_{>e}$  and boundary coefficients  $c \in MN$ . Thus the map to  $(l, w_{<e}, [c]X^e)$  has precisely that denominator as kernel. It is onto: any three displayed components give an admissible representative by their finite union. This proves the  $D$ -linear direct-sum formula. Tail multiplication can move a term of the open interval to its endpoint or past it, so the coefficientwise decomposition is not an  $A$ -module splitting.  $\square$

So the first Tor keeps a whole interval of scales and a boundary quotient  $k/MN$ , which can have torsion, while the second is a finite-rank coefficient module that does not see the scales at all; neither should be replaced by a leading-term approximation (21). In particular  $\text{Tor}_1 \neq 0$  always: the monomial  $X^{e/2}$  lies in  $\Pi_{(0,e)}$  and survives because the denominator has no support below  $e$ . This holds even if  $L_{g,h} = 0$  and  $\text{Tor}_2 = 0$ . Thus vanishing of  $\text{Tor}_2$  is never Tor-independence.

**Lemma 13.52** (The compositum [21]). *Let  $\mathfrak{o}_1, \mathfrak{o}_2 \subset k$  be orders over  $D$ : unital subrings, finite free over  $D$ , whose  $\text{Frac}(D)$ -spans  $K_1, K_2$  are fields. Then the  $\text{Frac}(D)$ -span of the product lattice  $\mathfrak{o}_1\mathfrak{o}_2$  is the compositum  $K_1K_2$ , so  $\text{rank}_D(\mathfrak{o}_1\mathfrak{o}_2) = [K_1K_2 : \text{Frac } D]$ .*

*Proof.* Put  $F = \text{Frac } D$  and let  $V$  be the  $F$ -span in  $k$  of all products  $xy$  with  $x \in K_1$  and  $y \in K_2$ . Products of finite  $F$ -bases of  $K_1$  and  $K_2$  span  $V$ , so it is finite dimensional. Since both fields contain 1,  $V$  contains  $K_1, K_2$  and 1. It is closed under multiplication: the product of  $xy$  and  $x'y'$  is  $(xx')(yy')$ , again a spanning product, and distributivity handles finite sums.

For  $0 \neq v \in V$ , multiplication by  $v$  is an injective  $F$ -linear endomorphism of  $V$ , because  $V$  lies in the field  $k$ . Finite dimensionality makes this map surjective. As  $1 \in V$ , some  $w \in V$  satisfies  $vw = 1$ . Thus  $V$  is a field containing  $K_1$  and  $K_2$ , and so it contains their compositum inside  $k$ . Conversely every spanning product belongs to that compositum, giving  $V = K_1K_2$ .

Each element of  $K_i = F\mathfrak{o}_i$  is a finite  $F$ -linear combination of elements of  $\mathfrak{o}_i$ . Expanding both factors shows that  $V = F(\mathfrak{o}_1\mathfrak{o}_2)$ , where the product lattice means the  $D$ -span of products. It is finitely generated and torsion free over the principal ideal domain  $D$ , hence free. Its rank is the dimension of this  $F$ -span, namely  $[K_1K_2 : F]$ .  $\square$

**Theorem 13.53** (Second Tor detects linear disjointness [21]). *For orders  $\mathfrak{o}_1, \mathfrak{o}_2$  as in Lemma 13.52 and all scales  $g, h$ ,*

$$\text{rank}_D \text{Tor}_2^A(A/J_g(\mathfrak{o}_1), A/J_h(\mathfrak{o}_2)) = [K_1 : \text{Frac } D][K_2 : \text{Frac } D] - [K_1K_2 : \text{Frac } D],$$

*and this  $\text{Tor}_2$  vanishes if and only if  $K_1$  and  $K_2$  (with their given embeddings in  $k$ ) are linearly disjoint over  $\text{Frac } D$ .*

*Proof.* Set  $F = \text{Frac } D$  and  $T = \mathcal{T}(\mathfrak{o}_1, \mathfrak{o}_2)$ . By Theorem 13.51(i), the second Tor group is  $T$  for every choice of scales. The ranks of the two orders are  $[K_1 : F]$  and  $[K_2 : F]$ , while Lemma 13.52 computes the rank of their product as  $[K_1K_2 : F]$ . Substituting these three ranks in the pair formula proves the displayed equality.

For the vanishing criterion, localize the exact sequence

$$0 \longrightarrow T \longrightarrow \mathfrak{o}_1 \otimes_D \mathfrak{o}_2 \xrightarrow{\text{mult}} \mathfrak{o}_1\mathfrak{o}_2 \longrightarrow 0$$

at  $D \setminus \{0\}$ . Localization is exact and commutes with tensor products, so it becomes

$$0 \longrightarrow F \otimes_D T \longrightarrow K_1 \otimes_F K_2 \xrightarrow{\text{mult}} K_1K_2 \longrightarrow 0.$$

The displayed multiplication map uses the given embeddings in  $k$ . Its injectivity is precisely linear disjointness in that common field. Thus this condition is equivalent to  $F \otimes_D T = 0$ .

Finally  $T$  is torsion free over  $D$ , as a submodule of the finite free module  $\mathfrak{o}_1 \otimes_D \mathfrak{o}_2$ . If  $F \otimes_D T = 0$ , each  $t \in T$  is killed by some nonzero  $d \in D$ ; torsion freeness forces  $t = 0$ . The converse is immediate. Hence localization loses no nonzero element relevant to this criterion, and  $T = 0$  exactly in the linearly disjoint case.  $\square$

For  $\mathfrak{o}_1 = \mathbb{Z}[\sqrt{2}]$  and  $\mathfrak{o}_2 = \mathbb{Z}[\sqrt{3}]$  this is 17's example after [Theorem 13.42](#) (rank zero), and for  $\mathfrak{o}_1 = \mathfrak{o}_2 = \mathbb{Z}[\sqrt{2}]$  the rank is two; 21 adds that the scales may be  $(1, 2)$  over  $T_{\mathbb{Q}}$  or  $(1, \omega)$  over the class ring, and that  $\text{Tor}_1 \neq 0$  in both cases. For  $D = \mathbb{Z}[i]$  the test is over  $\mathbb{Q}(i)$ . The criterion is a translation between two structures: it creates no extensions and does not decide linear disjointness without access to the arithmetic of the fields (21). Its diagonal counterpart, the maximal self-Tor rank detecting a field, is 17's [Proposition 13.37](#).

**The self-Tor algebra.** Take  $N = M$ ,  $h = g$ , so  $J = I$ , and write  $\mathbb{T}_j = \text{Tor}_j^A(A/I, A/I)$ :  $\mathbb{T}_0 = A/I$ ,  $\mathbb{T}_1 = I/I^2$  and  $\mathbb{T}_2 \cong \mathcal{T}(M, M)$ . The boundary map

$$\mathbf{b} : \mathbb{T}_1 \longrightarrow M, \quad [X^g(m + w)] \longmapsto m \quad (m \in M, w \in \Pi),$$

is well defined because  $I^2 = X^{2g}(M^2 + \Pi) \subseteq X^g\Pi$ . It is onto, by the classes  $[X^g m]$ , and its kernel is  $X^g\Pi/I^2$ . Since  $\Pi I = X^g\Pi$ , this kernel is exactly  $\Pi\mathbb{T}_1$ , giving  $\mathbb{T}_1/\Pi\mathbb{T}_1 \cong M$ . The target has the  $A$ -action through  $\text{ct}$ .

For the product we fix the identification of  $\mathbb{T}_2$  with  $\mathcal{T}(M, M)$  by sending the surviving coefficient vector  $(c_j)$  of [Theorem 13.34](#) to  $\sum_j c_j \otimes m_j$ . This is the flip of that proof's  $\sum_j m_j \otimes c_j$  convention. The flip preserves the multiplication kernel because multiplication in  $k$  is commutative; fixing it here determines the sign in the next formula.

**Theorem 13.54** (The self-Tor product [21]). *For  $x, y \in \mathbb{T}_1$ ,*

$$xy = \mathbf{b}(x) \otimes \mathbf{b}(y) - \mathbf{b}(y) \otimes \mathbf{b}(x) \in \mathcal{T}(M, M).$$

*All products of positive degrees summing to at least three vanish, and  $A$  acts on  $\mathbb{T}_2$  through  $\text{ct}$ . With the  $A/I$ -module  $\mathbb{T}_1$ , these rules determine the graded algebra  $\text{Tor}^A(A/I, A/I)$ .*

*Proof.* Use the algebra  $\mathbf{P}_{\bullet} \otimes_A A/I$  from [Theorem 13.50](#). Its degree-one differential is zero, because each  $X^g m_j$  belongs to  $I$ . Thus every vector of  $(A/I)^r$  is a cycle, and the degree-two differential has image the image of  $\text{Syz}_M$  in  $(A/I)^r$ . Consequently

$$H_1 = \frac{A^r}{\text{Syz}_M + IA^r} \xrightarrow{\sim} I/I^2, \quad [u] \longmapsto [X^g m(u)].$$

Indeed, the presentation map  $A^r \rightarrow I$  is onto with kernel  $\text{Syz}_M$ , and it takes  $IA^r$  onto  $I^2$ . This also proves the displayed map is independent of representatives.

Choose lifts  $u, v \in A^r$  for  $x, y$ , and put  $a = m(\text{ct } u)$  and  $b = m(\text{ct } v)$  in  $M$ . The boundary map just defined gives  $\mathbf{b}(x) = a$ ,  $\mathbf{b}(y) = b$ . Their product is represented by

$$z = X^g(m(u)v - m(v)u).$$



It belongs to  $\text{Syz}_M$  by the DGA calculation, and its coordinates lie in  $I$ , so it is a degree-two cycle. To apply the coefficient identification of [Theorem 13.51\(i\)](#), divide by  $X^g$  and take constants. The resulting entries are

$$c_j = a \text{ ct}(v)_j - b \text{ ct}(u)_j \in M.$$

They satisfy  $\sum_j m_j c_j = ab - ba = 0$ . Under the stipulated flipped identification, their tensor is

$$\begin{aligned} \sum_j c_j \otimes m_j &= \sum_j (a \text{ ct}(v)_j - b \text{ ct}(u)_j) \otimes m_j \\ &= a \otimes \sum_j \text{ ct}(v)_j m_j - b \otimes \sum_j \text{ ct}(u)_j m_j = a \otimes b - b \otimes a. \end{aligned}$$

Here the coefficients moved across the tensor sign belong to  $D$ . The formula depends only on  $\mathbf{b}(x)$  and  $\mathbf{b}(y)$ , so it is unaffected by the chosen lifts. With the unflipped coefficient convention it would have the opposite sign.

The complex has no terms above degree two, so all the stated higher products vanish. Its degree-zero algebra is  $A/I$ ; its degree-two action factors through  $\text{ct}$  by [Theorem 13.51\(i\)](#). Together with the specified  $A/I$ -module structure on  $\mathsf{T}_1$ , these actions and the computed degree-one product cover every pair of nonzero degrees, and hence determine the graded algebra.  $\square$

**Corollary 13.55** (Alternating and symmetric relations [21]). *Let  $\text{Ind}_2$  be  $\mathsf{T}_2$  modulo the  $D$ -span of the products  $\mathsf{T}_1 \mathsf{T}_1$ , and  $\ell = \text{rank}_D(M^2)$ . There is an exact sequence of  $D$ -modules*

$$0 \longrightarrow \bigwedge_D^2 M \longrightarrow \mathsf{T}_2 \longrightarrow \ker(\text{Sym}_D^2 M \rightarrow M^2) \longrightarrow 0,$$

so  $\text{Ind}_2 \cong \ker(\text{Sym}_D^2 M \rightarrow M^2)$ ,  $\text{rank } \mathsf{T}_2 = r^2 - \ell$  and  $\text{rank } \text{Ind}_2 = r(r+1)/2 - \ell$ . For an order  $(M^2 = M)$  the ranks are  $r(r-1)$  and  $r(r-1)/2$ : the products account for half of  $\mathsf{T}_2$ , and the other half consists of symmetric multiplication relations.

*Proof.* Write  $V = M \otimes_D M$  and let  $\mu : V \rightarrow M^2$  be multiplication. The alternating bilinear map  $(x, y) \mapsto x \otimes y - y \otimes x$  induces

$$\alpha : \bigwedge_D^2 M \longrightarrow V.$$

In a basis  $m_1, \dots, m_r$ , the basis vector  $m_i \wedge m_j$  for  $i < j$  maps to  $m_i \otimes m_j - m_j \otimes m_i$ . Its coefficient at  $m_i \otimes m_j$  distinguishes it from all other such images. Thus  $\alpha$  is injective; no division by two is involved. Commutativity gives  $\mu\alpha = 0$ , so its image lies in  $\ker \mu = \mathsf{T}_2$ .

Let  $q : V \rightarrow \text{Sym}_D^2 M$  be the quotient map. Its kernel is generated by the differences  $x \otimes y - y \otimes x$ , hence equals  $\text{im } \alpha$ . Multiplication factors as  $\mu = \bar{\mu}q$ , where  $\bar{\mu} : \text{Sym}_D^2 M \rightarrow M^2$  is onto. Restricting  $q$  to  $\ker \mu$  gives a map to  $\ker \bar{\mu}$  with kernel  $\text{im } \alpha$ . This restriction is onto: if  $w \in \ker \bar{\mu}$ , take any lift  $v \in V$  under  $q$ ; then  $\mu(v) = \bar{\mu}(w) = 0$ . This proves the claimed short exact sequence.

The boundary map  $\mathbf{b}$  is surjective, so for every pair  $a, b \in M$  choose degree-one classes mapping to them. [Theorem 13.54](#) says their product is  $\alpha(a \wedge b)$ . Conversely every degree-one product has this form. Thus the  $D$ -span of products is exactly  $\text{im } \alpha$ , and the asserted identification of  $\text{Ind}_2$  follows from the exact sequence.

The tensor and symmetric squares of the free rank- $r$  module have ranks  $r^2$  and  $r(r+1)/2$ , respectively. Both multiplication maps are onto the rank- $\ell$  module  $M^2$ . Localizing their kernel sequences at  $D \setminus \{0\}$  gives ranks  $r^2 - \ell$  and  $r(r+1)/2 - \ell$ . For an order, its unit and closure under multiplication give  $M^2 = M$ , so  $\ell = r$ . The product submodule has rank  $\binom{r}{2}$ , equal to the rank of the quotient, and their sum is  $r(r-1)$ , as asserted.  $\square$

Here “half” refers to ranks. The symmetric square is the quotient of the tensor square by alternating differences. The right-hand term is a submodule of that quotient, not a chosen submodule of symmetric tensors. Over the principal ideal domain  $D$ , this relation kernel is free, so the displayed sequence admits a  $D$ -linear splitting, but the statement specifies no such choice. In particular it does not use the averaging operators  $(1 \pm \text{flip})/2$ : two is not a unit in  $\mathbb{Z}$  or  $\mathbb{Z}[i]$ .

The rank  $r^2 - \ell$  and its range are 17’s [Corollary 13.36](#); new here are the product and the exact sequence separating alternating and symmetric relations (14’s [Theorem 13.20](#) computes exterior and symmetric powers of  $L(M)$ , not a Tor product). For a lattice that is not a ring the symmetric half can vanish: if  $M = D + D\alpha$  with  $1, \alpha, \alpha^2$  independent over  $\text{Frac } D$ , then  $\ell = 3$ ,  $\mathbb{T}_2 \cong D$  is spanned by the alternating tensor, and  $\text{Ind}_2 = 0$  (compare 17’s examples  $\sqrt[3]{2}$  and  $\tau$ ). So the minimal polynomial of an element of degree at least three cannot be read from a two-generator  $\mathbb{T}_2$  alone (21).

**Framed reconstruction of orders.** For an order  $\mathfrak{o} \subset k$  over  $D$  and  $I = J_g(\mathfrak{o})$ , the *framed boundary data* are the scale  $X^g$ , the identification  $\mathbb{T}_1/\text{IIT}_1 \cong \mathfrak{o}$  given by  $\mathbf{b}$ , under which the class of the generator  $X^g$  is  $1 \in \mathfrak{o}$ , and the resulting embedding  $\mathbb{T}_2 \cong \mathcal{T}(\mathfrak{o}, \mathfrak{o}) \subseteq \mathfrak{o} \otimes_D \mathfrak{o}$ . Equivalently, the two ideal sequences identify  $\mathbb{T}_2$  with the kernel of the multiplication  $I \otimes_A I \rightarrow I$ , and the coefficient boundaries give the embedding ([Theorem 13.34](#)); it is an explicit map, not a chosen isomorphism of free groups.

**Theorem 13.56** (Framed reconstruction [21]). *The  $D$ -module  $\mathfrak{o}$ , its element 1 and the submodule  $\mathcal{T}(\mathfrak{o}, \mathfrak{o}) \subseteq \mathfrak{o} \otimes_D \mathfrak{o}$  determine the multiplication of  $\mathfrak{o}$ :*

$$xy = z \iff x \otimes y - 1 \otimes z \in \mathcal{T}(\mathfrak{o}, \mathfrak{o}).$$

*A  $D$ -module isomorphism  $f : \mathfrak{o} \rightarrow \mathfrak{o}'$  between orders is a unital ring isomorphism if and only if  $f(1) = 1$  and  $(f \otimes f)(\mathcal{T}(\mathfrak{o}, \mathfrak{o})) = \mathcal{T}(\mathfrak{o}', \mathfrak{o}')$ .*

*Proof.* Let  $\mu : \mathfrak{o} \otimes_D \mathfrak{o} \rightarrow \mathfrak{o}$  be multiplication and  $T = \ker \mu$ . The map  $s : \mathfrak{o} \rightarrow \mathfrak{o} \otimes_D \mathfrak{o}$ ,  $s(z) = 1 \otimes z$ , satisfies  $\mu s = \text{id}$ . For any  $x, y, z \in \mathfrak{o}$ ,

$$\mu(x \otimes y - 1 \otimes z) = xy - z.$$

Thus the displayed membership in  $T$  is equivalent to  $z = xy$ . In particular there exists exactly one such  $z$ . To reconstruct the product from the framed data, use this unique element as its definition. The given multiplication verifies existence and uniqueness, so no additional choice of a quotient isomorphism is part of the reconstruction.

Suppose first that  $f$  is a unital ring isomorphism. It fixes the specified unit, and

$$\mu' \circ (f \otimes f) = f \circ \mu$$

on pure tensors and hence on their sums. This puts  $(f \otimes f)(T)$  in  $T' = \ker \mu'$ ; applying the same argument to  $f^{-1}$  gives equality. Conversely assume the two conditions in the statement. For  $z = xy$ , the tensor  $x \otimes y - 1 \otimes z$  belongs to  $T$ . Its image is  $f(x) \otimes f(y) - 1 \otimes f(z) \in T'$ , since  $f(1) = 1$ . The already proved membership criterion in  $\mathfrak{o}'$  gives  $f(z) = f(x)f(y)$ . The given  $D$ -module isomorphism is additive and bijective; this multiplicativity and the unit condition therefore make it a unital ring isomorphism.  $\square$

The reconstruction uses the located submodule  $\mathcal{T}(\mathfrak{o}, \mathfrak{o})$  and the distinguished unit, not just the abstract isomorphism type of the second Tor group. It differs from [Theorem 13.39](#), which recovers

the order as the constant quotient of  $\text{End}_A(J_\gamma(\mathfrak{o}))$ . The following basis description makes the framed reconstruction an integral calculation.

Every element of the finite free  $D$ -algebra  $\mathfrak{o}$  is integral over  $D$ , for example by the characteristic polynomial of its multiplication matrix. Since  $D$  is integrally closed,  $\mathfrak{o} \cap \text{Frac } D = D$ . Consequently  $\mathfrak{o}/D1$  is torsion free: if  $0 \neq d \in D$  and  $dx \in D1$ , then  $x \in \text{Frac } D \cap \mathfrak{o} = D1$ . This quotient is finitely generated over the principal ideal domain  $D$ , hence free. Lifting a basis of it, together with  $e_0 = 1$ , gives a basis  $e_0, e_1, \dots, e_{r-1}$  of  $\mathfrak{o}$ . In particular the unit is primitive in the integral lattice; a basis over the fraction field alone would not suffice.

For this basis put

$$t_{ij} = e_i \otimes e_j - e_0 \otimes (e_i e_j) \quad (1 \leq i \leq r-1, 0 \leq j \leq r-1).$$

Each  $t_{ij}$  belongs to  $\ker \mu$ . In the tensor basis, its coefficient at  $e_i \otimes e_j$  is one and all its other terms have first index zero. These distinct coordinates prove integral independence. Subtracting the corresponding multiples of the  $t_{ij}$  from any tensor removes all terms with first index positive, leaving  $e_0 \otimes z$ . If the original tensor was in the kernel, so is this remainder, and applying  $\mu$  gives  $z = 0$ . Thus the  $t_{ij}$  form a  $D$ -basis of the kernel, including the empty basis when  $r = 1$ .

The framed relations recover each product  $e_i e_j$ , hence its multiplication matrix. Taking traces yields the trace form  $(\text{Tr}_{\text{Frac}(D)\mathfrak{o}/\text{Frac } D}(e_i e_j))_{ij}$  and its determinant, the discriminant of this order in the chosen basis. For  $\mathbb{Z}[\sqrt{m}]$  with  $m > 0$  a nonsquare integer, the two relations are

$$e_1 \otimes e_0 - e_0 \otimes e_1, \quad e_1 \otimes e_1 - m e_0 \otimes e_0.$$

The first is, up to sign, a degree-one product; the second survives in  $\text{Ind}_2$ . The trace form is  $\text{diag}(2, 2m)$ , with discriminant  $4m$ . Thus  $\mathbb{Z}[\sqrt{2}]$  and  $\mathbb{Z}[\sqrt{3}]$  both have abstract second Tor group  $\mathbb{Z}^2$ , but their framed relations recover the distinct discriminants 8 and 12.

For  $\mathbb{Z}[\varphi]$ ,  $\varphi^2 = \varphi + 1$ , the square relation is  $e_1 \otimes e_1 - e_0 \otimes e_1 - e_0 \otimes e_0$ . The trace matrix is  $\begin{pmatrix} 2 & 1 \\ 1 & 3 \end{pmatrix}$ , with discriminant 5. For  $\mathbb{Z}[\sqrt[3]{2}]$ , use  $e_0 = 1$ ,  $e_1 = \sqrt[3]{2}$ ,  $e_2 = (\sqrt[3]{2})^2$ . The three alternating tensors, together with

$$e_1 \otimes e_1 - e_0 \otimes e_2, \quad e_1 \otimes e_2 - 2e_0 \otimes e_0, \quad e_2 \otimes e_2 - 2e_0 \otimes e_1,$$

form an integral basis of the six-dimensional relation module. To check spanning integrally, use the alternating tensors to replace  $e_1 \otimes e_0$ ,  $e_2 \otimes e_0$  and  $e_2 \otimes e_1$  by their flipped tensors, then use the three remaining relations. Every kernel element reduces to  $e_0 \otimes z$  and hence to zero. There are six generators of the free rank-six kernel, so they are a basis. The trace form and discriminant are

$$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 0 & 6 \\ 0 & 6 & 0 \end{pmatrix}, \quad -108.$$

Over  $\mathbb{Z}[i]$ , the order  $\mathbb{Z}[i][\sqrt{2}]$  has the same two quadratic relations and trace matrix as above, with discriminant 8. Complex conjugation takes  $J_g(M)$  to  $J_g(\overline{M})$  and transports these maps coefficientwise. In the Gaussian case this transport is semilinear with respect to conjugation on  $D = \mathbb{Z}[i]$ , rather than  $D$ -linear (21).

**Theorem 13.57** (The derived constant-term shadow [17; 21]). *Let  $E$  be a  $D$ -module on which  $A$  acts through  $\text{ct}$ . Then in the derived category of  $D$ -modules*

$$A/I \otimes_A^{\text{L}} E \cong E \oplus (M \otimes_D E)[1],$$

so  $\mathrm{Tor}_0^A(A/I, E) = E$ ,  $\mathrm{Tor}_1^A(A/I, E) \cong M \otimes_D E$  and  $\mathrm{Tor}_n^A(A/I, E) = 0$  for  $n \geq 2$ . For  $E = D$  the algebra  $\mathbf{P}_\bullet \otimes_A D$  is the square-zero differential graded algebra  $D \oplus M[1]$ .

*Proof.* The action on  $E$  factors through  $A/\Pi = D$ , so  $\Pi E = 0$ . Tensor associativity and the quotient formula give

$$\Pi \otimes_A E \cong (\Pi \otimes_A D) \otimes_D E \cong (\Pi/\Pi^2) \otimes_D E = 0,$$

because  $\Pi^2 = \Pi$ . Equivalently, writing any tail as a finite sum  $p = \sum_\nu a_\nu b_\nu$  with  $a_\nu, b_\nu \in \Pi$  gives  $p \otimes e = \sum_\nu a_\nu \otimes b_\nu e = 0$ . No flatness of  $E$  over  $D$  is required. Since  $\mathrm{Syz}_M \cong \Pi^{r-1}$ , its tensor product with  $E$  is zero as well.

Use the bounded flat resolution of [Theorem 13.50](#). After tensoring with  $E$ , it has  $E$  in degree zero,  $E^r$  in degree one, and zero elsewhere. Its degree-one differential has entries  $X^g m_j \in \Pi$ , so it acts as zero on  $E$ . The chosen lattice basis gives the  $D$ -linear isomorphism

$$E^r \longrightarrow M \otimes_D E, \quad (v_j) \longmapsto \sum_j m_j \otimes v_j.$$

Thus the tensor complex itself, not just its homology, is the direct sum of the two stated modules with zero differentials. Here [1] puts the second summand in homological degree one (cohomological degree minus one). This proves the derived isomorphism and all three assertions about  $\mathrm{Tor}$ .

For  $E = D$ , the tensor complex retains its DGA structure. The degree-zero multiplication and action are the usual ones on  $D$  and  $M$ ; every product of two degree-one terms is zero because the degree-two term has vanished. Under the displayed basis identification its multiplication is therefore

$$(d, m)(d', m') = (dd', dm' + d'm),$$

which is precisely the square-zero DGA  $D \oplus M[1]$ . The assertion for general  $E$  is about complexes of  $D$ -modules; no algebra structure on an arbitrary  $D$ -module  $E$  is assumed.  $\square$

The  $\mathrm{Tor}$  groups are 17's [Theorem 13.42](#) (over  $R_\kappa^{11}$ ), shifted by one degree; 21 proves them over  $T_{\mathbb{Q}}$  and over the class ring in the reading above, and the square-zero model is new. Consequently all orders of one rank, for instance those of discriminants 8, 12 and 5, have isomorphic derived constant-term shadows, although their framed self- $\mathrm{Tor}$  data recover them: derived constant-term reduction keeps the additive lattice and erases its multiplication (21). A  $D$ -module isomorphism between two rank- $r$  orders induces an isomorphism of these square-zero DGAs, even when it does not preserve the original order products. No equivalence of their other derived invariants is inferred.

For the literal class ring, every set-sized module acts through  $\mathrm{ct}$  by [Theorem 5.1\(iv\)](#). In the two-universe reading used here, this means modules with  $\mathcal{U}$ -small underlying sets, as specified in [Section 2.1](#). It does not mean every module formed in  $\mathcal{V}$ : the regular module  $A$  is a set there and has  $\Pi A = \Pi \neq 0$ . The shadow theorem itself only needs the stated hypothesis that  $E$  acts through  $\mathrm{ct}$ . On the finite-support core  $T_{\mathbb{Q}}$ , even a small target need not have this property: the augmentation sending every monomial to 1 is a map to  $k$  ([Proposition 13.2\(ii\)](#)), and gives a module on which the tail acts nontrivially. Thus the specified action hypothesis is essential in both readings (21).

### 13.11 Polyhedral cores: face topology and flat dimension

This subsection is manuscript 26's ([Section 1.3](#)). It replaces the coordinatewise cone  $\mathbb{Q}_{\geq 0}^d$  of 12's cores  $A_d$  by an arbitrary pointed full-dimensional rational polyhedral cone  $\mathcal{C} \subseteq \mathbb{R}^d$ , and the Boolean face lattice of [Theorem 13.4](#) by the face lattice of  $\mathcal{C}$ . Every face  $F$  gives a flat, idempotent,

non-finitely-generated ideal  $I_F$  of projective dimension one, a countable union of principal ideals, with  $I_F \otimes I_G \cong I_{F \vee G}$  (Theorems 13.61 and 13.62); the radical coefficient-full monomial ideals are exactly the sums  $\mathfrak{r}_{\mathcal{F}}$  over face antichains (Theorem 13.63); and the flat dimension of  $A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}$  is the top degree of the reduced homology of finite join complexes, witnessed by explicit cap modules (Theorem 13.70). For the quotient keeping one face  $F$  this is

$$\mathrm{fd}_{A_{\mathcal{C}}} A_{\mathcal{C}}(F) = \mu_{\mathcal{C}}(F) = \max\{\dim G : G \text{ a face of } \mathcal{C}, G \cap F = \{0\}\}$$

(Theorem 13.75; 26's Theorem A). Consequently  $\mathrm{fd} D = d$  for every such cone;  $\mathcal{C}$  is simplicial if and only if every facet quotient has flat dimension one (Theorem 13.79); cones over cubes have facet quotients of every flat dimension up to  $d-1$  (Theorem 13.81); every radical quotient is a homological epimorphism (Theorem 13.72); over a principal ideal domain  $D$ , modules acting through  $\mathrm{ct}$  have flat dimension 1 or  $d$  (Theorem 13.85); and the inclusion of a core of dimension at least two in a total-order core, or in **Oz**, fails the equational flatness criterion (Theorem 13.88). This answers, for the flat dimensions of radical coefficient-full monomial quotients, the non-simplicial-cone clause of Question 18.12; projective dimensions, arbitrary ideals, localizations and weak global dimensions stay open there. 26's Theorems A–D summarize Theorems 13.70, 13.72, 13.75, 13.77, 13.79, 13.81 and 13.88 and Proposition 13.58.

26 pins 7b256e0 (23 September 2026), whose text of this report is the one after the sixth merge (Git blob ef7aef1, the same as 24's and 25's pins); it read lines 1–210 and 3300–3520 of that text (12's resolution, Tor formula and flat dimensions, which it credits: its directly compared labels are Theorems 13.4 and 13.7, Corollaries 13.5 and 13.8, and Proposition 13.6), and it credits 21's exact projective dimension from the report's abstract without using it. 26 does not notice one reduction, which the merge adds: a rational simplicial cone core *is* a coordinate core. If  $\mathcal{C}$  is simplicial with rational ray vectors  $v_1, \dots, v_d$ , the coordinates in this basis are a rational linear isomorphism  $\mathbb{Q}^d \rightarrow \mathbb{Q}^d$  carrying  $\mathcal{C} \cap \mathbb{Q}^d$  onto  $\mathbb{Q}_{\geq 0}^d$ , hence a coefficient-preserving ring isomorphism  $A_{\mathcal{C}} \cong A_d$  sending the face spanned by the rays  $\{v_i : i \notin U\}$  to the coordinate face that 21's quotient  $F_U$  keeps [merge]. Under it: 26's Corollary 13.80 re-derives 21's Proposition 13.43, and 21's Corollary 13.47 gives the projective dimension  $d - \dim F + 1$  for proper faces  $F$ , the upper endpoint of Corollary 13.71; 26's Theorem 13.61 for coordinate faces is Lemma 13.3(i), (iii) with Proposition 13.12(i), (ii); its Proposition 13.76 for an orthant is Theorem 13.4; its Theorem 13.72 for  $\mathfrak{r}_{\mathcal{F}} = \Pi_{\mathcal{C}}$  is Corollary 13.5; its Proposition 13.84 is Proposition 13.2(i); and its Theorem 13.88 for an orthant is the nonflatness of Theorem 13.14, proved there by an intersection certificate over  $T_{\Gamma}$  and here by the equational criterion with a lexicographic embedding, 26 adding the failure of that criterion in **Oz** itself; it is kept as a second route. These are credited; the rest of 26 is new. 26 credits as antecedents the cellular resolutions of Bayer–Sturmfels [70], the lcm-lattice homology of Gasharov–Peeva–Welker [71], the real-exponent global dimensions of Geist–Miller [72] and Miller's real polyhedral theory [73], none of which it claims or reproves. The numbering of 26 cited here is that of its own manuscript.

**Notation and renamings.** In this subsection  $d \geq 1$ ,  $\mathcal{C} \subseteq \mathbb{R}^d$  is a pointed ( $\mathcal{C} \cap (-\mathcal{C}) = \{0\}$ ), full-dimensional, rational polyhedral cone,  $K$  is a field and  $D \subseteq K$  a unital subring, as in Section 13.1. Faces of  $\mathcal{C}$  include  $\{0\}$  and  $\mathcal{C}$ ; the letters  $F, G, H$  denote faces (not subsets of  $\Lambda$ , a field or a group),  $F \vee G$  is the least face containing both,  $<$  is strict inclusion, and a face is identified with its rational points where exponents are meant. Put

$$P_{\mathcal{C}} = \mathcal{C} \cap \mathbb{Q}^d, \quad \Pi_{\mathcal{C}} = \bigoplus_{0 \neq m \in P_{\mathcal{C}}} KX^m, \quad A_{\mathcal{C}} = D \oplus \Pi_{\mathcal{C}}, \quad (13.5)$$

finite sums,  $X^m X^n = X^{m+n}$ , graded by  $\mathbb{Q}^d$ . For  $\mathcal{C} = \mathbb{R}_{\geq 0}^d$  these are  $P_{[d]}$ ,  $\Pi_d$  and  $A_d$ . 26's symbols are renamed as follows.

- Its cone  $C$ , field  $k$ , ring  $A_D(C)$  (26's  $\mathcal{A}$ ; here  $\mathcal{A}$  is  $D + \Pi_k$ ), monoid  $M = C \cap \mathbb{Q}^d$  and tail  $I_+$  are  $\mathcal{C}$ ,  $K$ ,  $A_{\mathcal{C}}$ ,  $P_{\mathcal{C}}$  and  $\Pi_{\mathcal{C}}$  ( $C_p$  are 12's complexes and  $M$  a coefficient lattice); its face quotient  $A_D(F)$  is  $A_{\mathcal{C}}(F)$ .
- Its face ideal  $J_F$  is  $I_F$ : for a coordinate face on  $S$  it is 12's  $I_S$  of (13.3), whereas  $J_F = (X_i : i \in F)A$  is 12's ideal of Section 13.3. Its radical ideal  $I_{\mathcal{F}} = \sum_i J_{F_i}$  is  $\mathfrak{r}_{\mathcal{F}}$ , its face primes  $P_H$  are  $\mathfrak{p}_H$  ( $P_{\Lambda}$  is 12's monoid), and its smallest face  $H(m)$  containing  $m$  is  $\text{fc}(m)$  ( $H$  is a group,  $H_a$  the scale subgroup).
- Its  $V_G = \{i : F_i \subseteq G\}$  is  $\text{At}(G)$  ( $V_F$  are 12's test modules), its join complexes  $\Delta_{<G}$  are  $\mathbf{K}_{<G}$  ( $\Delta$  is 21's telescope map and the difference operator), its  $h(\mathcal{F}; k)$  is  $\text{len}(\mathcal{F})$ , and its complexes  $\mathcal{C}_*$ ,  $\mathcal{R}_*$ ,  $E_*$  are  $C_*^{\mathcal{F}}$ ,  $R_*^{\mathcal{F}}$  and  $C_*^{\text{cell}}$  (compare 21's  $C_*^U$ ).
- Its detecting module  $N_{G,b}$ , with ring  $S_G = k[G \cap \mathbb{Q}^d]$ , cap ideal  $L_{G,b}$  and facet functionals  $\eta_j$ , is  $\mathcal{W}_{G,b}$ , with  $K[P_G]$ ,  $\mathfrak{z}_{G,b}$  and  $\ell_j^G$  ( $N$  is a module,  $\mathfrak{c}$  the continuum,  $\eta$  18's translation); it plays the role of 12's test modules  $V_F$ . Its rays  $\mathcal{R}_F$  outside a face are written out ( $\mathcal{R}$  is 14's Rees algebra).
- Its embedding functional  $\lambda$  is  $\lambda_{\omega}$  ( $\lambda$  is a cardinal), and its total-order ring  $T_D(\lambda)$  is  $T_{\lambda_{\omega}}$ , the finite-support ring over the ordered group  $\lambda_{\omega}(\mathbb{Q}^d)$  (as  $T_{\Gamma}$  of (13.2), but with exponents in  $\mathbf{No}$ ).
- Its squarefree ring  $R = k[x_1, \dots, x_d]$  and ideal  $Q$  are  $K[\mathbf{x}]$  and  $\mathfrak{q}$ ; its module  $E$  acting through  $\text{ct}$ , with torsion  $T$  and quotient  $V$ , is  $N$  with  $N_{\text{tor}}$  and  $N/N_{\text{tor}}$ ; its cube cones  $C_n$  are  $\mathcal{C}_n$ ; its  $\text{wdim}$ ,  $\text{gldim}$  are  $\text{w.gl.dim}$ ,  $\text{gl.dim}$ .

The letters  $\beta_{p,G}$ ,  $\mu_{\mathcal{C}}$ ,  $u_F$  and  $\ell$  are kept. Four tempting readings are false. (1)  $I_F$  is not  $J_F$ :  $I_{\mathcal{C}}$  for the orthant is the relative-interior ideal, not  $\Pi_d$  unless  $d = 1$ , and  $J_{[d]} = (X_1, \dots, X_d)$  is neither. (2) “Core” does not mean integer part:  $A_{\mathcal{C}}$  embeds into  $\mathbf{Oz}$  for  $(D, K) = (\mathbb{Z}, \mathbb{R})$ , and a partial exponent cone must not be replaced by the positive cone of the ambient order (Theorem 13.88). (3) The numbers  $\beta_{p,G}$  count summands of a reduced *flat* resolution, not minimal free ranks, and the reduced resolution is not unique. (4) “All radical monomial quotients” means the coefficient-full geometric ones: arithmetic ideals such as  $pA_{\mathcal{C}}$  and ideals restricting the coefficient at an admitted exponent are excluded.

**The rings.** Pointedness means two nonzero exponents never add to zero, so  $\Pi_{\mathcal{C}}$  is an ideal and  $\text{ct}$  is a split surjection  $A_{\mathcal{C}} \rightarrow D$  with kernel  $\Pi_{\mathcal{C}}$ ;  $A_{\mathcal{C}} \subseteq K[P_{\mathcal{C}}]$  is a domain (compare least exponents in a total order of  $\mathbb{Q}^d$  compatible with addition). Every sum is finite; this is used in the common-shift argument, in degreewise exactness and in the arguments against finite generation, and is not shorthand for a Hahn completion (26). For  $m \in \mathcal{C}$ ,  $\text{fc}(m)$  is the smallest face containing  $m$ , so  $m \in \text{relint fc}(m)$ ,  $\text{fc}(0) = \{0\}$ , and a supporting functional gives

$$\text{fc}(m + n) = \text{fc}(m) \vee \text{fc}(n) \quad (m, n \in \mathcal{C}), \quad (13.6)$$

since a face contains  $m + n$  exactly when it contains  $m$  and  $n$ .

**Proposition 13.58** (Omnific embeddings [26]). *Choose a rational functional  $\ell_1$  strictly positive on  $\mathcal{C} \setminus \{0\}$ , extend it to a basis  $\ell_1, \dots, \ell_d$  of the rational dual of  $\mathbb{Q}^d$ , and put  $\lambda_{\omega}(m) = \ell_1(m)\omega^{d-1} + \ell_2(m)\omega^{d-2} + \dots + \ell_d(m)$ , an injective additive map  $\mathbb{Q}^d \rightarrow \mathbf{No}$ , positive on  $\mathcal{C} \setminus \{0\}$ . For  $(D, K) = (\mathbb{Z}, \mathbb{R})$  the rule  $c + \sum_m a_m X^m \mapsto c + \sum_m a_m \omega^{\lambda_{\omega}(m)}$  is an injective ring homomorphism  $A_{\mathcal{C}} \rightarrow \mathbf{Oz}$  preserving  $\text{ct}$ ; for  $(D, K) = (\mathbb{Z}[i], \mathbb{C})$  it is one into  $\mathbf{Oz}[i]$ .*



*Proof.* First choose rational facet functionals  $h_1, \dots, h_s$  with  $\mathcal{C} = \{x : h_j(x) \geq 0 \text{ for all } j\}$ . Their common kernel is  $\mathcal{C} \cap (-\mathcal{C}) = \{0\}$ , so  $\ell_1 = \sum_j h_j$  is strictly positive on  $\mathcal{C} \setminus \{0\}$ . In particular it is nonzero and extends to a rational dual basis. Additivity of  $\lambda_\omega$  follows from linearity of these functionals. If  $m \neq 0$ , at least one  $\ell_j(m)$  is nonzero; the first such coefficient is the leading coefficient of the finite normal form for  $\lambda_\omega(m)$ , so the latter is nonzero. This proves injectivity. If  $m \in \mathcal{C} \setminus \{0\}$ , its leading coefficient is already  $\ell_1(m) > 0$ , so  $\lambda_\omega(m) > 0$ . For  $d = 1$  this simply reads  $\lambda_\omega(m) = \ell_1(m)$ .

Now the monomial law  $\omega^{\lambda_\omega(m)} \omega^{\lambda_\omega(n)} = \omega^{\lambda_\omega(m+n)}$  makes the displayed rule a unital ring homomorphism. Its nonconstant exponents are distinct and positive, so uniqueness of finite Conway normal forms proves injectivity and preserves the constant term. Real coefficients at these positive exponents and an integer constant term give an element of  $\mathbf{Oz}$ . For  $(D, K) = (\mathbb{Z}[i], \mathbb{C})$ , apply the same construction to real and imaginary parts. The multiplication rule still holds, and the two real normal forms prove injectivity into  $\mathbf{Oz} + i\mathbf{Oz} = \mathbf{Oz}[i]$  with the specified Gaussian integer constant term.  $\square$

This is Lemma 13.1(ii) with a lexicographic, rather than a real, functional (26).

**Proposition 13.59** (A common fraction field [26]).  $\text{Frac } A_{\mathcal{C}} = \text{Frac } K[\mathbb{Q}^d]$ , independently of  $\mathcal{C}$  and of  $D \subseteq K$ .

*Proof.* Regard both fraction fields as subfields of  $\text{Frac } K[\mathbb{Q}^d]$  via  $A_{\mathcal{C}} \subseteq K[\mathbb{Q}^d]$ . Choose a nonzero rational point  $u \in \mathcal{C}$ , which exists since  $\mathcal{C}$  is full-dimensional and rational. For each  $a \in K$ , both  $aX^u$  and  $X^u$  belong to  $A_{\mathcal{C}}$ , and  $(aX^u)/X^u = a$ . Thus its fraction field contains all of  $K$ , even when  $D$  is a proper subring.

Choose a rational interior point  $v$  of  $\mathcal{C}$ . Such a point can be obtained by summing rational generators of all its rays. For rational facet functionals  $h_j$ , one has  $h_j(v) > 0$  for every  $j$ . Given  $g \in \mathbb{Q}^d$ , choose a positive integer  $N$  large enough that  $Nh_j(v) + h_j(g) > 0$  for every  $j$ . Then  $n = Nv$  and  $m = Nv + g$  are rational points of  $\mathcal{C}$ , and  $X^g = X^m/X^n$  lies in  $\text{Frac } A_{\mathcal{C}}$ . This field therefore contains every finite sum in  $K[\mathbb{Q}^d]$ , and hence its fraction field. The reverse containment was the initial inclusion.  $\square$

The fraction field forgets the cone, while the face-quotient flat dimensions detect whether it is simplicial (26). All rings and modules used to compute Tor and flat or projective dimension are set-sized, flat dimension is taken among all modules, and a nonzero homogeneous Tor component is a nonzero component of the ungraded Tor; filtered colimits of flat modules are flat [29, Lemma 10.39.3], flat resolutions compute Tor, and every projective-dimension bound below is proved by a telescope or by dimension shifting (26).

**Flat ideals indexed by faces.** For a nonzero face  $F$  fix a rational  $u_F \in \text{relint } F$ , for instance the sum of one rational vector on each ray of  $F$ .

**Lemma 13.60** (Face subtraction [26]). *For  $m \in \mathcal{C}$  the following are equivalent: (i)  $F \subseteq \text{fc}(m)$ ; (ii)  $m - \epsilon u_F \in \mathcal{C}$  for all sufficiently small positive rational  $\epsilon$ ; (iii)  $m - \epsilon u_F \in \mathcal{C}$  for some positive rational  $\epsilon$ . Then  $\text{fc}(m - \epsilon u_F) = \text{fc}(m)$  for all small  $\epsilon > 0$ .*

*Proof.* Write  $\mathcal{C} = \{x : \ell_j(x) \geq 0, 1 \leq j \leq s\}$  using facet functionals. The least face containing a point is determined by the facet inequalities that vanish there. Assume (i). If  $\ell_j(m) = 0$ , the corresponding facet contains  $\text{fc}(m)$  and hence  $F$ , so  $\ell_j(u_F) = 0$ . Consequently each index with

$\ell_j(u_F) > 0$  has  $\ell_j(m) > 0$ . Choose a real number  $\delta > 0$  smaller than all the finitely many positive ratios

$$\frac{\ell_j(m)}{\ell_j(u_F)} \quad (\ell_j(u_F) > 0).$$

This list is nonempty: otherwise  $u_F$  would be in the common kernel of the facet functionals, contradicting pointedness and  $u_F \neq 0$ . For every real  $0 < \epsilon < \delta$ , each inequality positive at  $m$  remains positive at  $m - \epsilon u_F$ , while each vanishing inequality still vanishes. Thus the difference lies in  $\mathcal{C}$  with the same least face as  $m$ . This proves (ii) and the final assertion.

Any interval of sufficiently small positive numbers contains a rational number, so (ii) implies (iii). Conversely, in (iii) both summands of  $m = (m - \epsilon u_F) + \epsilon u_F$  lie in the cone. The face property forces  $\text{fc}(m)$  to contain  $\epsilon u_F$ , whose least face is  $F$ , proving (i). Under these equivalent conditions that face is nonzero; therefore the sufficiently small differences just constructed are themselves nonzero.  $\square$

For a nonzero face  $F$  put

$$I_F = \bigoplus_{0 \neq m \in P_{\mathcal{C}}, F \subseteq \text{fc}(m)} KX^m \subseteq A_{\mathcal{C}}, \quad I_{\{0\}} = A_{\mathcal{C}} \text{ (a convention only)}. \quad (13.7)$$

Equation (13.6) makes this an ideal: adding any exponent of  $A_{\mathcal{C}}$  to an admitted exponent preserves the face-containment condition, and the coefficient space at every admitted exponent is all of  $K$ . The special convention for the zero face includes the constant ring  $D$ ; applying the displayed direct-sum rule to that face instead would give only  $\Pi_{\mathcal{C}}$ . In the orthant, the face spanned by the coordinate rays in  $S$  gives 12's  $I_S$ : the monomials positive at every index of  $S$ , possibly zero elsewhere (26).

**Theorem 13.61** (Principal exhaustion and dimension one [26]). *For every nonzero face  $F$ ,*

$$I_F = \bigcup_{n \geq 1} X^{2^{-n}u_F} A_{\mathcal{C}}, \quad (13.8)$$

*an increasing union. So  $I_F$  is flat, is not finitely generated, and  $\text{pd}_{A_{\mathcal{C}}} I_F = 1$ .*

*Proof.* Put  $P_n = X^{2^{-n}u_F} A_{\mathcal{C}}$ . Every exponent in an element of  $P_n$  is  $2^{-n}u_F + v$  for some  $v \in P_{\mathcal{C}}$ , so (13.6) gives  $P_n \subseteq I_F$ . The identity  $X^{2^{-n}u_F} = (X^{2^{-(n+1)}u_F})^2$  gives  $P_n \subseteq P_{n+1}$ . Conversely, let  $f = \sum_m a_m X^m \in I_F$ . By Lemma 13.60, a single sufficiently large  $n$  makes every  $m - 2^{-n}u_F$  in this finite support a nonzero point of  $P_{\mathcal{C}}$ . Hence

$$f = X^{2^{-n}u_F} \sum_m a_m X^{m-2^{-n}u_F} \in P_n.$$

The nonzero differences are essential: they place the remaining sum in  $\Pi_{\mathcal{C}}$ , where arbitrary coefficients in  $K$  are allowed, without requiring any  $a_m$  to belong to  $D$ . The zero element causes no exception. Since  $A_{\mathcal{C}}$  is a domain, multiplication by  $X^{2^{-n}u_F}$  identifies  $A_{\mathcal{C}}$  with  $P_n$ . Thus the displayed increasing union is a filtered colimit of free rank-one modules and is flat.

To rule out finite generation, suppose  $f_1, \dots, f_t$  generate  $I_F$ , discarding any zero generators. For a functional  $\ell$  strictly positive on  $\mathcal{C} \setminus \{0\}$ , let  $c > 0$  be its minimum on the union of their finite supports. Multiplication by an element of  $A_{\mathcal{C}}$  only adds exponents on which  $\ell \geq 0$ . Addition and cancellation cannot introduce a new exponent of smaller  $\ell$ -value. Thus every exponent in the generated ideal has value at least  $c$ , whereas  $X^{2^{-n}u_F} \in I_F$  has value  $2^{-n}\ell(u_F) < c$  for large  $n$ .

Finally let  $\iota_n : P_n \hookrightarrow P_{n+1}$  be inclusion. There is an exact sequence

$$0 \rightarrow \bigoplus_{n \geq 1} P_n \xrightarrow{\tau} \bigoplus_{n \geq 1} P_n \rightarrow I_F \rightarrow 0, \quad \tau(e_n x) = e_n x - e_{n+1} \iota_n(x), \quad (13.9)$$

where the last map sums the included coordinates. The least nonzero coordinate of a finitely supported vector survives under  $\tau$ , proving injectivity. Modulo its image each  $e_n x$  equals  $e_N x$  for  $N \geq n$ ; moving a finite sum to a common coordinate shows that a vector with augmentation zero is in the image. The augmentation is surjective by the union formula. Both direct sums are free, so  $\text{pd } I_F \leq 1$ .

For completeness, a nonzero projective ideal  $J$  in a domain has a dual basis  $(x_i, f_i)$ : each  $x \in J$  has only finitely many nonzero  $f_i(x)$  and equals  $\sum_i f_i(x)x_i$ . Fix  $0 \neq a \in J$ . Linearity gives  $af_i(x) = f_i(ax) = xf_i(a)$ ; if  $f_i(a) = 0$ , the domain property forces  $f_i = 0$ . Only finitely many  $f_i(a)$  are nonzero, so the corresponding  $x_i$  generate  $J$ . Applying this to  $I_F$ , which is not finitely generated, rules out projectivity and proves  $\text{pd } I_F = 1$ .  $\square$

This is Lemma 13.3(i), (iii) and Proposition 13.12(i), (ii) for arbitrary faces; the dual-basis step is Lemma 12.1.

**Theorem 13.62** (Intersections, products and tensors [26]). *For nonzero faces  $F, G$ ,*

$$I_F \cap I_G = I_{F \vee G} = I_F I_G, \quad I_F \otimes_{A_{\mathcal{C}}} I_G \xrightarrow{\sim} I_{F \vee G} \text{ (multiplication)}. \quad (13.10)$$

*In particular  $I_F^2 = I_F$ , and  $I_G \subseteq I_F$  if and only if  $F \subseteq G$ .*

*Proof.* Because the monomials form a direct sum, membership in  $I_F \cap I_G$  means that every exponent  $m$  in the support satisfies both  $F \subseteq \text{fc}(m)$  and  $G \subseteq \text{fc}(m)$ . This is equivalent to  $F \vee G \subseteq \text{fc}(m)$ , proving the intersection formula. Since  $I_F$  and  $I_G$  are ideals,  $I_F I_G \subseteq I_F \cap I_G$ .

For the reverse containment it suffices to factor each  $aX^m$  in  $I_{F \vee G}$ . Face subtraction supplies a positive rational  $q$  such that  $m - qu_F$  has the same nonzero least face as  $m$ . Then

$$aX^m = X^{qu_F} (aX^{m-qu_F}), \quad X^{qu_F} \in I_F, \quad aX^{m-qu_F} \in I_G.$$

Here the entire coefficient  $a \in K$  belongs to the second factor, whose nonzero exponent ensures that it is an element of  $A_{\mathcal{C}}$ . This proves the product formula without treating  $a$  itself as a scalar in the constant ring  $D$ .

Flatness of  $I_G$  now makes the map obtained by tensoring  $I_F \hookrightarrow A_{\mathcal{C}}$  injective. Under  $A_{\mathcal{C}} \otimes_{A_{\mathcal{C}}} I_G \cong I_G$ , this is multiplication, and its image is  $I_F I_G = I_{F \vee G}$ , giving the claimed isomorphism. Taking  $G = F$  proves idempotence. If  $F \subseteq G$ , the support condition immediately gives  $I_G \subseteq I_F$ ; conversely, that ideal inclusion puts  $X^{u_G}$  in  $I_F$ , forcing  $F \subseteq \text{fc}(u_G) = G$ .  $\square$

None of this needs  $D$  to be a field, Noetherian or a principal ideal domain; the constant ring is never replaced by  $K$  (26).

**Radical monomial ideals.** An ideal  $I \subseteq \Pi_{\mathcal{C}}$  is *geometric monomial* (26: coefficient-full) if it is a direct sum of some of the spaces  $KX^m$ ,  $m \neq 0$ , so that membership depends on exponents only and admits every coefficient of  $K$  at an admitted exponent.

**Theorem 13.63** (Radical classification [26]). *The nonzero radical geometric monomial ideals correspond bijectively to the nonempty upper sets of nonzero faces, equivalently to the nonempty antichains  $\mathcal{F}$  of nonzero faces, by*

$$\mathfrak{r}_{\mathcal{F}} = \sum_{F \in \mathcal{F}} I_F. \quad (13.11)$$

*Every  $\mathfrak{r}_{\mathcal{F}}$  is idempotent and not finitely generated.*

*Proof.* Suppose first that  $I$  is nonzero, radical and geometric monomial. If an exponent  $m$  is admitted, coefficient-fullness gives  $X^m \in I$ . Set  $F = \text{fc}(m)$ , and let  $n$  be any rational point of relint  $H$  for a face  $H \supseteq F$ . For each facet functional  $\ell_j$ , the equality  $\ell_j(n) = 0$  forces  $\ell_j(m) = 0$ , since that facet contains  $H$  and hence  $F$ . Choose a positive integer  $N$  large enough that  $N\ell_j(n) \geq \ell_j(m)$  at every remaining facet. Then  $Nn - m \in P_{\mathcal{C}}$ , and

$$(X^n)^N = X^m X^{Nn-m} \in I.$$

The second factor belongs to  $A_{\mathcal{C}}$  even if its exponent is zero, because its coefficient is  $1 \in D$ . Radicality gives  $X^n \in I$ , and coefficient-fullness gives  $KX^n \subseteq I$ . Thus admission is constant on each relative interior and upward closed in the face poset. The resulting nonempty upper set  $\mathcal{U}$  is determined by its minimal faces, a nonempty antichain  $\mathcal{F}$ , because the face poset is finite. Its ideal is precisely  $\sum_{F \in \mathcal{F}} I_F$ .

Conversely, let  $\mathcal{U}$  be a nonempty upper set of nonzero faces, and admit exactly the nonzero rational exponents whose least face is in  $\mathcal{U}$ , with all coefficients in  $K$ . Equation (13.6) makes their direct sum an ideal. To prove radicality for arbitrary finite sums, for each face  $H$  define

$$\mathfrak{p}_H = \bigoplus_{m \in P_{\mathcal{C}} \setminus H} KX^m, \quad A_{\mathcal{C}}(H) = D \oplus \bigoplus_{0 \neq m \in H \cap \mathbb{Q}^d} KX^m.$$

Deleting terms outside  $H$  is a surjective ring map to  $A_{\mathcal{C}}(H)$ : a sum of two cone exponents belongs to  $H$  if and only if both exponents do. Its kernel is  $\mathfrak{p}_H$ , and its target is a domain, being a subring of  $K[\mathbb{Q}^d]$ . This includes  $H = \{0\}$ , where the target is  $D$ . Hence each  $\mathfrak{p}_H$  is prime.

Take the complement  $\mathcal{V}$  of  $\mathcal{U}$  in the *full* face poset, including  $\{0\}$ . It is a nonempty lower set. A face outside  $\mathcal{U}$  lies below a maximal member of  $\mathcal{V}$ ; a face in  $\mathcal{U}$  lies below none of them. Reading supports now gives

$$\sum_{F \in \mathcal{F}} I_F = \bigcap_{H \in \max \mathcal{V}} \mathfrak{p}_H.$$

The right side is radical, and the two constructions recover exactly the admitted faces, proving the bijection. If all nonzero faces are admitted,  $\mathcal{V} = \{\{0\}\}$  and this formula reads  $\Pi_{\mathcal{C}} = \mathfrak{p}_{\{0\}}$ .

Each  $I_F$  is idempotent, so  $\sum_F I_F = \sum_F I_F^2 \subseteq (\sum_F I_F)^2$ ; the reverse inclusion holds for every ideal. Finally, finitely many generators would give a strictly positive lower bound for a positive functional on their supports, hence on all supports they generate, as in Theorem 13.61. For any  $F \in \mathcal{F}$ , the admitted monomials  $X^{2^{-n}u_F}$  violate that bound. Thus the ideal is not finitely generated.  $\square$

**Proposition 13.64** (Generators of a face prime [26]). *For a proper face  $F < \mathcal{C}$ ,  $\mathfrak{p}_F = \sum_{\rho} I_{\rho}$  over the rays  $\rho$  of  $\mathcal{C}$  not in  $F$ , which are the minimal faces of its upper set, and  $A_{\mathcal{C}}/\mathfrak{p}_F \cong A_{\mathcal{C}}(F)$ .*

*Proof.* For  $m \in P_{\mathcal{C}}$ , one has  $m \notin F$  if and only if  $\text{fc}(m) \not\subseteq F$ . A face  $H$  is generated by its rays, so  $H \not\subseteq F$  if and only if it contains a ray  $\rho$  not contained in  $F$ . Therefore  $m$  is admitted by  $\mathfrak{p}_F$  exactly when it is admitted by at least one such  $I_{\rho}$ . All admitted coefficient spaces are  $K$ , proving the ideal equality. Every face outside  $F$  contains one of these rays, and a ray has no nonzero proper subface, so these rays are exactly the minimal members of that upper set. The quotient is the face-deletion map constructed in the preceding proof; it fixes  $D$  and retains exactly the monomials on  $F$ .  $\square$

For  $F = \{0\}$  the generators are all rays and  $\mathfrak{p}_{\{0\}} = \Pi_{\mathcal{C}}$ . An orthant facet excludes one ray; a nonsimplicial facet can exclude several, and their joins raise the flat dimension (26).

**Finite flat resolutions and label reduction.** Fix a nonempty antichain  $\mathcal{F} = \{F_1, \dots, F_r\}$  of nonzero faces, and for  $S \subseteq [r]$  put  $G_S = \bigvee_{i \in S} F_i$ ,  $G_\emptyset = \{0\}$ .

**Theorem 13.65** (Face-labelled simplex resolution [26]). *The complex with  $C_0^{\mathcal{F}} = A_{\mathcal{C}}$  and*

$$C_p^{\mathcal{F}} = \bigoplus_{S \subseteq [r], |S|=p} I_{G_S} \quad (1 \leq p \leq r), \quad (13.12)$$

*whose differential sends the summand of  $S = \{i_1 < \dots < i_p\}$  to the summands of  $S \setminus \{i_j\}$  by  $(-1)^{j-1}$  times the inclusion, is a flat resolution of  $A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}$ .*

*Proof.* Augment the displayed complex by the quotient map  $C_0^{\mathcal{F}} = A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}$ . Deleting an index decreases the join of the faces, so Theorem 13.62 supplies the required inclusion of ideals. Deleting two different indices in the opposite order gives the same inclusion with the opposite sign; hence the differential squares to zero. The image in degree zero is  $\sum_i I_{F_i} = \mathfrak{r}_{\mathcal{F}}$ , as required by the augmentation.

All maps preserve the  $\mathbb{Q}^d$ -grading. Fix  $0 \neq m \in P_{\mathcal{C}}$  and put  $T(m) = \{i : F_i \subseteq \text{fc}(m)\}$ . The summand indexed by  $S$  has degree- $m$  coefficient space  $K$  precisely when  $S \subseteq T(m)$ , including  $S = \emptyset$  in homological degree zero. If  $T(m) \neq \emptyset$ , the quotient has zero degree- $m$  component, and the remaining complex is the augmented simplex on  $T(m)$ , with a  $p$ -element subset in homological degree  $p$ . Here is a contraction with the displayed sign convention. Let  $v$  be the least element of  $T(m)$ ; on the oriented basis, including the empty subset, set

$$h(e_S) = \begin{cases} e_{\{v\} \cup S}, & v \notin S, \\ 0, & v \in S. \end{cases}$$

When  $v \notin S$ , deleting  $v$  from  $\{v\} \cup S$  contributes  $e_S$  and the other terms cancel those in  $h d(e_S)$ . When  $v \in S$ , only the term deleting  $v$  survives under  $h d$ . Thus  $dh + hd = \text{id}$ , also at the empty subset.

If  $T(m) = \emptyset$ , all positive terms vanish and the augmentation is  $K \xrightarrow{\text{id}} K$  in that degree. At  $m = 0$  it is  $D \xrightarrow{\text{id}} D$ , and outside  $P_{\mathcal{C}}$  every component is zero. Thus the augmented complex is exact degree by degree. Since its modules are direct sums of their homogeneous components, each element has finite support and this proves exactness of the ungraded complex as well. The term  $A_{\mathcal{C}}$  is free, and every positive term is a finite direct sum of flat face ideals by Theorem 13.61. This proves the claimed flat resolution.  $\square$

This is a Taylor-type incidence construction, classical in spirit [70, 71], with non-finitely-generated flat summands instead of free ones; its length  $r$  is only a first bound, and  $r$  can exceed  $d$  (26). For  $\mathcal{C}$  an orthant and  $\mathcal{F}$  its rays it is 12's Theorem 13.4.

Let  $\mathcal{L} = \{G_S\}$ , ordered by inclusion, whose atoms are the  $F_i$ . For  $G \in \mathcal{L} \setminus \{\{0\}\}$  put

$$\text{At}(G) = \{i : F_i \subseteq G\}, \quad \mathbf{K}_{<G} = \{S \subseteq \text{At}(G) : G_S < G\}, \quad (13.13)$$

a simplicial complex containing the empty face, proper in the full simplex on  $\text{At}(G)$ ; augmented chains place the empty face in degree  $-1$ . The part of the differential of (13.12) keeping a label  $G$  is a finite complex of  $K$ -matrices with basis the  $p$ -subsets  $S$  with  $G_S = G$ .

**Lemma 13.66** (One diagonal block [26]). *Its homology in degree  $p \geq 1$  has dimension*

$$\beta_{p,G} = \dim_K \tilde{H}_{p-2}(\mathbf{K}_{<G}; K); \quad \beta_{1,F_i} = 1. \quad (13.14)$$

*Proof.* Let  $\Sigma_G$  be the full simplex on the nonempty vertex set  $\text{At}(G)$ . A subset  $S$  of these vertices has  $G_S \subseteq G$ ; it survives modulo  $K_{<G}$  exactly when  $G_S = G$ . Moreover, the relative boundary retains exactly those deletions with the same label  $G$ . Hence the block in homological degree  $p$  is  $\tilde{C}_{p-1}(\Sigma_G, K_{<G}; K)$ : its  $p$ -subsets have ordinary simplicial degree  $p-1$ .

The augmented chain complex of the nonempty simplex  $\Sigma_G$  is acyclic. The long exact sequence of its subcomplex  $K_{<G}$  therefore gives

$$H_p(\text{block at } G) \cong \tilde{H}_{p-1}(\Sigma_G, K_{<G}; K) \cong \tilde{H}_{p-2}(K_{<G}; K).$$

This explains both shifts in the formula. If  $G = F_i$ , the antichain condition gives  $\text{At}(G) = \{i\}$ . The proper-join complex then consists of the empty face alone: its augmented chain complex is  $K$  in degree  $-1$  and zero elsewhere. Thus  $\beta_{1,F_i} = 1$ ; no vertex is required for this reduced degree- $-1$  contribution.  $\square$

**Lemma 13.67** (Triangular cancellation [26]). *Splitting off finitely many contractible two-term complexes reduces (13.12) to a flat resolution  $R_*^{\mathcal{F}}$  of  $A_{\mathcal{F}}/\mathfrak{r}_{\mathcal{F}}$  with*

$$R_0^{\mathcal{F}} = A_{\mathcal{F}}, \quad R_p^{\mathcal{F}} = \bigoplus_{G \in \mathcal{L} \setminus \{\emptyset\}} I_G^{\beta_p, G} \quad (p \geq 1), \quad (13.15)$$

*whose positive differential has every nonzero component a scalar multiple of an inclusion strictly decreasing the face label.*

*Proof.* Give the summand  $A_{\mathcal{F}}$  label  $\{\emptyset\}$  and every positive summand its nonzero face label. Each nonzero initial matrix component goes from  $I_G$  to  $I_H$  with  $H \subseteq G$  and is a scalar multiple of the inclusion. This class of maps is closed under sums and composition. For nonzero  $G$ , multiplication by any  $a \in K$  preserves  $I_G$  and commutes with the  $A_{\mathcal{F}}$ -action. If  $a \neq 0$ , its inverse is multiplication by  $a^{-1}$ , even when neither scalar belongs to  $D$ . Scalar multiples of  $I_G \hookrightarrow A_{\mathcal{F}}$  are also valid: their image is in the coefficient-full tail. No scalar action on all of  $A_{\mathcal{F}}$  is being asserted.

Suppose a differential has a nonzero entry  $a$  from a copy of  $I_G$  in degree  $p$  to another copy in degree  $p-1$ , with column  $i$  and row  $j$ . Here  $p \geq 2$ , since the only degree-zero label is  $\{\emptyset\}$ . After putting this row and column first, write the differential as

$$\begin{pmatrix} a & b \\ c & e \end{pmatrix}.$$

The invertible changes of coordinates given on the target by  $\begin{pmatrix} 1 & 0 \\ -ca^{-1} & 1 \end{pmatrix}$  and on the source by  $\begin{pmatrix} 1 & -a^{-1}b \\ 0 & 1 \end{pmatrix}$  transform it to  $\text{diag}(a, e - ca^{-1}b)$ . In the original indices the remaining entries are

$$d'_{uv} = d_{uv} - d_{ui}a^{-1}d_{jv}; \quad (13.16)$$

the neighboring differentials are transformed by the inverse coordinate changes. Their components entering or leaving the isolated pair vanish because  $d^2 = 0$  and  $a$  is invertible. Thus this pair is a direct two-term contractible subcomplex, with contraction  $a^{-1}$ , and removing it is a chain-homotopy equivalence.

The correction term passes from its source label  $L$  through  $G$  to its target label  $H$ . A nonzero such term requires  $H \subseteq G \subseteq L$ , so label decrease is preserved. In particular, a diagonal block at another label  $H$  can change only if  $H \subseteq G \subseteq H$ , which forces  $G = H$ . The same containment condition applies to all compositions introduced in the neighboring maps. Consequently cancellation



at  $G$  is ordinary elimination in its finite complex of  $K$ -matrices and does not alter the diagonal homology at any other label. Each step removes two summands, so finitely many steps remove all nonzero equal-label entries. A complex of vector spaces with zero differential has its homology as its chain spaces; the surviving number of copies of  $I_G$  in degree  $p$  is therefore  $\beta_{p,G}$ .

The degree-zero term is never cancelled, the augmentation is preserved, and all remaining positive summands are flat face ideals. The resulting complex is the stated flat resolution, and every nonzero component of its positive differential strictly decreases the face label.  $\square$

“Reduced” means only that no equal-label contractible pair remains; the remaining maps are not unique, and no finite minimal-generator argument for the  $I_G$  is imported (26).

**Lemma 13.68** (Crosscut comparison [26]).  $K_{<G}$  is homotopy equivalent to the order complex of the open interval  $(\{0\}, G)$  of  $\mathcal{L}$  (empty complexes included), so  $\beta_{p,G} = 0$  for  $p > \dim G$ .

*Proof.* Let  $\mathcal{S}$  be the poset of nonempty faces of  $K_{<G}$  and let  $\mathcal{P} = (\{0\}, G)$  in  $\mathcal{L}$ . Define

$$f : \mathcal{S} \rightarrow \mathcal{P}, \quad f(S) = G_S, \quad g : \mathcal{P} \rightarrow \mathcal{S}, \quad g(H) = \{i : F_i \subseteq H\}.$$

For a nonempty  $S$ , its join is nonzero and is strictly below  $G$  by definition of  $K_{<G}$ , so  $f$  is well-defined. Every nonzero  $H \in \mathcal{L}$  is the join of a nonempty family of the  $F_i$ . Therefore  $g(H)$  is nonempty and its join is exactly  $H < G$ , so  $g$  is well-defined too. Both maps preserve order,  $f(g(H)) = H$ , and  $S \subseteq g(f(S))$ .

For clarity, if two order maps  $a, b$  satisfy  $a(x) \leq b(x)$  for all  $x$ , the map on the product with  $\{0 < 1\}$  taking  $(x, 0)$  to  $a(x)$  and  $(x, 1)$  to  $b(x)$  is order-preserving: when  $x \leq y$ ,  $a(x) \leq b(y)$ . Realizing the usual prism triangulation gives a homotopy between the induced maps on order complexes. Apply this to  $\text{id}_{\mathcal{S}} \leq gf$ , together with  $fg = \text{id}_{\mathcal{P}}$ . The order complex of  $\mathcal{S}$  is the barycentric subdivision of  $K_{<G}$ , so these maps prove the claimed homotopy equivalence. If  $\mathcal{P}$  is empty, then  $\mathcal{S}$  is empty as well and  $K_{<G} = \{\emptyset\}$ ; both sides have the same augmented degree-1 convention.

A strict chain of nonzero proper faces of  $G$  has at most  $\dim G - 1$  members, because their dimensions strictly increase through  $1, \dots, \dim G - 1$ . Thus the order complex has dimension at most  $\dim G - 2$ , with the empty complex assigned dimension  $-1$ . Its reduced homology vanishes in every larger degree. If  $p > \dim G$ , then  $p - 2 > \dim G - 2$ , and Lemma 13.66 gives  $\beta_{p,G} = 0$ .  $\square$

So the reduced resolution has length at most  $d$  (26).

**Face-isolating modules and the exact formula.** Fix a nonzero face  $G$  and a rational  $b \in \text{relint } G$ , choose facet functionals  $\ell_1^G, \dots, \ell_t^G$  of  $G$  in its linear span, so  $G = \{x : \ell_j^G(x) \geq 0\}$  and  $\ell_j^G(b) > 0$ , let  $K[P_G]$  ( $P_G = G \cap \mathbb{Q}^d$ ) be an  $A_{\mathcal{C}}$ -algebra by discarding the monomials outside  $G$  and including  $D$  in  $K$ , and put

$$\mathfrak{z}_{G,b} = \bigoplus_{m \in P_G, \ell_j^G(m) \geq \ell_j^G(b) \text{ for some } j} KX^m, \quad \mathcal{W}_{G,b} = K[P_G]/\mathfrak{z}_{G,b}; \quad (13.17)$$

$\mathfrak{z}_{G,b}$  is an ideal because the  $\ell_j^G$  are nonnegative on  $G$ . The surviving monomials lie below *every* cap:  $(\mathcal{W}_{G,b})_b = 0$ , while small inward translates of  $b$  survive. The coefficient space in degree zero is  $K$ . In fact,  $\mathcal{W}_{G,b}$  is finitely generated over  $A_{\mathcal{C}}$  if and only if  $K$  is finitely generated over  $D$ . Necessity follows by taking the constant-term quotient  $\mathcal{W}_{G,b} \twoheadrightarrow K$ , on which  $A_{\mathcal{C}}$  acts through  $D$ .

Conversely, finitely many  $D$ -module generators of  $K$ , together with 1, generate the cap module: every nonconstant surviving term  $aX^m$  is the action of that element of  $A_{\mathcal{C}}$  on 1. In particular it is cyclic when  $D = K$ . Detection of flat dimension allows all modules and needs no finite-generation hypothesis.

**Theorem 13.69** (Isolation of one face label [26]). *For every nonzero face  $H$ ,  $(I_H \otimes_{A_{\mathcal{C}}} \mathcal{W}_{G,b})_b$  is  $K$  if  $H = G$  and 0 otherwise.*

*Proof.* Let  $q_n = 2^{-n}$  and  $P_n = X^{q_n u_H} A_{\mathcal{C}}$ . The exhaustion  $I_H = \varinjlim_n P_n$  is an exhaustion by graded ideals. Each  $P_n$  is free on its generator of degree  $q_n u_H$ , so tensoring with  $\mathcal{W}_{G,b}$  and taking degree  $b$  gives

$$(I_H \otimes_{A_{\mathcal{C}}} \mathcal{W}_{G,b})_b \cong \varinjlim_n (\mathcal{W}_{G,b})_{b - q_n u_H}.$$

Tensor products commute with this filtered colimit, and its graded maps allow homogeneous components to be taken before the colimit. The inclusion  $P_n \hookrightarrow P_{n+1}$  sends its generator to  $X^{q_{n+1} u_H} X^{(q_n - q_{n+1}) u_H}$ ; hence the transition on the right is multiplication by  $X^{(q_n - q_{n+1}) u_H}$ . Its target degree is  $b - q_{n+1} u_H$ , as required. These shifts cannot be omitted.

If  $H \not\subseteq G$ , no  $b - q_n u_H$  lies in  $G$ . Indeed, otherwise  $b = (b - q_n u_H) + q_n u_H$  would be a sum of two cone points lying in the face  $G$ , forcing  $u_H \in G$  and hence  $H \subseteq G$ . Since  $\mathcal{W}_{G,b}$  is supported on  $P_G$ , every term of the colimit is therefore zero.

If  $H \subsetneq G$ , some facet of  $G$  contains  $H$ . For its functional  $\ell_j^G$  one has  $\ell_j^G(u_H) = 0$ , so  $\ell_j^G(b - q_n u_H) = \ell_j^G(b)$ . Whenever the shifted exponent lies in  $P_G$ , this equality puts its monomial in the cap ideal; otherwise there is no coefficient space at that exponent. Again every term is zero.

Finally let  $H = G$ . Both  $b$  and  $u_G$  are in the relative interior, so all their facet values are positive. Choose  $n_0$  such that

$$0 < q_n \ell_j^G(u_G) < \ell_j^G(b) \quad (j = 1, \dots, t, \ n \geq n_0).$$

Then  $b - q_n u_G \in P_G$  lies strictly below every cap, and its coefficient space is one copy of  $K$ . Each late transition sends  $aX^{b - q_n u_G}$  to  $aX^{b - q_{n+1} u_G}$ , a surviving monomial with the same coefficient. The cofinal tail of the system is thus  $K$  with identity maps, whose colimit is  $K$ .  $\square$

Faces not in  $G$  are excluded by support, proper subfaces of  $G$  by equality at a cap; only a move into the relative interior of  $G$  leaves all its facets (26).

**Theorem 13.70** (Join homology computes flat dimension [26]). *For  $\mathfrak{r}_{\mathcal{F}} \neq 0$  put  $\text{len}(\mathcal{F}) = \max\{p : \beta_{p,G} \neq 0 \text{ for some } G \in \mathcal{L} \setminus \{\{0\}\}\}$ . Then*

$$\text{fd}_{A_{\mathcal{C}}}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}) = \text{len}(\mathcal{F}), \quad \text{fd}_{A_{\mathcal{C}}} \mathfrak{r}_{\mathcal{F}} = \text{len}(\mathcal{F}) - 1, \quad 1 \leq \text{len}(\mathcal{F}) \leq d, \quad (13.18)$$

and for every nonzero  $G \in \mathcal{L}$  and  $p \geq 1$ ,

$$\text{Tor}_p^{A_{\mathcal{C}}}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}, \mathcal{W}_{G,b})_b \cong K^{\beta_{p,G}}. \quad (13.19)$$

*Proof.* Write  $\ell_0 = \text{len}(\mathcal{F})$ . The maximum is defined: there are finitely many labels and degrees, and  $\beta_{1,F_i} = 1$  for each member of the nonempty antichain. By Lemmas 13.67 and 13.68, the reduced flat resolution has no terms above degree  $\ell_0$ , with  $1 \leq \ell_0 \leq d$ . Forgetting the grading gives a flat resolution of the ordinary module, so its flat dimension is at most  $\ell_0$ .

Fix a nonzero  $G \in \mathcal{L}$  and  $b \in \text{relint } G \cap \mathbb{Q}^d$ . All coordinate changes in the reduced resolution preserve the exponent grading. Tensor it with  $\mathcal{W}_{G,b}$  and take degree  $b$ . The degree-zero homological

term becomes  $(\mathcal{W}_{G,b})_b = 0$ . By [Theorem 13.69](#), the term in each homological degree  $p \geq 1$  is  $K^{\beta_p, G}$ , contributed exactly by the summands labelled  $G$ . Every differential component either has a different label at one end, whose degree- $b$  tensor component is zero, or would keep label  $G$ , which is forbidden by the strict label decrease. Thus every differential in this degree is zero. The tensor complex computes Tor, proving [\(13.19\)](#).

Choose a label with  $\beta_{\ell_0, G} \neq 0$ . Its nonzero homogeneous  $\text{Tor}_{\ell_0}$  component is a direct summand of the ungraded homology: the tensor complex is a direct sum of its degree subcomplexes and the differentials preserve those degrees. Hence ordinary  $\text{Tor}_{\ell_0}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}, \mathcal{W}_{G,b})$  is nonzero. Every flat resolution of length less than  $\ell_0$  would make this group vanish, giving the required lower bound without any grading or finite-generation restriction on competing resolutions.

For the ideal itself, truncating the reduced resolution at its image  $\mathfrak{r}_{\mathcal{F}}$  gives a flat resolution of length  $\ell_0 - 1$ . If  $\ell_0 \geq 2$ , the short exact sequence  $0 \rightarrow \mathfrak{r}_{\mathcal{F}} \rightarrow A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}} \rightarrow 0$  gives, for every module  $N$ ,

$$\text{Tor}_j^{A_{\mathcal{C}}}(\mathfrak{r}_{\mathcal{F}}, N) \cong \text{Tor}_{j+1}^{A_{\mathcal{C}}}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}, N) \quad (j \geq 1).$$

Use the same detecting module and  $j = \ell_0 - 1$  for the lower bound. If  $\ell_0 = 1$ , the truncation identifies the nonzero ideal with the flat term  $R_1^{\mathcal{F}}$ , so its flat dimension is zero.  $\square$

A finite incidence calculation alone would give only an upper bound; the detecting module turns diagonal homology into an obstruction to every shorter flat resolution (26). This is 26's Theorem B.

**Corollary 13.71** (A one-step projective interval [26]). *With  $\ell_0 = \text{len}(\mathcal{F})$ ,  $\max\{2, \ell_0\} \leq \text{pd}_{A_{\mathcal{C}}}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}) \leq \ell_0 + 1$ ; a one-face quotient  $A_{\mathcal{C}}/I_F$  has flat dimension one and projective dimension two.*

*Proof.* Put  $B = A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}$ . Each positive term  $R_p^{\mathcal{F}}$  is a finite direct sum of face ideals, each with the free resolution of length one in [\(13.9\)](#); thus  $\text{pd } R_p^{\mathcal{F}} \leq 1$ . Set  $Z_0 = B$  and let  $Z_{p+1}$  be the kernel of the surjection  $R_p^{\mathcal{F}} \rightarrow Z_p$ . Exactness gives  $Z_{\ell_0} \cong R_{\ell_0}^{\mathcal{F}}$  and short exact sequences

$$0 \rightarrow Z_{p+1} \rightarrow R_p^{\mathcal{F}} \rightarrow Z_p \rightarrow 0 \quad (0 \leq p < \ell_0).$$

Their long exact Ext sequences give  $\text{pd } Z_p \leq \max\{\text{pd } R_p^{\mathcal{F}}, 1 + \text{pd } Z_{p+1}\}$ . Starting with  $\text{pd } Z_{\ell_0} \leq 1$  and using that  $R_0^{\mathcal{F}} = A_{\mathcal{C}}$  is free yields  $\text{pd } B \leq \ell_0 + 1$ .

Projective modules are flat, so  $\text{pd } B \geq \text{fd } B = \ell_0$ . If  $\text{pd } B \leq 1$ , the exact sequence with free middle term  $0 \rightarrow \mathfrak{r}_{\mathcal{F}} \rightarrow A_{\mathcal{C}} \rightarrow B \rightarrow 0$  would give  $\text{Ext}_{A_{\mathcal{C}}}^1(\mathfrak{r}_{\mathcal{F}}, N) = 0$  for every  $N$ , making the ideal projective. A projective ideal in this domain is finitely generated by the dual-basis argument in [Theorem 13.61](#), contradicting [Theorem 13.63](#). Hence  $\text{pd } B \geq 2$ . For a one-face antichain, the only nonzero multiplicity is  $\beta_{1,F} = 1$ , so  $\ell_0 = 1$ ; the two projective bounds coincide at two, while the flat dimension is one.  $\square$

For  $\ell_0 \geq 2$  the argument does not decide the endpoint, and 21's exact value for coordinate cores ([Corollary 13.47](#)) cannot be transferred to nonsimplicial quotients without an additional Ext argument (26). By the isomorphism  $A_{\mathcal{C}} \cong A_d$  for simplicial cones, the upper endpoint holds for every proper-face quotient of a simplicial cone; the identity quotient has projective dimension zero [merge].

**Theorem 13.72** (Homological epimorphisms [26]). *For every  $B = A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}$  and every  $B$ -module  $N$ ,  $B \otimes_{A_{\mathcal{C}}} N \cong N$  and  $\text{Tor}_j^{A_{\mathcal{C}}}(B, N) = 0$  for  $j > 0$ ; so  $A_{\mathcal{C}} \rightarrow B$  is a homological ring epimorphism. Yet  $B$  is not flat when  $\mathfrak{r}_{\mathcal{F}} \neq 0$ .*

*Proof.* The ideal  $\mathfrak{r}_{\mathcal{F}}$  annihilates any  $B$ -module  $N$ . The canonical map  $B \otimes_{A_{\mathcal{C}}} N \rightarrow N$ , given by  $\bar{a} \otimes n \mapsto \bar{a}n$ , has inverse  $n \mapsto \bar{1} \otimes n$ . Thus the claimed degree-zero isomorphism needs no flatness of  $N$ .

For a positive summand  $I_{G_S}$  in the simplex resolution and any  $i \in S$ , the containment  $F_i \subseteq G_S$  gives  $I_{F_i}I_{G_S} = I_{G_S}$  by [Theorem 13.62](#). Therefore  $\mathfrak{r}_{\mathcal{F}}I_{G_S} = I_{G_S}$ , and the standard balanced-tensor identification is

$$I_{G_S} \otimes_{A_{\mathcal{C}}} N \cong (I_{G_S}/\mathfrak{r}_{\mathcal{F}}I_{G_S}) \otimes_B N = 0.$$

Tensoring that flat resolution with  $N$  leaves  $N$  in homological degree zero and zero in every positive degree, proving all the asserted Tor vanishings. In particular this holds for  $N = B$ ; the quotient map is a ring epimorphism because it is surjective, so it is a homological ring epimorphism.

To check nonflatness directly, choose a nonzero monomial  $x \in \mathfrak{r}_{\mathcal{F}}$ . Multiplication by  $x$  gives an injection  $A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}$ , since this ring is a domain. Tensoring with  $B$  turns it into the zero map  $B \rightarrow B$ . The module  $B$  is nonzero because  $\mathfrak{r}_{\mathcal{F}} \subseteq \Pi_{\mathcal{C}}$  and its quotient still contains  $D$ . Hence this zero map is not injective, contradicting what flatness of  $B$  would require.  $\square$

Vanishing self-Tor and positive flat dimension do not conflict. The vanishing theorem tests modules killed by  $\mathfrak{r}_{\mathcal{F}}$ ; the detecting cap modules with  $G \in \mathcal{L} \setminus \{\{0\}\}$  are not killed by it. Indeed, such a  $G$  contains some  $F_i$ . For sufficiently small rational  $q > 0$ , the monomial  $X^{qu_{F_i}}$  survives every cap in  $\mathcal{W}_{G,b}$ . It belongs to  $I_{F_i} \subseteq \mathfrak{r}_{\mathcal{F}}$  and acts nontrivially on the class of 1.

More generally, a cap module on any nonzero face  $G$  is a  $B$ -module exactly when no  $F_i$  is contained in  $G$ . The forward obstruction is the same surviving monomial. For the converse, each exponent admitted by  $\mathfrak{r}_{\mathcal{F}}$  has least face containing some  $F_i$ ; if no such  $F_i$  lies in  $G$ , none of those exponents lies in  $G$ , so face deletion annihilates the ideal. Thus cap modules on other faces can be  $B$ -modules; the obstruction concerns the labels used for detection. For example, in the two-dimensional orthant with  $\mathcal{F}$  consisting of the first coordinate ray, cap modules on the second coordinate ray are  $B$ -modules, while those on the first ray detect nonzero  $\text{Tor}_1$ . For  $\mathfrak{r}_{\mathcal{F}} = \Pi_{\mathcal{C}}$  the vanishing theorem is 12's [Corollary 13.5](#).

## Face quotients.

**Lemma 13.73** (Deleting a face from a polytopal boundary [26]). *Let  $Q_0$  be a positive-dimensional compact convex polytope and  $Q_1$  a nonempty proper face. Then  $\partial Q_0 \setminus Q_1$  is contractible, and it is homotopy equivalent to the order complex of  $\{H : \emptyset \neq H < Q_0, H \not\subseteq Q_1\}$ .*

*Proof.* Work in the affine span of  $Q_0$ , choose  $c \in \text{relint } Q_0$  and  $q \in \text{relint } Q_1$ , and translate vectors relative to  $c$ . The cone  $\mathbf{k} = \{t(z - c) : t \geq 0, z \in Q_1\}$  is convex. A supporting functional exposing the proper face  $Q_1$ , normalized to have value one on  $Q_1 - c$ , is strictly positive on every nonzero vector of  $\mathbf{k}$ . Thus  $\mathbf{k}$  is pointed and  $q - c \neq 0$  belongs to it.

The radial map  $r$  from nonzero vectors to  $\partial Q_0$  is continuous. Indeed, normalize the facet inequalities so that  $Q_0 - c = \{v : a_j(v) \leq 1 \text{ for all } j\}$ . For  $v \neq 0$ ,  $\max_j a_j(v) > 0$ , since otherwise the whole ray through  $v$  would lie in the bounded polytope. Therefore

$$r(v) = c + \frac{v}{\max_j a_j(v)}.$$

Each ray from  $c$  meets the boundary once, and this formula shows that  $r(v) \in Q_1$  exactly when  $v \in \mathbf{k} \setminus \{0\}$ . For  $x \in \partial Q_0 \setminus Q_1$ , define

$$\mathbf{h}_t(x) = r((1-t)(x-c) + t(c-q)), \quad 0 \leq t \leq 1. \quad (13.20)$$

If the vector inside  $r$  belonged to  $k$  for  $t < 1$ , adding  $t(q - c) \in k$  would put  $(1 - t)(x - c)$  in  $k$ , and then  $x \in Q_1$ , a contradiction. At  $t = 1$  that vector is  $-(q - c)$ , which is outside  $k$  by pointedness. In particular the vector never vanishes. The homotopy is continuous, remains in  $\partial Q_0 \setminus Q_1$ , starts at  $x$  and ends at the fixed point  $r(c - q)$ . This proves contractibility, including when  $Q_0$  is a segment.

For the order-complex comparison, use the barycentric subdivision of  $\partial Q_0$ . Its vertices correspond to nonempty proper faces  $H$ . The subdivision of  $Q_1$  is the full subcomplex on vertices with  $H \subseteq Q_1$ . For a point outside  $Q_1$ , write its barycentric weights as  $\alpha_H$  and put  $w = \sum_{H \not\subseteq Q_1} \alpha_H > 0$ . Multiply the weights with  $H \subseteq Q_1$  by  $1 - t$ , leave the others unchanged, and divide all weights by  $w + (1 - t)(1 - w) > 0$ . The support stays in the same face chain, and these formulas agree on common simplices, giving a continuous homotopy. At  $t = 1$  only vertices outside  $Q_1$  remain; points already supported there are fixed throughout. This is a strong deformation retraction onto their full subcomplex, precisely the stated order complex.  $\square$

**Lemma 13.74** (The outside-vertex complex [26]). *Let  $Q_0$  be a convex polytope,  $Q_1$  a proper face (possibly empty),  $V$  the vertices of  $Q_0$  not in  $Q_1$ , and  $K$  the subsets of  $V$  lying in a proper face of  $Q_0$ . If  $Q_1 = \emptyset$  then  $K \simeq \partial Q_0$ ; if  $Q_1 \neq \emptyset$  then  $K$  is contractible. (For a point  $Q_0$  and empty  $Q_1$  both are  $\{\emptyset\}$ .)*

*Proof.* First assume  $Q_0$  has positive dimension. Let  $\mathcal{S}$  be the poset of nonempty simplices of  $K$ , and let  $\mathcal{P}$  be the poset of nonempty proper faces  $H$  of  $Q_0$  with  $H \not\subseteq Q_1$ . Define  $f : \mathcal{S} \rightarrow \mathcal{P}$  by taking the smallest face of  $Q_0$  containing the vertices of a simplex. That face is proper by the definition of  $K$ , and it is not contained in  $Q_1$  since each selected vertex is outside  $Q_1$ . Define  $g : \mathcal{P} \rightarrow \mathcal{S}$  by taking the vertices of  $H$  outside  $Q_1$ . This set is nonempty, since a face is the convex hull of its vertices and  $H \not\subseteq Q_1$ . It is a simplex of  $K$  because it lies in the proper face  $H$ .

Both maps preserve inclusion. The face generated by  $g(H)$  is contained in  $H$ , so  $fg \leq \text{id}_{\mathcal{P}}$ ; every vertex of  $\mathcal{S}$  occurs in  $g(f(S))$ , so  $\text{id}_{\mathcal{S}} \leq gf$ . The comparable-map homotopies from Lemma 13.68 make the induced maps on order complexes homotopy inverses. The order complex of  $\mathcal{S}$  is the barycentric subdivision of  $K$ . If  $Q_1 = \emptyset$ , the order complex of  $\mathcal{P}$  is the barycentric subdivision of the whole boundary  $\partial Q_0$ ; if  $Q_1 \neq \emptyset$ , it is contractible by Lemma 13.73.

If  $Q_0$  is a point, its only proper face is empty, so  $Q_1 = \emptyset$ . The singleton vertex set does not lie in a proper face, whereas the empty subset does. Consequently  $K = \{\emptyset\}$ , the same augmented complex as the boundary of a point. Its reduced homology is  $K$  in degree  $-1$ , with the convention used above.  $\square$

**Theorem 13.75** (Face-quotient multiplicities [26]). *For a face  $F < \mathcal{C}$ , use the antichain of rays outside  $F$  (Proposition 13.64) and extend  $\beta_{p,G}$  by zero. Then for every nonzero face  $G$  and  $p \geq 1$ ,*

$$\beta_{p,G} = 1 \text{ if } G \cap F = \{0\} \text{ and } p = \dim G, \quad \beta_{p,G} = 0 \text{ otherwise.} \quad (13.21)$$

*So  $A_{\mathcal{C}}(F)$  has a reduced flat resolution with  $\bigoplus_{G \cap F = \{0\}, \dim G = p} I_G$  in degree  $p \geq 1$ ,*

$$\text{fd}_{A_{\mathcal{C}}} A_{\mathcal{C}}(F) = \mu_{\mathcal{C}}(F) = \max\{\dim G : G \text{ a face, } G \cap F = \{0\}\}, \quad (13.22)$$

*the maximum including  $G = \{0\}$ , and  $\text{Tor}_{\dim G}^{A_{\mathcal{C}}}(A_{\mathcal{C}}(F), \mathcal{W}_{G,b})_b \cong K$  for every nonzero face  $G$  with  $G \cap F = \{0\}$ . In particular  $\text{fd } D = d$  and  $\text{fd } A_{\mathcal{C}} = 0$ .*

*Proof.* Fix a nonzero face  $G$ , and call its rays outside  $F$  active. If  $G \subseteq F$ , there are no active rays, so  $G$  is not a join label and its multiplicities are zero by the stated extension convention. Otherwise at least one ray is active. Let  $H_G$  be the join of all active rays in  $G$ . If  $H_G < G$ , no subfamily can

join to  $G$ , so again  $G$  is not a label and all its multiplicities are zero. In this case  $G \cap F \neq \{0\}$ : if the intersection were zero, every ray of  $G$  would be active and their join would be  $G$ . The auxiliary complex of active subsets contained in proper faces of  $G$  is then a nonempty full simplex, consistent with that vanishing. In particular no spurious degree  $-1$  class is assigned to a missing label.

It remains to treat  $H_G = G$ , when  $G \in \mathcal{L}$ . Choose a functional  $\ell$  strictly positive on  $\mathcal{C} \setminus \{0\}$  and put

$$Q_0 = G \cap \{\ell = 1\}, \quad Q_1 = (G \cap F) \cap \{\ell = 1\}.$$

The compact polytope  $Q_0$  has dimension  $\dim G - 1$ , and its vertices represent the rays of  $G$ . Its nonempty faces correspond to the nonzero faces of  $G$ . The active rays represent exactly the vertices outside the proper face  $Q_1$ . A subset of them has join strictly below  $G$  if and only if those vertices lie in a proper face of  $Q_0$ . Thus  $K_{<G}$  is precisely the outside-vertex complex of Lemma 13.74.

If  $G \cap F = \{0\}$ , then  $Q_1 = \emptyset$ , all rays are active, and their join is indeed  $G$ . The boundary of the polytope  $Q_0$  is a sphere of dimension  $\dim G - 2$ , as seen by radial projection from an interior point when  $\dim G \geq 2$ . Its reduced homology over  $K$  is one-dimensional in that degree and zero elsewhere. For a ray  $G$ ,  $Q_0$  is a point and the same assertion holds in augmented degree  $-1$ . If  $\{0\} < G \cap F < G$ , then  $Q_1$  is nonempty and proper, so the outside-vertex complex is contractible and all its reduced homology vanishes. Substitution into (13.14) proves every case of (13.21).

The face-prime quotient is  $A_{\mathcal{C}}(F)$  by Proposition 13.64. Apply Lemma 13.67 and Theorem 13.70 to obtain the stated resolution, the maximum formula and each detecting Tor component. For proper  $F$  there is a ray outside  $F$ , meeting it only at zero, so the maximum is positive. At  $F = \{0\}$  all faces are disjoint from  $F$ , the maximum is  $d$ , and the quotient is  $D$ . For completeness, at  $F = \mathcal{C}$  the quotient map is the identity, the only disjoint face is  $\{0\}$ , and both the displayed maximum and the flat dimension are zero; this case does not require a nonempty antichain.  $\square$

The equality involves the dimension of the disjoint face, not the number of its rays: a nonsimplicial face with many rays contributes one top class, which is what equal-label cancellation measures (26). This is 26's Theorem A.

**Proposition 13.76** (The face complex resolves  $D$  [26]). *Orient the faces of  $Q_0 = \mathcal{C} \cap \{\ell = 1\}$ . There is a flat resolution of  $D$  with  $C_0^{\text{cell}} = A_{\mathcal{C}}$ ,  $C_p^{\text{cell}} = \bigoplus_{\dim G=p} I_G$  ( $1 \leq p \leq d$ ), cellular incidence numbers times inclusions as differentials and  $\text{ct}$  as augmentation; hence  $\text{fd}_{A_{\mathcal{C}}} D = d$  independently of the cancellation.*

*Proof.* Write  $E_G = Q_0 \cap G$  for the cell corresponding to a nonzero face  $G$ ; its dimension is  $\dim G - 1$ . Choose positive orientations on the vertices and any orientations on the higher-dimensional faces. For  $p \geq 2$ , the component from  $I_G$  to  $I_H$ , with  $\dim G = p$  and  $\dim H = p - 1$ , is the oriented incidence number  $[E_G : E_H]$  times inclusion when  $H < G$ , and zero otherwise. In degree one use the sum of the inclusions from the ray ideals into  $A_{\mathcal{C}}$ . This is the augmented cellular incidence convention. The cellular boundary identity gives  $d^2 = 0$ , including at the augmentation degree of the cells, and  $\text{ct } d_1 = 0$  because all ray ideals lie in the tail.

Fix  $0 \neq m \in P_{\mathcal{C}}$  and put  $H = \text{fc}(m)$ . The degree- $m$  summands present are exactly the  $I_G$  with  $\{0\} < G \subseteq H$ , each contributing  $K$ . These are the cells of the nonempty convex polytope  $Q_0 \cap H$ , including its top cell. Their incidence maps form its augmented cellular chain complex, with cellular degree  $p - 1$  in homological degree  $p$  and the augmentation copy  $K = (A_{\mathcal{C}})_m$  in degree zero. This complex is exact because the polytope is contractible. At exponent zero the complex is  $D \xrightarrow{\text{id}} D$ ; outside  $P_{\mathcal{C}}$  every component is zero. Finite supports therefore give an exact ungraded complex as



well. Its positive terms are finite direct sums of flat face ideals, so it is a flat resolution of  $D$  of length  $d$ .

For the lower bound, tensor this resolution with  $\mathcal{W}_{\mathcal{E},b}$  and take exponent degree  $b$ . The homological degree-zero term is  $(\mathcal{W}_{\mathcal{E},b})_b = 0$ . Face isolation leaves only the single summand  $I_{\mathcal{E}}$  in homological degree  $d$ , contributing  $K$ . Consequently  $\mathrm{Tor}_d(D, \mathcal{W}_{\mathcal{E},b})_b \cong K \neq 0$ , also as part of ordinary ungraded Tor. This forces  $\mathrm{fd} D \geq d$  and proves equality without using equal-label cancellation.  $\square$

For a simplicial cone the numbers of summands are  $\binom{d}{0}, \dots, \binom{d}{d}$ , the Boolean pattern of [Theorem 13.4](#); for the cone over a square 1, 4, 4, 1, over a cube 1, 8, 12, 6, 1. The method is cellular in the sense of [70], but flatness comes from rational root exhaustion, not from free cellular summands (26).

### Flatness detects simpliciality.

**Theorem 13.77** (Only single-face ideals are flat [26]). *For an irredundant antichain  $\mathcal{F} = \{F_1, \dots, F_r\}$  the following are equivalent:  $\mathfrak{r}_{\mathcal{F}}$  is flat;  $\mathrm{fd}(A_{\mathcal{E}}/\mathfrak{r}_{\mathcal{F}}) = 1$ ;  $r = 1$ .*

*Proof.* Put  $J = \mathfrak{r}_{\mathcal{F}}$  and  $B = A_{\mathcal{E}}/J$ ; the antichain is nonempty, as fixed above. The sequence  $0 \rightarrow J \rightarrow A_{\mathcal{E}} \rightarrow B \rightarrow 0$  gives, for every module  $N$ ,

$$\mathrm{Tor}_1^{A_{\mathcal{E}}}(J, N) \cong \mathrm{Tor}_2^{A_{\mathcal{E}}}(B, N).$$

Thus  $J$  is flat if and only if  $\mathrm{fd} B \leq 1$ , by the Tor criterion for flat dimension. The quotient  $B$  is nonzero and nonflat by [Theorem 13.72](#), so this is equivalent to  $\mathrm{fd} B = 1$ . If  $r = 1$ , then  $J = I_{F_1}$  is flat by [Theorem 13.61](#).

Conversely, suppose  $r \geq 2$ . Choose  $G$  minimal under inclusion among the finitely many pairwise joins  $F_i \vee F_j$  with  $i \neq j$ . Let  $T = \{i : F_i \subseteq G\}$  and  $t = |T| \geq 2$ . No  $F_i$  with  $i \in T$  equals  $G$ : it would contain both generators of the chosen pair, contrary to the antichain condition. For distinct  $i, j \in T$ , the pairwise join  $F_i \vee F_j$  lies below  $G$  and is another member of the finite set in which  $G$  was minimal. It must therefore equal  $G$ . Every singleton from  $T$  has join strictly below  $G$ , and every subset of size at least two has join  $G$ . Hence  $K_{<G}$  consists of  $t$  isolated vertices and the empty face. Its augmented degree-zero boundary is the sum map  $K^t \rightarrow K$ , with kernel of dimension  $t - 1$ , and it has no degree-one chains. Consequently

$$\beta_{2,G} = \dim_K \tilde{H}_0(K_{<G}; K) = t - 1 > 0.$$

By [Theorem 13.70](#), this gives  $\mathrm{fd} B \geq 2$ , excluding flatness of  $J$ . This proves all three equivalences over every allowed coefficient field.  $\square$

**Corollary 13.78** (A marked face lattice [26]). *Among the nonzero radical geometric monomial ideals the flat ones are exactly the  $I_F$ ; with  $A_{\mathcal{E}} = I_{\{0\}}$ , their inclusion order is anti-isomorphic to the face lattice and tensor product corresponds to join.*

*Proof.* By [Theorem 13.63](#), every nonzero radical geometric monomial ideal has a unique nonempty antichain of minimal faces. By [Theorem 13.77](#), it is flat precisely when this antichain is a singleton. Thus  $F \mapsto I_F$  is a bijection from nonzero faces to the indicated flat ideals. Adjoin  $A_{\mathcal{E}} = I_{\{0\}}$ : this is a separate convention, since the ring contains 1 whereas all geometric monomial ideals under discussion lie in the tail.

For nonzero faces, [Theorem 13.62](#) gives  $I_G \subseteq I_F$  if and only if  $F \subseteq G$ . The same equivalence holds when exactly one face is zero, because  $A_{\mathcal{C}}$  contains every nonzero-face ideal and is contained in none of those proper tail ideals. If both faces are zero it is tautological. Hence the bijection reverses order, recovering the entire face lattice. Multiplication identifies  $I_F \otimes_{A_{\mathcal{C}}} I_G$  with  $I_{F \vee G}$  for nonzero faces; when one face is zero this is the usual unit isomorphism  $A_{\mathcal{C}} \otimes I_G \cong I_G$ . The claim uses the actual marked ideals and their inclusions, not merely their abstract module isomorphism classes.  $\square$

This reconstructs the face lattice from the ring *with its geometric monomial structure marked*; it does not show that an abstract ring isomorphism of cores preserves the monomials or the face lattice (26).

**Theorem 13.79** (A homological characterization of simplicial cones [26]). *The following are equivalent: (i)  $\mathcal{C}$  is simplicial; (ii) every facet omits exactly one ray; (iii)  $\mathfrak{p}_F$  is flat for every facet  $F$ ; (iv)  $\text{fd } A_{\mathcal{C}}(F) = 1$  for every facet  $F$ .*

*Proof.* If  $\mathcal{C}$  is simplicial, its  $d$  ray generators form a basis of  $\mathbb{R}^d$  and carry the cone to the nonnegative orthant. Its facets are therefore obtained by omitting one basis ray, proving (i) $\Rightarrow$ (ii). For each facet  $F$ , the rays outside  $F$  form the nonempty irredundant antichain defining  $\mathfrak{p}_F$  by [Proposition 13.64](#). The equivalences of [Theorem 13.77](#) then say exactly that this antichain has one member, that  $\mathfrak{p}_F$  is flat, and that  $\text{fd } A_{\mathcal{C}}(F) = 1$ . This proves the equivalence of (ii), (iii) and (iv).

Assume (ii), and choose one nonzero generator  $v_\rho$  of each ray. The intersection of all facets is  $\{0\}$ : if all facet functionals vanished on  $v$ , both  $v$  and  $-v$  would satisfy the cone inequalities, so pointedness would force  $v = 0$ . Consequently, for every ray  $\rho$  there is a facet  $F_\rho$  not containing it. Its nonnegative supporting functional  $\ell_\rho$  is positive on  $v_\rho$  and, by (ii), zero on every other ray generator. Applying it to any real linear relation  $\sum_\sigma a_\sigma v_\sigma = 0$  gives  $a_\rho \ell_\rho(v_\rho) = 0$ , hence  $a_\rho = 0$ . This holds for every ray, so all the ray generators are linearly independent. They span  $\mathbb{R}^d$  because the cone is full-dimensional; there are exactly  $d$  of them and the cone is simplicial. The argument includes  $d = 1$ , whose only facet is the zero face.  $\square$

The criterion concerns all facets. For example, let  $\mathcal{S} \subseteq \mathbb{R}^3$  be the cone over a square. The product  $\mathcal{S} \times \mathbb{R}_{\geq 0}$  has dimension four and five rays, so it is not simplicial. Yet its facet  $\mathcal{S} \times \{0\}$  contains the four square-cone rays and omits only the last coordinate ray. Its face-prime ideal is flat, and its face quotient has flat dimension one (26, with this explicit example).

**Corollary 13.80** (Simplicial face dimensions [26]). *If  $\mathcal{C}$  is simplicial,  $\text{fd } A_{\mathcal{C}}(F) = d - \dim F$  for every face  $F$ .*

*Proof.* Choose a basis of ray generators  $v_1, \dots, v_d$  for  $\mathcal{C}$ . Every face is the nonnegative span of a unique subset of these rays. Write  $F$  for the face on an index set  $S$ ; then  $\dim F = |S|$ . A face on  $T$  meets  $F$  only at zero if and only if  $T \cap S$  is empty, since coordinates in the ray basis are unique. Thus every face disjoint from  $F$  lies in the complementary face on  $[d] \setminus S$ , and that face itself is disjoint from  $F$  and has dimension  $d - |S|$ . The maximum in [Theorem 13.75](#) is therefore  $d - \dim F$ . This includes  $S = \emptyset$ , giving the quotient  $D$  of flat dimension  $d$ , and  $S = [d]$ , giving the identity quotient of flat dimension zero.  $\square$

Through  $A_{\mathcal{C}} \cong A_d$  this is 21's  $\text{fd } F_U = |U|$  ([Proposition 13.43](#)) with  $|U| = d - \dim F$ , and 21's [Corollary 13.47](#) adds  $\text{pd } A_{\mathcal{C}}(F) = d - \dim F + 1$  for  $F < \mathcal{C}$ . At  $F = \mathcal{C}$  the module is  $A_{\mathcal{C}}$  itself, so its projective dimension is zero [merge].

**Examples, sharpness and coefficients.** Let  $\mathcal{C} = \{(x, y, z) : z \geq |x|, z \geq |y|\}$ , the cone over a square, with rays through  $v_1 = (1, 1, 1)$ ,  $v_2 = (1, -1, 1)$ ,  $v_3 = (-1, -1, 1)$ ,  $v_4 = (-1, 1, 1)$  and adjacent two-faces  $F_{ij} = \rho_i \vee \rho_j$ . The augmentation has the flat resolution

$$0 \rightarrow I_{\mathcal{C}} \rightarrow I_{F_{12}} \oplus I_{F_{23}} \oplus I_{F_{34}} \oplus I_{F_{41}} \rightarrow \bigoplus_{i=1}^4 I_{\rho_i} \rightarrow A_{\mathcal{C}} \rightarrow D \rightarrow 0$$

of length three, where the unreduced simplex complex on four rays has length four. For the facet  $F_{12}$ ,  $\mathfrak{p}_{F_{12}} = I_{\rho_3} + I_{\rho_4}$ , whose join is the opposite facet  $F_{34}$ , disjoint from  $F_{12}$ , so

$$0 \rightarrow I_{F_{34}} \xrightarrow{u \mapsto (u, -u)} I_{\rho_3} \oplus I_{\rho_4} \rightarrow A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}(F_{12}) \rightarrow 0,$$

and a cap module in the interior of  $F_{34}$  gives  $\text{Tor}_2 \neq 0$ : a facet quotient of flat dimension two although its codimension is one. For the ray  $F = \rho_1$  the disjoint nonzero faces are  $\rho_2, \rho_3, \rho_4, F_{23}, F_{34}$ , the multiplicities are  $(1, 3, 2)$  and the flat dimension is two; the top cone meets  $F$  (26, with a figure of the square section).

**Theorem 13.81** (Cube cones [26]). *For  $n \geq 1$  let  $\mathcal{C}_n = \{(x_1, \dots, x_n, z) : z \geq |x_i| \ (1 \leq i \leq n)\}$  and  $F_{\pm} = \mathcal{C}_n \cap \{x_1 = \pm z\}$ . Then  $\dim \mathcal{C}_n = n + 1$ , the  $F_{\pm}$  are facets, and  $\text{fd}_{A_{\mathcal{C}_n}} A_{\mathcal{C}_n}(F_{\pm}) = n$ , in the omnific and in the Gaussian omnific cores.*

*Proof.* The inequalities imply  $z \geq 0$ , and at  $z = 0$  they force every  $x_i = 0$ . Thus  $z$  is strictly positive on nonzero cone points, and the section  $z = 1$  is the cube  $[-1, 1]^n$ . It has dimension  $n$ , so its cone has dimension  $n + 1$ . The sections  $x_1 = \pm 1$  are opposite cube facets of dimension  $n - 1$ , including two points when  $n = 1$ ; their cones  $F_{\pm}$  are consequently facets of dimension  $n$ .

On  $F_+ \cap F_-$  one has  $x_1 = z = -x_1$ , forcing  $z = 0$  and hence all coordinates zero. The disjoint-face formula Theorem 13.75 therefore gives a lower bound  $n$  using  $F_-$ . For the upper bound, the only face of dimension  $n + 1$  is the whole cone, which meets the nonzero face  $F_+$ ; every disjoint face has dimension at most  $n$ . The two bounds give equality.

This reasoning holds for every unital subring  $D \subseteq K$ . Specialize to  $(D, K) = (\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$  and use the embeddings of Proposition 13.58. They identify these rings with their omnific or Gaussian core images, preserving module dimensions. No extension of scalars to the entire ambient omnific ring is asserted.  $\square$

By Theorem 13.75, a quotient retaining a nonzero proper face has flat dimension at most  $d - 1$ : the full cone is not disjoint from that face. Cube cones attain this bound on a facet (26). The following table lists total summand counts by homological degree, beginning with the single  $A_{\mathcal{C}}$  in degree zero. They count flat face ideals, not ranks of free modules.

| Cone                    | Quotient                          | Summand counts                      | Flat dimension |
|-------------------------|-----------------------------------|-------------------------------------|----------------|
| $d$ -orthant            | $D$                               | $\binom{d}{0}, \dots, \binom{d}{d}$ | $d$            |
| Square cone             | $D$                               | $1, 4, 4, 1$                        | $3$            |
| Square cone             | Facet quotient                    | $1, 2, 1$                           | $2$            |
| Square cone             | Ray quotient                      | $1, 3, 2$                           | $2$            |
| Any cone                | $A_{\mathcal{C}}/I_{\mathcal{C}}$ | $1, 1$                              | $1$            |
| Cone over a 3-cube      | $D$                               | $1, 8, 12, 6, 1$                    | $4$            |
| Cone over a 3-cube      | Facet quotient                    | $1, 4, 4, 1$                        | $3$            |
| Cone over an octahedron | $D$                               | $1, 6, 12, 8, 1$                    | $4$            |

For  $D$ , the counts enumerate all section faces, including the top cell, by [Proposition 13.76](#). For a face quotient they enumerate the faces disjoint from the retained face, by [Theorem 13.75](#). In a cube, the faces disjoint from a fixed facet are precisely those in the opposite facet: a face fixes some coordinates to  $\pm 1$ , and avoids  $x_1 = 1$  exactly when it fixes  $x_1 = -1$ . This gives 1, 4, 4, 1 for a three-cube facet quotient. The single-ideal row is the resolution  $0 \rightarrow I_{\mathcal{C}} \rightarrow A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}/I_{\mathcal{C}} \rightarrow 0$ ; its exact dimension is one by [Theorem 13.77](#).

The quotient  $A_{\mathcal{C}}/I_{\mathcal{C}}$  keeps the boundary faces and is reduced. For  $d \geq 2$  it has zero divisors and is not a face-prime quotient: choose a facet  $H$ , a rational ray vector  $u$  outside  $H$  and a rational  $v \in \text{relint } H$ . Both vectors lie on the boundary, but  $u + v$  is interior, since every facet functional is positive on at least one of  $u, v$ . Thus  $X^u, X^v$  survive and their product vanishes. For  $d = 1$ , however,  $I_{\mathcal{C}} = \Pi_{\mathcal{C}}$  and the quotient is the domain  $D$ , the face-prime quotient retaining the zero face (26, with this dimension-one qualification).

**Corollary 13.82** (Squarefree homology survives rational roots [26]). *Let  $\mathfrak{q} \subsetneq K[\mathbf{x}] = K[x_1, \dots, x_d]$  be a nonzero squarefree monomial ideal with irredundant supports  $S_1, \dots, S_r$ , and in  $A_d$  let  $\mathfrak{r} = \sum_i I_{S_i}$ . Then  $\text{fd}_{A_d}(A_d/\mathfrak{r}) = \text{pd}_{K[\mathbf{x}]}(K[\mathbf{x}]/\mathfrak{q})$ , and the face-labelled multiplicities equal the multigraded Betti numbers of  $K[\mathbf{x}]/\mathfrak{q}$ .*

*Proof.* Put  $R = K[\mathbf{x}]$ . The minimal generators are the squarefree monomials  $x^{S_i} = \prod_{j \in S_i} x_j$ . Each  $S_i$  is nonempty because  $\mathfrak{q}$  is proper, and they form a nonempty antichain because the generators are irredundant and  $\mathfrak{q} \neq 0$ . For a subfamily  $T \subseteq [r]$ , their least common multiple is

$$\text{lcm}_{i \in T} x^{S_i} = x^{\bigcup_{i \in T} S_i},$$

with value 1 for  $T = \emptyset$ . Sending this monomial to the coordinate face on  $\bigcup_{i \in T} S_i$  identifies the lcm-lattice, ordered by divisibility, with the face-join semilattice  $\mathcal{L}$ . The minimal label 1 corresponds to the zero face.

For every nonzero label  $G$ , the open interval below its lcm monomial therefore has the same order complex as  $(\{0\}, G)_{\mathcal{L}}$ . The crosscut comparison [Lemma 13.68](#) identifies its reduced homology with that of  $K_{<G}$ . The lcm-lattice formula [71, Theorem 2.1] now gives, for  $p \geq 1$ ,

$$\beta_{p, x^{S_G}}^R(R/\mathfrak{q}) = \dim_K \tilde{H}_{p-2}(K_{<G}; K) = \beta_{p, G},$$

where  $S_G$  is the coordinate-ray set of  $G$ . At an atom the interval is empty and the augmented degree- $-1$  group is  $K$ , giving the minimal generator in homological degree one. Both resolutions have one degree-zero summand. Multidegrees outside the lcm-lattice contribute nothing: the Taylor resolution has only lcm-labelled free summands, and tensoring it with  $R/(x_1, \dots, x_d) = K$  computes the multigraded Tor groups.

Finally, the last nonzero Betti degree is  $\text{pd}_R(R/\mathfrak{q})$ . Indeed, cancelling homogeneous scalar units gives a finite minimal free resolution whose maps vanish after tensoring with  $K$ . Its last term supplies a nonzero Tor group and hence a lower bound for every projective resolution; its length supplies the upper bound. The same last degree is  $\text{fd}_{A_d}(A_d/\mathfrak{r})$  by [Theorem 13.70](#). Thus both the individual multiplicities and the stated dimensions agree, with ordinary projective dimension on the polynomial side and ordinary flat dimension on the core side.  $\square$

Projective dimension is not preserved: for one generator the ordinary quotient has projective dimension one, the root-face quotient two ([Corollary 13.71](#)); only the flat dimension and its face topology survive (26).

**Corollary 13.83** (Coefficient independence [26]). *For fixed  $\mathcal{C}$  and  $\mathcal{F}$ , the  $\beta_{p,G}$  and  $\text{fd}(A_{\mathcal{C}}/\mathfrak{t}_{\mathcal{F}})$  do not depend on  $D \subseteq K$ , and depend on  $K$  only through its characteristic; the real omnific and Gaussian omnific realizations agree.*

*Proof.* The finite complexes  $K_{<G}$  depend only on the cone and its antichain of faces. Choose orientations of their simplices. Every augmented boundary matrix then has integer entries 0, 1,  $-1$ , including the map from vertices to the empty-face copy of the field. Let  $k_0$  be the prime subfield of  $K$ , either  $\mathbb{Q}$  or  $\mathbb{F}_p$ . The rank of such a matrix over  $K$  equals its rank over  $k_0$ : a minor is nonzero over  $K$  exactly when its image in  $k_0$  is nonzero, since  $k_0 \hookrightarrow K$  is injective. Consequently

$$\dim_K \tilde{H}_j = \dim_K \tilde{C}_j - \text{rank}_K \partial_j - \text{rank}_K \partial_{j+1}$$

is determined by the characteristic, in every augmented degree. Equation (13.14) gives the same conclusion for all  $\beta_{p,G}$ . The ring  $D$  never enters these chain matrices. By Theorem 13.70, the flat dimension is the largest nonzero Betti degree for every unital  $D \subseteq K$ , proving its independence of  $D$  as well.

For the real and Gaussian realizations, the coefficient fields are  $\mathbb{R}$  and  $\mathbb{C}$ , both with prime field  $\mathbb{Q}$ ; the constant rings  $\mathbb{Z}$  and  $\mathbb{Z}[i]$  may differ without changing these invariants. This is independence within a fixed characteristic, not an assertion that reduction modulo every prime preserves the ranks.  $\square$

### Arithmetic modules.

**Proposition 13.84** (A nonzero scalar removes the tail [26]). *For  $0 \neq a \in D$ ,  $aA_{\mathcal{C}} = aD \oplus \Pi_{\mathcal{C}}$  and  $A_{\mathcal{C}}/aA_{\mathcal{C}} \cong D/aD$ ; if  $a \notin D^\times$ ,  $\text{fd}(D/aD) = \text{pd}(D/aD) = 1$ .*

*Proof.* Because  $a \neq 0$  in the field  $K$ , multiplication by  $a$  is a bijection on the  $K$ -vector space  $\Pi_{\mathcal{C}}$ . The direct sum  $A_{\mathcal{C}} = D \oplus \Pi_{\mathcal{C}}$  therefore gives  $aA_{\mathcal{C}} = aD \oplus \Pi_{\mathcal{C}}$ . Composing the constant map with  $D \rightarrow D/aD$  is surjective and has exactly this kernel, proving the quotient isomorphism.

The ring  $A_{\mathcal{C}}$  is a domain, so multiplication by  $a$  is injective and gives the free resolution

$$0 \longrightarrow A_{\mathcal{C}} \xrightarrow{a} A_{\mathcal{C}} \longrightarrow D/aD \longrightarrow 0.$$

If  $a$  is not a unit of  $D$ , then  $D/aD \neq 0$ . Tensoring the injection  $a : A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}$  with this module produces its zero endomorphism, which is not injective. Thus the quotient is not flat. The resolution gives  $\text{pd} \leq 1$ , and  $\text{fd} \leq \text{pd}$  together with nonflatness gives both dimensions equal to one. If  $a$  is a unit, the quotient is zero and no dimension-one assertion is made.  $\square$

This is 12's Proposition 13.2(i) for cones, and an elementary consequence of divisible tail coefficients, not a new finite-quotient theorem for **Oz** (26; compare Theorem 6.2).

**Theorem 13.85** (Arithmetic modules over a principal-ideal core [26]). *Let  $D$  be a principal ideal domain, not a field, and  $N \neq 0$  a  $D$ -module regarded as an  $A_{\mathcal{C}}$ -module through  $\text{ct}$ . Then  $\text{fd}_{A_{\mathcal{C}}} N = 1$  if  $N$  is torsion over  $D$  and  $\text{fd}_{A_{\mathcal{C}}} N = d$  otherwise. This applies to  $D = \mathbb{Z}$  and  $D = \mathbb{Z}[i]$ .*

*Proof.* Write  $T = N_{\text{tor}}$  and  $V = N/T$ . Every finitely generated submodule of  $T$  is, by the structure theorem over the principal ideal domain  $D$ , a finite direct sum of modules  $D/aD$  with  $a$  nonzero and noninvertible (or the zero module). Each such summand has flat dimension one over  $A_{\mathcal{C}}$  by Proposition 13.84. These submodules form a filtered system with union  $T$ . Tor commutes with

filtered colimits in its first variable, so  $\mathrm{Tor}_j(T, M) = 0$  for every module  $M$  and every  $j > 1$ ; hence  $\mathrm{fd} T \leq 1$ . If  $T \neq 0$ , choose  $0 \neq t \in T$  and  $0 \neq a \in D$  with  $at = 0$ . Multiplication by  $a$  is injective on  $A_{\mathcal{C}}$  but not on  $T$ , so  $T$  is not flat. This proves  $\mathrm{fd} N = 1$  when  $N = T \neq 0$ .

Now suppose  $V \neq 0$ . It is torsion free: if  $a(n + T) = 0$  with  $a \neq 0$ , some  $b \neq 0$  kills  $an$ , whence  $n \in T$  since  $ab \neq 0$ . Over a principal ideal domain a torsion-free module is flat, so  $-\otimes_D V$  preserves the cellular resolution of  $D$  from [Proposition 13.76](#) and its augmentation onto  $D \otimes_D V \cong V$ . All its terms are flat over  $A_{\mathcal{C}}$ . For degree zero, this follows from  $M \otimes_{A_{\mathcal{C}}} (A_{\mathcal{C}} \otimes_D V) \cong M \otimes_D V$ . For a positive term, associativity gives

$$I_G \otimes_D V \cong I_G \otimes_K (K \otimes_D V).$$

A  $K$ -basis of  $K \otimes_D V$  therefore expresses  $I_G \otimes_D V$  as a possibly infinite direct sum of copies of the flat ideal  $I_G$ . Thus this is a flat resolution of  $V$  of length  $d$ .

Choose  $b \in \mathrm{relint} \mathcal{C} \cap \mathbb{Q}^d$ , give  $V$  degree zero, and tensor the resolution over  $A_{\mathcal{C}}$  with  $\mathcal{W} = \mathcal{W}_{\mathcal{C}, b}$ . Tensor associativity identifies each term with the corresponding cellular term tensored first with  $\mathcal{W}$  and then over  $D$  with  $V$ . In exponent degree  $b$ , the degree-zero term vanishes because  $\mathcal{W}_b = 0$ . Face isolation leaves just  $K \otimes_D V$  in homological degree  $d$ . Consequently

$$\mathrm{Tor}_d^{A_{\mathcal{C}}}(V, \mathcal{W})_b \cong K \otimes_D V \neq 0. \quad (13.23)$$

For nonvanishing, put  $Q_D = \mathrm{Frac} D \subseteq K$ . The map  $V \rightarrow Q_D \otimes_D V$  is injective because  $V$  is torsion free, and  $Q_D \otimes_D V \neq 0$ . Extension of a nonzero vector space from  $Q_D$  to the field  $K$  remains nonzero, so  $K \otimes_D V \cong K \otimes_{Q_D} (Q_D \otimes_D V) \neq 0$ . This homogeneous Tor group is part of ordinary ungraded Tor and therefore proves  $\mathrm{fd} V = d$ .

It remains to transfer this lower bound to  $N$ , without assuming  $0 \rightarrow T \rightarrow N \rightarrow V \rightarrow 0$  splits. Tensor the two-term scalar resolution with the  $K$ -vector space  $\mathcal{W}$ . Its differential is multiplication by  $a \neq 0$ , a bijection on  $\mathcal{W}$ ; thus  $\mathrm{Tor}_j(D/aD, \mathcal{W}) = 0$  for every  $j \geq 0$ , including degree zero. Finite direct sums and filtered colimits give the same vanishing for  $T$ . The long exact Tor sequence now yields

$$\mathrm{Tor}_j(N, \mathcal{W}) \cong \mathrm{Tor}_j(V, \mathcal{W}) \quad (j \geq 1).$$

Degree-zero vanishing for  $T$  is what makes this valid also at  $j = 1$ , so the argument covers  $d = 1$ . The nonzero group in degree  $d$  gives  $\mathrm{fd} N \geq d$ . Finally, for any target module  $M$  and  $j > d$ , both  $\mathrm{Tor}_j(T, M)$  and  $\mathrm{Tor}_j(V, M)$  vanish, since  $d \geq 1$ . The same long exact sequence gives  $\mathrm{Tor}_j(N, M) = 0$ , hence  $\mathrm{fd} N \leq d$ . This proves the claimed dichotomy for arbitrary nonzero  $D$ -modules, with no finite-generation assumption.  $\square$

All these modules have  $\mathrm{Tor}_{>0}(D, N) = 0$  ([Theorem 13.72](#)); the theorem classifies modules acting through  $\mathrm{ct}$  only, not all modules over a core or the class ring (26). For the orthant the non-torsion value  $d$  contains 12's  $\mathrm{fd} D = d$  ([Corollary 13.8](#)).

*Remark 13.86* (Lattice modules over every cone [merge]). The proof of [Theorem 13.9\(ii\)](#) uses only that  $A = D \oplus \Pi$  with  $\Pi$  a  $K$ -space ideal of  $K \oplus \Pi$  inside a field and  $K \cap \Pi = 0$ ; so it applies to  $A_{\mathcal{C}}$  with  $\Pi = \Pi_{\mathcal{C}}$ , not finitely generated by [Theorem 13.63](#). For a coefficient lattice  $M$  of rank  $r \geq 2$  and  $J = X^g L(M)$ ,  $0 \neq g \in P_{\mathcal{C}}$ , the argument of [Theorem 13.11](#), with  $\mathrm{fd} \Pi_{\mathcal{C}} = d - 1$  ([Theorems 13.70](#) and [13.75](#) for the rays) and the cap module  $\mathcal{W}_{\mathcal{C}, b}$  in place of  $V_{[d]}$ , gives for  $d \geq 2$

$$\mathrm{fd}_{A_{\mathcal{C}}} L(M) = d, \quad \mathrm{fd}_{A_{\mathcal{C}}} (A_{\mathcal{C}}/J) = d + 1,$$

so  $\mathrm{w.gl.dim} A_{\mathcal{C}} \geq d + 1$  whenever  $K \neq \mathrm{Frac} D$  (take  $M = D + Dc$ ,  $c \in K \setminus \mathrm{Frac} D$ ). This improves the lower bound  $d$  of 26's Question 13.2, as 12 and 21 do for  $A_d$  ([Section 13.5](#) and [Theorem 13.48](#)). For  $d = 1$ ,  $A_{\mathcal{C}} \cong A_1 = T_{\mathbb{Q}}$ .



*Details of the lattice comparison.* Put  $A = A_{\mathcal{C}}$  and  $\Pi = \Pi_{\mathcal{C}}$ . A coefficient lattice here means a finite free  $D$ -submodule  $M \subseteq K$  with chosen basis  $m_1, \dots, m_r$ ; no principal-ideal hypothesis on  $D$  is needed. Set  $L(M) = M \oplus \Pi$ . Its generators  $m_j$  and the relations between them give the exact sequence

$$0 \longrightarrow \Pi^{r-1} \xrightarrow{\sigma} A^r \xrightarrow{\pi} L(M) \longrightarrow 0,$$

where  $\pi(a_1, \dots, a_r) = \sum_j a_j m_j$  and

$$\sigma(u_2, \dots, u_r) = \left( -\sum_{j=2}^r \frac{m_j}{m_1} u_j, u_2, \dots, u_r \right).$$

Indeed,  $m_1 \Pi = \Pi$  gives surjectivity. Writing  $a_j = d_j + u_j$ , a relation has constant part  $\sum_j d_j m_j = 0$ , so independence over  $D$  forces every  $d_j = 0$ . Its remaining tail equation is precisely the displayed formula for  $u_1$ . The ratios  $m_j/m_1$  preserve  $\Pi$ , making  $\sigma$  well-defined and  $A$ -linear; its last  $r-1$  coordinates give injectivity.

The antichain of all rays defines  $\Pi$ , so [Theorems 13.70](#) and [13.75](#) give  $\text{fd}_A D = d$  and  $\text{fd}_A \Pi = d-1$ . Splicing a flat resolution of  $\Pi^{r-1}$  into the displayed sequence gives  $\text{fd}_A L(M) \leq d$ . For  $0 \neq g \in P_{\mathcal{C}}$ , multiplication by  $X^g$  is an  $A$ -module isomorphism  $L(M) \rightarrow J$ . Its image is an ideal of  $A$ : all its exponents are nonzero cone points, so even coefficients in  $K$  remain allowed. The sequence  $0 \rightarrow J \rightarrow A \rightarrow A/J \rightarrow 0$  therefore gives  $\text{fd}_A(A/J) \leq d+1$ .

For every  $A$ -module  $N$  and  $j \geq 2$ , the free middle terms give

$$\text{Tor}_j^A(L(M), N) \cong \text{Tor}_{j-1}^A(\Pi, N)^{r-1} \cong \text{Tor}_j^A(D, N)^{r-1}, \quad \text{Tor}_{j+1}^A(A/J, N) \cong \text{Tor}_j^A(L(M), N).$$

These are ordinary ungraded isomorphisms; multiplication by  $X^g$  shifts exponent degrees when a grading is retained. For  $d \geq 2$ , choose  $N = \mathcal{W}_{\mathcal{C},b}$  with rational  $b \in \text{relint } \mathcal{C}$ . The group  $\text{Tor}_d^A(D, N)$  has degree- $b$  component  $K$  by [Proposition 13.76](#). Since  $r-1 \geq 1$ , the displayed comparisons supply nonzero Tor in degrees  $d$  and  $d+1$  for the two modules. This proves the matching lower bounds against all flat resolutions.

If  $K \neq \text{Frac } D$ , choose  $c \in K \setminus \text{Frac } D$ . The elements  $1, c$  are independent over  $D$ : a relation with nonzero coefficient of  $c$  would express  $c$  as a fraction from  $D$ . The rank-two lattice  $D + Dc$  therefore gives the stated weak-global-dimension lower bound. No integrality or quadratic equation for  $c$  is required.

For  $d = 1$ , a rational generator of the unique ray identifies  $P_{\mathcal{C}}$  with  $\mathbb{Q}_{\geq 0}$  and fixes the coefficients, giving  $A_{\mathcal{C}} \cong A_1 = T_{\mathbb{Q}}$ . The dimension-one case of [Theorem 13.11](#), using its antisymmetric tensor rather than the comparison for  $j \geq 2$ , gives  $\text{fd } L(M) = 1$  and  $\text{fd}(A/J) = 2$ . For completeness, if the lattice instead has rank one, then  $L(M) = m_1 A \cong A$  and  $J = m_1 X^g A$  is a nonzero proper principal ideal. Its quotient has flat dimension one: the principal free resolution gives the upper bound, and tensoring multiplication by  $m_1 X^g$  with the nonzero quotient gives a noninjective zero map.  $\square$

**Passage to the omnific ring is not flat.** The injection  $\lambda_{\omega}$  pulls back the order of **No** to a total order on  $\mathbb{Q}^d$  whose positive cone contains  $P_{\mathcal{C}} \setminus \{0\}$ . Put

$$T_{\lambda_{\omega}} = D \oplus \bigoplus_{g \in \mathbb{Q}^d, \lambda_{\omega}(g) > 0} KX^g \supseteq A_{\mathcal{C}}, \quad (13.24)$$

a set-sized ring with the same fraction field ([Proposition 13.59](#)) and a compatible ct.

**Lemma 13.87** (Two-ray syzygies have zero constants [26]). *If  $u, v$  are nonzero rational vectors on distinct rays and  $rX^u + sX^v = 0$  in  $A_{\mathcal{C}}$ , then  $\text{ct}(r) = \text{ct}(s) = 0$ .*

*Proof.* Suppose  $u - v \in \mathcal{C}$ . Then  $u = (u - v) + v$  is a sum of two cone points lying in the ray containing  $u$ . The face property forces both summands into that ray, in particular the nonzero point  $v$ , contrary to the rays being distinct. Interchanging  $u$  and  $v$  excludes  $v - u \in \mathcal{C}$  as well.

Write  $r = \sum_m r_m X^m$  and  $s = \sum_m s_m X^m$ , with finite supports in  $P_{\mathcal{C}}$  and constant coefficients in  $D$ . The coefficient of  $X^u$  in the relation is  $r_0 + s_{u-v} = r_0$ , since  $u - v$  is outside the support monoid. It is zero because the monomials are linearly independent in  $K[\mathbb{Q}^d]$ . Similarly, the coefficient of  $X^v$  is  $r_{v-u} + s_0 = s_0 = 0$ . Thus  $\text{ct}(r) = \text{ct}(s) = 0$  for arbitrary finite-sum coefficients  $r, s$ , not merely for monomial relations.  $\square$

**Theorem 13.88** (A two-monomial nonflatness certificate [26]). *If  $d \geq 2$ ,  $A_{\mathcal{C}} \hookrightarrow T_{\lambda_{\omega}}$  is not flat, and in the real and Gaussian realizations the inclusion into  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  fails the same finite equational criterion.*

*Proof.* A full-dimensional pointed cone of dimension  $d \geq 2$  has at least two distinct rays. Choose nonzero rational vectors  $u, v$  on them. Injectivity of  $\lambda_{\omega}$  makes their images unequal; after interchanging the vectors, assume  $\lambda_{\omega}(u) > \lambda_{\omega}(v)$ . Put  $T = T_{\lambda_{\omega}}$ . Then  $z = X^{u-v} \in T$ , since its image exponent is positive, and

$$X^u \cdot 1 + X^v \cdot (-z) = 0.$$

The previous lemma shows that  $u - v$  is outside  $\mathcal{C}$ , so  $z$  is a newly admitted monomial. Every old relation  $rX^u + sX^v = 0$  has both coordinates in  $\Pi_{\mathcal{C}}$ .

If  $T$  were flat over  $A_{\mathcal{C}}$ , the equational criterion [29, Tag 00HK] would express this new relation as

$$(1, -z) = \sum_{j=1}^m t_j(r_j, s_j), \quad t_j \in T, \quad r_j, s_j \in A_{\mathcal{C}}, \quad r_j X^u + s_j X^v = 0.$$

The constant map  $\text{ct} : T \rightarrow D$  extends that of the core. Its kernel is an ideal containing both coordinates of every old relation. Taking the constant coefficient of the first coordinate would give  $1 = \sum_j \text{ct}(t_j) \text{ct}(r_j) = 0$  in the nonzero ring  $D$ , a contradiction. Thus the finite equational criterion fails.

There is also an explicit tensor witness in these set-sized rings. Let  $I = (X^u, X^v)A_{\mathcal{C}}$ . The map

$$\theta : I \rightarrow D, \quad \theta(rX^u + sX^v) = \text{ct}(r),$$

is well-defined by Lemma 13.87 and is linear when  $D$  is an  $A_{\mathcal{C}}$ -module through  $\text{ct}$ . The balanced map  $(f, t) \mapsto \theta(f) \text{ct}(t)$  induces  $I \otimes_{A_{\mathcal{C}}} T \rightarrow D$ . It sends

$$\eta = X^u \otimes 1 - X^v \otimes z$$

to 1, so  $\eta \neq 0$ . But multiplication  $I \otimes_{A_{\mathcal{C}}} T \rightarrow T$  sends  $\eta$  to zero. Tensoring the injection  $I \hookrightarrow A_{\mathcal{C}}$  is therefore not injective, directly witnessing nonflatness.

For  $(D, K) = (\mathbb{Z}, \mathbb{R})$ , the embedded monomials are  $\omega^{\lambda_{\omega}(u)}$  and  $\omega^{\lambda_{\omega}(v)}$ . Their quotient  $\omega^{\lambda_{\omega}(u) - \lambda_{\omega}(v)}$  is an omnific monomial because the exponent is positive, so the same relation with first coordinate 1 holds in  $\mathbf{Oz}$ . Every old relation has coordinates in the purely infinite ideal, which remains an ideal in the whole target. Thus no finite combination with coefficients in  $\mathbf{Oz}$  can have first constant coefficient 1. The argument is identical for  $(\mathbb{Z}[i], \mathbb{C})$  and  $\mathbf{Oz}[i]$ , with constant map to  $\mathbb{Z}[i]$  and purely infinite ideal  $\Pi_{\mathbb{C}}$ . These full-ring assertions concern only the explicit finite equation and its possible finite trivializations; no tensor product of classes is needed.  $\square$

**Example 13.89** (An explicit pair in **Oz** [26]). For the orthant  $\mathbb{R}_{\geq 0}^2$  with  $\lambda_\omega(x, y) = (x + y)\omega + y$ , the coordinate monomials become  $b = \omega^\omega$  and  $a = \omega^{\omega+1}$ , and  $a - \omega b = 0$  in **Oz**; every syzygy of the pair in the core has zero constant terms, so this relation with first coefficient 1 does not descend.

*Verification of the explicit pair.* The first coordinate ray has  $\lambda_\omega(1, 0) = \omega$ , and the second has  $\lambda_\omega(0, 1) = \omega + 1$ . Thus the pair  $(a, b)$  in the example corresponds to  $(X^{(0,1)}, X^{(1,0)})$ , in that order. Their exponent difference is  $(-1, 1)$ , outside the orthant, but its  $\lambda_\omega$ -image is 1. The new ratio is consequently the surreal monomial  $\omega^1 = \omega$ , and multiplication of monomials gives  $\omega \omega^\omega = \omega^{\omega+1} = a$ . All core relations  $ra + sb = 0$  have  $\text{ct}(r) = \text{ct}(s) = 0$  by [Lemma 13.87](#). Any finite combination of them with omnific coefficients still has both constant coefficients zero, whereas  $(1, -\omega)$  has first constant coefficient 1. This verifies the claimed failure directly, including the ordering of the two generators.  $\square$

For the orthant the conclusion is the nonflatness in 12's [Theorem 13.14](#), proved there by an intersection certificate in  $T_\Gamma$  with a real embedding; 26's equational certificate, with a lexicographic embedding and the explicit failure in **Oz**, is a second route. Partial exponent order creates independent directions of divisibility, which the total order of **No** compares; so the flat dimensions of cores do not transfer to **Oz** by flat base change. The obstruction does not deny the embeddings, the fraction field or the retraction  $\text{ct}$ , and it says nothing about embeddings of one-dimensional cores; it is exactly the failure of the displayed finite test for  $d \geq 2$  (26).

**Dependencies and pitfalls (26).** Flat face ideals rest on facet subtraction, finite supports, principal exhaustion and filtered colimits; the reduced resolution on degreewise simplex exactness and same-label scalar cancellation; the exact dimension on the cap module and its degree-shifted colimit; the face formula on the outside-vertex complex and the explicit contraction of a polytopal boundary minus a face; simpliciality on a minimal pair join and facet normals; the arithmetic dichotomy on scalar resolutions, torsion theory over a principal ideal domain and the full-face cap; nonflatness on the absence of a positive difference between two rays. 26 records these obligations: a principal ideal  $X^q A_\mathcal{C}$  has coefficient space  $D$ , not  $K$ , at  $q$ , so the exhaustion always uses a strictly smaller shift; the  $I_G$  are flat but not projective, and a scalar cancellation does not make them free; the cap module is defined before use and its late transition maps are identities on  $K$ ; only nonzero joins are labels, and a face inside the retained face gets no spurious degree-one class, while a ray disjoint from it correctly contributes  $\tilde{H}_{-1}(\{\emptyset\}; K) = K$ ; and all rings, resolutions and test modules are sets, the statement about **Oz** being a finite relation. The main theorems use no unrefereed repository theorem as a premise.

## 14 Large internal fields and the binomial quotient

The universal theorem ([Theorem 5.1](#)) describes what a set-sized target can see of **Oz**: every purely infinite element dies, and only the constant term survives. This section, taken from manuscript 07 together with one proposition of manuscript 06, shows how much survives in a large target once a single purely infinite element becomes a *unit*. The answer is a great deal: the target then contains a real closed Hahn field into which all of **No** embeds as an ordered field. In the nonzero binomial quotient  $\mathcal{Q}_a = \mathbf{Oz}/(1 + \omega^a)$ , which has no nonzero set-sized ring image at all, the same mechanism produces an algebraically closed class field containing a copy of **No**[ $i$ ], exact finite cyclotomic algebras over it, and an atomless Boolean algebra of idempotents.

The dependence of the results is as follows (07). An ordered additive compression of  $\mathbf{No}$  and the closedness of set-sized Hahn fields give a copy of  $\mathbf{No}$  inside the compressed field  $\mathcal{F}_a$  (Proposition 14.1 and Theorem 14.4); localization transfers  $\mathcal{F}_a$  into every quotient in which a suitable purely infinite element is invertible (Theorem 14.7); a support argument gives exact binomial kernels (Theorem 14.15); and finite Chinese remainder decompositions give the internal field  $\mathcal{E}_a$  and the Cantor algebra (Corollary 14.17 and Theorem 14.19). The absence of small images of  $\mathcal{Q}_a$  comes from the other branch, Theorems 5.1 and 6.1. No factorization conjecture enters either chain. What manuscript 07 calls its Main Theorem B is Theorem 14.7 together with Theorem 14.4, and its Main Theorem C is the combination of Lemma 14.14, Corollary 14.17, and Theorem 14.19 with the absence of small images of  $\mathcal{Q}_a$  recorded before Lemma 14.14.

## 14.1 Compressed exponent fields

Fix  $a > 0$ . Recall from (2.4) and Lemma 4.2 that  $H_a = \{h \in \mathbf{No} : n|h| < a \text{ for all ordinary } n \geq 1\}$  is a convex divisible subgroup of  $(\mathbf{No}, +)$  containing a proper class of positive elements, and that  $\mathcal{F}_a(k) = k((X^{H_a}))$  is a field for every set field  $k$ . For  $k = \mathbb{R}$  and  $k = \mathbb{C}$  we realize these fields inside the surreals and surcomplex numbers,

$$\begin{aligned}\mathcal{F}_a &= \mathcal{F}_a(\mathbb{R}) = \{f \in \mathbf{No} : \text{supp } f \subseteq H_a\}, \\ \mathcal{F}_a(i) &= \mathcal{F}_a(\mathbb{C}) = \{f \in \mathbf{No}[i] : \text{supp } f \subseteq H_a\}.\end{aligned}\tag{14.1}$$

Taking real and imaginary parts of coefficients shows  $\mathcal{F}_a(i) = \mathcal{F}_a + i\mathcal{F}_a$ . Manuscript 07 calls them *compressed exponent fields*; the phrase refers to the restriction on exponents, not to finiteness of supports. Both are proper classes. Manuscript 07 checks directly that  $H_a$  is a subgroup by the inequality  $n|h + h'| \leq n|h| + n|h'| < a/2 + a/2$ , using the defining inequalities at  $2n$ ; this is part of Lemma 4.2.

**Proposition 14.1** (Closedness of the compressed fields [07, 24]). *The class  $\mathcal{F}_a$  is a real closed subfield of  $\mathbf{No}$ , and  $\mathcal{F}_a(i)$  is an algebraically closed subfield of  $\mathbf{No}[i]$ .*

*Proof.* Since  $H_a$  is a subgroup, sums and products of series supported in  $H_a$  are supported in  $H_a$ ; the inverse of a nonzero Hahn series has support in the subgroup generated by its own support (factor out the leading monomial and use the Hahn geometric series), so it stays in  $H_a$  as well. Hence  $\mathcal{F}_a$  and  $\mathcal{F}_a(i)$  are subfields (Lemma 4.2).

Let  $P$  be a nonconstant polynomial with coefficients in  $\mathcal{F}_a(i)$ . The union of the supports of its finitely many coefficients is a set; its rational span is a set-sized divisible ordered subgroup  $G \subseteq H_a$ , because  $H_a$  is a divisible subgroup. The series with support in  $G$  form the set-sized Hahn fields  $\mathbb{R}((\omega^G)) \subseteq \mathcal{F}_a$  and  $\mathbb{C}((\omega^G)) \subseteq \mathcal{F}_a(i)$ , whose operations are those of  $\mathbf{No}$  and  $\mathbf{No}[i]$ , and  $P$  has coefficients in  $\mathbb{C}((\omega^G))$ . For divisible  $G$  this field is algebraically closed by the Hahn-field theorem [27, Corollary 4] (stated there for well-ordered supports; the substitution  $t^g = \omega^{-g}$  converts). So  $P$  has a root in  $\mathbb{C}((\omega^G)) \subseteq \mathcal{F}_a(i)$ , and  $\mathcal{F}_a(i)$  is algebraically closed.

For the real field, note first that  $\mathbb{R}((\omega^G))$  is real closed. If  $0 < f \in \mathbb{R}((\omega^G))$ , write a square root of  $f$  in  $\mathbb{C}((\omega^G))$  as  $u + iv$  with  $u, v \in \mathbb{R}((\omega^G))$ ; then  $u^2 - v^2 = f$  and  $2uv = 0$ . If  $u = 0$  then  $f = -v^2 \leq 0$ , which is impossible; so  $v = 0$  and  $u^2 = f$ . If  $Q$  has odd degree and coefficients in  $\mathbb{R}((\omega^G))$ , coefficientwise conjugation is an automorphism of  $\mathbb{C}((\omega^G))$  fixing  $Q$ , so the nonreal roots of  $Q$  occur in conjugate pairs with equal multiplicities, and an odd number of roots counted with multiplicity must be real. An ordered field in which positive elements are squares and odd-degree polynomials have roots is real closed. Now let  $0 < f \in \mathcal{F}_a$ , or let  $Q$  be of odd degree over  $\mathcal{F}_a$ ;

choosing  $G$  as above for  $f$  or for the coefficients of  $Q$ , the required square root or root lies in  $\mathbb{R}((\omega^G)) \subseteq \mathcal{F}_a$ . Hence  $\mathcal{F}_a$  is real closed. Only set-sized instances of the classical theorem have been used.  $\square$

*Remark 14.2* (Bounded in the ambient field, yet a proper class [07]). Every positive  $f \in \mathcal{F}_a$  satisfies  $\omega^{-a} < f < \omega^a$ : its leading exponent  $h$  satisfies  $|h| < a/2$  and its leading coefficient is a positive real. A field contained in such an ambient interval need not be small, since the interval is itself a proper class;  $\mathcal{F}_a$  is a proper class, and by [Theorem 14.4](#) it even contains an ordered-field copy of **No**. Bounds in the surrounding field do not give bounds inside an isomorphic copy of **No**.

## 14.2 An ordered additive compression of the surreals

Knowing that  $\mathcal{F}_a$  contains Hahn fields of every set size ([Lemma 4.2](#)) would already show considerable largeness. Manuscript 07 does more: it embeds all of **No** into  $\mathcal{F}_a$  as an ordered field. Define

$$\vartheta(x) = \frac{1}{2} \left( \frac{x}{1+|x|} - 3 \right), \quad \vartheta : \mathbf{No} \longrightarrow (-2, -1). \quad (14.2)$$

For  $x \geq 0$  the inner fraction is  $x/(1+x) = 1 - 1/(1+x) \in [0, 1)$ , and for  $x < 0$  it is  $x/(1-x) = -1 + 1/(1-x) \in (-1, 0)$ . Each branch is strictly increasing and they agree at 0, so  $\vartheta$  is strictly increasing with values in  $(-2, -1)$ . For a real-coefficient normal form put

$$\Psi \left( \sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\vartheta(\gamma)}, \quad \Psi_a(x) = a\Psi(x). \quad (14.3)$$

(Manuscript 07 writes  $E_a$  for  $\Psi_a$ ; the letter  $\mathcal{E}_a$  is reserved here for the internal field of [Corollary 14.17](#).)

**Lemma 14.3** (Ordered additive compression [07]). *The map  $\Psi_a : (\mathbf{No}, +) \rightarrow (H_a, +)$  is an injective, order-preserving,  $\mathbb{R}$ -linear homomorphism of additive groups.*

*Proof.* A strictly increasing map sends a reverse well-ordered set to a reverse well-ordered set, so the support  $\vartheta(\text{supp } x)$  is a reverse well-ordered set and  $\Psi(x)$  exists. Since  $\vartheta$  is injective, distinct exponents stay distinct, and the coefficient of  $\Psi(x)$  at  $\vartheta(\gamma)$  is  $x_{\gamma}$ ; hence coefficientwise addition, with all cancellations, and multiplication by real scalars are preserved. If  $x \neq 0$ , the greatest exponent  $\gamma_0$  of  $\text{supp } x$  is sent to the greatest exponent  $\vartheta(\gamma_0)$  of  $\text{supp } \Psi(x)$  with the same coefficient; since the sign of a nonzero normal form is the sign of its leading coefficient,  $\Psi$  maps positive elements to positive elements, and being additive it is injective and order preserving. The leading exponent of a nonzero  $\Psi(x)$  lies in  $(-2, -1)$ , so  $\Psi(x)$  is infinitesimal and  $n|\Psi(x)| < 1$  for every ordinary  $n \geq 1$ . Multiplying by  $a > 0$  gives  $n|\Psi_a(x)| < a$ , that is,  $\Psi_a(x) \in H_a$ , and multiplication by the positive surreal  $a$  preserves additivity,  $\mathbb{R}$ -linearity and order.  $\square$

The map  $\vartheta$  is not additive, and  $\Psi$  is not asserted to be multiplicative. The construction deliberately has two levels: first an additive exponent group is compressed, and only then is the additive map used as a substitution on monomials.

**Theorem 14.4** (A full surreal field in every compressed field [07]). *The formula*

$$\Theta_a \left( \sum_{\gamma} r_{\gamma} \omega^{\gamma} \right) = \sum_{\gamma} r_{\gamma} \omega^{\Psi_a(\gamma)} \quad (14.4)$$

*defines an injective ordered-field homomorphism  $\Theta_a : \mathbf{No} \hookrightarrow \mathcal{F}_a$  fixing  $\mathbb{R}$ . It preserves every set-indexed strongly summable family. Extending coefficients,  $x + iy \mapsto \Theta_a(x) + i\Theta_a(y)$ , gives an injective field homomorphism  $\mathbf{No}[i] \hookrightarrow \mathcal{F}_a(i)$  fixing  $\mathbb{C}$ .*

*Proof.* The support of the right side of (14.4) is  $\Psi_a(\text{supp } x)$ , the image of a reverse well-ordered set under a strictly increasing map into  $H_a$  (Lemma 14.3); so the output exists and lies in  $\mathcal{F}_a$ . If  $(x_\lambda)$  is a strongly summable set-indexed family, the union of the supports is reverse well ordered and each exponent receives finitely many contributions; since  $\Psi_a$  is injective and strictly increasing, the same holds for the family  $(\Theta_a(x_\lambda))$  with the same contributions at  $\Psi_a(\gamma)$  as at  $\gamma$ , so  $\Theta_a(\sum_\lambda x_\lambda) = \sum_\lambda \Theta_a(x_\lambda)$ .

For multiplication, write  $x = \sum r_\gamma \omega^\gamma$  and  $y = \sum s_\delta \omega^\delta$ . Because  $\Psi_a$  is additive and injective,  $\Psi_a(\gamma) + \Psi_a(\delta) = \Psi_a(\varepsilon)$  holds exactly when  $\gamma + \delta = \varepsilon$ , and every sum  $\Psi_a(\gamma) + \Psi_a(\delta)$  lies in the image of  $\Psi_a$ . Hence the coefficient of  $\Theta_a(x)\Theta_a(y)$  at  $\Psi_a(\varepsilon)$  is  $\sum_{\gamma+\delta=\varepsilon} r_\gamma s_\delta = (xy)_\varepsilon$ , and its coefficients at exponents outside the image vanish; so  $\Theta_a(xy) = \Theta_a(x)\Theta_a(y)$ . Additivity is clear, and constants are fixed because  $\Psi_a(0) = 0$ . A nonzero  $x$  has a nonzero image whose leading coefficient is that of  $x$ , so  $\Theta_a$  is injective and preserves signs, hence order. It is an injective unital ring map between fields, hence a field embedding. The complex extension is the same coefficient-and-support argument with complex coefficients.  $\square$

**Example 14.5** ([07]). Since  $\vartheta(0) = -3/2$ , one has  $\Psi_a(1) = a\omega^{-3/2}$  and

$$\Theta_a(\omega) = \omega^{a\omega^{-3/2}}.$$

No preservation of the ordinal embedding, of birthdays or simplicity, or of the surreal exponential is asserted or needed:  $\Theta_a$  is an explicit ordered-field embedding with a controlled normal-form support image. Constructions of self-similar surreal subclasses and strongly linear maps have a substantial prior literature, notably the surreal substructures of Bagayoko and van der Hoeven [3]. Manuscript 07 uses the elementary construction above to make the transfer theorem independent of a general classification of surreal substructures, and does not claim that surreal self-similarity itself is new.

### 14.3 Transfer by inverting a purely infinite element

**Lemma 14.6** (Multiplication by a purely infinite denominator [07]). *Let  $0 \neq g \in \Pi$  and choose  $a > 0$  with  $a < \gamma$  for every  $\gamma \in \text{supp } g$  (possible by Lemma 3.1). If  $f \in \mathcal{F}_a$ , then  $gf \in \Pi$ . The same holds for  $0 \neq g \in \Pi_{\mathbb{C}}$  and  $f \in \mathcal{F}_a(i)$ .*

*Proof.* Every  $h \in \text{supp } f$  satisfies  $|h| < a/2$ . Every exponent occurring in the product before cancellation has the form  $\gamma + h$  with  $\gamma \in \text{supp } g$ , and  $\gamma + h > a - a/2 > 0$ . Hahn multiplication is defined and cancellation cannot create an exponent outside the sum of the supports, so  $gf$  has strictly positive support and real (respectively complex) coefficients.  $\square$

**Theorem 14.7** (Localization and quotient transfer [07]). *Under the hypotheses of Lemma 14.6, the subfield  $\mathcal{F}_a$  of  $\mathbf{No}$  lies in the localization  $\mathbf{Oz}[1/g] \subseteq \mathbf{No}$ . If  $q : \mathbf{Oz} \rightarrow B$  is a unital homomorphism to a nonzero class ring and  $q(g)$  is invertible in  $B$ , then*

$$\iota_g(f) = q(gf) q(g)^{-1} \quad (f \in \mathcal{F}_a) \quad (14.5)$$

*is an injective field homomorphism  $\mathcal{F}_a \hookrightarrow B$ , and  $\iota_g \circ \Theta_a$  is an injective field homomorphism  $\mathbf{No} \hookrightarrow B$ . For  $q : \mathbf{Oz}[i] \rightarrow B$  and  $0 \neq g \in \Pi_{\mathbb{C}}$  with  $q(g)$  invertible, the same formula gives an injective field homomorphism  $\mathcal{F}_a(i) \hookrightarrow B$ , and hence a copy of  $\mathbf{No}[i]$  in  $B$ . In either case  $B$  is a proper class.*



*Proof.* Inside  $\mathbf{No}$ ,  $f = (gf)/g$  with  $gf \in \Pi \subseteq \mathbf{Oz}$  by Lemma 14.6, so  $f \in \mathbf{Oz}[1/g]$ . The elements of  $q(\mathbf{Oz})$  commute pairwise, and  $q(g)^{-1}$  commutes with every element  $y$  commuting with  $q(g)$  (multiply  $q(g)y = yq(g)$  on both sides by  $q(g)^{-1}$ ). Hence  $q(\mathbf{Oz})$  and  $q(g)^{-1}$  generate a commutative subring of  $B$ , and all computations below take place there; no commutativity of  $B$  is needed. The map (14.5) is additive and  $\iota_g(1) = 1$ . For  $f, f' \in \mathcal{F}_a$  the elements  $gf, gf', gff'$  lie in  $\mathbf{Oz}$  and  $(gf)(gf') = g \cdot (gff')$ , so

$$\iota_g(f)\iota_g(f') = q(gf)q(gf')q(g)^{-2} = q(g)q(gff')q(g)^{-2} = \iota_g(ff').$$

(Equivalently,  $\iota_g$  is the restriction to  $\mathcal{F}_a$  of the extension of  $q$  to  $\mathbf{Oz}[1/g]$  given by the universal property of localization.) If  $f \neq 0$  and  $\iota_g(f) = 0$ , then  $1_B = \iota_g(f)\iota_g(f^{-1}) = 0$ , contradicting  $B \neq 0$ ; so  $\iota_g$  is injective. Composing with Theorem 14.4 gives the copy of  $\mathbf{No}$ . The complex case is identical, with Lemma 14.6 for  $\Pi_{\mathbb{C}}$  and the complex extension of  $\Theta_a$ . Finally, if  $B$  were set-sized, Theorem 5.1 would give  $q(g) = 0$ , which is not invertible in a nonzero ring.  $\square$

*Remark 14.8* ([07]). If the target  $B$  is noncommutative, the image of the localization still lies in a commutative subring, as the proof shows; no commutativity condition on all of  $B$  is needed. The hypothesis on  $a$  is always available in the full surreal exponent class, even when the exponents of  $g$  approach zero inside a smaller ordered group (Lemma 3.1). No inference of this kind is valid for an arbitrary fixed set-sized exponent group without an additional lower-bound hypothesis.

**Corollary 14.9** (Quotients by a unit-constant perturbation [07]). *If  $0 \neq g \in \Pi$ , then  $\mathbf{Oz}/(1+g)$  is nonzero and contains a class copy of  $\mathbf{No}$ . If  $0 \neq g \in \Pi_{\mathbb{C}}$ , then  $\mathbf{Oz}[i]/(1+g)$  is nonzero and contains a class copy of  $\mathbf{No}[i]$ . Neither quotient has a unital homomorphism to a nonzero set-sized ring.*

*Proof.* The element  $1+g$  is nonconstant, hence a nonunit by Lemma 2.4, so the quotients are nonzero. In the quotient  $g = -1$  is a unit, so Theorem 14.7 applies. Since  $\text{ct}(1+g) = 1$ , the ideal  $(1+g)$  has constant-term image all of  $\mathbb{Z}$  (respectively  $\mathbb{Z}[i]$ ), and Theorem 6.1 makes the universal set-sized image the zero ring; a unital map from the zero ring exists only to the zero ring.  $\square$

**Corollary 14.10** (A conditional statement about nonarithmetic primes [07]). *Let  $P$  be a prime ideal of  $\mathbf{Oz}$  such that  $P \cap \mathbb{Z} = 0$  and  $P \not\subseteq \Pi$ . Then  $\mathbf{Oz}/P$  contains a class copy of  $\mathbf{No}$  and is a  $\mathbb{Q}$ -algebra. Likewise, if  $P$  is a prime ideal of  $\mathbf{Oz}[i]$  with  $P \cap \mathbb{Z}[i] = 0$  and  $P \not\subseteq \Pi_{\mathbb{C}}$ , then  $\mathbf{Oz}[i]/P$  contains a class copy of  $\mathbf{No}[i]$  and is a  $\mathbb{Q}(i)$ -algebra.*

*Proof.* Choose  $f \in P$  with nonzero constant term  $c$ . The series  $f/c$  still lies in the omnific ring: its constant term is 1 and its other coefficients are real (respectively complex). Since  $f = c \cdot (f/c)$ ,  $P$  is prime and  $c \notin P$ , we get  $f/c \in P$ . Write  $f/c = 1+g$  with  $g$  purely infinite;  $g \neq 0$  because  $c \notin P$ . In the quotient  $g = -1$  is a unit, and Theorem 14.7 applies. For every nonzero ordinary coefficient  $d$  ( $d \in \mathbb{Z}$ , respectively  $d \in \mathbb{Z}[i]$ ),  $g/d$  lies in the ring, and in the quotient  $d \cdot (-g/d) = -g = 1$ ; so  $d$  is invertible, which gives the  $\mathbb{Q}$ -, respectively  $\mathbb{Q}(i)$ -, algebra structure.  $\square$

The corollary is deliberately conditional. Manuscript 07 does not assert that prime ideals with these properties exist; see Section 14.8. (Status (batch 34): with global choice they exist. Every maximal class ideal containing  $1 + \omega^a$  is one (Theorem 14.34 and Proposition 14.33), and 24 proves this corollary again, with a proper-class Hahn field in place of  $\mathbf{No}$  (Section 14.9).)

**Example 14.11** (Invertibility cannot be replaced by nonvanishing [07]). The quotient  $\mathbf{Oz}/(\omega^a)$  is nonzero, and the class of  $\omega^{a/2}$  is a nonzero element of square zero: it is nonzero because  $\omega^{a/2}/\omega^a = \omega^{-a/2} \notin \mathbf{Oz}$ . Since  $(\omega^a) \subseteq \Pi$ , this quotient still maps onto  $\mathbb{Z}$  by constant terms, and so

it cannot contain a unital copy of  $\mathbb{Q}$ : composing such a copy with that map would give a unital map  $\mathbb{Q} \rightarrow \mathbb{Z}$ . A surviving nilpotent and a surviving unit are therefore substantively different.

The same mechanism, with a fraction field in place of the quotient, applies to every prime ideal not containing the purely infinite ideal. Here  $X^g$  is the monomial of  $k((X^{\mathbf{No}}))$ , which is  $\omega^g$  when  $k = \mathbb{R}$  or  $k = \mathbb{C}$ .

**Proposition 14.12** (A scaled field in every non-arithmetic residue field [06]). *Let  $k$  be a set field,  $D \subseteq k$  a unital subring,  $A = \mathcal{A}_{D,k}$ , and let  $P$  be a prime class ideal of  $A$  with  $\Pi_k \not\subseteq P$ . There exists  $a > 0$  for which the class field  $\mathcal{F}_a(k) = k((X^{H_a}))$  embeds in  $\text{Frac}(A/P)$ .*

*Proof.* Choose  $f \in \Pi_k \setminus P$  and write  $f = X^g h$  with  $g > 0$  and  $h \in \Pi_k$  (Lemma 3.1 applied to  $\text{supp } f$ ). Then  $X^g \notin P$ , since otherwise  $f \in P$ . Put  $a = g/2$  and  $m = X^a$ . As  $m^2 = X^g \notin P$  and  $P$  is prime,  $\bar{m} \neq 0$  in the domain  $A/P$ . Since  $m\mathcal{F}_a(k) \subseteq \Pi_k \subseteq A$  (Lemma 4.2), we may define

$$\iota(x) = \frac{\overline{mx}}{\bar{m}} \in \text{Frac}(A/P) \quad (x \in \mathcal{F}_a(k)).$$

Additivity and  $\iota(1) = 1$  are immediate. Multiplicativity follows by reducing the identity  $(mx)(my) = m(mxy)$  in  $A$  and dividing by  $\bar{m}^2$ . For  $x \neq 0$ , the identity  $(mx)(m/x) = m^2$  in  $A$  shows  $\overline{mx} \neq 0$ , because  $\bar{m}^2 \neq 0$ . Thus the unital field homomorphism  $\iota$  is injective.  $\square$

This is the scaled-field collision mechanism of Lemma 4.1 in a fraction field. The conclusion says more than that the residue field is large: it contains a specific Hahn field built on all exponent scales infinitesimal relative to one positive  $a$ . It is a necessary structural condition on such residue fields; manuscript 06 states that it does not classify those prime ideals, assert that they exist, or assert that every field of this form is a residue field.

*Remark 14.13* (The composed embedding [merge]). Let  $P$  be a prime class ideal of  $\mathbf{Oz}$  with  $\Pi \not\subseteq P$ . Then there is an injective field homomorphism  $\mathbf{No} \hookrightarrow \text{Frac}(\mathbf{Oz}/P)$  whose first factor is an ordered-field embedding,

$$\mathbf{No} \xrightarrow{\Theta_a} \mathcal{F}_a \xrightarrow{\iota} \text{Frac}(\mathbf{Oz}/P).$$

Indeed, Proposition 14.12 with  $(D, k) = (\mathbb{Z}, \mathbb{R})$  gives  $a > 0$  and an injective field homomorphism  $\iota : \mathcal{F}_a \rightarrow \text{Frac}(\mathbf{Oz}/P)$ , and Theorem 14.4 gives  $\Theta_a$ ; a composite of injective field homomorphisms is one. With  $(D, k) = (\mathbb{Z}[i], \mathbb{C})$  and the complex extension of  $\Theta_a$ , every prime  $P$  of  $\mathbf{Oz}[i]$  with  $\Pi_{\mathbb{C}} \not\subseteq P$  gives  $\mathbf{No}[i] \hookrightarrow \text{Frac}(\mathbf{Oz}[i]/P)$ . Neither source states this composition.

The hypotheses of Proposition 14.12 and Corollary 14.10 are different. If  $P \cap \mathbb{Z} = 0$  and  $P \not\subseteq \Pi$ , then  $\Pi \not\subseteq P$ : otherwise, for  $x \in P$  we would have  $x - \text{ct}(x) \in \Pi \subseteq P$ , hence  $\text{ct}(x) \in P \cap \mathbb{Z} = 0$ , so  $P \subseteq \Pi$ . Conversely,  $\Pi \not\subseteq P$  forces  $P \cap \mathbb{Z} = 0$ , since a nonzero  $n \in P$  would give  $\Pi = n\Pi \subseteq P$  (Proposition 3.3), but it allows  $P \subseteq \Pi$ , for instance  $P = 0$ , where the statement is the trivial  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  (Proposition 3.4). Thus 07's hypothesis is the stronger one and yields the stronger conclusion (a copy of  $\mathbf{No}$  inside  $\mathbf{Oz}/P$  itself, and a  $\mathbb{Q}$ -algebra structure), while 06's hypothesis also covers primes contained in  $\Pi$ , with the conclusion only in the fraction field.

## 14.4 The binomial quotient and its localization

Fix  $a > 0$  and write

$$\mathcal{Q}_a = \mathbf{Oz}/(1 + \omega^a), \quad q_a : \mathbf{Oz} \longrightarrow \mathcal{Q}_a$$

for the quotient map (07's  $B_a$ ). The element  $1 + \omega^a$  is a nonunit (Lemma 2.4), so  $\mathcal{Q}_a \neq 0$ ; and since  $\text{ct}(1 + \omega^a) = 1$ , Theorem 6.1 shows that  $\mathcal{Q}_a$  has no unital homomorphism to a nonzero set-sized ring. A purely arithmetic calculation shows why: in a set-sized target  $\omega^a$  becomes 0 (Theorem 5.1), so  $1 + \omega^a$  becomes 1, and a target in which it vanishes has  $1 = 0$ . Its failure to have small ring images is therefore not caused by its being zero, a field, or a domain; it is none of these by Corollary 14.22. By Corollary 14.9 it contains a copy of  $\mathbf{No}$ ; the rest of this section identifies much more of its internal structure.

The support description of the localization

$$\mathbf{Oz}[\omega^{-a}] = \{f \in \mathbf{No} : \text{supp } f \subseteq [-Na, \infty) \text{ for some ordinary } N \geq 0\}$$

is Proposition 6.11; manuscripts 04 and 07 both prove it. (07 calls this ring  $L_a$ .)

**Lemma 14.14** (Localization of the binomial quotient [07]).

(i)  $\omega^a \mathcal{F}_a \subseteq \Pi$ ; in particular  $\mathcal{F}_a \subseteq \mathbf{Oz}[\omega^{-a}]$ .

(ii) There is a unique ring homomorphism  $\tilde{q}_a : \mathbf{Oz}[\omega^{-a}] \rightarrow \mathcal{Q}_a$  extending  $q_a$ . It is surjective with kernel  $(1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$ , so

$$\mathcal{Q}_a \cong \mathbf{Oz}[\omega^{-a}] / (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]. \quad (14.6)$$

(iii) The restriction of  $\tilde{q}_a$  to  $\mathcal{F}_a$  is the injective field homomorphism

$$\iota_a(f) = -q_a(\omega^a f) \quad (f \in \mathcal{F}_a). \quad (14.7)$$

*Proof.* (i) Every  $h \in H_a$  satisfies  $a + h > 0$ , so translating a support in  $H_a$  by  $a$  gives a strictly positive support (Lemma 4.2); then  $f = \omega^{-a}(\omega^a f)$ .

(ii) The subring of  $\mathbf{No}$  generated by  $\mathbf{Oz}$  and  $\omega^{-a}$  consists of the finite sums  $\sum_{j \leq N} h_j \omega^{-ja} = \omega^{-Na} \sum_{j \leq N} h_j \omega^{(N-j)a}$ , so every element has the form  $\omega^{-Na} h$  with  $N \geq 0$  and  $h \in \mathbf{Oz}$ . Any extension of  $q_a$  sends  $\omega^{-a}$  to  $q_a(\omega^a)^{-1} = -1$ , which proves uniqueness; define  $\tilde{q}_a(\omega^{-Na} h) = (-1)^N q_a(h)$ . This is well defined: if  $\omega^{-Na} h = \omega^{-Ma} h'$  with  $N \leq M$ , then  $h' = \omega^{(M-N)a} h$  and  $(-1)^M q_a(h') = (-1)^M (-1)^{M-N} q_a(h) = (-1)^N q_a(h)$ . Writing two elements over a common power  $\omega^{-Na}$  shows that  $\tilde{q}_a$  is a ring homomorphism; it extends  $q_a$  and is surjective. If  $\tilde{q}_a(\omega^{-Na} h) = 0$ , then  $q_a(h) = 0$ , so  $h = (1 + \omega^a)y$  with  $y \in \mathbf{Oz}$  and  $\omega^{-Na} h = (1 + \omega^a)\omega^{-Na} y \in (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$ . Conversely  $\tilde{q}_a((1 + \omega^a)\omega^{-Na} y) = (-1)^N q_a((1 + \omega^a)y) = 0$  for  $y \in \mathbf{Oz}$ .

(iii) By (i) and (ii),  $\tilde{q}_a(f) = \tilde{q}_a(\omega^{-a} \cdot \omega^a f) = -q_a(\omega^a f)$ . A unital ring homomorphism from a field to a nonzero ring is injective.  $\square$

Henceforth  $\mathcal{F}_a$  is identified with its image  $\iota_a(\mathcal{F}_a) \subseteq \mathcal{Q}_a$  whenever we work inside  $\mathcal{Q}_a$ ; all coefficient fields below are embedded with the identity of  $\mathcal{Q}_a$ . This identification is not constant-term extraction: a nonintegral real scalar  $r$  is represented in  $\mathcal{Q}_a$  by the omnific integer  $-r\omega^a$ .

## 14.5 Exact binomial kernels

For an ordinary integer  $n \geq 1$  put  $x_n = q_a(\omega^{a/n}) \in \mathcal{Q}_a$ . Certainly  $x_n^n = q_a(\omega^a) = -1$ . The substantive question is whether the full ambient omnific ring, with its arbitrary surreal exponents, creates additional polynomial relations over  $\mathcal{F}_a$ .

**Theorem 14.15** (Exact binomial kernel [07]). *For every  $n \geq 1$ , evaluation at  $x_n$  induces an injective ring homomorphism*

$$\mathcal{F}_a[T]/(T^n + 1) \hookrightarrow \mathcal{Q}_a. \quad (14.8)$$

*Equivalently, the kernel of  $\mathcal{F}_a[T] \rightarrow \mathcal{Q}_a$ ,  $T \mapsto x_n$ , is exactly  $(T^n + 1)$ .*

*Proof.* The polynomial  $T^n + 1$  lies in the kernel. Dividing by this monic polynomial over the field  $\mathcal{F}_a$ , it suffices to show that no nonzero remainder

$$R(T) = \sum_{j=0}^{n-1} r_j T^j \quad (r_j \in \mathcal{F}_a)$$

vanishes at  $x_n$ . Put  $\xi = \omega^{a/n} \in \mathbf{Oz}$ ; then  $R(\xi) \in \mathbf{Oz}[\omega^{-a}]$  by Lemma 14.14(i), and  $\tilde{q}_a(R(\xi)) = R(x_n)$ .

*The quotient in No.* The series  $U = \sum_{k \geq 0} (-1)^k \omega^{-(k+1)a}$  has the decreasing support  $\{-(k+1)a\}$ , and  $(1 + \omega^a)U = \sum_{k \geq 0} (-1)^k \omega^{-(k+1)a} + \sum_{k \geq 0} (-1)^k \omega^{-ka} = 1$ , each exponent receiving at most two contributions that cancel except at exponent 0. Hence  $U = (1 + \omega^a)^{-1}$  in  $\mathbf{No}$ , and multiplying the finite sum  $R(\xi) = \sum_j r_j \omega^{ja/n}$  by  $U$  gives

$$\frac{R(\xi)}{1 + \omega^a} = \sum_{k \geq 0} \sum_{j=0}^{n-1} (-1)^k r_j \omega^{(j/n - k - 1)a}, \quad (14.9)$$

provided the right side is a strongly summable family, which we now check together with the absence of cancellation between blocks.

*Block cosets.* The support of the block indexed by  $(j, k)$  lies in the coset  $c_{j,k} + H_a$  with centre  $c_{j,k} = (j/n - k - 1)a$ , because  $\text{supp } r_j \subseteq H_a$ . Distinct pairs give distinct cosets. Indeed,  $c_{j,k} - c_{j',k'} = (m/n)a$  with  $m = j - j' - n(k - k') \in \mathbb{Z}$ , and  $(m/n)a \in H_a$  only for  $m = 0$ , since for  $m \neq 0$  one has  $n \cdot |(m/n)a| = |m|a \geq a$ . From  $j - j' = n(k - k')$  and  $|j - j'| < n$  we get  $j = j'$  and  $k = k'$ . Cosets of  $H_a$  are disjoint, so distinct blocks have disjoint supports, and each exponent receives a contribution from at most one block.

*Reverse well-ordering.* As  $(j, k)$  runs over  $\{0, \dots, n-1\} \times \mathbb{N}$ , the integer  $j - n(k+1)$  runs bijectively over the integers  $m \leq -1$ , so the centres are the points  $(m/n)a$ ,  $m \leq -1$ , a strictly descending discrete progression with step  $a/n$ . Every offset  $h \in H_a$  satisfies  $4n|h| < a$ , that is,  $|h| < a/(4n)$ , so the block with centre  $(m/n)a$  lies in the open interval of radius  $a/(4n)$  about its centre; these intervals are pairwise disjoint and ordered as their centres. Given a nonempty subset  $S$  of the union of the supports, the set of  $m$  whose block meets  $S$  is a nonempty set of integers bounded above by  $-1$ , so it has a largest element  $m^*$ ; the elements of  $S$  in block  $m^*$  exceed all other elements of  $S$ , and since that block has reverse well-ordered support,  $S$  has a greatest element. All supports and the index family are sets. Hence the right side of (14.9) is a legitimate strongly summable Hahn expression, and (14.9) holds because Hahn multiplication distributes over strongly summable families. (Equivalently, multiplying the right side by  $1 + \omega^a$  cancels consecutive  $k$ -blocks and leaves exactly  $R(\xi)$ .)

*Unbounded-below support.* Choose  $j$  with  $r_j \neq 0$  and  $h \in \text{supp } r_j$ . For every  $k \geq 0$  the exponent  $h + (j/n - k - 1)a$  lies only in block  $(j, k)$ , so it occurs in (14.9) with the nonzero coefficient  $(-1)^k (r_j)_h$ . Since  $h < a$  and  $j/n < 1$ , this exponent is less than  $a + a - (k+1)a = (1-k)a$ , which is below  $-Na$  as soon as  $k \geq N+1$ . Thus the support of (14.9) is not contained in  $[-Na, \infty)$  for any ordinary  $N$ , and by Proposition 6.11 the quotient  $R(\xi)/(1 + \omega^a)$  does not belong to  $\mathbf{Oz}[\omega^{-a}]$ .

*Conclusion.* If  $R(x_n) = 0$ , then  $R(\xi) \in \ker \tilde{q}_a = (1 + \omega^a)\mathbf{Oz}[\omega^{-a}]$  by Lemma 14.14(ii), so  $R(\xi) = (1 + \omega^a)u$  with  $u \in \mathbf{Oz}[\omega^{-a}]$ . Division in the field  $\mathbf{No}$  identifies  $u$  with (14.9), a contradiction. Hence  $R(x_n) \neq 0$ .  $\square$

*Remark 14.16* ([07]). The proof does not restrict the exponents of an alleged ambient factor  $u$  to rational multiples of  $a$ ; the full support characterization of  $\mathbf{Oz}[\omega^{-a}]$  is exactly what rules out such hidden factors. A calculation solely in a finite group algebra (the span of the monomials  $\omega^{ja/n}$ ) would not establish the theorem.

**Corollary 14.17** (Internal algebraically closed field [07]). *Let  $u = x_2 = q_a(\omega^{a/2})$ . The subring  $\mathcal{E}_a = \mathcal{F}_a[u] \subseteq \mathcal{Q}_a$  is a field, isomorphic over  $\mathcal{F}_a$  to  $\mathcal{F}_a(i)$  by  $i \mapsto u$ . It is algebraically closed and contains a class copy of  $\mathbf{No}[i]$ .*

*Proof.* By Theorem 14.15 with  $n = 2$ ,  $\mathcal{E}_a \cong \mathcal{F}_a[T]/(T^2+1)$ . Since  $\mathcal{F}_a$  is real closed (Proposition 14.1),  $-1$  is not a square in it,  $T^2 + 1$  is irreducible, and the quotient is the field  $\mathcal{F}_a + i\mathcal{F}_a = \mathcal{F}_a(i)$ , which is algebraically closed by Proposition 14.1. The complex extension of  $\Theta_a$  (Theorem 14.4) embeds  $\mathbf{No}[i]$  in it.  $\square$

There is no order contradiction. Although  $\omega^{a/2}$  is positive in  $\mathbf{Oz}$ , its square becomes  $-1$  in  $\mathcal{Q}_a$ ; the order of  $\mathbf{Oz}$  does not descend to  $\mathcal{Q}_a$ . The corollary is a statement about rings and fields, not about ordered quotients.

More generally, finite real factorization and the Chinese remainder theorem give

$$\mathcal{F}_a[T]/(T^n + 1) \cong \begin{cases} \mathcal{E}_a^m, & n = 2m, \\ \mathcal{F}_a \times \mathcal{E}_a^m, & n = 2m + 1. \end{cases} \quad (14.10)$$

The roots of  $T^n + 1$  are the  $n$  distinct ordinary complex numbers  $\exp(i\pi(2k+1)/n)$ , which lie in  $\mathbb{C} \subseteq \mathcal{F}_a(i)$ . For even  $n$  none is real, and the  $m$  conjugate pairs give  $m$  distinct monic quadratic factors with real coefficients, irreducible over  $\mathcal{F}_a$  because their roots are not in  $\mathcal{F}_a$ . For odd  $n$  the only real root is  $-1$ , and the remaining roots give  $m$  such quadratics. All roots are simple, so the factors are pairwise coprime, and each quadratic quotient is isomorphic to  $\mathcal{F}_a(i) \cong \mathcal{E}_a$ . The isomorphisms in (14.10) need not identify a chosen imaginary unit in the same way at different stages; the dyadic tower below makes coherent choices.

## 14.6 The dyadic tower and its Cantor algebra

For  $r \geq 1$  define

$$\xi_r = q_a(\omega^{a/2^r}) = x_{2^r}, \quad \mathcal{S}_r = \mathcal{F}_a[\xi_r] \subseteq \mathcal{Q}_a$$

(07's  $\mathcal{S}_r$ ). Then  $\xi_r^2 = \xi_{r-1}$  for  $r \geq 2$ ,  $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$ , and  $\mathcal{S}_1 = \mathcal{E}_a$ , so each  $\mathcal{S}_r$  is an  $\mathcal{E}_a$ -algebra. In this subsection  $i$  denotes the element  $u = \xi_1$  of  $\mathcal{E}_a$ ; under  $\mathcal{E}_a \cong \mathcal{F}_a(i)$  the ordinary complex field  $\mathbb{C} \subseteq \mathcal{F}_a(i)$  corresponds to  $\mathbb{R} + \mathbb{R}u \subseteq \mathcal{E}_a$ , and ordinary complex numbers are read in  $\mathcal{E}_a$  through this identification.

**Lemma 14.18** (A single stage over the fixed complex field [07]). *Let  $m = 2^{r-1}$ . As  $\mathcal{E}_a$ -algebras,*

$$\mathcal{S}_r \cong \mathcal{E}_a[T]/(T^m - i), \quad T \mapsto \xi_r. \quad (14.11)$$

*Proof.* The  $\mathcal{E}_a$ -algebra map  $\mathcal{E}_a[T] \rightarrow \mathcal{S}_r$ ,  $T \mapsto \xi_r$ , is surjective, since  $\mathcal{E}_a[\xi_r] = \mathcal{F}_a[\xi_r^m, \xi_r] = \mathcal{S}_r$ , and it kills  $T^m - i$  because  $\xi_r^m = \xi_1 = i$ . For injectivity on the quotient, suppose  $\sum_{j < m} (a_j + b_j i) \xi_r^j = 0$  with  $a_j, b_j \in \mathcal{F}_a$ . Substituting  $i = \xi_r^m$  gives  $\sum_{j < m} a_j \xi_r^j + \sum_{j < m} b_j \xi_r^{j+m} = 0$ , a polynomial relation over  $\mathcal{F}_a$  of degree less than  $2m = 2^r$  satisfied by  $\xi_r = x_{2^r}$ . By Theorem 14.15 with  $n = 2^r$  all its coefficients vanish, so all  $a_j$  and  $b_j$  are zero.  $\square$

The roots of  $T^m - i$  can be chosen in the ordinary complex subfield of  $\mathcal{E}_a$ :

$$\zeta_{r,k} = \exp\left(i \frac{\pi/2 + 2\pi k}{2^{r-1}}\right) \quad (k \in \mathbb{Z}/2^{r-1}\mathbb{Z}). \quad (14.12)$$

The expression is evaluated in  $\mathbb{C}$  and then embedded in  $\mathcal{E}_a$ ; no exponential function on surcomplex arguments is assumed. These  $m$  roots are distinct, and

$$\zeta_{r+1,k}^2 = \zeta_{r,k \bmod 2^{r-1}} \quad (k \in \mathbb{Z}/2^r\mathbb{Z}).$$

Since the roots are distinct, the Chinese remainder theorem and (14.11) give an isomorphism of  $\mathcal{E}_a$ -algebras

$$\Phi_r : \mathcal{S}_r \longrightarrow \mathcal{E}_a^{\mathbb{Z}/2^{r-1}\mathbb{Z}}, \quad P(\xi_r) \longmapsto (P(\zeta_{r,k}))_k \quad (P \in \mathcal{E}_a[T]). \quad (14.13)$$

The inclusion  $\mathcal{S}_r \subseteq \mathcal{S}_{r+1}$  replaces  $\xi_r$  by  $\xi_{r+1}^2$ , so  $P(\xi_r) = P(\xi_{r+1}^2)$  has stage- $(r+1)$  coordinates  $P(\zeta_{r+1,k}^2) = P(\zeta_{r,k \bmod m})$ . In coordinates the transition map is therefore exactly

$$(c_k)_{k \bmod m} \longmapsto (c_{k \bmod m})_{k \bmod 2m},$$

which repeats each coordinate on its two lifts. This compatibility, and not merely the existence of separate finite product decompositions, is what identifies the direct limit.

**Theorem 14.19** (Cantor algebra theorem [07]). *Let  $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a)$  be the class of functions from the 2-adic integers  $\mathbb{Z}_2$  to  $\mathcal{E}_a$  that factor through some finite quotient  $\mathbb{Z}/2^N\mathbb{Z}$ , with pointwise operations; thus  $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a) = \bigcup_{N \geq 0} \mathcal{E}_a^{\mathbb{Z}/2^N\mathbb{Z}}$  with the transition maps repeating each coordinate on its two lifts. There is an explicit  $\mathcal{E}_a$ -algebra isomorphism*

$$\bigcup_{r \geq 1} \mathcal{S}_r \cong \text{LC}(\mathbb{Z}_2, \mathcal{E}_a), \quad (14.14)$$

under which  $\xi_r$  corresponds to the function  $z \mapsto \zeta_{r,z \bmod 2^{r-1}}$ . The left side is the subalgebra of  $\mathcal{Q}_a$  generated by  $\mathcal{F}_a$  and the classes of  $\omega^{a/2^r}$ ,  $r \geq 1$ .

*Proof.* Identify  $\mathcal{E}_a^{\mathbb{Z}/2^{r-1}\mathbb{Z}}$  with the functions  $\mathbb{Z}_2 \rightarrow \mathcal{E}_a$  factoring through  $\mathbb{Z}/2^{r-1}\mathbb{Z}$ , via  $c \mapsto (z \mapsto c_{z \bmod 2^{r-1}})$ . By the computation of the transition maps, the function attached to an element of  $\mathcal{S}_r$  by  $\Phi_r$  is the same as the one attached by  $\Phi_{r+1}$ , so the  $\Phi_r$  combine into a map  $\Phi$  on the increasing union; it is an  $\mathcal{E}_a$ -algebra homomorphism, and it is injective because each  $\Phi_r$  is. Conversely, every function factoring through  $\mathbb{Z}/2^N\mathbb{Z}$  lies in the image of the bijection  $\Phi_{N+1}$ , so  $\Phi$  is surjective. Evaluating  $T$  at the roots gives the function attached to  $\xi_r$ . Finally,  $\xi_{r'} = \xi_r^{2^{r-r'}} \in \mathcal{S}_r$  for  $r' \leq r$ , so the union of the  $\mathcal{S}_r$  is the subalgebra generated by  $\mathcal{F}_a$  and all  $\xi_r$ .  $\square$

The isomorphism is an algebraic statement about a specified subalgebra; it is not a classification of all of  $\mathcal{Q}_a$ .

*Remark 14.20* (Locally constant functions [07]). The topological terminology is justified: a locally constant function on the ordinary compact space  $\mathbb{Z}_2$  has a finite subcover of neighbourhoods of constancy and hence factors through one finite quotient, and conversely such a factorization is locally constant. This uses only the ordinary Cantor space; no topology is imposed on the proper-class field  $\mathcal{E}_a$ , and one can equally take the finite-factorization condition as the definition.



*Remark 14.21* (Evaluation at a point [07]). Evaluation at a point  $z \in \mathbb{Z}_2$  is a homomorphism from the subalgebra (14.14) onto  $\mathcal{E}_a$ . Its target is a proper-class field, not a set-sized ring, and no assertion is made that it extends to all of  $\mathcal{Q}_a$ . The construction is therefore fully compatible with the absence of small ring images of  $\mathcal{Q}_a$ . (Status (batch 34): with global choice, evaluation at  $z$  does extend, not onto  $\mathcal{E}_a$  but to a homomorphism of  $\mathcal{Q}_a$  onto a proper-class field containing  $\mathcal{E}_a$ , through the maximal ideal  $\mathfrak{M}_{a,z}$  of Theorem 14.46; see Remark 14.47.)

For  $m = 2^{r-1}$  and  $k \in \mathbb{Z}/m\mathbb{Z}$  define, in  $\mathcal{S}_r$ ,

$$e_{r,k} = \frac{1}{m} \sum_{\ell=0}^{m-1} \left( \frac{\xi_r}{\zeta_{r,k}} \right)^\ell. \quad (14.15)$$

At the root  $\zeta_{r,j}$  the sum has value 1 for  $j = k$  and 0 otherwise, by the finite geometric-sum identity for the  $m$ th root of unity  $\zeta_{r,j}/\zeta_{r,k}$ . Thus  $\Phi_r(e_{r,k})$  is the indicator of the coordinate  $k$ , and

$$e_{r,k}^2 = e_{r,k}, \quad e_{r,k}e_{r,j} = 0 \ (k \neq j), \quad \sum_{k \bmod m} e_{r,k} = 1, \quad (14.16)$$

$$e_{r,k} = e_{r+1,k} + e_{r+1,k+m}. \quad (14.17)$$

Each  $e_{r,k}$  is nonzero because  $\Phi_r$  is injective, which rests on Theorem 14.15. The refinement formula (14.17) follows from the coordinate transition, or by checking values at all roots of stage  $r+1$ .

The first nontrivial instance can be written without trigonometric notation. Put

$$y = \xi_2, \quad u = \xi_1 = y^2, \quad \zeta = \frac{1+u}{\sqrt{2}} \in \mathcal{E}_a.$$

Then  $u^2 = -1$  and  $\zeta^2 = (1+2u+u^2)/2 = u = y^2$ , so  $(y/\zeta)^2 = 1$ , and

$$e_+ = \frac{1}{2} \left( 1 + \frac{y}{\zeta} \right), \quad e_- = \frac{1}{2} \left( 1 - \frac{y}{\zeta} \right) \quad (14.18)$$

satisfy  $e_\pm^2 = e_\pm$ ,  $e_+e_- = \frac{1}{4}(1 - (y/\zeta)^2) = 0$  and  $e_+ + e_- = 1$ . Nonvanishing is not inferred from the suggestive formula: the roots at stage 2 are  $\zeta_{2,0} = \zeta$  and  $\zeta_{2,1} = -\zeta$ , so  $e_+ = e_{2,0}$ ,  $e_- = e_{2,1}$ , and  $\Phi_2$  identifies  $\mathcal{S}_2 \cong \mathcal{E}_a \times \mathcal{E}_a$  and sends  $e_+, e_-$  to  $(1, 0)$  and  $(0, 1)$ . This is why the exact-kernel theorem is an essential part of the argument.

**Corollary 14.22** (Boolean algebra and failure of finiteness conditions [07]). *The idempotents of  $\bigcup_r \mathcal{S}_r$  form precisely the Boolean algebra of clopen subsets of  $\mathbb{Z}_2$ , and this Boolean algebra is atomless. The ring  $\mathcal{Q}_a$  has an infinite sequence of pairwise orthogonal nonzero idempotents, is neither a domain nor a local ring, and fails the ascending chain condition on ideals.*

*Proof.* A field has only the idempotents 0 and 1, so an idempotent of  $\text{LC}(\mathbb{Z}_2, \mathcal{E}_a)$  is the indicator of a set that is a union of residue classes modulo some  $2^N$ , that is, of a clopen subset of  $\mathbb{Z}_2$ ; conversely, by compactness every clopen subset is a finite union of such residue classes, and its indicator is idempotent. Every nonempty clopen set contains a residue class modulo some  $2^N$ , which splits into two nonempty residue classes modulo  $2^{N+1}$ ; so there are no atoms.

For  $j \geq 0$  the sets  $C_j = \{z \in \mathbb{Z}_2 : z \equiv 2^j \pmod{2^{j+1}}\}$  (the  $z$  of 2-adic valuation exactly  $j$ ) are nonempty, clopen and pairwise disjoint. Their indicators give pairwise orthogonal nonzero idempotents  $f_j \in \mathcal{Q}_a$ . The ideals

$$(f_0) \subsetneq (f_0, f_1) \subsetneq (f_0, f_1, f_2) \subsetneq \cdots$$

of  $\mathcal{Q}_a$  strictly increase: if  $f_{n+1} = \sum_{j \leq n} c_j f_j$  with  $c_j \in \mathcal{Q}_a$ , multiplication by  $f_{n+1}$  would give  $f_{n+1} = 0$ . This proves the failure of the ascending chain condition without forming any class-sized generating family. A nontrivial idempotent  $e$  satisfies  $e(1 - e) = 0$ , so  $\mathcal{Q}_a$  is not a domain; and in a local ring (for every  $x$ , one of  $x$  and  $1 - x$  is a unit) either  $e$  or  $1 - e$  is a unit, which forces  $e = 1$  or  $e = 0$ .  $\square$

Grouping a sufficiently fine dyadic partition of  $\mathbb{Z}_2$  into any prescribed number  $d \geq 1$  of nonempty blocks yields a unital subalgebra of  $\mathcal{Q}_a$  isomorphic to  $\mathcal{E}_a^d$ . Manuscript 07 claims neither that all idempotents of  $\mathcal{Q}_a$  lie in the Cantor algebra (14.14) nor that  $\mathcal{Q}_a$  itself is a ring of locally constant functions. No infinite sum of the orthogonal sequence  $(f_j)$  is used.

## 14.7 Gaussian binomial quotients

Let

$$\mathcal{Q}_a^{\mathbb{C}} = \mathbf{Oz}[i]/(1 + \omega^a), \quad q_a^{\mathbb{C}} : \mathbf{Oz}[i] \rightarrow \mathcal{Q}_a^{\mathbb{C}}.$$

Since  $\mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz}$  and  $\omega^{-a}$  is real,  $\mathbf{Oz}[i][\omega^{-a}] = \mathbf{Oz}[\omega^{-a}] + i\mathbf{Oz}[\omega^{-a}]$ , so by Proposition 6.11 it consists of the surcomplex normal forms whose supports are bounded below by some  $-Na$ . It contains  $\mathcal{F}_a(i)$ , and the proof of Lemma 14.14 applies verbatim with  $\mathbf{Oz}[i]$  in place of  $\mathbf{Oz}$ : the extension  $\tilde{q}_a^{\mathbb{C}}$  of  $q_a^{\mathbb{C}}$  has kernel  $(1 + \omega^a)\mathbf{Oz}[i][\omega^{-a}]$ , and it embeds the coefficient field by

$$f \mapsto -q_a^{\mathbb{C}}(\omega^a f) \quad (f \in \mathcal{F}_a(i)).$$

For  $f = i$  this is the Gaussian imaginary unit of the quotient, since  $-q_a^{\mathbb{C}}(\omega^a i) = -q_a^{\mathbb{C}}(\omega^a)q_a^{\mathbb{C}}(i) = q_a^{\mathbb{C}}(i)$ .

**Theorem 14.23** (Complex exact kernels and dyadic algebra [07]). *For every  $n \geq 1$  there is an injective ring homomorphism*

$$\mathcal{F}_a(i)[T]/(T^n + 1) \hookrightarrow \mathcal{Q}_a^{\mathbb{C}}, \quad T \mapsto q_a^{\mathbb{C}}(\omega^{a/n}),$$

and its source is isomorphic to  $\mathcal{F}_a(i)^n$ . The increasing union of the dyadic stages  $n = 2^r$ ,  $r \geq 0$ , is isomorphic to  $\text{LC}(\mathbb{Z}_2, \mathcal{F}_a(i))$ . The ring  $\mathcal{Q}_a^{\mathbb{C}}$  is nonzero, has no unital homomorphism to a nonzero set-sized ring, and contains a class copy of  $\mathbf{No}[i]$ .

*Proof.* The proof of Theorem 14.15 uses only supports and the nonvanishing of coefficients, not their reality, so it applies verbatim over  $\mathcal{F}_a(i)$  and  $\mathbf{Oz}[i][\omega^{-a}]$ : a nonzero remainder  $R$  with coefficients in  $\mathcal{F}_a(i)$  gives the expansion (14.9), whose blocks lie in the distinct cosets  $(j/n - k - 1)a + H_a$ , whose union of supports is reverse well ordered, and whose support is not bounded below by any  $-Na$ ; so  $R(\omega^{a/n}) \notin (1 + \omega^a)\mathbf{Oz}[i][\omega^{-a}]$ . The polynomial  $T^n + 1$  has  $n$  distinct roots  $\exp(i\pi(2k + 1)/n)$  in  $\mathbb{C} \subseteq \mathcal{F}_a(i)$ , so the Chinese remainder theorem gives  $\mathcal{F}_a(i)[T]/(T^n + 1) \cong \mathcal{F}_a(i)^n$  by evaluation at the roots.

For  $n = 2^r$  label the roots  $\eta_{r,k} = \exp(i(\pi + 2\pi k)/2^r)$ ,  $k \in \mathbb{Z}/2^r\mathbb{Z}$ . Then  $\eta_{r,k}^2 = \eta_{r-1, k \bmod 2^{r-1}}$ , and the stage- $(r - 1)$  generator  $q_a^{\mathbb{C}}(\omega^{a/2^{r-1}})$  is the square of the stage- $r$  generator; so, exactly as in Theorem 14.19, the coordinate transitions repeat each entry on its two lifts, and the union is  $\text{LC}(\mathbb{Z}_2, \mathcal{F}_a(i))$ . The ring is nonzero because  $1 + \omega^a$  is not among the units  $\pm 1, \pm i$  of  $\mathbf{Oz}[i]$  (Lemma 2.4); it has no nonzero set-sized image by Theorem 6.1, since  $\text{ct}(1 + \omega^a) = 1$  generates  $\mathbb{Z}[i]$ ; and it contains a copy of  $\mathbf{No}[i]$  by the complex extension of  $\Theta_a$  (Theorem 14.4).  $\square$

Coefficientwise complex conjugation preserves  $\mathbf{Oz}[i]$  and, because  $1 + \omega^a$  is real, the ideal  $(1 + \omega^a)$ ; hence it descends to an involution of  $\mathcal{Q}_a^{\mathbb{C}}$ . It commutes with the coefficient-field embedding  $f \mapsto -q_a^{\mathbb{C}}(\omega^a f)$  because the multiplier  $-\omega^a$  is real. This supplies a compatible involution without requiring an order or an analytic interpretation of the quotient.

## 14.8 Boundaries of the large-field results

The small-image theorem and the large-field results concern different hypotheses and reinforce one another (07). A homomorphism to a set-sized ring kills every purely infinite element ([Theorem 5.1](#)); a homomorphism to a nonzero ring that makes even one purely infinite element invertible has a huge field in its target ([Theorem 14.7](#)). That target is necessarily a proper class, but the explicit field embedding gives considerably more than this size conclusion, and the Cantor algebra describes zero-divisor structure that no cardinality argument captures. Manuscript 07 insists on the following distinctions.

**A nonzero element versus a unit.** A nonzero nilpotent, as in [Example 14.11](#), does not produce a field. Invertibility is what allows the formula  $f = (gf)/g$  to be transported to a quotient. No cancellation in a ring with zero divisors has been used.

**A specified subalgebra versus the entire quotient.** The exact kernel theorem identifies the subalgebras  $\mathcal{F}_a[x_n]$ , and their dyadic union is a Cantor algebra. Surreal normal forms with other supports can still create further elements of  $\mathcal{Q}_a$ . Neither a classification of all idempotents of  $\mathcal{Q}_a$  nor a description of its prime ideals has been proved.

**Class ideals versus set-theoretic Zorn arguments.** Zorn's lemma applies to sets of candidates. The ring  $\mathcal{Q}_a$  is a proper class, so the usual existence argument for maximal ideals cannot simply be repeated over its class of ideals. Statements such as [Corollary 14.10](#) are deliberately conditional: it is not concluded that a proper-class nonarithmetic maximal ideal exists merely because  $(1 + \omega^a)$  is proper. Likewise (06) [Proposition 14.12](#) is a necessary condition on residue fields, not an existence theorem. (Status (batch 34): the existence question is settled with global choice. A proper class of ideals is not searched: 24's recursion ([Theorem 14.24](#)) enumerates the *elements* of the ring along a global well-order and stores a set of accepted generators at each stage, keeping finite witnesses of improperness, so  $(1 + \omega^a)$  does lie in maximal class ideals, all nonarithmetic ([Theorem 14.34](#)), and which monomials lie in a proper ideal containing  $1 + \omega^a$  is decided by [Theorem 14.37](#). 07's and 06's statements remain necessary conditions, now on ideals that exist; the prime ideals of  $\mathcal{Q}_a$  are still not described.)

**Strong sums versus limits.** Every infinite sum used above, namely the geometric series for  $(1 + \omega^a)^{-1}$  and the block expansion ([14.9](#)), has a set-sized reverse well-ordered support and finitely many contributions at each exponent, and is verified coefficientwise. Finite geometric remainders of the kind in [Lemma 4.3](#) do not tend to zero, and no limit argument is used.

**Finite checks.** The program accompanying manuscript 07 checks exact finite geometric identities, finite binomial remainders, the first nontrivial idempotents ([14.18](#)) and finite coordinate refinement maps (07 records that all 621 named checks passed). It does not represent arbitrary surreal normal forms, and it does not verify the ambient support argument, namely that the infinite expansion ([14.9](#))

lies outside  $\mathbf{Oz}[\omega^{-a}]$ ; that is a mathematical proof in the text. For a formalization, 07 recommends keeping the support characterization of  $\mathbf{Oz}[\omega^{-a}]$  and the exact binomial kernel separate from the routine finite Chinese remainder calculations; no corresponding Lean declarations are claimed.

## 14.9 Class residue fields beyond the constant term

This subsection is manuscript 24's (Section 1.3). Its subject is the maximal class ideals of  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  that the universal theorem cannot see. Working in Gödel–Bernays class theory with global choice, 24 proves that every proper class ideal of a commutative class ring lies in a maximal class ideal (Theorem 14.24), by a recursion that stores only sets of generators, and that the unrestricted form of this principle is equivalent to global choice (Theorem 14.30). Consequently  $1 + \omega^a$  lies in a characteristic-zero maximal ideal of  $\mathbf{Oz}$  and of  $\mathbf{Oz}[i]$  for every  $a > 0$  (Theorem 14.34): this answers, with global choice, the existence clause of Question 18.4, whose classification clause stays open. In the binomial quotient a monomial  $\omega^b$ ,  $b \geq 0$ , is zero exactly when  $a \ll b$  and is a unit otherwise (Theorem 14.37); separated scales are comaximal by an explicit Bézout identity (Corollary 14.42); every branch of 07's dyadic idempotent tree lies in its own maximal ideal, which gives at least  $2^{\aleph_0}$  maximal ideals above each  $1 + \omega^a$  and a family indexed by  $\mathbf{On} \times \mathbb{Z}_2$  across scales (Theorems 14.46 and 14.48); characteristic-zero maximal ideals separate the points of  $\mathbf{Oz}$ , and their residue fields embed in  $\mathbf{No}[i]$  (Theorems 14.50 and 14.53), the specializations that vanish on a binomial  $1 + \omega^a$  not preserving strong summability (Theorem 14.55); and the primes that are both term-closed and strongly sum-closed are classified (Theorem 14.57). Where the universal theorem says that every *set-sized* image of  $\mathbf{Oz}$  sees only ct, these class-sized field images together see every element.

24 pins 51c1cc7 (23 September 2026), whose text of this report is the one written after the sixth merge (it has 22's subsection, not 23's Section 15.10); it read this report's large-quotient section, the conditional Corollary 14.10 and the README sentence recording that the existence of nonarithmetic class primes is not claimed (Question 18.4, numbered 18.4 then as now). Its field core is already in this report, in a stronger form, and is printed once. 24's Lemma 7.1 is Proposition 14.1 together with Lemma 4.2(iii); its shifted embedding  $\Theta_a$  (its Theorem 7.2) is 07's  $\iota_a$  of Lemma 14.14(iii), extended to  $\mathcal{F}_a(i)$  by  $i \mapsto u$  in Corollary 14.17 and, for  $\mathbf{Oz}[i]$ , by the coefficient embedding before Theorem 14.23; and its field-reservoir Proposition 7.4 is Corollary 14.10, by the mechanism of Theorem 14.7 and Proposition 14.12. 24's Section 7 concludes only that these quotients contain a proper-class real closed or algebraically closed Hahn field, whereas Theorem 14.4 and Corollary 14.17 put a copy of  $\mathbf{No}$  in  $\mathcal{F}_a$  and a copy of  $\mathbf{No}[i]$  in  $\mathcal{E}_a \subseteq \mathcal{Q}_a$ . 24 is added as a proving source to Proposition 14.1, the one of these statements it proves in full, and is credited as an independent proof of the parts it proves of the others; its corollary on residue fields is printed below with the stronger conclusion (Corollary 14.44). 24's dyadic Lemma 8.1 is the constant-coefficient case  $n = 2^r$  of Theorems 14.15 and 14.23, and its idempotent tree is 07's tree (14.17); 24's degree proof, shorter for constant coefficients, is kept as a second route (Remark 14.45). The class back-and-forth behind 24's Corollary 9.3 is `saut:thm:acfhomogeneity` of the surcomplex-automorphism report [57] and `fkcsb:lem:backforth` of [46], which 24 does not cite. Set-sized analogues are older: 08 finds a characteristic-zero maximal ideal containing  $1 + X^{e_0}$  in  $A_\kappa^{08}$  (Theorem 11.16(vi)), and 09 has pairwise comaximal ideals  $(1 - X^{e_\ell})$  by the same geometric identity, field detection of every nonzero purely infinite element and  $\text{Jac} = 0$  (Theorem 11.20); those are set-sized rings with the ordinary maximal-ideal theorem. The recursion of Theorem 14.24, a greedy transfinite recursion that keeps finite witnesses, is the mechanism of 19's and 20's class-ultrafilter lemmas (Lemmas 15.48 and 15.78), there for Boolean algebras; the ring form is new here, and so is the equivalence with global choice (Theorem 14.30), which 24 says may be folklore. 24 credits as antecedents the greedy

Krull method, the polynomial compatibility argument of Hodges, Ern  and Entin [60, 61, 62], recursive constructions of maximal ideals in countable rings [63], the characteristic-zero Hahn-field theorem [64, 65], the exponent-rescaling automorphism, the standard universality and back-and-forth method for algebraically closed fields, the monomial-prime and convex-face mechanism of ordered monoid algebras, and this report’s field reservoirs and dyadic product algebras. The numbering of 24 cited here is that of its own manuscript.

**Foundations, notation and renamings.** Every statement of this subsection that asserts the existence of a maximal class ideal, or of a map built from one, is proved in GBC: G del–Bernays class theory GB, which here includes no choice principle, together with global choice GC. This is stronger than the base theory of Section 2.1, NBG with choice for sets, and 24 never substitutes choice for sets for global choice. Over GB, global choice is equivalent to a set-like global well-ordering of the universe and to an enumeration of every proper class in order type **On** [58]. A *class ring* has a class of set-valued elements and class functions as operations; a class ideal satisfies the finite ideal axioms; *maximal* means maximal among proper class ideals, which for a proper class ideal  $M$  of a commutative unital class ring  $R$  is equivalent to the finite criterion

$$x \notin M \implies 1 - rx \in M \text{ for some } r \in R, \quad (14.19)$$

since (14.19) follows from  $M + (x) = R$  and implies maximality; so no collection of class ideals is ever formed. A quotient  $R/I$  is coded in GBC by the least representative of each coset in a fixed set-like global well-order, with the operations computed on representatives and reduced, and over GB by Scott codes, the members of a coset of least rank, as in Section 2.1. Every recursion below has a *set* as its state at each ordinal stage: for each set ordinal, ordinary transfinite recursion with class parameters applies, uniqueness makes the runs of different lengths compatible, and elementary comprehension collects them into one class relation. No recursion with proper-class values, and no unrestricted elementary transfinite recursion for class-valued stages, is used; such principles need separate attention in class set theory [59].

Throughout,  $(D, k)$  is  $(\mathbb{Z}, \mathbb{R})$  or  $(\mathbb{Z}[i], \mathbb{C})$ ,  $A = \mathcal{A}_{D,k}$  is **Oz** or **Oz** $[i]$ , and  $a > 0$ . We write  $\mathcal{Q}_a^k = A/(1 + \omega^a)A$ , so that  $\mathcal{Q}_a^{\mathbb{R}} = \mathcal{Q}_a$  and  $\mathcal{Q}_a^{\mathbb{C}}$  are the rings of Sections 14.4 and 14.7, and  $q_a$  for the quotient map (07’s  $q_a^{\mathbb{C}}$  for the Gaussian pair). Every series has a set-sized support, even when its ambient ring or exponent group is a proper class.  $\text{cvx}(\mathbb{Z}a) = \{g \in \mathbf{No} : |g| \leq na \text{ for some ordinary } n \geq 1\}$  is the convex subgroup of **No** generated by  $a$ , the class form of  $\text{cvx}_{\Gamma}(G)$  of Proposition 16.86, and  $a \ll b$  is the relation of (2.4):  $b > na$  for every ordinary  $n \geq 1$ ; for  $b > 0$  this says  $b \notin \text{cvx}(\mathbb{Z}a)$ . 24’s symbols are renamed as follows.

- Its  $(D, K)$ ,  $\Pi_K$ ,  $A_D$ ,  $B_K = K \oplus \Pi_K$  and  $\mathcal{Q}_a$  are  $(D, k)$ ,  $\Pi_k$ ,  $\mathcal{A}_{D,k}$ ,  $\mathcal{A}_{k,k}$  and  $\mathcal{Q}_a^k$  ( $\mathcal{B}$  is the ring  $D + \mathfrak{m}_k$ ); its classes  $[x]$  are  $q_a(x)$ . Its relation  $b \gg a$  is  $a \ll b$ , the same relation written with the opposite symbol.
- Its convex subgroup  $C_a$  is  $\text{cvx}(\mathbb{Z}a)$  ( $C_j$  are clopen sets in Corollary 14.22,  $C_{d+1}$  a Gaussian grid and  $\mathbf{C}(G)$  a reduced fibre); its support ideal  $J_a$  is  $\Pi_k^{\gg a}$ , its truncation  $\tau_a$  is  $\text{tr}_a$  and its ring  $A_D(C_a)$  is  $\mathcal{A}_{D,k}(\text{cvx}(\mathbb{Z}a))$  ((14.27);  $J$  and  $\mathbf{J}$  are other ideals and  $\tau$  an exponent automorphism). For a convex subgroup  $H \subseteq \mathbf{No}$  its  $\Pi_K(H)$  is  $\Pi_k(H)$  of (2.1) and its support ideal  $P_H$  is  $\Pi_k^{\gg H}$  ((14.35)).
- Its geometric series  $q_{a,b}$  is  $\mathbf{g}_{a,b}$  ( $q_a$  is the quotient map); its scales  $a_\alpha = \omega^\alpha$  are written out, since  $a_\alpha$  denotes the small scales of (3.1), and its binomials  $f_\alpha$  are  $\mathbf{f}_\alpha$ ; its rescaling automorphism  $\sigma_{b/a}$  is  $\text{sc}_{b/a}$  ( $\sigma$  is an endomorphism of **Oz** in Section 5.2 and 20’s specialization).

- Its field  $F_a$  and group  $H_a$  are  $\mathcal{F}_a$  and  $H_a$ , the same objects (the Hahn field over the scale subgroup (2.4)), and its  $F_a[i]$  is  $\mathcal{F}_a(i)$ . Its shifted embedding  $\Theta_a(x) = -[\omega^a x]$  is 07's  $\iota_a$  of (14.7);  $\Theta_a$  is 07's compression  $\mathbf{No} \hookrightarrow \mathcal{F}_a$  of (14.4), a different map.
- Its dyadic classes  $u_n = [\omega^{a/2^n}]$  are 07's  $\xi_n$ , and  $u = u_1$  is 07's  $u = \xi_1$ ; 07's  $x_n = q_a(\omega^{a/n})$  has  $x_{2^n} = \xi_n$ . Its family  $z_n = \omega^{a/2^n}$  in Theorem 9.6 is written out ( $z_n$  are 23's branches and  $z$  is a 2-adic integer here). Its binary idempotent tree  $(e_s)$  is 07's  $(e_{r,k})$ , a branch  $\eta \in 2^{\mathbb{N}}$  being read as the 2-adic integer  $z = \sum_j \eta_j 2^j$ ; its maximal ideals  $M_{a,\eta}$ ,  $M_{\alpha,\eta}$ ,  $M_x$  and quotient maps  $\pi_{\beta,\eta}$  are  $\mathfrak{M}_{a,z}$ ,  $\mathfrak{M}_{\alpha,z}$ ,  $\mathfrak{M}_x$  and  $\varpi_{\beta,z}$  ( $M$  is a coefficient lattice and  $\pi_b$  a detecting map).
- Its multiplicative class  $T$  (its Proposition 3.3 and Lemma 4.3) is  $\Sigma$ , and its class poset  $X$ , with a variable for each element and the ideals  $I_x$ , is  $\mathcal{P}$ , with variables  $v_p$  and ideals  $I_p$  ( $X^g$  are formal monomials;  $T$  is a polynomial variable in Section 14.5 and the monomial  $\omega^a$  in Section 15.10).
- Its class field  $L$  and set subfield  $E_0$  in Theorem 9.2 are  $\mathbb{K}$  and  $\mathbb{K}_0$  ( $L$  is a set field or a lower set elsewhere and  $E$  an ambient field), and its specialization  $\varphi_a : \mathbf{Oz} \rightarrow \mathbf{No}[i]$  is  $\text{sp}_a$  ( $\varphi$  is a general map in Theorem 5.1 and a branch map in Section 15.9).

Five tempting readings are false. (1)  $a \ll b$  is not  $b > a$ :  $\omega^{2a}$  is infinitely larger than  $\omega^a$ , but  $q_a(\omega^{2a}) = 1$  is a unit; only  $a \ll b$  kills  $\omega^b$  (Theorem 14.37). (2)  $\text{cvx}(\mathbb{Z}a)$  is not  $H_a$ :  $H_a$  collects the exponents infinitesimal relative to  $a$ ,  $\text{cvx}(\mathbb{Z}a)$  those bounded by a multiple of  $a$ , and  $H_a \subsetneq \text{cvx}(\mathbb{Z}a)$ ; manuscript 25 writes  $H_a$  for  $\text{cvx}(\mathbb{Z}a)$  (Section 16.11). (3) A residue field containing an algebraically closed class subfield need not be algebraically closed (Question 18.30). (4)  $\text{sp}_a$  is a ring homomorphism, not an ordered or strongly additive map (Theorem 14.55). (5) “Nonarithmetical” is a predicate on individual ideals, and a family of maximal ideals is one class relation on parameters and ring elements; no class of all class ideals is formed.

### Maximal class ideals under global choice.

**Theorem 14.24** (Maximal class extensions [24]). *In GBC, let  $R$  be a nonzero commutative unital class ring and  $I \subsetneq R$  a class ideal. There is a maximal class ideal  $M$  of  $R$  containing  $I$ . With a fixed global well-order,  $M$  can be defined uniformly from the class parameters coding  $R$  and  $I$ .*

*Proof.* Choose an ordinal-indexed enumeration  $(r_\alpha)$  of  $R$  (for a set ring a set ordinal suffices) and define sets  $S_\alpha$  of accepted generators by  $S_0 = \emptyset$ ,

$$S_{\alpha+1} = \begin{cases} S_\alpha \cup \{r_\alpha\}, & 1 \notin I + (S_\alpha \cup \{r_\alpha\})R, \\ S_\alpha, & 1 \in I + (S_\alpha \cup \{r_\alpha\})R, \end{cases} \quad (14.20)$$

and  $S_\lambda = \bigcup_{\alpha < \lambda} S_\alpha$  at limits;  $S_\lambda$  is a set, being a set-indexed union of sets contained in the set of elements enumerated before  $\lambda$ . For a set  $T_0$ , the assertion  $1 \in I + T_0 R$  says that there are an ordinary  $m$ , elements  $t_1, \dots, t_m \in T_0$ , coefficients  $c_1, \dots, c_m \in R$  and  $z \in I$  with

$$1 = z + \sum_{j=1}^m c_j t_j; \quad (14.21)$$

this is an elementary formula with class parameters, all of whose quantified objects are sets, so the set-length recursions described above apply, and the usually proper-class ideal  $I + S_\alpha R$  is never stored. Let  $S$  be the class of all accepted generators and  $M = I + SR$ , defined by the same finite-sum formula; it is a class ideal. It is proper: an equation (14.21) with  $t_j \in S$  involves finitely many acceptance stages, all below one ordinal, and would already make an earlier  $I + S_\alpha R$  improper,



contrary to the recursion; at limit stages the same finite-witness argument preserves properness. If  $x \notin M$ , its occurrence in the enumeration was rejected, so  $1 \in I + (S_\alpha \cup \{x\})R \subseteq M + (x)$  at that stage, and (14.19) proves maximality. Every decision is definable from the global well-order and the class parameters, which proves uniformity.  $\square$

*Remark 14.25* (What has been made constructive [24]). The recursion is explicit as a class definition, not an effective algorithm: the finite-equation test quantifies over all ring elements and can be undecidable. The method is the familiar greedy proof of Krull's theorem with its size obligations checked; recursive approaches for set-sized rings are classical and appear in constructive work [63]. No new general algorithm for ideal membership is claimed (24).

**Proposition 14.26** (Prime extension avoiding a multiplicative class [24]). *In GBC, let  $\Sigma \subseteq R$  be a multiplicatively closed class containing 1 with  $I \cap \Sigma = \emptyset$ . There is a prime class ideal  $P \supseteq I$  disjoint from  $\Sigma$ .*

*Proof.* Accept  $r_\alpha$  exactly when  $(I + (S_\alpha \cup \{r_\alpha\})R) \cap \Sigma = \emptyset$ . A violation again has one element of  $\Sigma$  and a finite membership witness, so the resulting ideal  $P$  is disjoint from  $\Sigma$ , and rejection shows that it is maximal among ideals disjoint from  $\Sigma$ . If  $xy \in P$  with  $x, y \notin P$ , maximality gives  $t = p + rx \in \Sigma$  and  $u = p' + sy \in \Sigma$  with  $p, p' \in P$ , and then  $tu \in P \cap \Sigma$ , a contradiction. So  $P$  is prime; it is not asserted to be a maximal ideal of  $R$ .  $\square$

**Corollary 14.27** (Compatible prescribed equations [24]). *In GBC, if a class family  $(f_t)_{t \in T_0}$  of elements of  $R$  has no finite subfamily generating 1, then all the equations  $f_t = 0$  hold in some maximal class quotient of  $R$ .*

*Proof.* The ideal generated by the family is proper because its membership relation uses finite sums; apply Theorem 14.24 and code the quotient as above.  $\square$

*Remark 14.28* (Class ultrafilters as the Boolean case [merge]). For a Boolean algebra  $\mathcal{A}$  whose elements are sets, viewed as the Boolean ring with addition the symmetric difference, the ideal generated by finitely many complements  $\neg s_1, \dots, \neg s_n$  is principal, generated by  $\neg s_1 \vee \dots \vee \neg s_n$ , so it contains 1 exactly when  $s_1 \wedge \dots \wedge s_n = 0$ . A class family with the finite intersection property therefore has complements generating a proper ideal, and a maximal class ideal  $M$  containing them (Corollary 14.27) has residue field  $\mathbb{F}_2$ , so  $\{x : x \notin M\}$  is an ultrafilter containing the family. Thus 19's and 20's class-ultrafilter lemmas (Lemmas 15.48 and 15.78) are the Boolean-ring case of Corollary 14.27, and 20's lemma holds for a class family with the finite intersection property, not only a set. Their proofs decide  $b$  or  $\neg b$  at each stage; Theorem 14.24 accepts or rejects a generator. The foundation is the same.

**Definition 14.29** ([24]). The *class maximal ideal principle* CMIT asserts that every nonzero commutative unital class ring has a maximal class ideal. It quantifies over class rings, not only over rings whose carriers are sets.

**Theorem 14.30** (Class Krull and global choice [24]). *Over GB, CMIT is equivalent to GC. Equivalently, global choice is equivalent to the assertion that every proper class ideal of every commutative unital class ring has a maximal class extension.*

The reverse implication uses the finite-polynomial mechanism of the classical proofs that Krull's theorem implies choice, following the maximal-small-ideal argument in the corrected (third) version of Entin's exposition, with antecedents in Ern  and Hodges [62, 61, 60]. 24's contribution is the

passage to classes and the choice of a class poset encoding a *global* choice function; the finite algebraic lemma is not offered as new (24).

**Lemma 14.31** (Finite-polynomial compatibility [24]). *Assume CMIT. Every nonempty class poset  $\mathcal{P}$  has a subclass  $\mathcal{Y}$  maximal with the property that every finite subset of  $\mathcal{Y}$  has an upper bound in  $\mathcal{P}$ .*

*Proof.* Let  $R = \mathbb{F}_2[v_p : p \in \mathcal{P}]$  be the class polynomial ring with finite supports, and for  $p \in \mathcal{P}$  let  $I_p$  be the ideal generated by the variables  $v_q$  with  $q \leq p$ . Call a polynomial *small* if it lies in some  $I_p$ . Each  $I_p$  is a monomial prime ideal, so the nonsmall polynomials form a multiplicative class  $\Sigma$  with  $1 \in \Sigma$  and  $0 \notin \Sigma$ . A maximal ideal of  $\Sigma^{-1}R$ , which exists by CMIT, contracts to a prime ideal  $P$  of  $R$  maximal among ideals consisting of small polynomials; the localization uses finite fractions and Scott codes, so needs no global choice.

A small polynomial has zero constant term. If a monomial  $m$  occurs in some  $f \in P$ , choose a variable  $v$  dividing  $m$ ; given  $g \in P$ , take an ordinary  $d > \deg g$ . Then  $v^d f + g \in P$  is small, say in  $I_{p'}$ , and there is no cancellation between the terms of  $g$  and those of  $v^d f$  (the latter have degree above  $\deg g$ ). As  $I_{p'}$  is a monomial ideal,  $g \in I_{p'}$  and  $v^d m \in I_{p'}$ ; every variable dividing  $v^d m$  divides  $m$ , so  $m \in I_{p'}$  and  $g + hm$  is small for every polynomial  $h$ . So  $P + (m)$  consists of small polynomials, maximality gives  $m \in P$ , and primality puts a variable of  $m$  in  $P$ . Hence  $P = (\mathcal{Y})$  for the class  $\mathcal{Y}$  of  $p$  with  $v_p \in P$ .

A sum of finitely many distinct variables from  $\mathcal{Y}$  lies in  $P$ , hence in some  $I_p$ , so they have the common upper bound  $p$ . Conversely, if  $\mathcal{Y}' \supseteq \mathcal{Y}$  has the finite-upper-bound property, every element of the ideal  $(\mathcal{Y}')$  lies in some  $I_p$ ; maximality of  $P$  gives  $(\mathcal{Y}') = P$  and then  $\mathcal{Y}' = \mathcal{Y}$ .  $\square$

*Proof of Theorem 14.30.* Global choice implies the maximal-extension assertion by Theorem 14.24, and that assertion implies CMIT (take the zero ideal). Conversely assume CMIT and let  $\mathcal{P}$  be the class of all *finite* partial choice functions, finite functions  $p$  on sets of nonempty sets with  $p(B) \in B$  for  $B$  in their domain, ordered by extension; it contains the empty function, and a finite subset of  $\mathcal{P}$  has an upper bound exactly when its members agree on overlapping domains, the union then being a finite upper bound. Let  $\mathcal{Y}$  be given by Lemma 14.31. Pairwise compatibility makes  $h = \bigcup \mathcal{Y}$  a class function. If a nonempty set  $B$  were not in its domain, fix  $b \in B$ : the singleton function  $\{(B, b)\}$  is compatible with every finite subfamily of  $\mathcal{Y}$  (the union of such a subfamily and  $\{(B, b)\}$  is a finite upper bound), so it could be adjoined to  $\mathcal{Y}$ , and maximality would put it in  $\mathcal{Y}$ , contradicting the omission of  $B$ . Hence  $h(B) \in B$  for every nonempty set  $B$ , and  $h$  is a global choice function.  $\square$

*Remark 14.32* (The quantifier restriction matters [24]). Theorem 14.30 does *not* show that the existence of one nonarithmetic maximal ideal of  $\mathbf{Oz}$ , or even of one above every  $1 + \omega^a$ , implies global choice: the converse uses arbitrary, specially built class rings, and necessity for the restricted omnific problem is a separate question (Question 18.28). Likewise the ordinary maximal-ideal theorem for set rings yields choice for sets; it is the class-uniform version that yields global choice (24).

### Nonarithmetic maximal ideals exist.

**Proposition 14.33** (The arithmetic maximal ideals [24]). *A maximal class ideal  $M$  of  $A$  with  $M \cap D \neq 0$  is  $\text{ct}^{-1}(\mathfrak{p}) = \mathfrak{p}A$  for a nonzero prime ideal  $\mathfrak{p}$  of  $D$ , with quotient  $D/\mathfrak{p}$ . Every other maximal class ideal has  $M \cap D = 0$ , a residue field of characteristic 0, and  $M \not\subseteq \Pi_k$ .*

*Proof.* If  $0 \neq d \in M \cap D$ , then  $\Pi_k = d\Pi_k \subseteq M$  (Proposition 3.3), so  $M$  is the preimage of a maximal ideal of  $D$  (Theorem 6.2); for  $D = \mathbb{Z}$  or  $\mathbb{Z}[i]$  these are the nonzero prime ideals. Conversely each such

preimage is maximal, since its quotient is the field  $D/\mathfrak{p}$ , and coefficient division gives  $\text{ct}^{-1}(\mathfrak{p}) = \mathfrak{p}A$ . If  $M \cap D = 0$ , the quotient contains a copy of  $D$  and has characteristic 0; and  $M \subseteq \Pi_k$  would force  $M = \Pi_k$  by maximality, which is impossible because  $A/\Pi_k \cong D$  is not a field.  $\square$

For  $\mathbf{Oz}$  the arithmetic maximal ideals are the  $p\mathbf{Oz}$  of Corollary 6.3, and for  $\mathbf{Oz}[i]$  they are generated by the Gaussian primes. A maximal class ideal is *nonarithmetic* if it is not of this form; by Proposition 14.33 it then satisfies the hypotheses of Corollary 14.10. This defines a predicate on single ideals, not a set of ideals (24).

**Theorem 14.34** (Binomial maximal extensions [24]). *In GBC, for every  $a \in \mathbf{No}_{>0}$  there is a maximal class ideal  $M$  of  $A$  with  $1 + \omega^a \in M$  and  $M \cap D = 0$ . Every maximal class ideal containing  $1 + \omega^a$  is nonarithmetic, with a residue field of characteristic 0.*

*Proof.*  $1 + \omega^a$  is a nonunit (Lemma 2.4), so  $(1 + \omega^a)$  is proper and Theorem 14.24 gives  $M$ . If some maximal  $M \ni 1 + \omega^a$  met  $D$  nontrivially, Proposition 14.33 would give  $\omega^a \in \Pi_k \subseteq M$  and then  $1 \in M$ .  $\square$

**Corollary 14.35** (Unit-constant fibres [24]). *In GBC, every nonconstant  $f \in A$  with  $\text{ct}(f) \in D^\times$  lies in a maximal class ideal, and every maximal class ideal containing  $f$  has characteristic 0.*

*Proof.* Positive degree makes  $f$  a nonunit (Lemma 2.4); an arithmetic maximal ideal contains  $\Pi_k$ , hence  $\text{ct}(f)$ , and cannot contain an element with unit constant term. Apply Theorem 14.24.  $\square$

So the quotients  $\mathbf{Oz}/(1 + g)$  and  $\mathbf{Oz}[i]/(1 + g)$  of Corollary 14.9, which contain copies of  $\mathbf{No}$  and  $\mathbf{No}[i]$  and have no nonzero set-sized image, have characteristic-zero maximal class quotients. In 08's set-sized ring  $A_k^{08}$  the same element  $1 + X^{e_0}$  lies in a characteristic-zero maximal ideal by the ordinary maximal-ideal theorem, with a residue field of size  $\kappa$  (Theorem 11.16(vi) and the paragraph after it); 08 notes that this quotient map cannot extend to  $\mathbf{Oz}$  with a set-sized target, and Theorem 14.34 supplies class-sized targets instead.

**Lemma 14.36** (The same binomial quotient over  $D$  and over  $k$  [24]). *Inclusion induces an isomorphism of class rings*

$$\mathcal{A}_{D,k}/(1 + \omega^a)\mathcal{A}_{D,k} \xrightarrow{\sim} \mathcal{A}_{k,k}/(1 + \omega^a)\mathcal{A}_{k,k}. \quad (14.22)$$

*The quotient is nonzero, the natural map from  $k$  to it is injective, and  $k$  stays embedded in every proper quotient.*

*Proof.* For  $c \in k$  and  $s \in \Pi_k$ , the relation  $\omega^a = -1$  identifies the class of  $c + s$  with that of  $s - c\omega^a \in \mathcal{A}_{D,k}$ , which proves surjectivity. If  $x \in \mathcal{A}_{D,k}$  and  $x = (1 + \omega^a)y$  with  $y \in \mathcal{A}_{k,k}$ , then  $\text{ct}(y) = \text{ct}(x) \in D$  because  $\text{ct}$  is multiplicative on  $\mathcal{A}_{k,k}$ , so  $y \in \mathcal{A}_{D,k}$ ; this proves injectivity. The binomial is a nonunit of  $\mathcal{A}_{k,k}$ , so the quotient is nonzero, and a nonzero  $c \in k$  is a unit of  $\mathcal{A}_{k,k}$ , which stays a unit, hence nonzero, in every proper quotient.  $\square$

Both sides of (14.22) are  $\mathcal{Q}_a^k$ . The embedded scalar  $c \in k$  is represented in  $\mathcal{A}_{D,k}$  by  $-c\omega^a$ , as after Lemma 14.14; the embedding is a feature of the quotient and does not put  $k$  inside  $\mathcal{A}_{D,k}$ . Lemma 14.14(ii) presents the same ring as  $\mathbf{Oz}[\omega^{-a}]/(1 + \omega^a)$ , by the other inclusion.

**The exact monomial kernel at one scale.**

**Theorem 14.37** (Exact monomial dichotomy [24]). *Let  $a > 0$  and  $b \geq 0$ .*

(i) *If  $a \ll b$ , then  $q_a(\omega^b) = 0$ .*

(ii) *Otherwise  $b \leq Na$  for some ordinary  $N \geq 1$ , and  $q_a(\omega^b)$  is a unit of  $\mathcal{Q}_a^k$  with*

$$q_a(\omega^b)^{-1} = (-1)^N q_a(\omega^{Na-b}). \quad (14.23)$$

*Consequently, for every proper ideal  $J$  of  $A$  containing  $1 + \omega^a$ , prime or not,*

$$\omega^b \in J \iff a \ll b. \quad (14.24)$$

*Proof.* If  $a \ll b$ , the series

$$\mathbf{g}_{a,b} = \sum_{n \geq 0} (-1)^n \omega^{b-(n+1)a} \quad (14.25)$$

has strictly decreasing, strictly positive exponents, so  $\mathbf{g}_{a,b} \in \Pi_k$ , and multiplying by the two-term series  $1 + \omega^a$ , with at most two contributions at each exponent, gives

$$(1 + \omega^a) \mathbf{g}_{a,b} = \omega^b; \quad (14.26)$$

this is membership in the principal ideal, not only in its radical. If  $0 \leq b \leq Na$ , then  $\omega^{Na-b} \in A$  and  $q_a(\omega^b) (-1)^N q_a(\omega^{Na-b}) = (-1)^N q_a(\omega^a)^N = 1$ . A unit of a nonzero ring is not zero, and every proper quotient of  $\mathcal{Q}_a^k$  keeps both the zeros of (i) and the inverse equations of (ii), which gives (14.24).  $\square$

*Remark 14.38* (A second route through the localization [merge]). By Lemma 14.14(ii) and Proposition 6.11,  $q_a(\omega^b) = 0$  if and only if  $\omega^b/(1 + \omega^a) = \sum_{n \geq 0} (-1)^n \omega^{b-(n+1)a}$  has support bounded below by some  $-Na$ , that is,  $b - (n+1)a \geq -Na$  for every  $n$ , which holds exactly when  $a \ll b$ . This gives (i) and its converse without the inverse formula; the inverse (14.23) is the finer information of (ii).

**Example 14.39** (The fibre  $1 + \omega = 0$  [24]). For  $a = 1$  the relation is  $q_1(\omega) = -1$ . Every monomial  $\omega^b$  with  $0 \leq b \leq N$  for an ordinary  $N$  is a unit, for instance  $b = \sqrt{2}$ ,  $b = 1/\omega$  and every ordinary  $b \geq 0$ ; every monomial with positive infinite exponent vanishes, for instance  $q_1(\omega^\omega) = 0$  and  $q_1(\omega^{\sqrt{\omega}}) = 0$ . The square root  $u = q_1(\omega^{1/2})$  satisfies  $u^2 = -1$ , so  $\mathcal{Q}_1^k$  is not ordered by the order inherited from  $\mathbf{Oz}$ .

Truncation to  $\text{cvx}(\mathbb{Z}a)$  removes entire tails. Put

$$\Pi_k^{\gg a} = \{s \in \Pi_k : a \ll g \text{ for every } g \in \text{supp } s\}, \quad \text{tr}_a\left(\sum_g c_g \omega^g\right) = \sum_{g \in \text{cvx}(\mathbb{Z}a)} c_g \omega^g. \quad (14.27)$$

**Proposition 14.40** (Discarding all exponents above the scale [24]).  $\Pi_k^{\gg a} \subseteq (1 + \omega^a)A$ . The truncation  $\text{tr}_a : A \rightarrow \mathcal{A}_{D,k}(\text{cvx}(\mathbb{Z}a))$ , onto the elements of  $A$  supported in  $\text{cvx}(\mathbb{Z}a)$ , is a surjective ring homomorphism with kernel  $\Pi_k^{\gg a}$ , and

$$\mathcal{Q}_a^k \cong \mathcal{A}_{D,k}(\text{cvx}(\mathbb{Z}a)) / (1 + \omega^a) \mathcal{A}_{D,k}(\text{cvx}(\mathbb{Z}a)). \quad (14.28)$$

*Proof.* For  $s \in \Pi_k^{\gg a}$  multiply  $s$  in the ambient Hahn field by  $(1 + \omega^a)^{-1} = \sum_{n \geq 0} (-1)^n \omega^{-(n+1)a}$ ; the Hahn product is defined, and every exponent that can occur is  $g - (n+1)a$  with  $a \ll g$ , which is positive. So  $s/(1 + \omega^a) \in \Pi_k$ . If  $g, h \geq 0$  and  $g + h \in \text{cvx}(\mathbb{Z}a)$ , convexity gives  $g, h \in \text{cvx}(\mathbb{Z}a)$ , and if one of  $g, h$  lies outside, so does  $g + h$ ; so truncation is multiplicative on nonnegative supports, with the stated kernel and image. It fixes  $1 + \omega^a$  and induces the map (14.28), which is surjective, and injective because  $\Pi_k^{\gg a}$  is already in the binomial ideal: if  $x = (1 + \omega^a)y$  is supported in  $\text{cvx}(\mathbb{Z}a)$ , then  $x = \text{tr}_a(x) = (1 + \omega^a) \text{tr}_a(y)$ .  $\square$

*Remark 14.41* ([24]). The reduction does not replace the exponent group by a set:  $\text{cvx}(\mathbb{Z}a)$ , and the much smaller  $H_a$ , are proper classes. Nor does it classify the elements of the binomial ideal: inside  $\text{cvx}(\mathbb{Z}a)$  different terms can cancel modulo the binomial, and the dyadic idempotents show substantial branching (24). The ring  $\mathcal{A}_{D,k}(\text{cvx}(\mathbb{Z}a))$  is  $\mathcal{A}_{D,k}(G)$  of (2.2) for the class group  $G = \text{cvx}(\mathbb{Z}a)$ .

**Corollary 14.42** (A Bézout identity between separated scales [24]). *If  $a \ll b$ , then*

$$1 = (1 + \omega^b) - (1 + \omega^a) \mathfrak{g}_{a,b}, \quad (14.29)$$

*so the principal ideals  $(1 + \omega^a)$  and  $(1 + \omega^b)$  are comaximal and no proper ideal contains both.*

*Proof.* Subtract (14.26) from  $1 + \omega^b$ .  $\square$

For an ordinal  $\alpha$  put

$$\mathfrak{f}_\alpha = 1 + \omega^{\omega^\alpha}. \quad (14.30)$$

If  $\alpha < \beta$ , then  $\omega^\beta = \omega^\alpha \omega^{\beta-\alpha}$  with  $\omega^{\beta-\alpha}$  infinite, so  $\omega^\alpha \ll \omega^\beta$ , and the ideals  $(\mathfrak{f}_\alpha)$  are pairwise comaximal: a proper-class range of scales, obtained without counting a putative set of ideals (24). 09's comaximal ideals  $(1 - X^{e_\ell})$  in a set-sized ring are the same identity (Theorem 11.20(iv)).

**Proposition 14.43** (The abstract binomial ring does not depend on  $a$  [24]). *For  $a, b > 0$  the coefficient-fixing strong automorphism*

$$\text{sc}_{b/a} \left( \sum_g c_g \omega^g \right) = \sum_g c_g \omega^{(b/a)g}$$

*of No (of No[i] for  $k = \mathbb{C}$ ) preserves  $A$  and induces  $\mathcal{Q}_a^k \cong \mathcal{Q}_b^k$ .*

*Proof.* Multiplication by the positive surreal  $b/a$  is an order-preserving automorphism of the additive exponent group, with inverse multiplication by  $a/b$ ; it preserves reverse well-ordering and the finite-decomposition property of Hahn products, positive exponents and the constant coefficient, and it sends  $1 + \omega^a$  to  $1 + \omega^b$ .  $\square$

This is the standard Hahn rescaling, not a novelty claim; it shows that the monomial cutoff is an invariant of the embedded fibre at a chosen scale, not evidence that the abstract quotient rings are pairwise nonisomorphic (24).

**The field core.** As explained at the start of this subsection, 24's Lemma 7.1, Theorem 7.2 and Proposition 7.4 are printed once, as Proposition 14.1, Lemmas 4.2 and 14.14, Corollaries 14.10 and 14.17, and Theorem 14.23. 24's proofs are the same as 07's: the closure of  $\mathcal{F}_a$  and  $\mathcal{F}_a(i)$  is reduced polynomial by polynomial to set-sized divisible support groups, the shifted map is multiplicative because  $\omega^a xy(1 + \omega^a)$  lies in the binomial ideal,  $T^2 + 1$  is irreducible over the real closed  $\mathcal{F}_a$ , and a unital map from a field to a nonzero ring is injective. 24's reservoir argument for a

prime  $P$  with  $P \cap D = 0$  and  $P \not\subseteq \Pi_k$  takes  $c + s \in P$  with  $0 \neq c \in D$ , puts  $v = s/c$  and  $a$  below  $\text{supp } v$ , and embeds the Hahn field over  $H_a$  by  $x \mapsto -q(vx)$ , which is 07's transfer with  $g = v$ ; in the real case it asserts no square root of  $-1$  in  $A/P$ . Its purpose in 24, making the size statements independent of the universal theorem, is served equally by 07's versions.

**Corollary 14.44** (Residue fields above the binomial [24] [merge]). *Let  $M$  be a maximal class ideal of  $A$  containing  $1 + \omega^a$ , which exists in GBC (Theorem 14.34). Then  $A/M$  is a proper-class field of characteristic 0 containing an isomorphic copy of  $\mathcal{E}_a \cong \mathcal{F}_a(i)$ , an algebraically closed proper-class Hahn field, and hence of  $\mathbf{No}[i]$ . No maximal ideal containing  $1 + \omega^a$  has a set-sized residue field.*

*Proof.*  $A/M$  is a proper quotient of  $\mathcal{Q}_a^k$ . For  $k = \mathbb{R}$  it receives the field  $\mathcal{E}_a \subseteq \mathcal{Q}_a$  of Corollary 14.17; for  $k = \mathbb{C}$  it receives  $\mathcal{F}_a(i)$  by the coefficient embedding of Section 14.7. A unital map from a field to a nonzero ring is injective, and  $\mathbf{No}[i]$  embeds in  $\mathcal{F}_a(i)$  by Theorem 14.4. The characteristic is 0 by Theorem 14.34.  $\square$

24 states the corollary with “an algebraically closed proper-class Hahn subfield”; the copy of  $\mathbf{No}[i]$  is the merge's addition from 07's compression. Neither says that  $A/M$  is algebraically closed: a field can be a proper transcendental extension of an algebraically closed subfield (Question 18.30).

**Continuum branching at one scale.** 24's Lemma 8.1 states that  $\mathbb{R}[\xi_n] \cong \mathbb{R}[T]/(T^{2^n} + 1) \cong \mathbb{C}^{2^{n-1}}$  inside  $\mathcal{Q}_a$ , that  $\mathbb{R}[\xi_n] \subseteq \mathbb{R}[\xi_{n+1}]$  via  $\xi_n = \xi_{n+1}^2$  splits every primitive idempotent into two nonzero orthogonal ones, and that for  $\mathbf{Oz}[i]$  the corresponding rings are  $\mathbb{C}^{2^n}$  from  $n = 0$ . This is the restriction to constant coefficients of Theorem 14.15 (with (14.10), (14.17)) and Theorem 14.23.

*Remark 14.45* (A degree proof for constant coefficients [24]). Work in  $\mathcal{A}_{\mathbb{R},\mathbb{R}}/(1 + \omega^a)$ , which is  $\mathcal{Q}_a$  by Lemma 14.36. A nonzero remainder  $r(T) \in \mathbb{R}[T]$  of degree less than  $2^n$  gives  $\deg_\omega r(\omega^{a/2^n}) < a$ , while every nonzero multiple of  $1 + \omega^a$  in  $\mathcal{A}_{\mathbb{R},\mathbb{R}}$  has degree at least  $a$  (Lemma 2.4); so the kernel of  $T \mapsto \xi_n$  on  $\mathbb{R}[T]$  is exactly  $(T^{2^n} + 1)$ . The roots of  $T^{2^n} + 1$  are simple and form  $2^{n-1}$  conjugate pairs, and under squaring each pair has two pairs of preimages; over  $\mathbb{C}$  every root is its own factor with two square roots. This route needs no support description of  $\mathbf{Oz}[\omega^{-a}]$ , but it does not cover coefficients in  $\mathcal{F}_a$ , which have negative exponents; 07's proof does. At  $n = 2$ ,  $T^4 + 1 = (T^2 - \sqrt{2}T + 1)(T^2 + \sqrt{2}T + 1)$ , and 24's idempotents

$$e_{\pm} = \frac{1}{2} \pm \frac{\sqrt{2}}{4} (\xi_2 - \xi_2^3), \quad (\xi_2 - \xi_2^3)^2 = 2 \text{ modulo } \xi_2^4 + 1,$$

are 07's  $e_{\pm}$  of (14.18) [merge]: with  $y = \xi_2$  and  $\zeta = (1 + y^2)/\sqrt{2}$ ,  $y^4 = -1$  gives  $(1 + y^2)(1 - y^2) = 2$ , so  $y/\zeta = \frac{\sqrt{2}}{2}(y - y^3)$ .

A *branch* of 07's tree is a 2-adic integer  $z \in \mathbb{Z}_2$ ; its idempotent at stage  $r$  is  $e_{r,z \bmod 2^{r-1}}$ , and by (14.17) these form a decreasing chain of nonzero idempotents,  $e_{r+1,z \bmod 2^r} e_{r,z \bmod 2^{r-1}} = e_{r+1,z \bmod 2^r}$ . 24 indexes the same tree by finite binary strings, with  $e_{\emptyset} = 1$ ,  $e_s = e_{s0} + e_{s1}$ ,  $e_{s0}e_{s1} = 0$  and  $e_s \neq 0$ , rooted at the first complex factor in the real case and at the scalar field in the Gaussian case. For  $\mathbf{Oz}[i]$  the stages are those of Theorem 14.23,  $n = 2^r$  with  $r \geq 0$ , and the idempotent of  $z$  at stage  $r$  is the indicator of the residue class  $z \bmod 2^r$ ; the statements below hold verbatim with this indexing.

**Theorem 14.46** (Continuum many maximal extensions at a fixed scale [24]). *In GBC, for each  $a > 0$  there is a family  $(\mathfrak{M}_{a,z})_{z \in \mathbb{Z}_2}$  of pairwise distinct maximal class ideals of  $A$  containing  $1 + \omega^a$ , encoded by a single class membership relation on  $\mathbb{Z}_2 \times A$ , with  $e_{r,z \bmod 2^{r-1}} \equiv 1 \text{ modulo } \mathfrak{M}_{a,z}$  for every  $r \geq 1$ . Every residue field  $A/\mathfrak{M}_{a,z}$  has the properties of Corollary 14.44.*



*Proof.* For  $z \in \mathbb{Z}_2$  let  $I_z$  be the ideal of  $\mathcal{Q}_a^k$  generated by the  $1 - e_{r,z \bmod 2^{r-1}}$ ,  $r \geq 1$ . It is proper: an equation for 1 uses finitely many generators; multiplying by  $e_{N,z \bmod 2^{N-1}}$  for  $N$  at least all stages occurring annihilates every generator used, since  $e_{N,z \bmod 2^{N-1}}(1 - e_{r,z \bmod 2^{r-1}}) = 0$  for  $r \leq N$ , and leaves the nonzero idempotent  $e_{N,z \bmod 2^{N-1}}$  on the left. [Theorem 14.24](#) extends  $I_z$  to a maximal class ideal of  $\mathcal{Q}_a^k$ , whose preimage in  $A$  is  $\mathfrak{M}_{a,z}$ . For  $z \neq z'$  choose  $r$  with  $z \not\equiv z' \pmod{2^{r-1}}$ ; the two idempotents at stage  $r$  are orthogonal, and one maximal ideal containing both sets of branch equations would make their product both 0 and 1. With the global well-order the extension is a uniform class definition with the set parameters  $a, z$ , so the membership predicates form one class relation; no choice from a set of maximal class ideals is made.  $\square$

*Remark 14.47* (Evaluation at a point extends [merge]). For  $f$  in the Cantor subalgebra (14.14), factoring through  $\mathbb{Z}/2^N\mathbb{Z}$ , write  $f = \sum_k c_k e_{N+1,k}$  with  $c_k \in \mathcal{E}_a$ . Modulo  $\mathfrak{M}_{a,z}$  the idempotent  $e_{N+1,z \bmod 2^N}$  becomes 1 and the others, orthogonal to it, become 0, so  $f \equiv c_{z \bmod 2^N} = f(z)$ . Hence the quotient map  $\mathcal{Q}_a \rightarrow A/\mathfrak{M}_{a,z}$  restricts on the Cantor subalgebra to evaluation at  $z$  ([Remark 14.21](#)) followed by the embedding of  $\mathcal{E}_a$  in the residue field ([Corollary 14.44](#)). So with global choice every point evaluation of [Remark 14.21](#) extends to a homomorphism of all of  $\mathcal{Q}_a$  onto a field containing  $\mathcal{E}_a$ ; the extension is not unique or canonical, and its target is not  $\mathcal{E}_a$ . The same holds for  $\mathcal{Q}_a^C$  with  $\mathcal{F}_a(i)$ .

**Theorem 14.48** (Ordinal scales and binary branches together [24]). *In GBC there is a class relation encoding pairwise distinct maximal class ideals  $\mathfrak{M}_{\alpha,z}$  of  $A$ ,  $(\alpha, z) \in \mathbf{On} \times \mathbb{Z}_2$ , with  $\mathfrak{f}_\alpha \in \mathfrak{M}_{\alpha,z}$ , and with the quotient maps  $\varpi_{\beta,z}$ ,*

$$\varpi_{\beta,z}(\omega^{\omega^\alpha}) = \begin{cases} 0, & \alpha > \beta, \\ -1, & \alpha = \beta, \\ a \text{ nonzero unit}, & \alpha < \beta. \end{cases} \quad (14.31)$$

*Proof.* Apply [Theorem 14.46](#) with  $a = \omega^\alpha$ . Different branches at one scale give different ideals; at different scales the base ideals are comaximal ([Corollary 14.42](#)), so no proper ideal contains both. The pattern (14.31) is [Theorem 14.37](#) with the relation  $\omega^{\omega^\beta} = -1$ .  $\square$

*Remark 14.49* (Lower bounds, not a spectrum [24]). [Theorem 14.46](#) is a continuum lower bound within one fibre: it does not say that all maximal ideals of the fibre are indexed by  $\mathbb{Z}_2$ , nor that  $\mathcal{Q}_a^k$  is a ring of functions on Cantor space. [Theorem 14.48](#) is a uniform class-family statement. In two-sorted class theory the collection of *all* proper class ideals is not a class of set-valued elements, so no cardinality of it and no class-sized Zariski space with it as points is asserted (24). The ordinal family changes the embedded scale, and by [Proposition 14.43](#) the abstract fibres are isomorphic, so it gives no larger same-fibre bound ([Question 18.32](#)).

### Separation and realization in the surcomplex field.

**Theorem 14.50** (Characteristic-zero maximal separation [24]). *In GBC, for every  $0 \neq x \in A$  there is a characteristic-zero maximal class ideal  $\mathfrak{M}_x$  with  $x \notin \mathfrak{M}_x$  and*

$$1 - \omega x \in \mathfrak{M}_x. \quad (14.32)$$

*Its residue field is a proper class. So every nonzero element of  $A$  lies outside some nonarithmetic maximal class ideal: the nonarithmetic maximal ideals have zero intersection in this elementwise sense.*

*Proof.* As  $x \neq 0$  has nonnegative support,  $\omega x$  is a nonzero purely infinite element, so  $1 - \omega x$  is nonconstant with constant term 1 and [Corollary 14.35](#) gives a characteristic-zero maximal ideal containing it; if it contained  $x$  it would contain  $\omega x$  and 1. By [\(14.32\)](#) the image of  $\omega$  is the inverse of the image of  $x$ . The residue field contains a copy of **No** by [Corollary 14.10](#), since the ideal is nonarithmetic ([Proposition 14.33](#)); 24 cites its reservoir [Proposition 7.4](#).  $\square$

The intersection of the arithmetic maximal ideals is  $\Pi_k$ , since their membership conditions are divisibility conditions on the constant term and the nonzero primes of  $D$  intersect in 0; the tails invisible to every arithmetic residue field are separated by the nonarithmetic class residue fields. The Jacobson radical, expressed by the unit criterion without forming a collection of ideals, is

$$\text{Jac}(A) = \{x \in A : 1 - rx \text{ is a unit for every } r \in A\} = 0,$$

already because  $1 - \omega x$  is a nonunit for  $x \neq 0$  ([Lemma 2.4](#)); 24 regards this as an elementary consequence. [Theorem 14.50](#) realizes it by nonarithmetic maximal ideals, with the foundation stated. 09 proves the set-sized analogue for its rings ([Theorem 11.20](#)).

**Theorem 14.51** (Embedding a class field in **No**[ $i$ ] [24]). *In GBC, every set or class field  $\mathbb{K}$  of characteristic 0 has a unital field embedding into **No**[ $i$ ], and every embedding into **No**[ $i$ ] of a set-sized subfield  $\mathbb{K}_0 \subseteq \mathbb{K}$  extends to one of  $\mathbb{K}$ .*

*Proof.* Fix a set-like enumeration  $(x_\alpha)$  of  $\mathbb{K}$  and build compatible embeddings  $e_\alpha : \mathbb{K}_\alpha \rightarrow \mathbf{No}[i]$  of set-sized subfields, starting from the prescribed one (or from the canonical embedding of  $\mathbb{Q}$ ). At a successor stage do nothing if  $x_\alpha \in \mathbb{K}_\alpha$ . If  $x_\alpha$  is algebraic over  $\mathbb{K}_\alpha$ , transport its minimal polynomial by  $e_\alpha$  and send  $x_\alpha$  to the least root in **No**[ $i$ ] in the global well-order; **No**[ $i$ ] is algebraically closed. If  $x_\alpha$  is transcendental, send it to an element of **No**[ $i$ ] transcendental over  $e_\alpha(\mathbb{K}_\alpha)$ : the polynomials over a set field form a set, each nonzero one has finitely many roots, and the proper class **No**[ $i$ ] is not exhausted by this set of algebraic elements. At limits take unions; a set ordinal indexes a set of sets, and each  $\mathbb{K}_\alpha(x_\alpha)$  is a set, described by finite rational expressions. These are set-valued recursions as above, and the union is the required class embedding.  $\square$

This is the standard universality argument for an algebraically closed field with more than set-many transcendence resources; 24 includes it so as not to assume a classification of the residue fields or that they are algebraically closed.

**Corollary 14.52** (Algebraically closed class fields [24]). *In GBC, every proper-class algebraically closed field of characteristic 0 is isomorphic, as an abstract class field, to **No**[ $i$ ].*

*Proof.* Run the construction of [Theorem 14.51](#) back and forth along ordinal enumerations of both fields, alternately putting the next source element in the domain and the next target element in the range; both fields are algebraically closed and have elements transcendental over each set-sized subfield, every stage is a set partial isomorphism, and the union is bijective.  $\square$

This is the class back-and-forth of the surcomplex-automorphism report (`saut:thm:acfhomogeneity` [57], which extends every isomorphism between set-sized subfields of two proper-class algebraically closed fields of the same characteristic) and of `fkcsb:lem:backforth` [46]; 24's corollary is the case of prime subfields in characteristic 0.

**Theorem 14.53** (Residual surcomplexity [24]). *In GBC, every nonarithmetic maximal class ideal of  $A$  is the kernel of a unital class ring homomorphism  $A \rightarrow \mathbf{No}[i]$  whose image is a proper-class field, and these homomorphisms together separate the points of  $A$ . In particular, for every  $a > 0$  there is a unital homomorphism*

$$\mathrm{sp}_a : \mathbf{Oz} \longrightarrow \mathbf{No}[i], \quad \mathrm{sp}_a(\omega^a) = -1, \quad (14.33)$$

which can be chosen with

$$\mathrm{sp}_a(\omega^{a/2}) = i, \quad \mathrm{sp}_a(r\omega^a) = -r \quad (r \in \mathbb{R}). \quad (14.34)$$

*Proof.* Code the maximal quotient as a class field and compose the quotient map with an embedding from Theorem 14.51; the kernel is the given ideal, and the image is a proper class by Corollary 14.10. Theorem 14.50, applied to a nonzero difference, separates points. For (14.33) take a maximal ideal containing  $1 + \omega^a$ . For (14.34) note that its residue field contains the set-sized field  $\mathbb{C}$  formed from the embedded  $\mathbb{R}$  of Lemma 14.36 and the class  $u$  of  $\omega^{a/2}$ , with  $u^2 = -1$ ; prescribe its embedding into  $\mathbb{C} \subseteq \mathbf{No}[i]$  by  $u \mapsto i$  and the identity on  $\mathbb{R}$ , and use the extension clause of Theorem 14.51. Since the class of  $r\omega^a$  is the embedded scalar  $-r$ , all the identities follow.  $\square$

*Remark 14.54* (What “residually surcomplex” means [24]). The phrase means the separating family of Theorem 14.53. It is not residual finiteness, not a representation in a set-indexed product, and not a single injective homomorphism with maximal kernel; a product indexed by a proper class of class fields would need further foundations and is not used. It is not asserted that each residue field is isomorphic to  $\mathbf{No}[i]$ , nor that an embedding of a residue field is surjective (24).

**Theorem 14.55** (Failure to preserve Hahn summability [24]). *Let  $\varphi : A \rightarrow \mathbf{No}[i]$  be a unital ring homomorphism with  $\varphi(1 + \omega^a) = 0$  for some  $a > 0$ . Then  $\varphi$  does not send every strongly summable family in  $A$  to a strongly summable family in  $\mathbf{No}[i]$ .*

*Proof.* The family  $(\omega^{a/2^n})_{n \geq 1}$  is strongly summable: its supports are distinct points of a strictly decreasing sequence. But  $\varphi(\omega^{a/2^n})^{2^n} = -1$ , and  $T^{2^n} + 1$  splits over the ordinary complex numbers, so every  $\varphi(\omega^{a/2^n})$  is a nonzero ordinary complex number, with support  $\{0\}$ . The exponent 0 then occurs in infinitely many members of the image family.  $\square$

No sum in the source is missing, since  $\sum_{n \geq 1} \omega^{a/2^n}$  is a valid omnific integer; its image is simply not the strong sum of the images. So the specializations do not contradict rigidity theorems for strong maps: they have nonzero kernels and fail even to preserve summability. A homomorphism satisfying (14.33) cannot take values in an ordered field, since the image of  $\omega^{a/2}$  squares to  $-1$ ; the inclusion  $\mathbf{Oz} \hookrightarrow \mathbf{No}$  is a different homomorphism (24).

**Primes read term by term.** The monomial-prime and convex-face mechanism of ordered monoid algebras is standard; 24 uses it to delimit the support-controlled part of the prime ideals, not to suggest that every omnific prime is a monomial prime.

**Definition 14.56** ([24]). An ideal  $P \subseteq A$  is *term-closed* if for every  $x \in P$  every nonzero normal-form term of  $x$ , including its constant term, lies in  $P$ ; it is *strongly sum-closed* if the strong sum of every set-indexed strongly summable family of elements of  $P$  lies in  $P$ .

For a convex subgroup  $H \subseteq \mathbf{No}$  put

$$\Pi_k^{>H} = \{s \in \Pi_k : \text{supp } s \cap H = \emptyset\} = \{s \in \Pi_k : g > H \text{ for every } g \in \text{supp } s\}; \quad (14.35)$$

the two descriptions agree because a positive element outside a convex subgroup lies above it. The endpoints are  $\Pi_k^{>\{0\}} = \Pi_k$  and  $\Pi_k^{>\mathbf{No}} = 0$ , and  $\Pi_k^{\gg a} = \Pi_k^{>\text{cvx}(\mathbb{Z}a)}$ .

**Theorem 14.57** (Support-closed primes [24]). *The prime class ideals of  $A$  that are both term-closed and strongly sum-closed are exactly*

- (i) *the ideals  $\mathfrak{p}A$  for a nonzero prime ideal  $\mathfrak{p}$  of  $D$ , and*
- (ii) *the ideals  $\Pi_k^{>H}$  for a convex subgroup  $H \subseteq \mathbf{No}$ , with  $H$  unique.*

*None of the characteristic-zero primes in this list is maximal.*

*Proof.* If  $P$  meets  $D$  nontrivially,  $\Pi_k \subseteq P$  (Proposition 3.3),  $P \cap D$  is a nonzero prime of  $D$  and  $P = \text{ct}^{-1}(P \cap D) = (P \cap D)A$ ; such ideals have both closure properties, their only condition being divisibility of the constant term. Suppose  $P \cap D = 0$ . Term-closedness gives  $P \subseteq \Pi_k$ . For  $g > 0$  and  $r \in k^\times$  the identities  $(r\omega^g)^2 = \omega^g(r^2\omega^g)$  and  $(\omega^g)^2 = (r\omega^g)(r^{-1}\omega^g)$ , all of whose factors lie in  $A$ , and primality give

$$r\omega^g \in P \iff \omega^g \in P. \quad (14.36)$$

Let  $F = \{g \geq 0 : \omega^g \notin P\}$ . It contains 0, is closed under addition by primality, and is downward closed: if  $0 \leq h \leq g \in F$  and  $\omega^h \in P$ , then  $\omega^g = \omega^h\omega^{g-h} \in P$ . So  $H = F - F$  is a convex subgroup with  $H \cap \mathbf{No}_{\geq 0} = F$  (a nonnegative difference  $f_1 - f_2$  is at most  $f_1 \in F$ ). Term-closedness and (14.36) make every support in  $P$  avoid  $H$ ; conversely, if  $s \in \Pi_k$  has support avoiding  $H$ , its terms lie in  $P$  by (14.36), they form a strongly summable set-indexed family, and  $s \in P$ . So  $P = \Pi_k^{>H}$ . Conversely, deleting the exponents outside  $H$  is, as in Proposition 14.40, a ring retraction  $A \rightarrow D \oplus \Pi_k(H)$  onto a domain with kernel  $\Pi_k^{>H}$ , so  $\Pi_k^{>H}$  is prime, and both closure properties are clear from the support description.  $H$  is recovered from the nonnegative monomials outside the ideal. Finally  $\Pi_k^{>H} \subseteq \Pi_k$ , which is proper and not maximal since  $A/\Pi_k \cong D$ .  $\square$

**Corollary 14.58** (A maximal binomial kernel is not a support ideal [24]). *A maximal ideal containing  $1 + \omega^a$  is not term-closed: it contains the two-term series  $1 + \omega^a$  and neither of its terms. Its monomials are nevertheless exactly the  $\omega^b$  with  $a \ll b$  (Theorem 14.37).*

There is no tension: a monomial kernel records what happens to isolated scales, not how sums of surviving terms cancel, and the dyadic branch choices and the binomial equation live in that additional information (24).

**Dependencies and plausible but incorrect arguments (24).** The existence theorem is short only after the foundations are supplied; its chain is  $\text{GC} \Rightarrow$  a set-like enumeration  $\Rightarrow$  finite-witness recursion  $\Rightarrow$  maximal class extension, and normal-form degree (properness of the binomial ideal) and scalar divisibility of  $\Pi_k$  (no arithmetic extension) connect it to  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ . The monomial cutoff is independent of the maximal-ideal construction: it concerns the concrete principal ideal, and its only infinite calculation is the strongly summable identity (14.26). The continuum branching needs the finite dyadic algebras and one maximal extension per branch; the surcomplex realization needs algebraic closedness of  $\mathbf{No}[i]$  and set-valued field recursion; the support-prime classification uses primality, coefficient identities, convexity and the two closure hypotheses, and is not used to construct maximal ideals. 24 lists eight distinctions on which its proofs depend.

- (i) A proper ideal is not a set of generators:  $I$  and the ideals at the stages may be proper classes; the recursive state is the set  $S_\alpha$ , and membership is a finite formula with  $I$  as a class parameter.
- (ii) A class quotient is not a collection of literal cosets; least-representative coding in GBC, or Scott coding in GB, keeps its elements set-valued.
- (iii) The converse choice theorem has universal scope: it concerns every class ring and gives no lower bound for the choice strength of the omnific ring.
- (iv) Infinite telescoping is a support assertion: every exponent in (14.25) must stay positive, so  $b > a$  in place of  $a \ll b$  would invalidate it, and for  $b \leq Na$  the opposite holds, the monomial being a unit.
- (v) Finite product factors do not become prime by declaration: a dyadic branch first gives a proper ideal, and the class Krull construction extends it to a maximal one.
- (vi) Containing an algebraically closed field is not algebraic closedness.
- (vii) An algebraic homomorphism is not a strong homomorphism: the source sum in Theorem 14.55 is valid, its image family is not strongly summable, and no interchange of the map with the sum is allowed.
- (viii) Computations certify only finite examples: 24's checks test finite greedy constructions, polynomial identities, finite dyadic product algebras and finite-rank support examples, not class recursion, all surreal supports, global choice or priority.

## 15 Integral closure, complete integral closure, and invisible equations

The universal theorem uses exponent scales *below* the supports of purely infinite elements. This section uses a scale *above* all negative exponents, which yields one denominator for a whole set-generated algebra (03). The consequences separate three notions that coincide for Noetherian domains: membership in  $\mathbf{Oz}$ , integrality over  $\mathbf{Oz}$ , and almost integrality over  $\mathbf{Oz}$ . The complete integral closure of  $\mathbf{Oz}$  is all of  $\mathbf{No}$ , while its integral closure  $\mathcal{N}$  is a proper, fine-dense subring that no set of generators produces (03). A monic equation of 08 has no root in  $\mathbf{Oz}[i]$  although its image in every set-sized ring has the root 1; its positive root is an explicit element of  $\mathcal{N} \setminus \mathbf{Oz}$ . The section ends with 19's description of  $\mathcal{N}$  and  $\mathcal{N}_{\mathbb{C}}$  modulo the ideals generated by  $\Pi$ , which classifies their set-sized domain images (Section 15.8), and with 20's proof that the Boolean algebra of that description is atomless and a proper class (Section 15.9), and with 23's fixed cyclic workspace, whose extended ideal is not radical although the Boolean algebra of its reduced fibre is still atomless (Section 15.10), and with 27's henselization at the constant-term ideal of a set-sized omnific integer part, which has the maximal number of generic branches (Section 15.11).

**Standing notation.** As in Section 11.1,  $\kappa$  is an infinite cardinal with  $\mu = \text{cf}(\kappa) > \aleph_0$ ,  $\theta = \kappa^{\aleph_0}$ ,  $G_\kappa$  is the two-armed lexicographic group of 03,  $F_\kappa^{03} = H_\omega(\mathbb{R}, G_\kappa)$  is the field of countably supported Hahn series over  $G_\kappa$ , identified with its image in  $\mathbf{No}$ , and

$$A_\kappa^{03} = F_\kappa^{03} \cap \mathbf{Oz} = \mathbb{Z} + \Pi_\kappa^{03},$$

where  $\Pi_\kappa^{03}$  consists of the elements of  $F_\kappa^{03}$  with strictly positive support. We write  $X^g$  ( $g \in G_\kappa$ ) for the monomials of  $F_\kappa^{03}$ ; under the identification they are surreal monomials. The Gaussian versions

are  $F_\kappa^{03}[i] = H_\omega(\mathbb{C}, G_\kappa)$ ,  $A_\kappa^{03}[i] = \mathbb{Z}[i] + \Pi_\kappa^{03} + i\Pi_\kappa^{03}$  and the ideal  $\Pi_\kappa^{03} + i\Pi_\kappa^{03}$ . The field  $F_\kappa^{03}$  is real closed and  $F_\kappa^{03}[i]$  is algebraically closed (Section 11.1). We write  $\overline{\mathbb{Z}}$  for the ring of complex algebraic integers and  $\overline{\mathbb{Z}}_\mathbb{R} = \overline{\mathbb{Z}} \cap \mathbb{R}$  for the real ones. For an inclusion of rings  $A \subseteq R$  the *conductor* is  $(A : R) = \{a \in A : aR \subseteq A\}$ .

### 15.1 Uniform denominators

The clearing statement Proposition 3.4(i) clears one set of surreals. Applied to a well-chosen set it clears a whole algebra.

**Theorem 15.1** (One denominator for a set-generated algebra [03]). *For every set  $S \subseteq \mathbf{No}$  there is a positive monomial  $d = \omega^h \in \Pi$  with*

$$d \mathbf{Oz}[S] \subseteq \Pi. \quad (15.1)$$

*For every set  $S \subseteq \mathbf{No}[i]$  there is such a  $d$  with*

$$d \mathbf{Oz}[i][S] \subseteq \Pi_\mathbb{C}. \quad (15.2)$$

*In particular  $d$  is a nonzero element of the conductor  $(\mathbf{Oz} : \mathbf{Oz}[S])$ , respectively  $(\mathbf{Oz}[i] : \mathbf{Oz}[i][S])$ .*

*Proof.* Although  $\mathbf{Oz}[S]$  is in general a proper class, the ring  $\mathbb{Z}[S]$  is a set: it consists of the values of ordinary integer polynomials in finitely many members of  $S$ . By Proposition 3.4(i) there is  $h > 0$  with  $\omega^h \mathbb{Z}[S] \subseteq \Pi$ . Every element of  $\mathbf{Oz}[S]$  is a finite sum  $\sum_j a_j b_j$  with  $a_j \in \mathbf{Oz}$  and  $b_j \in \mathbb{Z}[S]$ , and

$$\omega^h \sum_j a_j b_j = \sum_j a_j (\omega^h b_j) \in \Pi$$

because  $\Pi$  is an ideal of  $\mathbf{Oz}$  (Lemma 2.2). The complex statement is the same argument with the set  $\mathbb{Z}[i][S]$ , the complex clause of Proposition 3.4(i) and the ideal  $\Pi_\mathbb{C}$  of  $\mathbf{Oz}[i]$ . Finally  $\Pi \subseteq \mathbf{Oz}$  and  $\Pi_\mathbb{C} \subseteq \mathbf{Oz}[i]$ .  $\square$

*Remark 15.2* (Conductors [03]). The theorem gives more than a nonzero element of the conductor  $(\mathbf{Oz} : \mathbf{Oz}[S])$ : it gives a monomial whose products with the whole algebra land in the strictly positive-support ideal  $\Pi$ . The real clearing statement and its proof are due to manuscript 05 (now in the sibling report [47]); the algebra formulation and the closure consequences below are 03's.

**Theorem 15.3** (The bounded-size version [03]). *Let  $A$  be  $A_\kappa^{03}$  or  $A_\kappa^{03}[i]$ , with ambient field  $F = F_\kappa^{03}$ , respectively  $F_\kappa^{03}[i]$ , constant ring  $D = \mathbb{Z}$ , respectively  $\mathbb{Z}[i]$ , and ideal  $P = \Pi_\kappa^{03}$ , respectively  $\Pi_\kappa^{03} + i\Pi_\kappa^{03}$ . For every set  $S \subseteq F$  with  $|S| < \mu = \text{cf}(\kappa)$  there is  $h > 0$  in  $G_\kappa$  with*

$$X^h A[S] \subseteq P. \quad (15.3)$$

*In particular  $\text{Frac}(A_\kappa^{03}) = F_\kappa^{03}$  and  $\text{Frac}(A_\kappa^{03}[i]) = F_\kappa^{03}[i]$ .*

*Proof.* The ring  $D[S]$  has cardinality at most  $\max(|S|, \aleph_0) < \mu$ , since  $\mu$  is uncountable. Each of its elements has countable support (in the complex case, take the union of the supports of the real and imaginary coordinates). The union  $U$  of these supports is a union of fewer than  $\mu$  countable sets, so  $|U| < \mu$  because  $\mu$  is regular and uncountable. By Lemma 11.2(i) the family  $\{-g : g \in U\} \cup \{0\}$ , which has fewer than  $\mu$  members, has an upper bound  $b \in G_\kappa$ ; put  $h = b + e$  for any  $e > 0$  in  $G_\kappa$ . Then  $h > 0$  and  $h > -g$  for every  $g \in U$ . Multiplication by  $X^h$  translates each support by  $h$ , so it maps every element of  $D[S]$  to a countably supported series with strictly positive support,



that is, into  $P$ . Every element of  $A[S]$  is a finite sum  $\sum_j a_j b_j$  with  $a_j \in A$  and  $b_j \in D[S]$ , and  $X^h \sum_j a_j b_j = \sum_j a_j (X^h b_j) \in P$  because  $P$  is an ideal of  $A$ .

For  $x \in F$  apply this to  $S = \{x\}$ : then  $x = (X^h x)/X^h$  is a quotient of elements of  $A$ , and  $X^h \in P \subseteq A$  is nonzero.  $\square$

This supplies the fraction-field clause  $\text{Frac}(A_\kappa^{03}) = F_\kappa^{03}$  of 03's cardinal realization (compare [Theorem 11.5](#)). Clearing one element is weaker than clearing *all powers* of that element by the same denominator; both theorems supply the latter as well, since  $x^n \in A[x]$  for every  $n$ .

## 15.2 Almost integrality and complete integral closure

**Definition 15.4** ([03]). Let  $A$  be a domain contained in a field  $F$ . An element  $x \in F$  is *almost integral* over  $A$  if there is  $0 \neq d \in A$  with  $dx^n \in A$  for every ordinary  $n \geq 0$ . When  $F = \text{Frac}(A)$ , the set  $A^*$  of these elements is the *complete integral closure* of  $A$ . The *integral closure*  $\text{IC}_F(A)$  consists instead of the roots in  $F$  of monic polynomials in  $A[T]$ .

These are the standard notions of commutative algebra; they differ in non-Noetherian settings, and we take the terminology from [4] without importing any result of that paper as a proof. The quantifier over powers ranges over ordinary finite powers, not over ordinal iterations or surreal exponentiation. Each individual witness  $d$  and each monic polynomial is set-sized data, so the definitions apply verbatim to the class rings  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ .

**Theorem 15.5** (Every ambient element is almost integral [03]). *Relative to their fraction fields,*

$$\mathbf{Oz}^* = \mathbf{No}, \quad \mathbf{Oz}[i]^* = \mathbf{No}[i], \quad (15.4)$$

$$(A_\kappa^{03})^* = F_\kappa^{03}, \quad (A_\kappa^{03}[i])^* = F_\kappa^{03}[i]. \quad (15.5)$$

More generally, every ring  $R$  with  $A \subseteq R \subseteq F$ , where  $(A, F)$  is one of the four pairs  $(\mathbf{Oz}, \mathbf{No})$ ,  $(\mathbf{Oz}[i], \mathbf{No}[i])$ ,  $(A_\kappa^{03}, F_\kappa^{03})$ ,  $(A_\kappa^{03}[i], F_\kappa^{03}[i])$ , has  $\text{Frac}(R) = F$  and  $R^* = F$ .

*Proof.* The fraction fields are the indicated ambient fields by [Proposition 3.4\(ii\)](#) and [Theorem 15.3](#), so there are no further elements to consider. For  $x$  in the ambient field apply [Theorem 15.1](#) or [Theorem 15.3](#) to  $S = \{x\}$ . The single nonzero denominator  $d$  supplied there satisfies  $dx^n \in dA[x] \subseteq A$  for every ordinary  $n \geq 0$ , so  $x$  is almost integral. If  $A \subseteq R \subseteq F$ , then  $F = \text{Frac}(A) \subseteq \text{Frac}(R) \subseteq F$ , and the same  $d \in A \subseteq R$  gives  $dx^n \in A \subseteq R$ .  $\square$

**Example 15.6** (Two explicit denominators [03]). The rational  $1/2$  is almost integral over  $\mathbf{Oz}$ , witnessed by  $d = \omega$ : indeed  $\omega/2^n \in \Pi$  for all  $n \geq 0$ . The monomial  $\omega^{-1}$  is almost integral, witnessed by  $d = \omega^\omega$ : indeed  $\omega^{\omega-n} \in \Pi$  for every ordinary  $n$ , since  $\omega - n > 0$ . Neither claim uses a limit as  $n$  tends to an infinite ordinal. Neither element is integral ([Lemma 15.7](#) below).

## 15.3 Integral closure: reciprocals, small units and density

The geometric statements are proved first for an arbitrary integer part of a real closed field; this isolates what is a consequence of integer-part geometry rather than of the proper-class size of  $\mathbf{No}$  (03).

**Lemma 15.7** (An integral reciprocal was already present [03]). *Let  $A$  be a domain in a field  $F$  and  $0 \neq a \in A$ . If  $a^{-1}$  is integral over  $A$ , then  $a^{-1} \in A$ . Hence the integral closure of  $A$  in  $F$  contains the inverse of no nonunit of  $A$ .*

*Proof.* Suppose  $(a^{-1})^n + c_{n-1}(a^{-1})^{n-1} + \cdots + c_0 = 0$  with  $c_j \in A$  and  $n \geq 1$ . Multiplying by  $a^{n-1}$  gives

$$a^{-1} = -c_{n-1} - c_{n-2}a - \cdots - c_0a^{n-1} \in A. \quad \square$$

Consequently  $1/2$  is not integral over  $\mathbf{Oz}$ ,  $A_\kappa^{03}$ ,  $\mathbf{Oz}[i]$  or  $A_\kappa^{03}[i]$ , despite [Example 15.6](#):  $2$  is a nonunit of each of these rings by [Lemma 2.4](#) (the units of  $A_\kappa^{03} \subseteq \mathbf{Oz}$  are among  $\mathbf{Oz}^\times = \{\pm 1\}$ , and similarly in the Gaussian case). More generally, for  $h > 0$  and  $0 \neq c \in \mathbb{R}$  the negative monomial  $c\omega^{-h}$  is not integral over  $\mathbf{Oz}$ , and for  $0 \neq c \in \mathbb{C}$  it is not integral over  $\mathbf{Oz}[i]$ : its reciprocal  $c^{-1}\omega^h$  lies in the purely infinite ideal  $\Pi$ , respectively  $\Pi_\mathbb{C}$ , and is a nonunit. The same holds for  $cX^{-h}$  over  $A_\kappa^{03}$  and  $A_\kappa^{03}[i]$  when  $h > 0$  in  $G_\kappa$ .

**Lemma 15.8** (Integer parts [03]). *Recall that a subring  $A$  of an ordered field  $F$  is an integer part if for every  $x \in F$  there is a unique  $\lfloor x \rfloor \in A$  with  $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ . Then  $\mathbf{Oz}$  is an integer part of  $\mathbf{No}$  and  $A_\kappa^{03}$  is an integer part of  $F_\kappa^{03}$ . Every nonzero element of  $\mathbf{Oz}$  has absolute value at least 1, and every nonzero element of  $\mathbf{Oz}[i]$  has surreal modulus  $|a + ib| = \sqrt{a^2 + b^2}$  at least 1.*

*Proof.* A nonzero element of  $\mathbf{Oz}$  is either a nonzero ordinary integer or has positive degree ([Lemma 2.4](#)); in the second case its absolute value exceeds every real number. Hence  $|a| \geq 1$  for  $0 \neq a \in \mathbf{Oz}$ , and for  $a + ib \in \mathbf{Oz}[i]$  with  $a, b \in \mathbf{Oz}$  not both zero,  $\sqrt{a^2 + b^2} \geq \max(|a|, |b|) \geq 1$ . This lower bound also gives uniqueness of floors: two elements of  $\mathbf{Oz}$  in a half-open interval of length 1 differ by an element of absolute value less than 1. For existence, write the normal form of  $x \in \mathbf{No}$  uniquely as  $x = P + r + \epsilon$  with  $P \in \Pi$ ,  $r \in \mathbb{R}$  and  $\text{supp } \epsilon$  strictly negative, so that  $\epsilon$  is zero or infinitesimal. Take  $\lfloor x \rfloor = P + \lfloor r \rfloor$ , except when  $r \in \mathbb{Z}$  and  $\epsilon < 0$ , where  $\lfloor x \rfloor = P + r - 1$ . In every case  $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$ . For  $x \in F_\kappa^{03}$  the same decomposition has  $P \in \Pi_\kappa^{03}$ , so the floor lies in  $A_\kappa^{03}$ .  $\square$

The exceptional case in the floor formula matters: discarding all negative exponents sends  $-\omega^{-1}$  to 0, which is not its floor.

**Why integral elements form a ring.** That the integral elements over a commutative ring  $A$  form a ring, and that integrality is transitive, is finite algebra, which we recall because it is applied to class rings. If  $x$  satisfies a monic equation of degree  $n$  over  $A$ , then  $A[x] = A + Ax + \cdots + Ax^{n-1}$  is a finitely generated  $A$ -module, by reducing powers with that equation. If  $x$  and  $y$  are integral,  $A[x, y]$  is generated by finitely many monomials  $w_1 = 1, w_2, \dots, w_N$  in  $x$  and  $y$ . For  $z \in A[x, y]$  write  $zw_k = \sum_l a_{kl}w_l$  with  $a_{kl} \in A$ ; the determinant trick shows that  $\det(zI - (a_{kl}))$  annihilates every  $w_l$ , in particular  $w_1 = 1$ , so it is zero, and this is a monic equation for  $z$ . The same argument, applied to the finitely many coefficients of a monic equation and to the element, proves transitivity. Only finitely many elements are involved at each step, so the arguments apply unchanged to  $\mathbf{Oz}$ ,  $\mathbf{Oz}[i]$  and their overrings in  $\mathbf{No}$  and  $\mathbf{No}[i]$  (03).

**Theorem 15.9** (Small integral units [03]). *Let  $F$  be a real closed ordered field,  $A \subseteq F$  an integer part, and  $N$  the integral closure of  $A$  in  $F$ . For every  $H \in A_{>0}$  the element  $u_H = \sqrt{H^2 + 1} - H$  is a positive unit of  $N$  satisfying*

$$u_H^2 + 2Hu_H - 1 = 0, \quad u_H^{-1} = u_H + 2H, \quad 0 < u_H < \frac{1}{2H}. \quad (15.6)$$

*For every  $\epsilon \in F_{>0}$  there is such a unit with  $0 < u_H < \epsilon$ .*

*Proof.* Since  $\sqrt{H^2 + 1} > H > 0$ , the element  $u_H$  is positive. Squaring  $u_H + H = \sqrt{H^2 + 1}$  gives the first equation, which is monic over  $A$ ; so  $u_H \in N$ . The same equation says  $u_H(u_H + 2H) = 1$ , and  $u_H + 2H \in N$  because  $N$  is a ring containing  $A$ ; so  $u_H$  is a unit of  $N$ . Rationalizing,

$$u_H = \frac{1}{\sqrt{H^2 + 1} + H} < \frac{1}{2H}.$$

An integer part is cofinal in  $F$ :  $\lfloor 1/(2\epsilon) \rfloor + 1 \in A$  is positive and exceeds  $1/(2\epsilon)$ . Taking  $H$  to be this element gives  $u_H < 1/(2H) < \epsilon$ .  $\square$

**Theorem 15.10** (Dense normalization of an integer part [03]). *Under the hypotheses of Theorem 15.9,  $N$  is dense in  $F$  for the order topology. More precisely, for every  $x \in F$  and every  $\epsilon \in F_{>0}$  there is  $y \in N$  with*

$$0 \leq x - y < \epsilon. \quad (15.7)$$

*Proof.* Choose a positive unit  $u < \epsilon$  of  $N$  by Theorem 15.9 and let  $n = \lfloor x/u \rfloor \in A$ . Multiplying  $n \leq x/u < n + 1$  by  $u > 0$  gives  $0 \leq x - un < u < \epsilon$ ; put  $y = un \in N$ . To meet a prescribed open interval  $(a, b)$ , apply this with  $x = (a + b)/2$  and  $\epsilon < (b - a)/2$ ; then  $y \in (x - \epsilon, x] \subseteq (a, b)$ .  $\square$

No completion and no sequential argument is used. The same proof is valid for the order topology of **No** (the fine topology), with all positive surreal radii. Results on dense subfields and integer parts in other settings provide context [22]; the object here is the integral closure of a specified integer part, not an arbitrary dense subfield, and the normalization theorem is not attributed to that work. The small-unit argument is a general elementary lemma about integer parts, and 03 makes no priority claim for it.

**The normalizations.** We write

$$\mathcal{N} = \text{IC}_{\mathbf{No}}(\mathbf{Oz}), \quad \mathcal{N}_\kappa = \text{IC}_{F_\kappa^{03}}(A_\kappa^{03}). \quad (15.8)$$

These are 03's  $B$  and  $B_\kappa$ . Manuscript 03 wrote  $\text{Int}_F(A)$  for the integral closure of  $A$  in  $F$ ; we write  $\text{IC}_F(A)$  because  $\text{Int}$  denotes rings of integer-valued polynomials, such as  $\text{Int}(\mathbb{Z}^r)$ , elsewhere in this report. Note that  $\mathcal{N}_\kappa$  is defined relative to  $F_\kappa^{03}$ ; only  $\mathcal{N}_\kappa \subseteq \mathcal{N} \cap F_\kappa^{03}$  is automatic, and 03 claims no descent equality. The equality  $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$  does hold: it follows from 20's support projection (Corollary 15.59, a merge result).

**Theorem 15.11** (Set-generated intermediate algebras are uniformly discrete [03]). *For every set  $S \subseteq \mathbf{No}$  the algebra  $C = \mathbf{Oz}[S]$  has a positive surreal separation bound  $r$ :*

$$|x - y| \geq r \quad \text{for all distinct } x, y \in C.$$

*It is a closed, uniformly discrete subclass of **No** in the fine order topology. The same holds for  $A_\kappa^{03}[S] \subseteq F_\kappa^{03}$ , in the order topology of  $F_\kappa^{03}$ , whenever  $|S| < \mu$ .*

*Proof.* Choose the positive monomial  $d$  of Theorem 15.1, respectively  $d = X^h$  of Theorem 15.3. For distinct  $x, y \in C$  the product  $d(x - y)$  is a nonzero element of  $\Pi$  (respectively  $\Pi_\kappa^{03}$ ), hence of the integer part, so  $|d(x - y)| \geq 1$  by Lemma 15.8. Thus  $|x - y| \geq 1/d =: r$ .

Uniform discreteness gives an isolating interval of radius  $r/3$  around each point. For closedness, let  $z \notin C$ . If no point of  $C$  lies within  $r/3$  of  $z$ , that neighbourhood misses  $C$ . Otherwise let  $c \in C$  be such a point; every other point of  $C$  is at distance greater than  $2r/3$  from  $z$  by the triangle inequality, so a radius smaller than both  $r/3$  and  $|z - c|$  gives a neighbourhood of  $z$  disjoint from  $C$ .  $\square$

This is a class-size statement about a proper-class base ring; it is not the familiar assertion that a finite extension of a discrete ring is discrete in an Archimedean topology (03).

**Theorem 15.12** (Proper dense normalization, zero conductor, no set of generators [03]). *For the full omnific ring,*

$$\mathbf{Oz} \subsetneq \mathcal{N} \subsetneq \mathbf{No}, \quad \overline{\mathcal{N}}^{\text{fine}} = \mathbf{No}, \quad (\mathbf{Oz} : \mathcal{N}) = 0, \quad (15.9)$$

and  $\mathcal{N}$  is not generated as an  $\mathbf{Oz}$ -algebra by any set of its elements. For the set-sized ring  $A_\kappa^{03}$ , the normalization  $\mathcal{N}_\kappa$  satisfies  $A_\kappa^{03} \subsetneq \mathcal{N}_\kappa \subsetneq F_\kappa^{03}$ , is dense in  $F_\kappa^{03}$ , has conductor  $(A_\kappa^{03} : \mathcal{N}_\kappa) = 0$ , and requires at least  $\mu$  algebra generators over  $A_\kappa^{03}$ .

*Proof.* The element  $\sqrt{2} \in \mathbb{R} \subseteq F_\kappa^{03}$  is integral, being a root of  $T^2 - 2$ , and it is in neither integer part, whose real elements are the ordinary integers; so the first inclusions are strict. The element  $1/2$  is not integral by Lemma 15.7, so the second inclusions are strict. The fields  $\mathbf{No}$  and  $F_\kappa^{03}$  are real closed and  $\mathbf{Oz}$ ,  $A_\kappa^{03}$  are integer parts (Lemma 15.8); density is Theorem 15.10.

For the conductor, fix  $a \neq 0$  in the integer part. Choose a positive unit  $u$  of the normalization with  $u < 1/|a|$  (Theorem 15.9). Then  $au$  lies in the normalization and  $0 < |au| < 1$ , so  $au$  is not in the integer part by Lemma 15.8. Hence  $a$  is not in the conductor, and this holds for every  $a \neq 0$ .

If  $\mathcal{N} = \mathbf{Oz}[S]$  for a set  $S$ , then Theorem 15.1 would give a nonzero element  $d$  of the conductor, a contradiction. (Equivalently,  $\mathcal{N}$  would be uniformly discrete by Theorem 15.11, contradicting density.) The same argument with Theorem 15.3 shows that  $\mathcal{N}_\kappa \neq A_\kappa^{03}[S]$  whenever  $|S| < \mu$ .  $\square$

Integral closure therefore changes the fine geometry of the ring, although  $\mathcal{N}$  still contains neither  $1/2$  nor any nonzero negative monomial  $c\omega^{-h}$ . For  $\kappa = \aleph_1$  the theorem says, without the continuum hypothesis, that the normalization cannot be obtained by adjoining countably many elements to the continuum-sized ring  $A_{\aleph_1}^{03}$  (here  $\theta = \aleph_1^{\aleph_0} = 2^{\aleph_0}$ ). For general  $\kappa$  the theorem gives only the *lower bound*  $\mu$  on the number of algebra generators of  $\mathcal{N}_\kappa$ , not an exact generation number; the exact count  $\mu$  of 03's cardinal realization (Theorem 11.5) concerns the ideal  $\Pi_\kappa^{03}$ , not the normalization.

*Remark 15.13* (Dense does not mean sequentially approximable [03]). The fine topology of  $\mathbf{No}$  has the size obstructions discussed in the foundations report [43]. The density statement chooses a point separately for each positive surreal radius; it does not produce a nonconstant convergent set-indexed net. Indeed, every fixed set of elements of  $\mathcal{N}$  lies in a single uniformly discrete algebra  $\mathbf{Oz}[S]$  by Theorem 15.11. The dense normalization requires access to more than any fixed set of its additional generators.

## 15.4 Constant slices, truncation, and a unit equation

The results above do not give a complete normal-form test for integrality. They do give exact tests on the nonnegative-support slice, and they show why leading-term tests fail (03). Recall  $\mathcal{A}_{\mathbb{R},\mathbb{R}} = \mathbb{R} + \Pi$  (03's  $P_\mathbb{R}$ ), on which  $\text{ct}$  is a ring homomorphism onto  $\mathbb{R}$  by Lemma 2.2, although  $\mathcal{A}_{\mathbb{R},\mathbb{R}}$  is not an integer part; its countable version is  $\mathbb{R} + \Pi_\kappa^{03}$ .

**Theorem 15.14** (The real constant and support slices [03]).

$$\mathcal{N} \cap \mathcal{A}_{\mathbb{R},\mathbb{R}} = \overline{\mathbb{Z}}_\mathbb{R} + \Pi = \mathcal{A}_{\overline{\mathbb{Z}}_\mathbb{R},\mathbb{R}}, \quad \mathcal{N} \cap \mathbb{R} = \overline{\mathbb{Z}}_\mathbb{R}, \quad \mathcal{N} \cap \mathbb{Q} = \mathbb{Z}. \quad (15.10)$$

The same formulas hold for  $\mathcal{N}_\kappa$ , with  $\mathcal{A}_{\mathbb{R},\mathbb{R}}$  and  $\Pi$  replaced by  $\mathbb{R} + \Pi_\kappa^{03}$  and  $\Pi_\kappa^{03}$  in the first formula.

*Proof.* Let  $x = r + P$  with  $r \in \mathbb{R}$  and  $P \in \Pi$ , and suppose  $x^n + a_{n-1}x^{n-1} + \dots + a_0 = 0$  with  $a_j \in \mathbf{Oz}$ . All terms lie in  $\mathcal{A}_{\mathbb{R}, \mathbb{R}}$ , and applying the ring homomorphism  $\text{ct}$  gives  $r^n + \text{ct}(a_{n-1})r^{n-1} + \dots + \text{ct}(a_0) = 0$ , a monic equation over  $\mathbb{Z}$ . So  $r \in \overline{\mathbb{Z}}_{\mathbb{R}}$ . Conversely, a real algebraic integer  $r$  is integral over  $\mathbb{Z} \subseteq \mathbf{Oz}$ , and  $P \in \Pi \subseteq \mathbf{Oz}$ , so  $r + P$  is integral over  $\mathbf{Oz}$  because integral elements form a ring. This is the first equality; intersecting with  $\mathbb{R}$  gives the second, and the third holds because a rational root of a monic integer polynomial is an integer. For  $\mathcal{N}_{\kappa}$  the coefficients lie in  $A_{\kappa}^{03} \subseteq \mathbf{Oz}$ , the element lies in  $\mathbb{R} + \Pi_{\kappa}^{03}$ , and the same argument applies; conversely  $\overline{\mathbb{Z}}_{\mathbb{R}} \subseteq \mathbb{R} \subseteq F_{\kappa}^{03}$  and  $\Pi_{\kappa}^{03} \subseteq A_{\kappa}^{03}$ .  $\square$

**Example 15.15** ([03]). The element  $\pi\omega + \sqrt{2}$  is integral over  $\mathbf{Oz}$ , whereas  $\pi\omega + \pi$  is not. The nature of a coefficient at a *positive* exponent is irrelevant to the test; only the ordinary constant coefficient is constrained. The element  $\omega + 1/2$  is almost integral ([Theorem 15.5](#)) but not integral, since  $1/2 \notin \overline{\mathbb{Z}}_{\mathbb{R}}$ .

**Proposition 15.16** (Normalization is not truncation closed [03]). *Let  $h > 0$  and  $H = \omega^h$  (or  $H = X^h$  with  $h > 0$  in  $G_{\kappa}$ ). The integral unit  $u_H$  of [Theorem 15.9](#) has the normal form*

$$u_H = \sum_{n \geq 1} \binom{1/2}{n} \omega^{-(2n-1)h} = \frac{1}{2}\omega^{-h} - \frac{1}{8}\omega^{-3h} + \frac{1}{16}\omega^{-5h} - \frac{5}{128}\omega^{-7h} + \frac{7}{256}\omega^{-9h} - \dots \quad (15.11)$$

*Its leading monomial  $\frac{1}{2}\omega^{-h}$  is not integral over the integer part. Therefore neither  $\mathcal{N}$  nor  $\mathcal{N}_{\kappa}$  is closed under leading normal-form truncation.*

*Proof.* By [Lemma 15.24](#) below with  $m = 2$  and  $\gamma = 2h$ , the series  $s = \omega^h \mathbf{b}_2(\omega^{-2h})$  satisfies  $s^2 = \omega^{2h}(1 + \omega^{-2h}) = H^2 + 1$ . Its leading term is  $\omega^h > 0$ , so  $s$  is the positive square root  $\sqrt{H^2 + 1}$  (an ordered field has at most one positive square root of a given element). Hence

$$u_H = s - \omega^h = \omega^h (\mathbf{b}_2(\omega^{-2h}) - 1) = \sum_{n \geq 1} \binom{1/2}{n} \omega^{h-2nh},$$

a series with countable, reverse well-ordered support  $\{-(2n-1)h : n \geq 1\}$ . The first five binomial coefficients are  $\binom{1/2}{1} = \frac{1}{2}$ ,  $\binom{1/2}{2} = -\frac{1}{8}$ ,  $\binom{1/2}{3} = \frac{1}{16}$ ,  $\binom{1/2}{4} = -\frac{5}{128}$  and  $\binom{1/2}{5} = \frac{7}{256}$ . In the countable model the same computation takes place in  $F_{\kappa}^{03}$ . The leading monomial  $\frac{1}{2}\omega^{-h}$  is the reciprocal of the nonunit  $2\omega^h$  of the integer part, so it is not integral by [Lemma 15.7](#).  $\square$

This is stronger than the statement that some integral elements have negative exponents: their negative tails can be essential, since deleting all but the leading term destroys integrality. In the opposite direction, [Theorem 15.14](#) gives a complete constant-term criterion when negative exponents are absent (03).

**Proposition 15.17** (A unit equation acquires solutions only after normalization [03]). *Let  $A$  be  $\mathbf{Oz}$  or  $A_{\kappa}^{03}$  and  $0 \neq H \in A$ . The equation*

$$U^2 + 2HU - 1 = 0 \quad (15.12)$$

*has no solution in  $A$ . For  $H > 0$  its two roots in the real closed ambient field are  $u_H$  and  $-2H - u_H$ ; both are integral units and belong to the normalization.*

*Proof.* A solution in  $A$  would satisfy  $U(U + 2H) = 1$ , so  $U$  would be a unit of  $A$ , hence  $U = \pm 1$  ([Lemma 2.4](#);  $A_{\kappa}^{03} \subseteq \mathbf{Oz}$ ). Substituting  $U = 1$  gives  $2H = 0$  and substituting  $U = -1$  gives  $-2H = 0$ , both contrary to  $H \neq 0$ . The two roots are  $u_H$  and  $-2H - u_H$  because their sum is  $-2H$ . Each is integral by the monic equation, and their product is  $-1$ , so each is a unit of the normalization.  $\square$

Thus membership, integrality and almost integrality give three different arithmetic environments even for a single quadratic equation with a parameter. This does not contradict constant-term transfer for systems with ordinary integer coefficients: here a possibly infinite coefficient  $H$  is fixed, and changing the ambient ring changes the allowed witnesses (03).

### 15.5 The surcomplex and Gaussian versions

The Gaussian omnific ring  $\mathbf{Oz}[i] = \mathbf{Oz} + i\mathbf{Oz} = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$  is the natural Gaussian analogue of  $\mathbf{Oz}$ ; it is not discretely ordered. It has the split Gaussian constant term  $\text{ct} : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$  (Lemma 2.2), its units are  $\pm 1, \pm i$  (Lemma 2.4), and each nonzero element has surreal modulus at least 1 (Lemma 15.8). Put

$$\mathcal{N}_{\mathbb{C}} = \text{IC}_{\mathbf{No}[i]}(\mathbf{Oz}[i]), \quad \mathcal{N}_{\kappa, \mathbb{C}} = \text{IC}_{F_{\kappa}^{03}[i]}(A_{\kappa}^{03}[i]), \quad (15.13)$$

03's  $C$  and  $C_{\kappa}$ . Since  $i$  is integral over  $\mathbf{Oz}$ , transitivity shows that these are also the integral closures of  $\mathbf{Oz}$  in  $\mathbf{No}[i]$  and of  $A_{\kappa}^{03}$  in  $F_{\kappa}^{03}[i]$ . Complex conjugation is a ring automorphism of  $\mathbf{No}[i]$  preserving  $\mathbf{Oz}[i]$  (and of  $F_{\kappa}^{03}[i]$  preserving  $A_{\kappa}^{03}[i]$ ), so it preserves  $\mathcal{N}_{\mathbb{C}}$  and  $\mathcal{N}_{\kappa, \mathbb{C}}$ . Consequently

$$\mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}, \quad \mathcal{N}_{\kappa, \mathbb{C}} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}. \quad (15.14)$$

Complexification nevertheless does not commute with normalization, and the failure is located at the prime 2.

**Proposition 15.18** (The complexification defect is killed by 2 [03]). *There are strict inclusions of additive subclasses*

$$\mathcal{N}[i] \subsetneq \mathcal{N}_{\mathbb{C}} \subsetneq \frac{1}{2}\mathcal{N}[i], \quad \mathcal{N}_{\mathbb{C}}[1/2] = \mathcal{N}[1/2][i]. \quad (15.15)$$

The last object of the chain is a fractional module, not asserted to be a ring. In particular  $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i]$  although  $\mathcal{N}_{\mathbb{C}} \neq \mathcal{N}[i]$ . The same inclusions hold for  $\mathcal{N}_{\kappa}$  and  $\mathcal{N}_{\kappa, \mathbb{C}}$ , and the set-sized quotient  $\mathcal{N}_{\kappa, \mathbb{C}}/\mathcal{N}_{\kappa}[i]$  is a nonzero  $\mathcal{N}_{\kappa}$ -module annihilated by 2.

*Proof.* Every element of  $\mathcal{N}[i]$  is integral over  $\mathbf{Oz}[i]$ , so  $\mathcal{N}[i] \subseteq \mathcal{N}_{\mathbb{C}}$ . If  $z = x + iy \in \mathcal{N}_{\mathbb{C}}$  with  $x, y \in \mathbf{No}$ , then  $\bar{z} \in \mathcal{N}_{\mathbb{C}}$ , so

$$2x = z + \bar{z} \in \mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}, \quad 2y = -i(z - \bar{z}) \in \mathcal{N}_{\mathbb{C}} \cap \mathbf{No} = \mathcal{N}$$

by (15.14). Hence  $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i]$ , that is,  $\mathcal{N}_{\mathbb{C}} \subseteq \frac{1}{2}\mathcal{N}[i]$ . Inverting 2 in  $2\mathcal{N}_{\mathbb{C}} \subseteq \mathcal{N}[i] \subseteq \mathcal{N}_{\mathbb{C}}$  gives  $\mathcal{N}_{\mathbb{C}}[1/2] = \mathcal{N}[i][1/2] = \mathcal{N}[1/2][i]$ .

The root of unity  $\zeta = (-1 + i\sqrt{3})/2$  satisfies  $\zeta^2 + \zeta + 1 = 0$ , so  $\zeta \in \mathcal{N}_{\mathbb{C}}$ ; its real coordinate  $-1/2$  is not in  $\mathcal{N}$  by Lemma 15.7, so  $\zeta \notin \mathcal{N}[i]$ . Finally  $1/2 \in \frac{1}{2}\mathcal{N}[i]$  is not in  $\mathcal{N}_{\mathbb{C}}$ , again by the reciprocal lemma, since 2 is a nonunit of  $\mathbf{Oz}[i]$ . The proof is unchanged in  $F_{\kappa}^{03}[i]$ , which contains  $\zeta$ ; there  $\zeta$  represents a nonzero class of  $\mathcal{N}_{\kappa, \mathbb{C}}/\mathcal{N}_{\kappa}[i]$ , and this quotient of  $\mathcal{N}_{\kappa}$ -modules is annihilated by 2 because  $2\mathcal{N}_{\kappa, \mathbb{C}} \subseteq \mathcal{N}_{\kappa}[i]$ .  $\square$

**Theorem 15.19** (Dense Gaussian normalization and its constant slice [03]). *The ring  $\mathcal{N}_{\mathbb{C}}$  is a proper dense subring of  $\mathbf{No}[i]$  for the surreal-modulus topology, its conductor  $(\mathbf{Oz}[i] : \mathcal{N}_{\mathbb{C}})$  is zero, and it cannot be generated over  $\mathbf{Oz}[i]$  by a set. Moreover, with  $\mathcal{A}_{\mathbb{C}, \mathbb{C}} = \mathbb{C} + \Pi_{\mathbb{C}}$ ,*

$$\mathcal{N}_{\mathbb{C}} \cap \mathcal{A}_{\mathbb{C}, \mathbb{C}} = \overline{\mathbb{Z}} + \Pi_{\mathbb{C}} = \mathcal{A}_{\overline{\mathbb{Z}}, \mathbb{C}}, \quad \mathcal{N}_{\mathbb{C}} \cap \mathbb{C} = \overline{\mathbb{Z}}. \quad (15.16)$$

For  $\mathcal{N}_{\kappa, \mathbb{C}}$  the analogous properness, density, slice (with  $\Pi_{\kappa}^{03} + i\Pi_{\kappa}^{03}$  in place of  $\Pi_{\mathbb{C}}$ ) and zero-conductor statements hold, and fewer than  $\mu$  algebra generators over  $A_{\kappa}^{03}[i]$  do not suffice.



*Proof.* Let  $z = x + iy \in \mathbf{No}[i]$  and  $\epsilon > 0$ . Choose a positive real integral unit  $u < \epsilon/2$  of  $\mathcal{N}$  (Theorem 15.9) and put

$$n = \lfloor x/u \rfloor, \quad m = \lfloor y/u \rfloor, \quad w = u(n + im).$$

The floors lie in  $\mathbf{Oz}$  and  $u \in \mathcal{N} \subseteq \mathcal{N}_{\mathbb{C}}$ , so  $w \in \mathcal{N}_{\mathbb{C}}$ . Each coordinate of  $z - w$  lies in  $[0, u)$ , hence  $|z - w| < \sqrt{2}u < \epsilon$ . This proves density. The element  $1/2$  is excluded by Lemma 15.7, which proves properness.

For  $0 \neq a \in \mathbf{Oz}[i]$  choose such a  $u$  with  $0 < u < 1/|a|$ . Then  $au \in \mathcal{N}_{\mathbb{C}}$  and  $0 < |au| < 1$ , so  $au \notin \mathbf{Oz}[i]$  by Lemma 15.8; thus the conductor is zero. If  $\mathcal{N}_{\mathbb{C}} = \mathbf{Oz}[i][S]$  for a set  $S$ , Theorem 15.1 would give a nonzero element of the conductor; for  $\mathcal{N}_{\kappa, \mathbb{C}}$  and fewer than  $\mu$  generators, Theorem 15.3 does the same. The two absorption theorems also show directly that every such small-generated algebra has a uniform modulus separation bound, as in Theorem 15.11.

Finally, let  $z = c + P$  with  $c \in \mathbb{C}$  and  $P \in \Pi_{\mathbb{C}}$ , integral over  $\mathbf{Oz}[i]$ . All terms of its monic equation lie in  $\mathcal{A}_{\mathbb{C}, \mathbb{C}}$ , and applying the ring homomorphism  $\text{ct} : \mathcal{A}_{\mathbb{C}, \mathbb{C}} \rightarrow \mathbb{C}$  shows that  $c$  is integral over  $\mathbb{Z}[i]$ ; since  $\mathbb{Z}[i]$  is integral over  $\mathbb{Z}$ ,  $c \in \overline{\mathbb{Z}}$ . Conversely such a  $c$  is integral over the base, and  $P$  already belongs to it. Restricting to constants gives the second equality. The same arguments work in  $F_{\kappa}^{03}[i]$ , using the real integer part  $A_{\kappa}^{03}$  of  $F_{\kappa}^{03}$  for the floors.  $\square$

The original integer-part rings have many finite quotients:  $\mathbf{Oz}/n\mathbf{Oz} \cong \mathbb{Z}/n\mathbb{Z}$  for ordinary  $n \geq 1$  by (3.2), and likewise  $A_{\kappa}^{03}/nA_{\kappa}^{03} \cong \mathbb{Z}/n\mathbb{Z}$ , because the purely infinite ideal is divisible by every nonzero ordinary integer (compare Theorem 6.2). Their normalizations behave oppositely.

**Theorem 15.20** (No nonzero finite ring targets [03]). *None of  $\mathcal{N}$ ,  $\mathcal{N}_{\kappa}$ ,  $\mathcal{N}_{\mathbb{C}}$ ,  $\mathcal{N}_{\kappa, \mathbb{C}}$  admits a unital homomorphism to a nonzero finite ring. In particular none of them has a proper ideal of finite index.*

*Proof.* Suppose such a homomorphism exists. Its image is a nonzero finite commutative ring, even if the target is not commutative, and a maximal ideal of the image gives a unital homomorphism onto a finite field  $\mathbb{F}_q$ . The ordinary monic polynomial  $T^q - T - 1$  is negative at 0 and positive at 2, so it has a real root  $\alpha \in (0, 2)$ . This  $\alpha$  is a real algebraic integer and belongs to all four normalizations by Theorems 15.14 and 15.19. Its image  $\bar{\alpha}$  would satisfy  $\bar{\alpha}^q - \bar{\alpha} - 1 = 0$  in  $\mathbb{F}_q$ , whereas every element of  $\mathbb{F}_q$  satisfies  $\bar{\alpha}^q - \bar{\alpha} = 0$ ; so  $1 = 0$  in  $\mathbb{F}_q$ , which is impossible. A proper ideal of finite index would give a unital homomorphism onto a nonzero finite ring.  $\square$

The proof uses only the real algebraic integer constants. Accordingly this finite-quotient obstruction is a general consequence of containing those constants, not a phenomenon specific to surreal supports. It says nothing about infinite set-sized homomorphic images of the normalizations, and 03 classifies none (03). Manuscript 19 classifies the domain images and factors every set-sized image through  $\mathcal{N}/\Pi\mathcal{N}$  (Theorem 15.46 and Corollary 15.45).

## 15.6 Set-valued valuations cannot contain the omnific ring

A *valuation* of a field  $F$  (possibly a proper class, such as  $\mathbf{No}$  or  $\mathbf{No}[i]$ ) is a map  $v : F^{\times} \rightarrow \Gamma$  to an ordered abelian group with  $v(xy) = v(x) + v(y)$  and  $v(x + y) \geq \min(v(x), v(y))$  whenever  $x + y \neq 0$ ; its valuation ring consists of 0 and the elements of nonnegative value. No compatibility with the ordering of  $F$  is assumed. The value group is *set-sized* if  $\Gamma$  is a set.

**Lemma 15.21** (No nonzero set-valued ordered quotient of  $(\mathbf{No}, +)$  [03]). *Every order-preserving additive class map  $w : (\mathbf{No}, +) \rightarrow \Gamma$  into a set-sized ordered abelian group is zero.*

*Proof.* Suppose  $w(x) \neq 0$  for some  $x$ . Replacing  $x$  by  $-x$  if necessary,  $w(x) > 0$ ; then  $x \neq 0$ , and  $x < 0$  would give  $w(x) \leq w(0) = 0$ , so  $h := x > 0$  and  $w(h) > 0$ . Choose a cardinal  $\lambda > |\Gamma|$  and consider the set of surreals  $\alpha h$  (surreal products) for ordinals  $\alpha < \lambda$ . If  $\alpha < \beta$ , then  $\beta \geq \alpha + 1$  in the surreal order, the ordinal successor being the surreal sum with 1. Hence  $\beta h - \alpha h \geq h$  and, by additivity and monotonicity,

$$w(\beta h) - w(\alpha h) = w(\beta h - \alpha h) \geq w(h) > 0.$$

So the  $\lambda$  elements  $\alpha h$  have pairwise distinct images in  $\Gamma$ , contradicting  $\lambda > |\Gamma|$ . Only a set of ordinals was used.  $\square$

**Theorem 15.22** (Full-class valuation obstruction [03]). *Let  $v : \mathbf{No}^\times \rightarrow \Gamma$  be a valuation with set-sized value group. If  $v(a) \geq 0$  for every  $0 \neq a \in \mathbf{Oz}$ , then  $v$  is trivial. Likewise, a valuation of  $\mathbf{No}[i]$  with set-sized value group whose valuation ring contains  $\mathbf{Oz}[i]$  is trivial.*

*Proof.* Define  $w(g) = v(\omega^g)$  for  $g \in \mathbf{No}$ . The monomial law  $\omega^{g+g'} = \omega^g \omega^{g'}$  makes  $w$  additive. If  $g > 0$  then  $\omega^g \in \Pi \subseteq \mathbf{Oz}$ , so  $w(g) \geq 0$ ; hence  $g \leq g'$  implies  $w(g') - w(g) = w(g' - g) \geq 0$ , and  $w$  is order preserving. By Lemma 15.21,  $w = 0$ .

For  $x \neq 0$ , clear the set  $\{x, x^{-1}\}$  by Proposition 3.4(i): there is a monomial  $d = \omega^h$  with  $dx, dx^{-1} \in \Pi \subseteq \mathbf{Oz}$ . Since  $v(d) = w(h) = 0$ ,

$$v(x) = v(dx) \geq 0, \quad -v(x) = v(x^{-1}) = v(dx^{-1}) \geq 0,$$

so  $v(x) = 0$ . The complex proof uses the same real monomials and the complex clause of Proposition 3.4(i), with  $dx, dx^{-1} \in \Pi_{\mathbb{C}} \subseteq \mathbf{Oz}[i]$ .  $\square$

*Remark 15.23* (Why the size restriction is essential [03]). The theorem does not say that  $\mathbf{No}$  has no interesting valuations. Its normal-form valuation  $x \mapsto -\max \text{supp } x$  takes values in the proper-class group  $(\mathbf{No}, +)$  and is negative on positive-exponent monomials, so it does not satisfy the containment hypothesis anyway. Nor is the theorem a consequence of complete integral closure alone: in the lexicographically ordered group  $\mathbb{Z} \oplus \mathbb{Z}$ , the values  $v(d) = (1, 0)$  and  $v(x) = (0, -1)$  would give  $v(dx^n) = (1, -n) > 0$  for every ordinary  $n$  although  $v(x) < 0$ . Almost integrality by itself cannot eliminate valuations of higher rank.

For ordinary set-sized domains the integral closure is characterized by valuation overrings with set-sized value groups [36, Lemmas 10.50.3 and 10.50.11]. That valuative criterion of integrality cannot be imported to a class ring while retaining only set-sized value groups:  $1/2$  is not integral over  $\mathbf{Oz}$ , yet Theorem 15.22 rules out every nontrivial valuation of that restricted kind whose ring contains  $\mathbf{Oz}$ . No assertion is made about the existence or classification of *proper-class-valued* valuation overrings separating nonintegral surreal elements.

## 15.7 An equation invisible to every set-sized image

Manuscript 08 writes  $B_m(t)$  for the binomial series; we write  $\mathbf{b}_m(t)$ .

**Lemma 15.24** (The binomial identity [08]). *For an ordinary integer  $m \geq 2$  the formal power series*

$$\mathbf{b}_m(t) = \sum_{n \geq 0} \binom{1/m}{n} t^n \in \mathbb{Q}[[t]]$$

satisfies  $\mathbf{b}_m(t)^m = 1 + t$ . For every  $\gamma \in \mathbf{No}_{>0}$  the substitution  $t = \omega^{-\gamma}$  is a valid Hahn substitution, and  $\mathbf{b}_m(\omega^{-\gamma})^m = 1 + \omega^{-\gamma}$  in  $\mathbf{No}$ . The same holds for  $t = X^{-\gamma}$  with  $\gamma > 0$  in any ordered abelian group, in particular in  $F_\kappa^{03}$  (this last sentence is 03's use of the binomial expansion in its Hahn fields, not 08's).

*Proof.* Write  $b_n = \binom{1/m}{n}$ ; then  $b_0 = 1$  and  $(n+1)b_{n+1} = (1/m - n)b_n$ . The coefficient of  $t^n$  in  $(1+t)\mathbf{b}'_m$  is  $(n+1)b_{n+1} + nb_n = b_n/m$ , so  $(1+t)\mathbf{b}'_m = \frac{1}{m}\mathbf{b}_m$ . Put  $c = \mathbf{b}_m^m = \sum_n c_n t^n$ . Then  $(1+t)c' = m\mathbf{b}_m^{m-1}(1+t)\mathbf{b}'_m = c$  and  $c_0 = 1$ . Comparing coefficients gives  $(n+1)c_{n+1} = (1-n)c_n$  for  $n \geq 0$ , so  $c_1 = 1$ ,  $c_2 = 0$  and hence  $c_n = 0$  for all  $n \geq 2$ . Thus  $c = 1 + t$ .

After substitution the support lies in  $\{-n\gamma : n \geq 0\}$ , which is reverse well ordered. In any fixed finite product each exponent  $-n\gamma$  receives finitely many contributions, because an ordinary integer  $n \geq 0$  has only finitely many ordered decompositions into a fixed number of nonnegative integers. Hence  $\sum_n c_n t^n \mapsto \sum_n c_n \omega^{-n\gamma}$  is a ring homomorphism  $\mathbb{Q}[[t]] \rightarrow \mathbf{No}$ , and the formal identity remains a Hahn identity.  $\square$

**Theorem 15.25** (Universal small-target failure of root detection [08]). *Fix  $\gamma \in \mathbf{No}_{>0}$  and an ordinary integer  $m \geq 2$ . The equation*

$$T^m = \omega^\gamma + 1 \tag{15.17}$$

*has no root in  $\mathbf{Oz}[i]$ , and therefore none in  $\mathbf{Oz}$ . Nevertheless, for every unital ring homomorphism  $\varphi : \mathbf{Oz} \rightarrow R$  with  $R$  set-sized, the image equation  $T^m = \varphi(\omega^\gamma) + 1_R$  has the solution  $T = 1_R$ . The same holds for unital homomorphisms with domain  $\mathbf{Oz}[i]$ .*

*Proof.* Put  $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma}) \in \mathbf{No}$ . By Lemma 15.24,  $r^m = \omega^\gamma(1 + \omega^{-\gamma}) = \omega^\gamma + 1$ . For every ordinary complex  $m$ th root of unity  $\zeta$ , the element  $\zeta r \in \mathbf{No}[i]$  is a root. These are  $m$  distinct roots (as  $r \neq 0$ ), so they exhaust the roots of the degree- $m$  polynomial  $T^m - \omega^\gamma - 1$  in the field  $\mathbf{No}[i]$ ; no algebraic-closedness argument is needed. The complex normal form of each root begins

$$\zeta r = \zeta \omega^{\gamma/m} + \frac{\zeta}{m} \omega^{\gamma/m - \gamma} + \sum_{n \geq 2} \zeta \binom{1/m}{n} \omega^{\gamma/m - n\gamma}. \tag{15.18}$$

The exponent  $\gamma/m - \gamma$  is strictly negative because  $m \geq 2$ , its coefficient  $\zeta/m$  is nonzero, and it differs from every other exponent  $\gamma/m - n\gamma$  ( $n \neq 1$ ) of the expansion. So this negative-exponent term cannot cancel, and no root lies in  $\mathbf{Oz}[i] = \mathbb{Z}[i] \oplus \Pi_{\mathbb{C}}$ , whose elements have nonnegative support.

On the other hand, by Theorem 5.1 every unital homomorphism  $\varphi$  from  $\mathbf{Oz}$  to a set-sized ring kills  $\Pi$ , so  $\varphi(\omega^\gamma) = 0$  and the image equation is  $T^m = 1_R$ , solved by  $1_R$ . For a unital homomorphism with domain  $\mathbf{Oz}[i]$ , the universal theorem with  $D = \mathbb{Z}[i]$  and  $k = \mathbb{C}$  (see Corollary 5.3 for the description of these maps) again gives  $\varphi(\omega^\gamma) = 0$ .  $\square$

The nonexistence of a root in  $\mathbf{Oz}[i]$  is also a case of 16's image-gap theorem (Remark 16.37(c)). This is not merely a failure of a local-global test by congruences modulo ordinary primes. Even allowing *every set-sized ring* as a target does not reveal the missing negative tail in (15.18). The obstruction belongs to the full normal-form arithmetic and survives in the proper-class ring itself (08).

**Corollary 15.26** ([08; 03]). *The ring  $\mathbf{Oz}$  is not integrally closed in its fraction field  $\mathbf{No}$ .*

*Proof.* The positive square root of  $\omega + 1$  lies in  $\mathbf{No} = \text{Frac}(\mathbf{Oz})$  (Proposition 3.4(ii)). It is integral over  $\mathbf{Oz}$ , being a root of the monic polynomial  $T^2 - (\omega + 1)$ , but it is not in  $\mathbf{Oz}$  by Theorem 15.25 with  $m = 2$  and  $\gamma = 1$ .  $\square$

The same corollary is contained in 03's [Theorem 15.12](#), with the witness  $\sqrt{2}$ , and manuscript 05 of the sibling report [\[47\]](#) proves it with the same witness  $\sqrt{2}$ . This elementary corollary is included to clarify the distinction between an integer part and an integral closure; it is not offered as a new factorization theorem (08).

*Remark 15.27* (Explicit elements of  $\mathcal{N} \setminus \mathbf{Oz}$  [merge]). Let  $r$  be the positive root of  $T^m - (\omega^\gamma + 1)$  in  $\mathbf{No}$ , for  $\gamma > 0$  and  $m \geq 2$ ; it is the element  $r = \omega^{\gamma/m} \mathbf{b}_m(\omega^{-\gamma})$  of the proof of [Theorem 15.25](#). Then

$$r \in \mathcal{N} \setminus \mathbf{Oz}, \quad \zeta r \in \mathcal{N}_{\mathbb{C}} \setminus \mathbf{Oz}[i] \quad (\zeta^m = 1).$$

Indeed  $r$  is a root of a monic polynomial with coefficients in  $\mathbf{Oz}$ , so it lies in  $\mathcal{N}$  by definition, and  $\zeta r$  lies in  $\mathcal{N}_{\mathbb{C}}$  for the same reason; neither lies in the respective integer ring by [Theorem 15.25](#). Thus 08's equation exhibits explicit elements of the normalization whose absence from  $\mathbf{Oz}$  is invisible to every set-sized image: under every unital homomorphism from  $\mathbf{Oz}$  to a set-sized ring the defining equation of  $r$  becomes  $T^m = 1$ , which already has the root 1. Equivalently, if  $\psi$  is a unital homomorphism from  $\mathcal{N}$  (or from  $\mathbf{Oz}[r]$ ) to a set-sized ring, then its restriction to  $\mathbf{Oz}$  kills  $\omega^\gamma$  by [Theorem 5.1](#), so  $\psi(r)^m = 1$ . Unlike  $\sqrt{2}$ , the elements  $r$  are not constants: by [\(15.18\)](#) they have infinite leading term  $\omega^{\gamma/m}$  and a nonzero negative tail. Nothing is claimed in this remark about the set-sized images of  $\mathcal{N}$  themselves; they are the subject of [Section 15.8](#), where [Remark 15.47](#) sharpens  $\psi(r)^m = 1$  for domain targets.

## 15.8 The arithmetic fibre of the normalization

This subsection is manuscript 19's. It studies the normalizations  $\mathcal{N}$  and  $\mathcal{N}_{\mathbb{C}}$  (19's  $N_{\mathbb{R}}$  and  $N_{\mathbb{C}}$ ) modulo the ideals that  $\Pi$  generates in them, and it answers part of [Question 18.11](#). We write  $\mathcal{N}_{\bullet}$  for  $\mathcal{N}_{\mathbb{R}} = \mathcal{N}$  or  $\mathcal{N}_{\mathbb{C}}$ ,  $\Pi\mathcal{N}_{\bullet}$  for the extended ideal of finite sums of products (19's  $K_{\mathbb{R}}, K_{\mathbb{C}}$ ), and

$$\mathbf{B}_{\bullet} = \mathcal{N}_{\bullet}/\Pi\mathcal{N}_{\bullet}, \quad \mathbf{B}_{\bullet}^{\mathbb{Q}} = \mathbf{B}_{\bullet} \otimes_{\mathbb{Z}} \mathbb{Q}$$

(19's  $B_{\epsilon}, E_{\epsilon}$ ; 03 had used  $B$  for the normalization itself, here  $\mathcal{N}$ , and the letters  $\mathcal{E}$  and  $\mathbf{E}$  are taken).  $\text{Idem}(R)$  is the Boolean algebra of idempotents of a commutative ring  $R$  (19's  $\mathcal{E}$ ),  $\overline{\mathbb{Z}}$  is the ring of all complex algebraic integers,  $\overline{\mathbb{Q}}$  its fraction field and  $\overline{\mathbb{F}_p}$  an algebraic closure of  $\mathbb{F}_p$ . Quotients by class ideals are read as in [Section 2.1](#). 19 did not see 03's text ([Section 1.3](#)), and four of its results are 03's, printed above and credited here: the constant slices  $\mathcal{N} \cap \mathbb{R} = \overline{\mathbb{Z}}_{\mathbb{R}}$  and  $\mathcal{N}_{\mathbb{C}} \cap \mathbb{C} = \overline{\mathbb{Z}}$  ([Theorems 15.14](#) and [15.19](#)), the absence of nonzero finite images ([Theorem 15.20](#); a different proof is [Theorem 15.46](#) below), the strict inclusion  $\mathcal{N}[i] \subsetneq \mathcal{N}_{\mathbb{C}}$  ([Proposition 15.18](#); 19's witness is  $w_{\pm}$  of [Proposition 15.43](#) in place of  $\zeta_3$ ), and infinitesimal integral units ([Theorem 15.9](#); 19's unit of [Proposition 15.34](#) has a different residue). Its reproof of the universal theorem for  $\mathbf{Oz}$  is credited in [Section 5.1](#).

**Lemma 15.28** (Contraction [19, 20]).  $\Pi\mathcal{N}_{\bullet} \cap \mathbf{Oz} = \Pi$ , and  $\Pi\mathcal{N}_{\bullet}$  is divisible by every nonzero ordinary integer. Hence  $\mathbf{B}_{\bullet}$  is a nonzero ring containing  $\mathbb{Z}$ , torsion free over  $\mathbb{Z}$ , and integral over  $\mathbb{Z}$ .

*Proof.* Suppose  $0 \neq n \in \mathbb{Z}$  lies in  $\Pi\mathcal{N}_{\bullet}$ , say  $n = \sum_{j \leq r} a_j b_j$  with  $a_j \in \Pi$  and  $b_j \in \mathcal{N}_{\bullet}$ . The ring  $C = \mathbf{Oz}[b_1, \dots, b_r]$  is generated as an  $\mathbf{Oz}$ -module by finitely many elements  $c_1 = 1, c_2, \dots, c_s$ , and  $nc_l = \sum_m h_{lm} c_m$  with all  $h_{lm} \in \Pi$  because  $n \in \Pi C$ . The adjugate of  $nI - (h_{lm})$  shows that  $\det(nI - (h_{lm}))$  kills every  $c_l$ , in particular  $c_1 = 1$ ; expanding,  $n^s + j_{s-1}n^{s-1} + \dots + j_0 = 0$  with  $j_\nu \in \Pi$ , and  $\text{ct}$  gives  $n^s = 0$ , which is absurd. If  $a \in \Pi\mathcal{N}_{\bullet} \cap \mathbf{Oz}$ , then  $\text{ct}(a) = a - (a - \text{ct}(a)) \in \Pi\mathcal{N}_{\bullet}$ , so  $\text{ct}(a) = 0$ . Divisibility:  $\sum_j a_j b_j / n = \sum_j (a_j / n) b_j$ . So  $nx \in \Pi\mathcal{N}_{\bullet}$  with  $x \in \mathcal{N}_{\bullet}$  forces  $x \in \Pi\mathcal{N}_{\bullet}$ , which is torsion freeness. Reducing a monic equation of  $x \in \mathcal{N}_{\bullet}$  over  $\mathbf{Oz}$  modulo  $\Pi\mathcal{N}_{\bullet}$  replaces each coefficient  $a$  by  $\text{ct}(a) \in \mathbb{Z}$ .  $\square$

A large normal overring need not preserve the contraction of an ideal; the determinant argument prevents  $\Pi\mathcal{N}_\bullet$  from containing 1, although it contains elements that differ from 1 by an infinitesimal (19).

**Theorem 15.29** (Radicality [19, 20]). *If  $x \in \mathcal{N}_\bullet$  and  $x^m \in \Pi\mathcal{N}_\bullet$  for some ordinary  $m \geq 1$ , then  $x \in \Pi\mathcal{N}_\bullet$ . Hence  $\mathcal{B}_\bullet$  and  $\mathcal{B}_\bullet^\mathbb{Q}$  are reduced,  $\mathcal{B}_\bullet \rightarrow \mathcal{B}_\bullet^\mathbb{Q}$  is injective, and  $\mathcal{B}_\bullet^\mathbb{Q}$  is algebraic over  $\mathbb{Q}$ .*

*Proof.* Write  $x^m = \sum_j a_j b_j$  with  $a_j \in \Pi$ ,  $b_j \in \mathcal{N}_\bullet$ . By Proposition 3.3(i) there is  $a > 0$  with every  $a_j \in \omega^a \Pi$ ; put  $d = \omega^{a/m}$ , so that  $a_j/d^m \in \Pi$  for all  $j$ . Then  $(x/d)^m = \sum_j (a_j/d^m) b_j \in \mathcal{N}_\bullet$ , so  $x/d$  is a root of a monic polynomial over  $\mathcal{N}_\bullet$  and lies in  $\mathcal{N}_\bullet$ ; hence  $x = d(x/d) \in \Pi\mathcal{N}_\bullet$ . Reducedness passes to the localization, injectivity is torsion freeness, and a rational multiple of an element integral over  $\mathbb{Z}$  is algebraic.  $\square$

**Theorem 15.30** (Rational saturation [19, 20]).  *$\mathcal{B}_\bullet$  is integrally closed in  $\mathcal{B}_\bullet^\mathbb{Q}$ , and  $\mathcal{B}_\bullet = \{y \in \mathcal{B}_\bullet^\mathbb{Q} : y \text{ is integral over } \mathbb{Z}\}$ .*

*Proof.* Represent  $y$  by  $x/n$  with  $x \in \mathcal{N}_\bullet$  and  $0 \neq n \in \mathbb{Z}$ , and let  $y$  satisfy a monic equation over  $\mathcal{B}_\bullet$ ; lift it to a monic  $f \in \mathcal{N}_\bullet[X]$ . The kernel of  $\mathcal{N}_\bullet \otimes \mathbb{Q} \rightarrow \mathcal{B}_\bullet^\mathbb{Q}$  is the localization of  $\Pi\mathcal{N}_\bullet$ , which is  $\Pi\mathcal{N}_\bullet$  itself by divisibility (Lemma 15.28); so  $c = f(x/n) \in \Pi\mathcal{N}_\bullet$ , and  $x/n$  is a root of the monic  $f(X) - c \in \mathcal{N}_\bullet[X]$ , hence lies in  $\mathcal{N}_\bullet$ . Elements of  $\mathcal{B}_\bullet$  are integral over  $\mathbb{Z}$  by Lemma 15.28, and an element of  $\mathcal{B}_\bullet^\mathbb{Q}$  integral over  $\mathbb{Z}$  is integral over  $\mathcal{B}_\bullet$ .  $\square$

**Proposition 15.31** (The extended ideal [19]).  *$\Pi\mathcal{N}_\bullet = (\Pi\mathcal{N}_\bullet)^2 = \bigcup_{\delta > 0} \omega^\delta \mathcal{N}_\bullet$ , and  $\Pi\mathcal{N}_\bullet$  is not generated by any set of its elements.*

*Proof.* For  $x = \sum_j a_j b_j$  choose a positive monomial  $d$  with  $a_j/d \in \Pi$  (Proposition 3.3(i)); then  $x/d \in \Pi\mathcal{N}_\bullet$  and  $x = d \cdot (x/d)$ . If a set  $G$  generated  $\Pi\mathcal{N}_\bullet$ , then for each  $g \in G$  the class  $D_g = \{\delta > 0 : g \in \omega^\delta \mathcal{N}_\bullet\}$  is nonempty and downward closed, and its elements of least birthday form a set; a positive  $\delta$  below the union of these sets lies in every  $D_g$ , so  $G \subseteq \omega^\delta \mathcal{N}_\bullet$ . Then  $\omega^{\delta/2} \in \Pi$  would lie in  $\omega^\delta \mathcal{N}_\bullet$ , so  $\omega^{-\delta/2}$  and hence  $\omega^{-\delta}$  would lie in  $\mathcal{N}_\bullet$ , and  $1 \in \omega^\delta \mathcal{N}_\bullet \subseteq \Pi\mathcal{N}_\bullet$ , contrary to Lemma 15.28.  $\square$

19 chooses one expression per generator with global choice; the least-birthday selection makes the argument choice free. This parallels 03's statement that  $\mathcal{N}$  is not set-generated as an **Oz**-algebra (Theorem 15.12), which concerns a different object.

**Theorem 15.32** (Real-rooted lifting [19]). *Let  $P(X) = X^d + \sum_{k < d} a_k X^k \in \mathcal{N}[X]$  with  $d \geq 1$ , let  $M \in \mathbf{No}$  with  $M \geq 1$  and  $M \geq |a_k|$  for all  $k$ , put  $\eta_d = \frac{1}{4}(\frac{3}{4})^{d-1}$  and  $C_d = d(d+1)^{d-1}$ , and let  $T \in \Pi$  be a positive monomial with  $T > C_d M / \eta_d$ . Then*

$$Q(X) = P(X) + \prod_{j=1}^d (X - jT) - X^d$$

*satisfies  $Q - P \in \Pi[X]$  and has exactly one root in each interval  $((j - \frac{1}{4})T, (j + \frac{1}{4})T)$ ,  $1 \leq j \leq d$ ; these are  $d$  distinct, positive, infinite roots, and they lie in  $\mathcal{N}$ .*

*Proof.* Put  $R_T = \prod_j (X - jT)$ . At an endpoint  $x = (j \pm \frac{1}{4})T$  the factor  $x - jT$  has modulus  $T/4$  and the others at least  $3T/4$ , so  $|R_T(x)| \geq \eta_d T^d$ , and  $R_T$  has opposite signs at the two endpoints of the  $j$ th interval. Since  $|x| \leq (d+1)T$  and  $T > 1$ ,  $|P(x) - x^d| \leq \sum_{k < d} |a_k| |x|^k \leq C_d M T^{d-1} < \eta_d T^d$ , so  $Q$  has the sign of  $R_T$  at every endpoint, and the intermediate value property of polynomials

over the real closed field  $\mathbf{No}$  gives a root in each of the  $d$  disjoint intervals; there are no others. The roots exceed  $\frac{3}{4}T$ , which is infinite; they are integral over  $\mathcal{N}$ , hence in  $\mathcal{N}$ . The coefficients of  $R_T - X^d$  are integer multiples of positive powers of  $T$ .  $\square$

**Corollary 15.33** (Absolute integral closedness [19]).  $\mathbf{B}_{\mathbb{R}}$ ,  $\mathcal{N}_{\mathbb{C}}$  and  $\mathbf{B}_{\mathbb{C}}$  are absolutely integrally closed: every monic polynomial over them splits into linear factors.  $\mathcal{N}$  itself is not, since  $X^2 + 1$  has no root in  $\mathcal{N} \subseteq \mathbf{No}$ .

*Proof.* Lift a monic polynomial over  $\mathbf{B}_{\mathbb{R}}$  to  $\mathcal{N}$ , apply Theorem 15.32 and reduce the factorization; the change of coefficients lies in  $\Pi[X]$ . Over  $\mathcal{N}_{\mathbb{C}}$  the roots in the algebraically closed field  $\mathbf{No}[i]$  are integral, hence in  $\mathcal{N}_{\mathbb{C}}$ ; reduce for  $\mathbf{B}_{\mathbb{C}}$ .  $\square$

The lifted roots need not stay distinct modulo  $\Pi\mathcal{N}$ : the lifting preserves the factorization, not the separation of residues (19).

**Proposition 15.34** (An infinitesimal unit whose residue squares to  $-1$  [19]). *Let  $T \in \Pi$  be a positive monomial and  $v_T = (T - \sqrt{T^2 - 4})/2 = 2/(T + \sqrt{T^2 - 4})$ . Then  $v_T^2 - Tv_T + 1 = 0$ ,  $v_T$  is a positive infinitesimal unit of  $\mathcal{N}$  with inverse  $T - v_T$ ,  $v_T \notin \mathbf{Oz}$ , and its image  $\mathbf{v} \in \mathbf{B}_{\mathbb{R}}$  satisfies  $\mathbf{v}^2 = -1$ . For  $T = \omega$ ,*

$$v_{\omega} = \omega^{-1} + \omega^{-3} + 2\omega^{-5} + 5\omega^{-7} + 14\omega^{-9} + \cdots = \sum_{n \geq 0} \frac{1}{n+1} \binom{2n}{n} \omega^{-(2n+1)}.$$

Consequently  $\mathcal{N} \rightarrow \mathbf{B}_{\mathbb{R}}$  is not compatible with any ordered-ring quotient, and  $\Pi\mathcal{N}$  is not convex in  $\mathbf{No}$ .

*Proof.* The equation is the quadratic formula; it is monic over  $\mathbf{Oz}$  and says  $v_T(T - v_T) = 1$ . The second expression gives  $0 < v_T < 2/T$ , and  $\mathbf{Oz}$  has no positive element below 1. Modulo  $\Pi\mathcal{N}$ ,  $v_T^2 + 1 = Tv_T \in \Pi\mathcal{N}$ . Writing  $v_{\omega} = \omega^{-1}C(\omega^{-2})$  turns the equation into  $C = 1 + zC^2$ , whose solution in  $\mathbb{Q}[[z]]$  has the Catalan coefficients; the substitution  $z = \omega^{-2}$  is a Hahn substitution as in Lemma 15.24, and the sum is a strong sum, not a limit of partial sums. Finally  $0 < 1 < Tv_T = 1 + v_T^2$  with  $Tv_T \in \Pi\mathcal{N}$  and  $1 \notin \Pi\mathcal{N}$ .  $\square$

03's unit  $u_H = \sqrt{H^2 + 1} - H$  with  $H \in \Pi$  (Theorem 15.9) has a residue  $\bar{u}_H$  with  $\bar{u}_H^2 = 1$ ; 19's residue squares to  $-1$ . Normalization adds a real infinitesimal whose arithmetic residue behaves like an imaginary unit; no imaginary surreal is inserted into the real field, and it is the quotient that loses the order (19).

**Lemma 15.35** (A coefficient copy of  $\overline{\mathbb{Q}}$  [19, 20]). *There is an embedding  $\overline{\mathbb{Q}} \hookrightarrow \mathbf{B}_{\bullet}^{\mathbb{Q}}$  whose restriction to  $\overline{\mathbb{Z}}$  lands in  $\mathbf{B}_{\bullet}$ .*

*Proof.* Real algebraic integers lie in  $\mathcal{N}$ . A nonzero one,  $a$ , satisfies  $ab = m$  for a real algebraic integer  $b$  and some  $0 \neq m \in \mathbb{Z}$ ; as  $\mathbb{Z} \subseteq \mathbf{B}_{\bullet}$  is torsion free, the image of  $a$  is a unit of  $\mathbf{B}_{\bullet}^{\mathbb{Q}}$ . So the field  $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$  of real algebraic numbers embeds. In the real case  $X \mapsto \mathbf{v}$  defines  $k_0[X]/(X^2 + 1) \rightarrow \mathbf{B}_{\mathbb{R}}^{\mathbb{Q}}$ , injective because its source is the field  $k_0(i) \cong \overline{\mathbb{Q}}$ ; in the complex case use  $i \in \mathcal{N}_{\mathbb{C}}$ . Algebraic integers map into  $\mathbf{B}_{\bullet}$  by Theorem 15.30.  $\square$

In the real case the embedding depends on the choice of  $\mathbf{v}$ ; it is an embedding of algebraic numbers into the quotient rationalization, not an embedding of  $\mathbb{C}$  into  $\mathbf{No}$ , whose existence is not asserted (19).



**Lemma 15.36** (Spectral idempotents [19, 20]). *Let  $E$  be a reduced algebraic  $\mathbb{Q}$ -algebra containing a unital copy of  $\overline{\mathbb{Q}}$ . Every  $x \in E$  is  $\sum_{j \leq r} \lambda_j e_j$  with distinct  $\lambda_j \in \overline{\mathbb{Q}}$  and orthogonal idempotents  $e_j$  with  $\sum_j e_j = 1$ , and finitely many elements have such expressions on one common partition.*

*Proof.* Let  $p$  be the squarefree part of a nonzero rational polynomial killing  $x$ ;  $p(x)$  is nilpotent, hence 0. Factor  $p = c \prod_j (X - \lambda_j)$  over  $\overline{\mathbb{Q}}$  and put  $e_j = L_j(x)$  with the Lagrange polynomials  $L_j = \prod_{k \neq j} (X - \lambda_k) / (\lambda_j - \lambda_k)$ ; modulo  $p$ ,  $L_j^2 \equiv L_j$ ,  $L_j L_k \equiv 0$ ,  $\sum_j L_j \equiv 1$  and  $X \equiv \sum_j \lambda_j L_j$ . For several elements multiply their partitions.  $\square$

**Definition 15.37** (Finite-partition Boolean powers [19, 20]). For a Boolean algebra  $\mathfrak{E}$  (possibly a proper class) and a commutative ring  $R$ , an element of  $R[\mathfrak{E}]$  is a finite partition  $1 = e_1 \vee \dots \vee e_r$  into disjoint elements of  $\mathfrak{E}$  with values  $d_1, \dots, d_r \in R$ ; values on zero parts are ignored, two such objects are equal when their values agree on every nonzero intersection of parts, and the ring operations are computed on a common refinement. For idempotents,  $e \wedge f = ef$ ,  $\neg e = 1 - e$  and  $e \vee f = e + f - ef$ . No infinite sums occur.

**Theorem 15.38** (The fibre is a Boolean power of the algebraic integers [19, 20]). *With the coefficient embedding of Lemma 15.35,  $\text{Idem}(\mathbf{B}_\bullet^\mathbb{Q}) = \text{Idem}(\mathbf{B}_\bullet)$ , and evaluation  $(\lambda_j, e_j)_j \mapsto \sum_j \lambda_j e_j$  gives isomorphisms*

$$\mathbf{B}_\bullet^\mathbb{Q} \cong \overline{\mathbb{Q}}[\text{Idem}(\mathbf{B}_\bullet)], \quad \mathbf{B}_\bullet \cong \overline{\mathbb{Z}}[\text{Idem}(\mathbf{B}_\bullet)].$$

*Every finite family of elements of  $\mathbf{B}_\bullet$  lies in a subring isomorphic to  $\overline{\mathbb{Z}}^r$ , embedded by an orthogonal partition of 1.*

*Proof.* Idempotents are roots of  $X^2 - X$ , so Theorem 15.30 puts those of  $\mathbf{B}_\bullet^\mathbb{Q}$  in  $\mathbf{B}_\bullet$ . Evaluation is onto by Lemma 15.36 and Theorem 15.29. On a nonzero common part  $e$  of two representations, equality gives  $(\lambda - \mu)e = 0$ , and a nonzero  $\lambda - \mu \in \overline{\mathbb{Q}}$  is a unit; so evaluation is injective. If  $x = \sum_j \lambda_j e_j \in \mathbf{B}_\bullet$  with every  $e_j \neq 0$  and  $f \in \mathbb{Z}[X]$  is monic with  $f(x) = 0$ , then  $f(\lambda_j)e_j = 0$ , so  $f(\lambda_j) = 0$  and  $\lambda_j \in \overline{\mathbb{Z}}$ ; conversely a finite sum of commuting integral elements is integral, and lies in  $\mathbf{B}_\bullet$  by Theorem 15.30. On a common partition with nonzero parts, the map from  $\overline{\mathbb{Z}}^r$  is injective.  $\square$

The Boolean description gives Corollary 15.33 again: refine the coefficients of a monic polynomial to one partition and split it over  $\overline{\mathbb{Z}}$  on each part.  $\mathbf{B}_\bullet$  is neither one finite product of copies of  $\overline{\mathbb{Z}}$  nor an unrestricted product; the finite-partition restriction is part of the theorem, and 19 does not determine  $\text{Idem}(\mathbf{B}_\mathbb{R})$  (19). 20 shows that it is nontrivial, atomless and a proper class with explicit independent families (Theorems 15.72 and 15.73); its isomorphism type is not determined.

**Theorem 15.39** (Finitely generated ideals of the fibre [19]). *Every finitely generated ideal of  $\overline{\mathbb{Z}}$  is principal (a classical fact [11]). Every finitely generated ideal  $I$  of  $\mathbf{B}_\bullet$  is principal,  $I = g\mathbf{B}_\bullet$ , and there is an idempotent  $e$  with  $\text{Ann}(g) = (1 - e)\mathbf{B}_\bullet$  and  $I \cong e\mathbf{B}_\bullet$ ; so  $I$  is projective and  $\mathbf{B}_\bullet$  is semihereditary.*

*Proof.* For  $I = (a_1, \dots, a_r)\overline{\mathbb{Z}} \neq 0$  choose a number field  $L$  containing the  $a_j$ , put  $I_0 = (a_1, \dots, a_r)O_L$ , and use finiteness of the class group [25, Theorem 4.4] to get  $h \geq 1$  and  $c \in O_L$  with  $I_0^h = cO_L$ . With  $t \in \overline{\mathbb{Z}}$ ,  $t^h = c$ , every  $x \in I_0$  has  $(x/t)^h = x^h/c \in O_L$ , so  $I \subseteq t\overline{\mathbb{Z}}$ ; and  $(I/t)^h = \overline{\mathbb{Z}}$  forces  $I/t = \overline{\mathbb{Z}}$ . For  $\mathbf{B}_\bullet$  put the generators on a common partition  $e_1, \dots, e_q$ , write  $x_i = \sum_j a_{ij}e_j$ , choose generators  $g_j$  of  $(a_{1j}, \dots, a_{rj})\overline{\mathbb{Z}}$  and put  $g = \sum_j g_j e_j$  and  $e = \sum_{g_j \neq 0} e_j$ . The coefficient identities on each part show  $I = g\mathbf{B}_\bullet$ ; since  $\overline{\mathbb{Z}}$  is a domain,  $\text{Ann}(g) = (1 - e)\mathbf{B}_\bullet$ , and multiplication by  $g$  identifies  $e\mathbf{B}_\bullet$  with  $I$ .  $\square$

No Bézout property of  $\mathbf{Oz}$  or of  $\mathcal{N}_\bullet$  is claimed; the theorem concerns their arithmetic fibre (19).

**Proposition 15.40** (Nilpotents modulo a prime [19]). *For every ordinary prime  $p$  and  $n \geq 2$ ,  $\mathbf{B}_\bullet/p\mathbf{B}_\bullet$  has an element of nilpotence index exactly  $n$ ; it is nonzero and not reduced.*

*Proof.* Take  $\alpha = p^{1/n} \in \overline{\mathbb{Z}} \subseteq \mathbf{B}_\bullet$ . If  $\alpha^k \in p\mathbf{B}_\bullet$  with  $1 \leq k < n$ , then  $\alpha^k/p \in \mathbf{B}_\bullet$  is integral over  $\mathbb{Z}$ , and so is its  $n$ th power  $p^{k-n} \in \mathbb{Q} \setminus \mathbb{Z}$ , which is absurd; the case  $k = 0$  is  $1 \notin p\mathbf{B}_\bullet$ .  $\square$

**Lemma 15.41** (Rational comparison [19, 20]).  $\mathcal{N}_\mathbb{C} \otimes \mathbb{Q} = (\mathcal{N} \otimes \mathbb{Q})[i]$  and  $\mathbf{B}_\mathbb{C}^\mathbb{Q} \cong \mathbf{B}_\mathbb{R}^\mathbb{Q}[X]/(X^2 + 1)$ ; the map  $\mathbf{B}_\mathbb{R} \rightarrow \mathbf{B}_\mathbb{C}$  is injective, that is,  $\Pi\mathcal{N}_\mathbb{C} \cap \mathcal{N} = \Pi\mathcal{N}$ .

*Proof.* For  $z \in \mathcal{N}_\mathbb{C}$ ,  $z + \bar{z}$  and  $(z - \bar{z})/i$  are real and integral, so lie in  $\mathcal{N}$ , and  $2z$  is in  $\mathcal{N}[i]$  (as in Proposition 15.18); with  $\mathcal{N}[i] \subseteq \mathcal{N}_\mathbb{C}$  this gives the first equality, and  $1, i$  are independent over  $\mathcal{N} \otimes \mathbb{Q}$ . Since  $\Pi\mathcal{N}$  is rationally divisible,  $\Pi\mathcal{N}_\mathbb{C} \otimes \mathbb{Q} = \Pi\mathcal{N} \otimes \mathbb{Q} + i(\Pi\mathcal{N} \otimes \mathbb{Q})$ , and the quotient is  $\mathbf{B}_\mathbb{R}^\mathbb{Q}[i]$ , into which  $\mathbf{B}_\mathbb{R}^\mathbb{Q}$  injects.  $\square$

With  $\mathbf{v}$  from Proposition 15.34,  $\mathbf{B}_\mathbb{R}^\mathbb{Q}$  contains  $\frac{1}{2}$  and the unit  $\mathbf{v}$ , and the Chinese remainder map

$$\Psi : \mathbf{B}_\mathbb{R}^\mathbb{Q}[i] \rightarrow \mathbf{B}_\mathbb{R}^\mathbb{Q} \times \mathbf{B}_\mathbb{R}^\mathbb{Q}, \quad a + ib \mapsto (a + \mathbf{v}b, a - \mathbf{v}b), \quad (15.19)$$

is an isomorphism with inverse  $(r, s) \mapsto \frac{1}{2}(r + s) + \frac{i}{2\mathbf{v}}(r - s)$ .

**Theorem 15.42** (Integral doubling [19, 20]).  $\Psi$  restricts to an isomorphism  $\mathbf{B}_\mathbb{C} \cong \mathbf{B}_\mathbb{R} \times \mathbf{B}_\mathbb{R}$  of  $\mathbf{B}_\mathbb{R}$ -algebras, with  $\mathbf{B}_\mathbb{R}$  acting diagonally and complex conjugation exchanging the factors. Hence  $\mathbf{B}_\mathbb{C}/m\mathbf{B}_\mathbb{C} \cong (\mathbf{B}_\mathbb{R}/m\mathbf{B}_\mathbb{R})^2$  for every ordinary  $m \geq 2$ .

*Proof.* By Theorem 15.30,  $\mathbf{B}_\mathbb{C}$  is the set of elements of  $\mathbf{B}_\mathbb{C}^\mathbb{Q}$  integral over  $\mathbb{Z}$ . A pair is integral over the diagonal  $\mathbb{Z}$  exactly when both coordinates are (multiply two monic annihilators), and the integral elements of  $\mathbf{B}_\mathbb{R}^\mathbb{Q}$  are  $\mathbf{B}_\mathbb{R}$ . The rest is read off from (15.19).  $\square$

Factoring  $X^2 + 1$  over  $\mathbf{B}_\mathbb{R}$  and applying the Chinese remainder theorem there would not prove this, because 2 need not be a unit; rationalization followed by the characterization of integral elements is what gives the integral statement (19).

**Proposition 15.43** (The integral idempotents [19]). *With  $v = v_T$  as in Proposition 15.34,  $w_\pm = (1 \mp iv)/2$  lie in  $\mathcal{N}_\mathbb{C}$ ; their residues are complementary nonzero idempotents of  $\mathbf{B}_\mathbb{C}$ , sent by  $\Psi$  to  $(1, 0)$  and  $(0, 1)$ . They are not in  $\mathcal{N}[i]$ .*

*Proof.* From  $v^2 + 1 = Tv$ :  $w_\pm^2 - w_\pm = -Tv/4$ ,  $w_+ + w_- = 1$  and  $w_+w_- = Tv/4$ , where  $Tv/4 = (T/4)v \in \Pi\mathcal{N}$ . So each  $w_\pm$  is integral over  $\mathcal{N}$ , hence over  $\mathbf{Oz}$ , and the residues are complementary idempotents;  $\Psi$  gives the coordinate vectors, using  $\mathbf{v}^2 = -1$ . The real coordinate of  $w_\pm$  is  $\frac{1}{2} \notin \mathcal{N}$  (Lemma 15.7).  $\square$

So  $w_\pm$  is a second witness for  $\mathcal{N}[i] \subsetneq \mathcal{N}_\mathbb{C}$  (03's is  $\zeta_3$ ): normalization adds integral elements written with a denominator 2, certified by  $Tv/4 \in \Pi\mathcal{N}$  (19).

**Corollary 15.44** (Characteristic two, trace and norm [19]). *Modulo 2 the residues of  $w_\pm$  still split  $\mathbf{B}_\mathbb{C}/2\mathbf{B}_\mathbb{C} \cong (\mathbf{B}_\mathbb{R}/2\mathbf{B}_\mathbb{R})^2$  into two nonzero factors, although  $i \mapsto (\mathbf{v}, -\mathbf{v})$  has equal coordinates modulo 2. Field trace and norm induce  $\text{Tr}, \text{Nm} : \mathbf{B}_\mathbb{C} \rightarrow \mathbf{B}_\mathbb{R}$ , equal to  $(r, s) \mapsto r + s$  and  $(r, s) \mapsto rs$ ; the fixed ring of conjugation is the diagonal  $\mathbf{B}_\mathbb{R}$ , and the trace is onto, also modulo 2.*

*Proof.* The factors are nonzero by [Proposition 15.40](#). For  $z \in \mathcal{N}_{\mathbb{C}}$ ,  $z + \bar{z}$  and  $z\bar{z}$  lie in  $\mathcal{N}$ , and changing  $z$  modulo  $\Pi\mathcal{N}_{\mathbb{C}}$  changes them by elements of  $\Pi\mathcal{N}_{\mathbb{C}} \cap \mathcal{N} = \Pi\mathcal{N}$  ([Lemma 15.41](#)). Conjugation swaps the coordinates, and  $(r, 0)$  has trace  $r$ .  $\square$

The mod-2 splitting is a product of two nonreduced rings, not of fields. The norm-one units of  $\mathbb{B}_{\mathbb{C}}$  form a group isomorphic to  $\mathbb{B}_{\mathbb{R}}^{\times}$  via  $r \mapsto (r, r^{-1})$ , on which conjugation acts by inversion; it is not asserted that such units lift to norm-one units of  $\mathcal{N}_{\mathbb{C}}$  (19).

**Corollary 15.45** (Set-sized images factor through the fibre [19, 20]). *Every unital homomorphism from  $\mathcal{N}_{\bullet}$  to a set-sized ring kills  $\Pi\mathcal{N}_{\bullet}$  and factors uniquely through  $\mathbb{B}_{\bullet}$ .*

*Proof.* Its restriction to  $\mathbf{Oz}$  kills  $\Pi$  ([Theorem 5.1](#)), hence it kills every finite sum of products in  $\Pi\mathcal{N}_{\bullet}$ .  $\square$

$\mathbb{B}_{\bullet}$  is not asserted to be a set: this is a factorization property for each set-sized target, not a universal object in a category of set-sized rings (19). In fact  $\mathbb{B}_{\bullet}$  is a proper class, and no set-indexed family of set-sized targets is jointly faithful on it (20; [Theorems 15.73](#) and [15.79](#) and [Corollaries 15.77](#) and [15.81](#)).

**Theorem 15.46** (Domain images [19]). *Let  $\varphi : \mathcal{N}_{\bullet} \rightarrow S$  be a unital homomorphism to a set-sized ring whose image  $R$  is a nonzero domain. Then  $R \cong \mathbb{Z}$  if  $\text{char } R = 0$  and  $R \cong \overline{\mathbb{F}}_p$  if  $\text{char } R = p > 0$ . In particular every such image is countable, no nonzero finite ring is a unital image of  $\mathcal{N}_{\bullet}$  ([Theorem 15.20](#)), and no set-sized formally real field is a unital target.*

*Proof.* By [Corollary 15.45](#) and [Theorem 15.38](#),  $\varphi$  factors through  $\mathbb{Z}[\text{Idem}(\mathbb{B}_{\bullet})]$ . In a domain each idempotent goes to 0 or 1, so exactly one part of each finite partition goes to 1, and  $R$  is the image of  $\mathbb{Z}$ . In characteristic 0 this map is injective: a nonzero  $\alpha \in \mathbb{Z}$  divides a nonzero ordinary integer. In characteristic  $p$ ,  $R$  is a domain algebraic over  $\mathbb{F}_p$ , hence a field, and it is algebraically closed because monic polynomials over  $\mathbb{Z}$  split; so  $R \cong \overline{\mathbb{F}}_p$ . A nonzero finite image, which is commutative, has a finite field as a quotient, and that would be a finite domain image; in a formally real field the image of  $v$  (or of  $i$ ) would square to  $-1$ .  $\square$

This answers, for domain targets, the clause of [Question 18.11](#) asking for the infinite set-sized images of  $\mathcal{N}$ , and it contains 03's absence of finite images with a different proof (03's uses a root of  $T^q - T - 1$ ). Images with zero divisors are not classified beyond the factorization through  $\mathbb{B}_{\bullet}$ , and it is not claimed that all set-sized images are countable (19).

*Remark 15.47* (Images of 08's invisible roots [merge]). Let  $r = \omega^{\gamma/m} b_m(\omega^{-\gamma}) \in \mathcal{N}$  be the root of  $T^m = \omega^{\gamma} + 1$  of [Remark 15.27](#). Its residue in  $\mathbb{B}_{\mathbb{R}}$  satisfies  $\bar{r}^m = 1$ , so under [Theorem 15.38](#) it is  $\sum_j \zeta_j e_j$  with  $\zeta_j^m = 1$  on every nonzero part; under every unital map from  $\mathcal{N}$  to a set-sized domain,  $r$  goes to an  $m$ th root of unity of  $\mathbb{Z}$  or  $\overline{\mathbb{F}}_p$ . This sharpens “ $\psi(r)^m = 1$ ” of [Remark 15.27](#) for domain targets.

**Lemma 15.48** (Class ultrafilters [19, 20]). *In Gödel–Bernays class theory with global choice, every Boolean algebra whose elements are sets, even a proper class, has an ultrafilter (a class) containing any prescribed nonzero element.*

*Proof.* Enumerate the algebra along a global well-order, by an ordinal or by the class of ordinals, starting from the prescribed element  $e$ . At stage  $\alpha$  add  $b_{\alpha}$  if the decisions so far together with  $b_{\alpha}$  have the finite intersection property, and otherwise  $\neg b_{\alpha}$ . If both failed, finite meets  $c, d$  of earlier

decisions would have  $c \wedge b_\alpha = 0 = d \wedge \neg b_\alpha$ , hence  $c \wedge d = 0$ , which is impossible; limit stages preserve the finite intersection property. The history at each stage is a set, so ordinary transfinite recursion with the global well-order as a class parameter defines the class of decisions, which generates an ultrafilter containing  $e$ .  $\square$

No class form of Zorn's lemma and no measurable cardinal is used; one part of each finite partition is chosen, without any countable completeness (19).

**Theorem 15.49** (Detection by maps to  $\overline{\mathbb{Z}}$  [19, 20]). *Assume global choice. For an ultrafilter  $U$  on  $\text{Idem}(\mathbf{B}_\bullet)$ ,  $\sum_j \lambda_j e_j \mapsto \lambda_{j_0}$  with  $e_{j_0} \in U$  is a surjective unital homomorphism  $\overline{\mathbb{Z}}[\text{Idem}(\mathbf{B}_\bullet)] \rightarrow \overline{\mathbb{Z}}$ . Every  $x \in \mathcal{N}_\bullet \setminus \Pi \mathcal{N}_\bullet$  is sent to a nonzero element by a surjective class homomorphism  $\mathcal{N}_\bullet \rightarrow \overline{\mathbb{Z}}$ . So  $\Pi \mathcal{N}_\bullet$  is the common kernel of the unital maps from  $\mathcal{N}_\bullet$  to countable domains, and already of those to  $\overline{\mathbb{Z}}$ , in the pointwise sense (no set or class of all such maps is formed).*

*Proof.* Exactly one part of a finite partition lies in  $U$ , and refinement preserves the selected value; so evaluation is a well-defined surjective homomorphism. If  $x \notin \Pi \mathcal{N}_\bullet$ , its image has a nonzero part  $e$  with a nonzero coefficient  $\alpha$ ; take  $U \ni e$  by Lemma 15.48, and  $x \mapsto \alpha$ . Conversely maps to countable domains kill  $\Pi \mathcal{N}_\bullet$  by Corollary 15.45.  $\square$

**Corollary 15.50** (Field targets [19]). *Assume global choice. A set-sized field  $L$  receives a unital homomorphism from  $\mathcal{N}_\bullet$  if and only if  $L$  contains an algebraic closure of its prime field.*

*Proof.* A map forces a copy of  $\overline{\mathbb{Z}}$ , hence of  $\overline{\mathbb{Q}}$ , or a copy of  $\overline{\mathbb{F}_p}$  (Theorem 15.46). Conversely compose an evaluation  $\mathcal{N}_\bullet \rightarrow \overline{\mathbb{Z}}$  with  $\overline{\mathbb{Z}} \subseteq \overline{\mathbb{Q}} \subseteq L$ , or with  $\overline{\mathbb{Z}} \rightarrow \overline{\mathbb{Z}/\mathfrak{p}} \cong \overline{\mathbb{F}_p}$  for a prime  $\mathfrak{p}$  of  $\overline{\mathbb{Z}}$  over  $p$  (lying over), followed by an inclusion.  $\square$

**Theorem 15.51** (A finite-axiom form [19]). *Let  $F$  be a real closed field (a set or a class) containing the real algebraic numbers and  $A \subseteq F$  a unital subring with a ring retraction  $c : A \rightarrow \mathbb{Z}$  and kernel  $J$  such that (i)  $J$  is divisible by every nonzero integer; (ii) for all  $a_1, \dots, a_r \in J$  and  $m \geq 1$  there is  $0 \neq d \in J$  with  $a_j/d^m \in A$  for all  $j$ ; (iii) the positive elements of  $J$  are cofinal in  $F_{>0}$ . Let  $N_{\mathbb{R}}, N_{\mathbb{C}}$  be the integral closures of  $A$  in  $F$  and  $F(i)$  and  $\mathbf{B}_\bullet = N_\bullet/JN_\bullet$ . Then Lemma 15.28 and Theorems 15.29, 15.30, 15.38 and 15.42 and Corollary 15.33 hold with  $(\mathbf{Oz}, \Pi, \text{ct})$  replaced by  $(A, J, c)$ , and Theorem 15.32 holds with  $T$  any sufficiently large positive element of  $J$ , the roots being positive and distinct but not necessarily infinite.*

*Proof.* The contraction uses only the retraction and finite integrality, torsion freeness uses (i), radicality uses exactly (ii), and saturation uses (i) and normality. By (iii) there is  $T \in J$  with  $T > 2$  for Proposition 15.34 and as large as needed for the lifting;  $v_T$  is a unit with residue square  $-1$  whether or not it is infinitesimal. Real algebraic integers are integral over  $\mathbb{Z} \subseteq A$ , so the coefficient embedding, the spectral idempotents, the Boolean description and the doubling apply unchanged.  $\square$

The theorem does not give that  $v_T$  is infinitesimal, nor that  $JN_\bullet$  has no set of generators (that needs the set-family divisibility of the full class), nor Corollary 15.45, which needs arbitrarily large sets of exponent scales below a fixed exponent, a cardinal condition absent from (i)–(iii). In a fixed real-exponent Hahn integer part the support  $\{1/n : n \geq 1\}$  blocks the monomial proof of (ii); this is a failure of the argument, not a counterexample to any conclusion (19). (Status (batch 32): the cyclic workspace  $\mathbb{Z} + \omega^a \mathbb{R}[\omega^a]$  lies outside the theorem, since its Laurent field is not real closed and (ii) fails there, and in it the conclusion of Theorem 15.29 does fail: the extended ideal is not radical,

[Theorem 15.86](#) and [Remark 15.119](#); for real closed fixed workspaces such as the Hahn field over  $\mathbb{Q}a$  the question stays open, [Question 18.21](#), while the bounded-denominator Puiseux ring satisfies (i)–(iii), [Remark 15.102](#).) For the reduced fibre  $\mathcal{C}(G)$  of the full Hahn field over a nonzero set group (defined in (15.20) below), 20 obtains the Boolean-power form without (ii) and without deciding radicality ([Proposition 15.76](#)).

*Remark 15.52* (Candidates for idempotents [merge]).  $\text{Idem}(\mathbb{B}_{\mathbb{R}})$  is nontrivial exactly when some  $x \in \mathcal{N}$  has  $x^2 - x \in \Pi\mathcal{N}$  while  $x, 1 - x \notin \Pi\mathcal{N}$ . For  $s = \sqrt{1 + \omega}$  and  $x = (1 + s)/2$  one has  $x^2 - x = \omega/4$ , so  $x \in \mathcal{N}$  (a root of  $X^2 - X - \omega/4$ ) and its residue is an idempotent. By [Proposition 15.31](#),  $x \in \Pi\mathcal{N}$  exactly when  $x/\omega^\delta \in \mathcal{N}$  for some  $\delta > 0$ ; so this residue is nontrivial exactly when neither  $x/\omega^\delta$  nor  $(1 - x)/\omega^\delta$  lies in  $\mathcal{N}$  for any  $\delta > 0$ , an instance of the open membership question. Likewise 03's unit gives the idempotent  $(1 + \bar{u}_H)/2$  of  $\mathbb{B}_{\mathbb{R}}$  ([Theorem 15.30](#)), nontrivial exactly when  $u_H \mp 1 \notin \Pi\mathcal{N}$ . Neither case is decided by this remark ([Question 18.17](#)).

This reduction to the membership question is superseded: 20 proves nontriviality, and much more, by support projection and two specializations, without a membership test ([Section 15.9](#)), and both candidates are nontrivial, the first being the second for  $H = \omega^{1/2}$  ([Proposition 15.68](#), a merge result).

## 15.9 Proper-class Boolean branching of the fibre

This subsection is manuscript 20's ([Section 1.3](#)). It answers [Question 18.17\(a\)](#):  $\text{Idem}(\mathbb{B}_{\mathbb{R}})$  is nontrivial and atomless, it is a proper class with no set-sized order-dense subset, it contains an explicit independent family indexed by all ordinals, and one idempotent splits every nonzero element of any prescribed set-sized subring of  $\mathbb{B}_{\mathbb{R}}$ . The tool is a support projection that makes integrality and radical membership local to the exponent group of an element, followed by two specializations of a quadratic branch algebra at one fresh scale. The splitting theorems use neither the Boolean-power theorem nor the universal theorem; only the statements about maps at the end of the subsection do (20).

20 read 19 as a preceding draft (it was not in the repository at 20's pin) and reproves what it needs of it, with the same constructions. Those results are printed once, above, and carry the tag [19, 20] where 20 proves the whole statement: the contraction, radicality and saturation ([Lemma 15.28](#) and [Theorems 15.29](#) and [15.30](#)), the coefficient copy, spectral idempotents and Boolean power ([Lemmas 15.35](#) and [15.36](#), [Definition 15.37](#), and [Theorem 15.38](#)), the rational comparison and integral doubling ([Lemma 15.41](#) and [Theorem 15.42](#)), the factorization of set-sized images, class ultrafilters and detection ([Corollary 15.45](#), [Lemma 15.48](#), and [Theorem 15.49](#)). 20 also proves most of [Proposition 15.34](#) (all but the Catalan expansion), of [Proposition 15.43](#) (all but  $w_{\pm} \notin \mathcal{N}[i]$ ) and of [Theorem 15.46](#) (the classification, not the consequences for finite and formally real targets), and the part of [Corollary 15.44](#) saying that  $w_{\pm}$  separate the two nonzero factors modulo 2. 20's set bounds and common monomial division are [Lemma 3.1](#) and [Proposition 3.3\(i\)](#), its constant term and  $\Pi^2 = \Pi$  are [Lemma 2.2](#) and [Proposition 3.3\(ii\)](#), and its proof of the universal theorem for  $\mathbf{Oz}$  is credited at [Theorem 5.1](#). Its dominant-scale lemma is 16's Laurent-block theorem, which was not in the repository at 20's pin; it is credited in [Lemma 15.62](#).

**Notation.** For a set-sized additive subgroup  $G \subseteq \mathbf{No}$  let  $K_G = \mathbb{R}((X^G)) \subseteq \mathbf{No}$  be the real Hahn field with supports in  $G$ , as in [Section 16.6](#) with  $k = \mathbb{R}$  (20's  $F_G$ ), and put

$$\begin{aligned} \mathcal{A}(G) &= \mathbf{Oz} \cap K_G, & \Pi(G) &= \Pi \cap K_G, \\ \mathcal{N}(G) &= \text{IC}_{K_G}(\mathcal{A}(G)), & \mathcal{C}(G) &= \mathcal{N}(G) / \sqrt{\Pi(G)\mathcal{N}(G)} \end{aligned} \tag{15.20}$$



(20's  $A_G, J_G, N_G, C_G$ ). Thus  $\mathcal{A}(G) = \mathcal{A}_{\mathbb{Z}, \mathbb{R}}(G)$  and  $\Pi(G) = \Pi_{\mathbb{R}}(G)$  in the notation of [Section 2.3](#); the parentheses keep  $\mathcal{N}(G)$  apart from 03's  $\mathcal{N}_\kappa$  and from  $\mathcal{N}_\bullet$ , and the letter  $C$  is 03's (for  $\mathbf{Oz}[S]$  and for its complex normalization). 20's  $N_{\mathbb{R}}, N_{\mathbb{C}}, \mathcal{J}, K_{\mathbb{R}}, K_{\mathbb{C}}, B_{\mathbb{R}}, B_{\mathbb{C}}, E, \mathcal{E}$  are  $\mathcal{N}, \mathcal{N}_{\mathbb{C}}, \Pi, \Pi\mathcal{N}, \Pi\mathcal{N}_{\mathbb{C}}, \mathcal{B}_{\mathbb{R}}, \mathcal{B}_{\mathbb{C}}, \mathcal{B}_{\mathbb{R}}^{\mathbb{Q}}, \text{Idem}(\mathcal{B}_{\mathbb{R}})$ ; its unit  $u$  and residue  $v$  are  $v_T$  and  $v$ . Its coefficient projection  $\pi_G$  is written  $\text{pr}_G$  ( $\pi_b$  is a detecting map in [Theorem 5.12](#)); its dominant exponent  $a$  and group  $H = G \oplus \mathbb{Z}a$  are 16's fresh exponent  $b$  and  $\tilde{G} = G \oplus \mathbb{Z}b$  ( $H_a$  is the scale subgroup (2.4), and  $H$  is 03's parameter in  $u_H$ ); its monomial  $T = \omega^a$  is  $t = \omega^b$ , with  $T$  the formal Laurent variable as in [Theorem 16.31](#); its branches  $s_0, s_1, s_r$  are  $\varrho_0, \varrho_1, \varrho_r$  ( $s$  is 16's centre and the root of [Remark 15.52](#)); its specializations  $\theta_{\pm}$  are  $\sigma_{\pm}$  ( $\theta = \kappa^{\aleph_0}$  in 03's construction); its splitting element  $w(T)$  is  $w_t$  ( $w_{\pm}$  are 19's idempotents). Square roots are the positive ones in  $\mathbf{No}$ , and extended ideals consist of finite sums of products.

**Lemma 15.53** (Monic radical certificates [20]). *Let  $D \subseteq C$  be an integral extension of commutative rings (sets or classes) and  $I \subseteq D$  an ideal. An element  $x \in C$  lies in  $\sqrt{IC}$  if and only if  $x^m + c_{m-1}x^{m-1} + \dots + c_0 = 0$  for some ordinary  $m \geq 1$  and  $c_0, \dots, c_{m-1} \in I$ .*

*Proof.* Let  $x^q = \sum_{l \leq r} a_l b_l$  with  $a_l \in I$  and  $b_l \in C$ . The ring  $C_0 = D[x, b_1, \dots, b_r]$  is generated as a  $D$ -module by finitely many elements  $c_1 = 1, c_2, \dots, c_s$ , and  $x^q c_\nu = \sum_l a_l (b_l c_\nu) \in IC_0 = \sum_\mu I c_\mu$ , so  $x^q c_\nu = \sum_\mu h_{\nu\mu} c_\mu$  with all  $h_{\nu\mu} \in I$ . The adjugate of  $x^q \mathbf{1}_s - (h_{\nu\mu})$  shows that  $P(x^q) = 0$  for the characteristic polynomial  $P$  of  $(h_{\nu\mu})$ , using  $c_1 = 1$ ; the coefficient of  $X^{s-j}$  in  $P$  ( $j \geq 1$ ) is a sum of products of  $j$  entries, hence in  $I$ . Then  $P(X^q)$  is a monic polynomial of degree  $qs$  killing  $x$  whose nonleading coefficients lie in  $I$  (or are 0). Conversely the displayed relation gives  $x^m = -\sum_j c_j x^j \in IC$ . Only finitely many elements occur, so the argument applies to class rings.  $\square$

**Corollary 15.54** (Exact membership certificate [20]). *For  $x \in \mathcal{N}_\bullet$ :  $x \in \Pi\mathcal{N}_\bullet$  if and only if  $x^m + \sum_{j < m} c_j x^j = 0$  for some  $m \geq 1$  and  $c_j \in \Pi$ .*

*Proof.*  $\mathcal{N}_\bullet$  is integral over  $\mathbf{Oz}$ ; apply [Lemma 15.53](#) with  $(D, C, I) = (\mathbf{Oz}, \mathcal{N}_\bullet, \Pi)$  and the radicality of  $\Pi\mathcal{N}_\bullet$  ([Theorem 15.29](#)).  $\square$

The certificate lets one project the coefficients of a vanishing relation without projecting a homomorphism or an ideal expression (20).

**Lemma 15.55** (Support projection [20]). *For a set-sized subgroup  $G \subseteq \mathbf{No}$  put  $\text{pr}_G(\sum_g r_g \omega^g) = \sum_{g \in G} r_g \omega^g$  on  $\mathbf{No}[i]$  (coefficients in  $\mathbb{C}$ ). Then  $\text{pr}_G$  is additive, fixes  $K_G[i]$  pointwise, and*

$$\text{pr}_G(yx) = \text{pr}_G(y) x \quad (y \in \mathbf{No}[i], x \in K_G[i]). \quad (15.21)$$

Moreover  $\text{pr}_G(\mathbf{Oz}) = \mathcal{A}(G)$  and  $\text{pr}_G(\Pi) = \Pi(G)$ .

*Proof.* A subset of a reverse well-ordered set is reverse well ordered, and additivity and the fixing are coefficientwise. A contribution to an exponent  $h \in G$  of  $yx$  comes from exponents  $g$  of  $y$  and  $s$  of  $x$  with  $g + s = h$ ; since  $s \in G$  and  $G$  is a group,  $g = h - s \in G$ . So exactly the terms of  $y$  kept by  $\text{pr}_G$  contribute to the exponents in  $G$ , and each coefficient of a Hahn product is a finite sum, so no exchange of infinite sums is needed. Selection keeps positive exponents and the constant coefficient, and it fixes  $\mathcal{A}(G)$  and  $\Pi(G)$ .  $\square$

*Remark 15.56* (Not a ring retraction [20]).  $\text{pr}_{\{0\}}(\omega) = \text{pr}_{\{0\}}(\omega^{-1}) = 0$  although  $\text{pr}_{\{0\}}(\omega \cdot \omega^{-1}) = 1$ ; only (15.21), with one factor supported in  $G$ , is used. This is the warning of [Remark 2.3](#) for  $G = \{0\}$ .



**Theorem 15.57** (Exact support descent [20]). *For set-sized subgroups  $G \subseteq G' \subseteq \mathbf{No}$ ,*

$$\mathcal{N} \cap K_G = \mathcal{N}(G), \quad \Pi\mathcal{N} \cap K_G = \sqrt{\Pi(G)\mathcal{N}(G)}, \quad (15.22)$$

$$\mathcal{N}(G') \cap K_G = \mathcal{N}(G), \quad \sqrt{\Pi(G')\mathcal{N}(G')} \cap K_G = \sqrt{\Pi(G)\mathcal{N}(G)}. \quad (15.23)$$

*Proof.* Let  $x \in K_G$  satisfy  $x^m + \sum_{j < m} a_j x^j = 0$  with  $a_j \in \mathbf{Oz}$ . Every  $x^j$  lies in  $K_G$ , so (15.21) gives  $x^m + \sum_{j < m} \text{pr}_G(a_j)x^j = 0$  with coefficients in  $\mathcal{A}(G)$ ; hence  $x \in \mathcal{N}(G)$ . The reverse inclusion is clear, and the same argument with  $a_j \in \mathcal{A}(G')$  gives  $\mathcal{N}(G') \cap K_G = \mathcal{N}(G)$ . If  $x \in \Pi\mathcal{N} \cap K_G$ , project the certificate of Corollary 15.54: its coefficients go to  $\Pi(G)$ ,  $x \in \mathcal{N}(G)$  by the first part, and Lemma 15.53 with  $(\mathcal{A}(G), \mathcal{N}(G), \Pi(G))$  gives  $x \in \sqrt{\Pi(G)\mathcal{N}(G)}$ . Conversely  $\Pi(G)\mathcal{N}(G) \subseteq \Pi\mathcal{N}$ , which is radical. The local radical equality is the same argument with Lemma 15.53 applied to  $(\mathcal{A}(G'), \mathcal{N}(G'), \Pi(G'))$  and then to  $(\mathcal{A}(G), \mathcal{N}(G), \Pi(G))$ .  $\square$

**Corollary 15.58** (Reduced fibres embed [20]). *For  $G \subseteq G'$  inclusion induces injections  $\mathbb{C}(G) \hookrightarrow \mathbb{C}(G') \hookrightarrow \mathbb{B}_{\mathbb{R}}$ . Larger surreal supports cannot annihilate a nonzero reduced-fibre class from a smaller support group.*

*Proof.* These are the contracted-ideal equalities of Theorem 15.57.  $\square$

So an integrality witness using arbitrary surreal scales can be replaced by one using only the exponent group of the element: in 19's membership question, which allows witnesses at larger scales, larger scales never help. Theorem 15.57 is a descent statement, not a decision procedure for  $\mathcal{N}(G)$  (20). The second equality in (15.22) deliberately keeps the radical: it is not asserted that  $\Pi(G)\mathcal{N}(G)$  is radical in a fixed workspace (20). (Status (batch 32): for  $G = \mathbb{Z}a$  it is not radical, so there the radical cannot be removed, Theorem 15.86 and Remark 15.119.) The same projection decides the descent asked in Question 18.11.

**Corollary 15.59** (Descent for 03's fields [merge]). *With the notation of Section 15,*

$$\mathcal{N} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}, \quad \Pi\mathcal{N} \cap F_{\kappa}^{03} = \sqrt{\Pi_{\kappa}^{03}\mathcal{N}_{\kappa}}, \quad \mathcal{N}_{\mathbb{C}} \cap F_{\kappa}^{03}[i] = \mathcal{N}_{\kappa, \mathbb{C}}, \quad \Pi\mathcal{N}_{\mathbb{C}} \cap F_{\kappa}^{03}[i] = \sqrt{\Pi_{\kappa, \mathbb{C}}^{03}\mathcal{N}_{\kappa, \mathbb{C}}},$$

*the radicals being taken in  $\mathcal{N}_{\kappa}$  and  $\mathcal{N}_{\kappa, \mathbb{C}}$ .*

*Proof.* Identify  $F_{\kappa}^{03}$  with its image in  $\mathbf{No}$  (Section 11.1): the series with countable support in the subgroup  $\iota(G_{\kappa})$ . Let  $x \in F_{\kappa}^{03}[i]$  be integral over  $\mathbf{Oz}$ , and let  $G'$  be the subgroup generated by the supports of its real and imaginary parts. It is countable and contained in  $\iota(G_{\kappa})$ , so every element of  $K_{G'}$  has countable support in  $\iota(G_{\kappa})$ :  $K_{G'} \subseteq F_{\kappa}^{03}$  and  $\mathcal{A}(G') \subseteq \mathbf{Oz} \cap F_{\kappa}^{03} = A_{\kappa}^{03}$ . Projecting a monic equation of  $x$  over  $\mathbf{Oz}$  by  $\text{pr}_{G'}$  (Lemma 15.55) gives one over  $\mathcal{A}(G') \subseteq A_{\kappa}^{03}$ , so  $x$  is integral over  $A_{\kappa}^{03}$ . By (15.8) and (15.13) (and the remark after it that  $\mathcal{N}_{\kappa, \mathbb{C}}$  is the integral closure of  $A_{\kappa}^{03}$  in  $F_{\kappa}^{03}[i]$ ) this proves the first and third equalities; the reverse inclusions hold because  $A_{\kappa}^{03} \subseteq \mathbf{Oz}$ . If moreover  $x \in \Pi\mathcal{N}_{\bullet}$ , the certificate of Corollary 15.54 has coefficients in  $\Pi$ , and its projection has coefficients in  $\Pi(G') \subseteq \Pi_{\kappa}^{03}$ ; Lemma 15.53, applied to  $A_{\kappa}^{03}$  and the ring  $\mathcal{N}_{\kappa}$  or  $\mathcal{N}_{\kappa, \mathbb{C}}$ , which is integral over it, puts  $x$  in the stated radical. Conversely  $\Pi_{\kappa}^{03}\mathcal{N}_{\kappa} \subseteq \Pi\mathcal{N}$  and  $\Pi_{\kappa, \mathbb{C}}^{03}\mathcal{N}_{\kappa, \mathbb{C}} \subseteq \Pi\mathcal{N}_{\mathbb{C}}$ , and these are radical.  $\square$

The proof uses only that  $F_{\kappa}^{03}$  contains the full Hahn field over the subgroup generated by the support of each of its elements; it is a descent, not a membership test for  $\mathcal{N}_{\kappa}$ .

**Theorem 15.60** (One-scale rational functions [20]). *For  $a > 0$  and  $T = \omega^a$ ,*

$$\mathcal{N} \cap \mathbb{R}(T) = \overline{\mathbb{Z}}_{\mathbb{R}} + T\mathbb{R}[T], \quad \Pi\mathcal{N} \cap \mathbb{R}(T) = T\mathbb{R}[T].$$

*Proof.* A rational function of  $T$  expands at infinity with support in  $\mathbb{Z}a$ , so  $\mathbb{R}(T) \subseteq K_{\mathbb{Z}a}$ . A reverse well-ordered subset of  $\{na : n \geq 0\}$  is finite, so  $\mathcal{A}(\mathbb{Z}a) = \mathbb{Z} + T\mathbb{R}[T]$  and  $\Pi(\mathbb{Z}a) = T\mathbb{R}[T]$ . If  $f \in \mathbb{R}(T)$  is integral over  $\mathbf{Oz}$ , it is integral over  $\mathbb{Z} + T\mathbb{R}[T] \subseteq \mathbb{R}[T]$  by [Theorem 15.57](#), hence lies in  $\mathbb{R}[T]$ , which is integrally closed (a monic equation for  $p/q$  in lowest terms, multiplied by  $q^m$ , forces  $q \mid p^m$ ). Evaluating the monic equation at  $T = 0$  makes  $f(0)$  a real algebraic integer. Conversely  $f(0) \in \overline{\mathbb{Z}}_{\mathbb{R}} \subseteq \mathcal{N}$  ([Theorem 15.14](#)) and  $f - f(0) \in T\mathbb{R}[T] \subseteq \Pi$ . If also  $f \in \Pi\mathcal{N}$ , project its certificate onto  $\mathbb{Z}a$ : the coefficients lie in  $T\mathbb{R}[T]$ , and evaluation at 0 gives  $f(0)^m = 0$ . Finally  $T\mathbb{R}[T] \subseteq \Pi \subseteq \Pi\mathcal{N}$ .  $\square$

**Example 15.61** ([20]).  $(\omega^2 - 1)^{-1}$  is not integral over  $\mathbf{Oz}$ , although a monic witness over  $\mathbf{Oz}$  could use coefficients at arbitrary surreal scales; 19's  $v_\omega = (\omega - \sqrt{\omega^2 - 4})/2$  ([Proposition 15.34](#)) is integral, and its infinite Laurent tail is algebraic but not rational in  $\omega$ , as [Theorem 15.60](#) requires.

**Lemma 15.62** (Laurent blocks exhaust the enlarged field [16, 20]). *Let  $G \subseteq \mathbf{No}$  be a set-sized subgroup,  $b > G$  fresh ([Lemma 16.30](#)),  $\tilde{G} = G \oplus \mathbb{Z}b$  and  $t = \omega^b$ . The map  $\iota_b$  of [Theorem 16.31](#) is a field isomorphism  $K_G((T^{-1})) \rightarrow K_{\tilde{G}}$  with  $\iota_b(T) = t$ , and*

$$\iota_b^{-1}(\mathcal{A}(\tilde{G})) = \mathcal{A}(G) + TK_G[T], \quad \iota_b^{-1}(\Pi(\tilde{G})) = \Pi(G) + TK_G[T].$$

*In particular a coefficient in  $\Pi(\tilde{G})$  is a polynomial in  $t$  over  $K_G$  whose value at  $T = 0$  lies in  $\Pi(G)$ .*

*Proof.* [Theorem 16.31](#) (16) gives an injective field homomorphism into  $K_{\tilde{G}}$ . It is onto (20): an exponent of  $\tilde{G}$  is uniquely  $g + jb$  with  $g \in G$ ,  $j \in \mathbb{Z}$ ; in a reverse well-ordered support the occurring  $j$  are bounded above, since increasing  $j$  gives larger exponents by freshness, and the part with a fixed  $j$  is  $\omega^{jb}$  times an element of  $K_G$ . The ring statements are [Corollary 16.32](#) with  $D = \mathbb{Z}$ :  $\iota_b(\sum_j c_j T^j) \in \mathbf{Oz}$  exactly when  $c_j = 0$  for  $j < 0$  and  $c_0 \in \mathbf{Oz}$ , and the positive-index blocks are purely infinite, so the constant term is  $\text{ct}(c_0)$ .  $\square$

**Lemma 15.63** (Independent quadratic branches [20]). *Let  $k$  be an ordered field and, in  $k((T^{-1}))$ , let  $\varrho_0 = T(1 - T^{-2})^{1/2}$  and  $\varrho_1 = \frac{T}{2}(1 - 4T^{-2})^{1/2}$  be the binomial branches. Then*

$$k[T, \varrho_0, \varrho_1] \cong k[T, U, V]/(U^2 - T^2 + 1, V^2 - T^2/4 + 1),$$

*a free  $k[T]$ -module with basis  $1, \varrho_0, \varrho_1, \varrho_0\varrho_1$ , and there are  $k$ -algebra homomorphisms*

$$\sigma_{\pm} : k[T, \varrho_0, \varrho_1] \rightarrow k[i] = k[X]/(X^2 + 1), \quad T \mapsto 0, \quad \varrho_0 \mapsto i, \quad \varrho_1 \mapsto \pm i.$$

*Proof.* The binomial series exist in characteristic 0, and  $\varrho_0^2 = d_0 = T^2 - 1$ ,  $\varrho_1^2 = d_1 = T^2/4 - 1$ . At  $T = 1$ ,  $d_0$  has a simple zero and  $d_1 \neq 0$ ; at  $T = 2$  the roles are reversed. So  $d_0$ ,  $d_1$  and  $d_0d_1$  are not squares in  $k(T)$ , since a square has even order at every linear factor. If  $d_1$  were a square  $(x + y\varrho_0)^2$  in  $k(T)(\varrho_0)$ , then  $2xy = 0$ , and  $y = 0$  or  $x = 0$  would make  $d_1$  or  $d_0d_1$  a square in  $k(T)$ . Hence  $[k(T)(\varrho_0, \varrho_1) : k(T)] = 4$ . The presented algebra is free over  $k[T]$  on the four monomials, by reduction with the two monic relations, and their images are linearly independent over  $k(T)$ ; so the map is injective. The substitutions respect both relations, which defines  $\sigma_{\pm}$ .  $\square$

**Remark 15.64** (Specializations, not evaluations [20]).  $\sigma_{\pm}$  are specializations of a finite branch algebra whose presentation has been proved. They are not evaluations of Laurent series at  $T = 0$ , where these have no meaning, nor maps  $K_{\tilde{G}} \rightarrow k[i]$ , and no evaluation map on all surreal numbers is used.

For a fresh  $b$  and  $t = \omega^b$  put, with  $\varrho_0, \varrho_1$  read through  $\iota_b$  (they are the positive roots in  $\mathbf{No}$ , having positive leading coefficients),

$$\varrho_0 = \sqrt{t^2 - 1}, \quad \varrho_1 = \sqrt{t^2/4 - 1}, \quad \mathbf{w}_t = \frac{1 - \varrho_0\varrho_1}{2}, \quad \mathbf{w}_t^2 - \mathbf{w}_t = \frac{t^4 - 5t^2}{16} \in \Pi. \quad (15.24)$$

So  $\mathbf{w}_t \in \mathcal{N}$ , and  $\mathbf{e}_t = [\mathbf{w}_t]$  is an idempotent of  $\mathbf{B}_{\mathbb{R}}$ . The element  $\mathbf{w}_t = -t^2/4 + 9/8 + 9t^{-2}/32 + \dots$  has an infinite Laurent tail, but its certificate is a polynomial in  $t$  with zero constant coefficient (20).

**Theorem 15.65** (One-scale uniform splitting [20]). *Let  $G \subseteq \mathbf{No}$  be a set-sized subgroup,  $b > G$  fresh and  $t = \omega^b$ . For every  $x \in \mathcal{N} \cap K_G$ ,*

$$x\mathbf{w}_t \in \Pi\mathcal{N} \iff x \in \Pi\mathcal{N} \iff x(1 - \mathbf{w}_t) \in \Pi\mathcal{N}.$$

*Equivalently, multiplication by  $\mathbf{e}_t$  and by  $1 - \mathbf{e}_t$  kills no nonzero element of  $\mathbf{C}(G) \subseteq \mathbf{B}_{\mathbb{R}}$ .*

*Proof.* The implications from the middle hold because  $\Pi\mathcal{N}$  is an ideal. Suppose  $z = x\mathbf{w}_t \in \Pi\mathcal{N}$  and take a certificate  $z^m + \sum_{j < m} c_j z^j = 0$ ,  $c_j \in \Pi$  (Corollary 15.54). Since  $z \in K_{\tilde{G}}$ , Lemma 15.55 gives  $z^m + \sum_{j < m} p_j z^j = 0$  with  $p_j = \text{pr}_{\tilde{G}}(c_j) \in \Pi(\tilde{G})$ , and by Lemma 15.62 each  $p_j$  is  $\iota_b$  of a polynomial in  $\Pi(G) + TK_G[T]$ . So the relation holds in  $\iota_b(K_G[T, \varrho_0, \varrho_1])$ , and Lemma 15.63 with  $k = K_G$  lets us apply  $\sigma_+$ . It fixes  $x$  and sends  $\mathbf{w}_t$  to  $(1 - i^2)/2 = 1$ , so  $x^m + \sum_{j < m} p_j(0)x^j = 0$  in  $K_G$ , with  $p_j(0) \in \Pi(G) \subseteq \Pi$ ; hence  $x \in \Pi\mathcal{N}$  by Corollary 15.54. If instead  $x(1 - \mathbf{w}_t) \in \Pi\mathcal{N}$ , use  $\sigma_-$ , which sends  $\mathbf{w}_t$  to  $(1 - i(-i))/2 = 0$  and  $1 - \mathbf{w}_t$  to 1.  $\square$

**Theorem 15.66** (Uniform splitting [20]). *For every set-sized unital subring  $R \subseteq \mathbf{B}_{\mathbb{R}}$  there is an idempotent  $e \in \mathbf{B}_{\mathbb{R}}$  with  $re \neq 0$  and  $r(1 - e) \neq 0$  for every  $0 \neq r \in R$ . Consequently  $R \times R \rightarrow \mathbf{B}_{\mathbb{R}}$ ,  $(r, s) \mapsto re + s(1 - e)$ , is an injective unital homomorphism extending the diagonal copy of  $R$ .*

*Proof.* Represent the elements of  $R$  by elements of  $\mathcal{N}$  (a set of representatives exists, for instance the least-rank ones of Section 2.1), let  $G$  be the group generated by the union of their supports, choose  $b > G$  and put  $e = \mathbf{e}_{\omega^b}$ . Theorem 15.65 gives  $re \neq 0 \neq r(1 - e)$ . The map is a unital homomorphism because  $e(1 - e) = 0$ , and if  $re + s(1 - e) = 0$ , multiplying by  $e$  and by  $1 - e$  gives  $re = 0 = s(1 - e)$ , so  $r = s = 0$ .  $\square$

This splits all nonzero elements of a prescribed set at once, with no exception caused by zero divisors of  $R$  (20).

**Example 15.67** (A nontrivial idempotent at  $t = \omega$  [20]). With  $G = \{0\}$  and  $t = \omega$ , the residue  $[(1 - \sqrt{\omega^2 - 1})\sqrt{\omega^2/4 - 1}]/2$  is neither 0 nor 1: apply Theorem 15.65 to  $x = 1 \notin \Pi\mathcal{N}$ . The identity (15.24) proves idempotence; the two specializations prove the separately needed nontriviality.

One radical suffices, and it can be 03's. This decides the two candidates of Remark 15.52.

**Proposition 15.68** (03's unit splits, and the candidates are nontrivial [merge]). *Let  $G, b, t$  be as in Theorem 15.65, let  $h \in TK_G[T]$  be nonzero with positive leading coefficient, and put  $H = \iota_b(h)$ , a positive element of  $\Pi$ . With 03's unit  $u_H = \sqrt{H^2 + 1} - H$  (Theorem 15.9), the element  $y_H = (1 + u_H)/2$  lies in  $\mathcal{N}$ , its residue  $e_H$  is an idempotent of  $\mathbf{B}_{\mathbb{R}}$ , and for every  $x \in \mathcal{N} \cap K_G$*

$$xy_H \in \Pi\mathcal{N} \iff x \in \Pi\mathcal{N} \iff x(1 - y_H) \in \Pi\mathcal{N}.$$

*In particular, for every  $a > 0$  and every positive  $H \in \Pi$  with  $\text{supp } H \subseteq \mathbb{Z}a$ , the idempotent  $(1 + \bar{u}_H)/2$  of Remark 15.52 is neither 0 nor 1, that is,  $u_H \mp 1 \notin \Pi\mathcal{N}$ ; and  $x = (1 + \sqrt{1 + \omega})/2$  has a residue that is neither 0 nor 1.*

*Proof.* From  $u_H^2 + 2Hu_H - 1 = 0$ ,  $y_H^2 - y_H = (u_H^2 - 1)/4 = -Hu_H/2$ , and  $H/2 \in \Pi$ ,  $u_H \in \mathcal{N}$ ; so  $y_H$  is a root of  $X^2 - X + Hu_H/2 \in \mathcal{N}[X]$ , hence in  $\mathcal{N}$ , and  $e_H$  is idempotent. Put  $\varrho = \iota_b(h(1+h^{-2})^{1/2}) = \sqrt{H^2 + 1}$  (the binomial series in  $h^{-2} \in T^{-2}K_G[[T^{-1}]]$ ). The polynomial  $h^2 + 1$  is not a square in  $K_G(T)$ : a square root would lie in  $K_G[T]$ , and  $(g-h)(g+h) = 1$  would make  $g \pm h$  constant and  $h$  constant. So  $1, \varrho$  are linearly independent over  $K_G(T)$ ,  $K_G[T, \varrho] \cong K_G[T, U]/(U^2 - h^2 - 1)$  is free over  $K_G[T]$  with basis  $1, U$ , and since  $h(0) = 0$  there are  $K_G$ -algebra maps  $\sigma'_\pm : K_G[T, \varrho] \rightarrow K_G$  with  $T \mapsto 0$ ,  $\varrho \mapsto \pm 1$ ; they send  $u_H = \varrho - h$  to  $\pm 1$  and  $y_H$  to  $1$ , respectively  $0$ . The proof of [Theorem 15.65](#) now applies verbatim with  $w_t$  replaced by  $y_H$  and  $\sigma_\pm$  by  $\sigma'_\pm$ . For the candidates take  $G = \{0\}$  and  $b = a$ , so that  $H = h(\omega^a)$  with  $h \in T\mathbb{R}[T]$ , and  $x = 1$ : then  $e_H \neq 0, 1$ , and  $e_H = (1 + \bar{u}_H)/2$  differs from  $0$  and  $1$  exactly when  $u_H \mp 1 \notin \Pi\mathcal{N}$  (both  $\Pi\mathcal{N}$  and  $\mathcal{B}_\mathbb{R}$  are torsion free, [Lemma 15.28](#)). Finally  $(1 + \sqrt{1 + \omega})/2 = y_H + H/2$  for  $H = \omega^{1/2}$ , and  $H/2 \in \Pi$ , so its residue is  $e_H$ .  $\square$

The two candidates are therefore one idempotent (for  $H = \omega^{1/2}$ ), and [Remark 15.52](#)'s reduction to the membership question is superseded: nontriviality follows from support projection and two specializations, without a membership test. For  $H \in \Pi$  whose support is not contained in one cyclic group the method needs  $H$  to be a polynomial in a fresh monomial over a group containing the rest of its support, and general  $H$  is not decided here. 20's two-radical splitter ([15.24](#)) is not minimal in this sense, which is not an error.

**Lemma 15.69** (Multiquadratic extensions [20, 23]). *Over a field  $L$  of characteristic different from 2, classes  $d_1, \dots, d_n$  independent in  $L^\times/L^{\times 2}$  give an extension  $L(\sqrt{d_1}, \dots, \sqrt{d_n})$  of degree  $2^n$ , with the square-root monomials as a basis and all  $2^n$  independent sign automorphisms.*

*Proof.* Induction on  $n$ . If  $y^2 = d_n$  in the extension by the first  $n-1$  roots, every sign automorphism sends  $y$  to  $\pm y$ , so  $y$  is a common eigenvector; the monomial basis has distinct sign characters, so  $y$  is a scalar multiple of one monomial, and squaring expresses the class of  $d_n$  as a product of earlier classes, a contradiction. So the last step has degree 2 and supplies the new basis elements and sign change.  $\square$

**Theorem 15.70** (Rank extension of set-sized Hahn fibres [20]). *Let  $G$  be a set-sized ordered abelian group,  $K_G = \mathbb{R}((X^G))$ , and define  $\mathcal{A}(G) = \mathbb{Z} + \Pi_\mathbb{R}(G)$ ,  $\Pi(G) = \Pi_\mathbb{R}(G)$ ,  $\mathcal{N}(G)$  and  $\mathcal{C}(G)$  as in ([15.20](#)). Let  $\tilde{G} = G \oplus \mathbb{Z}$  with the new coordinate dominant, and make the same definitions for  $\tilde{G}$ . Then  $\mathcal{C}(G) \rightarrow \mathcal{C}(\tilde{G})$  is injective, there is a diagonal-extending injection  $\mathcal{C}(G) \times \mathcal{C}(G) \hookrightarrow \mathcal{C}(\tilde{G})$ , and for every set  $P \subseteq K_G^{>0} \setminus \{1\}$  there is an injective  $\mathcal{C}(G)$ -algebra map*

$$\mathcal{C}(G)[X_r : r \in P]/(X_r^2 - X_r : r \in P) \hookrightarrow \mathcal{C}(\tilde{G}).$$

*So one dominant coordinate gives a free Boolean family of cardinality  $|K_G|$  over the smaller reduced fibre.*

*Proof.* A monic  $\Pi(G)$ -certificate for  $1$  ([Lemma 15.53](#)) would give  $1 = 0$  after ct; so  $\mathcal{C}(G) \neq 0$ . The proofs of [Theorem 16.31](#) and [Lemmas 15.55](#) and [15.62](#) apply verbatim to  $G \subseteq \tilde{G}$ , with  $T$  the monomial of the new coordinate and  $k = K_G$ ; the local equality ([15.23](#)) then gives the first injection, using radical ideals only and no global radicality. The second is proved exactly as [Theorem 15.65](#). For the third put  $\varrho_0 = \sqrt{T^2 - 1}$ ,  $\varrho_r = \frac{T}{r}(1 - r^2T^{-2})^{1/2}$  and  $w_r = (1 - \varrho_0\varrho_r)/2$  ( $r \in P$ ), binomial branches with positive leading coefficients, so that

$$w_r^2 - w_r = \frac{T^4/r^2 - (1 + r^{-2})T^2}{4} \in Tk[T] \subseteq \Pi(\tilde{G}),$$

and  $w_r \in \mathcal{N}(\tilde{G})$  has an idempotent residue  $e_r$ . For finite  $P_0 \subseteq P$  the radicands  $T^2 - 1$  and  $T^2/r^2 - 1$  ( $r \in P_0$ ) have the simple zeros  $\pm 1$  and  $\pm r$ , pairwise disjoint because the  $r$  are positive, distinct and  $\neq 1$ ; so every nonempty product of them has a simple zero and is not a square, and [Lemma 15.69](#) gives degree  $2^{1+|P_0|}$ . As in [Lemma 15.63](#), the branch algebra has only its monic quadratic relations, and  $T \mapsto 0$ ,  $\varrho_0 \mapsto i$ ,  $\varrho_r \mapsto \pm i$  can be chosen independently in  $k[i]$ , realizing every assignment of 0 and 1 to the  $e_r$ . If  $0 \neq \bar{x} \in \mathbb{C}(G)$  and  $p$  is a finite word in the  $e_r$  and their complements, a certificate for  $xp \in \sqrt{\Pi(\tilde{G})\mathcal{N}(\tilde{G})}$  specialized with  $p \mapsto 1$  would be a  $\Pi(G)$ -certificate for  $x$ ; so  $\bar{x}p \neq 0$ . An element of the source uses finitely many variables and has a unique expansion on their Boolean atoms, and multiplying a vanishing image by each atom kills its coefficient; so the map is injective. The positive part of an infinite ordered field minus one point has the cardinality of the field.  $\square$

Neither  $K_G$  nor  $K_{\tilde{G}}$  need be real closed, and  $\Pi(G)\mathcal{N}(G)$  is not assumed radical; the theorem concerns reduced fibres. For  $G = \{0\}$ ,  $K_G = \mathbb{R}$  and  $\mathbb{C}(G) = \overline{\mathbb{Z}}_{\mathbb{R}}$ , so one Laurent scale already carries continuum many independent idempotents over  $\overline{\mathbb{Z}}_{\mathbb{R}}$  (20). 23 shows that  $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$  is atomless, of cardinality and order density exactly  $2^{\aleph_0}$ , with another explicit independent family ([Corollary 15.110](#) and [Theorem 15.111](#)).

**Theorem 15.71** (Relative Boolean freeness [20]). *For every set-sized unital subring  $R \subseteq \mathbb{B}_{\mathbb{R}}$  and every set  $I$  there is an injective  $R$ -algebra homomorphism*

$$R[X_i : i \in I]/(X_i^2 - X_i : i \in I) \hookrightarrow \mathbb{B}_{\mathbb{R}}.$$

*Proof.* Well-order  $I$  as an ordinal  $\kappa$ . At stage  $\alpha < \kappa$  let  $R_\alpha$  be the ring generated by  $R$  and the  $e_\beta$ ,  $\beta < \alpha$ ; it is a set, and [Theorem 15.66](#) gives a splitter  $e_\alpha$  for it. For  $0 \neq r \in R$  the product of  $r$  with any finite choice of  $e_\alpha$  or  $1 - e_\alpha$ , in increasing order of indices, stays nonzero, since each partial product lies in the ring present before the next index. In finitely many variables the polynomials  $p_\varepsilon = \prod_j X_j^{\varepsilon_j} (1 - X_j)^{1-\varepsilon_j}$  ( $\varepsilon \in \{0, 1\}^n$ ) are orthogonal idempotents summing to 1, and every element is uniquely  $\sum_\varepsilon r_\varepsilon p_\varepsilon$  (evaluate at zero-one vectors); a vanishing image, multiplied by the image of each  $p_\varepsilon$ , forces every  $r_\varepsilon = 0$ . A polynomial in infinitely many variables uses finitely many.  $\square$

For  $|I| = n$  the source is  $R^{2^n}$ ; for infinite  $I$  it is the directed union of these finite products, not the unrestricted product  $R^{\{0,1\}^I}$  (20).

**Theorem 15.72** (An explicit independent class [20]). *For every ordinal  $\alpha$  put  $t_\alpha = \omega^{\omega^\alpha}$  and  $e_\alpha = \mathbf{e}_{t_\alpha} = \left[ (1 - \sqrt{t_\alpha^2 - 1} \sqrt{t_\alpha^2/4 - 1})/2 \right]$ . Every finite Boolean word in distinct  $e_\alpha$  and their complements is nonzero in  $\mathbb{B}_{\mathbb{R}}$ .*

*Proof.*  $w_{t_\alpha}$  has support in  $\mathbb{Z}\omega^\alpha$ . The group generated by the supports for  $\beta < \alpha$  lies in the rational span of  $\{\omega^\beta : \beta < \alpha\}$  in the exponent field, whose members are finite combinations with leading exponent below  $\alpha$ ; so  $\omega^\alpha$  exceeds them all and is fresh over that group. [Theorem 15.65](#), applied inductively along increasing indices, proves every finite word nonzero. Only individual earlier supports are used, never an infinite Boolean product, and the family is defined by a formula, without class-valued recursion.  $\square$

**Theorem 15.73** (Atomless, a proper class, no set-sized dense core [20]).  *$\text{Idem}(\mathbb{B}_{\mathbb{R}})$  is atomless and a proper class, and no set  $\mathcal{D} \subseteq \text{Idem}(\mathbb{B}_{\mathbb{R}}) \setminus \{0\}$  has the property that every nonzero idempotent lies above a member of  $\mathcal{D}$ .*



*Proof.* For  $f \neq 0$  split the set-sized ring  $\mathbb{Z}[f]$ :  $fe$  and  $f(1 - e)$  are nonzero orthogonal idempotents with sum  $f$ . If  $\text{Idem}(\mathbb{B}_{\mathbb{R}})$  were a set, the ring it generates would be a set; a splitter  $e$  of that ring lies in it and is nonzero, and splitting  $r = e$  would give  $e(1 - e) \neq 0$ . (Alternatively [Theorem 15.72](#) injects the ordinals.) If  $\mathcal{D}$  were such a set, a splitter  $e$  of the ring it generates is nonzero, and no  $d \in \mathcal{D}$  has  $d \leq e$ , since  $d(1 - e) \neq 0$ .  $\square$

With [Theorem 15.38](#) this sharpens 19's description:  $\mathbb{B}_{\mathbb{R}} \cong \overline{\mathbb{Z}}[\text{Idem}(\mathbb{B}_{\mathbb{R}})]$  with the ring of all complex algebraic integers as coefficients and an atomless proper-class Boolean parameter with no set-sized dense core; the complexity lies in how arithmetic branches are organized, not in new algebraic numbers on one branch (20). Free subalgebras of every set size do not make the whole class algebra free or complete, nor show that it realizes every set-sized Boolean type (20).

**Corollary 15.74** (Idempotent matrices split [20]). *Every nonzero idempotent matrix  $P \in M_n(\mathbb{B}_{\mathbb{R}})$  is  $eP + (1 - e)P$  with  $eP$ ,  $(1 - e)P$  nonzero orthogonal idempotents, for a scalar idempotent  $e$ .*

*Proof.* Split the set-sized ring generated by the entries of  $P$ ; a nonzero entry survives multiplication by  $e$  and by  $1 - e$ , and centrality of  $e$  with  $P^2 = P$  gives idempotence and orthogonality.  $\square$

In the idempotent-matrix description of finitely generated projective modules, every nonzero one therefore decomposes nontrivially. This is a finite matrix statement over a class ring; no category of class-sized modules is formed, and no projective or global dimension is computed (20).

*Remark 15.75* (No global choice in the splitting results [merge]). 20 works in Gödel–Bernays class theory with global choice, but [Theorems 15.66](#) and [15.71](#) to [15.73](#) and [Corollary 15.74](#) need only choice for sets (for well-ordering  $I$  in [Theorem 15.71](#)). The splitter can be chosen canonically: take as representatives of the elements of  $R$  all their members of least rank (a set), let  $G$  be the group generated by all their supports and  $b$  the simplest surreal above  $G \cup \{0\}$ ; [Theorem 15.65](#) applies to every representative. The recursion in [Theorem 15.71](#) then makes no choices, and [Theorem 15.72](#) is explicit. [Theorem 15.79](#) below uses no global choice either; [Theorem 15.80](#), like [Theorem 15.49](#), uses class ultrafilters.

**Proposition 15.76** (Reduced fibres of fixed Hahn fields [20]). *For a nonzero set-sized ordered abelian group  $G$ , with the notation of [Theorem 15.70](#),  $\mathbb{C}(G) \cong \overline{\mathbb{Z}}[\text{Idem}(\mathbb{C}(G))]$ . For  $G = \{0\}$ ,  $\mathbb{C}(G) = \overline{\mathbb{Z}}_{\mathbb{R}}$ .*

*Proof.* Let  $I = \Pi(G)\mathcal{N}(G)$  and  $L = \sqrt{I}$ .  $I$  is divisible by nonzero integers, and so is  $L$ : if  $x^m \in I$ , then  $(x/n)^m \in I \subseteq \mathcal{N}(G)$ , so  $x/n \in \mathcal{N}(G)$  by normality and then  $x/n \in L$ . A monic  $\Pi(G)$ -certificate for a nonzero integer, followed by  $\text{ct}$ , would make a nonzero power of it vanish; so  $L \cap \mathbb{Z} = 0$ , and  $\mathbb{C}(G)$  is nonzero, reduced, torsion free and integral over  $\mathbb{Z}$ . The proof of [Theorem 15.30](#) applies unchanged. A positive monomial  $T = \omega^g$ ,  $0 < g \in G$ , gives  $v_T \in K_G$  by its binomial Laurent expansion, whether or not  $K_G$  is real closed, with  $v_T^2 + 1 = Tv_T \in I$ ; so the coefficient embedding and the spectral argument of [Lemmas 15.35](#) and [15.36](#) and [Theorem 15.38](#) apply. For  $G = \{0\}$ :  $K_G = \mathbb{R}$ ,  $\mathcal{A}(G) = \mathbb{Z}$  and  $\Pi(G) = 0$ .  $\square$

This complements [Theorem 15.51](#), which describes the unreduced  $N_{\bullet}/JN_{\bullet}$  under the divisibility axiom (ii) that a fixed Hahn field may fail. No atomlessness of  $\text{Idem}(\mathbb{C}(G))$  for a fixed  $G$  follows: the proof for **No** gains a fresh scale beyond a prescribed set of supports, which a fixed workspace need not supply, and reduction by  $\sqrt{\Pi(G)\mathcal{N}(G)}$  cannot silently be replaced by reduction by  $\Pi(G)\mathcal{N}(G)$  (20). (Status (batch 32): for cyclic  $G$  both points are decided by 23 without a fresh scale:  $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$  is atomless, by ramification at unused finite places, and the replacement is impossible because  $\Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$  is not radical, [Corollary 15.110](#) and [Theorem 15.86](#); see also [Remark 15.114](#).)



**Corollary 15.77** (The surcomplex fibre [20]). *Every set-sized subring of  $B_{\mathbb{C}}$  has a uniform splitter, and Theorems 15.71 and 15.73 and Corollary 15.74 hold for  $B_{\mathbb{C}}$ .*

*Proof.* Identify  $B_{\mathbb{C}} = B_{\mathbb{R}} \times B_{\mathbb{R}}$  (Theorem 15.42). For a set-sized subring  $S$  let  $R$  be the ring generated by both coordinate projections of  $S$  and  $e$  a splitter of  $R$ ; the diagonal idempotent  $(e, e)$  splits every nonzero pair, since some coordinate is nonzero. The other proofs use only uniform splitting.  $\square$

Atomlessness does not give  $B_{\mathbb{R}} \cong B_{\mathbb{R}} \times B_{\mathbb{R}}$ , which would need more information about the global Boolean algebra; the integral doubling is  $B_{\mathbb{C}} \cong B_{\mathbb{R}} \times B_{\mathbb{R}}$  (20).

**Maps to set-sized rings.** The remaining results use the universal theorem through Corollary 15.45, and the last one uses class ultrafilters. For a class ultrafilter  $U$  on  $\text{Idem}(B_{\mathbb{R}})$ , the evaluation  $\text{ev}_U$  of Theorem 15.49 is a surjective unital map  $B_{\mathbb{R}} \rightarrow \overline{\mathbb{Z}}$  fixing the coefficient copy; 20 calls it a *branch map*.

**Lemma 15.78** (Class ultrafilters through a prescribed set [19, 20]). *With global choice, a set of elements of a Boolean algebra (possibly a proper class) with the finite intersection property is contained in an ultrafilter (a class).*

*Proof.* As for Lemma 15.48, adding the prescribed set to the decisions: at stage  $\alpha$  add  $b_\alpha$  if the prescribed set and the earlier decisions together with  $b_\alpha$  keep the finite intersection property, and  $\neg b_\alpha$  otherwise; if both failed, finite meets  $c, d$  from the prescribed set and the earlier decisions would satisfy  $c \wedge b_\alpha = 0 = d \wedge \neg b_\alpha$ , hence  $c \wedge d = 0$ .  $\square$

**Theorem 15.79** (An idempotent invisible to every set of small targets [20]). *Let  $(\varphi_i)_{i \in I}$  be a set-indexed family, given by one class relation, of unital homomorphisms  $\varphi_i : \mathcal{N} \rightarrow S_i$  to set-sized rings. There is  $x \in \mathcal{N}$  whose residue is a nonzero idempotent of  $B_{\mathbb{R}}$  with  $\varphi_i(x) = 0$  for all  $i$ . Hence  $\bigcap_i \ker \varphi_i \supsetneq \Pi \mathcal{N}$ , and  $B_{\mathbb{R}}$  does not embed in a set-indexed product of set-sized rings.*

*Proof.* The maps factor through  $B_{\mathbb{R}}$  (Corollary 15.45), and together give  $\psi : B_{\mathbb{R}} \rightarrow S = \prod_i S_i$ , a set. Take a cardinal  $\kappa > |S|$  and the independent family  $(e_\alpha)_{\alpha < \kappa}$  of Theorem 15.72; two members  $e_\alpha \neq e_\beta$  have the same image  $p$ , an idempotent. Their symmetric difference  $d = e_\alpha + e_\beta - 2e_\alpha e_\beta = e_\alpha(1 - e_\beta) + (1 - e_\alpha)e_\beta$  is an idempotent, nonzero by independence, and  $\psi(d) = 2p - 2p^2 = 0$  in every characteristic. Any lift  $x \in \mathcal{N}$  of  $d$  works.  $\square$

Each nonzero residue is detected by a map to the countable domain  $\overline{\mathbb{Z}}$  (Theorem 15.49), so in the elementwise sense of Theorem 5.12  $B_{\mathbb{R}}$  is residually set-sized [merge]; but no set-indexed family is jointly faithful. The obstruction concerns the size of the index set, not of any one target, and it exhibits a nonzero idempotent in every such common kernel (20).

**Theorem 15.80** (Many branches through prescribed data [20]). *Assume global choice. Fix a coefficient copy  $\overline{\mathbb{Z}} \subseteq B_{\mathbb{R}}$  and a branch map  $\varphi : B_{\mathbb{R}} \rightarrow \overline{\mathbb{Z}}$  fixing it. For every set  $S \subseteq B_{\mathbb{R}}$  and every cardinal  $\kappa$  there are at least  $2^\kappa$  branch maps agreeing with  $\varphi$  on  $S$ .*

*Proof.* Choose finite-partition representations of the elements of  $S$ , let  $\mathfrak{E}_0$  be the set-sized Boolean algebra generated by the parts used and  $B_0 = \overline{\mathbb{Z}}[\mathfrak{E}_0]$ , a set-sized subring containing  $S$  and the coefficient copy;  $\varphi$  restricted to  $\mathfrak{E}_0$  is an ultrafilter  $U_0$ . Take  $(e_\alpha)_{\alpha < \kappa}$  free over  $B_0$  (Theorem 15.71). For  $\eta : \kappa \rightarrow \{0, 1\}$ , the set  $U_0 \cup \{e_\alpha : \eta(\alpha) = 1\} \cup \{1 - e_\alpha : \eta(\alpha) = 0\}$  has the finite intersection property: a finite meet from  $U_0$  is a nonzero idempotent of  $B_0$ , and its product with any finite

assignment word stays nonzero by relative freeness. Extend it to a class ultrafilter (Lemma 15.78) and evaluate. The map selects the same part as  $\varphi$  in every partition from  $\mathfrak{E}_0$ , so it agrees with  $\varphi$  on  $B_0$ ; different  $\eta$  give maps differing on some  $e_\alpha$ . The recursion is uniform in the set parameter  $\eta$ , so this is a set-indexed class family of maps.  $\square$

Fixing the coefficient copy only removes an irrelevant automorphism of  $\overline{\mathbb{Z}}$ : an arbitrary characteristic-zero domain map can be normalized by composing with the inverse of the automorphism it induces on the copy (20).

**Corollary 15.81** (Surcomplex maps [20]). *Theorems 15.79 and 15.80 hold for  $\mathcal{N}_{\mathbb{C}}$ ,  $\Pi\mathcal{N}_{\mathbb{C}}$  and  $\mathbf{B}_{\mathbb{C}}$ ; in the extension statement the factor of  $\mathbf{B}_{\mathbb{C}} \cong \mathbf{B}_{\mathbb{R}} \times \mathbf{B}_{\mathbb{R}}$  selected by the given branch can be kept fixed.*

*Proof.* Corollary 15.77 gives uniform splitting and relative freeness in  $\mathbf{B}_{\mathbb{C}}$ , and the proof of Theorem 15.79 is unchanged. A domain map out of a product selects one factor; keep it, project the prescribed set into it, and apply Theorem 15.80.  $\square$

$\mathcal{N} \subseteq \mathbf{No}$  has no idempotents other than 0, 1, while  $\mathbf{B}_{\mathbb{R}}$  has a proper class of them and contains the residue  $\mathbf{v}$  with  $\mathbf{v}^2 = -1$ ; this is possible because  $\Pi\mathcal{N}$  is not convex (Proposition 15.34). The quotient is not standard part, and an idempotent residue does not say that its chosen representative is idempotent (20). What 20 leaves open is recorded in Questions 18.17 and 18.18 and its limitations in Section 18.2.

## 15.10 Ramification in a cyclic workspace

This subsection is manuscript 23's (Section 1.3). It settles the fixed-group clause of Question 18.17(b) for every nonzero cyclic exponent group  $G = \mathbb{Z}a$ , with opposite answers for its two parts: the extended ideal  $\Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$  is *not* radical, since the branch  $\mathbf{z}_n = T(1 + T^{1-n})^{1/n}$  has nilpotency index exactly  $n$  modulo it (Theorem 15.86), whereas  $\text{Idem}(\mathbf{C}(\mathbb{Z}a))$  is atomless, of cardinality and order density  $\mathfrak{c} = 2^{\aleph_0}$  (Corollary 15.110 and Theorem 15.111). It describes  $\mathcal{N}(\mathbb{Z}a)$  completely by a residue test at the finitely many branches over  $T = 0$  of each finite subextension (Theorems 15.91 and 15.93); this answers the fixed-group clause of Question 18.11 for cyclic groups, as a criterion and not as an algorithm. The mechanism is a distinction between two places. The elements are Laurent series at  $T = \infty$ , where  $T = \omega^a$  is infinitely large, while integrality and nilpotence modulo the extended ideal are governed by the algebraic places over  $T = 0$ ; a series at infinity is never evaluated term by term at zero (23). Atomlessness needs no fresh dominant scale, which 20's argument uses and a fixed workspace lacks (the paragraph after Proposition 15.76): the new idempotents come from ramification at unused finite places of  $k(T)$ . 23 adds exact thresholds for the refinement  $T = U^m$ , a reduced bounded-denominator Puiseux limit with no reduced finite stage, the local structure of the coefficient normalization, the associated graded ring and the completion, a periodic free resolution over the coefficient fibre that fails over the arithmetic fibre, and a countable fragment whose Boolean algebra is the Cantor algebra. Every result holds for the real pair and for its Gaussian counterpart.

23 pins 343dc2c (23 September 2026), which contains the writing of 20's subsection but not that of 22's; it read this report's normalization, fibre and branching material (its audit names Questions 18.11 and 18.17 and Theorems 15.57 and 15.60). Printed once, under the existing label and with 23 added as a proving source, is 20's multiquadratic lemma (Lemma 15.69; 23's Lemma 7.4, with the same proof). 23's Corollary 4.5 (Corollary 15.94) contains, for the real pair, the first equality of 20's Theorem 15.60, which 23 credits and recovers as a check; its Gaussian form and its statement for the fixed ideal are new. At its pin this report also contained, after Theorem 15.70,

the remark that one Laurent scale carries continuum many independent idempotents over  $\overline{\mathbb{Z}}_{\mathbb{R}}$ : with  $G = \{0\}$ , [Theorem 15.70](#) gives an independent family of size  $\mathfrak{c}$  in  $\text{Idem}(\mathbb{C}(\mathbb{Z}a))$  (20). 23 does not cite it. The lower-bound half of 23's cardinality statement for the reduced fibre is therefore 20's; 23's family, its independence in the unreduced and coefficient fibres, the exact value  $\mathfrak{c}$ , atomlessness, order density, uniform splitting and the count  $2^{\mathfrak{c}}$  are new. The radicality of the Puiseux limit also follows from 19's finite-axiom theorem [Theorem 15.51](#), which was in the report at 23's pin and which 23 does not cite ([Remark 15.102](#), a merge observation); 23's refinement proof is kept as the first route, and the stage-by-stage statements of [Theorem 15.100](#) are new. The numbering of 23 cited here is that of its own manuscript.

**Notation and renamings.** Throughout,  $(k, D)$  is  $(\mathbb{R}, \mathbb{Z})$  or  $(\mathbb{C}, \mathbb{Z}[i])$ ,  $a > 0$  is a fixed surreal exponent, and  $T = \omega^a$  is the actual monomial, as in [Theorem 15.60](#).  $K_{\mathbb{Z}a} = k((X^{\mathbb{Z}a}))$  is 16's Hahn field with coefficients  $k$  and supports in  $\mathbb{Z}a$  (for  $k = \mathbb{R}$  it is 20's  $K_{\mathbb{Z}a}$ ); by [Lemma 15.62](#) with  $G = \{0\}$  and  $b = a$  it is the Laurent field  $k((T^{-1}))$ , the Laurent variable being identified with  $\omega^a$ , so its elements are actual surreal, respectively surcomplex, numbers, and  $\mathfrak{c} = |K_{\mathbb{Z}a}|$ . For the real pair (15.20) gives  $\mathcal{A}(\mathbb{Z}a) = \mathbb{Z} + T\mathbb{R}[T]$ ,  $\Pi(\mathbb{Z}a) = T\mathbb{R}[T]$ ,  $\mathcal{N}(\mathbb{Z}a) = \text{IC}_{K_{\mathbb{Z}a}}(\mathcal{A}(\mathbb{Z}a))$  and  $\mathbb{C}(\mathbb{Z}a)$ . For the Gaussian pair the same symbols denote  $\mathcal{A}(\mathbb{Z}a) = \mathbf{Oz}[i] \cap K_{\mathbb{Z}a}$ ,  $\Pi(\mathbb{Z}a) = \Pi_{\mathbb{C}} \cap K_{\mathbb{Z}a}$ , the integral closure  $\mathcal{N}(\mathbb{Z}a)$  of  $\mathcal{A}(\mathbb{Z}a)$  in  $K_{\mathbb{Z}a}$ , and  $\mathbb{C}(\mathbb{Z}a)$  defined by (15.20) verbatim; the pair is fixed and not shown in the notation. Put

$$\begin{aligned} \mathcal{B} &= \text{IC}_{K_{\mathbb{Z}a}}(k[T]), & \mathfrak{l} &= \Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a), & \mathfrak{J} &= \sqrt{T\mathcal{B}} \subseteq \mathcal{B}, \\ \mathbf{B}(\mathbb{Z}a) &= \mathcal{N}(\mathbb{Z}a)/\mathfrak{l}, & \mathbf{B}^{\text{co}}(\mathbb{Z}a) &= \mathcal{B}/T\mathcal{B}, & \mathbf{C}^{\text{co}}(\mathbb{Z}a) &= \mathcal{B}/\mathfrak{J}, \end{aligned} \quad (15.25)$$

the radical  $\mathfrak{J}$  being taken in  $\mathcal{B}$ . [Theorems 15.91](#) and [15.93](#) show  $\mathfrak{J} \subseteq \mathcal{N}(\mathbb{Z}a)$ ,  $\mathfrak{l} = T\mathcal{B}$  and  $\mathbb{C}(\mathbb{Z}a) = \mathcal{N}(\mathbb{Z}a)/\mathfrak{J}$ .  $\mathbf{B}(\mathbb{Z}a)$  is the *arithmetic fibre*, the fixed-workspace analogue of 19's  $\mathbf{B}_{\bullet}$ , and  $\mathbb{C}(\mathbb{Z}a)$  is its reduction;  $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$  is the *coefficient fibre*, with reduction  $\mathbf{C}^{\text{co}}(\mathbb{Z}a)$ . The coefficient fibre is the arithmetic fibre of the pair  $(k, k)$ , since  $k[T] = \mathcal{A}_{k,k}(\mathbb{Z}a)$  and  $T\mathcal{B} = \Pi_k(\mathbb{Z}a)\mathcal{B}$ .  $K^{\text{alg}}$  is the subfield of  $K_{\mathbb{Z}a}$  of elements algebraic over  $k(T)$ . For a finite extension  $k(T) \subseteq L \subseteq K^{\text{alg}}$  put  $\mathcal{B}_L = \text{IC}_L(k[T])$ ; for a maximal ideal  $\mathfrak{q}$  of  $\mathcal{B}_L$  write  $\mathfrak{q} \mid T$  if  $\mathfrak{q} \cap k[T] = (T)$ ,  $\ell_{\mathfrak{q}} = \mathcal{B}_L/\mathfrak{q}$  for its residue field,  $f \bmod \mathfrak{q}$  for the residue of  $f$ , and

$$v_{\mathfrak{q}}(f) = \frac{\text{ord}_{\mathfrak{q}}(f)}{\text{ord}_{\mathfrak{q}}(T)}, \quad v_{\mathfrak{q}}(T) = 1. \quad (15.26)$$

For an integer  $m \geq 1$ ,  $U = \omega^{a/m}$ , so  $T = U^m$ , and  $\mathbb{Z}a/m$  is the group generated by  $a/m$ ; the objects defined with  $U$  in place of  $T$  are  $K_{\mathbb{Z}a/m}$ ,  $\mathcal{A}(\mathbb{Z}a/m)$ ,  $\Pi(\mathbb{Z}a/m)$ ,  $\mathcal{N}(\mathbb{Z}a/m)$ ,  $\mathfrak{l}(\mathbb{Z}a/m)$  and  $\mathbf{B}(\mathbb{Z}a/m)$ , so that  $\mathfrak{l} = \mathfrak{l}(\mathbb{Z}a)$ . 23's symbols are renamed as follows.

- Its  $F, A, P, N, B, I, E$  are  $K_{\mathbb{Z}a}, \mathcal{A}(\mathbb{Z}a), \Pi(\mathbb{Z}a), \mathcal{N}(\mathbb{Z}a), \mathcal{B}, \mathfrak{l}, K^{\text{alg}}$  ( $\mathcal{B}$  is the ring  $D + \mathfrak{m}_k$  and  $\mathbf{B}_{\bullet}$  the full-class fibre;  $I$  and  $J$  are local ideals in [Proposition 15.76](#) and [Theorem 15.51](#);  $\mathcal{E}$  is 07's field and  $E$  an ambient field). Its arithmetic fibre  $\mathcal{R} = \mathcal{N}/\mathfrak{l}$ , its reduction  $\mathcal{R}_{\text{red}}$ , its coefficient fibre  $\mathcal{C} = \mathcal{B}/T\mathcal{B}$  and  $\mathcal{C}_{\text{red}}$  are  $\mathbf{B}(\mathbb{Z}a), \mathbb{C}(\mathbb{Z}a), \mathbf{B}^{\text{co}}(\mathbb{Z}a)$  and  $\mathbf{C}^{\text{co}}(\mathbb{Z}a)$  ( $\mathcal{R}$  is 14's Rees algebra). Its  $J$  and its nilradical  $\mathfrak{n} = J/T\mathcal{B}$  are  $\mathfrak{J}$  and  $\mathfrak{J}/T\mathcal{B}$ .
- Its  $B_L$  and residue field  $\kappa(\mathfrak{q})$  are  $\mathcal{B}_L$  and  $\ell_{\mathfrak{q}}$  ( $\kappa$  is a cardinal); its minimum order  $\mu_T$  and truncated order  $h_T$  are  $\nu_T$  and  $\hat{\nu}_T$  ( $\mu$  is a cardinal letter,  $h$  an exponent); its filtration  $\mathcal{F}^r \mathcal{R}$  is  $\text{Fil}^r$  ( $\mathcal{F}$  is a family in [Proposition 3.3](#)); its  $V = B_{\mathfrak{q}}$  with maximal ideal  $\mathfrak{m}$  is  $\mathcal{B}_{\mathfrak{q}}$  with maximal ideal  $\mathfrak{q}\mathcal{B}_{\mathfrak{q}}$  ( $\mathfrak{m}$  is the negative-support ideal).
- Its branches  $z_n, h_c, e_c$  and  $w$  are  $\mathbf{z}_n, \varsigma_c, \mathbf{y}_c$  and 20's  $\varrho_0$  of [Lemma 15.63](#) (a sequence  $z_n$  occurs in the proof of [Theorem 13.45](#);  $e_r, \mathbf{e}_t, e_{\alpha}, e_H, w_{\pm}$  and  $\mathbf{w}_t$  are 19's and 20's idempotents and splitters); the residue of  $\mathbf{y}_c$  is  $\bar{\mathbf{y}}_c$ .

- Its refined rings  $F_m, A_m, P_m, N_m, I_m, \mathcal{R}_m$  are the objects over  $\mathbb{Z}a/m$  above; its Puiseux rings  $F_{\text{Pui}}, A_{\text{Pui}}, P_{\text{Pui}}, N_{\text{Pui}}, I_{\text{Pui}}$  are  $K_{\text{Pui}}, \mathcal{A}_{\text{Pui}}, \Pi_{\text{Pui}}, \mathcal{N}_{\text{Pui}}, \mathcal{I}_{\text{Pui}}$ ; its  $F_{\mathbb{Q}}, A_{\mathbb{Q}}, P_{\mathbb{Q}}$  are  $K_{\mathbb{Q}a}, \mathcal{A}(\mathbb{Q}a), \Pi(\mathbb{Q}a)$ .
- Its  $D_k^{\text{int}}$  is  $\overline{\mathbb{Z}}_{\mathbb{R}}$  or  $\overline{\mathbb{Z}}$ ; its coefficient fragments  $k_r, k_c$  are 19's  $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$  and  $\overline{\mathbb{Q}}$ ; its module  $M = \mathcal{C}/(x)$  is  $\mathcal{M}_n$  ( $M$  is a lattice); its graded variable  $X$  is  $W$  ( $X^g$  are formal monomials).

23's finite extensions  $L$  keep their letter (as the set-sized field of [Corollary 15.50](#); not 14's module  $L(M)$ , 16's lower set or 03's  $L_c(F)$ ), and so do its primes  $\mathfrak{q}$  (not 14's kernels  $\mathfrak{q}_M$ ). The letter  $c$  is a positive real parameter,  $\mathfrak{c}$  the continuum. Four tempting readings are false. (1) The Laurent constant coefficient is not a residue at a place over zero:  $z_2 = T + \frac{1}{2} - \dots$  lies in  $\mathcal{N}(\mathbb{Z}a)$  although  $\frac{1}{2}$  is not an algebraic integer ([Example 15.87](#)). (2) The valuations  $v_{\mathfrak{q}}$  have  $v_{\mathfrak{q}}(T) = 1$ ; the Laurent valuation at infinity has value  $-1$  at  $T$ , and neither extends the other ([Remark 15.85](#)). (3)  $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$  is a finite  $k$ -linear span: it does not say  $k \subseteq \mathcal{N}(\mathbb{Z}a)$ , and  $\mathcal{I} = T\mathcal{B}$  is larger than  $T\mathcal{N}(\mathbb{Z}a)$ . (4) The injection  $\mathcal{C}(\mathbb{Z}a) \hookrightarrow \mathcal{B}_{\mathbb{R}}$  of [Corollary 15.58](#) concerns the reduced fibre; on  $\mathcal{B}(\mathbb{Z}a)$  the map to  $\mathcal{B}_{\mathbb{R}}$  kills exactly the nilradical ([Remark 15.119](#)).

### The workspace and its finite normalizations.

**Proposition 15.82** (The cyclic integer part [23]). **Oz**  $\cap K_{\mathbb{Z}a} = \mathbb{Z} + T\mathbb{R}[T]$  for the real pair and **Oz** $[i] \cap K_{\mathbb{Z}a} = \mathbb{Z}[i] + T\mathbb{C}[T]$  for the Gaussian pair. In both cases  $\Pi(\mathbb{Z}a) = Tk[T]$  and  $\text{Frac}(\mathcal{A}(\mathbb{Z}a)) = k(T)$ .

*Proof.* An omnific normal form has no negative exponents and an integer constant coefficient, and a reverse well-ordered subset of  $\{na : n \geq 1\}$  is finite; so the positive part is a polynomial in  $T$  without constant term (as in the proof of [Theorem 15.60](#)). The Gaussian case applies this to both coordinates. Finally  $T$  and  $cT$  lie in  $\mathcal{A}(\mathbb{Z}a)$  for  $c \in k$ , so  $c = (cT)/T \in \text{Frac}(\mathcal{A}(\mathbb{Z}a))$  and  $k(T) \subseteq \text{Frac}(\mathcal{A}(\mathbb{Z}a))$ ; the reverse inclusion is clear.  $\square$

For the real pair  $\mathcal{A}(\mathbb{Z}a)$  is an integer part of  $K_{\mathbb{Z}a}$ , although  $K_{\mathbb{Z}a}$  is not real closed: keep the positive powers of  $f$ , round its real constant down, and subtract one more if the constant was an integer and the negative-power tail is negative; this is the unique  $u \in \mathcal{A}(\mathbb{Z}a)$  with  $u \leq f < u + 1$ . It does not extend  $\text{ct}$  from  $\mathcal{A}(\mathbb{Z}a)$  to  $\mathcal{N}(\mathbb{Z}a)$  (23).

For every finite extension  $k(T) \subseteq L \subseteq K^{\text{alg}}$  (separable, in characteristic 0),  $\mathcal{B}_L$  is a finite  $k[T]$ -module and a Dedekind domain whose nonzero prime localizations are discrete valuation rings, its residue fields are finite over those of  $k[T]$ , and every prime of  $k[T]$  has a prime above it [37, 38, 40]. There are finitely many  $\mathfrak{q} \mid T$ , and  $\ell_{\mathfrak{q}}$  is  $\mathbb{C}$  if  $k = \mathbb{C}$  and is  $\mathbb{R}$  or  $\mathbb{C}$  if  $k = \mathbb{R}$ .

**Lemma 15.83** (Finite detection and descent [23]). *The rings  $\mathcal{B}_L$  form a directed system with  $\mathcal{B} = \bigcup_L \mathcal{B}_L$ ,  $\mathcal{B} \cap L = \mathcal{B}_L$  and  $\text{Frac}(\mathcal{B}) = K^{\text{alg}}$ . For  $f \in \mathcal{B}_L$  and an integer  $r \geq 1$ ,*

$$f \in T^r \mathcal{B} \iff f \in T^r \mathcal{B}_L \iff v_{\mathfrak{q}}(f) \geq r \text{ for every } \mathfrak{q} \mid T. \quad (15.27)$$

*Proof.* An integral element is algebraic and lies in a finite extension, and integrality over  $k[T]$  is a property of the element, so  $\mathcal{B} \cap L = \mathcal{B}_L$ . Clearing the denominators of a monic equation, a nonzero polynomial multiple of any element of  $K^{\text{alg}}$  is integral over  $k[T]$ . Next,  $f \in T^r \mathcal{B}$  means  $f/T^r \in \mathcal{B} \cap L = \mathcal{B}_L$ . A Dedekind domain is the intersection of its localizations at nonzero primes; at primes not above  $T$  the element  $T$  is a unit, and at the primes above  $T$  regularity of  $f/T^r$  is the stated inequality.  $\square$

**Lemma 15.84** (Finite branch locus [23]). *A finite extension  $L/k(T)$  is unramified at all but finitely many nonzero primes of  $k[T]$ .*

*Proof.* Multiply a primitive element by a nonzero polynomial to make it integral over  $k[T]$ . Its minimal polynomial is monic and separable, with nonzero discriminant  $\Delta$ . After inverting  $\Delta$  the finite free algebra it generates is étale over  $k[T][\Delta^{-1}]$ , hence normal, hence the normalization there; only the finitely many prime divisors of  $\Delta$  can ramify [39].  $\square$

*Remark 15.85* (Two unrelated valuations [23]). The Laurent valuation at infinity has value  $-1$  at  $T$ , the valuations (15.26) have value 1. The latter are valuations of finite algebraic function fields, not continuous extensions of the Laurent topology to an evaluation at  $T = 0$ . An expansion with infinitely many negative powers can be regular at a place over zero, and an element finite in the surreal order, such as  $z_2/T$  below, can have a pole there. Confusing the two would invalidate every argument below (23).

**Explicit nilpotents of every finite order.** For an ordinary integer  $n \geq 2$  put

$$z_n = T \sum_{j \geq 0} \binom{1/n}{j} T^{j(1-n)}, \quad z_n^n = T^n + T = T(1 + T^{n-1}). \quad (15.28)$$

The exponents are integers decreasing to  $-\infty$ , the formal binomial identity holds in characteristic 0, and the coefficients are rational; in  $K_{\mathbb{Z}a}$  for  $k = \mathbb{R}$  this is the positive root with leading term  $T$ , and the same branch is used for  $k = \mathbb{C}$ .

**Theorem 15.86** (Exact cyclic nilpotency [23]).  $z_n \in \mathcal{N}(\mathbb{Z}a)$ ,  $z_n^n \in \mathfrak{l}$ , and  $z_n^j \notin \mathfrak{l}$  for  $1 \leq j < n$ . Hence  $\mathfrak{l} = \Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$  is not radical, and the residue of  $z_n$  in  $\mathcal{B}(\mathbb{Z}a)$  has nilpotency index exactly  $n$ .

*Proof.* (15.28) is a monic equation over  $\mathcal{A}(\mathbb{Z}a)$  with right side in  $\Pi(\mathbb{Z}a)$ , so  $z_n \in \mathcal{N}(\mathbb{Z}a)$  and  $z_n^n \in \mathfrak{l}$ . Suppose  $z_n^j = \sum_{\ell \leq r} p_\ell b_\ell$  with  $1 \leq j < n$ ,  $p_\ell \in Tk[T]$  and  $b_\ell \in \mathcal{N}(\mathbb{Z}a)$ . Choose a finite  $L$  containing  $z_n$  and all  $b_\ell$ ; each  $b_\ell$  is integral over  $k[T]$ , hence in  $\mathcal{B}_L$ . At any  $\mathfrak{q} \mid T$  every summand has value at least 1, while  $1 + T^{n-1}$  is a unit and (15.28) gives  $v_{\mathfrak{q}}(z_n^j) = j/n < 1$ , a contradiction.  $\square$

This proof uses only a finite ideal expression, the normalization of one finite function field and one valuation; it does not use the identity  $\mathfrak{l} = T\mathcal{B}$  of Theorem 15.93, and it is a self-contained negative answer to fixed-cyclic radicality (23).

**Example 15.87** (The first obstruction [23]). For  $n = 2$ ,

$$z_2 = \sqrt{T^2 + T} = T + \frac{1}{2} - \frac{1}{8T} + \frac{1}{16T^2} - \frac{5}{128T^3} + \cdots,$$

and its residue in  $\mathcal{B}(\mathbb{Z}a)$  is nonzero with square zero. Although  $z_2$  is integral over  $\mathcal{A}(\mathbb{Z}a)$ , its Laurent constant coefficient  $\frac{1}{2}$  is not an algebraic integer, so integrality over  $\mathcal{A}(\mathbb{Z}a)$  cannot be tested on the constant coefficient at infinity. And  $z_2/T$  is infinitesimally close to 1 but not integral over  $\mathcal{A}(\mathbb{Z}a)$ : it has value  $-\frac{1}{2}$  at every place over zero. In the full class  $z_2 = \omega^{a/2}\sqrt{1+\omega^a}$  with  $\sqrt{1+\omega^a} \in \mathcal{N}$ , so  $z_2 \in \Pi\mathcal{N}$ , in accordance with Theorem 15.29 [merge].

**Corollary 15.88** (Unbounded nilpotence [23]). *The nilradical of  $\mathcal{B}(\mathbb{Z}a)$  is not nilpotent and not finitely generated;  $\mathcal{B}(\mathbb{Z}a)$  and  $\mathcal{N}(\mathbb{Z}a)$  are not Noetherian.*



*Proof.* The nilpotency indices of [Theorem 15.86](#) are unbounded. A nil ideal generated by  $x_1, \dots, x_s$  with  $x_i^{d_i} = 0$  has vanishing  $(1 + \sum_i (d_i - 1))$ th power, since every monomial of that degree contains some  $x_i^{d_i}$ . A quotient of a Noetherian ring is Noetherian.  $\square$

**Theorem 15.89** (Exact order after refinement [23]). *For  $n \geq 2$ ,  $m \geq 1$  and  $j \geq 1$ ,  $z_n^j \in \mathcal{I}(\mathbb{Z}a/m)$  if and only if  $mj \geq n$ . So under  $\mathcal{B}(\mathbb{Z}a) \rightarrow \mathcal{B}(\mathbb{Z}a/m)$ , induced by  $\mathcal{A}(\mathbb{Z}a) \subseteq \mathcal{A}(\mathbb{Z}a/m)$ , the residue of  $z_n$  has index  $\lceil n/m \rceil$ .*

*Proof.* At a place above  $U = 0$ , normalized by  $v(U) = 1$ , the relation  $z_n^n = U^m(1 + U^{m(n-1)})$  gives  $v(z_n^j) = mj/n$ , while every element of  $\mathcal{I}(\mathbb{Z}a/m)$  has value at least 1, as in [Theorem 15.86](#). Conversely, if  $mj \geq n$ , then  $y = z_n^j/U$  satisfies  $y^n = U^{mj-n}(1 + U^{m(n-1)})^j$ , a polynomial with constant coefficient 0 (if  $mj > n$ ) or 1 (if  $mj = n$ ), which lies in  $\mathcal{A}(\mathbb{Z}a/m)$ ; so  $y$  is integral over  $\mathcal{A}(\mathbb{Z}a/m)$  and  $z_n^j = Uy \in \mathcal{I}(\mathbb{Z}a/m)$ .  $\square$

For  $z_{12}$  and  $m = 1, 2, 3, 4, 6, 12$  the indices are 12, 6, 4, 3, 2, 1; index 1 means that the residue has become 0. Every refined workspace still has nilpotents of unbounded index ( $z_n$  with  $n$  large), so no single refinement removes all nilpotents; that each one is removed by some refinement ([Theorem 15.99](#)) has the other order of quantifiers (23).

**The residue criterion.**  $\mathcal{I}$  is in general larger than  $T\mathcal{N}(\mathbb{Z}a)$ :  $\Pi(\mathbb{Z}a)$  contains arbitrary coefficients in  $k$ , which need not lie in  $\mathcal{N}(\mathbb{Z}a)$ . The following lemma accounts for them.

**Lemma 15.90** (Scaling [23]).  $T\mathcal{B} \subseteq \mathcal{N}(\mathbb{Z}a)$ .

*Proof.* If  $b^d + p_{d-1}(T)b^{d-1} + \dots + p_0(T) = 0$  with  $p_j \in k[T]$ , multiplying by  $T^d$  gives  $(Tb)^d + Tp_{d-1}(T)(Tb)^{d-1} + \dots + T^d p_0(T) = 0$ , a monic equation with nonleading coefficients in  $Tk[T] \subseteq \mathcal{A}(\mathbb{Z}a)$ .  $\square$

**Theorem 15.91** (Residue criterion [23]). *Let  $k(T) \subseteq L \subseteq K^{\text{alg}}$  be finite and  $f \in \mathcal{B}_L$ . Then  $f \in \mathcal{N}(\mathbb{Z}a)$  if and only if  $f \bmod \mathfrak{q} \in \ell_{\mathfrak{q}}$  is integral over  $D$  for every  $\mathfrak{q} \mid T$ . In particular  $\sqrt{T\mathcal{B}_L} \subseteq \mathcal{N}(\mathbb{Z}a)$  and  $\mathcal{J} \subseteq \mathcal{N}(\mathbb{Z}a)$ .*

*Proof.* If  $f$  satisfies a monic equation over  $\mathcal{A}(\mathbb{Z}a)$ , reduce it modulo  $\mathfrak{q} \mid T$ : a coefficient  $d + Tp(T)$  becomes  $d \in D$ . Conversely, for each of the finitely many  $\mathfrak{q} \mid T$  choose a monic  $Q_{\mathfrak{q}} \in D[Y]$  killing  $f \bmod \mathfrak{q}$ , and let  $Q$  be their product; then  $Q(f) \in \mathfrak{q}$  for all  $\mathfrak{q} \mid T$ . With  $T\mathcal{B}_L = \prod_{\mathfrak{q} \mid T} \mathfrak{q}^{e_{\mathfrak{q}}}$  and  $r \geq \max e_{\mathfrak{q}}$ ,  $Q(f)^r = Tb$  with  $b \in \mathcal{B}_L$ . By [Lemma 15.90](#)  $Tb$  is integral over  $\mathcal{A}(\mathbb{Z}a)$ ,  $Y^r - Tb$  shows that  $Q(f)$  is integral over  $\mathcal{A}(\mathbb{Z}a)[Tb]$ , and  $Q(Y) - Q(f)$  is monic over  $\mathcal{A}(\mathbb{Z}a)[Q(f)]$  and kills  $f$ ; transitivity gives  $f \in \mathcal{N}(\mathbb{Z}a)$ . Elements of  $\sqrt{T\mathcal{B}_L}$  have zero residues. An element of  $\mathcal{J}$  lies in some  $\mathcal{B}_L$  with a power in  $T\mathcal{B}$ , hence in  $T\mathcal{B}_L$  by [Lemma 15.83](#).  $\square$

The proof holds for every coefficient field of characteristic 0 and every unital  $D$  in it; the real and complex hypotheses enter only in [Theorem 15.93](#), through the small dimensions of the residue fields (23).

**Remark 15.92** (A finite certificate, not an oracle [23]). The sufficiency proof yields an explicit monic certificate once the normalization and the residues are known: if  $b = Q(f)^r/T$  satisfies a monic equation of degree  $d$  over  $k[T]$ , [Lemma 15.90](#) gives a monic  $H$  of degree  $d$  over  $\mathcal{A}(\mathbb{Z}a)$  killing  $Q(f)^r$ , and  $H(Q(Y)^r)$  kills  $f$  and has degree  $dr \deg Q$ . This is a finite algebraic criterion for a represented algebraic Laurent element, not a decision procedure for arbitrary real coefficients or arbitrary normal forms: an implementation must represent  $L$ , identify the chosen branch at infinity, normalize at the



finite places and decide integrality of the residue constants, and substituting  $T = 0$  in a possibly singular defining equation need not separate the branches of its normalization (23).

**Theorem 15.93** (Coefficient span and the extended ideal [23]).  $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$  (finite  $k$ -linear span) and  $\mathfrak{l} = T\mathcal{B}$ . Hence  $\mathcal{B}(\mathbb{Z}a) = \mathcal{N}(\mathbb{Z}a)/T\mathcal{B} \hookrightarrow \mathcal{B}^{\text{co}}(\mathbb{Z}a)$ , the two fibres have the same nilradical  $\mathfrak{J}/T\mathcal{B}$ , and  $\mathcal{C}(\mathbb{Z}a) = \mathcal{N}(\mathbb{Z}a)/\mathfrak{J} \hookrightarrow \mathcal{C}^{\text{co}}(\mathbb{Z}a)$ .

*Proof.* Clearly  $k\mathcal{N}(\mathbb{Z}a) \subseteq \mathcal{B}$ . Let  $b \in \mathcal{B}_L$ . By the Chinese remainder theorem  $\mathcal{B}_L/\sqrt{T\mathcal{B}_L} \cong \prod_{\mathfrak{q}|T} \ell_{\mathfrak{q}}$ , a finite product of copies of  $\mathbb{C}$  (if  $k = \mathbb{C}$ ) or of  $\mathbb{R}$  and  $\mathbb{C}$  (if  $k = \mathbb{R}$ ). The coordinate vectors, together for  $k = \mathbb{R}$  with the vector equal to  $i$  in one complex factor and 0 elsewhere, form a  $k$ -basis whose entries are integral over  $D$ . Lift them to  $b_1, \dots, b_s \in \mathcal{B}_L$ ; they lie in  $\mathcal{N}(\mathbb{Z}a)$  by Theorem 15.91. Choose  $c_j \in k$  with  $b - \sum_j c_j b_j \in \sqrt{T\mathcal{B}_L} \subseteq \mathcal{N}(\mathbb{Z}a)$ ; so  $b \in k\mathcal{N}(\mathbb{Z}a)$ . As  $T \in \mathcal{N}(\mathbb{Z}a)$ ,  $k[T]\mathcal{N}(\mathbb{Z}a) = k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$  and  $\mathfrak{l} = Tk[T]\mathcal{N}(\mathbb{Z}a) = T\mathcal{B}$ . The injection is immediate,  $\mathfrak{J} \subseteq \mathcal{N}(\mathbb{Z}a)$  by Theorem 15.91, and nilpotence modulo  $T\mathcal{B}$  means the same in both rings; the radical of  $\mathfrak{l}$  in  $\mathcal{N}(\mathbb{Z}a)$  is  $\mathfrak{J} \cap \mathcal{N}(\mathbb{Z}a) = \mathfrak{J}$ .  $\square$

**Corollary 15.94** (Rational functions [23]).  $\mathcal{N}(\mathbb{Z}a) \cap k(T) = \overline{\mathbb{Z}}_k + Tk[T]$  and  $\mathfrak{l} \cap k(T) = Tk[T]$ , where  $\overline{\mathbb{Z}}_k$  is  $\overline{\mathbb{Z}}_{\mathbb{R}}$  for the real pair and  $\overline{\mathbb{Z}}$  for the Gaussian pair.

*Proof.*  $k[T]$  is integrally closed in  $k(T)$ , and its only place over  $T$  is  $T = 0$ , so Theorem 15.91 asks exactly that  $f(0)$  be integral over  $D$ . For the ideal use  $\mathfrak{l} = T\mathcal{B}$  and  $\mathcal{B} \cap k(T) = k[T]$ .  $\square$

For the real pair the first equality is 20's Theorem 15.60, credited by 23 and recovered as a check on the residue criterion. 20's second equality concerns the full-class ideal,  $\Pi\mathcal{N} \cap \mathbb{R}(T) = T\mathbb{R}[T]$ ; by Theorem 15.57 this is  $\mathfrak{J} \cap \mathbb{R}(T)$ , and 23's  $\mathfrak{l} \cap \mathbb{R}(T) = T\mathbb{R}[T]$  is the same set because  $T\mathbb{R}[T] \subseteq \mathfrak{l} \subseteq \mathfrak{J}$ . The Gaussian pair is new.

### The ramification filtration and exact refinement.

**Definition 15.95** (Minimum finite-place order [23]). For  $0 \neq f \in \mathcal{B}$  choose a finite  $L \ni f$  and put  $\nu_T(f) = \min_{\mathfrak{q}|T} v_{\mathfrak{q}}(f) \in \mathbb{Q}_{\geq 0}$ ; put  $\nu_T(0) = +\infty$ .

**Lemma 15.96** (Invariance and inequalities [23]).  $\nu_T(f)$  does not depend on  $L$ . For  $f, g \in \mathcal{B}$  and  $r \geq 1$ ,  $\nu_T(f + g) \geq \min\{\nu_T(f), \nu_T(g)\}$ ,  $\nu_T(fg) \geq \nu_T(f) + \nu_T(g)$  and  $\nu_T(f^r) = r\nu_T(f)$ ; and

$$f \in T^r \mathcal{B} \iff \nu_T(f) \geq r, \quad f \in \mathfrak{J} \iff \nu_T(f) > 0. \quad (15.29)$$

*Proof.* If  $L \subseteq L'$ , every  $\mathfrak{q} | T$  of  $\mathcal{B}_L$  has a prime of  $\mathcal{B}_{L'}$  above it, and with both valuations normalized at  $T$  the larger restricts to the smaller; so the set of values gains repetitions but no new values. Compare two choices in a compositum. The inequalities hold at each prime of one finite field containing both elements. The ideal criterion is (15.27); a positive minimum becomes at least 1 after multiplication by a large integer, a zero minimum never does.  $\square$

**Definition 15.97** (The truncated order and its filtration [23]). For  $\bar{f} \in \mathcal{B}(\mathbb{Z}a)$  put  $\hat{\nu}_T(\bar{f}) = \min\{1, \nu_T(f)\} \in [0, 1] \cap \mathbb{Q}$ , and for rational  $0 \leq r \leq 1$  put  $\text{Fil}^r = \{x \in \mathcal{B}(\mathbb{Z}a) : \hat{\nu}_T(x) \geq r\}$ .

Representatives differing by an element of  $T\mathcal{B}$  have the same values below 1, so  $\hat{\nu}_T$  is well defined, and  $\hat{\nu}_T(\bar{f}) = 1$  exactly when  $\bar{f} = 0$ . Moreover  $\hat{\nu}_T(x + y) \geq \min\{\hat{\nu}_T(x), \hat{\nu}_T(y)\}$  and  $\hat{\nu}_T(xy) \geq \min\{1, \hat{\nu}_T(x) + \hat{\nu}_T(y)\}$ ; the  $\text{Fil}^r$  are ideals with  $\text{Fil}^0 = \mathcal{B}(\mathbb{Z}a)$ ,  $\text{Fil}^1 = 0$ ,  $\text{Fil}^r \text{Fil}^s \subseteq \text{Fil}^{\min(1, r+s)}$  and  $\text{Nil}(\mathcal{B}(\mathbb{Z}a)) = \bigcup_{0 < r \leq 1} \text{Fil}^r$ . This is a rationally indexed filtration, not a valuation: the product inequality can be strict because different branches realize the two minima, and nontrivial complementary idempotents (constructed below) have order 0 while their product has order 1 (23).

**Corollary 15.98** (The index of any nilpotent [23]). *For  $0 \neq f \in \mathcal{N}(\mathbb{Z}a)$  with  $\nu_T(f) > 0$  the residue of  $f$  has nilpotency index  $\lceil 1/\nu_T(f) \rceil$ ; if  $\nu_T(f) = 0$  it is not nilpotent.*

*Proof.* Its  $r$ th power vanishes exactly when  $r\nu_T(f) \geq 1$ , by [Theorem 15.93](#) and (15.29).  $\square$

**Theorem 15.99** (General refinement law [23]). *For  $m \geq 1$  and  $f \in \mathcal{N}(\mathbb{Z}a)$ , with  $\nu_U$  normalized by  $v(U) = 1$ ,*

$$\nu_U(f) = m \nu_T(f), \quad f \in \mathfrak{l}(\mathbb{Z}a/m) \iff \nu_T(f) \geq \frac{1}{m}; \quad (15.30)$$

*equivalently  $\hat{\nu}_U(\bar{f}) = \min\{1, m\hat{\nu}_T(\bar{f})\}$  under  $\mathcal{B}(\mathbb{Z}a) \rightarrow \mathcal{B}(\mathbb{Z}a/m)$ . Every nilpotent residue is killed by some refinement, and a non-nilpotent residue by none.*

*Proof.* Let  $L \ni f$  be finite and  $L' = Lk(U) \subseteq K_{\mathbb{Z}a/m}$ . The integral closures of  $k[T]$  and of  $k[U]$  in  $L'$  coincide ( $k[U]$  is integral over  $k[T]$ ), every prime of  $\mathcal{B}_L$  above  $T$  extends, and the extensions are exactly the primes above  $U = 0$ . A valuation normalized by  $v(U) = 1$  sends  $T$  to  $m$ , so it is  $m$  times the old normalization on  $L$ ; every old prime occurs, so the minima scale by  $m$ . The second formula is [Theorem 15.93](#) over  $\mathbb{Z}a/m$  with (15.29). If  $\nu_T(f) > 0$  take  $m \geq 1/\nu_T(f)$ .  $\square$

This is an exact finite certificate for the disappearance of a represented nilpotent: compute its positive minimum order at the finitely many branches over zero and take a root refinement at least its reciprocal. The bound is a threshold for ideal membership, not a ramification index or a common multiple of ramification indices; for  $\mathbf{z}_n$  every branch has order  $1/n$  (23).

**The bounded-denominator Puiseux limit.** For  $m \mid m'$  the inclusions  $K_{\mathbb{Z}a/m} \subseteq K_{\mathbb{Z}a/m'}$  form a directed system. Put

$$\begin{aligned} K_{\text{Pui}} &= \bigcup_m K_{\mathbb{Z}a/m}, & \mathcal{A}_{\text{Pui}} &= \bigcup_m \mathcal{A}(\mathbb{Z}a/m), & \Pi_{\text{Pui}} &= \bigcup_m \Pi(\mathbb{Z}a/m), \\ \mathcal{N}_{\text{Pui}} &= \text{IC}_{K_{\text{Pui}}}(\mathcal{A}_{\text{Pui}}), & \mathfrak{l}_{\text{Pui}} &= \Pi_{\text{Pui}} \mathcal{N}_{\text{Pui}}. \end{aligned} \quad (15.31)$$

Each element of  $K_{\text{Pui}}$  has exponents with a bounded common denominator, so  $K_{\text{Pui}}$  is strictly smaller than the Hahn field  $K_{\mathbb{Q}a}$ :  $\sum_{n \geq 1} T^{1/n}$  has reverse well-ordered support in  $\mathbb{Q}a$  and no common denominator.

**Theorem 15.100** (Reduction by rational scales [23]).  $\mathcal{N}_{\text{Pui}} = \bigcup_m \mathcal{N}(\mathbb{Z}a/m)$ ,  $\mathfrak{l}_{\text{Pui}} = \bigcup_m \mathfrak{l}(\mathbb{Z}a/m)$  and  $\mathcal{N}_{\text{Pui}}/\mathfrak{l}_{\text{Pui}} \cong \text{colim}_m \mathcal{B}(\mathbb{Z}a/m)$ . *The ideal  $\mathfrak{l}_{\text{Pui}}$  is radical, and for each  $m$  the kernel of  $\mathcal{B}(\mathbb{Z}a/m) \rightarrow \mathcal{N}_{\text{Pui}}/\mathfrak{l}_{\text{Pui}}$  is exactly  $\text{Nil}(\mathcal{B}(\mathbb{Z}a/m))$ . Finitely many nilpotent residues are killed at one common later stage, and no finite stage is reduced.*

*Proof.* An element of  $K_{\text{Pui}}$  and the finitely many coefficients of an integrality equation over  $\mathcal{A}_{\text{Pui}}$  lie in a common stage; so do the finitely many factors of an extended-ideal expression. This gives the two unions and the colimit. If  $f^r \in \mathfrak{l}_{\text{Pui}}$ , move  $f$  and the expression to a common stage  $m$ ; the residue of  $f$  in  $\mathcal{B}(\mathbb{Z}a/m)$  is nilpotent, so [Theorem 15.99](#) (over  $\mathbb{Z}a/m$ ) kills it at a later stage and  $f \in \mathfrak{l}_{\text{Pui}}$ . A residue that dies in the colimit dies at a finite later stage, so its minimum order was positive by (15.30) and it was nilpotent. For finitely many residues take a common refinement above all thresholds. Each finite stage is a cyclic workspace and contains the family of [Theorem 15.86](#).  $\square$

*Remark 15.101* (A set-sized positive case [23]). For  $k = \mathbb{R}$ ,  $K_{\text{Pui}}$  is real closed and  $K_{\text{Pui}}[i]$  is the complex Puiseux field, algebraically closed by the Newton–Puiseux theorem [26]; the real case also follows from the characterization of real closed fields by algebraic closedness after adjoining  $i$ .  $\mathcal{A}_{\text{Pui}}$  is its truncation integer part. So radicality is restored in a set-sized real closed surreal subfield, without a proper class of scales. This concerns the bounded-denominator support class only: it decides nothing for the Hahn field  $K_{\mathbb{Q}a}$  (Question 18.21), and it does not say that the colimit fibre is isomorphic to a reduced finite-stage fibre; the kernel on each stage is known, but the image may grow (23).

*Remark 15.102* (A second route through the finite-axiom theorem [merge]). For the real pair, the field  $K_{\text{Pui}}$ , the ring  $\mathcal{A}_{\text{Pui}} = \mathbb{Z} + \Pi_{\text{Pui}}$ , the retraction  $\text{ct}$  and its kernel  $\Pi_{\text{Pui}}$  (the  $F$ ,  $A$ ,  $c$  and  $J$  of Theorem 15.51) satisfy its hypotheses:  $K_{\text{Pui}}$  is real closed and contains the real algebraic numbers; (i)  $\Pi_{\text{Pui}}$  is divisible by nonzero integers; (ii) if  $x_1, \dots, x_r \in \Pi(\mathbb{Z}a/n)$  and  $m \geq 1$ , then  $d = T^{1/(2mn)} \in \Pi_{\text{Pui}}$  and every  $x_j/d^m$  has  $T$ -exponents at least  $\frac{1}{n} - \frac{1}{2n} > 0$  with denominator dividing  $2n$ , so lies in  $\mathcal{A}_{\text{Pui}}$ ; (iii) the powers  $T^j$  are cofinal. So 19’s theorem gives the radicality of  $\text{I}_{\text{Pui}}$  for the real pair again, and more:  $\mathcal{N}_{\text{Pui}}/\text{I}_{\text{Pui}}$  is integrally closed in its rationalization and is the Boolean power  $\mathbb{Z}[\text{Idem}(\mathcal{N}_{\text{Pui}}/\text{I}_{\text{Pui}})]$ , and the fibre of the integral closure  $\mathcal{N}'$  of  $\mathcal{A}_{\text{Pui}}$  in  $K_{\text{Pui}}[i]$  is absolutely integrally closed and splits as the square of the real one (Theorems 15.30, 15.38 and 15.42 and Corollary 15.33). The Gaussian pair is covered too:  $\mathbb{Z}[i] + \Pi_{\text{Pui}}[i] = \mathcal{A}_{\text{Pui}}[i]$  is integral over  $\mathcal{A}_{\text{Pui}}$ , so its integral closure in  $K_{\text{Pui}}[i]$  is  $\mathcal{N}'$ , and its extended ideal  $\Pi_{\text{Pui}}[i]\mathcal{N}' = \Pi_{\text{Pui}}\mathcal{N}'$  is the ideal of Theorem 15.51. Its Boolean algebra is atomless of cardinality  $\mathfrak{c}$ : an idempotent of the colimit is the residue of an idempotent of some  $\text{B}(\mathbb{Z}a/m)$ , which Theorem 15.109 (over  $\mathbb{Z}a/m$ ) splits into two nonzero idempotents; these are not nilpotent and so survive by Theorem 15.100, and likewise the independent family of Theorem 15.111 survives, while  $|K_{\text{Pui}}| = \mathfrak{c}$ . This is an instance of the second clause of Question 18.17(b): real-rooted lifting gives a Boolean-power fibre here, with the monomial divisibility (ii), although set-family divisibility fails ( $T^{1/n} \in \Pi_{\text{Pui}}$  for all  $n$  and no  $T^\delta$  divides all of them in  $\Pi_{\text{Pui}}$ ). The cyclic workspace itself is outside Theorem 15.51:  $K_{\mathbb{Z}a}$  is not real closed and (ii) fails there.

**Local structure and the nilradical.** The valuation facts used are standard [36].

**Theorem 15.103** (Local structure of the coefficient normalization [23]).  *$\mathcal{B}$  is a Prüfer domain. For every maximal ideal  $\mathfrak{q} \ni T$  of  $\mathcal{B}$ , the valuation domain  $\mathcal{B}_{\mathfrak{q}}$  has value group  $\mathbb{Q}$ , normalized by  $v(T) = 1$ , and residue field isomorphic to  $\mathbb{C}$ , for both pairs; it has rank one and is not a discrete valuation ring.*

*Proof.* A maximal ideal  $\mathfrak{q}$  of  $\mathcal{B}$  contracts to a maximal ideal of  $k[T]$ , as  $\mathcal{B}$  is integral over  $k[T]$ , and with  $\mathfrak{q}_L = \mathfrak{q} \cap \mathcal{B}_L$  one has  $\mathcal{B}_{\mathfrak{q}} = \bigcup_L (\mathcal{B}_L)_{\mathfrak{q}_L}$ , a directed union of discrete valuation rings with common fraction field  $K^{\text{alg}}$ : for a nonzero element put numerator and denominator in one stage, where it or its inverse is regular. So  $\mathcal{B}_{\mathfrak{q}}$  is a valuation domain. If  $T \in \mathfrak{q}$ , its normalized value group is a union of groups  $\frac{1}{e_L}\mathbb{Z}$ , so lies in  $\mathbb{Q}$ , and  $v(\mathbf{z}_n) = 1/n$  for all  $n \geq 2$  by (15.28); so it is  $\mathbb{Q}$ , Archimedean and not discrete. The residue field is algebraic over  $k$ , the union of the finite residue fields; for  $k = \mathbb{R}$ , 20’s  $\varrho_0 = T(1 - T^{-2})^{1/2} \in \mathcal{B}$  has  $\varrho_0^2 = T^2 - 1$ , so its residue squares to  $-1$  and the residue field is  $\mathbb{C}$ .  $\square$

*Remark 15.104* (A real domain with a nonreal arithmetic fibre [23]).  $\varrho_0$  is even integral over  $\mathcal{A}(\mathbb{Z}a)$ , since the constant coefficient of  $T^2 - 1$  is  $-1 \in D$ , and its residue in  $\text{B}(\mathbb{Z}a)$  squares to  $-1$ . So the real arithmetic fibre is not formally real although  $\mathcal{N}(\mathbb{Z}a)$  lies in an ordered field;  $\text{I}$  is not order-convex, and the quotient is not a standard-part construction. 23 notes that the full-class normalization

shows the same phenomenon; its witness there is 19's  $v_T$  (Proposition 15.34), which for  $T = \omega^a$  also lies in  $K_{\mathbb{Z}a}$  [merge], while  $\varrho_0$  is a second witness inside the cyclic workspace.

**Theorem 15.105** (The nilradical is idempotent [23]).  $J^2 = J$ , as an ideal of  $\mathcal{B}$  and as an ideal of  $\mathcal{N}(\mathbb{Z}a)$ . So the common nilradical  $J/T\mathcal{B}$  of  $\mathcal{B}(\mathbb{Z}a)$  and  $\mathcal{B}^{\text{co}}(\mathbb{Z}a)$  is nonzero and equal to its square, and it is neither nilpotent nor finitely generated.

*Proof.* At a maximal ideal not containing  $T$  both  $J$  and  $J^2$  localize to the unit ideal. At  $\mathfrak{q} \ni T$  the value group is  $\mathbb{Q}$ , so  $\sqrt{T\mathcal{B}_{\mathfrak{q}}} = \mathfrak{q}\mathcal{B}_{\mathfrak{q}}$ ; for  $0 \neq b$  in it choose rational  $0 < \delta < v(b)$  and  $u$  with  $v(u) = \delta$ , so that  $b = u \cdot (b/u)$  is a product of two elements of the maximal ideal, which is therefore idempotent. Equality of ideals is local, so  $J^2 = J$  in  $\mathcal{B}$ . As  $J \subseteq \mathcal{N}(\mathbb{Z}a)$ , a coefficient from  $\mathcal{B}$  in a finite sum of products can be absorbed into one factor, which stays in  $J$ ; so the same holds over  $\mathcal{N}(\mathbb{Z}a)$ . The rest is Theorem 15.86 and Corollary 15.88.  $\square$

This does not conflict with Nakayama's lemma, since the ideal is not finitely generated, and an idempotent ideal need not consist of idempotents: here all its elements are nilpotent (23).

**Theorem 15.106** (The associated graded ring [23]). For every  $r \geq 1$ ,  $\mathfrak{l}^r = T^r\mathcal{B}$  and  $\mathfrak{l}^r/\mathfrak{l}^{r+1} \cong \mathcal{B}^{\text{co}}(\mathbb{Z}a)$  as  $\mathcal{B}(\mathbb{Z}a)$ -modules, the action being through the injection of Theorem 15.93. Multiplication gives a graded isomorphism

$$\text{gr}_1(\mathcal{N}(\mathbb{Z}a)) \cong \mathcal{B}(\mathbb{Z}a) \oplus W \mathcal{B}^{\text{co}}(\mathbb{Z}a)[W] = \{g \in \mathcal{B}^{\text{co}}(\mathbb{Z}a)[W] : g(0) \in \mathcal{B}(\mathbb{Z}a)\}, \quad (15.32)$$

and in particular  $\mathfrak{l}/\mathfrak{l}^2 \cong \mathcal{B}^{\text{co}}(\mathbb{Z}a) \neq 0$  and  $\text{Tor}_1^{\mathcal{N}(\mathbb{Z}a)}(\mathcal{B}(\mathbb{Z}a), \mathcal{B}(\mathbb{Z}a)) \cong \mathcal{B}^{\text{co}}(\mathbb{Z}a)$ .

*Proof.*  $\mathfrak{l} = T\mathcal{B}$  gives  $\mathfrak{l}^r \subseteq T^r\mathcal{B}$ , and  $T^rb = (Tb)T \cdots T$  is a product of  $r$  elements of  $\mathfrak{l}$ . Multiplication by  $T^r$  identifies  $\mathcal{B}/T\mathcal{B}$  with  $T^r\mathcal{B}/T^{r+1}\mathcal{B}$  ( $\mathcal{B}$  is a domain),  $\mathcal{N}(\mathbb{Z}a)$  acts through its image in  $\mathcal{B}/T\mathcal{B}$ , and products of these identifications multiply classes in  $\mathcal{B}^{\text{co}}(\mathbb{Z}a)$ . Tensoring  $0 \rightarrow \mathfrak{l} \rightarrow \mathcal{N}(\mathbb{Z}a) \rightarrow \mathcal{B}(\mathbb{Z}a) \rightarrow 0$  with  $\mathcal{B}(\mathbb{Z}a)$ , the map  $\mathfrak{l} \otimes \mathcal{B}(\mathbb{Z}a) \cong \mathfrak{l}/\mathfrak{l}^2 \rightarrow \mathcal{B}(\mathbb{Z}a)$  is zero, which gives the  $\text{Tor}_1$ .  $\square$

The conormal module is a further reason not to transfer identities from the full-class ideal, which satisfies  $\Pi\mathcal{N} = (\Pi\mathcal{N})^2$  (Proposition 15.31), to a cyclic workspace:  $\mathfrak{l}$  has a genuine descending sequence of powers, although the nilradical of its quotient is idempotent (23).

**Corollary 15.107** (Adic completion as a pullback [23]). With  $\widehat{\mathcal{B}} = \varprojlim_r \mathcal{B}/T^r\mathcal{B}$  and  $\widehat{\mathcal{N}} = \varprojlim_r \mathcal{N}(\mathbb{Z}a)/\mathfrak{l}^r$ ,  $\widehat{\mathcal{N}} \cong \widehat{\mathcal{B}} \times_{\mathcal{B}^{\text{co}}(\mathbb{Z}a)} \mathcal{B}(\mathbb{Z}a)$ , and  $\mathcal{B} \rightarrow \widehat{\mathcal{B}}$  and  $\mathcal{N}(\mathbb{Z}a) \rightarrow \widehat{\mathcal{N}}$  are injective.

*Proof.* Since  $T\mathcal{B} \subseteq \mathcal{N}(\mathbb{Z}a)$ , the image of  $\mathcal{N}(\mathbb{Z}a)/\mathfrak{l}^r$  in  $\mathcal{B}/T^r\mathcal{B}$  is the inverse image of  $\mathcal{B}(\mathbb{Z}a) \subseteq \mathcal{B}^{\text{co}}(\mathbb{Z}a)$ ; pass to the limit. A nonzero element of  $\mathcal{B}$  has finite  $\nu_T$ , so lies in only finitely many  $T^r\mathcal{B}$ .  $\square$

**Boolean splitting without a new scale.** For a real  $c > 0$  put

$$\varsigma_c = T(1 + cT^{-2})^{1/2}, \quad y_c = \frac{1 + \varsigma_c/\sqrt{c}}{2}, \quad \varsigma_c^2 = T^2 + c, \quad y_c^2 - y_c = \frac{T^2}{4c}. \quad (15.33)$$

The last equation is monic over  $\mathcal{A}(\mathbb{Z}a)$ , so  $y_c \in \mathcal{N}(\mathbb{Z}a)$  and  $\bar{y}_c$  is an idempotent of  $\mathcal{B}(\mathbb{Z}a)$ , for both pairs. Its nontriviality and independence from given data are proved by choosing  $c$  outside the finite branch locus of the data.

**Lemma 15.108** (A quadratic branch splitting every zero-place branch [23]). *Let  $k(T) \subseteq L \subseteq K^{\text{alg}}$  be finite. For all but finitely many real  $c > 0$ : (i)  $T^2 + c$  is not a square in  $L$ , and  $[L(\varsigma_c) : L] = 2$ ; (ii) every  $\mathfrak{q} \mid T$  of  $\mathcal{B}_L$  has extensions to  $\mathcal{B}$  at which the residue of  $\varsigma_c$  is  $\sqrt{c}$ , and extensions at which it is  $-\sqrt{c}$ ; (iii) at these, with  $v(T) = 1$ ,  $(v(y_c), v(1 - y_c))$  is  $(0, 2)$ , respectively  $(2, 0)$ .*

*Proof.* Only finitely many primes of  $k[T]$  ramify in  $L$  (Lemma 15.84); choose  $c$  so that the prime factors of  $T^2 + c$  avoid them, which excludes finitely many  $c$  (for  $k = \mathbb{R}$  the  $T^2 + c$  are distinct irreducible quadratics, for  $k = \mathbb{C}$  their pairs of roots are disjoint). Then  $T^2 + c$  has order 1 at every prime of  $\mathcal{B}_L$  above one of its factors, so it is not a square in  $L$ . Hence  $\mathcal{B}_L[\varsigma_c]$  is free over  $\mathcal{B}_L$  on  $1, \varsigma_c$ , and modulo  $\mathfrak{q} \mid T$  it is  $\ell_{\mathfrak{q}}[Y]/(Y^2 - c) \cong \ell_{\mathfrak{q}} \times \ell_{\mathfrak{q}}$ , since  $\sqrt{c} \in k$  and  $2\sqrt{c} \neq 0$ . Both primes of this algebra extend to  $\mathcal{B}$  by lying over, in set-sized integral extensions. There  $y_c$  has residue 1, respectively 0, and  $y_c(1 - y_c) = -T^2/(4c)$  fixes the other value as 2.  $\square$

Irreducibility is forced at a previously unused finite place, while the two branches occur over zero, and the Laurent expansion at infinity is kept throughout: a new Boolean coordinate with the exponent group still exactly  $\mathbb{Z}a$  (23).

**Theorem 15.109** (Uniform relative splitting [23]). *Let  $S$  be a unital subring of  $\mathbf{B}(\mathbb{Z}a)$  with  $|S| < \mathfrak{c}$ . There is a real  $c > 0$  such that, for  $e = \bar{y}_c$ , the ring homomorphism*

$$S \times S \rightarrow \mathbf{B}(\mathbb{Z}a), \quad (s_0, s_1) \mapsto s_0(1 - e) + s_1e, \quad (15.34)$$

*is injective. The same holds with  $\mathbf{B}(\mathbb{Z}a)$  replaced by  $\mathbf{C}(\mathbb{Z}a)$ ,  $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$  or  $\mathbf{C}^{\text{co}}(\mathbb{Z}a)$ .*

*Proof.* Choose a representative  $f_s \in \mathcal{N}(\mathbb{Z}a)$  of each  $s \in S$ . The finite subsets of representatives generate at most  $\max(|S|, \aleph_0) < \mathfrak{c}$  finite extensions of  $k(T)$ , each excluding finitely many  $c$  in Lemma 15.108; choose  $c > 0$  outside their union. (No regularity of  $\mathfrak{c}$  is used.) The map is a ring homomorphism because  $e$  is idempotent. Let  $(s_0, s_1) \neq (0, 0)$ , say  $s_j \neq 0$ , and put both representatives in one of the chosen fields  $L$ . As  $f_{s_j} \notin T\mathcal{B}$ , some  $\mathfrak{q} \mid T$  has  $v_{\mathfrak{q}}(f_{s_j}) < 1$ . At an extension of Lemma 15.108 where the idempotent multiplying  $s_j$  has value 0 and the other value 2, the summand of  $s_j$  has value below 1 and the other at least 2, so the sum is not in  $T\mathcal{B}$ . For  $\mathbf{C}(\mathbb{Z}a)$  a nonzero class has a branch of value exactly 0, and the sum has value 0, so it is not in  $\mathbf{J}$ . For the coefficient fibres take representatives in  $\mathcal{B}$ , whose values are still nonnegative.  $\square$

If all representatives of a subring lie in one finite  $L$ , one finite exceptional set suffices, whatever the size of the subring; this is used for countable coefficient fields in Theorem 15.118 (23).

**Corollary 15.110** (Atomlessness [23]). *The Boolean algebras  $\text{Idem}(\mathbf{B}(\mathbb{Z}a))$ ,  $\text{Idem}(\mathbf{C}(\mathbb{Z}a))$ ,  $\text{Idem}(\mathbf{B}^{\text{co}}(\mathbb{Z}a))$  and  $\text{Idem}(\mathbf{C}^{\text{co}}(\mathbb{Z}a))$  are atomless.*

*Proof.* A nonzero idempotent  $b$  generates a countable unital ring; Theorem 15.109 gives  $e$  with  $be$  and  $b(1 - e)$  nonzero, disjoint and with sum  $b$ .  $\square$

**Theorem 15.111** (Independent idempotents and cardinal invariants [23]). *The family  $(\bar{y}_c)_{c \in \mathbb{R}_{>0}}$  is Boolean independent in each of the four fibres of Corollary 15.110. Their Boolean algebras have cardinality  $\mathfrak{c}$  and order density  $\mathfrak{c}$ : no set of fewer than  $\mathfrak{c}$  nonzero idempotents has a member below every nonzero idempotent.*

*Proof.* For distinct  $c_1, \dots, c_r > 0$  the square classes of the  $T^2 + c_j$  in  $k(T)$  are independent, since a nonempty product has odd order at a prime factor of only one of them; by [Lemma 15.69](#) adjoining the  $\varsigma_{c_j}$  gives degree  $2^r$ . So  $k[T, \varsigma_{c_1}, \dots, \varsigma_{c_r}]$  is free over  $k[T]$  on the square-root monomials (the monic relations span, the degree gives independence), and modulo  $T$  it is the product of  $2^r$  copies of  $k$ , one for each assignment  $\varsigma_{c_j} \mapsto \pm\sqrt{c_j}$ . Every such prime extends to  $\mathcal{B}$ , and the corresponding Boolean word in the  $y_{c_j}$  and  $1 - y_{c_j}$  has residue 1 there, so it is not in  $J$  and is nonzero in all four fibres. As  $K^{\text{alg}}$  is algebraic over  $k(T)$ ,  $|K^{\text{alg}}| \leq |k(T)|\aleph_0 = \mathfrak{c}$ , which bounds the fibres. If  $\mathcal{D}$  were order dense of size  $< \mathfrak{c}$ , the ring  $S$  it generates has  $|S| < \mathfrak{c}$ , and a splitter  $e$  of  $S$  is nonzero while  $d(1 - e) \neq 0$  for every  $0 \neq d \in \mathcal{D}$ , so no  $d \leq e$ .  $\square$

**Corollary 15.112** (The number of zero-place branches [23]).  *$\mathcal{B}$  has exactly  $2^{\mathfrak{c}}$  maximal ideals containing  $T$ , hence at least  $2^{\mathfrak{c}}$  distinct valuation domains as in [Theorem 15.103](#).*

*Proof.* For  $\eta : \mathbb{R}_{>0} \rightarrow \{0, 1\}$ , the ideal generated by  $T$  and the elements  $y_c - \eta(c)$  is proper, since every finite subset of these relations holds at a prime of the proof of [Theorem 15.111](#); a maximal ideal containing it determines  $\eta$ . There are at most  $2^{|\mathcal{B}|} = 2^{\mathfrak{c}}$  subsets of  $\mathcal{B}$ , and distinct maximal ideals give distinct localizations.  $\square$

*Remark 15.113* (What these invariants do not determine [23]). An independent family of size  $\mathfrak{c}$  does not make the algebra free on it; uniform splitting adds one independent coordinate over a prescribed small subring and does not realize arbitrary infinite Boolean types; neither completeness nor infinite joins follow, and the isomorphism type for  $k = \mathbb{R}$  or  $\mathbb{C}$  is open ([Question 18.22](#)). The nilradical does not obstruct comparison with the reduced fibre: in a commutative ring idempotents lift uniquely across a nil ideal (if  $b^2 - b = t$  is nilpotent,  $2b - 1$  is a unit since  $(2b - 1)^2 = 1 + 4t$ , and  $b - t/(2b - 1)$  squares the error up to a unit; two idempotents differing by a nilpotent have nilpotent idempotent cross terms  $e(1 - f)$ ,  $f(1 - e)$ , hence zero ones). This standard fact is not claimed (23).

*Remark 15.114* (The Boolean-power form of the cyclic reduced fibre [merge]). For the real pair [Proposition 15.76](#) gives  $\mathbb{C}(\mathbb{Z}a) \cong \overline{\mathbb{Z}}[\text{Idem}(\mathbb{C}(\mathbb{Z}a))]$ , and by [Corollary 15.110](#) and [Theorem 15.111](#) the Boolean parameter is atomless of cardinality and order density  $\mathfrak{c}$ , with  $2^{\mathfrak{c}}$  branches over zero. The same proof of [Proposition 15.76](#) applies verbatim over the coefficient field  $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$  of [Theorem 15.118](#): it uses normality, divisibility of the ideal by nonzero integers,  $v_T$  (whose Laurent coefficients are rational) and the real algebraic integers, all present there. So the reduced fibre of the  $k_0$ -fragment is  $\overline{\mathbb{Z}}[\mathfrak{B}]$  with  $\mathfrak{B}$  the Cantor algebra, the ring of locally constant functions from Cantor space  $\{0, 1\}^{\mathbb{N}}$  to the discrete ring  $\overline{\mathbb{Z}}$ . By [Remark 15.113](#) the idempotents of the unreduced fibres are the same.

### Exact zero divisors and periodic homology.

**Theorem 15.115** (Exact annihilators in the coefficient fibre [23]). *Fix  $n \geq 2$  and let  $\bar{z}_n$  be the residue of  $z_n$  in  $\mathbb{B}^{\text{co}}(\mathbb{Z}a)$ . For  $1 \leq j < n$ ,  $\text{Ann}_{\mathbb{B}^{\text{co}}(\mathbb{Z}a)}(\bar{z}_n^j) = \bar{z}_n^{n-j}\mathbb{B}^{\text{co}}(\mathbb{Z}a)$ .*

*Proof.* The right side annihilates since  $\bar{z}_n^n = 0$ . If  $y \in \mathcal{B}$  represents an annihilator,  $yz_n^j = Tb$  with  $b \in \mathcal{B}$ ; multiplying by  $z_n^{n-j}$  and cancelling  $T$  in the domain  $\mathcal{B}$  gives  $y(1 + T^{n-1}) = bz_n^{n-j}$ , and  $1 + T^{n-1} \equiv 1$  modulo  $T\mathcal{B}$ .  $\square$

Locally, in  $\mathcal{B}_q/T\mathcal{B}_q$ , annihilation by  $\bar{z}_n^j$  means value at least  $(n - j)/n$ , which is the principal ideal of  $\bar{z}_n^{n-j}$ ; the cancellation proof needs no local-to-global theorem (23).



**Theorem 15.116** (A periodic resolution [23]). *With  $B^{\text{co}} = B^{\text{co}}(\mathbb{Z}a)$  and  $\mathcal{M}_n = B^{\text{co}}/(\bar{z}_n)$ , the complex*

$$\dots \xrightarrow{\bar{z}_n} B^{\text{co}} \xrightarrow{\bar{z}_n^{n-1}} B^{\text{co}} \xrightarrow{\bar{z}_n} B^{\text{co}} \longrightarrow \mathcal{M}_n \longrightarrow 0, \quad (15.35)$$

*with the two maps alternating, is a free resolution,  $\text{Tor}_q^{B^{\text{co}}}(\mathcal{M}_n, \mathcal{M}_n) \cong \mathcal{M}_n \neq 0$  for all  $q \geq 0$ ,  $\text{pd } \mathcal{M}_n = \text{fd } \mathcal{M}_n = \infty$ , and  $B^{\text{co}}(\mathbb{Z}a)$  has infinite global and weak global dimension.*

*Proof.* Consecutive maps compose to multiplication by  $\bar{z}_n^n = 0$ , and [Theorem 15.115](#) (for  $j = 1$  and  $j = n - 1$ ) identifies every kernel with the preceding image. Tensoring with  $\mathcal{M}_n$  makes every differential zero, so the homology is  $\mathcal{M}_n$  in every degree, and  $\mathcal{M}_n \neq 0$  because the nilpotent  $\bar{z}_n$  is not a unit of the nonzero ring  $B^{\text{co}}$ . Nonvanishing Tor in all degrees excludes every finite flat, hence every finite projective, resolution.  $\square$

Periodicity from a pair of exact zero divisors is classical and related to matrix factorizations [15]; the content here is the explicit pair in these non-Noetherian fibres, proved without a theorem that needs a regular Noetherian local ring (23).

**Proposition 15.117** (Arithmetic annihilators [23]). *Under  $B(\mathbb{Z}a) \hookrightarrow B^{\text{co}}(\mathbb{Z}a)$ , for  $1 \leq j < n$ ,*

$$\text{Ann}_{B(\mathbb{Z}a)}(\bar{z}_n^j) = \bar{z}_n^{n-j} B^{\text{co}}(\mathbb{Z}a) \subseteq B(\mathbb{Z}a), \quad \bar{z}_n^{n-j} B(\mathbb{Z}a) \subsetneq \bar{z}_n^{n-j} B^{\text{co}}(\mathbb{Z}a).$$

*So (15.35) with  $B(\mathbb{Z}a)$  in place of  $B^{\text{co}}(\mathbb{Z}a)$  is not exact.*

*Proof.* For  $b \in \mathcal{B}$  and  $1 \leq r < n$  every value of  $z_n^r b$  over zero is at least  $r/n > 0$ , so  $z_n^r b \in J \subseteq \mathcal{N}(\mathbb{Z}a)$  and  $\bar{z}_n^r B^{\text{co}}(\mathbb{Z}a) \subseteq B(\mathbb{Z}a)$ ; intersect [Theorem 15.115](#) with  $B(\mathbb{Z}a)$ . If  $\bar{z}_n^{n-j}/2 = \bar{z}_n^{n-j} u$  with  $u \in B(\mathbb{Z}a)$ , then  $u - \frac{1}{2}$  kills  $\bar{z}_n^{n-j}$  in  $B^{\text{co}}(\mathbb{Z}a)$ , so lies in  $\bar{z}_n^j B^{\text{co}}(\mathbb{Z}a)$ , and a representative of  $u$  has residue  $\frac{1}{2}$  at every place over zero; but  $\frac{1}{2}$  is integral over neither  $\mathbb{Z}$  nor  $\mathbb{Z}[i]$ , contrary to [Theorem 15.91](#).  $\square$

For  $n = 2$ : the residue of  $z_2/2$  is killed by that of  $z_2$  but is not in the principal arithmetic ideal it generates; the missing coefficient  $\frac{1}{2}$  is the obstruction. 23 computes the weak global dimension of  $B^{\text{co}}(\mathbb{Z}a)$  and deliberately leaves that of  $B(\mathbb{Z}a)$  open ([Question 18.24](#)).

**A countable fragment with a determined Boolean algebra.** Let  $k_0 = \overline{\mathbb{Q}} \cap \mathbb{R}$  (as in [Lemma 15.35](#)), a real closed field, and  $\overline{\mathbb{Q}}$ . Repeat the definitions of  $K_{\mathbb{Z}a}$ ,  $\mathcal{A}(\mathbb{Z}a)$ ,  $\mathcal{N}(\mathbb{Z}a)$ ,  $\mathcal{B}$ ,  $\mathfrak{l}$  and the four fibres with  $k$  replaced by  $k_0$  (and  $D = \mathbb{Z}$ ), respectively by  $\overline{\mathbb{Q}}$  (and  $D = \mathbb{Z}[i]$ ). These are smaller coefficient fragments, not the workspaces above; their Laurent fields are still uncountable, but the relative algebraic subfields are countable.

**Theorem 15.118** (Algebraic-coefficient fragments [23]). *In either fragment: (i)  $\mathcal{N}(\mathbb{Z}a)$  and  $\mathcal{B}$  are countable,  $\mathfrak{l} = T\mathcal{B}$ , and [Theorems 15.86](#) and [15.99](#) hold; (ii) every localization of  $\mathcal{B}$  at a maximal ideal containing  $T$  has value group  $\mathbb{Q}$  and residue field  $\overline{\mathbb{Q}}$ ; (iii) the Boolean algebras of idempotents of the four fibres are each isomorphic to the algebra of clopen subsets of Cantor space  $\{0, 1\}^{\mathbb{N}}$ , the countable atomless Boolean algebra.*

*Proof.* There are countably many polynomials over the countable field of rational functions, each with finitely many roots, so the relative algebraic closure is countable, and it contains  $\mathcal{N}(\mathbb{Z}a)$  and  $\mathcal{B}$ . The normalization and residue arguments use characteristic 0 and, for the coefficient span, that finite residue extensions are the field itself or its extension by  $i$  (real case) and the field itself (algebraically closed case); the basis vectors with entries  $0, 1, i$  are integral over  $D$ . So  $\mathfrak{l} = T\mathcal{B}$

and the criterion hold.  $z_n$  and all refinement certificates have rational coefficients;  $z_n$  supplies the values  $1/n$  and  $\varrho_0$  a square root of  $-1$ , giving the residue field  $k_0(i) = \overline{\mathbb{Q}}$ , respectively  $\overline{\mathbb{Q}}$ . To split a nonzero idempotent put a representative in a finite  $L$  and restrict the parameters of [Lemma 15.108](#) to positive rational squares, of which only finitely many are excluded. So all four algebras are countable and atomless. Two countable atomless Boolean algebras are isomorphic: an isomorphism of finite subalgebras extends to one more prescribed element, since atomlessness allows exactly the needed splits of atoms on the other side, and alternating in both directions along enumerations gives an isomorphism. The finite unions of cylinder sets of  $\{0, 1\}^{\mathbb{N}}$  form a countable atomless algebra, and by compactness they are all the clopen sets.  $\square$

The uniqueness of the countable atomless Boolean algebra is classical; the application is to these fibres, and the uncountable algebras of the full-coefficient workspaces are not classified ([Question 18.22](#)). The fragment is a concrete countable model in which nilpotence, refinement and Boolean splitting coexist and can be approached computationally (23).

### Comparison with the full class.

*Remark 15.119* (What larger scales kill [merge]). By [Theorem 15.57](#) and [Theorem 15.93](#),  $\Pi\mathcal{N} \cap K_{\mathbb{Z}a} = \sqrt{I} = J$  for the real pair. So the natural map  $B(\mathbb{Z}a) \rightarrow B_{\mathbb{R}}$  has kernel exactly the nilradical  $J/T\mathcal{B}$ , nonzero by [Theorem 15.86](#), and the radical in the second equality of (15.22) cannot be removed for  $G = \mathbb{Z}a$ ; the reduced fibre  $C(\mathbb{Z}a)$  embeds ([Corollary 15.58](#)). The comparison of ideals is:  $\Pi\mathcal{N}$  is radical with  $\Pi\mathcal{N} = (\Pi\mathcal{N})^2$  ([Theorem 15.29](#) and [Proposition 15.31](#));  $I$  is not radical and  $I/I^2 \neq 0$  ([Theorems 15.86](#) and [15.106](#)), while its radical  $J$  satisfies  $J^2 = J$  ([Theorem 15.105](#)); and  $I_{\text{Pui}}$  is radical ([Theorem 15.100](#)). So the conclusion that [Theorem 15.29](#) proves for the full class fails in a fixed workspace; it does not contradict [Theorem 15.51](#), whose hypotheses the cyclic workspace fails ([Remark 15.102](#)), and it says nothing about real closed fixed workspaces such as  $K_{\mathbb{Q}a}$  ([Question 18.21](#)).

23 records the resolution as follows: for every  $G = \mathbb{Z}a$  the extended ideal is not radical and its fibre has nilpotents of every order, which is not a classification for arbitrary ordered groups or support classes; the cyclic fibre is atomless with exact cardinality and order density  $\mathfrak{c}$ , without its isomorphism type; the residue criterion is a finite-place test for algebraic Laurent elements, not an oracle for arbitrary coefficients or Hahn input; in every cyclic workspace  $I \subsetneq \sqrt{I}$ , so removing the radical from [Theorem 15.57](#) would give a stronger, false statement there, and that theorem is compared with, not used by, the cyclic proofs; the bounded-denominator Puiseux normalization has radical extended ideal, and the full rational-exponent Hahn field is left open. Its dependencies are: finite normalization gives  $z_n$  and the exact refinement; scaling and the residue criterion give  $k\mathcal{N}(\mathbb{Z}a) = \mathcal{B}$ , hence  $I = T\mathcal{B}$ , hence the filtration and the general refinement law; the finite ramification locus gives the fresh quadratic splitting and the Boolean results; and  $I = T\mathcal{B}$  gives the exact annihilators and the periodic homology over  $B^{\text{co}}(\mathbb{Z}a)$ , with deliberately no arrow to  $B(\mathbb{Z}a)$ , which [Proposition 15.117](#) disproves. All rings and modules in these proofs are sets; the surreal  $a$  only names the exponent group; composita, primes, localizations and branches are taken in set-sized rings and fields, with ordinary choice (the maximal-ideal theorem) and no class Zorn lemma, class module category or measurable cardinal. Neither the continuum hypothesis nor regularity of  $\mathfrak{c}$  is used. The Newton–Puiseux theorem is used only to identify the limit field as real closed or algebraically closed; the reducedness of its fibre does not need it (23).

### 15.11 Henselian branching at an omnific augmentation

This subsection is manuscript 27’s (Section 1.3). It moves 20’s splitters from the normalization fibre into étale neighbourhoods. For every infinite cardinal  $\kappa$  and every set  $S_0 \subseteq \mathbf{No}$  of at most  $\kappa$  surreals, 27 builds a real closed field  $\mathbb{E} \supseteq S_0$  of cardinality  $\kappa$  whose *actual* omnific integer part  $B_{\mathbb{E}} = \mathbb{E} \cap \mathbf{Oz}$  has an idempotent constant-term ideal, so that its local ring  $\mathcal{L}$  at that ideal has residue field  $\mathbb{Q}$ ,  $\mathfrak{n}^2 = \mathfrak{n}$  and completion  $\mathbb{Q}$ ; yet the henselization  $\mathcal{L}^h$  has exactly  $2^\kappa$  minimal primes, the most a ring of cardinality  $\kappa$  can have, each with fraction field  $\mathbb{E}[i]$ , and its generic fibre is the algebra  $\mathrm{LC}(X, \mathbb{E}[i])$  of locally constant functions on a Stone space with  $|X| = 2^\kappa$  (Theorem 15.120). The branches are forced to be complex, the completion forgets all of them, strict henselization adds a factor  $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$  and the Gaussian ring doubles the generic fibre (Theorem 15.121); finite projectives, finite étale algebras and the centred Riemann–Zariski space are determined (Theorem 15.140, Proposition 15.141, and Corollary 15.142). All rings, spectra and henselizations are sets. No numbered question of this report is answered.

27 pins fc27d87 (23 September 2026), whose text of this report is again blob ef7aef1 (the sixth merge, without 23’s Section 15.10 and without 24’s to 26’s subsections), and re-fetched there the file 20-boolean-branching-PROOF\_AUDIT.md (blob feea0ee), from which it reconstructs 20’s splitter; it credits the constant-term retraction, monomial clearing, idempotence of the tail and the Boolean branching mechanism to the repository. Printed once, under the existing labels, are: its two monomial operations (its Lemma 2.1: Lemma 3.1 and Proposition 3.3), its omnific floor (its Lemma 2.2: Lemma 15.8), its support projection and dominant scale (its Lemmas 2.3, 2.5: Lemmas 15.55 and 15.62), its monic certificate (its Lemma 3.1: Lemma 15.53), its nontriviality and integer-torsion lemma (its Lemma 3.2: Lemma 15.28 with Theorem 15.29), its quadratic splitter (its Lemma 3.3: Theorem 15.65, the same element  $w_t$  of (15.24)) and its independent splitters (its Proposition 3.5: Theorem 15.72 for  $\alpha < \kappa$ , the same  $t_\alpha = \omega^{\omega^\alpha}$ ); 27 is credited as an independent re-derivation, and its proof of the splitter through the quadratic subalgebra alone, which it calls “why one radical suffices”, is kept as a second route (Remark 15.122). 27 works with  $\sqrt{\Pi N}$  and says it does not need  $\Pi N$  to be radical; this report proves that it is (Theorem 15.29), so 27’s  $\sqrt{\Pi N}$  is  $\Pi N$  [merge]. Its hull (Proposition 15.123) has a near-antecedent in 08’s  $A_\kappa^{08} = F_\kappa^{08} \cap \mathbf{Oz}$  (Theorem 11.16), which 27 does not cite; 27 adds a prescribed seed, the equality  $\mathrm{Frac} B_{\mathbb{E}} = \mathbb{E}$  and the factor pairs giving an idempotent ideal. Its local ring (Corollary 15.124) is the set-sized analogue of  $\mathbf{Oz}_\Pi$  of the Diophantine report (odg:frac:prop:localization [47]), also local with residue field  $\mathbb{Q}$ , and its forced complexification (Proposition 15.128) is the henselian counterpart of that report’s odg:ex:hensel:  $T^2 - T - \omega$  has simple residue roots and no root in  $\mathbf{Oz}$ ; henselization supplies roots, and the branches over the generic point are forced complex. 27’s SOURCES\_AND\_SCOPE says that a repository search for “hensel” returned no match and that this is not an exhaustive absence proof. *Correction:* at its pin the word occurs in sixteen article.tex files, among them the Diophantine report (odg:ex:hensel, “Hensel-style intuition fails for  $\mathbf{Oz} \rightarrow \mathbb{Z}$ ”) and this report (the bibliography entry [22]); the construction here is not among them. 23’s cyclic workspace (Section 15.10) is a different ring; the branch counts are compared in Remark 15.143. 27 credits as standard the henselization and permanence results of the Stacks Project [74, 75, 76, 77], whose statements it checked to need no Noetherian hypothesis, and imports de Felipe–Teissier’s theorem [78] for the valuation components. The numbering of 27 cited here is that of its own manuscript.

**Notation and renamings.** Throughout,  $\kappa$  is an infinite cardinal and  $S_0 \subseteq \mathbf{No}$  a set with  $|S_0| \leq \kappa$ . Exponents follow [Section 2.2](#) (27's growth convention). The rings are

$$B_{\mathbb{E}} = \mathbb{E} \cap \mathbf{Oz}, \quad \Pi_{\mathbb{E}} = \ker(\text{ct}|_{B_{\mathbb{E}}}) = B_{\mathbb{E}} \cap \Pi, \quad \mathfrak{L} = (B_{\mathbb{E}})_{\Pi_{\mathbb{E}}}, \quad \mathfrak{n} = \Pi_{\mathbb{E}} \mathfrak{L}, \quad \mathcal{G} = \mathbb{E} \otimes_{\mathfrak{L}} \mathfrak{L}^{\text{h}}, \quad (15.36)$$

with  $\mathfrak{L}^{\text{h}}$  and  $\mathfrak{L}^{\text{sh}}$  the henselization and strict henselization. 27's symbols are renamed as follows.

- Its field  $E$ , seed  $\mathcal{S}$ , integer part  $B$ , ideal  $J$ , local ring  $A$  with maximal ideal  $\mathfrak{m}$ , henselization  $H$ , field  $C = E[i]$  and generic fibre  $G = E \otimes_A H$  are  $\mathbb{E}$ ,  $S_0$ ,  $B_{\mathbb{E}}$  (a cofinal discretely ordered subring, as in [Section 16.11](#)),  $\Pi_{\mathbb{E}}$ ,  $\mathfrak{L}$  with  $\mathfrak{n}$ ,  $\mathfrak{L}^{\text{h}}$ ,  $\mathbb{E}[i]$  and  $\mathcal{G}$  ( $E$  is the ambient field and  $\mathcal{E}_a$  07's field;  $\mathfrak{m}$  is the negative-support ideal;  $H$  is a group or  $H_a$ ;  $G$  is a set-sized subgroup;  $C(G)$  a reduced fibre;  $A$  a ring such as  $\mathcal{A}_{D,k}$  or  $A_{\kappa}^{\text{08}}$ ;  $J$  and  $\mathbf{J}$  are other ideals). Its Stone space  $X$  is  $\mathbf{X}$  ( $X^g$  are monomials), and its  $\mathcal{G}_{\mathbb{Q}}$  is  $\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ .
- Its  $F_{\Gamma}, O_{\Gamma}, \Pi_{\Gamma}, \pi_{\Gamma}$  are  $K_G, \mathcal{A}(G), \Pi(G)$  and  $\text{pr}_G$  of [Section 15.9](#); its  $\mathcal{N}$  is this report's  $\mathcal{N}$  (the same letter), and its  $\mathcal{K} = \sqrt{\Pi \mathcal{N}}$  is  $\Pi \mathcal{N}$  ([Theorem 15.29](#)); its integral closures  $N_B, N_A$  of  $B, A$  in  $E$  are  $\text{IC}_{\mathbb{E}}(B_{\mathbb{E}})$  and  $\text{IC}_{\mathbb{E}}(\mathfrak{L})$ .
- Its polynomial  $D(T) = (T^2 - 1)(T^2 - 4)$  is  $\mathbf{q}(T)$  ( $D$  is the constant ring), its  $a(T) = (T^4 - 5T^2)/16$  and  $w(T)$  are  $\mathbf{a}(T)$  and  $\mathbf{w}(T)$ , so  $\mathbf{w}(t) = \mathbf{w}_t$  of [\(15.24\)](#); its  $T_{\alpha} = \omega^{b_{\alpha}}, b_{\alpha} = \omega^{\alpha}$ , are 20's  $t_{\alpha}$  (here  $T = \omega^a$  elsewhere), its  $a_{\alpha} = \mathbf{a}(T_{\alpha})$  are  $\mathbf{a}_{\alpha}$  ( $a_{\alpha}$  are the small scales of [\(3.1\)](#)), and its  $w_{\alpha}$  are  $\mathbf{w}_{t_{\alpha}}$ .
- Its roots  $u_{\alpha} \in \mathfrak{m}H$ , generic idempotents  $e_{\alpha}$  and imaginary unit  $j$  are  $\tilde{u}_{\alpha}, \hat{e}_{\alpha}$  and  $j$  ( $u_n$  and  $e_{\alpha}$  are 21's and 20's elements,  $e_{\alpha} = \mathbf{e}_{t_{\alpha}}$  being a residue in  $B_{\mathbb{E}}$ ); its pointed neighbourhoods  $D_0, D = (D_0)_{\mathfrak{n}}$  are  $\mathfrak{U}_{F,0}$  and  $\mathfrak{U}_F$  with residue-point ideal  $\mathfrak{u}$ ; its minimal primes  $P$  are  $\mathfrak{Q}$ ; its  $h$  with  $h > 1/4$  is written  $h$ .
- In the hull, its clearing pairs  $d_r, d_r r$  and factor pairs  $e_x, x/e_x$  are  $m_r, m_r r$  and  $f_x, x/f_x$ , and its  $R_n$  are  $\mathbb{E}_n$ .

Four tempting readings are false. (1)  $\mathfrak{n}$  is not the ideal of order-infinitesimals: it contains the positive infinite  $\omega$ , so the local topology is not the valuation topology of  $\mathbb{E}$ , and henselian lifting need not respect the order. (2)  $\mathfrak{L}^{\text{h}}$  is local and has no nontrivial idempotents; the  $2^{\kappa}$  branches live in  $\mathcal{G}$ , and  $\mathfrak{L}^{\text{h}}$  is not a product of branches. (3)  $j^2 = -1$  holds in  $\mathcal{G}$ ;  $\mathfrak{L}^{\text{h}}$  contains no square root of  $-1$ , since its residue field is  $\mathbb{Q}$ . (4) The specializations  $T \mapsto 0$  act on finite algebras, not on Laurent series, and projection is linear, not a ring map ([Remarks 15.56](#) and [15.64](#)).

**Theorem 15.120** (Maximal branching at an omnific augmentation [27]). *There is a real closed field  $\mathbb{E} \subseteq \mathbf{No}$  with the following properties.*

- (i)  $S_0 \subseteq \mathbb{E}$ ;  $|B_{\mathbb{E}}| = |\mathbb{E}| = |\mathfrak{L}| = |\mathfrak{L}^{\text{h}}| = \kappa$ ;  $B_{\mathbb{E}}$  is an integer part of  $\mathbb{E}$  with  $\text{Frac } B_{\mathbb{E}} = \mathbb{E}$ ;  $\Pi_{\mathbb{E}}$  and  $\mathfrak{n}$  are nonzero and idempotent;  $\mathfrak{L}$  is a local domain with residue field  $\mathbb{Q}$ .
- (ii)  $\mathfrak{L}^{\text{h}}$  is reduced, local and faithfully flat over  $\mathfrak{L}$ , with maximal ideal  $\mathfrak{n}\mathfrak{L}^{\text{h}}$  and residue field  $\mathbb{Q}$ ;  $\widehat{\mathfrak{L}}_{\mathfrak{n}} = \mathbb{Q} = \widehat{\mathfrak{L}^{\text{h}}}$  and  $\mathfrak{n}/\mathfrak{n}^2 = 0$ .
- (iii)  $\mathcal{G}$  contains an  $\mathbb{E}$ -algebra copy of  $\mathbb{E}[i]$ , and for the copy of [Proposition 15.128](#) there is an  $\mathbb{E}[i]$ -algebra isomorphism  $\mathcal{G} \cong \text{LC}(\mathbf{X}, \mathbb{E}[i])$  with  $\mathbf{X}$  a Stone space,  $|\mathbf{X}| = 2^{\kappa}$ ,  $\dim_{\mathbb{E}[i]} \mathcal{G} = \kappa$ , where LC means locally constant functions to the discrete field (so of finite image);  $\text{Idem}(\mathcal{G})$  contains a free Boolean algebra on  $\kappa$  generators.
- (iv)  $|\text{Min } \mathfrak{L}^{\text{h}}| = 2^{\kappa}$ ,  $\text{Frac}(\mathfrak{L}^{\text{h}}/\mathfrak{Q}) \cong_{\mathbb{E}} \mathbb{E}[i]$  for every minimal prime  $\mathfrak{Q}$ , and there is no  $\mathfrak{L}$ -algebra map from  $\mathfrak{L}^{\text{h}}$  to  $\mathbb{E}$  or to any ordered extension field of  $\mathbb{E}$ .

No regularity of  $\kappa$ , cardinal-arithmetic hypothesis or large cardinal is used.

“Maximal” is exact: a ring of cardinality  $\kappa$  has at most  $2^\kappa$  ideals; it does not mean that  $\mathbf{X}$  is classified (27). The proof occupies the rest of the subsection (Propositions 15.123, 15.128 and 15.131 to 15.133 and Corollary 15.124).

**Theorem 15.121** (Arithmetic and Gaussian factors [27]). Choose  $\overline{\mathbb{Q}} \hookrightarrow \mathbb{E}[i]$ . Then  $\mathfrak{L}^{\text{sh}} \cong \mathfrak{L}^{\text{h}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$  and  $\mathbb{E} \otimes_{\mathfrak{L}} \mathfrak{L}^{\text{sh}} \cong \text{LC}(\mathbf{X} \times \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \mathbb{E}[i])$ , the Galois group acting on the second factor with fixed ring  $\mathcal{G}$ . Also  $\mathfrak{L}[i]$  is local with residue field  $\mathbb{Q}(i)$ ,  $(\mathfrak{L}[i])^{\text{h}} \cong \mathfrak{L}^{\text{h}}[i]$ , and  $\mathbb{E}[i] \otimes_{\mathfrak{L}[i]} (\mathfrak{L}[i])^{\text{h}} \cong \mathcal{G} \times \mathcal{G}$  as rings and  $\mathbb{E}$ -algebras, the two factors carrying conjugate  $\mathbb{E}[i]$ -structures; the Gaussian generic spectrum is  $\mathbf{X} \sqcup \mathbf{X}$ .

This is Propositions 15.135 and 15.138 and Theorems 15.137 and 15.139 (27).

### The splitter, a second route.

*Remark 15.122* (One radical suffices [27]). Put  $\mathbf{q}(T) = (T^2 - 1)(T^2 - 4)$ ,  $\mathbf{a}(T) = (T^4 - 5T^2)/16$  and  $\mathbf{w}(T) = \frac{1}{2} - \frac{1}{4}\sqrt{\mathbf{q}(T)}$  (positive root). Since  $\mathbf{q}(t) = 4(t^2 - 1)(t^2/4 - 1)$ ,  $\mathbf{w}(t) = \mathbf{w}_t$  of (15.24), and  $\mathbf{w}(T)^2 - \mathbf{w}(T) = \mathbf{a}(T)$ . 27 proves Theorem 15.65 with the quadratic subalgebra instead of 20’s biquadratic one:  $\mathbf{q}$  has the four simple zeros  $\pm 1, \pm 2$ , so it is not a square in  $k(T)$  ( $k = K_G$ ) and  $k[T, W]/(W^2 - W - \mathbf{a}(T)) \rightarrow k((T^{-1}))$ ,  $W \mapsto \mathbf{w}(T)$ , is injective; the two  $k$ -algebra specializations  $T \mapsto 0, W \mapsto 0$  and  $T \mapsto 0, W \mapsto 1$  exist because  $\mathbf{a}(0) = 0$ . A monic certificate for  $x\mathbf{w} \in \Pi\mathcal{N}$ , projected by Lemma 15.55 and read in this finite algebra through Lemma 15.62, specializes under  $W \mapsto 1$  to a certificate for  $x$ ; for  $x(1 - \mathbf{w})$  use  $W \mapsto 0$ . The Laurent expansion is

$$\mathbf{w}(T) = -\frac{T^2}{4} + \frac{9}{8} + \frac{9}{32}T^{-2} + \frac{45}{64}T^{-4} + \frac{981}{512}T^{-6} + \frac{5715}{1024}T^{-8} + O(T^{-10}),$$

an expansion in  $T^{-1}$ , not a series for small  $T$ ; its coefficient  $\frac{9}{8}$  shows that integral elements need not be omnific, and the specializations are not evaluations of it (27). In the independent family of Theorem 15.72, for  $\alpha < \kappa$  the support of  $\mathbf{w}_{t_\beta}$  lies in  $\mathbb{Z}\omega^\beta$ ,  $\omega^\alpha$  dominates the group generated by the earlier  $\omega^\beta$ , and applying the splitter in increasing order keeps every finite atom  $\prod_{\alpha \in F} \mathbf{w}_{t_\alpha}^{\epsilon_\alpha} (1 - \mathbf{w}_{t_\alpha})^{1 - \epsilon_\alpha}$  outside  $\Pi\mathcal{N}$ ; the family is defined by a formula, with no class ultrafilter (27, as 20).

### The omnific hull.

**Proposition 15.123** (A cardinal-preserving omnific hull [27]). There are a real closed field  $\mathbb{E} \subseteq \mathbf{No}$  and  $B_{\mathbb{E}} = \mathbb{E} \cap \mathbf{Oz}$  with:  $S_0 \subseteq \mathbb{E}$ ;  $|B_{\mathbb{E}}| = |\mathbb{E}| = \kappa$  and  $B_{\mathbb{E}}$  an integer part of  $\mathbb{E}$  with  $\text{Frac } B_{\mathbb{E}} = \mathbb{E}$ ; every  $t_\alpha$  and  $\mathbf{a}_\alpha$  ( $\alpha < \kappa$ ) in  $B_{\mathbb{E}}$ ; and every nonzero element of  $\Pi_{\mathbb{E}}$  a product of two elements of  $\Pi_{\mathbb{E}}$ , so that  $\Pi_{\mathbb{E}}^2 = \Pi_{\mathbb{E}} \neq 0$ .

*Proof.* For  $0 \neq s \in S_0$  choose a monomial  $m_s \in \Pi$  with  $m_s s \in \Pi$  (Lemma 3.1). Let  $B_0$  be the subring of  $\mathbf{Oz}$  generated by  $\mathbb{Z}$ , all  $t_\alpha, \mathbf{a}_\alpha$  and all  $m_s, m_s s$  ( $\mathbf{a}_\alpha$  is purely infinite, so omnific despite the coefficient  $\frac{1}{16}$ );  $|B_0| = \kappa$ . Given  $B_n$ , let  $\mathbb{E}_n$  be the relative real algebraic closure of  $\text{Frac } B_n$  in  $\mathbf{No}$ , of cardinality  $\kappa$ , and adjoin to  $B_n$ :  $\lfloor r \rfloor$  for every  $r \in \mathbb{E}_n$  (Lemma 15.8); a clearing pair  $m_r, m_r r \in \Pi$  for every  $0 \neq r \in \mathbb{E}_n$ ; and a factor pair  $f_x, x/f_x \in \Pi$  for every  $0 \neq x \in B_n \cap \Pi$  (Proposition 3.3). Put  $B_{\mathbb{E}} = \bigcup_n B_n$  and  $\mathbb{E} = \text{Frac } B_{\mathbb{E}} = \bigcup_n \text{Frac } B_n$ . The clearing pairs give  $\mathbb{E}_n \subseteq \text{Frac } B_{n+1}$ , so  $\mathbb{E}$  is real closed, and  $S_0 \subseteq \mathbb{E}$ . Every  $r \in \mathbb{E}$  has  $\lfloor r \rfloor \in B_{\mathbb{E}}$ , so  $B_{\mathbb{E}}$  is an integer part, and  $r \in \mathbf{Oz}$  equals its floor, so  $\mathbb{E} \cap \mathbf{Oz} \subseteq B_{\mathbb{E}}$ . The factor pairs give  $\Pi_{\mathbb{E}} \subseteq \Pi_{\mathbb{E}}^2$ , and  $\omega = t_0 \in \Pi_{\mathbb{E}}$ . Countable unions keep the cardinal  $\kappa$ , even for singular  $\kappa$ .  $\square$



A convenient subring of  $\mathbf{Oz}$  need not be the integer part of its fraction field; closure under the floor makes  $B_{\mathbb{E}} = \mathbb{E} \cap \mathbf{Oz}$  an equality, and the monomial operations are performed in  $\mathbf{No}$ , before the small hull is formed, because a fixed Hahn group need not admit fresh bounds (27).  $08$ 's integer part  $A_{\kappa}^{08} = F_{\kappa}^{08} \cap \mathbf{Oz}$  (Theorem 11.16) is built in the same spirit, for truncation and support control, without a seed [merge].

**Corollary 15.124** (The local augmentation ring [27]).  $\Pi_{\mathbb{E}}$  is prime,  $\mathcal{L} \subseteq \mathbb{E}$ ,  $\text{Frac } \mathcal{L} = \mathbb{E}$ ,  $\mathcal{L}/\mathfrak{n} = \mathbb{Q}$ ,  $|\mathcal{L}| = \kappa$  and  $\mathfrak{n}^2 = \mathfrak{n} \neq 0$ .

*Proof.*  $\text{ct} : B_{\mathbb{E}} \rightarrow \mathbb{Z}$  is onto with kernel  $\Pi_{\mathbb{E}}$ ; localizing inverts the nonzero integers, giving  $\mathbb{Q}$ ; localize  $\Pi_{\mathbb{E}}^2 = \Pi_{\mathbb{E}}$ .  $\square$

**From normalization to henselian branches.** Put  $\Sigma_0 = B_{\mathbb{E}} \setminus \Pi_{\mathbb{E}}$ . Then  $\text{IC}_{\mathbb{E}}(\mathcal{L}) = \Sigma_0^{-1} \text{IC}_{\mathbb{E}}(B_{\mathbb{E}})$ : an element integral over  $\Sigma_0^{-1} B_{\mathbb{E}}$  times one common denominator of its monic equation is integral over  $B_{\mathbb{E}}$ . Every  $w_{t_{\alpha}}$  lies in  $\mathbb{E}$  (real closed and containing  $t_{\alpha}$ ) and in  $\text{IC}_{\mathbb{E}}(B_{\mathbb{E}})$ .

**Lemma 15.125** (Finite atoms survive local normalization [27]). For finite  $F \subseteq \kappa$  and  $\epsilon \in \{0, 1\}^F$ ,  $p_{\epsilon} = \prod_{\alpha \in F} w_{t_{\alpha}}^{\epsilon_{\alpha}} (1 - w_{t_{\alpha}})^{1 - \epsilon_{\alpha}}$  is a nonzero idempotent in  $\text{IC}_{\mathbb{E}}(\mathcal{L})/\sqrt{\mathfrak{n} \text{IC}_{\mathbb{E}}(\mathcal{L})}$ .

*Proof.* The quadratic equations make the  $w_{t_{\alpha}}$  idempotent modulo  $\mathfrak{n} \text{IC}_{\mathbb{E}}(\mathcal{L})$ . If  $p_{\epsilon}$  were in the radical, clearing denominators gives  $sp_{\epsilon}^q \in \Pi_{\mathbb{E}} \text{IC}_{\mathbb{E}}(B_{\mathbb{E}}) \subseteq \Pi \mathcal{N}$  with  $s \in \Sigma_0$ ; with  $s = n + z$ ,  $n = \text{ct}(s) \neq 0$ ,  $z \in \Pi$ , also  $zp_{\epsilon}^q \in \Pi \mathcal{N}$ , so  $np_{\epsilon}^q \in \Pi \mathcal{N}$  and  $p_{\epsilon} \in \Pi \mathcal{N}$  (Lemma 15.28 and Theorem 15.29), against Remark 15.122.  $\square$

$\mathcal{L}^{\text{h}}$  is henselian, local, faithfully flat and ind-étale over  $\mathcal{L}$  with the same residue field, without Noetherian hypotheses [74, 75]. For each  $\alpha$  there is a unique  $\tilde{u}_{\alpha} \in \mathfrak{n} \mathcal{L}^{\text{h}}$  with  $\tilde{u}_{\alpha}^2 - \tilde{u}_{\alpha} - \mathfrak{a}_{\alpha} = 0$  (simple root 0 modulo  $\mathfrak{n}$ , derivative  $-1$ ). For finite  $F = \{\alpha_1, \dots, \alpha_n\}$  put  $\mathfrak{U}_{F,0} = \mathcal{L}[U_1, \dots, U_n]/(U_j^2 - U_j - \mathfrak{a}_{\alpha_j})$ ,  $\mathfrak{u} = (\mathfrak{n}, U_1, \dots, U_n)$  and  $\mathfrak{U}_F = (\mathfrak{U}_{F,0})_{\mathfrak{u}}$ : essentially étale over  $\mathcal{L}$  (the Jacobian entries  $2U_j - 1$  are units at  $\mathfrak{u}$ ), residue field  $\mathbb{Q}$ , and  $\mathfrak{U}_F \rightarrow \mathcal{L}^{\text{h}}$ ,  $U_j \mapsto \tilde{u}_{\alpha_j}$ , identifies  $\mathcal{L}^{\text{h}}$  with  $\mathfrak{U}_F^{\text{h}}$ , so it is faithfully flat [74, 75].

**Lemma 15.126** (Every finite generic branch survives localization [27]).  $\mathbb{E} \otimes_{\mathcal{L}} \mathfrak{U}_F \cong \mathbb{E}^{\{0,1\}^F}$ , and the map from this product into  $\mathcal{G}$  is injective.

*Proof.* Before localizing, the Chinese remainder theorem gives  $\mathbb{E} \otimes \mathfrak{U}_{F,0} \cong \prod_{\{0,1\}^F} \mathbb{E}$ , each  $U_j$  evaluated at  $w_{t_{\alpha_j}}$  or  $1 - w_{t_{\alpha_j}}$ , distinct because  $1 - 2w = \sqrt{q}/2 \neq 0$ . Fix  $v_j \in \{w_{t_{\alpha_j}}, 1 - w_{t_{\alpha_j}}\}$ . By Lemma 15.125 the product  $\prod_j (1 - v_j)$  is nonzero in the set-sized ring  $R = \text{IC}_{\mathbb{E}}(\mathcal{L})/\sqrt{\mathfrak{n} \text{IC}_{\mathbb{E}}(\mathcal{L})}$ , so some prime of  $R$  omits it; in its residue domain the idempotents  $1 - v_j$  are 1 and the  $v_j$  are 0. Lifting gives a prime  $\mathfrak{p} \supseteq \mathfrak{n} \text{IC}_{\mathbb{E}}(\mathcal{L})$  containing every  $v_j$ , contracting to  $\mathfrak{n}$ . A representative of  $g \in \mathfrak{U}_{F,0} \setminus \mathfrak{u}$  has nonzero residue  $g(0, \dots, 0) \in \mathbb{Q}$ , and its value at  $(v_j)$  reduces modulo  $\mathfrak{p}$  to the same residue, so it is nonzero in  $\mathbb{E}$ : every inverted element is nonzero on every factor, and localization keeps the product. Faithful flatness survives base change, so the map to  $\mathcal{G}$  is injective.  $\square$

A split generic polynomial need not keep every branch after localization at one residue point: a denominator outside the residue prime may vanish on a whole generic component. The prime chosen above excludes this for every denominator and every finite branch pattern; this is 27's central new step, and the reason the support certificates are needed (27).



**Proposition 15.127** (A free Boolean family in the generic fibre [27]). *The elements*

$$\hat{e}_\alpha = \frac{\tilde{u}_\alpha - \mathbf{w}_{t_\alpha}}{1 - 2\mathbf{w}_{t_\alpha}} \in \mathcal{G} \quad (\alpha < \kappa) \quad (15.37)$$

are idempotents freely generating a Boolean subalgebra of  $\text{Idem}(\mathcal{G})$ .

*Proof.* Over  $\mathbb{E}$ ,  $U^2 - U - \mathbf{a}_\alpha = (U - \mathbf{w}_{t_\alpha})(U - 1 + \mathbf{w}_{t_\alpha})$ , and the affine change (15.37) sends the roots to 0, 1. By Lemma 15.126 every finite root assignment is a factor of a product injecting into  $\mathcal{G}$ , and each Boolean atom is 1 on one factor.  $\square$

The  $\mathbf{w}_{t_\alpha}$  are available only after passing to the generic fibre, and the  $\hat{e}_\alpha$  are not idempotents of the local ring  $\mathcal{L}^h$  (27). For  $\alpha = 0, 1$ ,  $t_0 = \omega$ ,  $t_1 = \omega^\omega$ , the four atoms  $\mathbf{w}_\omega \mathbf{w}_{\omega^\omega}, \dots, (1 - \mathbf{w}_\omega)(1 - \mathbf{w}_{\omega^\omega})$  survive modulo  $\Pi\mathcal{N}$  (the support of  $\mathbf{w}_\omega$  is in  $\mathbb{Z}$  and  $\omega$  dominates the integers), and the four orthogonal idempotents  $\hat{e}_0 \hat{e}_1, \dots, (1 - \hat{e}_0)(1 - \hat{e}_1)$  of  $\mathcal{G}$  sum to 1 although the neighbourhood and the henselization are local; the roots  $\tilde{u}_0, \tilde{u}_1$  reducing to 0 are not the real roots  $\mathbf{w}_{t_0}, \mathbf{w}_{t_1}$  (27).

### Forced complexification and the Boolean power.

**Proposition 15.128** (Forced complexification [27]). *Let  $A$  be a local domain with real closed fraction field  $\mathbb{F}$  and maximal ideal  $\mathfrak{m}_A$ , and  $h \in \mathfrak{m}_A$  with  $h > \frac{1}{4}$  in  $\mathbb{F}$ . Then  $\mathbb{F} \otimes_A A^h$  contains an  $\mathbb{F}$ -algebra copy of  $\mathbb{F}[i]$ , and no  $A$ -algebra map takes  $A^h$  to an ordered extension field of  $\mathbb{F}$ .*

*Proof.*  $U^2 - U + h$  reduces to  $U(U - 1)$ , so henselization gives a root  $u \in \mathfrak{m}_A A^h$ . With  $c = \sqrt{4h - 1} > 0$  in  $\mathbb{F}$  put  $j = (2u - 1)/c$ ; then  $j^2 = (4u^2 - 4u + 1)/(4h - 1) = -1$ . The generic fibre is nonzero by faithful flatness, so  $\mathbb{F}[i] \rightarrow \mathbb{F} \otimes_A A^h$ ,  $i \mapsto j$ , is injective. An  $A$ -algebra map to an ordered field extending  $\mathbb{F}$  would send  $u$  to a root of a quadratic with negative discriminant.  $\square$

Take  $h = \omega$ : algebraically small for  $\mathfrak{n}$ , order-theoretically large in  $\mathbb{E}$ . The root lands in the new algebra, not in  $\mathbb{E}$ , so this is no failure of real closedness; and  $j$  lives in  $\mathcal{G}$ , where division by  $c \in \mathbb{E}^\times$  is allowed, not in  $\mathcal{L}^h$  (27). In  $\mathbf{Oz}$  itself  $T^2 - T - \omega$  has simple residue roots 0, 1 and no root (`odg:ex:hensel`); here the henselization supplies them.

**Lemma 15.129** (Boolean powers from finite products [27]). *Let  $\mathbb{F}$  be real closed,  $\mathcal{A}$  a nonzero ind-étale  $\mathbb{F}$ -algebra with a chosen embedding  $\mathbb{F}[i] \hookrightarrow \mathcal{A}$ . Then  $\mathcal{A}$  is the directed union of finite  $\mathbb{F}[i]$ -subalgebras isomorphic to finite products of  $\mathbb{F}[i]$ ; it is reduced and von Neumann regular, every prime is maximal and minimal with residue field  $\mathbb{F}[i]$ ;  $\text{Spec } \mathcal{A}$  is a Stone space  $\mathbf{X}$ ; and residue evaluation is an isomorphism  $\mathcal{A} \cong \text{LC}(\mathbf{X}, \mathbb{F}[i])$ .*

*Proof.* An étale algebra over a field is a finite product of finite separable extensions [76], over a real closed field  $\mathbb{F}$  or  $\mathbb{F}[i]$ ; finitely many elements with the chosen  $i$  lie in the image of one such algebra, a quotient of a finite product of fields, hence a finite product of fields, none equal to  $\mathbb{F}$  since it contains a root of  $X^2 + 1$ . In a finite product each  $g$  has  $y$  with  $g = g^2 y$  (reciprocal on nonzero coordinates), a relation that persists: principal ideals are generated by idempotents, prime quotients are fields, nilpotents already vanish in a finite stage, and each residue field is the union of images of  $\mathbb{F}[i]$ . Basic opens  $D(e)$  of idempotents are clopen and separate primes, so with quasi-compactness  $\mathbf{X}$  is compact, Hausdorff and zero-dimensional. Each  $g$  is constant on the parts of a finite clopen partition, so evaluation is locally constant, injective by reducedness, and onto because a locally constant function has finite image with clopen fibres, each a finite union of basic clopens, hence given by an idempotent.  $\square$

**Lemma 15.130** (Size of the henselization [27]). *An infinite local ring of cardinality  $\kappa$  with residue field  $\mathbb{Q}$  has a henselization of cardinality  $\kappa$ .*

*Proof.* There are at most  $\kappa$  finite presentations of étale algebras, each of size at most  $\kappa$ , and each has finitely many points over the closed point with residue field  $\mathbb{Q}$ ; the pointed-neighbourhood colimit has at most  $\kappa$  objects of size at most  $\kappa$ , and  $A \rightarrow A^h$  is injective.  $\square$

**Proposition 15.131** (Exact generic size [27]). *With  $X = \text{Spec } \mathcal{G}$ ,  $|X| = 2^\kappa$ ,  $|\mathcal{G}| = \kappa$  and  $\dim_{\mathbb{E}[i]} \mathcal{G} = \kappa$ .*

*Proof.*  $\mathcal{G}$  is a localization of  $\mathfrak{L}^h$  containing  $\mathbb{E}$ , so  $|\mathcal{G}| = \kappa$  (Lemma 15.130) and  $|X| \leq 2^\kappa$ . The map  $X \rightarrow \{0, 1\}^\kappa$ ,  $\mathfrak{p} \mapsto (\hat{e}_\alpha \bmod \mathfrak{p})$ , is continuous and onto: finite parts of an assignment are realized because the atoms are nonzero, and compactness realizes the whole. A relation  $\sum_j c_j \hat{e}_{\alpha_j} = 0$  with  $c_j \in \mathbb{E}[i]$ , multiplied by the atom with  $\hat{e}_{\alpha_j} = 1$  and the others 0, gives  $c_j = 0$ ; so the  $\hat{e}_\alpha$  are independent, and  $|\mathcal{G}|$  bounds the dimension.  $\square$

**Proposition 15.132** (Henselian minimal primes [27]). *Localization is a bijection  $\text{Min } \mathfrak{L}^h \rightarrow \text{Spec } \mathcal{G}$ , and for  $\Omega \in \text{Min } \mathfrak{L}^h$  the fraction field of  $\mathfrak{L}^h/\Omega$  is the corresponding residue field of  $\mathcal{G}$ , which is  $\mathbb{E}[i]$  over  $\mathbb{E}$ .*

*Proof.*  $\mathfrak{L}^h$  is reduced (reducedness passes to henselizations [75]). Flat going down sends a minimal prime to a minimal prime of the domain  $\mathfrak{L}$ , that is to 0, so it survives in  $\mathcal{G}$ . A prime contracting to 0 contains, and localizes to, a minimal one, and all primes of  $\mathcal{G}$  are minimal (Lemma 15.129), so the two coincide.  $(\mathfrak{L}^h/\Omega) \otimes \mathbb{E}$  is the corresponding prime quotient  $\mathbb{E}[i]$  of  $\mathcal{G}$ , a localization of the domain  $\mathfrak{L}^h/\Omega$  that is already a field.  $\square$

### Completion sees only the residue.

**Proposition 15.133** (Adic invisibility [27]).  *$\mathfrak{L}/\mathfrak{n}^n = \mathbb{Q} = \mathfrak{L}^h/(\mathfrak{n}\mathfrak{L}^h)^n$  for every  $n \geq 1$ ; both completions are the discrete field  $\mathbb{Q}$ , with kernels  $\mathfrak{n}$  and  $\mathfrak{n}\mathfrak{L}^h$ ; neither completion map is flat; neither maximal ideal is finitely generated, and  $\mathfrak{L}$ ,  $\mathfrak{L}^h$  are not Noetherian.*

*Proof.*  $\mathfrak{n}^2 = \mathfrak{n}$  makes both inverse systems constant.  $h = \omega$  is a nonzerodivisor on  $\mathfrak{L}$ , and on the flat  $\mathfrak{L}^h$ , but acts by zero on  $\mathbb{Q}$ . A finitely generated idempotent maximal ideal is zero by Nakayama's lemma, and both contain  $h \neq 0$ .  $\square$

**Corollary 15.134** (No nilpotent local deformation [27]). *For a local  $\mathbb{Q}$ -algebra  $(R_0, \mathfrak{r})$  with nilpotent maximal ideal and residue field  $\mathbb{Q}$ , every local  $\mathbb{Q}$ -algebra map from  $\mathfrak{L}$  or  $\mathfrak{L}^h$  to  $R_0$  factors through the residue field; in particular the residue-preserving maps to  $\mathbb{Q}[\varepsilon]/(\varepsilon^n)$  are unique.*

*Proof.* If  $\mathfrak{r}^n = 0$ , the image of  $\mathfrak{n} = \mathfrak{n}^n$  lies in  $\mathfrak{r}^n = 0$ .  $\square$

This is the idempotent-ideal mechanism of Proposition 13.6 (27 credits it to the repository). Both tangent spaces vanish, yet  $\mathcal{G}$  has  $2^\kappa$  points;  $\mathbb{E} \otimes_{\mathfrak{L}} \widehat{\mathfrak{L}^h} = 0$  since  $h$  is inverted on the left and kills  $\mathbb{Q}$ , while  $\mathbb{E} \otimes_{\mathfrak{L}} \mathfrak{L}^h = \text{LC}(X, \mathbb{E}[i]) \neq 0$ : completion destroys every generic branch at once.  $\text{Spec } \mathfrak{L}^h$  is connected, all minimal primes lying in one maximal ideal, and the clopen structure appears only after inverting  $\mathfrak{L} \setminus \{0\}$ ;  $\mathfrak{L}^h$  must not be replaced by a product of branches (27). No infinitesimal lifting criterion for smoothness applies, the rings being neither Noetherian nor of finite type (27).

### Strict henselization and the Galois factor.

**Proposition 15.135** (Strict henselization by constants [27]). *With  $\overline{\mathbb{Q}}$  an algebraic closure of the residue field,  $\mathfrak{L}^{\text{sh}} \cong \mathfrak{L}^{\text{h}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$  as local  $\mathfrak{L}$ -algebras; its maximal ideal is generated by  $\mathfrak{n}$ , its residue field is  $\overline{\mathbb{Q}}$  and its completion the discrete field  $\overline{\mathbb{Q}}$ .*

*Proof.* For finite  $L/\mathbb{Q}$  inside  $\overline{\mathbb{Q}}$ ,  $\mathfrak{L}^{\text{h}} \otimes_{\mathbb{Q}} L$  is finite étale over  $\mathfrak{L}^{\text{h}}$ , local with residue field  $L$  (finite over a local ring with a field as residue algebra), hence henselian [77]. The union is henselian, local, ind-étale over  $\mathfrak{L}$  with residue field  $\overline{\mathbb{Q}}$ , which identifies it with  $\mathfrak{L}^{\text{sh}}$  [74]; the maximal ideal stays idempotent.  $\square$

**Lemma 15.136** (Splitting the arithmetic tensor product [27]). *For an algebraically closed field  $\mathbb{F}'$  of characteristic zero containing  $\overline{\mathbb{Q}}$ ,  $\mathbb{F}' \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} \cong \text{LC}(\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \mathbb{F}')$  by  $c \otimes a \mapsto (\sigma \mapsto c\sigma(a))$ , the Galois group profinite and  $\mathbb{F}'$  discrete.*

*Proof.* For finite Galois  $L/\mathbb{Q}$ ,  $\mathbb{F}' \otimes_{\mathbb{Q}} L \cong \prod_{\text{Gal}(L/\mathbb{Q})} \mathbb{F}'$ , compatibly with inflation; a locally constant function on the profinite group has finite image and factors through a finite quotient.  $\square$

**Theorem 15.137** (The strict generic fibre [27]). *With  $\overline{\mathbb{Q}} \hookrightarrow \mathbb{E}[i]$  chosen,  $\mathbb{E} \otimes_{\mathfrak{L}} \mathfrak{L}^{\text{sh}} \cong \text{LC}(\mathbf{X}, \mathbb{E}[i]) \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} \cong \text{LC}(\mathbf{X} \times \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}), \mathbb{E}[i])$ ;  $1 \otimes \tau$  acts by  $f(x, \sigma) \mapsto f(x, \sigma\tau)$  with fixed ring  $\mathcal{G}$ ; and  $\mathfrak{L}^{\text{sh}}$  has cardinality  $\kappa$  and exactly  $2^{\kappa}$  minimal primes.*

*Proof.* The first isomorphism is base change of Proposition 15.135; the second holds on each finite stage by Lemma 15.136, and a locally constant function on the compact zero-dimensional product is given by finitely many clopen rectangles. Right translation is transitive, so invariants ignore the second coordinate.  $\overline{\mathbb{Q}}$  is countable; the generic spectrum  $\mathbf{X} \times \text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$  has cardinality  $2^{\kappa}$ , as  $|\text{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})| \leq 2^{\aleph_0} \leq 2^{\kappa}$ ; and the minimal primes correspond to generic points as in Proposition 15.132.  $\square$

The product depends on the chosen algebraic closure and embedding; it separates two sources of branching,  $\mathbf{X}$  recording normalization and henselian choices and the Galois group recording arithmetic embeddings of the residue closure (27).

**Gaussian omnific integers.** Put  $B_{\mathbb{E}}[i] \subseteq \mathbb{E}[i]$ , with constant term in  $\mathbb{Z}[i]$  and kernel  $\Pi_{\mathbb{E}} + i\Pi_{\mathbb{E}}$ .

**Proposition 15.138** (Gaussian localization [27]). *The localization of  $B_{\mathbb{E}}[i]$  at  $\Pi_{\mathbb{E}} + i\Pi_{\mathbb{E}}$  is  $\mathfrak{L}[i]$ , a local domain with residue field  $\mathbb{Q}(i)$  and fraction field  $\mathbb{E}[i]$ .*

*Proof.* If  $b+ci$  has nonzero Gaussian constant term,  $(b+ci)^{-1} = (b-ci)/(b^2+c^2)$  with  $\text{ct}(b)^2 + \text{ct}(c)^2 \neq 0$ , already inverted in  $\mathfrak{L}$ .  $\square$

**Theorem 15.139** (Henselization and doubling [27]).  *$(\mathfrak{L}[i])^{\text{h}} \cong \mathfrak{L}^{\text{h}}[i]$ , and  $\mathbb{E}[i] \otimes_{\mathfrak{L}[i]} \mathfrak{L}^{\text{h}}[i] \cong \mathbb{E}[i] \otimes_{\mathbb{E}} \mathcal{G} \cong \mathcal{G} \times \mathcal{G}$  as rings and  $\mathbb{E}$ -algebras; its spectrum is  $\mathbf{X} \sqcup \mathbf{X}$ , it has  $2^{\kappa}$  generic points, and the completion of  $\mathfrak{L}^{\text{h}}[i]$  is  $\mathbb{Q}(i)$ .*

*Proof.*  $\mathfrak{L}^{\text{h}}[i]$  is finite étale over  $\mathfrak{L}^{\text{h}}$ , local with residue field  $\mathbb{Q}(i)$ , henselian and ind-étale over  $\mathfrak{L}[i]$  with the same residue field, which gives the first identification. In the generic fibre  $Z^2 + 1 = (Z-j)(Z+j)$  with  $2j$  a unit, so  $a + bZ \mapsto (a + bj, a - bj)$  is an isomorphism  $\mathcal{G}[Z]/(Z^2 + 1) \rightarrow \mathcal{G} \times \mathcal{G}$  with inverse  $(r, s) \mapsto \frac{r+s}{2} + \frac{r-s}{2j}Z$ ; the external  $i$  acts as  $(j, -j)$ , so the factors carry conjugate scalar structures.  $\square$

The doubling decomposes the generic fibre;  $\mathcal{L}^h[i]$  itself is local, and  $X \not\cong X \sqcup X$  is not asserted (27).

### Projective modules, étale algebras and valuations.

**Theorem 15.140** (Finite projectives on the generic fibre [27]). *Isomorphism classes of finitely generated projective  $\mathcal{G}$ -modules correspond to locally constant  $r : X \rightarrow \mathbb{N}$  (pointwise rank);  $\text{Pic } \mathcal{G} = 0$ ,  $K_0(\mathcal{G}) \cong \text{LC}(X, \mathbb{Z})$  as rings, and constant rank  $d$  means free of rank  $d$ .*

*Proof.* An idempotent matrix over  $\mathcal{G}$  is constant on the parts of a finite clopen partition, where it is an idempotent matrix over  $\mathbb{E}[i]$  with image of some rank; the local free identifications assemble along the central idempotents, and every locally constant rank function is realized by free summands on clopen pieces. Direct sum and tensor product are addition and multiplication of ranks.  $\square$

This is a general consequence of the Boolean-power description, not special to surreal coefficients (27).

**Proposition 15.141** (Finite étale algebras [27]). *Reduction gives an equivalence  $\text{Fin}\acute{\text{Et}}(\mathcal{L}^h) \simeq \text{Fin}\acute{\text{Et}}(\mathbb{Q})$ , the lift of a number field  $L$  being  $\mathcal{L}^h \otimes_{\mathbb{Q}} L$ , with generic fibre  $\mathcal{G} \otimes_{\mathbb{Q}} L \cong \mathcal{G}^{[L:\mathbb{Q}]}$ ; finite étale  $\mathcal{G}$ -algebras are classified up to isomorphism by locally constant ranks  $X \rightarrow \mathbb{N}$ , and constant rank  $d$  means  $\mathcal{G}^d$ .*

*Proof.* The first is the henselian finite étale equivalence [77] with Proposition 15.135. A finite étale  $\mathcal{G}$ -algebra descends to a finite stage  $\mathbb{E}[i]^r$  [76], where it is a product of copies of the algebraically closed field on each factor; ranks at residue points determine it.  $\mathbb{E}[i] \otimes_{\mathbb{Q}} L$  is  $[L : \mathbb{Q}]$  copies of  $\mathbb{E}[i]$ .  $\square$

The rank classification is of isomorphism classes, not of morphisms, and the equivalence for  $\mathcal{L}^h$  says nothing about  $B_{\mathbb{E}}$  or  $\mathbf{Oz}$  (27). Let  $\text{RZ}(\mathcal{L})$  be the valuation rings of  $\mathbb{E}$  dominating  $\mathcal{L}$ , with the Zariski topology of finite-subset containment.

**Corollary 15.142** (Centred valuation components [27]).  *$\text{RZ}(\mathcal{L})$  has exactly  $2^{\kappa}$  connected components.*

*Proof.* De Felipe and Teissier prove, for local domains without Noetherian hypotheses, that these components correspond to the minimal primes of the henselization [78, Corollary 4.2]; apply Propositions 15.131 and 15.132.  $\square$

The external theorem is imported; the count is the application, and nothing is said about the components themselves (singletons, openness, a fixed valuation space) or about the quotient topology compared with  $X$  (27).

*Remark 15.143* (Branch counts compared [merge]). Three maximal branch counts now appear in the report, for different rings. 20's full-class fibre  $B_{\mathbb{R}}$  has a proper class of idempotents (Theorem 15.73). 23's cyclic coefficient fibre has exactly  $2^{\mathfrak{c}}$  maximal ideals over  $T = 0$  for the ring  $\mathcal{B}$  of cardinality  $\mathfrak{c} = |K_{\mathbb{Z}a}|$  (Corollary 15.112), and its Boolean algebras have cardinality  $\mathfrak{c}$ . 27's  $\mathcal{L}^h$  has  $2^{\kappa}$  minimal primes for a ring of cardinality  $\kappa$ , every  $\kappa$  allowed; for  $\kappa = \aleph_0$  a countable ring has continuum many minimal primes, which is no paradox,  $\mathcal{G}$  containing only finite-image locally constant functions (27). In all three the new idempotents come from quadratic branches at a scale unseen by the given data; 23 needs no fresh scale (ramification at unused finite places), while 20 and 27 use 20's splitter at dominant scales. 27's local ring  $\mathcal{L}$  is not a cyclic workspace: its exponent group contains the  $\omega^{\alpha}$ ,  $\alpha < \kappa$ , and its fraction field is real closed.

**Seeds and dependencies (27).** The seed may consist of infinite scales, infinitesimals or earlier surreal values, all placed in  $\mathbb{E}$  with their floors in  $B_{\mathbb{E}}$  without exceeding  $\kappa$ ; for surcomplex values seed the real and imaginary parts, and the Gaussian statements apply. This is seed containment, not preservation of the algebraic or analytic operations of a prior workspace without further closure. The dependencies are: normal forms and monomial bounds; monic certificates and support projection; the dominant-scale splitter and finite Boolean atoms; the actual hull and survival in the local normalization; survival of every finite generic branch under localization; free generic idempotents and forced complexification; the Boolean power, exact cardinality and minimal primes. Completion is a separate consequence of the idempotent ideal; the strict and Gaussian results depend on the generic fibre, and the valuation corollary also on de Felipe–Teissier. 27 lists ten failure points it addresses: a convenient subring need not be the integer part (the floor closure proves  $B_{\mathbb{E}} = \mathbb{E} \cap \mathbf{Oz}$ ); a fixed Hahn group need not admit fresh bounds (the splitters are built in **No** first); projection is not a ring map; a Laurent series cannot be evaluated at zero; the displayed identity does not prove nontriviality (arbitrary radical witnesses are excluded by certificates and projection); localization may delete generic factors (Lemma 15.126); a local ring has no nontrivial idempotents; the generic imaginary unit is not a residue-field extension; Noetherian finiteness theorems for branches do not apply; and finite tests are not transfinite verification.

## 16 Arithmetic of polynomial maps on the omnific lattices

This section is manuscript 10’s arithmetic of finite-degree polynomial and rational maps on the real lattice  $\mathbf{Oz} \subset \mathbf{No}$  and the Gaussian lattice  $\mathbf{Oz}[i] \subset \mathbf{No}[i]$ , followed by manuscript 16’s fresh-scale results for  $D + \Pi_k$  with any unital constant ring  $D$  (Sections 16.6 and 16.7) and by manuscript 18’s ordinary-output rigidity (Section 16.8), with manuscript 22’s single-scale certificates at the end (Section 16.10). It uses from the core only the universal theorem Theorem 5.1, the quotient list Theorem 6.2 with the identity (3.2), the constant-term lemma Lemma 2.2, the clearing of supports Proposition 3.4(i) and the unit lists of Lemma 2.4.

**Renamings.** Manuscript 10’s symbols are renamed as follows. Its ideals  $\mathfrak{P}$  and  $\mathfrak{P}_{\mathbb{C}}$  are  $\Pi$  and  $\Pi_{\mathbb{C}}$ ; its constant-term map  $\pi$  is  $\text{ct}$ ; its projection  $\Pi$  on polynomial rings (the *shadow*) is  $\text{ct}_*$ , the coefficientwise constant term; its Gaussian lattice is  $\mathbf{Oz}[i]$  inside  $\mathbf{No}[i]$ ; its ring  $R(D, K)$  is  $\mathcal{A}_{D,K}$ ; its binomial polynomials  $B_k(X)$  are written  $\binom{X}{k}$ ; its  $L_k = \text{lcm}(1, \dots, k)$  is written  $\text{lcm}_k$ ; and its rings  $S_r$  and  $I_r$  are written  $\mathcal{N}_r(\mathbf{Oz})$  and  $\text{Int}(\mathbb{Z}^r)$  (the letter  $I_r$  would collide with ideals). The numbered results keep 10’s content.

**Notation.** Throughout,  $r \geq 1$  is an ordinary integer,  $\mathbf{X} = (X_1, \dots, X_r)$ , and bold letters denote  $r$ -tuples. For an ordinary  $k \geq 0$ ,

$$\binom{X}{k} = \frac{X(X-1) \cdots (X-k+1)}{k!}, \quad \binom{X}{0} = 1,$$

and for a multi-index  $\mathbf{k} = (k_1, \dots, k_r)$  we put  $\binom{\mathbf{X}}{\mathbf{k}} = \prod_{j=1}^r \binom{X_j}{k_j}$ . We write  $\Delta_j F(\mathbf{X}) = F(\mathbf{X} + \mathbf{e}_j) - F(\mathbf{X})$ , and  $\Delta = \Delta_1$  when  $r = 1$ . Put  $\text{lcm}_0 = 1$  and  $\text{lcm}_k = \text{lcm}(1, \dots, k)$  for  $k \geq 1$ . Recall  $\mathcal{A}_{\mathbb{Q}, \mathbb{R}} = \mathbb{Q} + \Pi$  and  $\mathcal{A}_{\mathbb{Q}(i), \mathbb{C}} = \mathbb{Q}(i) + \Pi_{\mathbb{C}}$ ; by Lemma 2.2 the constant term  $\text{ct}$  is a ring homomorphism on both, with kernels  $\Pi$  and  $\Pi_{\mathbb{C}}$ , so  $\Pi$  is an ideal of  $\mathbb{Q} + \Pi$  and  $\Pi_{\mathbb{C}}$  is an ideal of  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ . Its coefficientwise extension

$$\text{ct}_* : (\mathbb{Q} + \Pi)[\mathbf{X}] \rightarrow \mathbb{Q}[\mathbf{X}], \quad \text{ct}_* : (\mathbb{Q}(i) + \Pi_{\mathbb{C}})[\mathbf{X}] \rightarrow \mathbb{Q}(i)[\mathbf{X}]$$

is therefore a ring homomorphism with kernel  $\Pi[\mathbf{X}]$ , respectively  $\Pi_{\mathbb{C}}[\mathbf{X}]$ . Define

$$\begin{aligned}\mathcal{N}_r(\mathbf{Oz}) &= \{F \in \mathbf{No}[\mathbf{X}] : F(\mathbf{Oz}^r) \subseteq \mathbf{Oz}\}, & \text{Int}(\mathbb{Z}^r) &= \{f \in \mathbb{Q}[\mathbf{X}] : f(\mathbb{Z}^r) \subseteq \mathbb{Z}\}, \\ \mathcal{N}_r(\mathbf{Oz}[i]) &= \{F \in \mathbf{No}[i][\mathbf{X}] : F(\mathbf{Oz}[i]^r) \subseteq \mathbf{Oz}[i]\}, & \text{Int}(\mathbb{Z}[i]^r) &= \{f \in \mathbb{Q}(i)[\mathbf{X}] : f(\mathbb{Z}[i]^r) \subseteq \mathbb{Z}[i]\}.\end{aligned}$$

These are rings under pointwise operations. All degrees, finite sums and factorials are ordinary.

## 16.1 Lattice separation and an omitted congruence type

**Lemma 16.1** (Lattice separation and cofinality [10]). *For  $\Lambda = \mathbf{Oz}$  and for  $\Lambda = \mathbf{Oz}[i]$ , every nonzero  $x \in \Lambda$  satisfies  $|x| \geq 1$ . The finite elements of  $\mathbf{Oz}$  are exactly the ordinary integers. The positive elements of  $\mathbf{Oz}$  are cofinal in  $\mathbf{No}$ . The units are  $\mathbf{Oz}^\times = \{\pm 1\}$  and  $\mathbf{Oz}[i]^\times = \{\pm 1, \pm i\}$  (Lemma 2.4).*

*Proof.* A nonzero real purely infinite series has a positive leading exponent, so its absolute value exceeds every positive real. Hence  $n + u$  with  $n \in \mathbb{Z}$  and  $0 \neq u \in \Pi$  is infinite, the finite elements of  $\mathbf{Oz}$  are the ordinary integers, and every nonzero element of  $\mathbf{Oz}$  has absolute value at least 1. In the Gaussian case write  $x = a + ib$  with  $a, b \in \mathbf{Oz}$ , not both zero; the surcomplex modulus dominates either coordinate,  $|x| = \sqrt{a^2 + b^2} \geq \max(|a|, |b|) \geq 1$ .

Given  $y \in \mathbf{No}$ , choose an exponent  $g > 0$  larger than the leading exponent of  $y$  (any  $g > 0$  if  $y = 0$ ). Then  $\omega^g \in \mathbf{Oz}$  is positive and exceeds  $n|y|$  for every ordinary  $n$ , in particular  $\omega^g > |y|$ . This proves cofinality. The unit lists are Lemma 2.4.  $\square$

*Remark 16.2* (Modulus and standard part [10]). The surcomplex modulus needs only real closedness of  $\mathbf{No}$ :  $|a + ib| = \sqrt{a^2 + b^2}$  for  $a, b \in \mathbf{No}$ , and it is multiplicative and satisfies the triangle inequality. The standard part of a finite surcomplex number is taken coordinatewise in  $\mathbb{C}$ ; on the finite rings  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$  it is the map  $\text{st}$  of Lemma 2.2.

**Proposition 16.3** (A finitely satisfiable countable congruence type [10]). *Fix an ordinary prime  $p$  and  $\alpha \in \mathbb{Z}_p \setminus \mathbb{Z}$ . For  $e \geq 1$  let  $a_e$  be the representative of  $\alpha \bmod p^e$  in  $\{0, \dots, p^e - 1\}$ . The countable family of ring formulas*

$$\exists y (x - a_e = p^e y), \quad e = 1, 2, \dots, \quad (16.1)$$

*is finitely satisfiable in  $\mathbb{Z} \subset \mathbf{Oz}$  but has no realization in  $\mathbf{Oz}$ . An explicit choice is*

$$\alpha = \sum_{j \geq 0} p^{2^j} \in \mathbb{Z}_p. \quad (16.2)$$

*Proof.* For a finite subfamily let  $E$  be its largest index and take  $x = a_E$ . Since the residues are compatible,  $a_E \equiv a_e \pmod{p^e}$  for  $e \leq E$ , which gives witnesses  $y \in \mathbb{Z}$  for every formula of the subfamily. If  $x \in \mathbf{Oz}$  realized all formulas, applying the ring homomorphism  $\text{ct}$  would give an ordinary integer  $\text{ct}(x)$  with  $\text{ct}(x) \equiv a_e \equiv \alpha \pmod{p^e}$  for every  $e$  (by (3.2),  $x - a_e \in p^e \mathbf{Oz}$  means  $\text{ct}(x) - a_e \in p^e \mathbb{Z}$ ). In  $\mathbb{Z}_p$  this forces  $\text{ct}(x) = \alpha$ , contradicting  $\alpha \notin \mathbb{Z}$ .

For (16.2): the powers  $p^{2^j}$  are distinct, so the base- $p$  digits of  $\alpha$  are 1 at the positions  $2^j$  and 0 elsewhere. They are not eventually zero, so  $\alpha$  is not a nonnegative integer; and they are not eventually  $p - 1$  (for  $p = 2$  the positions  $2^j + 1 < 2^{j+1}$  with  $j \geq 1$  carry the digit 0), so  $\alpha$  is not a negative integer.  $\square$

*Remark 16.4* (Ring types, not saturation [10]). This elementary omitted type concerns the ring structure. It shows that the order-theoretic richness of the full surreal class does not give a corresponding realization property for countable ring types. No use of the term “saturated model” for a proper-class structure is needed, and none is made.



## 16.2 Polynomials preserving the omnific lattices

**Lemma 16.5** (Omnific binomial closure [10; 22]). *For every  $x \in \mathbf{Oz}$  and every ordinary  $k \geq 0$ ,  $\binom{x}{k} \in \mathbf{Oz}$ . More generally, if  $f \in \mathbb{Q}[\mathbf{X}]$  and  $f(\mathbb{Z}^r) \subseteq \mathbb{Z}$ , then  $f(\mathbf{Oz}^r) \subseteq \mathbf{Oz}$ .*

*Proof.* Write  $x_j = n_j + u_j$  with  $n_j \in \mathbb{Z}$  and  $u_j \in \Pi$ . Expanding each monomial of  $f$  shows that  $f(\mathbf{n} + \mathbf{u}) - f(\mathbf{n})$  is a finite sum of terms, each a rational multiple of a product of ordinary integers and of powers of the  $u_j$  containing at least one factor  $u_j$ . Since  $\Pi$  is an ideal of  $\mathbb{Q} + \Pi$ , every such term lies in  $\Pi$ ; thus

$$f(\mathbf{n} + \mathbf{u}) - f(\mathbf{n}) \in \Pi.$$

As  $f(\mathbf{n}) \in \mathbb{Z}$ , we get  $f(\mathbf{x}) \in \mathbb{Z} + \Pi = \mathbf{Oz}$ . The binomial case is the case  $f = \binom{X}{k}$ , which takes integer values at all ordinary integers, including negative ones:  $\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}$ .  $\square$

Thus  $\mathbf{Oz}$  is a binomial ring in the usual torsion-free sense: it is  $\mathbb{Z}$ -torsion free and closed under  $x \mapsto \binom{x}{k}$  [16]. The terminology is classical; the support argument above explains its application here without importing an abstract free-ring theorem.

**Theorem 16.6** (Exact finite-grid test [10; 16; 22]). *Let  $F \in \mathbf{No}[\mathbf{X}]$  with  $\deg_{X_j} F \leq d_j$  for  $j = 1, \dots, r$ . The following are equivalent:*

- (i)  $F(\mathbf{Oz}^r) \subseteq \mathbf{Oz}$ ;
- (ii)  $F(n_1, \dots, n_r) \in \mathbf{Oz}$  for all ordinary integers  $0 \leq n_j \leq d_j$ ;
- (iii) in the unique expansion

$$F(\mathbf{X}) = \sum_{\mathbf{k} \leq \mathbf{d}} c_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}} \quad (16.3)$$

all coefficients  $c_{\mathbf{k}}$  belong to  $\mathbf{Oz}$ .

Explicitly,

$$c_{\mathbf{k}} = (\Delta_1^{k_1} \cdots \Delta_r^{k_r} F)(\mathbf{0}) = \sum_{\ell \leq \mathbf{k}} (-1)^{\sum_j (k_j - \ell_j)} \left( \prod_{j=1}^r \binom{k_j}{\ell_j} \right) F(\ell). \quad (16.4)$$

*Proof.* The polynomials  $\binom{X}{0}, \dots, \binom{X}{d}$  form a basis of the univariate polynomials of degree at most  $d$  over the field  $\mathbf{No}$ : their degrees are  $0, \dots, d$  and their leading coefficients are nonzero. Taking products, the  $\binom{\mathbf{X}}{\mathbf{k}}$  with  $\mathbf{k} \leq \mathbf{d}$  form a basis of the polynomials with  $\deg_{X_j} \leq d_j$ , which gives existence and uniqueness of (16.3). Since  $\Delta_m^X = \binom{X}{m-1}$  for  $m \geq 1$  and  $\Delta_0^X = 0$ , one has  $\Delta_j^{k_j} \binom{X_j}{m_j} = \binom{X_j}{m_j - k_j}$  for  $m_j \geq k_j$  and 0 otherwise; as  $\binom{0}{m} = 0$  for  $m > 0$ , applying  $\Delta_1^{k_1} \cdots \Delta_r^{k_r}$  to (16.3) and evaluating at  $\mathbf{0}$  leaves exactly  $c_{\mathbf{k}}$ . Writing  $\Delta_j = \tau_j - 1$  with the shift  $\tau_j$  and expanding by the binomial theorem gives the last expression in (16.4).

(i) $\Rightarrow$ (ii) is restriction to ordinary grid points. Under (ii), the last expression in (16.4) is an integral linear combination of grid values in  $\mathbf{Oz}$ , which proves (iii). Under (iii), each factor  $\binom{x_j}{k_j}$  with  $x_j \in \mathbf{Oz}$  lies in  $\mathbf{Oz}$  by Lemma 16.5, and evaluating (16.3) proves (i).  $\square$

The test is finite for a given finite-degree polynomial, although the preservation property quantifies over the entire class  $\mathbf{Oz}^r$ . It is a mathematical test with exact surreal input; no claim is made that arbitrary surreal coefficients have an effective finite encoding.

**Theorem 16.7** (The real numerical-polynomial ring [10; 16; 18; 22]). *Every  $F \in \mathcal{N}_r(\mathbf{Oz})$  has a unique decomposition*

$$F = U + f, \quad U \in \Pi[\mathbf{X}], \quad f \in \text{Int}(\mathbb{Z}^r), \quad (16.5)$$

*so that  $\mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$  additively. The coefficientwise constant term  $\text{ct}_*(F) = f$  is a surjective ring homomorphism  $\mathcal{N}_r(\mathbf{Oz}) \rightarrow \text{Int}(\mathbb{Z}^r)$  with kernel  $\Pi[\mathbf{X}]$ , split by the inclusion  $\text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$ . For  $\mathbf{x} \in \mathbf{Oz}^r$ ,*

$$\text{ct}(F(\mathbf{x})) = \text{ct}_*(F)(\text{ct}(x_1), \dots, \text{ct}(x_r)). \quad (16.6)$$

*Proof.* Let  $F \in \mathcal{N}_r(\mathbf{Oz})$ . By Theorem 16.6 its Newton coefficients  $c_{\mathbf{k}}$  lie in  $\mathbf{Oz}$ ; write  $c_{\mathbf{k}} = m_{\mathbf{k}} + u_{\mathbf{k}}$  with  $m_{\mathbf{k}} \in \mathbb{Z}$  and  $u_{\mathbf{k}} \in \Pi$ , and put

$$U = \sum_{\mathbf{k}} u_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}}, \quad f = \sum_{\mathbf{k}} m_{\mathbf{k}} \binom{\mathbf{X}}{\mathbf{k}}.$$

The binomials have rational coefficients and  $\Pi$  is a  $\mathbb{Q}$ -vector space, so  $U \in \Pi[\mathbf{X}]$ ; products of binomials are integer valued on  $\mathbb{Z}^r$ , so  $f \in \text{Int}(\mathbb{Z}^r)$ . Conversely, a polynomial in  $\Pi[\mathbf{X}]$  takes values in  $\Pi$  on  $\mathbf{Oz}^r$ , because  $\Pi$  is an ideal of  $\mathbb{Q} + \Pi \supseteq \mathbf{Oz}$ , and  $\text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$  by Lemma 16.5; hence  $\Pi[\mathbf{X}] + \text{Int}(\mathbb{Z}^r) \subseteq \mathcal{N}_r(\mathbf{Oz})$ . The sum is direct because a coefficient that is both rational and purely infinite is zero.

By (16.5) all coefficients of  $F$  lie in  $\mathbb{Q} + \Pi$ , so  $\mathcal{N}_r(\mathbf{Oz})$  is a subring of  $(\mathbb{Q} + \Pi)[\mathbf{X}]$  and  $\text{ct}_*$  restricts to a ring homomorphism on it. It sends  $U$  to 0 and fixes  $f$ , so its restriction is the projection onto the second summand; its kernel is  $\Pi[\mathbf{X}]$  and the inclusion of  $\text{Int}(\mathbb{Z}^r)$  is a ring section. Finally write  $x_j = n_j + u_j$  with  $n_j = \text{ct}(x_j)$ . Then  $U(\mathbf{x}) \in \Pi$  and  $f(\mathbf{x}) - f(\mathbf{n}) \in \Pi$  by the proof of Lemma 16.5, while  $f(\mathbf{n}) \in \mathbb{Z}$ . Hence  $\text{ct}(F(\mathbf{x})) = f(\mathbf{n})$ , which is (16.6).  $\square$

The decomposition is an additive direct sum whose first summand is an ideal; it is *not* a direct product of rings.

**Example 16.8** ([10]). The polynomial

$$F(X) = 3 \binom{X}{2} + \omega^{\sqrt{2}} X^3 + \left( \sum_{m \geq 1} \omega^{1/m} \right) (X - 1)$$

is omnific-integer valued and has shadow  $\text{ct}_*(F) = 3 \binom{X}{2}$ . By contrast  $\omega^{-1} X$  fails already at  $X = 1$ . No negative-exponent coefficient can be hidden in a finite-degree lattice-preserving polynomial: by Theorem 16.7 every coefficient of an element of  $\mathcal{N}_r(\mathbf{Oz})$  lies in  $\mathbb{Q} + \Pi$ .

**Theorem 16.9** (Gaussian numerical polynomials [10; 16; 18; 22]). *For every ordinary  $r \geq 1$ ,*

$$\mathcal{N}_r(\mathbf{Oz}[i]) = \Pi_{\mathbb{C}}[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}[i]^r) \quad (16.7)$$

*additively. The coefficientwise constant term  $\text{ct}_*$  is a surjective ring homomorphism  $\mathcal{N}_r(\mathbf{Oz}[i]) \rightarrow \text{Int}(\mathbb{Z}[i]^r)$  with kernel  $\Pi_{\mathbb{C}}[\mathbf{X}]$ , split by inclusion, and for  $\mathbf{x} \in \mathbf{Oz}[i]^r$ , with  $\text{ct} : \mathbf{Oz}[i] \rightarrow \mathbb{Z}[i]$  the Gaussian constant term,*

$$\text{ct}(F(\mathbf{x})) = \text{ct}_*(F)(\text{ct}(x_1), \dots, \text{ct}(x_r)). \quad (16.8)$$

*Proof.* Let  $F \in \mathcal{N}_r(\mathbf{Oz}[i])$  with  $\deg_{X_j} F \leq d_j$ . The Newton expansion (16.3) and formula (16.4) hold over the field  $\mathbf{No}[i]$  by the same basis argument. The ordinary grid points lie in  $\mathbf{Oz}[i]^r$ , so (16.4) shows  $c_{\mathbf{k}} \in \mathbf{Oz}[i] = \mathbb{Z}[i] + \Pi_{\mathbb{C}}$ . Expanding the binomials, each monomial coefficient of  $F$  is a  $\mathbb{Q}$ -linear combination of the  $c_{\mathbf{k}}$ , hence lies in  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ . Put  $f = \text{ct}_*(F) \in \mathbb{Q}(i)[\mathbf{X}]$  and  $U = F - f \in \Pi_{\mathbb{C}}[\mathbf{X}]$ . For an ordinary Gaussian tuple  $\mathbf{z} \in \mathbb{Z}[i]^r$  we have  $U(\mathbf{z}) \in \Pi_{\mathbb{C}}$ , because  $\Pi_{\mathbb{C}}$  is an ideal of  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ , and  $F(\mathbf{z}) = w + v$  with  $w \in \mathbb{Z}[i]$ ,  $v \in \Pi_{\mathbb{C}}$ . Then  $f(\mathbf{z}) - w = v - U(\mathbf{z})$  lies in  $\mathbb{Q}(i) \cap \Pi_{\mathbb{C}} = 0$ , so  $f(\mathbf{z}) = w \in \mathbb{Z}[i]$ . Thus  $f \in \text{Int}(\mathbb{Z}[i]^r)$ .

Conversely write a Gaussian omnific input as  $\mathbf{z} + \mathbf{u}$  with  $\mathbf{z} \in \mathbb{Z}[i]^r$  and  $u_j \in \Pi_{\mathbb{C}}$ . For  $f \in \text{Int}(\mathbb{Z}[i]^r)$  the expansion argument of Lemma 16.5 gives  $f(\mathbf{z} + \mathbf{u}) - f(\mathbf{z}) \in \Pi_{\mathbb{C}}$ , so  $f(\mathbf{z} + \mathbf{u}) \in \mathbb{Z}[i] + \Pi_{\mathbb{C}} = \mathbf{Oz}[i]$ ; and a polynomial in  $\Pi_{\mathbb{C}}[\mathbf{X}]$  takes values in  $\Pi_{\mathbb{C}}$  on  $\mathbf{Oz}[i]^r$ . This proves (16.7); the sum is direct since  $\mathbb{Q}(i) \cap \Pi_{\mathbb{C}} = 0$ . As in Theorem 16.7,  $\mathcal{N}_r(\mathbf{Oz}[i])$  is a subring of  $(\mathbb{Q}(i) + \Pi_{\mathbb{C}})[\mathbf{X}]$ , on which  $\text{ct}_*$  is a ring homomorphism; this gives multiplicativity, the kernel and the section. The same computation,  $\text{ct}(F(\mathbf{z} + \mathbf{u})) = f(\mathbf{z})$ , proves (16.8).  $\square$

*Remark 16.10* (Why the real Newton test cannot simply be copied [10]). The ordinary binomial polynomials need not be Gaussian-integer valued:

$$\binom{i}{2} = \frac{i(i-1)}{2} = \frac{-1-i}{2} \notin \mathbb{Z}[i].$$

Consequently values on an ordinary integer grid do not by themselves certify preservation of  $\mathbf{Oz}[i]$  (the grid values are necessary, not sufficient). The Gaussian theorem retains the ordinary ring  $\text{Int}(\mathbb{Z}[i]^r)$  rather than imposing a false binomial basis on it; the arithmetic of  $\text{Int}(\mathbb{Z}[i]^r)$  is not replaced by the simpler real Newton basis.

**Theorem 16.11** (Small-target reflection for numerical polynomials [10]). *Every class ring homomorphism  $\varphi$  from  $\mathcal{N}_r(\mathbf{Oz})$  to a set-sized ring  $A$  factors uniquely through*

$$\text{ct}_* : \mathcal{N}_r(\mathbf{Oz}) \longrightarrow \text{Int}(\mathbb{Z}^r).$$

*The analogous assertion holds for  $\mathcal{N}_r(\mathbf{Oz}[i])$  with universal image  $\text{Int}(\mathbb{Z}[i]^r)$ . Therefore the set-sized quotient presentations of either numerical-polynomial ring are precisely the quotient presentations of its ordinary shadow pulled back along  $\text{ct}_*$ : their kernels are the class ideals  $\text{ct}_*^{-1}(\mathfrak{a})$  for ideals  $\mathfrak{a}$  of  $\text{Int}(\mathbb{Z}^r)$ , respectively  $\text{Int}(\mathbb{Z}[i]^r)$ , with quotients  $\text{Int}(\mathbb{Z}^r)/\mathfrak{a}$ , respectively  $\text{Int}(\mathbb{Z}[i]^r)/\mathfrak{a}$ .*

*Proof.* The restriction of  $\varphi$  to the ideal  $\Pi$  of constant polynomials is a (possibly nonunital) class ring homomorphism  $\Pi \rightarrow A$ , hence zero by Theorem 5.1. Every element of  $\Pi[\mathbf{X}]$  is a finite sum of products  $u \mathbf{X}^{\mathbf{m}}$  with  $u \in \Pi$  and  $\mathbf{X}^{\mathbf{m}} \in \mathcal{N}_r(\mathbf{Oz})$ , so  $\varphi(u \mathbf{X}^{\mathbf{m}}) = \varphi(u) \varphi(\mathbf{X}^{\mathbf{m}}) = 0$  and  $\varphi$  kills  $\Pi[\mathbf{X}] = \ker \text{ct}_*$  (Theorem 16.7). Hence  $\varphi = \psi \circ \text{ct}_*$  with  $\psi$  the restriction of  $\varphi$  to the section  $\text{Int}(\mathbb{Z}^r)$ ;  $\psi$  is unique because  $\text{ct}_*$  is surjective. The Gaussian case is identical, with  $\Pi_{\mathbb{C}}$  and Theorem 16.9.

If  $\varphi$  is surjective onto a set-sized ring with kernel  $J$ , then  $\psi$  is surjective and  $J = \text{ct}_*^{-1}(\ker \psi)$ . Conversely, for an ideal  $\mathfrak{a}$  of the shadow ring, the composite of  $\text{ct}_*$  with the quotient map onto the set  $\text{Int}(\mathbb{Z}^r)/\mathfrak{a}$  is a set-sized quotient presentation with kernel  $\text{ct}_*^{-1}(\mathfrak{a})$ . This is the kernel argument of Theorem 6.2.  $\square$

### 16.3 Congruences modulo every omnific integer

**Definition 16.12** ([10]). A function  $F : \mathbf{Oz} \rightarrow \mathbf{Oz}$  is *congruence preserving* if  $a - b$  divides  $F(a) - F(b)$  in  $\mathbf{Oz}$  for all  $a, b \in \mathbf{Oz}$ ; equivalently, for every  $m \in \mathbf{Oz}$ ,  $a \equiv b \pmod{m\mathbf{Oz}}$  implies

$F(a) \equiv F(b) \pmod{m\mathbf{Oz}}$ . We write  $\text{CP}(\mathbf{Oz})$  for the congruence-preserving elements of  $\mathcal{N}_1(\mathbf{Oz})$  and  $\text{CP}(\mathbb{Z})$  for the elements  $f \in \text{Int}(\mathbb{Z})$  such that  $a - b$  divides  $f(a) - f(b)$  in  $\mathbb{Z}$  for all  $a, b \in \mathbb{Z}$ .

The two formulations agree: the forward implication is transitivity of divisibility, and the reverse one takes  $m = a - b$ . When  $a = b$  the condition reads  $0 \mid 0$ , which holds. Differences and finite sums of congruence-preserving functions are congruence preserving.

**Lemma 16.13** (Integral divided differences of weighted binomials [10]). *For every ordinary  $k \geq 1$ , the identity*

$$\frac{\binom{X+T}{k} - \binom{X}{k}}{T} = \sum_{j=1}^k \binom{X}{k-j} \frac{1}{j} \binom{T-1}{j-1} \quad (16.9)$$

*holds in  $\mathbb{Q}[X, T]$ . In particular  $\text{lcm}_k \binom{X}{k}$  is congruence preserving on  $\mathbf{Oz}$ .*

*Proof.* Vandermonde's identity gives  $\binom{X+T}{k} = \sum_{j=0}^k \binom{X}{k-j} \binom{T}{j}$ . For  $j \geq 1$  the falling-factorial formula gives  $\binom{T}{j} = \frac{T}{j} \binom{T-1}{j-1}$ . Subtracting the term  $j = 0$  and cancelling the polynomial factor  $T$  gives (16.9).

After multiplication by  $\text{lcm}_k$ , every scalar  $\text{lcm}_k / j$  with  $1 \leq j \leq k$  is an ordinary integer, so by Lemma 16.5 the right side evaluates into  $\mathbf{Oz}$  for all  $X, T \in \mathbf{Oz}$ . Taking  $X = b$  and  $T = a - b$  gives  $\text{lcm}_k \binom{a}{k} - \text{lcm}_k \binom{b}{k} = (a - b)w$  with  $w \in \mathbf{Oz}$ , including for nonzero infinite  $a - b$ ; the case  $a = b$  is trivial.  $\square$

**Theorem 16.14** (All-modulus Newton criterion [10]). *Let  $F \in \mathcal{N}_1(\mathbf{Oz})$  have Newton expansion  $F = \sum_{k=0}^d c_k \binom{X}{k}$ ,  $c_k \in \mathbf{Oz}$ . The following are equivalent:*

- (i)  $F$  is congruence preserving on  $\mathbf{Oz}$ ;
- (ii)  $c_k \in \text{lcm}_k \mathbf{Oz}$  for every  $0 \leq k \leq d$ ;
- (iii)  $\text{ct}(c_k) \in \text{lcm}_k \mathbb{Z}$  for every  $0 \leq k \leq d$ ;
- (iv) the ordinary numerical polynomial  $\text{ct}_*(F)$  is congruence preserving on  $\mathbb{Z}$ .

*Consequently, within the respective numerical-polynomial rings,*

$$\text{CP}(\mathbf{Oz}) = \Pi[X] \oplus \text{CP}(\mathbb{Z}). \quad (16.10)$$

*Proof.* (ii) $\Rightarrow$ (i). Write  $c_k = \text{lcm}_k d_k$  with  $d_k \in \mathbf{Oz}$ . Each summand  $d_k \text{lcm}_k \binom{X}{k}$  is congruence preserving by Lemma 16.13 (multiply the divisibility by  $d_k$ ), and a finite sum of congruence-preserving functions is congruence preserving.

(i) $\Rightarrow$ (ii). We show  $c_k \in \text{lcm}_k \mathbf{Oz}$  by induction on  $k$ . For  $k = 0$  it is automatic since  $\text{lcm}_0 = 1$ . Suppose it holds for all  $j < k$ . Then  $\sum_{j < k} c_j \binom{X}{j}$  is congruence preserving by the previous paragraph, so

$$G(X) = F(X) - \sum_{j < k} c_j \binom{X}{j} = \sum_{j \geq k} c_j \binom{X}{j}$$

is congruence preserving. Since  $\binom{n}{j} = 0$  for ordinary  $0 \leq n < j$  and  $\binom{k}{k} = 1$ , we get  $G(0) = \dots = G(k-1) = 0$  and  $G(k) = c_k$ . For each  $1 \leq m \leq k$ , comparing the inputs  $k$  and  $k - m$  gives  $m \mid G(k) - G(k - m) = c_k$  in  $\mathbf{Oz}$ . By (3.2), the ordinary integer  $\text{ct}(c_k)$  is divisible by every  $m \leq k$ , hence by  $\text{lcm}_k$ , and again by (3.2),  $c_k \in \text{lcm}_k \mathbf{Oz}$ .

(ii) $\Leftrightarrow$ (iii) is (3.2). For (iii) $\Leftrightarrow$ (iv), note that  $\text{ct}_*(F) = \sum_k \text{ct}(c_k) \binom{X}{k}$  is the Newton expansion of  $\text{ct}_*(F)$ , since the binomials have rational coefficients. The same two arguments over  $\mathbb{Z}$  (the identity

(16.9) with  $X, T \in \mathbb{Z}$ , and the same induction with divisibility in  $\mathbb{Z}$  show that an  $f \in \text{Int}(\mathbb{Z})$  with Newton coefficients  $m_k \in \mathbb{Z}$  is congruence preserving on  $\mathbb{Z}$  if and only if  $m_k \in \text{lcm}_k \mathbb{Z}$  for all  $k$ . Applied to  $m_k = \text{ct}(c_k)$ , this is (iii) $\Leftrightarrow$ (iv).

For (16.10), decompose  $F = U + f$  by Theorem 16.7. The Newton coefficients of  $U \in \Pi[X]$  are integral combinations of values  $U(\ell) \in \Pi$ , and  $\Pi \subseteq \text{lcm}_k \mathbf{Oz}$  by (3.2); so  $U \in \text{CP}(\mathbf{Oz})$  by (ii). An  $f \in \text{Int}(\mathbb{Z})$  lies in  $\text{CP}(\mathbf{Oz})$  exactly when it lies in  $\text{CP}(\mathbb{Z})$ , by (i) $\Leftrightarrow$ (iv) applied to  $f = \text{ct}_*(f)$ . Since  $\text{ct}_*(F) = f$ , the equivalence (i) $\Leftrightarrow$ (iv) for  $F$  now gives  $F \in \text{CP}(\mathbf{Oz})$  if and only if  $f \in \text{CP}(\mathbb{Z})$ .  $\square$

The divisibility condition in (i) includes infinite differences and therefore every infinite omnific modulus; no extra coefficient condition appears at those moduli. The ordinary least-common-multiple pattern is classical, and related finite-ring formulations are proved in [9]; Newton bases and integer-valued polynomials are treated in [8]. The content specific to the present domain is that the pattern suffices for differences at every surreal scale, not only for congruences modulo ordinary integers.

**Example 16.15** (Two kinds of denominators [10]). The polynomial  $12\binom{X}{4}$  is congruence preserving ( $\text{lcm}_4 = 12$ ), although its leading coefficient  $1/2$  is not an ordinary integer. On the other hand

$$F(X) = \binom{X}{2} + \omega^{\sqrt{2}} X^3$$

is integer valued on all of  $\mathbf{Oz}$  but not congruence preserving:  $F(2) - F(0) = 1 + 8\omega^{\sqrt{2}}$  has odd constant term and so is not divisible by 2 in  $\mathbf{Oz}$ , by (3.2). The purely infinite term does not repair the odd constant term.

*Remark 16.16* (What Theorem 16.14 does not say [10]). The theorem does not say that reduction modulo an infinite omnific integer is an ordinary finite congruence; such a quotient need not have any set-sized presentation (by Theorem 6.2 it has one only when it contains  $\Pi$ ). It says that the polynomial operation respects those congruences, with an explicit quotient supplied by (16.9). It also does not classify all congruence-preserving class functions  $\mathbf{Oz} \rightarrow \mathbf{Oz}$ : the finite Newton expansion is essential to the statement and the proof, and no infinite interpolation formula or full-class summation is implicit.

## 16.4 $p$ -adic characters and completions

In this subsection  $p$  is an ordinary prime,  $e \geq 1$  and  $r \geq 1$  are ordinary integers, and  $C(\mathbb{Z}_p^r, M)$  denotes the continuous functions from  $\mathbb{Z}_p^r$  with its  $p$ -adic topology to  $M$  ( $M = \mathbb{Z}/p^e\mathbb{Z}$  or  $\mathbb{F}_p$  discrete, or  $M = \mathbb{Z}_p$ ). All  $p$ -adic fields, rings and function spaces here are ordinary set-sized structures.

**Lemma 16.17** (Continuous extension [10]). *Every  $f \in \text{Int}(\mathbb{Z}^r)$  extends, by its rational-coefficient formula, to a continuous function  $\mathbb{Z}_p^r \rightarrow \mathbb{Z}_p$ , and this continuous extension is unique.*

*Proof.* A rational polynomial defines a continuous function  $\mathbb{Q}_p^r \rightarrow \mathbb{Q}_p$ . Ordinary integer tuples are dense in  $\mathbb{Z}_p^r$  and take values in  $\mathbb{Z} \subseteq \mathbb{Z}_p$ ; since  $\mathbb{Z}_p$  is closed in  $\mathbb{Q}_p$ , all values on  $\mathbb{Z}_p^r$  lie in  $\mathbb{Z}_p$ . Density also gives uniqueness.  $\square$

Reduction of values modulo  $p^e$  is therefore a ring homomorphism  $\text{Int}(\mathbb{Z}^r) \rightarrow C(\mathbb{Z}_p^r, \mathbb{Z}/p^e\mathbb{Z})$ . Its surjectivity needs an approximation argument; we prove it explicitly by indicator polynomials

rather than assume it, so no unproved approximation assertion is used. For  $0 \leq a < p^k$  write  $a = a_0 + a_1p + \cdots + a_{k-1}p^{k-1}$  with digits  $0 \leq a_j < p$ , and put

$$E_{a,k}(X) = \prod_{j=0}^{k-1} \left[ 1 - \left( \binom{X}{p^j} - a_j \right)^{p-1} \right] \in \text{Int}(\mathbb{Z}), \quad I_{a,k,e}(X) = E_{a,k}(X)^{p^{e-1}}. \quad (16.11)$$

**Lemma 16.18** (Digit and indicator congruences [10]). *For every  $x \in \mathbb{Z}_p$ , with  $\mathbf{1}_{a,k}(x) = 1$  if  $x \equiv a \pmod{p^k}$  and  $\mathbf{1}_{a,k}(x) = 0$  otherwise,*

$$E_{a,k}(x) \equiv \mathbf{1}_{a,k}(x) \pmod{p}, \quad I_{a,k,e}(x) \equiv \mathbf{1}_{a,k}(x) \pmod{p^e}.$$

*Proof.* First let  $x$  be a nonnegative ordinary integer with base- $p$  digits  $x_j$ . In  $\mathbb{F}_p[T]$  the Frobenius identity gives

$$(1+T)^x = \prod_{j \geq 0} (1+T)^{x_j p^j} = \prod_{j \geq 0} (1+T^{p^j})^{x_j}.$$

The coefficient of  $T^{p^j}$  on the right is  $x_j$ : the factors of index below  $j$  have combined degree at most  $\sum_{i < j} (p-1)p^i = p^j - 1$ , and factors of index above  $j$  contribute only exponents 0 or at least  $p^{j+1}$ . Hence  $\binom{x}{p^j} \equiv x_j \pmod{p}$ . Both sides are continuous functions  $\mathbb{Z}_p \rightarrow \mathbb{F}_p$  of  $x$  (Lemma 16.17; the  $j$ th digit depends only on  $x \bmod p^{j+1}$ ), and the nonnegative integers are dense in  $\mathbb{Z}_p$ , so the congruence holds for all  $x \in \mathbb{Z}_p$  with  $x_j$  the  $j$ th  $p$ -adic digit. By Fermat,  $t^{p-1} = 1$  for  $0 \neq t \in \mathbb{F}_p$ , so the  $j$ th factor of  $E_{a,k}(x)$  is  $\equiv 1$  if  $x_j = a_j$  and  $\equiv 0$  otherwise. The product is the indicator of  $x_j = a_j$  for all  $j < k$ , that is, of  $x \equiv a \pmod{p^k}$ .

For the lift, let  $z \in \mathbb{Z}_p$ . If  $z \equiv 0 \pmod{p}$  then  $z^{p^{e-1}} \equiv 0 \pmod{p^e}$ , because  $p^{e-1} \geq e$ . If  $z = 1 + pw$ , then  $(1 + pw)^{p^{e-1}} \equiv 1 \pmod{p^e}$  by induction from  $(1 + p^s v)^p \equiv 1 \pmod{p^{s+1}}$  for  $s \geq 1$ : in  $(1 + p^s v)^p = 1 + p^{s+1}v + \sum_{i=2}^p \binom{p}{i} p^{si} v^i$  the middle binomial coefficients are divisible by  $p$  and  $sp \geq s+1$ ; this is also valid for  $p = 2$ . Applying these two cases to  $z = E_{a,k}(x)$  gives the second congruence.  $\square$

**Theorem 16.19** (Exact finite-level function rings [10]). *The coefficientwise constant term and  $p$ -adic evaluation induce natural ring isomorphisms*

$$\mathcal{N}_r(\mathbf{Oz})/p^e \mathcal{N}_r(\mathbf{Oz}) \simeq \text{Int}(\mathbb{Z}^r)/p^e \text{Int}(\mathbb{Z}^r) \simeq C(\mathbb{Z}_p^r, \mathbb{Z}/p^e \mathbb{Z}). \quad (16.12)$$

*Proof.* By Theorem 16.7,  $\mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus \text{Int}(\mathbb{Z}^r)$  additively, and  $p^e \Pi[\mathbf{X}] = \Pi[\mathbf{X}]$  because  $\Pi$  is a  $\mathbb{Q}$ -vector space. Hence  $p^e \mathcal{N}_r(\mathbf{Oz}) = \Pi[\mathbf{X}] \oplus p^e \text{Int}(\mathbb{Z}^r)$ , and the ring homomorphism  $\text{ct}_*$  induces the first isomorphism.

For the second, the map sends  $f$  to  $x \mapsto f(x) \bmod p^e$  (Lemma 16.17). If  $f = p^e g$  with  $g \in \text{Int}(\mathbb{Z}^r)$ , its reduction is zero. Conversely, if  $f$  vanishes modulo  $p^e$  at every  $p$ -adic tuple, its values at ordinary integer tuples are divisible by  $p^e$ , so  $f/p^e \in \text{Int}(\mathbb{Z}^r)$ . The kernel is exactly  $p^e \text{Int}(\mathbb{Z}^r)$ .

Let  $g \in C(\mathbb{Z}_p^r, \mathbb{Z}/p^e \mathbb{Z})$ . The target is discrete, so every point has a residue box  $\mathbf{y} + p^k \mathbb{Z}_p^r$  on which  $g$  is constant; by compactness of  $\mathbb{Z}_p^r$  finitely many of these boxes cover it. Let  $k$  be the largest of their levels. Every level- $k$  box  $\mathbf{y} + p^k \mathbb{Z}_p^r$  lies in the covering box that contains  $\mathbf{y}$ , so  $g$  is constant on every residue box modulo  $p^k$ . For  $\mathbf{a} \in \{0, \dots, p^k - 1\}^r$  choose an ordinary integer  $c_{\mathbf{a}}$  lifting the value of  $g$  on the box of  $\mathbf{a}$ , and put

$$f(\mathbf{X}) = \sum_{\mathbf{a}} c_{\mathbf{a}} \prod_{j=1}^r I_{a_j, k, e}(X_j) \in \text{Int}(\mathbb{Z}^r).$$

By Lemma 16.18, at  $\mathbf{x}$  in the box of  $\mathbf{a}$  the product indexed by  $\mathbf{a}$  is  $\equiv 1$  and all others are  $\equiv 0$  modulo  $p^e$ , so  $f(\mathbf{x}) \equiv g(\mathbf{x}) \pmod{p^e}$ . This proves surjectivity.  $\square$



**Corollary 16.20** (The  $p$ -adic completion [10]). *There is a canonical isomorphism of topological rings*

$$\varprojlim_{e \geq 1} \mathcal{N}_r(\mathbf{Oz})/p^e \mathcal{N}_r(\mathbf{Oz}) \simeq C(\mathbb{Z}_p^r, \mathbb{Z}_p), \quad (16.13)$$

where the left side carries the inverse-limit topology of the discrete quotients and the right side the uniform  $p$ -adic topology. The canonical map  $\mathcal{N}_r(\mathbf{Oz}) \rightarrow C(\mathbb{Z}_p^r, \mathbb{Z}_p)$  sends  $F$  to the continuous extension of  $\text{ct}_*(F)$ , has dense image, and has kernel  $\Pi[\mathbf{X}]$ .

*Proof.* The isomorphisms of Theorem 16.19 commute with reduction of values from level  $e + 1$  to level  $e$ . A coherent sequence of functions  $g_e : \mathbb{Z}_p^r \rightarrow \mathbb{Z}/p^e\mathbb{Z}$  determines a unique function  $g : \mathbb{Z}_p^r \rightarrow \mathbb{Z}_p$  pointwise, and  $g$  is continuous exactly when every  $g_e = g \bmod p^e$  is continuous, since the sets  $y + p^e\mathbb{Z}_p$  form a base of the topology of  $\mathbb{Z}_p$ . This identifies the inverse limit with  $C(\mathbb{Z}_p^r, \mathbb{Z}_p)$  as rings. The kernel of the projection to level  $e$  corresponds to the functions that vanish modulo  $p^e$  everywhere, which are the basic neighbourhoods of 0 in the uniform  $p$ -adic topology; so the topologies agree. Surjectivity at every finite level gives density of the image.

If the continuous extension of  $f \in \text{Int}(\mathbb{Z}^r)$  is zero, then  $f$  vanishes on  $\mathbb{Z}^r$ , and a rational polynomial vanishing on  $\mathbb{Z}^r$  is zero (write it as a polynomial in  $X_r$  with coefficients in  $\mathbb{Q}[X_1, \dots, X_{r-1}]$ ; for each ordinary tuple  $(n_1, \dots, n_{r-1})$  the resulting univariate polynomial has infinitely many roots, so all coefficients vanish on  $\mathbb{Z}^{r-1}$ ; induct on  $r$ ). With Theorem 16.7 the kernel on  $\mathcal{N}_r(\mathbf{Oz})$  is exactly  $\Pi[\mathbf{X}]$ . Equivalently  $\bigcap_e p^e \text{Int}(\mathbb{Z}^r) = 0$ , because an ordinary integer divisible by every  $p^e$  is zero.  $\square$

*Remark 16.21* (Not a profinite ring completion [10]). For  $r \geq 1$  the characteristic functions of residue boxes at increasing depths are infinitely many distinct elements of  $C(\mathbb{Z}_p^r, \mathbb{F}_p)$ . Therefore  $\mathcal{N}_r(\mathbf{Oz})/p\mathcal{N}_r(\mathbf{Oz})$  is infinite (and discrete in its quotient topology), as are all the rings in (16.12). The completion (16.13) is a  $p$ -adic completion, not an inverse limit of finite rings. This differs from the genuine profinite completion of  $\mathbf{Oz}$ : its finite quotient presentations are the rings  $\mathbb{Z}/n\mathbb{Z}$  with kernels  $n\mathbf{Oz}$  (Theorem 6.2), so  $\varprojlim_n \mathbf{Oz}/n\mathbf{Oz} \simeq \widehat{\mathbb{Z}}$ .

The ordinary polynomial-approximation content is part of the classical theory of integer-valued polynomials [8]. The explicit proof above also identifies exactly which omnific coefficients disappear, rather than only invoking density for the ordinary subring. The compactness of  $\mathbb{Z}_p^r$  used here is ordinary  $p$ -adic compactness, not compactness of an omnific interval.

**Theorem 16.22** ( $p$ -adic classification of characters [10]). *Every unital class ring homomorphism  $\chi : \mathcal{N}_r(\mathbf{Oz}) \rightarrow \mathbb{F}_p$  has the form*

$$\chi_\alpha(F) = \text{ct}_*(F)(\alpha) \bmod p \quad (16.14)$$

for a unique  $\alpha \in \mathbb{Z}_p^r$ , and every  $\alpha \in \mathbb{Z}_p^r$  defines such a character. The prime ideals of  $\mathcal{N}_r(\mathbf{Oz})$  containing  $p\mathcal{N}_r(\mathbf{Oz})$  are exactly the kernels  $\ker \chi_\alpha$ , with distinct  $\alpha$  giving distinct kernels; every such prime ideal is maximal with quotient  $\mathbb{F}_p$ .

*Proof.* The map  $\chi_\alpha$  is a composite of ring homomorphisms (Theorem 16.7 and Lemma 16.17), so it is a character. Conversely, a unital  $\chi$  kills  $p\mathcal{N}_r(\mathbf{Oz})$ , so by Theorem 16.19 with  $e = 1$  it is  $\psi \circ \rho$ , where  $\rho(F) = \text{ct}_*(F) \bmod p$  and  $\psi : C(\mathbb{Z}_p^r, \mathbb{F}_p) \rightarrow \mathbb{F}_p$  is a unital ring homomorphism. At level  $k$ , the indicators of the  $p^{kr}$  residue boxes modulo  $p^k$  are pairwise orthogonal idempotents with sum 1. Their images are idempotents of  $\mathbb{F}_p$ , hence 0 or 1, pairwise orthogonal with sum 1, so exactly one box of level  $k$  is selected. A box indicator is the sum of the indicators of its children at level  $k + 1$ , and a child indicator  $c$  satisfies  $c = c \cdot b$  for its parent  $b$ ; so the selected boxes are nested. They are nonempty compact sets with diameters tending to 0, so by completeness of  $\mathbb{Z}_p^r$  they have exactly

one common point  $\alpha$ . Every  $h \in C(\mathbb{Z}_p^r, \mathbb{F}_p)$  is constant on the residue boxes of some level  $k$  (as in the proof of [Theorem 16.19](#)), so  $h$  is an  $\mathbb{F}_p$ -linear combination of box indicators and  $\psi(h)$  is the value of  $h$  on the selected box, that is,  $h(\alpha)$ . Hence  $\chi = \chi_\alpha$ . Distinct points are separated by a residue-box indicator, which is  $\rho(F)$  for some  $F$  by surjectivity; so  $\alpha$  is unique.

The prime ideals of  $\mathcal{N}_r(\mathbf{Oz})$  containing  $p\mathcal{N}_r(\mathbf{Oz})$  correspond, through the explicit set-sized presentation  $\rho$ , to the prime ideals of  $C(\mathbb{Z}_p^r, \mathbb{F}_p)$ . Every element  $z$  of this ring satisfies  $z^p = z$  pointwise. The quotient by a prime ideal is a commutative domain in which every element is a root of  $X^p - X$ ; such a domain has at most  $p$  elements, and it contains the  $p$  distinct images of the constants, so it is  $\mathbb{F}_p$ . Thus every such prime is maximal with quotient  $\mathbb{F}_p$ , and the quotient map is a character, hence some  $\chi_\alpha$ . A unital surjection onto  $\mathbb{F}_p$  is determined by its kernel, because  $\mathbb{F}_p$  has only the identity automorphism; with the uniqueness of  $\alpha$  this shows that distinct parameters give distinct kernels.  $\square$

**Corollary 16.23** (Most characters are not omnific evaluations [10]). *A character  $\mathcal{N}_r(\mathbf{Oz}) \rightarrow \mathbb{F}_p$  obtained by evaluating at a tuple  $\mathbf{x} \in \mathbf{Oz}^r$  and then applying a unital ring homomorphism  $\mathbf{Oz} \rightarrow \mathbb{F}_p$  has parameter  $\alpha = (\text{ct}(x_1), \dots, \text{ct}(x_r)) \in \mathbb{Z}^r \subset \mathbb{Z}_p^r$ . Conversely every parameter in  $\mathbb{Z}^r$  arises in this way. Parameters in  $\mathbb{Z}_p^r \setminus \mathbb{Z}^r$  give characters that are not evaluation at any omnific tuple.*

*Proof.* By [Theorem 5.1](#) the unital homomorphism  $\mathbf{Oz} \rightarrow \mathbb{F}_p$  is uniquely  $\text{ct}$  followed by reduction modulo  $p$ . By (16.6), its composite with evaluation at  $\mathbf{x}$  is  $F \mapsto \text{ct}_*(F)(\text{ct}(\mathbf{x})) \bmod p$ , the character with parameter  $\text{ct}(\mathbf{x})$ . An ordinary integer tuple is itself an allowed input, which gives the converse. The last assertion follows from the uniqueness of the parameter in [Theorem 16.22](#).  $\square$

For  $r = 1$  the lacunary  $p$ -adic integer (16.2) of the omitted type in [Proposition 16.3](#) gives a concrete nonevaluation character. There are uncountably many parameters and only countably many integer ones. The enormous class of omnific inputs does not enlarge the evaluation parameters beyond the ordinary integers, because the constant term of every omnific input is an ordinary integer.

## 16.5 Rational self-maps and gaps at arbitrary surreal scales

Throughout this subsection  $(E, \Lambda)$  is one of the pairs  $(\mathbf{No}, \mathbf{Oz})$  or  $(\mathbf{No}[i], \mathbf{Oz}[i])$ ; in the second the absolute value is the surreal-valued modulus. The input and output lattices are the full classes, not a fixed set-sized truncation. We write  $\mathcal{N}_1(\Lambda)$  for  $\mathcal{N}_1(\mathbf{Oz})$ , respectively  $\mathcal{N}_1(\mathbf{Oz}[i])$ , and  $E(X)$  for the rational functions in one variable over the class field  $E$ . We use repeatedly the following estimate, valid in  $E$  by the properties of the modulus: if  $Q(X) = \sum_{j \leq n} q_j X^j \in E[X]$  has  $q_n \neq 0$  and  $|y| \geq R_Q := 1 + 2 \sum_{j < n} |q_j/q_n|$ , then

$$\frac{1}{2}|q_n||y|^n \leq |Q(y)| \leq \frac{3}{2}|q_n||y|^n, \quad (16.15)$$

because  $|\sum_{j < n} q_j y^j| \leq \sum_{j < n} |q_j||y|^{n-1} \leq \frac{1}{2}|q_n||y|^n$ .

**Lemma 16.24** (Finite differences of a proper rational function [10]). *Let  $r \in E(X)$  be nonzero and proper, and let  $\Delta r(X) = r(X+1) - r(X)$ . For every ordinary  $k \geq 0$ ,  $\Delta^k r$  is nonzero and proper. If the leading Laurent term of  $r$  at infinity is  $cX^{-m}$  with  $m \geq 1$ , then the leading term of  $\Delta^k r$  is*

$$(-1)^k c m(m+1) \cdots (m+k-1) X^{-m-k}, \quad (16.16)$$

*the empty product for  $k = 0$  being 1.*

*Proof.* Compute in the field of formal Laurent series in  $X^{-1}$  over a set-sized subfield of  $E$  containing the finitely many coefficients involved; the rational functions over that subfield embed in it, a rational function is proper exactly when its expansion has only negative exponents, and the shift  $X \mapsto X + 1$  becomes the substitution  $X^{-1} \mapsto X^{-1}(1 + X^{-1})^{-1}$ . The binomial series gives

$$(X + 1)^{-n} - X^{-n} = X^{-n}((1 + X^{-1})^{-n} - 1) = -nX^{-n-1} + \text{terms of lower degree in } X.$$

Since  $r = cX^{-m} + (\text{terms } X^{-n}, n > m)$ , and  $\Delta$  sends each such lower term to terms of degree at most  $-n - 1 < -m - 1$ , the leading term of  $\Delta r$  is  $-cmX^{-m-1}$ . Induction on  $k$  gives (16.16). The coefficient is nonzero in characteristic zero and the exponent is negative, which proves both assertions.  $\square$

**Theorem 16.25** (Polynomiality of rational lattice self-maps [10; 16; 18]). *Let  $f \in E(X)$  satisfy  $f(x) \in \Lambda$  for every  $x \in \Lambda$  at which  $f$  is defined. Then  $f \in E[X]$ . In particular its reduced rational presentation has no finite poles anywhere in  $E$ .*

*Proof.* Divide the numerator by the denominator to write  $f = P + r$  with  $P \in E[X]$  and  $r$  proper, and suppose  $r \neq 0$ . Choose an ordinary  $k \geq 1$  larger than  $\deg P$  (any  $k \geq 1$  if  $P = 0$ ). Then  $\Delta^k P = 0$ , so  $g := \Delta^k f = \Delta^k r$  is nonzero and proper by Lemma 16.24. Write  $f$  and  $g$  in reduced form,  $f = N_f/D_f$  and  $g = N_g/D_g$ , with  $\deg N_g < \deg D_g$ .

We find  $B \in \mathbf{No}$  such that for every  $x \in \mathbf{No}$  with  $x > B$  all of  $f(x), \dots, f(x + k)$  are defined,  $g(x)$  is defined, and

$$0 < |g(x)| < 1. \quad (16.17)$$

By (16.15),  $D_f(y) \neq 0$  for  $|y| \geq R_{D_f}$ ,  $N_g(y) \neq 0$  for  $|y| \geq R_{N_g}$ , and for  $|y| \geq \max(1, R_{N_g}, R_{D_g})$ ,

$$|g(y)| \leq \frac{\frac{3}{2}|\ell(N_g)||y|^{\deg N_g}}{\frac{1}{2}|\ell(D_g)||y|^{\deg D_g}} \leq 3 \frac{|\ell(N_g)|}{|\ell(D_g)|} |y|^{-1},$$

where  $\ell(\cdot)$  is the leading coefficient. Take  $B$  larger than 1, than  $R_{D_f}, R_{N_g}, R_{D_g}$  and than  $3|\ell(N_g)/\ell(D_g)|$ ; for real  $x > B$  the points  $x, \dots, x + k$  are real and larger than  $B$ , so every requirement holds. These are finitely many inequalities in the ordered field  $\mathbf{No}$ .

By Lemma 16.1 choose  $x \in \mathbf{Oz}$  with  $x > B$ . Then  $x + j \in \mathbf{Oz} \subseteq \Lambda$  for  $0 \leq j \leq k$ ,  $f$  is defined at these points, and

$$g(x) = \sum_{j=0}^k (-1)^{k-j} \binom{k}{j} f(x + j) \in \Lambda.$$

This contradicts (16.17) and the lattice separation of Lemma 16.1. Therefore  $r = 0$  and  $f = P$ .  $\square$

The finite-difference argument is a general discrete-lattice method. It is included as a supporting result, not asserted to be a new ordinary theorem about rational functions on  $\mathbb{Z}$ . 16 proves the theorem for every unital constant ring with one fresh input (Theorem 16.42); the two routes are kept.

Recall that  $y \in E$  is finite if  $|y|$  is bounded by an ordinary integer and infinitesimal if  $|y| < 1/n$  for every ordinary  $n \geq 1$ ; the standard part of a finite element is the unique real (respectively complex) number at infinitesimal distance, taken coordinatewise in  $\mathbf{No}[i]$  (Remark 16.2).

**Lemma 16.26** (Finite roots and reduction [10]). *A root in  $E$  of a monic polynomial with finite coefficients is finite. The standard part is a ring homomorphism on the finite elements of  $E$  and carries such a root to a root of the coefficientwise standard-part polynomial. For finite  $z \in \mathbf{No}[i]$ ,*

$$\text{st}(|z|) = |\text{st}(z)|.$$

*Proof.* Let  $z^d + \sum_{j < d} c_j z^j = 0$  with  $|c_j| \leq C$  for an ordinary integer  $C$ . If  $|z| > 1 + dC$ , then

$$\left| \sum_{j < d} c_j z^j \right| \leq dC|z|^{d-1} < |z|^d,$$

which excludes a root; so every root is finite. Sums and products of finite elements with infinitesimal differences have infinitesimal differences, so  $\text{st}$  is a ring homomorphism on the finite elements (Lemma 2.2 for  $\mathcal{O}_{\mathbb{R}}$  and  $\mathcal{O}_{\mathbb{C}}$ ) and commutes with evaluation of polynomials; applying it to the equation gives the second assertion. For finite  $z \in \mathbf{No}[i]$ ,  $|z| \leq |\Re z| + |\Im z|$  is finite; taking standard parts in  $|z|^2 = (\Re z)^2 + (\Im z)^2$  gives  $\text{st}(|z|)^2 = |\text{st}(z)|^2$ , and  $\text{st}(|z|) \geq 0$ , so uniqueness of nonnegative real square roots gives the identity.  $\square$

For an ordinary  $d \geq 2$  define the ordinary integer

$$q_{E,d} = \begin{cases} 1, & E = \mathbf{No} \text{ and } d \text{ odd,} \\ 2, & E = \mathbf{No} \text{ and } d \text{ even,} \\ d, & E = \mathbf{No}[i]. \end{cases} \quad (16.18)$$

**Theorem 16.27** (A bounded number of lattice preimages at every scale [10]). *Let  $F \in \mathcal{M}_1(\Lambda)$  have degree  $d \geq 2$ , and let  $H \in \mathbf{No}_{>0}$ . There is a positive surreal  $T_0$  such that, for every  $T \in \mathbf{Oz}$  with  $T > T_0$ , the class*

$$\{u \in \Lambda : |F(u) - F(T)| \leq H\} \quad (16.19)$$

*has at most  $q_{E,d}$  elements. In particular  $F(\Lambda)$  has at most  $q_{E,d}$  points in the closed ball of radius  $H$  about  $F(T)$ .*

*Proof.* Write  $F(X) = a_d X^d + \sum_{j < d} a_j X^j$  with  $a_d \neq 0$ , and put  $T_0 = \omega \cdot \max(1, H/|a_d|, |a_j/a_d| \ (j < d))$ . For  $T > T_0$  we have  $T > 1$  and, for every ordinary  $n \geq 1$ ,  $T > n|a_j/a_d|$  and  $T > nH/|a_d|$ . Since  $d - j \geq 1$  and  $d - 1 \geq 1$ , it follows that

$$b_j := \frac{a_j}{a_d T^{d-j}} \ (j < d) \quad \text{and} \quad \frac{H}{|a_d| T^{d-1}} \quad \text{are infinitesimal.} \quad (16.20)$$

Fix such a  $T$  and an element  $u$  of (16.19), and put  $z = u/T$ . Dividing  $F(u) - F(T)$  by  $a_d T^d$  gives

$$z^d + \sum_{j < d} b_j z^j = 1 + \sum_{j < d} b_j + \delta, \quad |\delta| \leq \frac{H}{|a_d| T^d} \leq \frac{H}{|a_d| T^{d-1}}, \quad (16.21)$$

so  $\delta$  is infinitesimal. Thus  $z$  is a root of a monic polynomial with finite coefficients, hence finite by Lemma 16.26, and taking standard parts gives  $\text{st}(z)^d = 1$ . There are at most  $d$  such residues in  $\mathbb{C}$ ; among real residues only 1 is possible for odd  $d$ , and only  $\pm 1$  for even  $d$ . So there are at most  $q_{E,d}$  possible residues  $\text{st}(u/T)$ .

It remains to show that each residue belongs to at most one lattice input. Let  $u, v$  be elements of (16.19) with  $\text{st}(u/T) = \text{st}(v/T) = \zeta$ , and put  $z = u/T$ ,  $w = v/T$ . Write  $F(u) - F(v) = (u - v)Q_F(u, v)$  with  $Q_F(u, v) = \sum_{j=1}^d a_j \sum_{\ell=0}^{j-1} u^{j-1-\ell} v^\ell$ . Direct expansion gives

$$\frac{Q_F(u, v)}{a_d T^{d-1}} = \sum_{\ell=0}^{d-1} z^{d-1-\ell} w^\ell + \sum_{j=1}^{d-1} b_j \sum_{\ell=0}^{j-1} z^{j-1-\ell} w^\ell. \quad (16.22)$$

Its standard part is  $d\zeta^{d-1}$ , of modulus  $d$  because  $\zeta^d = 1$ ; by Lemma 16.26 the standard part of its modulus is  $d$ , so

$$|Q_F(u, v)| > \frac{d}{2}|a_d|T^{d-1}.$$

If  $u \neq v$ , the triangle inequality gives  $|F(u) - F(v)| \leq 2H$ , hence

$$|u - v| = \frac{|F(u) - F(v)|}{|Q_F(u, v)|} < \frac{4H}{d|a_d|T^{d-1}} < 1,$$

the middle quantity being infinitesimal by (16.20). This contradicts lattice separation for  $0 \neq u - v \in \Lambda$  (Lemma 16.1). So  $u = v$ , and (16.19) has at most  $q_{E,d}$  elements. Every image point in the ball is the image of an element of (16.19), which gives the last assertion.  $\square$

The proof uses the finite residue polynomial  $Z^d - 1$  only as a label for possible clusters. It assumes no convergence of an iterative root-finding scheme, no compactness of an omnific interval, and no existence theorem for new roots in the full field.

**Theorem 16.28** (Arbitrary-scale holes in nonlinear polynomial images [10]). *Let  $F \in \mathcal{N}_1(\Lambda)$  have degree  $d \geq 2$ . For every  $H > 0$  and every  $B \geq 0$  in  $\mathbf{No}$  there is  $c \in \Lambda$  with  $|c| > B$  such that*

$$\{y \in E : |y - c| \leq H\} \cap F(\Lambda) = \emptyset. \quad (16.23)$$

*For the real lattice this is an empty closed interval; for the Gaussian lattice it is an empty closed surcomplex disk.*

*Proof.* Put  $q = q_{E,d}$  and choose, by Lemma 16.1, a positive  $M \in \mathbf{Oz}$  with  $M > H$ . Let  $L = M + 1$  and  $R = 3qL + M$ . Apply Theorem 16.27 with radius  $R$  to get  $T_0$ , and choose  $T \in \mathbf{Oz}$  with  $T > T_0$  and also

$$|F(T)| > B + 3qL.$$

This is possible: by (16.15), for  $T \geq \max(1, R_F)$  one has  $|F(T)| \geq \frac{1}{2}|a_d|T^d \geq \frac{1}{2}|a_d|T$ , so any  $T \in \mathbf{Oz}$  exceeding  $T_0$ ,  $1$ ,  $R_F$  and  $2(B + 3qL)/|a_d|$  works (Lemma 16.1).

Set  $y = F(T) \in \Lambda$  and consider the  $q + 1$  lattice points  $c_j = y + 3jL$ ,  $j = 0, \dots, q$ . The closed balls of radius  $M$  about them are pairwise disjoint, since distinct centres are at distance at least  $3L > 2M$ . Each lies in the closed ball of radius  $R$  about  $y$ , since  $|x - y| \leq |x - c_j| + 3jL \leq M + 3qL$ . That ball contains at most  $q$  points of  $F(\Lambda)$  by Theorem 16.27, so one of the  $q + 1$  smaller balls contains none. Its centre  $c = c_j$  satisfies  $|c| \geq |y| - 3qL > B$ , and since  $H < M$  the ball of radius  $H$  about  $c$  is disjoint from  $F(\Lambda)$  as well.  $\square$

Failure of surjectivity is therefore not a single small defect near zero: the gaps exceed any prescribed infinite surreal radius and occur beyond any prescribed bound in modulus, that is, at arbitrarily large moduli.

**Corollary 16.29** (All rational bijections of the two lattices [10; 16; 18]). *A rational function  $F \in E(X)$  defining a bijection  $\Lambda \rightarrow \Lambda$  has the form*

$$F(X) = uX + b, \quad b \in \Lambda, \quad u \in \Lambda^\times,$$

*and every such map is a bijection. Thus the rational bijections of  $\mathbf{Oz}$  are  $X \mapsto \pm X + b$  with  $b \in \mathbf{Oz}$ , and those of  $\mathbf{Oz}[i]$  are  $X \mapsto uX + b$  with  $b \in \mathbf{Oz}[i]$  and  $u \in \{\pm 1, \pm i\}$ .*

*Proof.* By [Theorem 16.25](#),  $F$  is a polynomial, so  $F \in \mathcal{N}_1(\Lambda)$ . A constant is not a bijection of  $\Lambda$ . Degree at least two is excluded by [Theorem 16.28](#), because the centre  $c$  there is itself a lattice point missing from the image. Hence  $F(X) = aX + b$  with  $b = F(0) \in \Lambda$  and  $a = F(1) - F(0) \in \Lambda$ . Surjectivity gives  $x \in \Lambda$  with  $F(x) = b + 1$ , so  $ax = 1$  and  $a$  is a unit of  $\Lambda$ . Conversely, an affine map with slope  $u \in \Lambda^\times$  has the lattice-preserving inverse  $X \mapsto u^{-1}(X - b)$ . The unit lists are those of [Lemma 2.4](#).  $\square$

Here “rational” means a rational function in one variable over the ambient field  $E$ . Arbitrary functions  $\Lambda \rightarrow \Lambda$ , and maps such as coordinatewise conjugation on  $\mathbf{Oz}[i]$ , are not being classified. 16 shows that surjectivity alone already forces the affine form, for every unital constant ring ([Theorem 16.43](#)).

## 16.6 Fresh scales: image gaps, fibers and surjective self-maps

This subsection and the next are manuscript 16’s. They work over an arbitrary constant ring and replace the lattice-separation mechanism of [Section 16.5](#) by a *fresh scale*: an exponent above an entire set-sized group of old exponents. Throughout,  $k \in \{\mathbb{R}, \mathbb{C}\}$ ,  $E_k = k((X^{\mathbf{No}}))$  (so  $E_{\mathbb{R}} = \mathbf{No}$ ,  $E_{\mathbb{C}} = \mathbf{No}[i]$ ),  $D \subseteq k$  is a unital subring, and  $\mathcal{A}_{D,k} = D + \Pi_k$  as in [Section 2.3](#);  $\mathcal{A}_{\mathbb{Z},\mathbb{R}} = \mathbf{Oz}$  and  $\mathcal{A}_{\mathbb{Z}[i],\mathbb{C}} = \mathbf{Oz}[i]$ . The ring  $\mathcal{A}_{k,k} = k + \Pi_k$  of normal forms without negative exponents (16’s  $\mathcal{I}_k$ ) is *not* the ring  $\mathcal{O}_k$  of finite surreals:  $\omega \in \mathcal{A}_{k,k}$  and  $\omega^{-1} \notin \mathcal{A}_{k,k}$ . 16’s  $\mathcal{P}_k$  and  $\mathcal{A}_D$  are  $\Pi_k$  and  $\mathcal{A}_{D,k}$ ; its defect polynomial  $D_{\zeta,c}$  is  $d_{\zeta,c}$  (the letter  $D$  is the constant ring), its centre  $\mu$  is  $s$ , its affine part  $L_\zeta$  is  $a_\zeta$  and its forbidden coefficient  $A_\zeta$  is  $c_\zeta$ . For a set-sized additive subgroup  $G \subseteq \mathbf{No}$ ,  $K_G = k((X^G))$  is the Hahn field with supports in  $G$ , identified with its normal-form image in  $E_k$  ( $X^g \mapsto \omega^g$ ).

**Lemma 16.30** (Fresh exponents [16; 22]). *Every set-sized family of elements of  $E_k$  lies in  $K_G$  for some set-sized subgroup  $G \subseteq \mathbf{No}$ . There is  $b > 0$  with  $b > g$  for all  $g \in G$  (a fresh exponent), and then  $jb + g > 0$  for  $j > 0$  and  $jb + g < 0$  for  $j < 0$  ( $j \in \mathbb{Z}$ ,  $g \in G$ ); in particular  $G \cap \mathbb{Z}b = \{0\}$ .*

*Proof.* Let  $G$  be the subgroup generated by the union of the supports; it is a set, and  $K_G$  is a set-sized Hahn field. Choose  $b$  strictly above the set  $G \cup \{0\}$  ([Lemma 3.1](#)); since  $G = -G$ ,  $b + g > 0 > -b + g$ , and the same follows for all nonzero integral multiples of  $b$ .  $\square$

Freshness dominates the whole group, not only finitely many leading exponents: to dominate  $G = \mathbb{Q}$  one may take  $b = \omega$ , while no ordinary real suffices (16).

**Theorem 16.31** (Laurent blocks [16; 22]). *Let  $b > G$  be fresh and  $t = \omega^b$ . The map  $\iota_b : K_G((T^{-1})) \rightarrow E_k$ ,  $\sum_{j \leq m} c_j T^j \mapsto \sum_{j \leq m} c_j \omega^{jb}$ , is an injective field homomorphism, the identity on  $K_G$ , with  $\iota_b(T) = t$ . Each nonzero  $c_j$  occupies the support block  $jb + \text{supp } c_j$ ; blocks with different  $j$  are disjoint and ordered by  $j$ . In particular  $t$  is transcendental over  $K_G$ .*

*Proof.* For  $j < \ell$  and  $g, h \in G$ ,  $(\ell b + h) - (jb + g) = (\ell - j)b + (h - g) > 0$ . A nonempty subset of the union of the blocks has a greatest element: take the greatest outer index of a block meeting it (the indices are bounded above), then the greatest element in that block. So the image is a legitimate Hahn sum even when the  $c_j$  have infinite supports. Addition is coefficientwise; in a product a fixed outer index receives contributions from finitely many pairs of outer indices, each an ordinary Hahn product. A nonzero coefficient gives a nonempty block that cannot cancel, so  $\iota_b$  is injective; a unital injective ring map from a field preserves inverses, and injectivity on  $K_G[T]$  is transcendence.  $\square$



**Corollary 16.32** (Exact support test [16]). *For  $L = \sum_{j \leq m} c_j T^j$ ,  $\iota_b(L) \in \mathcal{A}_{D,k}$  if and only if  $c_j = 0$  for all  $j < 0$  and  $c_0 \in \mathcal{A}_{D,k}$ ; the same holds with  $\mathcal{A}_{k,k}$  in place of  $\mathcal{A}_{D,k}$ . Positive-index blocks are purely infinite.*

*Proof.* Negative-index blocks consist of negative exponents and cannot cancel; the block of index 0 is  $c_0$ ; positive-index blocks do not affect the negative terms or the constant term.  $\square$

**Theorem 16.33** (One fresh value forces polynomiality [16]). *Let  $F \in K_G(T)$ . Its denominator does not vanish at  $t$ , and  $F(t) \in \mathcal{A}_{D,k}$  if and only if  $F \in K_G[T]$  and the constant coefficient of  $F$  lies in  $\mathcal{A}_{D,k}$ . The same holds with  $t$  replaced by  $ut$  for any  $u \in K_G^\times$ ; these inputs lie in  $\Pi_k$ . If the proper part of  $F$  begins  $aT^{-n} + O(T^{-n-1})$  with  $a \neq 0$ ,  $n \geq 1$ , then  $F(t)$  has the nonzero block  $-nb + \text{supp } a$ , and if the constant coefficient of the polynomial part has no negative terms, the leading negative exponent of  $F(t)$  is  $\max \text{supp } a - nb$ , with the leading coefficient of  $a$ .*

*Proof.* Transcendence gives the first assertion. Expand  $F$  at infinity by polynomial division and a geometric inverse; its image under  $\iota_b$  is  $F(t)$ , and a rational function has no negative Laurent coefficients exactly when it is a polynomial. Apply [Corollary 16.32](#). Substituting  $uT$  multiplies  $c_j$  by  $u^j$  and fixes  $c_0$ ;  $ut$  occupies the positive block  $b + G$ . The last assertion is the block order of [Theorem 16.31](#).  $\square$

One fresh value detects negative Laurent tails, not integer-valuedness:  $F(X) = X/2$  passes the test but fails at  $X = 1$  (16).

**Lemma 16.34** (Branches of a shifted value [16]). *Let  $d \geq 2$ , let  $K$  be a field of characteristic 0 containing the  $d$ th roots of unity, and  $f = a_d X^d + \dots + a_0 \in K[X]$  with  $a_d \neq 0$ . For every  $c \in K$  the equation  $f(Y) = f(T) + c$  has exactly  $d$  distinct roots in  $K((T^{-1}))$ , indexed by  $\zeta^d = 1$ , of the form*

$$Y_{\zeta,c} = \zeta T + (\zeta - 1)s + O(T^{-1}), \quad s = \frac{a_{d-1}}{da_d}.$$

*Proof.* With  $u = T^{-1}$  and  $W = uY$ , multiplying by  $u^d$  gives  $\sum_j a_j u^{d-j}(W^j - 1) - cu^d = 0$ , which at  $u = 0$  is  $a_d(W^d - 1) = 0$  with simple roots  $\zeta$  (the  $W$ -derivative there is  $da_d \zeta^{d-1} \neq 0$ ). The coefficient of  $u^n$  is then solved uniquely step by step, giving  $W_\zeta \in K[[u]]$  with  $W_\zeta(0) = \zeta$ , and  $Y = TW_\zeta(T^{-1})$ ; the  $d$  distinct leading coefficients give  $d$  distinct roots, which exhaust the roots of a polynomial of degree  $d$ . Comparing coefficients of  $T^{d-1}$  in  $f(\zeta T + w + \dots) = f(T) + c$  gives  $da_d \zeta^{d-1} w + a_{d-1} \zeta^{d-1} = a_{d-1}$ , that is  $w = (\zeta - 1)s$ .  $\square$

Put  $\mathbf{a}_\zeta(T) = \zeta T + (\zeta - 1)s$  and  $\mathbf{d}_{\zeta,c}(T) = f(\mathbf{a}_\zeta(T)) - f(T) - c$ . The choices of  $\zeta$  and  $s$  cancel the terms of degrees  $d$  and  $d - 1$ , so  $\deg \mathbf{d}_{\zeta,c} \leq d - 2$  unless  $\mathbf{d}_{\zeta,c} = 0$ . Since  $\mathbf{a}_\zeta$  fixes  $-s$ ,

$$\mathbf{d}_{\zeta,c}(-s) = -c : \tag{16.24}$$

a nonzero output shift cannot be produced by an affine change of input, even if the coefficients of  $f$  are infinite or infinitesimal (16). Degrees and leading coefficients lc below refer to the formal variable  $T$ .

**Theorem 16.35** (Forbidden layers [16]). *In [Lemma 16.34](#) let  $c \neq 0$ . For each  $\zeta^d = 1$  put  $q_\zeta = \deg \mathbf{d}_{\zeta,c}$ ,  $r_\zeta = d - 1 - q_\zeta$  and  $\mathbf{c}_\zeta = -\text{lc}(\mathbf{d}_{\zeta,c})/(da_d \zeta^{d-1})$ . Then  $\mathbf{d}_{\zeta,c} \neq 0$ ,  $1 \leq r_\zeta \leq d - 1$ ,  $\mathbf{c}_\zeta \neq 0$  and*

$$Y_{\zeta,c} = \mathbf{a}_\zeta(T) + \mathbf{c}_\zeta T^{-r_\zeta} + O(T^{-r_\zeta-1}); \quad \text{in particular} \quad Y_{1,c} = T + \frac{c}{da_d} T^{1-d} + O(T^{-d}),$$

so the bound  $d - 1$  is attained for every  $c \neq 0$ .

*Proof.* By (16.24) the defect is nonzero, of degree between 0 and  $d-2$ . The remainder  $R = Y_{\zeta,c} - \mathbf{a}_\zeta$  lies in  $T^{-1}K[[T^{-1}]]$  and is nonzero; let  $\chi T^m$  ( $m \leq -1$ ) be its first term. Taylor expansion gives  $0 = \mathbf{d}_{\zeta,c} + f'(\mathbf{a}_\zeta)R + \sum_{j \geq 2} f^{(j)}(\mathbf{a}_\zeta)R^j/j!$ . The linear term has degree  $d-1+m$  and leading coefficient  $da_d\zeta^{d-1}\chi$ , while the  $j$ th term has degree at most  $d-j+jm < d-1+m$  because  $(j-1)(1-m) > 0$ . Hence  $q_\zeta = d-1+m$  and  $\text{lc}(\mathbf{d}_{\zeta,c}) + da_d\zeta^{d-1}\chi = 0$ . For  $\zeta = 1$ ,  $\mathbf{a}_1 = T$  and  $\mathbf{d}_{1,c} = -c$ .  $\square$

**Theorem 16.36** (Uniform image gap [16]). *Let  $f \in K_G[X]$  have degree  $d \geq 2$ , let  $b > G$  be fresh and  $t = \omega^b$ . Then*

$$f(\mathcal{A}_{k,k}) \cap (f(t) + K_G) = \{f(t)\}, \quad \text{hence} \quad f(\mathcal{A}_{D,k}) \cap (f(t) + K_G) = \{f(t)\}.$$

*For every  $0 \neq c \in K_G$ , every root in  $\mathbf{No}[i]$  of  $f(Y) = f(t) + c$  has a nonzero block  $-r_\zeta b + \text{supp } c_\zeta$  of negative exponents, with  $1 \leq r_\zeta \leq d-1$  as in Theorem 16.35.*

*Proof.* Work over  $\mathbb{C}((X^G))$  if  $k = \mathbb{R}$  (the roots of unity are ordinary constants, so no new exponents arise). Lemma 16.34 and Theorem 16.31 give  $d$  distinct roots of  $f(Y) - f(t) - c$  in  $\mathbf{No}[i]$ , hence all of them. For  $c \neq 0$  each has, by Theorem 16.35, a nonzero coefficient in a block of negative index, which lies below exponent 0 and cannot cancel against the affine constant or another block; so no root lies in  $\mathcal{A}_{\mathbb{C},\mathbb{C}}$ , and none in  $\mathcal{A}_{k,k} \supseteq \mathcal{A}_{D,k}$ . For  $c = 0$ ,  $t \in \Pi_k$  gives the value  $f(t)$ .  $\square$

The same  $b$  works for every  $c \in K_G$  at once, and  $G$  need not be Archimedean, countable or divisible (16).

*Remark 16.37* (Special cases printed elsewhere [16; 08]). Three nonsolvability statements proved separately are instances of Theorem 16.36 with  $G = \{0\}$ , where every  $b > 0$  is fresh and  $K_G = k$ . (a)  $x^2 = \omega^2 + 1$  has no solution in  $\mathbf{Oz}$ : take  $f = X^2$ ,  $t = \omega$ ,  $c = 1$ ; this is an example of the Diophantine report [47]. (b) For  $a \in \mathbb{R}_{>0}$ ,  $\gamma > 0$ ,  $0 \neq \beta \in \mathbb{Z}$  and  $n \geq 2$ ,  $X^n = a\omega^\gamma + \beta$  has no solution in  $\mathbf{Oz}$ : put  $X = a^{1/n}Y$  with  $Y \in \mathcal{A}_{\mathbb{R},\mathbb{R}}$ ,  $f = aY^n$ ,  $t = \omega^{\gamma/n}$ ,  $c = \beta$ ; this is the two-term root obstruction of the Diophantine report [47], whose short proof by the binomial expansion is kept there. (c) For  $\gamma > 0$  and  $m \geq 2$ ,  $X^m = \omega^\gamma + 1$  has no root in  $\mathbf{Oz}[i]$ : take  $k = \mathbb{C}$ ,  $f = X^m$ ,  $t = \omega^{\gamma/m}$ ,  $c = 1$ ; this is 08's invisible equation (Theorem 15.25), whose explicit expansion (15.18) exhibits the forbidden block of layer  $r_1 = m-1$ . Theorem 16.36 is also consistent with 10's local count (Theorem 16.27): for fresh  $t$  the fiber of  $f(t)$  has at most  $e(f) \leq d$  points in the complex case and at most one ( $d$  odd) or two ( $d$  even) in the real case (Theorem 16.40 below), within 10's bound  $q_{E,d}$ .

**Corollary 16.38** (Omnific non-solvability [16]). *If  $f$  maps  $\mathcal{A}_{D,k}$  into itself and has degree at least two, then with  $K_G$  and  $t$  as above,  $f(Y) = f(t) + c$  has no solution  $Y \in \mathcal{A}_{D,k}$  for any  $0 \neq c \in K_G \cap \mathcal{A}_{D,k}$ , although its right side lies in  $\mathcal{A}_{D,k}$ . For  $f \in k[X]$  one may take  $G = \{0\}$  and any  $b > 0$ .*

**Definition 16.39** (Centered polynomial [16]). For  $f$  of degree  $d \geq 2$  with centre  $s = a_{d-1}/(da_d)$  write  $h(U) = f(U - s) = \sum_{j=0}^d h_j U^j$  (so  $h_{d-1} = 0$ ) and  $e(f) = \gcd\{j \geq 1 : h_j \neq 0\}$ .

**Theorem 16.40** (Exact fresh fibers [16]). *For  $\zeta^d = 1$ ,  $f(\mathbf{a}_\zeta(T)) = f(T)$  if and only if  $h(\zeta U) = h(U)$  if and only if  $\zeta^{e(f)} = 1$ ; if this fails, the branch  $Y_{\zeta,0}$  has a nonzero negative Laurent term, and if it holds,  $Y_{\zeta,0} = \mathbf{a}_\zeta$ . Consequently, for  $f \in K_G[X]$  of degree at least two and fresh  $t$ ,*

$$\{y \in \mathcal{A}_{D,k} : f(y) = f(t)\} = \{\zeta t + (\zeta - 1)s : \zeta \in k, \zeta^{e(f)} = 1, (\zeta - 1)s \in \mathcal{A}_{D,k}\},$$

*distinct  $\zeta$  giving distinct points. If  $s$  has a nonzero negative-exponent part the fiber is  $\{t\}$ ; if  $s \in \Pi_k$ , all  $e(f)$  points occur when  $k = \mathbb{C}$ , and one or two when  $k = \mathbb{R}$ , according as  $e(f)$  is odd or even.*

*Proof.*  $a_\zeta(T) + s = \zeta(T + s)$  gives the first equivalence; the second is  $\zeta^j = 1$  for all supported  $j$ , that is  $\zeta^{e(f)} = 1$  by Bézout. If the defect vanishes,  $a_\zeta$  is a root with leading coefficient  $\zeta$ , hence the branch by uniqueness of the simple-root lift; otherwise the remainder is nonzero and the argument of [Theorem 16.35](#) applies. Every root is a branch; nonsymmetric branches have a forbidden block; a symmetric one is  $\zeta t + (\zeta - 1)s$ , whose fresh term cannot affect the old negative terms or the constant, so it lies in  $\mathcal{A}_{D,k}$  exactly when  $(\zeta - 1)s$  does (and  $\zeta \in k$ : a nonreal  $\zeta$  gives a nonreal leading coefficient). Distinct  $\zeta$  have distinct fresh leading coefficients; multiplying a nonzero negative part of  $s$  by  $\zeta - 1 \neq 0$  keeps it; the count is that of roots of unity in  $k$ .  $\square$

**Example 16.41** (Worked fibers and defects [16]). (a) For  $f = X^d$ ,  $s = 0$  and  $e(f) = d$ : the fresh fiber of  $t^d$  in  $\mathbf{Oz}[i]$  is  $\{\zeta t : \zeta^d = 1\}$  (the constant term of  $\zeta t$  is 0, so  $\zeta$  need not be a Gaussian integer), and for  $0 \neq c \in k$  the shifted roots are  $\zeta t(1 + ct^{-d})^{1/d} = \zeta t + \frac{\zeta c}{d} t^{1-d} + \frac{\zeta(1-d)c^2}{2d^2} t^{1-2d} + \dots$ , with first negative layer exactly  $d - 1$  (a formal binomial expansion followed by the block embedding). (b)  $f = X^3 + X^2$  has  $s = \frac{1}{3}$ ,  $h = U^3 - \frac{1}{3}U + \frac{2}{27}$  and  $e(f) = 1$ , so  $f(t)$  has the single preimage  $t$  in  $\mathbf{Oz}$  and in  $\mathbf{Oz}[i]$ ; this does not make  $f$  injective ( $f(0) = f(-1)$ ). (c)  $f = \omega X^2 + X$  has the infinitesimal centre  $1/(2\omega)$ ; at  $t = \omega^\omega$  the other algebraic preimage of  $f(t)$  is  $-t - \omega^{-1}$ , and  $f$  is in fact injective on  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , since  $f(x) = f(y)$ ,  $x \neq y$ , forces  $x + y = -\omega^{-1}$ . (d) For  $f = X^3 + \lambda X + \nu$  ( $s = 0$ ),  $d_{\zeta,c} = \lambda(\zeta - 1)T - c$ : the identity branch first detects the shift at layer two ( $\frac{c}{3}T^{-2}$ ), a nontrivial branch with  $\lambda \neq 0$  at layer one ( $-\lambda(\zeta - 1)T^{-1}/(3\zeta^2)$ ). (e) For  $f = X^4 + pX^2 + qX + r$ ,  $d_{\zeta,c} = p(\zeta^2 - 1)T^2 + q(\zeta - 1)T - c$ ; the first negative terms are  $cT^{-3}/4$  at  $\zeta = 1$ ,  $-qT^{-2}/2$  at  $\zeta = -1$  ( $q \neq 0$ ) and  $ipT^{-1}/2$  at  $\zeta = i$  ( $p \neq 0$ ); when a displayed coefficient vanishes the next term of the finite defect decides. (f) Freshness cannot be weakened casually: with  $G = \mathbb{Q}$ ,  $F(T) = \omega/T$  takes the value 1 at the non-fresh  $T = \omega$ ; and  $f = X^2$ ,  $t = \omega$  with the shift  $2\omega + 1$ , which is not below a fresh exponent, gives  $(t + 1)^2$ , an omnific square.

**Theorem 16.42** (Rational self-maps over any constant ring [16]; second route to [Theorem 16.25](#)). *Let  $F \in E_k(X)$  be a rational function, in lowest terms, whose values lie in  $\mathcal{A}_{D,k}$  at every point of  $\mathcal{A}_{D,k}$  where it is defined. Then  $F$  is a polynomial. Hence the rational self-maps of  $\mathcal{A}_{D,k}$  are exactly its integer-valued polynomials.*

*Proof.* Put the finitely many coefficients in some  $K_G$  ([Lemma 16.30](#)) and take a fresh  $t = \omega^b \in \Pi_k \subseteq \mathcal{A}_{D,k}$ . The denominator does not vanish at  $t$ , and  $F(t) \in \mathcal{A}_{D,k}$ , so  $F$  is a polynomial by [Theorem 16.33](#).  $\square$

For  $D = \mathbb{Z}, \mathbb{Z}[i]$  this is 10's [Theorem 16.25](#); the two proofs are different routes. 10's uses finite differences and lattice separation ( $|x| \geq 1$  for  $0 \neq x \in \mathbf{Oz}$ ), which fails for  $D = \mathbb{Q}$  or  $D = k$ ; 16's uses one fresh input and works for every  $D$ . The bare statement that integer-valued rational functions are polynomials is not new in principle: it is the D-ring phenomenon of Gunji–McQuillan [21], as recalled by Liu [24, Proposition 1.12]; for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ , whose unit groups are finite, a nonconstant polynomial cannot take all its values in the unit group, which is the link with that criterion. 16 avoids importing a theorem about set-sized domains into a proper-class ring and supplies an explicit fresh witness (16). 22 credits the same antecedent.

**Theorem 16.43** (Surjective rational self-maps [16]; second route to [Corollary 16.29](#)). *A rational function over  $E_k$  that is defined at every point of  $\mathcal{A}_{D,k}$  and maps  $\mathcal{A}_{D,k}$  into itself is surjective onto  $\mathcal{A}_{D,k}$  if and only if it is  $X \mapsto uX + a$  with  $a \in \mathcal{A}_{D,k}$  and  $u \in D^\times$ ; every such map is bijective, and under composition they form the affine group  $\mathcal{A}_{D,k} \rtimes D^\times$ ,  $(a, u)(a', u') = (a + ua', uu')$ . For  $\mathbf{Oz}$  these are  $X \mapsto \pm X + a$ , and for  $\mathbf{Oz}[i]$  they are  $X \mapsto uX + a$  with  $u \in \{\pm 1, \pm i\}$ .*

*Proof.* By [Theorem 16.42](#) the map is a polynomial; constants are not surjective. If its degree is at least two, then for a fresh  $t$  the element  $F(t) + 1 \in \mathcal{A}_{D,k}$  is not a value, by [Theorem 16.36](#). So  $F = uX + a$  with  $a = F(0)$  and  $u = F(1) - F(0)$  in  $\mathcal{A}_{D,k}$ , and surjectivity at  $a + 1$  gives  $ux = 1$  with  $x \in \mathcal{A}_{D,k}$ , so  $u \in D^\times$  ([Lemma 2.4](#)). The inverse is  $X \mapsto u^{-1}(X - a)$ .  $\square$

For  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  this contains 10's classification of the rational bijections; 16 assumes only surjectivity and allows every unital  $D \subseteq k$ , and its proof uses the image gap where 10's uses holes of every radius ([Theorem 16.28](#)). Both are kept.

**Corollary 16.44** (Omitted values form a proper class [16]). *If  $F$  is an integer-valued polynomial of  $\mathcal{A}_{D,k}$  of degree at least two, then  $\mathcal{A}_{D,k} \setminus F(\mathcal{A}_{D,k})$  is a proper class; changing  $F$  on a set of inputs cannot make it surjective.*

*Proof.* Given a set  $S \subseteq \mathcal{A}_{D,k}$ , enlarge  $G$  to contain the supports of  $S$  and of the coefficients of  $F$  and take  $b > G$ . The omitted value  $F(\omega^b) + 1$  has leading exponent  $db + g_d$  ( $g_d$  the leading exponent of the leading coefficient), which is not in  $G$ , so it is not in  $S$ . Changing  $F$  on a set adds at most a set of values.  $\square$

Complex conjugation bijects  $\mathbf{Oz}[i]$  but is not a rational function of one surcomplex variable (a rational function agreeing with  $X$  on the ordinary integers is  $X$ ); admitting  $F(\bar{X})$  adds exactly the maps  $u\bar{X} + a$ . The theorem is one-variable:  $(x, y) \mapsto (x + y^2, y)$  is a nonlinear polynomial bijection of  $\mathcal{A}_{D,k}^2$  (16).

## 16.7 Finite tests, fresh probes and sampling

For a unital  $D \subseteq k$  and  $r \geq 1$  put  $\text{Int}(D^r) = \{g \in \text{Frac}(D)[\mathbf{X}] : g(D^r) \subseteq D\}$  and  $\mathcal{N}_r(\mathcal{A}_{D,k}) = \{F \in E_k[\mathbf{X}] : F(\mathcal{A}_{D,k}^r) \subseteq \mathcal{A}_{D,k}\}$ , so that  $\mathcal{N}_r(\mathbf{Oz})$  and  $\mathcal{N}_r(\mathbf{Oz}[i])$  are those of [Section 16](#).

**Proposition 16.45** (Constant-ring transfer [16]; 10 for  $D = \mathbb{Z}, \mathbb{Z}[i]$ ). *For every unital  $D \subseteq k$  and  $r \geq 1$ ,  $\mathcal{N}_r(\mathcal{A}_{D,k}) = \text{Int}(D^r) \oplus \Pi_k[\mathbf{X}]$  additively, the second summand an ideal, and the coefficientwise constant term is a ring homomorphism onto  $\text{Int}(D^r)$ , split by inclusion.*

*Proof.* Let  $F \in \mathcal{N}_r(\mathcal{A}_{D,k})$  have degree at most  $d$  in each variable. The Newton coefficients (16.4) are integral combinations of values at  $\{0, \dots, d\}^r$ , hence lie in  $\mathcal{A}_{D,k}$ , and  $(d!)^r \binom{\mathbf{X}}{\mathbf{k}} \in \mathbb{Z}[\mathbf{X}]$ , so every monomial coefficient of  $F$  lies in  $\text{Frac}(D) + \Pi_k$ . Split  $F = g + p$  with  $g \in \text{Frac}(D)[\mathbf{X}]$ ,  $p \in \Pi_k[\mathbf{X}]$ . The constant term is a ring homomorphism on  $\text{Frac}(D) + \Pi_k$  ([Lemma 2.2](#)), so for  $\mathbf{a} \in D^r$ ,  $g(\mathbf{a}) = \text{ct}(F(\mathbf{a})) \in D$ ; thus  $g \in \text{Int}(D^r)$ . Conversely, for  $\mathbf{x} = \mathbf{a} + \mathbf{z}$  with  $\mathbf{a} \in D^r$ ,  $\mathbf{z} \in \Pi_k^r$ , every term of  $g(\mathbf{a} + \mathbf{z}) - g(\mathbf{a})$  contains a factor  $z_j$ , so it lies in the ideal  $\Pi_k$  of  $\mathcal{A}_{k,k}$ , and  $p(\mathbf{x}) \in \Pi_k$ ; hence  $F(\mathbf{x}) \in g(\mathbf{a}) + \Pi_k$ . Uniqueness is  $k \cap \Pi_k = 0$ , and the section is the inclusion.  $\square$

For  $D = \mathbb{Z}$  and  $\mathbb{Z}[i]$  this is [Theorems 16.7](#) and [16.9](#) (10, and independently 18 in  $n$  variables); 16 proves it for  $r = 1$  and for several variables with any  $D$ . The finite-difference identity needs no convergence argument: both sides agree at the grid points (16).

**Theorem 16.46** (Optimal point tests [16]; 10, 18, 22). *Let  $L \subseteq \mathbb{N}^n$  be a finite nonempty lower set ( $\alpha \in L$  and  $\beta \leq \alpha$  coordinatewise imply  $\beta \in L$ ) and  $V_L = \text{span}_{\mathbf{No}}\{\mathbf{X}^\alpha : \alpha \in L\}$ . For  $F \in V_L$  the following are equivalent:  $F(\mathbf{Oz}^n) \subseteq \mathbf{Oz}$ ;  $F(\alpha) \in \mathbf{Oz}$  for all  $\alpha \in L$ ; all coefficients of the expansion  $F = \sum_{\alpha \in L} c_\alpha \binom{\mathbf{X}}{\alpha}$  lie in  $\mathbf{Oz}$ . Moreover  $|L|$  is the least size of a set of points of  $\mathbf{Oz}^n$ , arbitrary omnific points allowed, whose values test membership for all of  $V_L$ . In particular, for one variable and degree at most  $d$ , the  $d + 1$  points  $0, \dots, d$  suffice and no  $d$  points do.*

*Proof.* The  $\binom{\mathbf{X}}{\alpha}$ ,  $\alpha \in L$ , form a basis of  $V_L$ : each has leading monomial  $\mathbf{X}^\alpha / \prod_j \alpha_j!$  and its other monomials have smaller indices, which stay in  $L$ . In a linear extension of the coordinatewise order the evaluation matrix  $((\binom{\beta}{\alpha}))$  on  $L$  is integral and unitriangular, so  $c_\alpha = \sum_{\beta \leq \alpha} (-1)^{|\alpha| - |\beta|} \prod_j \binom{\alpha_j}{\beta_j} F(\beta)$ . Each  $\binom{X_j}{\alpha_j}$  preserves  $\mathbf{Oz}$  (Lemma 16.5). If fewer than  $|L|$  points are proposed, their evaluation maps have a common nonzero kernel element  $h \in V_L$ ; a polynomial vanishing on  $\mathbb{Z}^n$  is zero, so  $h(\mathbf{n}) \neq 0$  for some  $\mathbf{n} \in \mathbb{Z}^n$ , and  $h/(2h(\mathbf{n}))$  passes all proposed tests but takes the value  $\frac{1}{2}$  at  $\mathbf{n}$ .  $\square$

For rectangular  $L$  the equivalences are 10's Theorem 16.6; general lower sets (for instance total degree at most  $d$ , with  $\binom{n+d}{n}$  points) and the optimality statement are 16's; 18 proves the total-degree case and the binomial basis independently. So the binomial polynomials form a basis of the  $\mathbf{Oz}$ -module  $\mathcal{N}_1(\mathbf{Oz})$ , each element having finitely many nonzero coordinates, and the countable set  $\mathbb{Z}$  tests all of  $\mathcal{N}_1(\mathbf{Oz})$ ; no topological density is asserted (16).

**Proposition 16.47** (A finite Gaussian test [16]). *Let  $F \in \mathbf{No}[i][\mathbf{X}]$  have total degree at most  $d$ ,  $M = d!$  and  $C_M = \{a + bi : a, b \in \mathbb{Z}, 0 \leq a, b < M\}$ . Then  $F$  preserves  $\mathbf{Oz}[i]$  if and only if its values lie in  $\mathbf{Oz}[i]$  on  $L_d \cup C_M^n$ , where  $L_d = \{\alpha \in \mathbb{N}^n : |\alpha| \leq d\}$ . This is an explicit finite certificate, not a claim of minimal size.*

*Proof.* The values on  $L_d$  give  $F = g + p$  with  $g \in M^{-1}\mathbb{Z}[i][\mathbf{X}]$  and  $p \in \Pi_{\mathbb{C}}[\mathbf{X}]$  (the denominators  $\prod_j \alpha_j!$  divide  $d!$ ). With  $h = Mg \in \mathbb{Z}[i][\mathbf{X}]$ ,  $g(\mathbf{z}) \in \mathbb{Z}[i]$  means  $h(\mathbf{z}) \equiv 0 \pmod{M\mathbb{Z}[i]}$ , which depends only on the residues of the coordinates, and  $C_M$  is a complete residue system. So  $g \in \text{Int}(\mathbb{Z}[i]^n)$ ; apply Proposition 16.45.  $\square$

The ordinary grid alone is not enough over  $\mathbf{Oz}[i]$ :  $\binom{i}{2} = (-1 - i)/2$  (Remark 16.10).

**Lemma 16.48** (Independent fresh coordinates [16]). *For a set-sized subgroup  $G$  there are positive  $b_1, \dots, b_n$  whose classes are linearly independent over  $\mathbb{Q}$  modulo  $G$ , and the  $t_j = \omega^{b_j}$  are algebraically independent over  $K_G$ .*

*Proof.* Choose  $b_j$  above the set-sized rational span of  $G, b_1, \dots, b_{j-1}$ ; a nontrivial relation would contradict the freshness of the last participating  $b_j$ . Distinct monomials  $\prod_j t_j^{\alpha_j}$  have exponents in distinct cosets of  $G$ , so their  $K_G$ -multiples have disjoint supports.  $\square$

With such  $b_j$ , let  $H = G + \sum_j \mathbb{Q}b_j$ , choose  $B > H$ , and let  $\xi^{(i)} \in \mathcal{A}_{D,k}^n$  have  $i$ th coordinate  $\omega^B$  and  $j$ th coordinate  $\omega^{b_j}$  for  $j \neq i$ .

**Theorem 16.49** (Coordinate-dominant probes [16]). *If  $F \in K_G(X_1, \dots, X_n)$  and  $F(\xi^{(i)}) \in \mathcal{A}_{D,k}$  for  $1 \leq i \leq n$ , then  $F \in K_G[X_1, \dots, X_n]$ . Consequently every rational function mapping  $\mathcal{A}_{D,k}^n$  into  $\mathcal{A}_{D,k}$  at its points of definition is a polynomial.*

*Proof.* Fix  $i$  and view  $F$  as a rational function of  $X_i$  over  $E_i = K_G(X_j : j \neq i)$ . Substituting  $X_j = t_j$  ( $j \neq i$ ) embeds  $E_i$  in  $K_H$  (Lemma 16.48); a reduced numerator and denominator stay coprime (their Bézout identity survives) and keep their degrees. As  $\omega^B$  is fresh over  $K_H$ , Theorem 16.33 makes the denominator constant in  $X_i$ , so  $F \in E_i[X_i]$  for each  $i$ . If  $F = P/Q$  in lowest terms in the unique factorization domain  $K_G[\mathbf{X}]$  had nonconstant  $Q$ , an irreducible factor of positive degree in some  $X_i$  would remain a denominator over  $E_i$  (Gauss's lemma).  $\square$

**Corollary 16.50** (Workspace-uniform rational certificates [16]). *Fix  $K_G$  and  $d \geq 0$ . For rational functions over  $K_G$  with numerator of total degree at most  $d$  in lowest terms, being a self-map of*



$\mathbf{Oz}^n \rightarrow \mathbf{Oz}$  is tested by  $L_d$  together with the  $n$  probes  $\xi^{(i)}$  (at most  $\binom{n+d}{n} + n$  values;  $d+2$  for  $n=1$ ), and of  $\mathbf{Oz}[i]^n \rightarrow \mathbf{Oz}[i]$  by  $L_d \cup C_d^n$  together with the probes.

*Proof.* The probes force a polynomial, of total degree at most  $d$ ; then apply [Theorem 16.46](#) or [Proposition 16.47](#). Necessity is clear since all test points lie in the domain.  $\square$

These certificates are uniform for a whole coefficient workspace  $K_G$ ; they are relative to exact coefficient data, zero tests and a certificate of freshness, and are not an algorithm for arbitrary presentations of surreal coefficients (16). For  $n=1$  the certificate is also 22's; 22's certificate in several variables, with one fresh exponent, is compared with this one in [Remark 16.71](#).

**Theorem 16.51** (No set-sized sample tests all rational maps [16; 22]). *Let  $\Lambda$  be  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  and  $S \subseteq \Lambda$  a set. There is  $\beta > 0$  such that  $R_S(X) = \omega^\beta / (X - \frac{1}{2})$  is defined on all of  $\Lambda$ , satisfies  $R_S(S) \subseteq \Pi_k$ , and  $R_S(\omega^{2\beta}) = \sum_{m \geq 0} 2^{-m} \omega^{-(2m+1)\beta} \notin \Lambda$ .*

*Proof.*  $\frac{1}{2} \notin \Lambda$ . Clear the set of supports of the  $(s - \frac{1}{2})^{-1}$ ,  $s \in S$  ([Proposition 3.4\(i\)](#)) to get  $\beta$ . At  $x = \omega^{2\beta}$ ,  $R_S(x) = \omega^{-\beta} (1 - \frac{1}{2} \omega^{-2\beta})^{-1}$  is a nonzero infinitesimal.  $\square$

Thus no set of points, with all values known, detects all rational self-maps, even among functions of numerator degree 0 and denominator degree 1 with no pole on  $\Lambda$ : the single function  $\omega / (X - \frac{1}{2})$  takes purely infinite values at every ordinary and Gaussian integer, yet  $\omega / (\omega^2 - \frac{1}{2}) = \omega^{-1} + \frac{1}{2} \omega^{-3} + \dots \notin \mathbf{Oz}$ . For polynomials, by contrast, the ordinary integers suffice in every degree ([Theorem 16.46](#)) (16).

**Theorem 16.52** (Co-small integrality [16]). *If  $F \in E_k(X)$  is defined and  $\mathcal{A}_{D,k}$ -valued outside a set-sized subset of  $\mathcal{A}_{D,k}$ , then  $F$  is an integer-valued polynomial of  $\mathcal{A}_{D,k}$ : a rational function that is not a self-map fails on a proper class of inputs (poles counted as failures).*

*Proof.* Let  $S$  contain all failures and put its supports and the coefficients of  $F$  into  $K_G$ . If  $F$  is not a polynomial, a fresh  $t \notin S$  gives  $F(t) \notin \mathcal{A}_{D,k}$  ([Theorem 16.33](#)). If  $F$  is a polynomial with  $F(a) \notin \mathcal{A}_{D,k}$ , enlarge  $G$  by the supports of  $a$ ; the constant coefficient of  $F(a + T)$  is  $F(a)$ , so  $F(a + t) \notin \mathcal{A}_{D,k}$  by [Corollary 16.32](#), and  $a + t \notin S$ .  $\square$

[Theorems 16.51](#) and [16.52](#) do not conflict: the first builds a bad function adapted to a set of successful tests, the second says its failures form a proper class; neither provides one set meeting the failures of every function (16). 22 proves [Theorem 16.51](#) again with another witness ([Remark 16.73](#)), and [Theorem 16.52](#) for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  in every number of variables ([Theorem 16.75](#)).

**Proposition 16.53** (Hulls [16]). *For  $S \subseteq \mathcal{A}_{D,k}$  let  $\text{Hull}_{\text{rat}}(S)$  be the set of  $a \in \mathcal{A}_{D,k}$  such that every rational function over  $E_k$  defined and  $\mathcal{A}_{D,k}$ -valued on  $S$  is defined and  $\mathcal{A}_{D,k}$ -valued at  $a$  (poles off  $S$  allowed), and define  $\text{Hull}_{\text{pol}}$  with polynomials. Then  $\text{Hull}_{\text{rat}}(S) = S$  for every set  $S$ , while  $\text{Hull}_{\text{pol}}(\mathbb{Z}) = \mathbf{Oz}$  and  $\text{Hull}_{\text{pol}}(\mathbb{Z}[i]) = \mathbf{Oz}[i]$ .*

*Proof.* For  $a \notin S$ , clearing the supports of the  $(s - a)^{-1}$  gives  $\beta$  with  $\omega^\beta / (X - a)$  purely infinite on  $S$  and a pole at  $a$ . The polynomial statements follow from [Proposition 16.45](#): values on  $D$  force the ordinary part into  $\text{Int}(D)$ , and the purely infinite part is harmless.  $\square$

The pole allowed in the hull is not what drives [Theorem 16.51](#), whose functions have no pole on the ring (16). In a fixed smaller Hahn field whose value group has no exponent above the coefficient group, the fresh-scale argument cannot be reused; this is one of 16's questions ([Question 18.14](#)), and 22 states exactly when a fixed Hahn field has such an exponent ([Proposition 16.86](#)).



## 16.8 Ordinary outputs, canonical translations and composition

This subsection is manuscript 18's new material. Its description of  $\mathcal{N}_r(\mathbf{Oz})$  and  $\mathcal{N}_r(\mathbf{Oz}[i])$ , its simplex test for  $\mathbf{Oz}^n$  (the lower set  $L_d$  of Theorem 16.46) with the binomial basis, its rational collapse and its classification of surjective rational self-maps are 10's and 16's results proved again independently; they are credited at Theorems 16.7, 16.9, 16.25 and 16.46 and Corollary 16.29. Throughout,  $(E, \Lambda, D)$  is  $(\mathbf{No}, \mathbf{Oz}, \mathbb{Z})$  or  $(\mathbf{No}[i], \mathbf{Oz}[i], \mathbb{Z}[i])$ ,  $\Pi_\Lambda$  is  $\Pi$  or  $\Pi_{\mathbb{C}}$ , and  $F_0 = \text{Frac } D$ . 18's ideal  $\mathcal{J}$ , ring  $A$ , field  $K$ , field  $k_0$  and class  $\text{Int}(A)$  are  $\Pi_\Lambda$ ,  $\Lambda$ ,  $E$ ,  $F_0$  and  $\mathcal{N}_1(\Lambda)$ ; its reduction  $\bar{f}$  is  $\text{ct}_*(f)$ ; its  $\text{pi}(a) = a - \text{ct}(a)$  is written  $\hat{a}$  (for  $a \in F_0 + \Pi_\Lambda$ ); and its *ordinary-output locus*  $E_f = f^{-1}(D)$  and target loci  $E_b(f) = f^{-1}(b + D)$  are  $\mathcal{L}_f$  and  $\mathcal{L}_{f,b}$  (the letter  $E$  is the field). The coefficients of every  $f \in \mathcal{N}_1(\Lambda)$  lie in  $F_0 \oplus \Pi_\Lambda$  (Theorems 16.7 and 16.9), so  $\text{ct}_*(f) \in \text{Int}(D)$  and  $\hat{a}$  are defined for them.

The first results hold for an abstract *discretely ordered ring*  $B$ , an ordered domain in which 1 is the least positive element. Put  $(R_*, D_*) = (B, \mathbb{Z})$  or  $(B[i], \mathbb{Z}[i])$ ; an element of an ordered field extension of  $\text{Frac } B$ , or of its quadratic extension with the modulus  $|x + iy| = \sqrt{x^2 + y^2}$  computed in a real closure, is *finite* if its modulus is bounded by an ordinary integer and *infinitesimal* if it is below every  $1/n$ .

**Lemma 16.54** (Discrete lattices [18]). *The finite elements of  $B$  are the ordinary integers, and distinct elements of  $B$  are at distance at least 1; the same holds for  $B[i]$  with  $\mathbb{Z}[i]$  and the modulus.  $B_{\mathbb{Q}} = \{b/n : b \in B, n \geq 1\}$  and  $B[i]_{\mathbb{Q}}$  contain no nonzero infinitesimal. For  $B = \mathbf{Oz}$  this is Lemma 16.1.*

*Proof.* A finite  $b$  lies between two consecutive ordinary integers or equals one, and discreteness excludes the first case; apply this to differences. Both coordinates of a finite Gaussian element are finite, and a nonzero difference has a coordinate of modulus at least 1. A nonzero  $b/n$  has modulus at least  $1/n$ .  $\square$

**Lemma 16.55** (Coefficient gap and one cluster [18]).

- (i) *If  $\deg f \leq d$  and  $f(0), \dots, f(d) \in R_*$ , then all coefficients of  $f$  lie in  $(R_*)_{\mathbb{Q}}$ ; a nonzero leading coefficient is not infinitesimal.*
- (ii) *If  $f$  has degree  $d \geq 1$  and a leading coefficient that is not infinitesimal, and  $x_0, \dots, x_d \in R_*$  are distinct with all  $f(x_j)$  finite, then all differences  $x_i - x_j$  are finite.*

*Proof.* (i) Lagrange interpolation at  $0, \dots, d$  has rational basis coefficients; apply Lemma 16.54. (ii) If the nodes fell into at least two classes of finite distance, then for each  $i$  some  $|x_i - x_j|$  would be infinite and the others at least 1, so every summand of

$$\text{lc}(f) = \sum_{i=0}^d \frac{f(x_i)}{\prod_{j \neq i} (x_i - x_j)},$$

the coefficient of  $X^d$  in Lagrange interpolation, would be infinitesimal, and so would  $\text{lc}(f)$ .  $\square$

**Theorem 16.56** (Sharp ordinary-output dichotomy [18]). *Let  $f$  be a polynomial of degree  $d \geq 1$  with  $f(R_*) \subseteq R_*$ , and  $\mathcal{L}_f = \{x \in R_* : f(x) \in D_*\}$ . Exactly one of the following holds:*

- (i)  $\#\mathcal{L}_f \leq d$ ;
- (ii)  $f(X) = g(X - a)$  for some  $a \in R_*$  and  $g \in \text{Int}(D_*)$  of degree  $d$ , and then  $\mathcal{L}_f = a + D_*$ , a coset determined by  $f$ .

*Proof.* Let  $\mathcal{L}_f$  contain  $d + 1$  points. By Lemma 16.55(i) the leading coefficient is not infinitesimal, and by (ii) the points lie in one class of finite distance: they are  $a + n_j$  with  $n_j \in D_*$  (Lemma 16.54). Then  $g(X) = f(X + a)$  takes values in  $D_*$  at the  $d + 1$  points  $n_j$ , so  $g \in \text{Frac}(D_*)[X]$ ; for every  $n \in D_*$ ,  $g(n) = f(a + n) \in R_*$  is an ordinary rational (or Gaussian rational) element of  $R_*$ , hence finite, hence in  $D_*$ . So  $g \in \text{Int}(D_*)$  and  $a + D_* \subseteq \mathcal{L}_f$ . If  $x \in \mathcal{L}_f$  and  $y = x - a$  were infinite, the ordinary coefficients of  $g$  would give  $|g(y)| \geq \frac{1}{2}|\text{lc}(g)||y|^d$ , which is infinite, although  $g(y) = f(x) \in D_*$ ; so  $y$  is finite and lies in  $D_*$ . The alternatives exclude each other because  $D_*$  is infinite.  $\square$

The clustering and the descent to  $\text{Frac}(D_*)[X]$  use only  $f(0), \dots, f(d) \in R_*$  and  $d + 1$  ordinary outputs; the step  $g \in \text{Int}(D_*)$  uses preservation of all of  $R_*$ . For **Oz** that global hypothesis is itself finitely testable (Theorem 16.46); for a general discretely ordered ring this must not be assumed (18).

**Corollary 16.57** ([18]).  *$f(R_*) \cap D_*$  has at most  $d$  elements or equals  $g(D_*)$  with  $g$  as in Theorem 16.56. If  $R_* \neq D_*$ , no nonconstant polynomial preserving  $R_*$  maps  $R_*$  into  $D_*$ .*

*Proof.* Apply  $f$  to  $\mathcal{L}_f$ . If  $\mathcal{L}_f = R_*$ , then  $R_*$  is infinite and equals a coset  $a + D_*$  containing 0, so  $R_* = D_*$ .  $\square$

**Proposition 16.58** (The bound is sharp [18]). *If  $B$  has a positive infinite element  $H$ , then for  $d \geq 2$  the polynomial  $\prod_{j=1}^d (X - jH)$ , and for  $d = 1$  the polynomial  $HX$ , preserves  $B$  and  $B[i]$ , and its ordinary-output locus consists of exactly  $d$  points ( $H, \dots, dH$ , respectively 0).*

*Proof.* For  $d \geq 2$  the  $d$  roots lie in different classes of finite distance, so by Theorem 16.56 there is no further point. If  $Hx$  is finite and  $x \neq 0$ , then  $|x| \geq 1$  and  $|Hx|$  is infinite.  $\square$

The degree-one replacement is necessary:  $X - H$  has the locus  $H + D_*$ . The bound concerns inputs, not distinct outputs: all  $d$  roots above go to 0 (18).

**Theorem 16.59** (Canonical translation [18]). *Let  $f \in \mathcal{N}_1(\Lambda)$  have degree  $d \geq 1$ . Either  $\#f^{-1}(D) \leq d$ , or there is a unique  $\eta \in \Pi_\Lambda$  with*

$$f(X) = \text{ct}_*(f)(X - \eta),$$

*and then  $\deg \text{ct}_*(f) = d$  and  $f^{-1}(D) = \eta + D$ .*

*Proof.* **Oz** is discretely ordered (Lemma 16.1); apply Theorem 16.56. In its second case write  $a = c + \eta$  with  $c \in D$  and  $\eta \in \Pi_\Lambda$ , so  $f(X) = h(X - \eta)$  with  $h(Y) = g(Y - c) \in \text{Int}(D)$ . The coefficients of  $h(X - \eta)$  are polynomials in  $\eta$  with coefficients in  $F_0$ , and  $\text{ct}$  is a ring homomorphism on  $F_0 + \Pi_\Lambda$  with  $\text{ct}(\eta) = 0$ , so  $\text{ct}_*(f) = h$ . If  $\eta + D = \eta' + D$  then  $\eta - \eta' \in D \cap \Pi_\Lambda = 0$ .  $\square$

**Theorem 16.60** (The two highest coefficients [18]). *Let  $f = \sum_{j \leq d} a_j X^j \in \mathcal{N}_1(\Lambda)$  with  $a_d \neq 0$ .*

(i) *If  $a_d \notin F_0$ , then  $\#f^{-1}(D) \leq d$ .*

(ii) *If  $a_d \in F_0$ , put  $\eta_f = -\hat{a}_{d-1}/(da_d) \in \Pi_\Lambda$ . Then  $f^{-1}(D)$  is infinite if and only if  $f(X + \eta_f) = \text{ct}_*(f)(X)$ ; otherwise it has at most  $d$  elements.*

*Proof.* In the canonical form the leading coefficient is that of  $\text{ct}_*(f)$ , which is in  $F_0$ ; this proves (i). Comparing coefficients of  $X^{d-1}$  in  $f(X) = \text{ct}_*(f)(X - \eta)$  gives  $a_{d-1} = \text{ct}(a_{d-1}) - da_d\eta$ , so  $\eta = \eta_f$ ; this holds for  $d = 1$  too. The converse is the identity itself and Theorem 16.59.  $\Pi_\Lambda$  is closed under division by the nonzero ordinary scalar  $da_d$ .  $\square$

The test is a finite algebraic certificate relative to exact coefficient operations (two purely infinite parts, one translation, finitely many comparisons), not an algorithm for arbitrary presentations of surreal numbers (18). For example,  $f = \binom{X-\eta}{3}$  with  $\eta = \omega + \omega^{1/2}/3$  has  $\text{ct}_*(f) = \binom{X}{3}$  and  $f^{-1}(\mathbb{Z}) = \eta + \mathbb{Z}$ , and  $\eta_f$  recovers  $\eta$ ; whereas  $\omega X^d + X$  ( $d \geq 2$ ) has  $\text{ct}_*(f) = X$  but, its leading coefficient being infinite, at most  $d$  ordinary outputs: coefficientwise reduction controls congruences, the canonical form controls exact values (18). In the notation of Definition 16.39,  $\eta_f$  is minus the purely infinite part of 16's centre  $s = a_{d-1}/(da_d)$ : indeed  $s = \text{ct}(a_{d-1})/(da_d) + \mathring{a}_{d-1}/(da_d)$  when  $a_d \in F_0$ .

**Theorem 16.61** (At most one rich target [18]). *Let  $f \in \mathcal{N}_1(\Lambda)$  have degree  $d \geq 2$  and put  $\mathcal{L}_{f,b} = f^{-1}(b + D)$  for  $b \in \Lambda$ . The following are equivalent: (i) some  $\mathcal{L}_{f,b}$  has more than  $d$  elements; (ii)  $f(X) = \text{ct}_*(f)(X - \eta) + \beta$  for some  $\eta, \beta \in \Pi_\Lambda$ . Then  $\eta$  and  $\beta$  are unique,  $\beta = f(\eta) - \text{ct}(f(\eta))$ , and  $\mathcal{L}_{f,b} = \eta + D$  if  $\mathring{b} = \beta$ , while  $\#\mathcal{L}_{f,b} \leq d$  otherwise.*

*Proof.* If  $\mathcal{L}_{f,b}$  has more than  $d$  points, Theorem 16.59 applied to  $f - b$  gives  $f(X) - b = (\text{ct}_*(f) - \text{ct}(b))(X - \eta)$ , which is (ii) with  $\beta = \mathring{b}$ . Conversely (ii) and the growth of  $\text{ct}_*(f)$  give  $f^{-1}(\beta + D) = \eta + D$ . Since  $d \geq 2$ , subtracting a constant does not change  $a_{d-1}$ , so Theorem 16.60 gives the same  $\eta$  for every rich target, and then  $\beta = f(\eta) - \text{ct}(f(\eta))$  because  $\text{ct}_*(f)(0) \in D$ .  $\square$

(Status (batch 34): the Diophantine report [47] reaches the same normal forms for monic polynomials from a different hypothesis, a nonzero constant discriminant instead of many ordinary values. Let  $k$  have characteristic zero,  $G$  be a set and  $\mathfrak{o} \subseteq k$  unital. A monic  $P \in \mathcal{A}_{\mathfrak{o},k}(G)[X]$  of degree  $n \geq 1$  has  $\text{Disc}(P) \in \mathfrak{o} \setminus \{0\}$  if and only if  $\text{ct}_*(P)$  is squarefree and  $P(X) = \text{ct}_*(P)(X - \eta)$  with  $\eta = -\mathring{a}_{n-1}/n \in \Pi_k(G)$ , which is  $\eta_P$  of Theorem 16.60 (`odg:disc:thm:translation`); and a monic  $f \in \mathcal{A}_{\mathfrak{o},k}(G)[X]$  of degree  $d \geq 2$  whose critical polynomial  $f'/d$  has nonzero constant discriminant is  $\text{ct}_*(f)(X - \eta) + \beta$  with  $\eta, \beta \in \Pi_k(G)$  and  $\beta = f(\eta) - \text{ct}_*(f)(0)$ , the form (ii) above (`odg:disc:thm:critical`). Over  $\mathbf{Oz}$  these hypotheses give the locus  $\eta + \mathbb{Z}$  and the rich target  $\beta + \mathbb{Z}$ ; the converses fail, since for  $0 \neq h \in \Pi$  both  $(X - h)^2$  and  $(X - h)^3$  send all of  $h + \mathbb{Z}$  to  $\mathbb{Z}$ , while the discriminant of the first and the critical discriminant of the second are 0. Those theorems need no discrete order.)

**Corollary 16.62** (One test for all targets [18]). *For  $d \geq 2$  and  $a_d \in F_0$ , let  $Q(X) = f(X + \eta_f)$ . A rich target exists if and only if every nonconstant coefficient of  $Q$  lies in  $F_0$ ; then  $Q = \text{ct}_*(f) + \beta$  with  $\beta = Q(0) - \text{ct}(Q(0))$ .*

*Proof.* This is (ii) of Theorem 16.61 translated back, using  $\text{ct}(\eta_f) = 0$ , so that  $\text{ct}_*(Q) = \text{ct}_*(f)$ .  $\square$

Degree one is different: for  $f = aX + c$  with  $0 \neq a \in D$  every target  $\beta + D$  has the infinite preimage  $(\beta - \mathring{c})/a + D$ , and an infinite linear coefficient gives at most one preimage in each target. For  $(X - \omega)^2 + \omega^2$  the unique rich target is  $\omega^2 + \mathbb{Z}$ , with preimage  $\omega + \mathbb{Z}$ ; for  $X^2$  it is  $\mathbb{Z}$ , and the target  $\omega + \mathbb{Z}$  has exactly the two preimages  $\pm\omega^{1/2}$  (18). This agrees with 16's fresh fibre of  $f(t) = \omega$  at  $t = \omega^{1/2}$  (Theorem 16.40) and with 10's bound  $q_{\mathbf{No},2} = 2$  (Theorem 16.27); 18 counts preimages of whole  $D$ -cosets, 10 of balls at large scale, 16 of fresh values.

**Theorem 16.63** (Composition alignment [18]). *Let  $f = p(X - \eta) + \beta$  and  $g = q(X - \xi) + \gamma$  with  $p, q \in \text{Int}(D)$  of degrees  $m, n \geq 2$  and  $\eta, \beta, \xi, \gamma \in \Pi_\Lambda$ . Then  $h = f \circ g$  has a target with more than  $mn$  preimages if and only if  $\gamma = \eta$ . In that case  $h(X) = (p \circ q)(X - \xi) + \beta$  and  $h^{-1}(\beta + D) = \xi + D$ ; otherwise  $\#h^{-1}(b + D) \leq mn$  for every  $b \in \Lambda$ .*

*Proof.* Put  $\delta = \gamma - \eta$  and  $Y = X - \xi$ , so  $h(Y + \xi) = p(q(Y) + \delta) + \beta$ . If  $\delta = 0$  the assertions are immediate. Let  $\delta \neq 0$ ,  $a = \text{lc}(p)$  and  $c = \text{lc}(q)$ . Expanding  $p(q + \delta) - p(q) = \sum_{j \geq 1} p^{(j)}(q) \delta^j / j!$ , every  $\delta$ -dependent term has degree at most  $(m-1)n \leq mn-2$ ; so the leading coefficient  $ac^m$  and the coefficient of  $Y^{mn-1}$  are ordinary, and [Theorem 16.60](#) gives the input translation  $\xi$  for  $h$ . In degree  $(m-1)n \geq 1$  the only  $\delta$ -dependent contribution is the leading term of  $p'(q)\delta$ , so that coefficient has purely infinite part

$$mac^{m-1}\delta \neq 0.$$

Since this coefficient is not constant, [Corollary 16.62](#) fails, and [Theorem 16.61](#) applies to the degree- $mn$  polynomial  $h$ .  $\square$

In particular, if  $f = p(X - \eta)$  and  $g = q(X - \xi)$  both have infinite ordinary-output loci,  $f \circ g$  has one exactly when  $\eta = 0$ . For example  $(X - \omega)^2$  and  $X^2$  both have infinite loci, but  $(X^2 - \omega)^2$  has the ordinary-output locus  $\{\omega^{1/2}, -\omega^{1/2}\}$ . The hypotheses  $m, n \geq 2$  are needed: a linear inner map absorbs an input translation and a linear outer map moves an output translation without creating a nonconstant obstruction (18).

**Proposition 16.64** (A Gaussian simplex test [18]; comparison [merge]). *A polynomial  $F \in \mathbf{No}[i][Z_1, \dots, Z_n]$  of total degree at most  $d$  preserves  $\mathbf{Oz}[i]^n$  if and only if  $F(\mathbf{a} + i\mathbf{b}) \in \mathbf{Oz}[i]$  for all  $\mathbf{a}, \mathbf{b} \in \mathbb{N}^n$  with  $|\mathbf{a}| + |\mathbf{b}| \leq d$ , a set of  $\binom{2n+d}{2n}$  points. For  $d \geq 3$  it has fewer points than the certificate  $L_d \cup C_{d!}^n$  of [Proposition 16.47](#), which contains the  $(d!)^{2n}$  points of  $C_{d!}^n$ ; for  $d = 1$ , and for  $(n, d) = (1, 2)$ , 16's certificate is the smaller one. More generally, every set  $S \subseteq \mathbb{Z}[i]$  that tests membership in  $\text{Int}(\mathbb{Z}[i])$  for polynomials over  $\mathbb{Q}(i)$  of degree at most  $d$  also tests preservation of  $\mathbf{Oz}[i]$  for polynomials over  $\mathbf{No}[i]$  of degree at most  $d$ .*

*Proof.*  $G(\mathbf{U}, \mathbf{V}) = F(\mathbf{U} + i\mathbf{V})$  is a polynomial of total degree at most  $d$  in  $2n$  variables, so by the basis argument of [Theorem 16.46](#) for the lower set  $L_d \subseteq \mathbb{N}^{2n}$  its coefficients in the binomial basis  $\binom{\mathbf{U}, \mathbf{V}}{\alpha}$  are integral combinations of its values on  $L_d$ , that is of  $F$  at the stated points; they lie in  $\mathbf{Oz}[i]$ , and  $\binom{u}{k} \in \mathbf{Oz}$  for  $u \in \mathbf{Oz}$ , so  $F(\mathbf{u} + i\mathbf{v}) \in \mathbf{Oz}[i]$  for  $\mathbf{u}, \mathbf{v} \in \mathbf{Oz}^n$ . Necessity is clear. For the comparison,  $\binom{2n+d}{2n} < 2^{2n+d}$ , while  $(d!)^{2n} = 36^n > 8 \cdot 4^n = 2^{2n+3}$  for  $d = 3$ , and for  $d \geq 4$ ,  $d! \geq 2^d$  gives  $(d!)^{2n} \geq 2^{2nd} > 2^{2n+d}$  since  $2nd - 2n - d = (2n-1)(d-1) - 1 \geq 2$ . For  $n = 1$  the certificates of 16 and 18 have 2 and 3 points when  $d = 1$ , and 5 and 6 when  $d = 2$ . For the last assertion, interpolate  $F$  at  $d+1$  points of  $S$  (a testing set has at least  $d+1$  points): its coefficients lie in  $\mathbb{Q}(i) + \Pi_{\mathbb{C}}$ , so  $F = g + p$  with  $g \in \mathbb{Q}(i)[X]$  and  $p \in \Pi_{\mathbb{C}}[X]$ , and  $g(s) = \text{ct}(F(s)) \in \mathbb{Z}[i]$  for  $s \in S$ ; hence  $g \in \text{Int}(\mathbb{Z}[i])$  and  $F$  preserves  $\mathbf{Oz}[i]$  by [Proposition 16.45](#).  $\square$

The comparison of the two certificates is the merge's; neither is claimed minimal. The ordinary test  $\{0, \dots, d\}$  fails over  $\mathbf{Oz}[i]$  ( $\binom{i}{2} \notin \mathbb{Z}[i]$ ). By Volkov and Petrov [41], as cited by 18, one-variable testing sets of size  $O(d)$  exist for  $\text{Int}(\mathbb{Z}[i])$ , while  $d+1$  points do not suffice for large  $d$ ; by the last assertion they test  $\mathbf{Oz}[i]$  as well. These are classical inputs about  $\mathbb{Z}[i]$ , not claims of 18. 22 proves the converse of the last assertion and extends it to  $\mathbb{Z}$  and to every number of variables ([Theorem 16.82](#)).

**Theorem 16.65** (Matrices and jets [18]). *For  $s \geq 1$  let  $\mathcal{M}_s(\Lambda) = \{f \in E[X] : f(M) \in \Lambda^{s \times s} \text{ for all } M \in \Lambda^{s \times s}\}$  (scalar coefficients) and  $\mathcal{M}_s(D) = \{g \in F_0[X] : g(D^{s \times s}) \subseteq D^{s \times s}\}$ ; let  $\Lambda[\varepsilon]_s = \Lambda[\varepsilon]/(\varepsilon^{s+1})$ ,  $\text{Jet}_s(\Lambda) = \{f \in E[X] : f(\Lambda[\varepsilon]_s) \subseteq \Lambda[\varepsilon]_s\}$  and  $\text{Jet}_s(D)$  likewise with  $F_0$  and  $D$ . Then*

$$\mathcal{M}_s(\Lambda) = \Pi_{\Lambda}[X] \oplus \mathcal{M}_s(D), \quad \text{Jet}_s(\Lambda) = \Pi_{\Lambda}[X] \oplus \text{Jet}_s(D), \quad \bigcap_s \mathcal{M}_s(\Lambda) = \bigcap_s \text{Jet}_s(\Lambda) = \Lambda[X].$$

Moreover  $f \in \text{Jet}_s(\Lambda)$  if and only if the Hasse derivatives  $f^{[r]} = \sum_{j \geq r} \binom{j}{r} a_j X^{j-r}$ ,  $r \leq s$ , lie in  $\mathcal{N}_1(\Lambda)$ , and for  $\deg f \leq d$  the single nilpotent Jordan block  $N$  of size  $d+1$  forces all coefficients into  $\Lambda$ , since  $f(N)_{1,j+1} = a_j$ .

*Proof.* Scalar matrices give  $f \in \mathcal{N}_1(\Lambda)$ , so  $f = h + g$  with  $h \in \Pi_\Lambda[X]$  and  $g = \text{ct}_*(f) \in \text{Int}(D)$ . Entrywise  $\text{ct}$  is a ring homomorphism on matrices over  $F_0 + \Pi_\Lambda$  and gives  $\text{ct}(f(M)) = g(\text{ct}(M))$ ; so  $g \in \mathcal{M}_s(D)$ . Conversely  $h(M)$  has entries in  $\Pi_\Lambda$  and  $g(M)$  has entries with constant terms those of  $g(\text{ct}(M)) \in D^{s \times s}$ . For jets,  $f(a+u) = \sum_{r \leq s} f^{[r]}(a)u^r$  for  $a \in \Lambda$  and nilpotent  $u$  with coefficients in  $\Lambda$ , and  $\text{ct}$  commutes with Hasse derivatives, which preserve  $\Pi_\Lambda[X]$ ; apply [Theorems 16.7 and 16.9](#). The Jordan block and evaluation at  $\varepsilon$  in  $\Lambda[\varepsilon]_d$  read off the coefficients.  $\square$

For example  $\begin{pmatrix} X \\ 2 \end{pmatrix}$  preserves  $\mathbf{Oz}$  but  $\begin{pmatrix} N \\ 2 \end{pmatrix} = -\frac{1}{2}N$  for  $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ , and  $(X^2 - X)/(1+i)$  preserves  $\mathbf{Oz}[i]$  (as  $x^2 \equiv x$  modulo  $1+i$  on  $\mathbb{Z}[i]$ ) but has the entry  $-1/(1+i)$  at  $N$ . These are applications of the constant-term structure and of nilpotent evaluation; the all-size coefficient test is a general algebraic observation, not a new matrix theorem, and the scalar-coefficient convention is one of those surveyed by Frisch [\[19\]](#) (18).

**Theorem 16.66** (Rational functions on discretely ordered rings [\[18\]](#)). *Let  $B$  be a discretely ordered ring and  $R_* = B$  or  $B[i]$ . A rational function in one variable, over an ordered field extension of  $\text{Frac } B$  or its quadratic extension, that takes values in  $R_*$  at all but finitely many points of  $R_*$  where it is defined, is a polynomial and maps all of  $R_*$  into  $R_*$ . A rational function in finitely many variables that is defined and  $R_*$ -valued on all of  $R_*^n$  is a polynomial.*

*Proof. Descent.* If  $r = P/Q \in L(X)$ ,  $\deg P \leq m$ ,  $\deg Q \leq n$ , takes values in a subfield  $F$  at infinitely many points of  $F$ , choose  $m+n+1$  such nodes  $t_j$  with values  $v_j$ ; the  $m+n+1$  linear equations  $A(t_j) = v_j B(t_j)$  in the  $m+n+2$  coefficients of  $A, B$  have a nonzero solution over  $F$ ,  $B \neq 0$  (else  $A$  would have too many roots), and  $PB - AQ$  vanishes at the  $m+n+1$  nodes, so  $r = A/B \in F(X)$ . *Collapse.* Write  $r = q + u$  with  $q$  a polynomial and  $u$  proper over  $\text{Frac}(R_*)$ , and suppose  $u \neq 0$ . For an ordinary  $s > \deg q$ ,  $\Delta^s r = \Delta^s u$  is nonzero and proper ([Lemma 16.24](#));  $B$  is cofinal in  $\text{Frac } B$  ( $|a/b| \leq |a|$ ), so  $|\Delta^s u(x)| < 1$  for all sufficiently large  $x \in B$ . Avoiding the finitely many exceptions, their shifts by  $0, \dots, s$  and the poles,  $\Delta^s r(x) = \sum_j (-1)^{s-j} \binom{s}{j} r(x+j) \in R_*$  has modulus below 1, hence is 0, for infinitely many  $x$ , which is impossible for the nonzero  $\Delta^s u$ . *Filling.* For  $a \in R_*$  choose  $N$  with  $a+N, \dots, a+N+\deg r$  outside the exceptions; Newton interpolation of  $r(a+N+X)$  at  $0, \dots, \deg r$ , evaluated at  $X = -N$ , writes  $r(a)$  as an integral combination of allowed values. *Several variables.* If  $r = P/Q$  in lowest terms had a denominator of positive degree in  $X_1$ , a specialization of the other variables to ordinary integers avoiding a nonzero resultant condition keeps the degrees and the coprimality, and gives a nonpolynomial one-variable function that is  $R_*$ -valued on  $R_*$ , contrary to the first part.  $\square$

For  $\Lambda$  this is [Theorem 16.25](#) (10), [Theorems 16.42 and 16.49](#) (16), and 16's [Theorem 16.52](#), which allows a set-sized exceptional set, is stronger there; 18 adds the abstract discretely ordered ring and the finitely many exceptions. The collapse is not new in principle: it is the D-ring phenomenon of Gunji–McQuillan [\[21\]](#), as recalled by Liu [\[24, Proposition 1.12\]](#), since a discretely ordered ring has unit group  $\{\pm 1\}$  and a nonconstant polynomial cannot be unit-valued; 18's direct proof avoids importing maximal-ideal arguments into a proper-class ring. 18 then reproves the classification of surjective rational self-maps ([Corollary 16.29](#) and [Theorem 16.43](#)) by a third route: a surjective  $f$  has an infinite ordinary-output locus, so  $f = \text{ct}_*(f)(X - \eta)$  by [Theorem 16.59](#),  $\text{ct}_*(f)$  is surjective on  $D$ , and counting lattice points forces  $\text{ct}_*(f) = uX + c$  with  $u \in D^\times$ . Injectivity alone is weaker:  $X^3$  is injective on  $\mathbf{Oz}$  but not surjective.



**Corollary 16.67** (Fixed Hahn workspaces [18]). *Let  $\Gamma$  be a set-sized ordered abelian group, not necessarily divisible,  $J_\Gamma$  the series in  $\mathbb{R}((T^\Gamma))$  with support in  $\Gamma_{>0}$  (large monomials) and  $A_\Gamma = \mathbb{Z} \oplus J_\Gamma$ , with the Gaussian version  $\mathbb{Z}[i] \oplus (J_\Gamma + iJ_\Gamma)$ . Then Theorems 16.56, 16.59 to 16.61, 16.63, 16.65 and 16.66, the description of the integer-valued polynomials, the finite tests and the classification of surjective rational self-maps hold for these rings.*

*Proof.*  $A_\Gamma$  is discretely ordered (a nonzero element of  $J_\Gamma$  is infinite),  $\text{ct}$  is a ring homomorphism on  $\mathbb{Q} \oplus J_\Gamma$ , and  $J_\Gamma$  is closed under ordinary scalars; the proofs use nothing else, the descent step supplying coefficients in the full Hahn field, and complex moduli being computed in a real closure.  $\square$

So 18's arguments, which use only the discrete gap, survive in a fixed Hahn field, where 16's fresh-scale arguments cannot be reused (Section 16.9). For  $\Gamma = \{0\}$  the dichotomy is its ordinary case, and the sharp examples need an infinite element. Several variables behave differently:  $XY$  preserves  $\Lambda^2$ , but its ordinary-output locus contains both axes, proper classes not contained in one translate of  $D^2$ ; what survives is the linewise statement that  $t \mapsto f(\mathbf{a} + t\mathbf{v})$  obeys the one-variable dichotomy. The discrete gap fails in **No** and in the finite surreals, and finite testing concerns polynomials of known degree only: no finite sample certifies rational functions (compare Theorem 16.51), and entire Hahn series and expressions of nonstandard degree are outside the theorems (18). (Status (batch 32): entire series in a fixed workspace are treated in the report on entire functions at arbitrary rank [55]: for  $k = \mathbb{R}$  or  $\mathbb{C}$ , an entire Hahn series in finitely many variables over  $k((T^\Gamma))$  that maps a Zariski-dense set of ordinary points into  $k$  plus the series supported in  $\Gamma_{>0}$  is a polynomial (`ent:as:thm:sampling`), and the entire series mapping the omnific part of the workspace into itself are exactly the integer-valued polynomials plus the polynomials with purely infinite coefficients (`ent:as:thm:integer`), which this section's classification gives for polynomials. Expressions of nonstandard degree remain outside.)

## 16.9 Scope of the polynomial arithmetic

Two mechanisms act in this section and have different hypotheses. The reflection theorem Theorem 16.11 transports the size-sensitive divisibility mechanism of Theorem 5.1 through finite polynomial algebra; that mechanism applies to arbitrary set coefficient fields in the class Hahn construction. The results of Section 16.5 use instead *finite reduction plus lattice separation*: at a large scale  $T$ , polynomial preimages of a prescribed output ball reduce to roots of  $Z^d - 1$ , and a nonzero reduced divided difference would force two inputs of one cluster to lie less than 1 apart, which the lattice forbids. This gives Theorems 16.27 and 16.28 without a count of points in a surreal interval or a real-valued asymptotic density; it uses the real or surcomplex modulus, the standard part and the unit-spaced lattices. Neither mechanism should be used as a substitute for the other. 16 adds a third: *fresh scales* (Sections 16.6 and 16.7). A monomial above an entire set-sized exponent group separates formal Laurent powers into disjoint support blocks, so negative Laurent tails cannot be hidden by cancellation; this needs no lattice separation, works for every unital  $D \subseteq k$ , and is a support argument, not a convergence argument in the fine topology. It needs the full surreal exponent class: in a fixed Hahn field whose value group has no exponent above the coefficient group it cannot be reused (16). 18 adds a fourth, *finite interpolation in a discretely ordered ring* (Section 16.8): the Lagrange formula for the leading coefficient, whose summands become infinitesimal when ordinary outputs occur in two infinitely separated clusters, while a coefficient that is a rational combination of lattice values cannot be infinitesimal. It uses neither the modulus estimates at a large scale nor fresh exponents, and it survives in every fixed Hahn workspace (Corollary 16.67). 22 (Section 16.10) combines 16's fresh scale with a resultant specialization at



ordinary integer coordinates, so that one fresh exponent suffices in every number of variables, and states exactly when a fixed Hahn workspace supplies that exponent (Proposition 16.86).

Manuscript 10 proposes Theorem 16.11 and the arbitrary-scale bound Theorem 16.27 as new; the other results of this section are explicit consequences or transfers of classical arithmetic facts (Newton interpolation, binomial-ring facts [16], the least-common-multiple pattern [9],  $p$ -adic approximation by integer-valued polynomials [8]), proved here to make the scope complete. The results do not classify all class ideals of  $\mathbf{Oz}$  or of  $\mathcal{N}_r(\mathbf{Oz})$ , all automorphisms of  $\mathbf{Oz}$ , or the factorization of arbitrary omnific integers, and they neither construct nor use a global surcomplex exponential. They do not assert a general transfer principle from  $\mathbb{Z}$  to  $\mathbf{Oz}$ ; the omitted type of Proposition 16.3 is a concrete warning against such a principle. The Gaussian theorem Theorem 16.9 does not replace the arithmetic of  $\text{Int}(\mathbb{Z}[i]^r)$  by a binomial basis, and no Gaussian analogue of the congruence criterion Theorem 16.14 is given. For  $r \geq 1$  the completion of Corollary 16.20 is a  $p$ -adic, not a profinite, completion. The classification of Corollary 16.29 concerns rational functions only. Manuscript 10 records a question suggested by this material: for the full omnific integers, determine how much an automorphism can do inside the ideal  $\Pi$  while fixing the ordinary shadow  $\text{ct}$  (every ring endomorphism of  $\mathbf{Oz}$  preserves  $\text{ct}$ , by the uniqueness in Theorem 5.1). Manuscript 16 presents the image-gap, forbidden-layer and fresh-fiber package and the two sampling statements as its proposed contributions, credits rational collapse to the D-ring theory and Newton interpolation to the classical literature, and proves only one-variable surjectivity rigidity. Manuscript 18 presents the ordinary-output dichotomy, the canonical translation, the uniqueness of a rich target and composition alignment as its proposed contributions, and treats the description of  $\mathcal{N}_r$ , the matrix and jet reductions and the rational collapse as applications or classical material. Manuscript 22 presents the single-scale detector, the multivariate certificate and its common-scale uniformity, the pole-free construction defeating every set of tests, the removal of set-sized exceptions and their joint real and Gaussian formulation as its proposed contributions, and credits the D-ring collapse and the binomial basis; several of these were already 16's (Section 16.10).

## 16.10 Finite tests at a single new scale

This subsection is manuscript 22's (Section 1.3). 22 certifies that a rational function of  $m$  variables preserves  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  by its values on an ordinary box and on finitely many points that have one coordinate at a single fresh monomial and ordinary integers elsewhere. It adds the removal of set-sized exceptions in every number of variables, a tensor description of the numerical polynomials, an exact transfer of ordinary constant test sets, finite generators of image ideals, a birational consequence and a criterion for fresh scales in a fixed Hahn workspace. 22 did not read this report, which at its pin already contained 16's and 18's subsections (Section 1.4), and about half of it re-derives them. Printed once, under the existing labels and with 22 added as a proving source, are: the fresh exponent and the Laurent blocks (Lemma 16.30 and Theorem 16.31; 22's Lemma 3.1, with the fresh exponent built as  $\omega$  times a surreal bound of the supports, and Theorem 3.2); the one-point detector (Corollary 16.32 and Theorem 16.33, for  $\mathcal{A}_{k,k}$ ,  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$ ; its Corollary 3.3); omnific binomial closure, with the constant term of  $\binom{a}{n}$  equal to  $\binom{\text{ct}(a)}{n}$ , the grid test, and its corollary that  $F(\mathbf{Oz}^m) \subseteq \mathbf{Oz}$  if and only if  $F(\mathbb{Z}^m) \subseteq \mathbf{Oz}$  (Lemma 16.5 and Theorem 16.6; its Lemma 4.1, Theorem 4.2, Corollary 4.3); the optimal number  $\prod_j (d_j + 1)$  of fixed polynomial tests (Theorem 16.46 for a rectangular lower set; its Theorem 4.4); the decomposition of the numerical polynomials with its split specialization and evaluation formula (Theorems 16.7 and 16.9 and Proposition 16.45; its Proposition 4.5 and the decomposition in its Theorem 7.3), and the binomial basis of  $\mathcal{N}_m(\mathbf{Oz})$  over  $\mathbf{Oz}$  (Theorem 16.6(iii)); multivariate rational collapse (the last

assertion of [Theorem 16.49](#), and [Theorem 16.66](#); its Corollary 5.3); the case  $m = 1$  of its certificate, with the  $d + 2$  values  $0, \dots, d, t$  ([Corollary 16.50](#)); the absence of a set-sized test for rational maps ([Theorem 16.51](#); its Theorem B, with a second witness, [Remark 16.73](#)); the case  $m = 1$  of its removal of exceptions ([Theorem 16.52](#), there for every  $D$ ); the Gaussian denominators  $\binom{i}{2} \notin \mathbb{Z}[i]$  and  $(Z^2 - Z)/(1 + i) \in \text{Int}(\mathbb{Z}[i])$  ([Remark 16.10](#) and the paragraph after [Theorem 16.65](#); its Example 7.6); the warnings on ct ([Remark 2.3](#); its Remark 2.1); fractions and units ([Proposition 3.4](#) and [Lemma 2.4](#); its Lemma 2.2); and the D-ring antecedent (after [Theorem 16.42](#)). 22's Theorem 7.1 is the coordinate-degree twin of 18's simplex test ([Theorem 16.78](#)). 22 reads class statements as [Section 2.1](#) does, in a class theory carrying **No** or with a lower and an upper universe; every support, coefficient family and group it generates is a set, and a fresh exponent is chosen only after such a set has been assembled (its Remark 2.3). The numbering of 22 cited here is that of its own manuscript.

**Notation and renamings.** Throughout,  $(D, k, \Lambda, E)$  is  $(\mathbb{Z}, \mathbb{R}, \mathbf{Oz}, \mathbf{No})$  or  $(\mathbb{Z}[i], \mathbb{C}, \mathbf{Oz}[i], \mathbf{No}[i])$ ,  $F_0 = \text{Frac } D$ , and  $m \geq 1$  is the number of variables (10 writes  $r$ , 16 and 18 write  $n$ ). A rational function is written in lowest terms over  $E$ ; it is *defined* at a point if its reduced denominator does not vanish there, so a removable factor of a displayed expression is never a pole, and it *preserves*  $\Lambda$  if it is defined and  $\Lambda$ -valued on all of  $\Lambda^m$ . 22's symbols are renamed as follows.

- Its positive-support classes  $\mathcal{J}_{\mathbb{R}}, \mathcal{J}_{\mathbb{C}}$  are  $\Pi, \Pi_{\mathbb{C}}$  (together  $\Pi_k$ ); its  $\mathcal{A}_k$  is  $\mathcal{A}_{k,k}$ ; its Gaussian ring **Og** is  $\mathbf{Oz}[i]$ ; and its  $B \in \{\mathbf{Oz}, \mathbf{Oz}[i]\}$  is  $\Lambda$  ( $B$  is 18's discretely ordered ring and  $\mathcal{B}$  the ring  $D + \mathfrak{m}_k$ ).
- Its  $\text{Int}(B^m)$ , the ambient-coefficient polynomials preserving  $B$ , is  $\mathcal{N}_m(\Lambda)$ , and its  $\text{Int}^{\mathbf{R}}(B^m)$ , the rational functions preserving  $B$ , is  $\mathcal{N}_m^{\text{rat}}(\Lambda)$ ; the letter  $\text{Int}$  is kept for the ordinary rings  $\text{Int}(D^m) = \{f \in F_0[\mathbf{X}] : f(D^m) \subseteq D\}$ . Its  $E = \text{Frac } D$  is  $F_0$  ( $E$  is the ambient field, as in [Section 16.8](#)), and its binomial products  $B_{\nu}$  are  $\binom{\mathbf{X}}{\nu}$ .
- Its coefficient group  $\Gamma$  and Hahn field  $K_k(\Gamma)$  are 16's  $G$  and  $K_G$ ; its exponent  $\beta$  with  $\Gamma \ll \beta$ , its monomial  $T = \omega^{\beta}$  and its embedding  $\mathcal{E}_T$  are 16's fresh  $b > G$ ,  $t = \omega^b$  and  $\iota_b$  ([Lemma 16.30](#) and [Theorem 16.31](#)).
- Its field  $L$  of characteristic 0 is  $K$  ( $L$  is a lower set in [Theorem 16.46](#)); its bad-specialization polynomial  $H_j$  is  $h_j$  ( $H$  is a group in [Theorem 16.49](#)); and its degree bound  $D = 2d^2 + 2d$  is  $\mathbf{N}_d$  ( $D$  is the constant ring, as in 22's own Sections 7–9).
- Its box  $\mathcal{B}_{m,d}$ , remote grid  $\mathcal{H}_{m,D}(T)$  and test set  $\mathcal{T}_{m,d}(T)$  are  $\square_{m,d}$ ,  $\mathbf{R}_{m,d}(t)$  and  $\text{Test}_{m,d}(t)$  ( $\mathcal{B}$  is  $D + \mathfrak{m}_k$ ,  $\mathcal{T}(M, N)$  a multiplication kernel and  $\mathbf{T}_j$  21's self-Tor groups); its Gaussian grid  $G_d$  is  $C_{d+1}$  in the notation of [Proposition 16.47](#).
- Its exceptional set  $E$  is  $S$ ; its deceiver's  $\tau$ ,  $\eta = \omega\beta$ ,  $A = \omega^{\eta}$ ,  $\lambda$  and  $F_S$  are  $t$ ,  $\omega b$ ,  $\omega^{\omega b}$ ,  $a$  and  $\tilde{R}_S$  ( $\eta$  is 18's translation and  $R_S$  16's witness); the ambient field  $K$  of its Corollary 8.2 is  $E$ .
- In its Section 9, its value group  $\Lambda$  and coefficient subgroup  $\Gamma \subseteq \Lambda$  are  $\Gamma$  and  $G$ , as in [Corollary 16.67](#); its ring  $B_{\Lambda} = D \oplus \mathcal{J}_k(\Lambda)$  is  $\mathcal{A}_{D,k}(\Gamma)$  of (2.2); and its convex subgroup  $C(\Gamma)$  is  $\text{cvx}_{\Gamma}(G)$  ( $C_M$  is 16's residue grid and  $C$  is 03's letter).

Three tempting readings are false. (1) 22's  $\Gamma \ll \beta$ , which says  $|\gamma| < \beta$  for every  $\gamma \in \Gamma$ , is freshness  $b > G$ ; it is not the relation  $h \ll a$  of (2.4), which compares two positive exponents up to ordinary multiples. (2) Passing at  $t$  does not certify integer-valuedness:  $X/2$  is **Oz**-valued at  $t$  but not at 1; the fresh point detects denominators, and the ordinary box does the arithmetic. (3)  $C_{d+1}$  is a grid of interpolation nodes for coordinate degree  $d$ , whereas 16's  $C_d!$  is a complete residue system; the first works because Newton interpolation needs  $d + 1$  nodes in each real coordinate, the second because integrality of  $h/d!$  depends only on residues.

**Lemma 16.68** (Rectangular grids [22]). *Let  $K$  be a field of characteristic 0 and  $0 \neq h \in K[Y_1, \dots, Y_r]$  with  $\deg_{Y_i} h \leq e_i$ . Then  $h$  does not vanish at some point of  $\prod_{i=1}^r \{0, 1, \dots, e_i\}$ .*

*Proof.* For  $r = 0$  a nonzero constant is nonzero, and for  $r = 1$  a nonzero polynomial of degree at most  $e_1$  has at most  $e_1$  roots. For  $r \geq 2$  expand in  $Y_r$ , choose a nonzero coefficient, find by induction a point of the first  $r - 1$  grids at which it does not vanish, and apply the case  $r = 1$  to the specialized polynomial in  $Y_r$ .  $\square$

Let  $P, Q \in K[X_1, \dots, X_m]$  be nonzero and coprime,  $p_i = \deg_{X_i} P$ ,  $q_i = \deg_{X_i} Q$ , and choose  $j$  with  $q_j > 0$ . By Gauss's lemma  $P$  and  $Q$  stay coprime in  $K(X_i : i \neq j)[X_j]$ , so the resultant  $\text{Res}_{X_j}(P, Q)$  is a nonzero polynomial in the other variables. Put

$$h_j = \text{lc}_{X_j}(P) \text{lc}_{X_j}(Q) \text{Res}_{X_j}(P, Q). \quad (16.25)$$

The leading coefficients make the degrees survive specialization, and the resultant prevents a common factor from appearing afterwards.

**Lemma 16.69** (Good specializations [22]). *For  $i \neq j$ ,*

$$\deg_{X_i} h_j \leq (q_j + 1)p_i + (p_j + 1)q_i. \quad (16.26)$$

*If  $\mathbf{a}$  assigns values in  $K$  to the variables other than  $X_j$  and  $h_j(\mathbf{a}) \neq 0$ , then the specializations  $P_{\mathbf{a}}, Q_{\mathbf{a}} \in K[X_j]$  have degrees  $p_j, q_j$  and are coprime, so  $P_{\mathbf{a}}/Q_{\mathbf{a}}$  is not a polynomial.*

*Proof.* The Sylvester determinant is homogeneous of degree  $q_j$  in the coefficients of  $P$  and of degree  $p_j$  in those of  $Q$ , which have degree at most  $p_i$ , respectively  $q_i$ , in  $X_i$ ; so  $\deg_{X_i} \text{Res}_{X_j}(P, Q) \leq q_j p_i + p_j q_i$ , and the two leading coefficients add at most  $p_i + q_i$ . This covers  $p_j = 0$ , where the resultant is  $P^{q_j}$ . Where  $h_j \neq 0$  both leading coefficients survive, so the specialized Sylvester matrix is that of  $P_{\mathbf{a}}, Q_{\mathbf{a}}$ , and its nonzero determinant is their resultant; they are coprime, and the denominator has degree  $q_j > 0$ .  $\square$

For  $m, d \geq 1$  put  $N_d = 2d^2 + 2d$  and, for a fresh monomial  $t = \omega^b$ ,

$$\begin{aligned} \square_{m,d} &= \{0, 1, \dots, d\}^m, & R_{m,d}(t) &= \bigcup_{j=1}^m \{0, \dots, N_d\}^{j-1} \times \{t\} \times \{0, \dots, N_d\}^{m-j}, \\ \text{Test}_{m,d}(t) &= \square_{m,d} \cup R_{m,d}(t). \end{aligned} \quad (16.27)$$

For  $m = 1$  this is  $\{0, 1, \dots, d, t\}$ .

**Theorem 16.70** (Single-scale finite certificate [22]). *Let  $F = P/Q \in \mathbf{No}(X_1, \dots, X_m)$  be in lowest terms with  $\deg_{X_i} P \leq d$  and  $\deg_{X_i} Q \leq d$  for all  $i$ , where  $d \geq 1$ . Let  $G \subseteq \mathbf{No}$  be a set-sized subgroup containing the supports of all coefficients of  $P$  and  $Q$ ,  $b > G$  fresh and  $t = \omega^b$ . Then  $F$  preserves  $\mathbf{Oz}$  if and only if it is defined and  $\mathbf{Oz}$ -valued at every point of  $\text{Test}_{m,d}(t)$ , a set of at most*

$$(d + 1)^m + m(2d^2 + 2d + 1)^{m-1}$$

*points. The same  $t$  and the same test set serve every rational function over  $K_G$  with these degree bounds.*

*Proof.* Necessity is clear; assume that the tests pass and that  $P \neq 0$ . The coefficients lie in  $K_G$ , and  $P, Q$  are coprime over  $K_G$ , since a common factor there would be one over  $\mathbf{No}$ . Suppose  $Q$  is not constant and choose  $j$  with  $q_j > 0$ . By (16.26),  $\deg_{X_i} h_j \leq (d+1)d + (d+1)d = \mathbf{N}_d$  for  $i \neq j$ , so Lemma 16.68 gives ordinary integers  $\mathbf{a} \in \{0, \dots, \mathbf{N}_d\}^{m-1}$  with  $h_j(\mathbf{a}) \neq 0$ , and  $F_{\mathbf{a}} = P_{\mathbf{a}}/Q_{\mathbf{a}} \in K_G(X_j)$  is not a polynomial (Lemma 16.69). Its denominator does not vanish at  $t$ , and  $F_{\mathbf{a}}(t) \notin \mathcal{A}_{\mathbb{R}, \mathbb{R}} \supseteq \mathbf{Oz}$  (Theorem 16.33). This is the value of  $F$  at the point of  $\mathbf{R}_{m,d}(t)$  whose  $j$ th coordinate is  $t$  and whose other coordinates are  $\mathbf{a}$ , a failed test. So  $Q$  is constant,  $F$  is a polynomial with  $\deg_{X_i} F \leq d$ , and the tests on  $\square_{m,d}$  give  $F(\mathbf{Oz}^m) \subseteq \mathbf{Oz}$  by Theorem 16.6. The box has  $(d+1)^m$  points and each of the  $m$  remote grids has  $(\mathbf{N}_d + 1)^{m-1}$ . Only  $K_G$ , the degree bounds and  $b > G$  were used.  $\square$

The certificate is relative to exact field operations and exact membership in  $\mathbf{Oz}$ ; it is not an algorithm for arbitrary descriptions of surreal numbers, and choosing  $b$  above an arbitrary support is a construction, not an effective operation (22). When the degree data are known, the grid in direction  $j$  may use  $\{0, \dots, (q_j + 1)p_i + (p_j + 1)q_i\}$  in coordinate  $i \neq j$ , only directions with  $q_j > 0$  are needed, and the box may use  $\max(p_i, q_i)$  in coordinate  $i$ , or the actual degrees once the denominator is known to be constant; this refinement is available only after the presentation has been inspected and does not change the coefficient-uniform statement. The bound is  $d + 2$  for  $m = 1$  and  $(d+1)^2 + 2(2d^2 + 2d + 1) = 5d^2 + 6d + 3$  for  $m = 2$ , that is 3 and 14 for  $d = 1$ . These count exact ring-valued evaluations at particular surreal points, not floating-point samples or bit operations, and the bound is not claimed optimal (22).

*Remark 16.71* (Comparison with 16's probes [merge]). For  $m = 1$ ,  $\text{Test}_{1,d}(t) = \{0, \dots, d, t\}$  is 16's certificate (Corollary 16.50). For  $m \geq 2$ , 16's coordinate-dominant probes (Theorem 16.49) force a polynomial without any bound on the denominator, and together with the rectangular lower set  $\square_{m,d}$  of Theorem 16.46 they test the functions of Theorem 16.70 with  $(d+1)^m + m$  values, fewer than  $(d+1)^m + m(2d^2 + 2d + 1)^{m-1}$ . 16's probes use  $m + 1$  new exponents ( $b_1, \dots, b_m$  independent modulo  $G$ , and one exponent above all of them) and put a fresh monomial in every coordinate; 22's points use one fresh exponent, in one coordinate, with ordinary integers elsewhere, and a failed test comes with an explicit specialization  $\mathbf{a}$  at which the one-variable restriction has a negative block. Both are sizes of explicit certificates; neither is claimed minimal.

**Corollary 16.72** (Common tests for a set of functions [22]). *Every set-sized family of rational functions in  $m$  variables over  $\mathbf{No}$  has one countable set of points testing preservation of  $\mathbf{Oz}$  for all its members, and one finite set of the form  $\text{Test}_{m,d}(t)$  if the coordinate degrees of the reduced presentations are bounded by  $d$ .*

*Proof.* Put all coefficient supports of the family into one set-sized  $G$  (Lemma 16.30) and take  $t = \omega^b$  with  $b > G$ . In the bounded case apply Theorem 16.70; in general the countable set  $\mathbb{Z}^m \cup \bigcup_{j=1}^m \mathbb{Z}^{j-1} \times \{t\} \times \mathbb{Z}^{m-j}$  contains every grid needed for every member.  $\square$

So Theorem 16.51 is sharp with respect to the quantification over coefficients: every set of functions has common tests, and the class of all functions has no set of common tests (22). For bounded degree over one workspace  $K_G$  this is also the uniformity of Corollary 16.50.

*Remark 16.73* (A second witness for Theorem 16.51 [22]). Let  $S \subseteq \Lambda$  be a set,  $G$  a set-sized group containing the supports of its elements,  $b > G$  and  $t = \omega^b$ , and put

$$\tilde{R}_S(X) = \frac{\omega^{\omega b}}{X + t + \frac{1}{2}} = \frac{2\omega^{\omega b}}{2X + 2t + 1}, \quad (16.28)$$

where  $\omega b$  is a product in  $\mathbf{No}$  and  $\omega^{\omega b}$  a monomial. Numerator and denominator have coefficients in  $\mathbf{Oz}$  and are coprime, and the only pole  $-t - \frac{1}{2}$  has constant coefficient  $-\frac{1}{2} \notin D$ , so it is not in  $\Lambda$ . For  $s \in S$ ,  $u = (s + \frac{1}{2})/t$  is infinitesimal (its exponents are  $g - b < 0$ ), and  $\tilde{R}_S(s) = \omega^{\omega b} t^{-1} \sum_{n \geq 0} (-u)^n$  has its  $n$ th summand supported in  $\omega b - (n + 1)b + G$ , whose elements exceed  $(\omega - n - 2)b > 0$ ; the blocks are ordered as in [Theorem 16.31](#), so  $\tilde{R}_S(s) \in \Pi_k$ . For  $a > 0$  above the group generated by  $G$ ,  $b$  and  $\omega b$ , and  $x = \omega^a \in \mathbf{Oz}$ , the value  $\tilde{R}_S(x)$  is nonzero with all exponents at most  $\omega b - a < 0$ , so it is not in  $\Lambda$ . For  $S \subseteq \Lambda^m$  apply this to the first coordinates and let the function depend on  $X_1$  only; so [Theorem 16.51](#) holds in every number of variables and for  $\mathbf{Oz}[i]$ , still with numerator degree 0, denominator degree 1, coefficients in  $\mathbf{Oz}$  and no pole on  $\Lambda^m$ . 16's  $R_S = \omega^\beta / (X - \frac{1}{2}) = 2\omega^\beta / (2X - 1)$  also has coefficients in  $\mathbf{Oz}$ ; its pole is at  $\frac{1}{2}$ , 22's at the fresh scale. The failure of either witness is not caused by a high degree or by a pole at an untested input, but by coefficients at scales above the prescribed inputs (22).

**Example 16.74** (Every ordinary integer passes [22]).  $F(X) = \omega^\omega / (X + \omega + \frac{1}{2}) = 2\omega^\omega / (2X + 2\omega + 1)$  is defined on  $\mathbf{Oz}$ , and for  $n \in \mathbb{Z}$ ,  $F(n) = \sum_{r \geq 0} (-n - \frac{1}{2})^r \omega^{\omega - r - 1} \in \Pi$ ; but  $F(\omega^{\omega^2})$  has leading exponent  $\omega - \omega^2 < 0$ . The ordinary integers, an exact test set for polynomials, miss this obstruction, and  $\omega^{\omega^2}$  is the remote test for the group generated by 1 and  $\omega$ . 16's  $\omega / (X - \frac{1}{2})$  (after [Theorem 16.51](#)) is an example of the same kind.

**Theorem 16.75** (Removal of set-sized exceptions in several variables [22]; 16 for  $m = 1$ ). *Let  $F \in E(X_1, \dots, X_m)$  be defined and  $\Lambda$ -valued at every point of  $\Lambda^m$  outside a set  $S \subseteq \Lambda^m$ . Then  $F$  preserves  $\Lambda$ . Equivalently, the class of points of  $\Lambda^m$  at which  $F$  is undefined or not  $\Lambda$ -valued is empty or a proper class.*

*Proof.* Let  $F = P/Q \neq 0$  in lowest terms. If  $Q$  is not constant, choose  $j$  and ordinary integers  $\mathbf{a}$  with  $h_j(\mathbf{a}) \neq 0$  ([Lemmas 16.68](#) and [16.69](#); only characteristic 0 is used, so this works over  $\mathbf{No}[i]$ ). Put the supports of the coefficients and of all coordinates of all points of  $S$  into one set-sized  $G$  and take  $b > G$ ,  $t = \omega^b$ . The point with  $j$ th coordinate  $t$  and other coordinates  $\mathbf{a}$  is not in  $S$ , because the exponent  $b$  lies outside  $G$ , and there  $F$  is defined with a value outside  $\mathcal{A}_{k,k} \supseteq \Lambda$  ([Theorem 16.33](#)), a contradiction. So  $F$  is a polynomial. If  $F(\mathbf{a}) \notin \Lambda$  for some  $\mathbf{a} \in \Lambda^m$ , write  $F(a_1 + X, a_2, \dots, a_m) = F(\mathbf{a}) + \sum_{n=1}^e c_n X^n$  and enlarge  $G$  by the supports of the coefficients of  $F$ , of  $\mathbf{a}$  and of the points of  $S$ . For fresh  $t$  the value at  $X = t$  differs from  $F(\mathbf{a})$  by an element of  $\Pi_k$  ([Corollary 16.32](#)), which cannot move a nonmember of the additive group  $\Lambda$  into it, and  $(a_1 + t, a_2, \dots, a_m) \notin S$ , since its first coordinate has the new leading exponent  $b$ .  $\square$

[Theorem 16.75](#) and [Theorem 16.51](#) are compatible for the reason given after [Theorem 16.52](#): one prescribes good values on a set, the other assumes good values off a set. The two steps of the proof use different mechanisms (22). A denominator is detected by a negative Laurent block at a new scale. A polynomial failure is propagated by adding a positive-support block, which cannot alter an old negative tail or an old nonintegral constant coefficient; this step does not assume that the bad value is finite or that it has finitely many exponents.

**Remark 16.76** (Specialization and composition [22]). The coefficientwise constant term of [Theorem 16.7](#) respects composition: if  $F \in \mathcal{N}_m(\mathbf{Oz})$  and  $F_1, \dots, F_m \in \mathcal{N}_n(\mathbf{Oz})$ , then  $F(F_1, \dots, F_m) \in \mathcal{N}_n(\mathbf{Oz})$  and  $\text{ct}_*(F(F_1, \dots, F_m)) = \text{ct}_*(F)(\text{ct}_*(F_1), \dots, \text{ct}_*(F_m))$ , because all coefficients lie in the ring  $\mathbb{Q} + \Pi$ , on which  $\text{ct}$  is a ring homomorphism, and substitution is built from finitely many sums and products; the same holds for  $\mathbf{Oz}[i]$ . The decomposition  $F = \text{ct}_*(F) + (F - \text{ct}_*(F))$  is an arithmetic classification, not a statement that the purely infinite part is negligible: a coefficient such as  $\omega^\alpha$  dominates every ordinary coefficient, and preservation places no condition on it beyond positive support (22).

**Example 16.77** (Leading exponents are not enough [22]). If  $0 < b < c$ , the coefficient of  $(1 + \omega^{-c})X$  has leading exponent 0, but the value  $\omega^b + \omega^{b-c}$  at  $\omega^b$  has a negative exponent. A scale chosen above leading exponents only does not give the conclusion of [Corollary 16.32](#) for polynomials;  $G$  must contain the complete supports, remote tails included (compare the remark after [Lemma 16.30](#)).

**Theorem 16.78** (Gaussian coordinate-degree grid [22]). *Let  $F \in \mathbf{No}[i][Z_1, \dots, Z_m]$  with  $\deg_{Z_j} F \leq d_j$ , and  $C_{d_j+1} = \{a + bi : a, b \in \{0, \dots, d_j\}\}$ . Then  $F(\mathbf{Oz}[i]^m) \subseteq \mathbf{Oz}[i]$  if and only if  $F(\prod_j C_{d_j+1}) \subseteq \mathbf{Oz}[i]$ ; for a uniform bound  $d$  these are  $(d+1)^{2m}$  tests.*

*Proof.*  $H(\mathbf{U}, \mathbf{V}) = F(\mathbf{U} + i\mathbf{V})$  has degree at most  $d_j$  in each of  $U_j$  and  $V_j$ , and the tests are its values on the box  $\prod_j \{0, \dots, d_j\}^2$ . By (16.4) its Newton coefficients lie in  $\mathbf{Oz}[i]$ , and binomials of elements of  $\mathbf{Oz}$  lie in  $\mathbf{Oz}$  ([Lemma 16.5](#)), so  $H(\mathbf{Oz}^{2m}) \subseteq \mathbf{Oz}[i]$ ; every element of  $\mathbf{Oz}[i]$  is  $x + iy$  with  $x, y \in \mathbf{Oz}$ . The converse is restriction.  $\square$

This is the coordinate-degree twin of 18's simplex test ([Proposition 16.64](#)), which uses the total degree and the  $\binom{2m+d}{2m}$  points  $\mathbf{a} + i\mathbf{b}$  with  $|\mathbf{a}| + |\mathbf{b}| \leq d$ , a subset of  $C_{d+1}^m$ ; for polynomials of total degree at most  $d$  the simplex is the smaller test, and neither is claimed minimal. The ordinary box is not enough over  $\mathbf{Oz}[i]$  ([Remark 16.10](#)).

**Theorem 16.79** (Gaussian single-scale certificate [22]). *Let  $F = P/Q \in \mathbf{No}[i](Z_1, \dots, Z_m)$  be in lowest terms with all coordinate degrees of  $P$  and  $Q$  at most  $d \geq 1$ , let  $G$  contain the supports of the real and imaginary parts of all coefficients,  $b > G$  and  $t = \omega^b$ . Then  $F$  preserves  $\mathbf{Oz}[i]$  if and only if it is defined and  $\mathbf{Oz}[i]$ -valued on  $C_{d+1}^m \cup \mathbf{R}_{m,d}(t)$ , a set of at most  $(d+1)^{2m} + m(2d^2 + 2d + 1)^{m-1}$  points. Every rational function preserving  $\mathbf{Oz}[i]$  is a polynomial.*

*Proof.* Run the proof of [Theorem 16.70](#) over  $K_G$  with  $k = \mathbb{C}$ : a nonconstant denominator gives a nonpolynomial one-variable restriction at ordinary integer values of the other coordinates, whose value at  $t$  has a negative block. So the remote tests force a constant denominator, and [Theorem 16.78](#) applies. The last assertion is also [Theorem 16.49](#).  $\square$

The remote points need no Gaussian coordinates: ordinary integers already avoid a nonzero polynomial in characteristic 0, and only the arithmetic stage needs the Gaussian grid (22). [Theorems 16.51](#) and [16.75](#) hold for  $\mathbf{Oz}[i]$  as stated.

**Theorem 16.80** (Tensor base change [22]; the decomposition is 10's, 16's and 18's).  $\mathcal{N}_m^{\text{rat}}(\Lambda) = \mathcal{N}_m(\Lambda) = \text{Int}(D^m) \oplus \Pi_k[\mathbf{X}]$ , and the multiplication map

$$\Lambda \otimes_D \text{Int}(D^m) \longrightarrow \mathcal{N}_m(\Lambda), \quad \lambda \otimes f \longmapsto \lambda f, \quad (16.29)$$

is a ring isomorphism, the tensor product being read through finite sums of pure tensors, or in a containing universe in which  $\Lambda$  is a set; it does not assert that  $\Lambda$  is a set in the original universe. The coefficientwise constant term is a split retraction onto  $\text{Int}(D^m)$ , and evaluation commutes with it.

*Proof.* The first equality is the polynomiality of preserving rational functions ([Theorem 16.49](#); 22's [Corollary 5.3](#) and [Theorem 7.2](#)), and the decomposition is [Theorems 16.7](#) and [16.9](#) and [Proposition 16.45](#); 22 proves it as 16 does, without a basis of  $\text{Int}(D^m)$ : values on a large ordinary box put the Newton coefficients, hence the monomial coefficients, in  $F_0 + \Pi_k$ ; constant terms at points of  $D^m$  put the ordinary part in  $\text{Int}(D^m)$ ; and  $f(\mathbf{z} + \mathbf{u}) - f(\mathbf{z}) \in \Pi_k$  for  $\mathbf{z} \in D^m$ ,  $\mathbf{u} \in \Pi_k^m$ . Put  $M = \text{Int}(D^m)$ . As  $D$ -modules  $\Lambda = D \oplus \Pi_k$ , so  $\Lambda \otimes_D M \cong M \oplus (\Pi_k \otimes_D M)$ .



Since  $D[\mathbf{X}] \subseteq M \subseteq F_0[\mathbf{X}]$  and every polynomial over  $F_0$  has a common denominator in  $D \setminus \{0\}$ , localization gives  $F_0 \otimes_D M \cong F_0[\mathbf{X}]$  (injective because  $M$  lies in the torsion-free module  $F_0[\mathbf{X}]$ );  $\Pi_k$  is an  $F_0$ -vector space, so  $\Pi_k \otimes_D M \cong \Pi_k \otimes_{F_0} (F_0 \otimes_D M) \cong \Pi_k[\mathbf{X}]$ . Under these identifications the multiplication map is the direct-sum map of the decomposition. The assertions on the constant term follow from its multiplicativity on  $F_0 + \Pi_k$  (Lemma 2.2).  $\square$

**Definition 16.81** ([22]). For a coordinate degree bound  $\mathbf{d}$ , a finite set  $S \subseteq D^m$  is *universal* in that degree space if every  $f \in F_0[\mathbf{X}]$  with  $\deg_{X_j} f \leq d_j$  and  $f(S) \subseteq D$  satisfies  $f(D^m) \subseteq D$ .

**Theorem 16.82** (Lifting constant test sets [22]). *A finite set  $S \subseteq D^m$  is universal in a coordinate degree space if and only if every  $F \in E[\mathbf{X}]$  in that degree space with  $F(S) \subseteq \Lambda$  satisfies  $F(\Lambda^m) \subseteq \Lambda$ .*

*Proof.* Let  $S$  be universal. Its evaluation matrix on the monomials of the degree space has full column rank over  $F_0$ : otherwise a nonzero  $h \in F_0[\mathbf{X}]$  in the space vanishes on  $S$ , Lemma 16.68 gives  $\mathbf{c} \in \mathbb{Z}^m \subseteq D^m$  with  $h(\mathbf{c}) \neq 0$ , and  $h/(2h(\mathbf{c}))$  is  $D$ -valued on  $S$  but takes the value  $\frac{1}{2} \notin D$  at  $\mathbf{c}$ . Inverting a square invertible submatrix writes every coefficient of  $F$  as an  $F_0$ -linear combination of values in  $\Lambda = D \oplus \Pi_k$ , so  $F = f + g$  with  $f \in F_0[\mathbf{X}]$  and  $g \in \Pi_k[\mathbf{X}]$  in the same space. Constant terms at the points of  $S$  give  $f(S) \subseteq D$ , so  $f \in \text{Int}(D^m)$ , and  $F$  preserves  $\Lambda$  by Theorem 16.80. Conversely, if  $S$  tests  $\Lambda$  and  $f \in F_0[\mathbf{X}]$  has  $f(S) \subseteq D$ , then  $f(\Lambda^m) \subseteq \Lambda$  and  $f(D^m) \subseteq \Lambda \cap F_0 = D$ .  $\square$

So among test points in the ordinary lattice  $D^m$ , minimal polynomial testing is the same problem before and after adjoining the purely infinite part (22). This turns the last assertion of Proposition 16.64, the forward direction for  $D = \mathbb{Z}[i]$  in one variable (18), into an equivalence in every number of variables, for  $\mathbb{Z}$  and for  $\mathbb{Z}[i]$ . In one Gaussian variable the ordinary problem is the classical universal-set problem connected with simultaneous  $p$ -orderings (Byszewski, Frączyk and Szumowicz [7], with the work of Volkov and Petrov [41] discussed there); 22 does not determine the optimal sets, and supplies only the transfer and the grid of Theorem 16.78. The theorem bears on Question 18.14(a), not on Question 18.16(c): its test points lie in  $D^m$ , and whether points of  $\mathbf{Oz}[i]$  outside  $\mathbb{Z}[i]^m$  can do better is not addressed.

**Theorem 16.83** (Finite generators of image ideals [22]). *Let  $F \in \mathcal{N}_m(\mathbf{Oz})$  have  $\deg_{X_j} F \leq d_j$  and Newton coefficients  $c_{\mathbf{k}}$  (Theorem 16.6), and let  $I_F$  be the ideal of  $\mathbf{Oz}$  generated by  $F(\mathbf{Oz}^m)$ , the class of finite  $\mathbf{Oz}$ -linear combinations of values. Then*

$$I_F = (c_{\mathbf{k}} : \mathbf{k} \leq \mathbf{d}) = \left( F(\boldsymbol{\mu}) : \boldsymbol{\mu} \in \prod_j \{0, \dots, d_j\} \right). \quad (16.30)$$

*Hence for every ideal  $I$  of  $\mathbf{Oz}$ ,  $F(\mathbf{Oz}^m) \subseteq I$  if and only if the finitely many box values lie in  $I$ .*

*Proof.* The Newton expansion writes each value  $F(\mathbf{x})$  as an  $\mathbf{Oz}$ -linear combination of the  $c_{\mathbf{k}}$  (Lemma 16.5); (16.4) writes each  $c_{\mathbf{k}}$  as an integral combination of box values; and box values are values.  $\square$

A finite generating set avoids set-size questions about an ideal generated by a class, and no principal generator or greatest common divisor is assumed or inferred (22). The Gaussian version, with the box  $\prod_j C_{d_j+1}$ , follows in the same way from the  $2m$ -variable polynomial  $H$  in the proof of Theorem 16.78: the ideal of  $\mathbf{Oz}[i]$  generated by all values on  $\mathbf{Oz}[i]^m$  is generated by the finitely many box values.

**Corollary 16.84** (Arithmetic birational rigidity [22]). *If  $\Phi, \Psi : E^m \dashrightarrow E^m$  are rational maps inverse to each other as rational maps, and both are defined on  $\Lambda^m$  with values in  $\Lambda^m$ , then all coordinates of  $\Phi$  and  $\Psi$  are polynomials in  $\mathcal{N}_m(\Lambda)$ , and  $\Phi, \Psi$  are mutually inverse polynomial automorphisms of  $E^m$ .*

*Proof.* Each coordinate preserves  $\Lambda$ , so it is a polynomial (Theorems 16.49 and 16.79); the rational identities  $\Psi \circ \Phi = \text{id}$  and  $\Phi \circ \Psi = \text{id}$  are then polynomial identities.  $\square$

The rational inverse is a hypothesis: nothing is asserted about a bijective polynomial map of  $\Lambda^m$  without one, no Jacobian-conjecture claim is made, and the coordinates need not have coefficients in  $\Lambda$  (22).

**Example 16.85** ([22]).  $\Phi(X, Y) = (X + \frac{1}{2}Y(Y - 1), Y)$  and  $\Psi(X, Y) = (X - \frac{1}{2}Y(Y - 1), Y)$  are inverse polynomial maps that induce inverse bijections of  $\mathbf{Oz}^2$ , with the coefficient  $\frac{1}{2} \notin \mathbf{Oz}$ ; the Gaussian analogue uses  $Y(Y - 1)/(1 + i)$ . So arithmetic polynomial maps form a larger class than maps with coordinates in  $\mathbf{Oz}[X, Y]$ . 16's shear  $(x, y) \mapsto (x + y^2, y)$  (after Theorem 16.43) has coordinates in  $\mathbf{Oz}[X, Y]$ .

Three conditions on a polynomial are distinct (22): coefficients in  $\Lambda$ , values in  $\Lambda$  on  $\Lambda^m$ , and values in  $\Lambda$  at sampled points. The first implies the second but not conversely ( $\binom{X}{2}$ ); for bounded degree the tests above make the second and third equivalent, while for rational functions no coefficient-independent set of samples does (Theorem 16.51). Coefficients in  $\mathbf{Oz}$  for numerator and denominator do not make the denominator invertible at every omnific input (Example 16.74).

**Proposition 16.86** (Fresh scales in a fixed Hahn workspace [22]). *Let  $\Gamma$  be a set-sized ordered abelian group,  $G \subseteq \Gamma$  a subgroup containing the coefficient supports, and  $\text{cvx}_\Gamma(G) = \{x \in \Gamma : |x| \leq g \text{ for some } g \in G\}$  the convex subgroup it generates. A positive  $b \in \Gamma$  with  $b > G$  exists if and only if  $\text{cvx}_\Gamma(G) \neq \Gamma$ . When it exists, the Laurent blocks, the detector and the one-point test (Theorems 16.31 and 16.33 and Corollary 16.32) and the certificates of Theorems 16.70 and 16.79 hold in  $k((X^\Gamma))$ , for  $\mathcal{A}_{D,k}(\Gamma)$  and coefficients supported in  $G$ .*

*Proof.* If  $b > G$ , then  $b \notin \text{cvx}_\Gamma(G)$ . Conversely a positive  $b \notin \text{cvx}_\Gamma(G)$  satisfies  $b > |g|$  for all  $g \in G$ , since  $b \leq |g|$  would put it in the convex subgroup. The remaining arguments use only supports, block separation, finite polynomial algebra and the split ring  $D \oplus \Pi_k(\Gamma)$ ; the exponents  $jb + g$  stay in  $\Gamma$ , and the countable union of the blocks is a legitimate support.  $\square$

For an Archimedean  $\Gamma$  every nonzero  $G$  is cofinal, and no fresh exponent exists in  $\Gamma$ ; this is a failure of the method's hypothesis, not a theorem that every finite test fails there, and one may enlarge  $\Gamma$  by a lexicographic factor or work in **No** (22). Theorem 16.51 cannot hold for a set-sized ring, which is itself a set of tests; its strength comes from the unbounded supply of surreal scales, and Corollary 16.72 explains how set-sized computations coexist with it (22). Where no fresh exponent exists, rational collapse still holds by 18's discrete-gap argument, which uses none (Theorem 16.66 and Corollary 16.67); what 22's method loses there is the finite rational certificate, and whether another finite certificate exists is open (Question 18.14(c)) [merge].

**Example 16.87** (One new coordinate exposes a denominator [22]). For  $F(X, Y) = (X + Y)/(XY + 1)$  over  $\mathbb{Q}$ , the leading coefficients in  $X$  are 1 and  $Y$  and  $\text{Res}_X(X + Y, XY + 1) = 1 - Y^2$ , so  $\mathbf{h}_X = Y(1 - Y^2)$ . At  $Y = 2$ ,  $F(X, 2) = (X + 2)/(2X + 1) = \frac{1}{2} + \frac{3}{2(2X+1)}$ , whose second term has a nonzero negative block at every fresh  $t$ . The resultant excludes exactly the values of  $Y$  at which the denominator loses degree or a cancellation occurs.

For every  $u \in \Pi$ , even of infinite support,  $\binom{X}{3} - 4\binom{Y}{2} + uX^2Y$  preserves  $\mathbf{Oz}$  with specialization  $\binom{X}{3} - 4\binom{Y}{2}$ , while  $\omega^{-1}X$  fails at  $X = 1$  and  $\sqrt{2} + \omega \notin \mathbf{Oz}$ : what matters is negative support or a nonintegral constant coefficient, not magnitude, and adding positive support cannot repair a nonintegral constant coefficient, which is the invariance behind the polynomial step of [Theorem 16.75](#) (22; compare [Example 16.8](#)).

A construction of the certificate reduces  $P/Q$  over the coefficient field, collects every exponent of every coefficient support, forms the group they generate, chooses  $b$  above it, and tests exact values on the box and the remote grids. An effective version needs an implementation contract for equality, polynomial gcd, support bounds and membership in  $\Lambda$ , none of which can be presumed decidable for arbitrary notations of surreal numbers; when the coefficients can be inspected exactly, checking that the reduced denominator is constant and then the Newton coefficients may be cheaper than evaluating the grid. The theorem is an exact value certificate, not a complexity improvement over symbolic algebra. Once a nonzero resultant specialization is found it is also a witness generator: it names a failing point without numerical approximation, since the first nonzero negative Laurent coefficient contributes a whole block that cannot cancel (22).

*Remark 16.88* (Other constant rings [merge]). The conclusion of [Theorem 16.51](#) holds for  $\mathcal{A}_{D,k}$  with every unital  $D \subsetneq k$ , in every number of variables. Take  $c \in k \setminus D$ . The only pole of  $\omega^\beta/(X_1 - c)$  is  $c \notin \mathcal{A}_{D,k}$ ; if  $\beta > 0$  exceeds the absolute values of all exponents in the supports of the  $(s_1 - c)^{-1}$ ,  $s \in S$  (a set), the values on  $S$  lie in  $\Pi_k$ , while at a point with first coordinate  $\omega^{2\beta}$  the value is the nonzero infinitesimal  $\omega^{-\beta}(1 - c\omega^{-2\beta})^{-1}$ . This is 16's proof with  $\frac{1}{2}$  replaced by  $c$ ; for  $D = k$  it does not apply.

22 summarizes its dependencies as follows: the certificate uses the fresh scale, the resultant specialization and the ordinary binomial test; the no-set-test theorem uses the fresh scale and a geometric inverse, not the resultant or the binomial machinery; the removal of exceptions combines denominator detection with invariance under positive-support additions; and the Gaussian versions change the constant arithmetic and keep the support mechanism. Its proofs need neither algebraic closedness of  $\mathbf{No}[i]$ , nor differentiation, nor a topology, and  $\omega^\gamma$  is the normal-form monomial, not an analytic exponential.

### 16.11 Semialgebraic preservers: smooth rigidity and bounded-scale flexibility

This subsection is manuscript 25's ([Section 1.3](#)). It answers the first clause of [Question 18.20\(a\)](#) for globally smooth maps: over any real closed field  $E$  with a cofinal discretely ordered subring  $B$ , a globally  $C^\infty$  semialgebraic map  $E^r \rightarrow E^{r'}$  maps  $B^r$  into  $B^{r'}$  if and only if its coordinates are integer-valued polynomials in  $\mathcal{N}_r(B)$  ([Theorem 16.102](#)), and for a scalar graph of degree at most  $D$  global  $C^{D^2}$  regularity suffices in every dimension; for  $\mathbf{Oz}$  and  $\mathbf{Oz}[i]$  this gives finite binomial expansions with omnific coefficients ([Corollary 16.103](#) and [Theorem 16.115](#)). Without smoothness every semialgebraic preserver is still polynomial on each of its two tails ([Theorem 16.95](#)), but on a bounded interval it can be far from polynomial:  $\omega^b\sqrt{x}$  is omnific for every omnific  $x \in [0, \omega^a]$  exactly when  $a \ll b$  ([Theorem 16.108](#)), the same threshold as [Theorem 14.37](#); for every ordinary  $k$  an explicit  $C^k$  bump sends  $\mathbf{Oz}$  into  $\Pi$  and agrees with no finite family of polynomials on  $\mathbf{Oz}$  ([Theorem 16.112](#)); and no set-sized sample tests preservation, even for the globally  $C^\infty$  branch of an irreducible quadratic graph ([Theorem 16.117](#)). So the answer to the first clause of [Question 18.20\(a\)](#) is complete for globally  $C^\infty$  semialgebraic maps in every dimension and negative, as far as polynomial or finitely piecewise polynomial descriptions go, below that regularity; the certificate clause is answered negatively for samples chosen independently of the coefficients, and coefficient-adapted certificates stay open.

25 pins 738a390 (23 September 2026), whose text of this report is the same blob as at 24’s pin, the text after the sixth merge. It read selected parts: the numerical polynomials, the rational-map results and 22’s non-claim (10), which leaves algebraic and semialgebraic maps unclassified and names  $|X|$ . Printed once, under 10’s labels, are the decomposition of the numerical polynomials with its Newton coefficients (25’s Proposition 5.3, [Theorems 16.6](#) and [16.7](#)) and its Gaussian form in one variable (25’s Proposition 9.3, [Theorem 16.9](#) with  $r = 1$ ); 25 proves the additive decompositions again, with 10’s arguments, and credits them to this report. 25’s two-tail theorem ([Theorem 16.95](#)) generalizes the *Collapse* and *Filling* steps of 18’s [Theorem 16.66](#) from rational functions over a discretely ordered ring to semialgebraic functions over a real closed field in which the ring is cofinal; the mechanism, a finite difference below 1 at lattice points that the discrete gap makes exactly 0, is 18’s, which 25 credits only as “rational-collapse context”. The sufficiency half of its absorption theorem ([Theorem 16.105](#)) contains [Lemma 4.2\(ii\)](#), 07’s [Lemma 14.6](#) and the freshness of 16’s [Lemma 16.30](#); its quadratic no-set-test theorem refines [Theorem 16.51](#) and [Remark 16.73](#), whose witnesses are rational functions with a pole in **No**; and its removal of exceptions extends [Theorems 16.52](#) and [16.75](#) from rational functions to globally  $C^\infty$  semialgebraic ones. 25 credits as antecedents the definable integer-valued rigidity of Jones–Thomas–Wilkie and of Bhardwaj–McCulloch–Ramachandran–Woo [66, 67], the finite-difference strategy, finite-order detection of smoothness and nonflatness of tame functions [68, 69], Hahn-field real closedness [23], Newton interpolation and this report’s numerical-polynomial arithmetic and fresh-scale sampling obstructions. It names as its principal candidate contributions the combination of the cofinal smooth classification with the exact bounded-scale obstruction, the explicit family separating every finite order of smoothness from  $C^\infty$ , and the fixed-degree quadratic sampling obstruction; its bounded-window theorem controls a proper class of inputs at once, not only the denominators of one set of values. The numbering of 25 cited here is that of its own manuscript.

**Conventions and renamings.**  $E$  is a real closed ordered field and  $B \subseteq E$  a unital subring that is *discretely ordered*, 1 being its least positive element, as for 18’s rings ([Section 16.8](#)), and *cofinal*: every element of  $E$  lies below some element of  $B$ . Every integer part of a real closed field satisfies both conditions, and the proofs use no floor function;  $\mathbb{Z}$  is not cofinal in a non-Archimedean field. **Oz** is a cofinal discretely ordered subring of **No**: a nonconstant omnific integer has a positive leading exponent and so exceeds every ordinary real in absolute value, and a monomial above the leading scale of a given surreal is omnific (25). A function  $E^r \rightarrow E$  is *semialgebraic* if its graph is defined by a finite ordered-field formula with finitely many parameters from  $E$ , which by quantifier elimination may be taken quantifier-free; the predicate  $x \in B$  (or  $x \in \mathbf{Oz}$ ) is *not* allowed in the formula, and preservation of  $B$  is a separate hypothesis. Differentiability is taken in the order topology of  $E$ , all positive elements of  $E$  being tolerances;  $C^k$  means continuous partial derivatives through order  $k$ , and global  $C^\infty$  means  $C^k$  for every ordinary  $k$ . Exponents follow [Section 2.2](#) (25’s growth convention). For a divisible ordered subgroup or subclass  $H \subseteq \mathbf{No}$ ,

$$K_H = \{x \in \mathbf{No} : \text{supp } x \subseteq H\}, \quad H^\uparrow = \{\gamma \in \mathbf{No} : \gamma > h \text{ for every } h \in H\}; \quad (16.31)$$

for a set  $H$  this is 16’s Hahn field  $K_H$  with  $k = \mathbb{R}$ , and  $K_{H_a} = \mathcal{F}_a$ .  $T = \omega^a$  is the actual monomial, as in [Sections 15.9](#) and [15.10](#), and  $a \ll b$  is the relation of (2.4). For an ordinary  $D \geq 1$ , a *degree-D graph description* of  $f : E^r \rightarrow E$  is a quantifier-free Boolean formula defining the graph of  $f$  whose polynomial atoms all have total degree at most  $D$ ; the number of atoms is finite but not bounded, and the coefficients are arbitrary elements of  $E$ . 25’s symbols are renamed as follows.

- Its real closed field  $F$  and cofinal discrete ring  $R$  are  $E$  (the ambient field, as in [Sections 16.8](#) and [16.10](#)) and  $B$  (18’s letter;  $\mathcal{B}$  is the ring  $D + \mathfrak{m}_k$ ); its input and output dimensions  $n, m$  are

$r, r'$ , as in  $\mathcal{N}_r$ ; the order  $m$  of its finite differences is  $s$ , as in the proof of [Theorem 16.66](#).

- Its graph degree  $D$  is  $\mathbf{D}$  and its smoothness order  $k$  is  $\mathbf{k}$  ( $D$  is the constant ring and  $k$  the coefficient field); so its  $C^{D^2}$ ,  $C^k$  and  $s(D)$  are  $C^{\mathbf{D}^2}$ ,  $C^{\mathbf{k}}$  and  $\mathbf{s}(\mathbf{D})$ .  $C^{\mathbf{k}}$  is a regularity class, not a fibre  $\mathbf{C}(G)$  or a set  $C_j$ .
- Its convex subgroup  $H_a = \{\gamma : |\gamma| \leq na \text{ for some } n\}$  is  $\text{cvx}(\mathbb{Z}a)$ , as in [Section 14.9](#); it is the *opposite* of this report's  $H_a$  of (2.4), which collects the exponents infinitesimal relative to  $a$ . Its  $K_H$  keeps its name, extended to classes, and its  $H^+$  is  $H^\uparrow$  ( $\kappa^+$  is a successor cardinal).
- Its multiplier  $L = \omega^b$  is written  $\omega^b$  ( $L(M)$  is 14's module and  $L_h$  03's field); its bumps  $B_k$  are  $\mathbf{b}_k$  ( $\mathbf{B}$  is the negative-support ring); its ring  $\mathcal{E}_R$  of semialgebraic preservers and ideal  $\mathcal{B}_R$  of bounded-region ones are  $\text{SA}(B)$  and  $\text{SA}_b(B)$  ( $\mathcal{E}_a$  is 07's field and  $\mathbf{B}_{\mathbb{R}}$  19's fibre).
- Its test function  $G_b$  is  $\mathbf{r}_b$  ( $G$  is a group), its shears  $S_k$  are  $\text{Sh}_k$  ( $S$  is a set of points), and in proofs its polynomial  $A(x, h)$ , bound  $M$  and Lagrange basis  $\ell_j$  are  $\mathbf{u}(x, h)$ ,  $M_0$  and  $\mathbf{l}_j$ .

Five tempting readings are false. (1) 25's  $H_a$  is  $\text{cvx}(\mathbb{Z}a) \supsetneq H_a$ ; its threshold “ $b > na$  for every  $n$ ” is  $a \ll b$ , not  $b \in H_a$ . (2) Ordinary shifts are not cofinal:  $\mathbb{Z}$  is not cofinal in a non-Archimedean  $E$ , so extrapolation must use a large step in  $B$  ((16.32)). (3) A finite difference below 1 is an estimate; it becomes 0 only at lattice points, by the discrete gap, and semialgebraicity then spreads the zero set to a ray. (4) Rolle's theorem, the mean-value theorem and the identity principle are used only for semialgebraic functions and finite derivative orders, never for arbitrary functions on **No** ([Remark 16.90](#)). (5) The multiplier acts on supports:  $|\omega^b \phi(x)| > 1$  does not make  $\omega^b \phi(x)$  omnific, since an infinite surreal may still have negative support.

### Polynomial rigidity on the two tails.

**Lemma 16.89** (Polynomial and reciprocal growth [25]). *If  $g : (c_0, \infty) \rightarrow E$  is semialgebraic, there are an ordinary  $d \geq 0$  and  $C > 0$  in  $E$  with  $|g(x)| \leq Cx^d$  for all sufficiently large  $x$ . If a semialgebraic  $h$  is nonzero on  $(c, c + \delta)$ , then, after shrinking  $\delta$ ,  $|h(c + t)| \geq Ct^N$  for  $0 < t < \delta$ , some  $C > 0$  and an ordinary  $N \geq 0$ ; similarly on the left of  $c$ .*

*Proof.* A graph has empty interior, so in a quantifier-free description at every graph point some nonzero defining polynomial vanishes (otherwise all signs are locally constant and the formula contains an open box). There are finitely many of them, so after a finite semialgebraic partition, on a final interval  $Q(x, g(x)) = 0$  for a nonzero  $Q \in E[X, Y]$  of positive degree in  $Y$ , whose leading coefficient in  $Y$  is nonzero for large  $x$ . The root bound  $|y| \leq 1 + \sum_{j < q} |a_j/a_q|$  for  $a_q y^q + \dots + a_0 = 0$  (compare the leading term with the others when  $|y| > 1$ ) bounds  $g(x)$  by finitely many rational functions of  $x$ , hence by  $Cx^d$ . For the second assertion apply the first to  $u \mapsto 1/|h(c + 1/u)|$  on a final ray and substitute  $u = 1/t$ . No Archimedean assumption and no convergent real sequence is used.  $\square$

The same graph argument shows semialgebraic smoothness on small intervals: take a squarefree irreducible factor of  $Q$  involving  $Y$ ; its resultant with its  $Y$ -derivative is a nonzero polynomial in  $X$ , away from whose finitely many zeros a branch is a simple algebraic root, differentiable to every finite order by the semialgebraic implicit-function theorem; monotonicity follows after further finite partition (25). One-variable semialgebraic functions are eventually monotone and eventually  $C^k$  for each fixed  $\mathbf{k}$ , their derivatives are semialgebraic, and Rolle's and the mean-value theorem hold for them; these, quantifier elimination and a semialgebraic subset of the line being a finite union of points and intervals are the standard facts of real algebraic geometry used [69].



*Remark 16.90* (What is not transferred [25]). The mean-value theorem is not asserted for arbitrary functions on  $\mathbf{No}$ ; it is used only for semialgebraic functions and a fixed finite derivative order, where it holds over every real closed field by real-algebraic methods. This excludes the known pathologies of unconstrained surreal-valued calculus (25).

**Lemma 16.91** (A derivative lowers the growth [25]). *If  $g$  is semialgebraic and eventually  $C^1$  with  $|g(x)| \leq Cx^j$  eventually,  $j \in \mathbb{Z}$ , then  $|g'(x)| \leq C'x^{j-1}$  eventually for some  $C' > 0$ .*

*Proof.* Eventually  $|g'|$  is monotone. If it is nondecreasing, the mean-value theorem on  $[x, 2x]$  gives  $|g'(x)| \leq |g'(\xi)| = |g(2x) - g(x)|/x \leq C(2^j + 1)x^{j-1}$ ; if it is nonincreasing,  $[x/2, x]$  gives  $|g'(x)| \leq 2C(1 + 2^{-j})x^{j-1}$ . The constants lie in  $E$ ; no real uniformity is needed.  $\square$

**Lemma 16.92** (Small high differences [25]). *For a semialgebraic  $f : E \rightarrow E$  there is an ordinary  $s \geq 1$  such that  $\Delta^s f(x) = \sum_{j=0}^s (-1)^{s-j} \binom{s}{j} f(x+j)$  satisfies  $|\Delta^s f(x)| < 1$  for all sufficiently large  $x$ .*

*Proof.* Take  $|f(x)| \leq Cx^d$  (Lemma 16.89) and  $s = d + 1$ . On a final ray  $f$  is  $C^s$ , and Lemma 16.91 applied  $s$  times gives  $|f^{(s)}(x)| \leq C_s/x$ . The interpolating polynomial of  $f$  at  $x, x+1, \dots, x+s$  has leading coefficient  $\Delta^s f(x)/s!$ , and repeated Rolle applied to the difference gives  $\Delta^s f(x) = f^{(s)}(\xi)$  with  $x < \xi < x+s$ ; take  $x > C_s$ .  $\square$

**Lemma 16.93** (A semialgebraic difference kernel [25]). *A semialgebraic function with  $\Delta^s f = 0$  on a final ray agrees on a final ray with a polynomial of degree at most  $s - 1$ .*

*Proof.* A semialgebraic  $g$  with  $g(x+1) = g(x)$  on a final ray is constant there, being eventually constant or strictly monotone. Applied to  $\Delta^{s-1} f$  this gives a constant  $c$ , and since  $\Delta \binom{X}{j} = \binom{X}{j-1}$ ,  $f - c \binom{X}{s-1}$  has vanishing  $(s-1)$ st difference on a final ray; induce on  $s$ .  $\square$

**Lemma 16.94** (Tail values determine membership [25]). *Let  $B \subseteq E$  be cofinal and  $P \in E[X]$  of degree at most  $d$ . If  $P(a) \in B$  for all sufficiently large  $a \in B$ , then  $P(B) \subseteq B$ .*

*Proof.* Fix  $a \in B$ . By cofinality there is a positive  $h \in B$  such that all  $a + jh$ ,  $1 \leq j \leq d+1$ , lie in the allowed tail, and the vanishing of the  $(d+1)$ st difference with step  $h$  gives

$$P(a) = \sum_{j=1}^{d+1} (-1)^{j+1} \binom{d+1}{j} P(a + jh) \in B. \quad (16.32)$$

The step  $h$  may be an infinite element of  $B$ ; an ordinary integer step would not reach the tail in a non-Archimedean field.  $\square$

**Theorem 16.95** (Two-tail polynomiality [25]). *Let  $B$  be a cofinal discretely ordered subring of the real closed field  $E$ . If  $f : E \rightarrow E$  is semialgebraic and  $f(B) \subseteq B$ , there are unique  $P_-, P_+ \in \mathcal{N}_1(B)$  and  $H > 0$  with  $f = P_-$  on  $(-\infty, -H)$  and  $f = P_+$  on  $(H, \infty)$ . For  $P_+$  alone it suffices that  $f(a) \in B$  for all sufficiently large  $a \in B$ .*

*Proof.* By Lemma 16.92,  $|\Delta^s f| < 1$  on a final ray. At lattice points  $a$  far enough out, all  $f(a+j)$  lie in  $B$ , so  $\Delta^s f(a) \in B$  has modulus below 1 and is 0. The zero set of  $\Delta^s f$  is semialgebraic and contains a cofinal subset of  $E$ , and a finite union of points and intervals with that property contains a final ray. Lemma 16.93 gives a polynomial  $P_+$  on a final ray and Lemma 16.94 puts it in  $\mathcal{N}_1(B)$ ; apply the same to  $x \mapsto f(-x)$  for  $P_-$ . Two distinct polynomials do not agree on a ray.  $\square$



This isolates the mechanism: semialgebraicity turns infinitely many discrete zero values into an identity on a tail, and the lattice gap supplies those zeros; it is a non-Archimedean adaptation of a classical finite-difference idea, not an asymptotic expansion (25). For rational functions over a discretely ordered ring these are the *Collapse* and *Filling* steps of [Theorem 16.66](#) (18).

Let  $\text{SA}(B)$  be the ring of semialgebraic functions  $E \rightarrow E$  preserving  $B$ , and  $\text{SA}_b(B)$  its ideal of those vanishing outside a bounded interval.

**Corollary 16.96** (The two-end quotient [25]). *There is a split exact sequence of rings*

$$0 \longrightarrow \text{SA}_b(B) \longrightarrow \text{SA}(B) \xrightarrow{(P_-, P_+)} \mathcal{N}_1(B) \times \mathcal{N}_1(B) \longrightarrow 0, \quad (16.33)$$

so  $\text{SA}(B)/\text{SA}_b(B) \cong \mathcal{N}_1(B)^2$ .

*Proof.* Uniqueness of the tail polynomials makes the map a ring homomorphism, with kernel  $\text{SA}_b(B)$ ; a splitting sends  $(P, Q)$  to the function equal to  $P$  on  $x < 0$  and to  $Q$  on  $x \geq 0$ .  $\square$

“Bounded” is order-theoretic; no compactness of  $[-H, H]$  is asserted. For  $E = \mathbf{No}$  the function rings are read through finite formula codes with finite parameter tuples, statements about their operations pointwise; no set of arbitrary class functions is formed (25).

### Global smoothness and a degree bound.

**Lemma 16.97** (Smooth identity principle [25]). *Let  $I \subseteq E$  be an interval and  $h : I \rightarrow E$  semialgebraic and  $C^\infty$ . If  $h$  vanishes on a nonempty open subinterval, it vanishes on  $I$ .*

*Proof.* Otherwise, since the zero set is a finite union of points and intervals, there is an interior point  $c$  where a zero interval meets an adjacent interval on which  $h \neq 0$ ; all derivatives of  $h$  vanish at  $c$ , by continuity from the zero side. On the other side [Lemma 16.89](#) gives  $|h(c+t)| \geq C|t|^N$  for small  $t \neq 0$ . If  $N = 0$  continuity is contradicted; if  $N \geq 1$ , Taylor’s formula with Rolle remainder,  $h(c+t) = h^{(N)}(\xi_t)t^N/N!$  with  $h^{(N)}(\xi_t) \rightarrow 0$ , gives  $h(c+t) = o(|t|^N)$  in the order topology, a contradiction.  $\square$

This is the elementary semialgebraic nonflatness argument, with classical and more general antecedents [68, 69]; 25 uses it to propagate an arithmetic tail identity, not as a new identity theorem.

**Theorem 16.98** (One-variable smooth classification [25]). *A globally  $C^\infty$  semialgebraic  $f : E \rightarrow E$  preserves  $B$  if and only if  $f$  is a polynomial in  $\mathcal{N}_1(B)$ .*

*Proof.*  $f = P_+$  on a final ray ([Theorem 16.95](#)); apply [Lemma 16.97](#) to  $f - P_+$ . The converse is clear.  $\square$

**Corollary 16.99** (One irreducible graph equation [25]). *If  $f : E \rightarrow E$  is semialgebraic, preserves  $B$  and satisfies  $Q(x, f(x)) = 0$  for all  $x$  for one irreducible nonconstant  $Q \in E[X, Y]$ , then  $f \in \mathcal{N}_1(B)$ , without any smoothness hypothesis.*

*Proof.* On a final ray  $f = P$  ([Theorem 16.95](#)), so  $Q(X, P(X))$  has infinitely many roots and vanishes; division in  $Y$  shows  $Y - P(X) \mid Q$ , and irreducibility makes  $Q$  a scalar multiple of  $Y - P(X)$ , so  $f = P$  everywhere.  $\square$

The hypothesis is one global irreducible equation; an arbitrary semialgebraic graph may switch between components, as the bumps below do at the ends of their support (25).

**Theorem 16.100** (An explicit smoothness threshold [25]). *Let  $f : E \rightarrow E$  be semialgebraic with a degree- $D$  graph description, and suppose  $f$  agrees with a polynomial  $P$  on a nonempty open interval. Put  $d = \max(0, \deg P)$  ( $d = 0$  also for  $P = 0$ ) and  $N = D \max(1, d)$ . Then  $d \leq D$ , and if  $f$  is globally  $C^N$ , then  $f = P$  everywhere. In particular a globally  $C^{D^2}$  semialgebraic preserver of  $B$  with a degree- $D$  graph description lies in  $\mathcal{N}_1(B)$ .*

*Proof.* At every graph point some nonzero atom vanishes. Factoring the atoms, on a subinterval of agreement one irreducible factor  $Q$  vanishes along  $(x, P(x))$ , so  $Y - P(X)$  divides  $Q$  and  $\max(1, d) \leq D$ . Suppose  $h = f - P \neq 0$ . Its semialgebraic zero set contains an open interval, so it has a boundary point  $c$  with  $h = 0$  on one side and  $h \neq 0$  on an adjacent interval on the other, which we shrink so that one irreducible factor  $Q$  of an atom vanishes along the graph of  $f$  there.  $Q$  is not a scalar multiple of  $Y - P(X)$ , since  $f \neq P$  there, so  $q(X) = Q(X, P(X)) \neq 0$  with  $\deg q \leq D \max(1, d) = N$ , and  $q(c) = Q(c, f(c)) = 0$  by continuity. If  $q$  vanishes to order  $j_0$  at  $c$ , then  $1 \leq j_0 \leq N$  and  $|q(x)| \geq C|x - c|^{j_0}$  near  $c$ . On the other hand  $Q(x, P(x) + h(x)) - Q(x, P(x)) = h(x)u(x, h(x))$  for a polynomial  $u$ , and the first term is 0; as  $h$  is continuous with  $h(c) = 0$ ,  $|u(x, h(x))| \leq M_0$  near  $c$ , so  $|h(x)| \geq (C/M_0)|x - c|^{j_0}$ . All derivatives of  $h$  through order  $N$  vanish at  $c$ , and the Taylor argument of Lemma 16.97 through order  $j_0 \leq N$  contradicts this bound. For preservers, Theorem 16.95 supplies the initial agreement.  $\square$

The bound  $D^2$  is an elementary sufficient bound, not claimed optimal; it does not depend on the size of the surreal parameters, the number of atoms or the ring  $B$ . Finite-differentiability-to-smoothness phenomena have classical antecedents [68]; the arithmetic contact estimate is proved here explicitly (25). The degree is that of a quantifier-free description; a degree bound for a formula with unresolved quantifiers is a different notion and is not covered.

### Several variables and the numerical polynomials.

**Lemma 16.101** (Uniform degree of polynomial slices [25]). *Let  $f : E^r \rightarrow E$  have a degree- $D$  graph description. If fixing all but one coordinate gives a polynomial in the remaining one, its degree is at most  $D$ .*

*Proof.* Specialize the fixed coordinates and discard atoms that become zero. At every point of the specialized graph some remaining nonzero atom vanishes, otherwise the formula would contain an open box; so one of them vanishes at infinitely many points  $(t, P(t))$ , hence vanishes after  $Y = P(X)$  and is divisible by  $Y - P(X)$ . Its degree is at most  $D$ ; a constant slice is included.  $\square$

**Theorem 16.102** (Finite-dimensional smooth classification [25]). *Let  $B$  be a cofinal discretely ordered subring of the real closed field  $E$ . If  $f : E^r \rightarrow E$  is globally  $C^\infty$ , semialgebraic and maps  $B^r$  into  $B$ , then  $f \in \mathcal{N}_r(B)$ . If its graph has a degree- $D$  description, global  $C^{D^2}$  regularity suffices, independently of  $r$ , and  $f$  has total degree at most  $D$ . Applied coordinatewise: a globally  $C^\infty$  semialgebraic map  $E^r \rightarrow E^{r'}$  maps  $B^r$  into  $B^{r'}$  if and only if its coordinates lie in  $\mathcal{N}_r(B)$ .*

*Proof.* It suffices to assume global  $C^{D^2}$  regularity for a degree- $D$  description. Fixing all but one coordinate at elements of  $B$  gives a semialgebraic  $C^{D^2}$  function preserving  $B$  with a graph of degree

at most  $D$ , so by [Theorems 16.95 and 16.100](#) a polynomial of degree at most  $D$ . With the Lagrange basis  $l_j$  of the nodes  $\{0, 1, \dots, D\}$  put

$$P(\mathbf{X}) = \sum_{\mathbf{j} \in \{0, \dots, D\}^r} f(\mathbf{j}) \prod_{i=1}^r l_{j_i}(X_i); \quad (16.34)$$

repeated one-variable interpolation gives  $f = P$  on  $B^r$ . A lattice is not order-dense, so equality off the lattice needs an argument. Put  $h = f - P$ . For fixed  $x_2, \dots, x_r \in B$ ,  $t \mapsto h(t, x_2, \dots, x_r)$  vanishes on  $B$ , so its zero set contains a final ray, and as the slice of  $P$  has degree at most  $D$ , the propagation of [Theorem 16.100](#) makes it zero on  $E$ :  $h = 0$  on  $E \times B^{r-1}$ . Fixing the first coordinate in  $E$  and the last  $r - 2$  in  $B$ , the same argument in the second coordinate gives  $h = 0$  on  $E^2 \times B^{r-2}$ , and so on; at each stage the slice of  $f$  is  $C^{D^2}$  with graph degree at most  $D$  and the slice of  $P$  has degree at most  $D$ . So  $f = P$ , and its values on  $B^r$  put  $P \in \mathcal{N}_r(B)$ . For the total degree, with the nonzero atoms  $Q_j(\mathbf{X}, Y)$  of the description,  $\prod_j Q_j(\mathbf{X}, P(\mathbf{X}))$  vanishes on  $E^r$ , so some  $Q_j(\mathbf{X}, P(\mathbf{X}))$  is identically zero,  $Y - P(\mathbf{X})$  divides  $Q_j$ , and  $\deg P \leq D$ .  $\square$

No analyticity, no ordinary-real coefficients and no density of lattice points is used; the uniform complexity bound supplies one tensor interpolant instead of unrelated polynomials on different lines. For a map  $E^r \rightarrow E^{r'}$  the bound  $C^{D^2}$  applies when each scalar coordinate graph has a degree- $D$  description; a degree bound for the full vector graph must first be converted into bounds for the coordinate graphs (25).

For  $\mathbf{Oz}$ , 25's Proposition 5.3 is [Theorem 16.7](#) with the Newton coefficients of [Theorem 16.6](#): every  $P \in \mathcal{N}_r(\mathbf{Oz})$  is a unique finite sum  $\sum_{\alpha} c_{\alpha} \binom{\mathbf{X}}{\alpha}$  with  $c_{\alpha} = (\Delta_1^{\alpha_1} \cdots \Delta_r^{\alpha_r} P)(\mathbf{0}) \in \mathbf{Oz}$ , and  $\mathcal{N}_r(\mathbf{Oz}) = \text{Int}(\mathbb{Z}^r) \oplus \Pi[\mathbf{X}]$  additively. 25's proof is 10's (binomial closure, the ordinary Newton expansion and the splitting  $c = m + p$ ); it credits this report.  $\binom{\mathbf{X}}{2}$  is numerical although  $\frac{1}{2} \notin \mathbf{Oz}$ , which is why “polynomial” alone would be an incomplete arithmetic answer (25).

**Corollary 16.103** (An explicit smooth omnific classification [25]). *A globally  $C^{\infty}$  semialgebraic map  $f : \mathbf{No}^r \rightarrow \mathbf{No}^{r'}$  preserves  $\mathbf{Oz}^r$  if and only if each coordinate has a finite expansion*

$$f_j(\mathbf{X}) = \sum_{\alpha} c_{j,\alpha} \prod_{i=1}^r \binom{X_i}{\alpha_i}, \quad c_{j,\alpha} \in \mathbf{Oz}. \quad (16.35)$$

*Its constant-term action at a lattice input is obtained by applying ct to every coefficient and every input.*

*Proof.* Combine [Theorem 16.102](#) with [Theorems 16.6 and 16.7](#); the last assertion is (16.6), since ct is multiplicative on nonnegative supports.  $\square$

The expansion is an exact algebraic description, not an algorithm on arbitrary symbolic descriptions of surreal numbers: equality, support membership and exact coefficient operations are not assumed decidable (25).

**Convex support fields and absorption.** Tail rigidity says nothing inside a bounded interval; there the relevant object is a convex portion of the exponent group, not a set of values (25).

**Lemma 16.104** (Real closed support fields [25]). *For a divisible ordered subgroup or subclass  $H \subseteq \mathbf{No}$ ,  $K_H$  is a real closed subfield of  $\mathbf{No}$ , and a semialgebraic function defined over  $K_H$  takes an argument in  $K_H$  to a value in  $K_H$  whenever it is defined.*

*Proof.* Hahn operations keep supports in the group generated by the original supports, so  $K_H$  is a field. For a polynomial over  $K_H$  the union of the supports of its coefficients is a set whose divisible hull  $G \subseteq H$  is a set-sized divisible group;  $K_G$  is real closed [23] and embeds in  $\mathbf{No}$  by normal forms, so positive elements have square roots and odd-degree polynomials have roots in  $K_H$ . An inclusion of real closed fields is elementary, so for parameters and input in  $K_H$  the unique output lies in  $K_H$ ; for a class  $H$  this is a pointwise argument inside a set-sized support field containing the finitely many parameters and the input.  $\square$

This is [Proposition 14.1](#) for  $H = H_a$ , and the same set-sized reduction.

**Theorem 16.105** (Exact support-field absorption [25]). *For  $c \in \mathbf{No}$  and a divisible  $H \subseteq \mathbf{No}$ ,*

$$cK_H \subseteq \Pi \iff \text{supp } c \subseteq H^\uparrow. \quad (16.36)$$

*In particular  $\omega^b K_H \subseteq \Pi$  whenever  $b > h$  for every  $h \in H$ . If  $H^\uparrow$  is empty, the only such multiplier is  $c = 0$ .*

*Proof.* If  $cK_H \subseteq \Pi$ , then for each  $h \in H$  the monomial  $\omega^{-h} \in K_H$  translates  $\text{supp } c$  without cancellation, so every  $\gamma \in \text{supp } c$  has  $\gamma - h > 0$ ; hence  $\gamma \in H^\uparrow$ . Conversely, if  $\text{supp } c \subseteq H^\uparrow$  and  $x \in K_H$ , every exponent that can occur in  $cx$  is  $\gamma + \eta$  with  $\gamma \in \text{supp } c$ ,  $\eta \in H$ , and  $-\eta \in H$  gives  $\gamma + \eta > 0$ ; cancellation removes exponents but cannot create a nonpositive one.  $\square$

The sufficiency half contains [Lemma 4.2\(ii\)](#) ( $H = H_a$  and  $c = X^{c'}$  with  $2c' \geq a$ , since  $H_a < a/2$ ), [07's Lemma 14.6](#) ( $H = H_a$  and  $\text{supp } g > a$ ) and the freshness of [Lemma 16.30](#) ( $H$  a set group  $G$  and  $b > G$ ) [merge]. The theorem is a direct support calculation, not a union-of-supports argument over a proper class:  $K_{\text{cvx}(\mathbb{Z}a)}$  contains every omnific input of a bounded box and every semialgebraic value defined over it (25).

**Lemma 16.106** (A bounded omnific window lies in one support field [25]). *If  $x \in \mathbf{Oz}$  and  $|x| \leq T = \omega^a$ , then  $\text{supp } x \subseteq [0, a] \subseteq \text{cvx}(\mathbb{Z}a)$ .*

*Proof.* Omnific supports are nonnegative, and a leading exponent above  $a$  would make  $|x| > \omega^a$  whatever its nonzero real coefficient.  $\square$

**Theorem 16.107** (Bounded-window absorption [25]). *Let  $a \ll b$ ,  $T = \omega^a$ , and let  $\phi : \mathbf{No}^r \rightarrow \mathbf{No}^{r'}$  be semialgebraic with parameters in  $K_{\text{cvx}(\mathbb{Z}a)}$ . Then*

$$\Phi(\mathbf{x}) = \begin{cases} \omega^b \phi(\mathbf{x}), & |x_j| \leq T \text{ for every } j, \\ 0, & \text{otherwise,} \end{cases} \quad (16.37)$$

*is semialgebraic and  $\Phi(\mathbf{Oz}^r) \subseteq \Pi^{r'}$ . The same holds for a semialgebraic partial map, set to 0 outside its domain and the box.*

*Proof.* The defining conditions are semialgebraic. An omnific input in the box has coordinates in  $K_{\text{cvx}(\mathbb{Z}a)}$  ([Lemma 16.106](#)), so the outputs of  $\phi$  lie there ([Lemma 16.104](#)), and  $a \ll b$  means  $b > \text{cvx}(\mathbb{Z}a)$ , so [Theorem 16.105](#) moves them into  $\Pi$ .  $\square$

A multiplier always exists:  $b = \omega a$ , a product in  $\mathbf{No}$ , satisfies  $a \ll b$ , although  $\text{cvx}(\mathbb{Z}a)$  need not be a set; here  $\omega^b$  is a normal-form monomial, not an analytic exponential (25). So no rule forces a semialgebraic lattice preserver to be locally polynomial throughout an infinite bounded

interval: a radical, an algebraic implicit branch or any finite semialgebraic construction survives there after multiplication by a higher scale, and every ordinary constant-term output is 0 although the underlying geometry is nontrivial. The theorem asserts no regularity at the boundary of the box (the bumps below prescribe finite regularity), and it gives no holomorphic cutoff: the surcomplex reading is through real coordinates (25).

### The square-root threshold.

**Theorem 16.108** (Sharp square-root threshold [25]). *Let  $a > 0$ ,  $b \in \mathbf{No}$  and  $T = \omega^a$ . The following are equivalent:*

- (i)  $\omega^b \sqrt{x} \in \mathbf{Oz}$  for every  $x \in \mathbf{Oz}$  with  $0 \leq x \leq T$ ;
- (ii)  $\omega^b \sqrt{T-1} \in \mathbf{Oz}$ ;
- (iii)  $a \ll b$ .

Then all the values in (i) lie in  $\Pi$ .

*Proof.* (i) $\Rightarrow$ (ii) because  $T-1$  is a positive omnific integer, and (iii) $\Rightarrow$ (i), with values in  $\Pi$ , by Lemmas 16.104 and 16.106 and Theorem 16.105. For (ii) $\Rightarrow$ (iii) use the formal binomial expansion

$$\omega^b \sqrt{T-1} = \sum_{j \geq 0} (-1)^j \binom{1/2}{j} \omega^{b+(1/2-j)a}, \quad (16.38)$$

a Hahn series, not a limit in the fine topology: its exponents strictly decrease, its square is  $\omega^{2b}(T-1)$  by multiplying the formal power series, and its leading coefficient is positive. Every coefficient is nonzero. If  $b \leq na$  for an ordinary  $n$ , then  $b + (1/2 - j)a < 0$  for  $j > n + 1$ , and that exponent cannot cancel with another, so the value has negative support.  $\square$

**Corollary 16.109** (Other rational powers [25]). *If  $r_0 \in \mathbb{Q} \setminus \mathbb{Z}_{\geq 0}$  and  $x^{r_0}$  is taken in the positive real closed sense on  $x > 0$ , then  $\omega^b x^{r_0} \in \mathbf{Oz}$  for every  $x \in \mathbf{Oz}$  with  $1 \leq x \leq \omega^a$  if and only if  $a \ll b$ .*

*Proof.* Sufficiency is Theorem 16.105, positive rational powers being algebraic and so in  $K_{\text{cvx}(\mathbb{Z}a)}$ . For necessity expand  $\omega^b (T-1)^{r_0} = \sum_{j \geq 0} (-1)^j \binom{r_0}{j} \omega^{b+(r_0-j)a}$ ; no binomial coefficient vanishes for  $r_0 \notin \mathbb{Z}_{\geq 0}$ , and without  $a \ll b$  an exponent becomes negative for large ordinary  $j$ .  $\square$

**Example 16.110** (An ordinary power of the scale never suffices [25]). For  $a = 1$  no  $\omega^M$  with ordinary  $M$  makes  $\omega^M \sqrt{x}$  omnific for all omnific  $x \in [0, \omega]$ : already  $x = \omega - 1$  leaves a nonzero negative tail.  $\omega^\omega$  works for every such  $x$  at once.

The threshold is an Archimedean-rank statement about the *exponent*:  $\omega^b$  must have an exponent above every ordinary multiple of the exponent of  $T$ , not merely be larger than  $T$  or a few of its powers (25).

*Remark 16.111* (One threshold, three readings [merge]). For  $a > 0$  and  $b \geq 0$  the following are equivalent:  $a \ll b$ ;  $q_a(\omega^b) = 0$  in the binomial quotient  $\mathcal{Q}_a$  (Theorem 14.37, 24);  $\omega^b \sqrt{\omega^a - 1} \in \mathbf{Oz}$  (Theorem 16.108, 25);  $\omega^b K_{\text{cvx}(\mathbb{Z}a)} \subseteq \Pi$  (Theorem 16.105); and  $\omega^b/(1 + \omega^a) \in \mathbf{Oz}$ . All rest on one computation, the positivity of every exponent  $b - ja$  of a binomial expansion in  $\omega^{-a}$ ; 24's ideal  $\Pi^{\gg a}$  ((14.27)) is the set of  $c \in \Pi$  with  $\text{supp } c \subseteq \text{cvx}(\mathbb{Z}a)^\uparrow$ , the multipliers absorbing  $K_{\text{cvx}(\mathbb{Z}a)}$  into  $\Pi$ . The two sources state the threshold independently, 24 as  $b \gg a$  and 25 as “ $b > na$  for every  $n$ ”.

**Counterexamples at every finite order of differentiability.**

**Theorem 16.112** (Finite-smoothness flexibility [25]). *Let  $a \ll b$  and  $T = \omega^a$ . For an ordinary  $k \geq 0$  put*

$$\mathbf{b}_k(x) = \begin{cases} \omega^b [x(T-x)]^k \sqrt{x(T-x)}, & 0 \leq x \leq T, \\ 0, & x < 0 \text{ or } x > T. \end{cases} \quad (16.39)$$

- (i)  $\mathbf{b}_k$  is semialgebraic, globally  $C^k$  and not  $C^{k+1}$ ;
- (ii)  $\mathbf{b}_k(\mathbf{Oz}) \subseteq \Pi$ ;
- (iii) no finite family  $P_1, \dots, P_s \in \mathbf{No}[X]$  has  $\mathbf{b}_k(x) \in \{P_1(x), \dots, P_s(x)\}$  for every  $x \in \mathbf{Oz}$ .

*Proof.* On  $(0, T)$  the function is a positive algebraic branch of

$$Y^2 = \omega^{2b} [X(T-X)]^{2k+1}, \quad Y > 0, \quad (16.40)$$

which with the zero branches outside  $[0, T]$  is a finite semialgebraic description. Near 0 on the right,  $\mathbf{b}_k(x) = \omega^b x^{k+1/2} (T-x)^{k+1/2}$ , the second factor being smooth near 0 with value  $T^{k+1/2} \neq 0$ ; derivatives of order  $j \leq k$  have smallest power  $x^{k+1/2-j}$ , which tends to 0, and the same at  $T$  with  $T-x$ , so all derivatives through order  $k$  match the zero branches (for  $k=0$  this is continuity). The  $k$ th derivative is  $A_0 x^{1/2} + o(x^{1/2})$  with  $A_0 = \omega^b T^{k+1/2} (k + \frac{1}{2})(k - \frac{1}{2}) \cdots \frac{3}{2} \neq 0$  (the empty product being 1 for  $k=0$ ), so its difference quotient at 0 is unbounded as  $x \downarrow 0$  and  $\mathbf{b}_k$  is not  $C^{k+1}$ ; these are order-topological statements, valid with infinite  $T$  and  $\omega^b$  since arbitrarily small surreal radii may be chosen for a prescribed surreal tolerance. For omnific  $x \in [0, T]$ ,  $x$  and  $T-x$  lie in  $K_{\text{cvx}(\mathbb{Z}a)}$ , and so do their product, its square root and its powers; multiplication by  $\omega^b$  moves the value into  $\Pi$  (Theorem 16.105), and outside  $[0, T]$  the value is 0. For (iii),  $P(X)^2 - \omega^{2b} [X(T-X)]^{2k+1} \neq 0$  for every  $P \in \mathbf{No}[X]$ , since the second term has odd multiplicity  $2k+1$  at the distinct roots  $0, T$  and a square has even multiplicities; so  $P$  agrees with  $\mathbf{b}_k$  at finitely many points of  $(0, T)$ , and finitely many such sets cannot cover the ordinary positive integers, which all lie in  $(0, T)$ .  $\square$

With  $a = 1$  and  $b = \omega$  this is 25's overview Theorem 1.2. Assertion (iii) is stronger than failing to be piecewise polynomial on a finite semialgebraic partition: even with arbitrary pieces, finitely many polynomial formulas do not suffice. The family shows that no coefficient-independent finite regularity order forces global polynomiality; it does not contradict Theorem 16.100, because the degree of (16.40) grows with  $k$  (25).

**Corollary 16.113** (Bounds for the least regularity order [25]). *Let  $s(D)$  be the least  $s_0 \geq 0$  such that every globally  $C^{s_0}$  semialgebraic preserver  $\mathbf{No} \rightarrow \mathbf{No}$  of  $\mathbf{Oz}$  with a degree- $D$  graph description is a polynomial. Then  $s(1) = 1$ , and  $1 + \lfloor (D-2)/4 \rfloor \leq s(D) \leq D^2$  for every  $D \geq 2$ .*

*Proof.* The upper bound is Theorem 16.100. For  $D = 1$ ,  $|X|$  is a continuous nonpolynomial preserver with a degree-one description. For  $D \geq 2$  put  $k = \lfloor (D-2)/4 \rfloor$ : the description of  $\mathbf{b}_k$  has total degree at most  $4k+2 \leq D$ , its only high-degree atom being (16.40) and its cutoff and sign conditions linear, so Theorem 16.112 gives  $s(D) > k$ .  $\square$

**Corollary 16.114** (Finite-smoothness shears [25]). *For  $k \geq 1$ ,  $\text{Sh}_k(x, y) = (x, y + \mathbf{b}_k(x))$  is a semialgebraic  $C^k$  diffeomorphism of  $\mathbf{No}^2$  with Jacobian determinant 1 that restricts to a bijection of  $\mathbf{Oz}^2$ , and no finite family of polynomial maps represents it on  $\mathbf{Oz}^2$ .*



*Proof.* The inverse  $(x, y) \mapsto (x, y - \mathbf{b}_k(x))$  has the same regularity and preserves  $\mathbf{Oz}^2$ ; the Jacobian matrix is  $\begin{pmatrix} 1 & 0 \\ \mathbf{b}'_k(x) & 1 \end{pmatrix}$ . A finite polynomial representation, restricted to the line  $y = 0$  and to the second coordinate, would contradict [Theorem 16.112\(iii\)](#).  $\square$

These are maps of the lattice as a set, not ring automorphisms, and the Jacobian statement is a triangular calculation, not a claim about the Jacobian conjecture. Under  $z = x + iy$ ,  $z \mapsto x + i(y + \mathbf{b}_k(x))$  is a real-semialgebraic bijection of  $\mathbf{Oz}[i]$  that is not holomorphic in general (25). They are not polynomial automorphisms, so they do not bear on [Question 18.20\(c\)](#), and 16's and 22's shears ([Example 16.85](#)) are polynomial.

### Surcomplex consequences.

**Theorem 16.115** (Surcomplex smooth classification [25]). *Identify  $\mathbf{No}[i]^r$  with  $\mathbf{No}^{2r}$ . A globally  $C^\infty$  real-semialgebraic map  $\mathbf{No}[i]^r \rightarrow \mathbf{No}[i]^{r'}$  preserves  $\mathbf{Oz}[i]^r$  if and only if every real and imaginary coordinate lies in  $\mathcal{N}_{2r}(\mathbf{Oz})$ , that is, has a finite binomial expansion (16.35) with omnific coefficients in the  $2r$  real coordinates.*

*Proof.* [Theorem 16.102](#) and [Corollary 16.103](#) under  $\mathbf{Oz}[i]^r = \mathbf{Oz}^{2r}$ .  $\square$

The statement concerns the actual surcomplex lattice but is deliberately real-algebraic: it presupposes no global surreal exponential, no global sine and no topology in which all real analytic series converge (25).

**Corollary 16.116** (Holomorphic polynomial rigidity [25]). *If such a map also satisfies the Cauchy–Riemann equations in every complex input coordinate, each coordinate is a polynomial in  $z_1, \dots, z_r$  alone, with coefficients in  $\mathbf{No}[i]$ , and is  $\mathbf{Oz}[i]$ -valued on  $\mathbf{Oz}[i]^r$ .*

*Proof.* By [Theorem 16.115](#) a coordinate is a polynomial in the  $x_j, y_j$ ; the invertible linear substitution  $x_j = (z_j + \bar{z}_j)/2$ ,  $y_j = (z_j - \bar{z}_j)/(2i)$  makes it a polynomial in the  $z_j, \bar{z}_j$ , whose formal  $\bar{z}_j$ -derivatives vanish by the Cauchy–Riemann equations; in characteristic 0 all positive powers of the  $\bar{z}_j$  then have zero coefficients.  $\square$

In one variable the coefficient ring is [Theorem 16.9](#) with  $r = 1$ : the polynomials  $P \in \mathbf{No}[i][Z]$  with  $P(\mathbf{Oz}[i]) \subseteq \mathbf{Oz}[i]$  are  $\text{Int}(\mathbb{Z}[i]) \oplus \Pi_{\mathbb{C}}[Z]$  additively. This is 25's Proposition 9.3, whose proof (interpolation at  $0, \dots, d$  and coefficientwise constant terms) is 10's, and 25 credits it; it is not a Gaussian binomial-basis theorem, ordinary binomial polynomials need not be  $\mathbb{Z}[i]$ -valued on  $\mathbb{Z}[i]$  ([Remark 16.10](#)), and the smooth classification must be combined with this Gaussian coefficient condition (25).

**Sampling and exceptional sets.** For polynomials of known degree finite interpolation gives exact tests; smooth algebraic branches of fixed degree behave differently (25).

**Theorem 16.117** (No coefficient-independent set of quadratic tests [25]). *For every set  $S \subseteq \mathbf{Oz}$  there is  $b > 0$  such that the globally  $C^\infty$  semialgebraic function*

$$\mathbf{r}_b(x) = \omega^b \sqrt{x^2 + 1} \quad (16.41)$$

*satisfies  $\mathbf{r}_b(S) \subseteq \Pi$  but does not preserve  $\mathbf{Oz}$ . Its graph is one branch of an irreducible polynomial equation of total degree 2.*

*Proof.* The supports of the  $\sqrt{s^2 + 1}$ ,  $s \in S$ , form a set; take  $b > 0$  with  $b + \gamma > 0$  for every exponent  $\gamma$  in their union (Lemma 3.1), so  $r_b(S) \subseteq \Pi$ . At the omnific monomial  $x_b = \omega^{b+1}$  the formal binomial expansion gives

$$r_b(x_b) = \omega^{2b+1} + \frac{1}{2}\omega^{-1} - \frac{1}{8}\omega^{-2b-3} + \dots, \quad (16.42)$$

with strictly decreasing exponents since  $b > 0$ ; the coefficient of  $\omega^{-1}$  is  $\frac{1}{2} \neq 0$  and cannot cancel, so  $r_b(x_b) \notin \mathbf{Oz}$ . The graph is the positive branch of  $Y^2 - \omega^{2b}(X^2 + 1) = 0$ , which is irreducible in  $\mathbf{No}[X, Y]$ : a factorization of this monic quadratic in  $Y$  would make  $X^2 + 1$  a square in  $\mathbf{No}(X)$ , contrary to the simple roots  $\pm i$  in  $\mathbf{No}[i]$ . Since  $x^2 + 1 > 0$  everywhere,  $r_b$  is semialgebraic and  $C^\infty$  on  $\mathbf{No}$ .  $\square$

The adversary is severe: a globally smooth branch of an irreducible quadratic graph, whose degree stays 2 while the scale of its coefficient changes with the test set (25). There is no contradiction with Corollary 16.99, which assumes preservation at every lattice point, nor with coefficient-adapted tests: once  $b$  is known, (16.42) gives a bad point. 16's and 22's witnesses for Theorem 16.51,  $\omega^\beta/(X - \frac{1}{2})$  and  $\omega^{\omega b}/(X + t + \frac{1}{2})$  (Remark 16.73), are rational functions of degree one with a pole in  $\mathbf{No}$  (outside  $\Lambda$ );  $r_b$  has no pole, is globally smooth, and has graph degree 2.

**Corollary 16.118** (Removal of set-sized exceptions [25]). *Let  $f : \mathbf{No} \rightarrow \mathbf{No}$  be semialgebraic and globally  $C^\infty$ . If  $\{x \in \mathbf{Oz} : f(x) \notin \mathbf{Oz}\}$  is a set, it is empty and  $f \in \mathcal{N}_1(\mathbf{Oz})$ ; a globally smooth semialgebraic function that fails to preserve  $\mathbf{Oz}$  fails on a proper class of omnific inputs.*

*Proof.* A set of surreals is bounded above, so the exceptions miss a final ray; the one-sided form of Theorem 16.95 gives a numerical polynomial on a positive tail, and Lemma 16.97 extends the equality to  $\mathbf{No}$ .  $\square$

**Corollary 16.119** (Removal of exceptions in several variables [25]). *Let  $f : \mathbf{No}^r \rightarrow \mathbf{No}^{r'}$  be globally  $C^\infty$  and semialgebraic. If the set of bad lattice inputs is a set,  $f$  is a coordinatewise numerical polynomial and has no bad lattice input.*

*Proof.* The coordinate projections of the exceptional set are sets, hence bounded. On each coordinate line with the other coordinates fixed in  $\mathbf{Oz}$ , all sufficiently large positive lattice inputs are good, and the one-sided tail theorem with the identity principle makes each output coordinate on that line a numerical polynomial; evaluating at the chosen coordinate shows that every lattice point is good. Apply Theorem 16.102.  $\square$

The two results are complementary: every fixed set of witnesses is fooled by a new coefficient scale, yet every nonpreserving smooth semialgebraic function has more bad inputs than a set can hold. This is a proper-class quantifier phenomenon, not a statement about running time (25). For rational functions the same alternative is Theorem 16.52 (one variable, 16) and Theorem 16.75 (several, 22).

*Remark 16.120* (Smoothness cannot be dropped [merge]). For every ordinary  $k \geq 0$  the function  $f(x) = \frac{1}{2}(1 - 4x^2)^{k+1}$  on  $|x| \leq \frac{1}{2}$ , and 0 elsewhere, is semialgebraic and globally  $C^k$  (its derivatives through order  $k$  vanish at  $\pm\frac{1}{2}$ ), and its set of bad omnific inputs is  $\{0\}$ , since  $\mathbf{Oz} \cap [-\frac{1}{2}, \frac{1}{2}] = \{0\}$ . So Corollary 16.118 fails for  $C^k$  in place of  $C^\infty$ , and nonempty set-sized exceptional sets occur.

**Set-sized models and the role of rank.** The bounded-window construction is not an artifact of proper classes; the no-set-test theorem, by contrast, uses scales beyond an arbitrary set (25).

**Proposition 16.121** (A set-sized realization [25]). *Let  $\Gamma = \mathbb{Q} \times \mathbb{Q}$  with the lexicographic order,  $a = (0, 1)$ ,  $b = (1, 0)$ ,  $E = \mathbb{R}((X^\Gamma))$  with reverse well-ordered supports and  $X^g$  written  $\omega^g$ , and  $B = \mathbb{Z} \oplus \{x \in E : \text{supp } x \subseteq \Gamma_{>0}\}$ . Then  $E$  is real closed,  $B$  is an integer part of  $E$ , and the functions (16.39) with  $T = \omega^a$  and multiplier  $\omega^b$  satisfy (i)–(iii) of Theorem 16.112 with  $E, B$  in place of  $\text{No}, \text{Oz}$ .*

*Proof.*  $\Gamma$  is divisible, so  $E$  is real closed; truncating positive support and flooring the constant coefficient, with the correction for a negative infinitesimal remainder, gives an integer part. For every ordinary  $n$ ,  $b > na$ , and the convex subgroup generated by  $a$  is  $\{0\} \times \mathbb{Q}$ , which is divisible; the absorption, the endpoint calculation and the square obstruction apply unchanged.  $\square$

**Proposition 16.122** (The rank-one obstruction to the multiplier mechanism [25]). *Let  $\Gamma \neq 0$  be an Archimedean divisible ordered group and  $a \in \Gamma_{>0}$ . In  $\mathbb{R}((X^\Gamma))$  with its normal-form integer part, no monomial  $\omega^b$ ,  $b \in \Gamma$ , sends  $\sqrt{x}$  into the integer part for every integer-part input  $x \in [0, \omega^a]$ .*

*Proof.* For every  $b \in \Gamma$  some ordinary  $n$  has  $b \leq na$ , and the expansion (16.38) at  $x = \omega^a - 1$  has negative support.  $\square$

This limits the uniform monomial multiplier only; it does not classify the semialgebraic preservers in rank one, and nonmonomial multipliers or other input domains need separate analysis (25). It is the analogue of the fixed-workspace remark after Proposition 16.86: in an Archimedean group the hypothesis of the method fails, which is not a proof that nonpolynomial preservers do not exist. The rigidity half, Theorems 16.95 and 16.102, holds in every such Hahn field, since it needs only a cofinal discretely ordered subring of a real closed field; this matches 18's observation that discrete-gap arguments survive in fixed workspaces (Corollary 16.67).

For **No** itself every normal-form support and every formula used is set-sized; the possibly proper-class  $K_{\text{cvx}(\mathbb{Z}a)}$  is used only through its individual elements, finitely many at a time, and real closedness is reduced to the set-sized divisible hull of a finite support union. The absorption theorem is checked on individual Hahn products. No class Zorn lemma, no set of all class ideals and no class-indexed product is used. In Theorem 16.117 there is a set-indexed support union, so an exponent above it exists; in Theorem 16.107 the inputs form a proper class in general, and their exponents are controlled by the explicitly bounded convex group  $\text{cvx}(\mathbb{Z}a)$ . Confusing the two arguments would leave a foundational gap, which is one reason to state the absorption theorem separately (25).

**Dependencies and tempting errors (25).** The argument has two largely independent halves. Rigidity (tails, smooth classification, the  $D^2$  threshold) uses real closed semialgebraic calculus, cofinality and the discrete gap, and needs surreal normal forms only for the final numerical-polynomial reading; it needs neither lattice density nor infinite Taylor series. Flexibility (window absorption, the  $C^k$  examples) uses normal forms, real closedness of support fields and a larger Archimedean exponent scale, not the tail theorem and not a set of all window inputs; the quadratic no-set theorem uses a set support union and one explicit negative term, no undecidability assumption. 25 lists six places where an incorrect argument is tempting.

- (i) Ordinary shifts are not cofinal: (16.32) uses a large element of  $B$  as its step, and moving an infinite negative point into a positive tail by an ordinary integer would fail.

- (ii) Finite differences vanish only after the discrete-gap argument: an estimate tending to zero is not an identity; the difference lies in  $B$  and has modulus below 1 at lattice points, and semialgebraicity then extends the zero set to a ray.
- (iii) The smooth identity principle is not available for arbitrary functions; the reciprocal-growth argument is why a semialgebraic function cannot be infinitely flat where it stops vanishing.
- (iv) The multiplier acts on supports, not on numerical approximations:  $\omega^b K_{\text{cvx}(\mathbb{Z}a)} \subseteq \Pi$  because every shifted exponent is positive, not because  $|\omega^b \phi(x)| > 1$ .
- (v) The radical tail is an exact Hahn object: the proof at  $T - 1$  uses an infinite binomial series with distinct exponents and nonzero coefficients; checking many terms cannot prove the threshold.
- (vi) Set-sized and class-sized families differ: a support union over a sample set is a set, over all inputs of a bounded window it need not be, and the latter case is controlled by  $\text{cvx}(\mathbb{Z}a)$ , not by a union over a proper class.

## 17 Natural ordinal arithmetic is not a quotient of the omnific integers

This section is manuscript 11's. Let  $\mathcal{G}_{\text{nat}}$  be the Grothendieck ring of the ordinals under Hessenberg's natural sum and natural product. Every ordinal has a finite Cantor normal form  $\sum_j n_j \omega^{\gamma_j}$ , each  $\gamma_j$  has its own finite Cantor normal form  $\sum_\alpha m_{j,\alpha} \omega^\alpha$ , and under natural multiplication  $\omega^{\gamma_j} = \prod_\alpha (\omega^{\omega^\alpha})^{m_{j,\alpha}}$ . Uniqueness of these forms gives the free commutative semiring  $\mathbb{N}[T_\alpha : \alpha \in \mathbf{On}]$ , and adjoining additive inverses gives

$$\mathcal{G}_{\text{nat}} \cong \mathbb{Z}[T_\alpha : \alpha \in \mathbf{On}], \quad (17.1)$$

with  $T_\alpha$  represented by  $\omega^{\omega^\alpha}$  under the natural embedding in  $\mathbf{No}$ . Natural operations, not the usual noncommutative ordinal operations, are essential. The description (17.1) appears in the question [18]; the argument above fixes the operations.

**Theorem 17.1** (The ordinal quotient question [11, 13]). *Every unital ring map  $h : \mathbf{Oz} \rightarrow \mathcal{G}_{\text{nat}}$  has image exactly the ordinary integral constants. In particular  $\mathcal{G}_{\text{nat}}$  is not isomorphic to a quotient ring of  $\mathbf{Oz}$ . More generally, no ring map  $\mathbf{Oz} \rightarrow \mathbb{Z}[X]$  with  $X \neq \emptyset$  is surjective.*

*Proof. First proof.*  $\mathbb{Z}[X]$  is residually set-sized by Theorem 5.12 (for any class  $X$  of indeterminates), so  $h$  kills  $\Pi$  and factors through  $\text{ct}$ ; its image is  $\mathbb{Z}$ , a proper subring since  $T_0 \notin \mathbb{Z}$ .

*Second proof, specific to this target.* The additive group of  $\Pi$  is divisible ( $s/n \in \Pi$  for every ordinary  $n \geq 1$ ), while no nonzero element of  $\mathbb{Z}[T_\alpha]$  is divisible by every  $n$  (each of its finitely many integral coefficients would be). So every additive map from  $\Pi$  to  $\mathbb{Z}[X]$  is zero, and a unital ring map from  $\mathbf{Oz} = \mathbb{Z} + \Pi$  has image  $\mathbb{Z}$ .  $\square$

The question was posted by Jesse Elliott as the final addition to MathOverflow question 188430, “Transcendence degree of the surreals over the subfield generated by the ordinals” (30 November 2014) [18]; the displayed answer, by Eric Wofsey, concerns the transcendence degree, a different question. The answer to the quotient question is negative. 11 gives the short additive proof precisely so that the answer is not mistaken for a difficult new factorization theorem: no claim is made that the elementary obstruction was unknown to specialists or absent from every other source. The first proof illustrates the general residual theorem, and the second isolates the weaker mechanism that

suffices for this target. The obstruction is not a cardinal comparison, since both rings may be proper classes; it is their divisibility and representation properties (11).

*Remark 17.2* (An independent answer [13]). Manuscript 13, dated a day after 11 and pinned to a repository commit that precedes 11’s placement, answers the same addendum negatively with the same two proofs: its lemma that every unital map from  $\mathbf{Oz}$  into a ring  $B$  with  $\bigcap_n nB = 0$  factors through  $\text{ct}$  is the second proof above, and its route through the maps  $q_F : \mathcal{G}_{\text{nat}} \rightarrow \mathbb{Z}[T_\alpha : \alpha \in F]$  ( $F$  finite) and the universal theorem is the first (its corollary for rings “separated by set-sized ring targets” is [Theorem 5.12](#)). 13 adds that the inclusion  $\mathcal{G}_{\text{nat}} \subset \mathbf{Oz}$  has no unital ring retraction, which is immediate since every unital map has image  $\mathbb{Z}$ , and derives (17.1) from Cantor normal forms in the same way. 13 records that the question writes the ring as  $K[\Omega]$ , with  $K$  a Grothendieck construction; that notation was not rechecked for this merge. 13 does not claim to answer the original transcendence-degree question, and states that its priority as a first answer is not established.

## 18 Boundaries, non-claims and open questions

### 18.1 What the results do not say

The universal theorem is a statement about homomorphisms into set-sized rings and about set-sized modules. It is not an isomorphism  $\mathbf{Oz} \cong \mathbb{Z}$ ; it does not classify class ideals, class modules, automorphisms or factorizations of  $\mathbf{Oz}$ ; it concerns multiplicative maps and says nothing against additive or valuation-theoretic observations; and it fails verbatim for every fixed set-sized exponent group, whose identity map is a set-sized target. Size alone does not prove it (finite supports have proper-class many monomials and an augmentation), nor does divisibility alone (only targets of positive characteristic are handled). The set-sized thresholds are exact for the five constructions and the two fixed-group examples treated; no classification of exponent groups or support restrictions is claimed, and generator counts follow no general law stated here. The large-quotient results describe specified subalgebras of  $\mathbf{Oz}/(1 + \omega^a)$  and conditional statements about primes; no existence theorem for nonarithmetic class primes is claimed. (Status (batch 34): the eighth-batch addition proves one with global choice, [Theorem 14.34](#); without global choice, and for  $\mathbf{Oz}$  specifically, the question of [Question 18.28](#) remains, and the nonarithmetic primes and maximal ideals are not classified.) The later additions do not change this. The flat and projective dimensions of [Section 13](#) are those of set-sized finite-support cores, of coefficient-lattice modules and of the rings of [Section 12](#); they say nothing about the homological dimensions of the class ring  $\mathbf{Oz}$ , and a flat dimension computed for one subring of  $\mathbf{Oz}$  bounds nothing for another ([Remark 13.16](#)). The finite-support cores are not invisible to set-sized rings, only to finite ones ([Proposition 13.2](#)). The fresh-scale theorems concern polynomials and rational functions with finitely many coefficients and use the whole surreal exponent class, or a fixed Hahn workspace in which the coefficient group generates a proper convex subgroup ([Proposition 16.86](#)). Nor do the third-batch additions: the relation counts and Tor groups of [Section 13.9](#) concern the lattice ideals  $J_g(M)$  and constant matrices at one common scale, not arbitrary finitely generated ideals; the ordinary-output dichotomy is a one-variable statement; and the fibre theorem of [Section 15.8](#) gives no membership test for  $\mathcal{N}$ . The fourth-batch addition determines of  $\text{Idem}(\mathcal{N}/\Pi\mathcal{N})$  that it is nontrivial, atomless and a proper class with independent families of every set size, not its isomorphism type, and its descent theorem is not a decision procedure; its fixed-workspace statements concern reduced fibres and assert neither radicality of  $\Pi(G)\mathcal{N}(G)$  nor atomlessness of  $\text{Idem}(\mathcal{C}(G))$  ([Section 15.9](#); status of batch 32: for cyclic  $G$  the seventh-batch addition decides both, the first negatively and the second positively, [Section 15.10](#)). Nor does the fifth-batch addition: its exact projective dimensions are those of the set-sized cores

$A_d$  and of specified modules over them, not global or weak global dimensions and not those of  $\mathbf{Oz}$ ; its Tor statements over the class ring are read in two universes; and its reconstruction of an order needs the framed boundary data (Section 13.10). Nor does the sixth-batch addition: its certificates are relative to exact arithmetic, its rational bound and its Gaussian grid are not claimed optimal, its birational corollary assumes a rational inverse, and its fixed-workspace criterion says when its method applies, not that finite tests fail otherwise (Section 16.10). Nor does the seventh-batch addition: it concerns one cyclic exponent group, its root refinements and their bounded-denominator union, not arbitrary groups or support classes or the Hahn field over  $\mathbb{Q}a$ ; its residue criterion is a criterion, not an algorithm; of the uncountable Boolean algebras it determines atomlessness, cardinality and order density, not the isomorphism type; and its periodic resolution and infinite dimensions concern the coefficient fibre, not the arithmetic fibre or  $\mathbf{Oz}$  (Section 15.10). Nor do the eighth-batch additions. 24's maximal ideals exist in GBC, not in the base theory of Section 2.1; they are defined relative to a global well-order, not canonically, their membership may be undecidable, and neither the maximal ideals above  $1 + \omega^a$  nor their residue fields are classified: the branching count  $2^{\aleph_0}$  is a lower bound, and the residue fields embed in  $\mathbf{No}[i]$  without being identified with it (Section 14.9). 25 classifies the globally smooth semialgebraic preservers only; finitely smooth and nonsmooth ones are controlled on their tails, not on bounded intervals, and its sampling theorem excludes coefficient-independent tests, not coefficient-adapted certificates (Section 16.11). 26 computes flat dimensions of the geometric radical quotients of set-sized cone cores, not their projective dimensions in general, not arbitrary ideals and not any dimension of  $\mathbf{Oz}$ , and its flat dimensions do not transfer to  $\mathbf{Oz}$ , the passage being nonflat (Section 13.11). 27's henselization and spectra are those of set-sized local rings  $\mathfrak{L}$  of its hulls, not of  $\mathbf{Oz}$ ; it counts the minimal primes and the generic branches but does not classify the Stone space or the Boolean algebra (Section 15.11).

As an internal ring,  $\mathbf{Oz}$  has all surreal scales available through its fraction field, a nonzero proper-class ideal  $\Pi$ , nonzero class-valued derivations and nonarithmetic prime quotients; as a source of representations on sets it carries no more scalar information than  $\mathbb{Z}$ . The gap is caused not by ignoring limits but by finite identities that become available once the source contains the appropriate Hahn sums (06). Adapting 06's list of hypotheses that must remain visible to all the constructions, the proofs use four different size restrictions, each for its own reason: individual supports are sets, so that a common positive gap exists; the full class of smaller scales supplies arbitrarily large collision families; in the set-sized models an uncountable bound admits the countable witnesses, and regularity (or cofinality, or the finite-coordinate invariant) bounds unions of fewer than the relevant number of supports; and cardinal arithmetic enters only in computing the size of the ring, which is the exact threshold. Dropping any of them needs a new argument: finite supports lack the witnesses (Section 9), full supports over a fixed set group need not have gaps (Remark 11.21), and singular support bounds may preserve field closure but not the simultaneous gap property.

## 18.2 Non-claims of the sources

Every limitation stated by a source survives. They are listed and numbered by source; several recur in the text where the corresponding result is proved.

**03.** (1) Not peer reviewed; no Lean declarations. (2) “Developed here” is relative to the inspected material; no certified priority. (3) Neither Conway's general refinement problem nor the general factorization problems for omnific integers are settled. (4) No classification of exponent groups or support restrictions; the cardinal theorem is a realization theorem for its family. (5) cf  $\kappa = \aleph_0$  is excluded. (6) Only the lower bound  $\mu$  on the number of algebra generators of  $\mathcal{N}_\kappa$  is proved.



(7) No complete integrality test for mixed supports. (8) No descent equality  $\mathcal{N} \cap F_\kappa^{03} = \mathcal{N}_\kappa$  (the equality holds by [Corollary 15.59](#), a merge result from 20's support projection). (9) No statement on proper-class-valued valuations. (10) Infinite set-sized images of the normalizations are not classified, and the absence of finite images is a general consequence of containing the real algebraic integers (19 has since classified the domain images, [Theorem 15.46](#); images with zero divisors remain unclassified). (11) No class tensor products and no class prime-ideal theorem; all Tor is over set-sized rings. (12) The finite checks do not verify the cardinal or class arguments. (13) Restricted Hahn fields, truncation integer parts, complete-integral-closure terminology and the small-unit lemma are standard or elementary, with no priority claim; Kuhlmann is context only. (14) The full-class theorem, common divisors and clearing are credited to the companion manuscript (05). (15) Density produces no convergent set-indexed net: every set-generated intermediate algebra is uniformly discrete, so the dense normalization needs more than any set of generators. (16)  $\frac{1}{2}\mathcal{N}[i]$  is not asserted to be a ring. (17) The flatness result is an application of standard module theory, not a new general flatness theorem. (18) Density does not impose continuity on abstract ring homomorphisms. (19) No claim about unrestricted surcomplex analytic continuation is made.

**04.** (1) Unrefereed; no Lean formalization; the universal theorems are candidate contributions without certified priority. (2) No classification of all class ideals, no class Zorn argument, no claim that proper class ideals extend to maximal ones (24 has since proved the extension with global choice, [Theorem 14.24](#); batch 34). (3) The module equivalence covers set-sized modules only, not class modules, group representations of  $\mathrm{GL}_n(\mathbf{Oz})$ , or additive homomorphisms. (4) It does not say that  $P(\mathrm{ct}(x)) = 0$  implies  $P(x) = 0$ . (5) It does not classify infinite points of curves; the earlier project's curve questions are not declared solved. (6) Nothing is inferred about fraction fields of the bounded-support rings. (7) The fixed-workspace results are lower bounds only. (8) No named open conjecture is solved. (9) Additive coefficient maps escape the theorem and are not counterexamples. (10) The telescope has a direct precursor [[23](#), Remark 3.4.4]; the constant term, finite quotients and common shifts come from the prior manuscript (02 of [[47](#)]). (11) The existence of standard part is classical and in the repository's infrastructure. (12) The support-threshold theorem differs from fixed-group completion or omitted-type questions. (13) The finite checks do not prove normal forms, infinite identities, class factorizations or novelty. (14) The quotient, module, derivation and localization results are a connected theorem package deduced from the universal theorems, not unrelated discoveries.

**06.** (1) Unrefereed, not Lean-verified; priority not established by the targeted searches. (2)  $\mathrm{pd}\mathbb{Z}$  is shown only to be at least 2. (3) Residue fields of nonarithmetic primes are not classified; the prime-reservoir proposition is a necessary condition, not an existence theorem. (4) The primality of  $\omega^{\sqrt{2}} + \omega + 1$  is imported; no factorization, refinement, GCD or unique-factorization conjecture is used or claimed. (5) No automorphisms, class ideals or factorization patterns are described. (6) No elementarity claim for  $A_\kappa^{06} \subseteq \mathbf{Oz}$ ; no compactness argument for a proper-class theory. (7) The Euler derivation is not asserted to be the Berarducci–Mantova derivation or compatible with exponentiation. (8) Real closedness of bounded-support Hahn fields is not a novelty claim, and regularity is not claimed necessary for field closure. (9) The idea of a universal set-sized quotient is not new to the repository. (10) The module correspondence is not a Morita equivalence of unrestricted module theories. (11) The invisible prime quotient does not say that a particular maximal class ideal fails to exist. (12) The polynomial-test theorem does not assert that a proper-class coordinate ring defines an ordinary scheme. (13) The finite-congruence observation is elementary and not a strong novelty claim. (14) Numerical tests and a successful build are not proofs; the formalization route was not

executed.

**07.** (1) Conway’s refinement conjecture is neither settled nor used (the Abramov development [1] is cited for its claim and caveats only); no historical priority; no referee; no Lean. (2) The Cantor algebra is a specified subalgebra; neither the idempotents nor the prime ideals of  $\mathcal{Q}_a$  are classified, and  $\mathcal{Q}_a$  is not claimed to be a function ring. (3) The statement on nonarithmetic primes is conditional; no existence is asserted (24 has since proved existence with global choice, [Theorem 14.34](#); batch 34). (4) Surreal self-similarity is credited to Bagayoko–van der Hoeven. (5)  $\Theta_a$  is not claimed to preserve the ordinal embedding, birthdays, simplicity or the exponential;  $\vartheta$  is not additive and  $\Psi$  not multiplicative. (6) Quotient, completion and module classifications are consequences, not separate major discoveries. (7) The small-image theorem does not say that no set-valued invariant sees positive exponents. (8) The module equivalence is not a homological-dimension statement over class modules. (9) The global theorem is not a theorem about small approximations of  $\mathbf{Oz}$ . (10) The finite checks do not verify the transfinite or ambient-support arguments. (11) The CRT isomorphisms at different stages need not be coherent; evaluation at a point of  $\mathbb{Z}_2$  is not claimed to extend to  $\mathcal{Q}_a$  (with global choice it extends to a map onto a larger field, [Remark 14.47](#); batch 34); no topology on  $\mathcal{E}_a$  is used. (12) Class prime ideals are not classified, and no Zorn argument for a proper class of ideals is used (24 has since shown, with global choice and a recursion on elements rather than a Zorn argument, that maximal class ideals exist, [Theorem 14.24](#); they are still not classified; batch 34). (13) No exponential function on surcomplex arguments is assumed.

**08.** (1) Candidate-original, not certified priority, not refereed, no Lean; the broad factorization questions for  $\mathbf{Oz}$  are not addressed. (2)  $\text{Frac}(A_\kappa^{08}) = F_\kappa^{08}$  is not claimed. (3) Residue-field isomorphism types are not classified. (4) No closure under derivations or exponentials, no initial-segment or birthday condition on the embedding. (5) The collision mechanism has published precedents ([23, Proposition 8.2.1]); the constant retraction and finite reductions are not principal novelty claims. (6) Fixed Hahn workspaces lie outside the theorem (an Archimedean group has no small scales at all). (7) The non-integral-closure corollary is not a factorization theorem. (8) Class-sized modules are not classified; the theorem is not an isomorphism with  $\mathbb{Z}$ , and individual monomials can be represented faithfully in small rings. (9) The Hahn derivations  $D_\eta$  are not an exponential-compatible calculus. (10) No maximal-ideal theorem for proper-class rings; no assertion that proper class ideals lie in maximal ones (24 has since proved one with global choice, [Theorem 14.24](#); batch 34). (11) The quantitative criterion gives sufficient conditions only. (12) The finite checks prove neither the class theorem, nor summability, nor real closedness. (13) The primality of  $\omega^{\sqrt{2}} + \omega + 1$  is an imported published theorem.

**09.** (1) Novelty provisional after a targeted comparison; no referee; no proof-assistant verification. (2) No named factorization conjecture solved; primality imported. (3) The number of maximal ideals and the isomorphism types of the residue fields of  $A_{\lambda,D}^{09}$  are not determined. (4) No assertion about the unrestricted (uncountable-support) ring over  $\Gamma_\lambda^{09}$ . (5) No class Zorn argument; the Jacobson statements concern set-sized rings. (6)  $\mathbf{Oz}[i]$  is a subring of the surcomplex field, not the field. (7) The checks do not verify infinite supports, cardinal arithmetic or the existence of maximal ideals. (8) The geometric identity is elementary with precedents; profinite, derivation and flatness statements are consequences; the Gaussian 2-torsion computation is standard. (9) No global surcomplex exponential, sine, integration or topology is used; the arithmetic quotient is not an ordered quotient. (10) The rotation quotient is a conceptual predecessor, not imported. (11) No

new exponential derivation is constructed and no uniqueness of the Berarducci–Mantova derivation is claimed. (12)  $I/I^2 = 0$  is not a finite-presentation or smoothness claim.

**10.** (1) Unrefereed and not machine-checked; targeted review only; no named factorization conjecture solved. (2) For  $r \geq 1$  the  $p$ -adic completion is not a profinite completion. (3) No classification of all class ideals, automorphisms or factorizations. (4) No transfer principle from  $\mathbb{Z}$  to **Oz**. (5)  $\text{Int}(\mathbb{Z}[i]^r)$  is not replaced by a binomial basis. (6) The derivation  $D_h$  is not the Berarducci–Mantova derivation. (7) The universe-relative reading must carry the size restriction in its statement. (8) The rational classification is not a classification of arbitrary functions or of coordinatewise conjugation. (9) The congruence theorem does not make reduction modulo an infinite omnific integer an ordinary finite congruence, and does not classify congruence-preserving class functions. (10) Classical ingredients (Newton interpolation, binomial rings, the least-common-multiple pattern,  $p$ -adic approximation) are credited, not claimed. (11) The grid test assumes exact surreal input; no effective encoding is claimed. (12) The finite checks do not test class functions, support lemmas or priority.

**11.** (1) The main theorem is relative to one fixed universe, not to fragments viewed in a larger universe. (2) No classification of prime or maximal class ideals. (3) No class module of Kähler differentials. (4) The presentation theorem is not quantifier elimination or a reconstruction of class-valued solutions. (5) Sharpness of the coinitality theorem is stated under  $\kappa^{<\kappa} = \kappa$  and  $|K| \leq \kappa$  as explicit extra assumptions (the unconditional value is a merge result, [Proposition 11.23](#)). (6) Necessary closure conditions for intermediate rings are not given. (7) The ordinal answer is not claimed hard or unknown to specialists. (8) Bounded-support coefficient rings and lower-scale geometric division have precedents in [\[23, Section 8\]](#). (9) Targeted novelty assessment only; the finite checks prove no infinite-support or class-size statement. (10) The rotation-group theme is acknowledged, not originated. (11) The only answer to the MathOverflow question concerns transcendence degree. (12) No primality or refinement conjecture is newly settled; the ideals, class-valued representations and factorization theory of quotients invisible to set-sized rings are not determined, and ordinary factorization questions about **Oz** remain outside the conclusions.

**12.** (1) Nothing is claimed about the homological dimensions of the full proper-class ring **Oz**; all rings are set-sized and all embeddings use finite normal forms. (2) The weak global dimension of  $A_d$  is not computed; only the lower bound  $d + 1$  is proved. (3) Projective dimensions are given only as bounds (the exact values in rank one are merge results; 21 has since determined them for every  $d$ , [Corollary 13.47](#) and [Theorem 13.48](#)). (4) Arbitrary non-simplicial cones, arbitrary ideals and all homological localizations of the cores are not classified (26 has since computed the flat dimensions of radical monomial quotients over every rational polyhedral cone, [Section 13.11](#); batch 34). (5) The cores are not integer parts of **No**, not initial in the simplicity order, not preserved by all surreal automorphisms and not Hahn rings; the countable-rank ring still consists of finite sums. (6)  $\hat{A}$  is the profinite completion, not a Hahn completion or a completion in a surreal-valued topology. (7) The  $D + M$  framework (with the pullback literature, for instance [\[42\]](#)), filtered-colimit flatness, coherence criteria and the idempotent-kernel mechanism of formal étaleness are classical and credited; the proposed contributions are the Boolean Tor computation, its dimension consequences, the countable-rank realization and the ordered-cone comparison. (8) No named conjecture is claimed solved; the originality check was targeted, not an exhaustive priority audit. (9) Not refereed and not checked in Lean. (10) The finite audit uses  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  on a five-point grid and proves nothing over  $\mathbb{Z}$  or for all exponents. (11) The Gaussian version uses only the quadratic extension **No** $[i]$ ; no

analytic continuation and no topology on a proper class is used. (12) “Nilpotent” is a test condition on algebras; the cores are domains.

**13.** (1) The theorems are relative to size; nothing is claimed about all maps out of a fixed set-sized Hahn ring, whose identity is a counterexample ( $\mathbb{Z} + X\mathbb{R}[X]$  with evaluation at 1 is 13’s example). (2) No classification of class quotients, class residue objects or class ideals, and no maximal-ideal theorem or Zorn argument for class rings. (3) The ordinal result answers only the quotient addendum; the original transcendence-degree question has Wofsey’s answer; 13’s priority as a first answer is not established. (4) The integer divisibility of the purely infinite part is known (L’Innocente–Mantova); equation-only transfer along a retraction, filtered-colimit flatness and the algebra of idempotent quotient ideals are standard, and the homological and lifting statements are standard consequences, with no étaleness claimed under finite presentation. (5) The fiber theorem (printed in [47]) classifies only nonzero fibers of decomposable polynomials; there is no general classification of fibers, no Thue or Baker finiteness, and no enumeration of the ordinary solutions of its cubic. (6) The cardinal criterion is a necessary condition, not an existence theorem for detecting maps. (7) The statement on Hilbert’s tenth problem is a transfer of the classical theorem, with no complexity claim for Gaussian or number-field problems. (8) No claim about Conway’s refinement conjecture; no refinement, primality or unique-factorization property is used. (9) In the two-universe reading “not generated by a set” means not generated by a family from the smaller universe. (10) Unrefereed; no Lean; the finite checks verify identities only; the literature and repository searches were bounded, and the repository was neither audited nor built.

**14.** (1) Arbitrary finitely generated omnific ideals, with generators at several support levels, are not classified. (2) The rank-one modules with double dual  $B$  are not all listed. (3) No coefficient ring other than a principal ideal domain is treated. (4) The exact weak global dimension of  $A_0$  is not claimed. (5) The ideal-class group under  $\odot$  is not a subgroup of  $\text{Pic}(A)$ , and its identity is  $L(\mathfrak{o})$ , not  $A$ . (6) Full-class tensor products are explicit carriers only, and ordinary flat dimensions are asserted only over  $A_0$ . (7) No novelty is attached to noncoherence; the  $D + M$  context is credited to Dobbs–Papick, and no valuation-ring hypothesis is used. (8) The number field spanned by a multiplier ring need not be totally real. (9) The unimodular-direction criterion concerns coordinate representatives, not all points of a projective scheme over a ring, and its example  $[1 : \sqrt{2}]$  is not new. (10) The matrix theorem concerns constant matrices at one common monomial scale. (11) Constant-term arithmetic, flat idempotent support ideals and cardinal obstructions are credited to companion manuscripts and are not presented as new. (12) No Conway factorization or refinement claim; no peer review and no new Lean verification; priority is provisional; the finite checks are supplementary and certify no arbitrary support, class universal property, flatness statement or Eisenstein criterion in all degrees.

**15.** (1) The main theorem is for the full class only; the finite-support ring and fixed Hahn workspaces are counterexamples. (2) Its local cardinal criterion is sufficient only. (3) No classification of class prime or maximal ideals; its prime classification carries a set-sized residue hypothesis. (4) The primality of  $\omega^{\sqrt{2}} + \omega + 1$  is imported from L’Innocente–Mantova and is not a new solution of Gonshor’s question. (5) No claim about Conway’s refinement problem; the repository `gaearon/conway-refinement` was noticed but neither built, audited nor relied on. (6) Only set-sized modules are classified; this is not a Morita statement. (7) Universe parameters must stay separate. (8) Geometric factorization and the congruence calculations are classical or elementary, the categorical deductions are not new techniques, and the rotation-group theorem is a precedent,

not a dependency. (9) Blocking one extension of the finite-support augmentation is weaker than the theorem. (10) Embeddings of set-sized rings into  $\mathbf{Oz}$  are not excluded, and  $\mathbf{Oz}$  and  $\mathbb{Z}$  are not claimed to have the same first-order theory. (11) Candidate originality rests on a bounded search; no referee; no Lean; the checks cover finite identities only.

**16.** (1) Rational-to-polynomial collapse is not new in principle (D-rings: Gunji–McQuillan, as recalled by Liu; related framework in [10]). (2) Surjectivity rigidity covers one variable and rational functions only; conjugation, multivariate triangular bijections, class functions, definable maps and formal power series are excluded, and not every bijection of  $\mathbf{Oz}$  is claimed affine. (3) The fresh-scale arguments do not transfer to a fixed Hahn field without dominant exponents. (4) The certificates are relative to exact coefficient data, zero tests and a certificate of freshness; they are not a decision procedure for arbitrary surreal inputs. (5) No convergence, analytic continuation, surreal exponential, Berarducci–Mantova derivation, fine-topological completeness or factorization is used. (6) Newton interpolation and binomial bases are classical, and the Gaussian test is not claimed minimal. (7) The fiber theorem classifies the particular fresh fiber, not every fiber. (8) Priority rests on a bounded search; no referee; no Lean; the finite checks are not proofs.

**17.** (1) No Lean formalization and no peer review; no named factorization conjecture is resolved, and no factorization property is inferred from noncoherence. (2) The kernel theorem classifies constant matrices after one common monomial scaling, not matrices with several leading scales or positive tails; ideals not of the form  $J_g(M)$  are not classified. (3) No upper bound on the weak global dimension of  $R_\kappa^{11}$ , no global dimension, and no projective dimension of  $\Pi_\kappa^{11}$  or  $J_g(M)$  is claimed (the last two are asked); Dobbs’s valuation-pullback theorem does not apply because  $K + \Pi_\kappa^{11}$  is not a valuation ring. (4) The power-growth law counts generators only; the lattices  $M^n$  need not stabilize when their ranks do, and it is not a reconstruction theorem. (5) Over the class ring, Hom means scalar codes, not a class of class maps; ordinary Tor, flat resolutions and weak dimensions are taken over set-sized models only, and no full-class Tor is inferred from the statement on modules killed by the tail. (6) Neither inversion closure of  $R_\kappa^{11}$  nor its equality with a Hahn field is asserted; nothing is said about unrestricted Hahn series over an arbitrary divisible group, and the coinitality of the positive cone is not the cofinality of the group. (7) The appendix proof of the universal theorem is background, not a novelty candidate. (8) The  $D + M$  noncoherence mechanism and the relation module  $M^{r-1}$  are classical (Dobbs, Dobbs–Papick); the pullback literature may contain antecedents of the Hom and tensor formulas; the audit of the repository and of the literature was targeted, not exhaustive. (9) The rank algorithm assumes an exact description of the coefficient span, not floating-point data. (10) The checks are finite arithmetic; kernel bases computed over  $\mathbb{Q}$  are not integral lattice bases, and no proper-class argument, regularity statement, Hahn support or homological theorem is checked. (11) A field  $\text{Frac}(D)M$  in the field test does not make  $M$  a ring. (12) The support-gap argument does not transfer to unrestricted Hahn series with real exponents.

**18.** (1) The dichotomy is one-variable:  $XY$  defeats any bound in several variables, only the linewise statement is proved, and how the exceptional cosets on different lines fit together is open. (2) The coefficient test is an algebraic certificate relative to exact coefficient operations, not an algorithm for arbitrary surreal representations. (3) Rational collapse is not new (D-rings: Gunji–McQuillan, recalled by Liu); the description of the integer-valued polynomials and the matrix and jet reductions are applications, and the all-size matrix test is a general algebraic observation. (4) Finite testing concerns polynomials of known degree only; no finite sample tests rational functions, and entire Hahn series and nonstandard degrees are excluded (for entire series see the batch-32 status note



after [Corollary 16.67](#)). (5) The Gaussian simplex test is not claimed minimal; the results of Volkov and Petrov are classical input. (6) Surjections are classified among rational maps of one variable only, not arbitrary bijections, automorphisms of the surreal field or polynomial automorphisms in several variables; injectivity alone is weaker. (7) Nothing is claimed about  $\mathbf{No}$  or the finite surreals, where the discrete gap fails, and no factorization conjecture is addressed. (8) The checks cover finite-support models  $\mathbb{Q}[H]$  and  $\mathbb{Q}(i)[H]$  only and prove neither the cluster lemma, nor the proper-class reading, nor novelty. (9) Novelty rests on a bounded search; no referee; no Lean. (10) The sharp examples need an infinite element, and for a general discretely ordered ring finite testability of preservation must not be assumed.

**19.** (1)  $\text{Idem}(\mathbf{B}_{\mathbb{R}})$  is not determined: its size, its atoms, whether it is trivial and whether it is a set (20 has since shown that it is nontrivial, atomless and a proper class, [Theorem 15.73](#); its isomorphism type is still not determined). (2) There is no membership criterion for  $\mathcal{N}$  by normal forms (20 adds support descent and the one-scale rational case, [Theorems 15.57](#) and [15.60](#), but no general criterion). (3)  $\mathbf{B}_{\bullet}$  is neither one finite product nor an unrestricted product of copies of  $\overline{\mathbb{Z}}$ . (4) The real-case coefficient embedding depends on the choice of  $v$ ; no embedding of  $\mathbb{C}$  into  $\mathbf{No}$  is asserted. (5) No Bézout property of  $\mathbf{Oz}$  or of the normalizations is claimed. (6) The characteristic- $p$  fibres are neither reduced nor fields; the mod-2 splitting is a product of nonreduced rings. (7) Set-sized images with zero divisors are classified only up to the factorization through  $\mathbf{B}_{\bullet}$ ; not all set-sized images are claimed countable;  $\mathbf{B}_{\bullet}$  is not asserted to be a set (20 has since proved it a proper class), and no universal object in a category of set-sized rings is claimed. (8) The kernel detection uses an explicit foundation (class theory with global choice, no class Zorn lemma, no measurable cardinals), and its intersections are pointwise kernel characterizations, not sets of class maps. (9) In the finite-axiom form,  $v_T$  need not be infinitesimal, non-set-generation needs the full class, the small-image theorem needs a cardinal condition absent from the axioms, and nothing transfers to a fixed Hahn field, where the support  $\{1/n\}$  blocks the monomial proof without being a counterexample (20's reduced-fibre result, [Proposition 15.76](#), avoids the axioms and decides no radicality; 23 has since shown that radicality fails in the cyclic workspace, which is outside the axioms, [Remark 15.119](#)). (10) Integrality, absolute integral closure, Boolean powers and the Bézout property of  $\overline{\mathbb{Z}}$  are standard, and the small-image theorem is credited; priority is not established. (11)  $\mathcal{N}$  itself is not absolutely integrally closed, lifted roots need not stay distinct modulo  $\Pi\mathcal{N}$ , and norm-one units of the fibre are not asserted to lift. (12) No named factorization conjecture is resolved; no referee; no Lean; the checks are finite identities and rational inequalities.

**20.** (1) The class isomorphism type of  $\text{Idem}(\mathbf{B}_{\mathbb{R}})$  is not determined: atomlessness and free subalgebras of every set size do not make the class algebra free, complete or saturated for arbitrary Boolean types, and give no arbitrary infinite joins; no isomorphism  $\mathbf{B}_{\mathbb{R}} \cong \mathbf{B}_{\mathbb{R}} \times \mathbf{B}_{\mathbb{R}}$  is claimed (the proved doubling is  $\mathbf{B}_{\mathbb{C}} \cong \mathbf{B}_{\mathbb{R}} \times \mathbf{B}_{\mathbb{R}}$ ). (2) The fixed-workspace statements use  $\sqrt{\Pi(G)\mathcal{N}(G)}$ ;  $\Pi(G)\mathcal{N}(G)$  is not asserted radical in any fixed workspace, no atomlessness of  $\text{Idem}(\mathbf{C}(G))$  for a fixed  $G$  is asserted, and the rank extension needs a new dominant coordinate, which a fixed workspace need not supply (23 has since decided both for every cyclic  $G$ , without a new coordinate:  $\Pi(G)\mathcal{N}(G)$  is not radical and  $\text{Idem}(\mathbf{C}(G))$  is atomless, [Theorem 15.86](#) and [Corollary 15.110](#)). (3) Support descent is not a decision algorithm; effective membership for general infinite supports is open, and the exact rational criterion concerns one monomial scale. (4) The matrix corollary computes no projective or global dimension and forms no category of class-sized modules; the higher projective dimensions of this report ([Question 18.3](#)) are not addressed. (5)  $\sigma_{\pm}$  are specializations of a finite branch algebra, not evaluations of Laurent series at  $T = 0$  or maps on the Laurent field;  $\text{pr}_G$  is linear, not a ring



retraction; no evaluation map on all surreal numbers is used. (6) An idempotent residue does not say that its representative is idempotent; the quotient is not standard part. (7) The free algebra over an infinite index set is a directed union of finite products, not an unrestricted product. (8) The foundation is class theory with global choice; no set of all class homomorphisms or prime class ideals is formed, no set-sized Zorn argument is applied to a proper class, set-indexed families of class maps are single uniform class relations, and statements about all maps are pointwise. (9) The coefficient copy of  $\overline{\mathbb{Z}}$  is not canonical. (10) Radicality, saturation, the Boolean power, integral doubling and domain-map detection are 19's and the small-image theorem is this report's, reproved and not claimed; general Boolean powers, multiquadratic extensions, the determinant trick, coefficient projection as a module map and ultrafilter methods are not new; it is not asserted that 19 was in the repository at 20's pin. (11) The questions resolved are those of 19, not a named conjecture; no Gonsior prime-element, refinement or other named factorization problem is solved. (12) The source comparison was targeted; historical priority is not established. (13) The finite checks verify polynomial identities, branch specializations, finite Boolean tables, finite support projections and initial Laurent coefficients, not class or set arguments, Hahn support theorems, arbitrary supports, integrality of arbitrary surreals or novelty. (14) No Lean formalization, no repository build, no peer review; Python and SymPy versions other than those recorded (3.13.5 and 1.14.0) were not tested.

**21.** (1) Unrefereed and AI-assisted; no Lean or other proof-assistant check; priority is not certified: the dimension question is answered relative to the inspected repository version, the literature search was targeted, and unintegrated companion manuscripts and historical sources were not exhaustively audited. (2) No major named factorization conjecture is claimed solved. (3) Arbitrary omnific ideals are not classified. (4) The value  $d + 1$  concerns the specific set-sized core  $A_d$ , not  $\text{pd } \mathbb{Z}$  over the class ring; no exact global or weak global dimension of  $\mathbf{Oz}$  or of  $A_d$  is claimed by 21; the stronger core lower bounds  $d + 2$  and  $d + 1$  use  $K \neq \text{Frac } D$ . (5) The class rings are read under a two-universe convention, not as literal proper-class rings of ordinary set theory; the core theorems and all examples are ordinary set-sized mathematics. (6) The finite checks do not prove the infinite Ext nonvanishing, exactness in unbounded degree or the normal-form foundations, and replace no proof assistant. (7) Filtered colimits of flat modules, flat models of derived tensor products, countable telescopes (Positselski [28]) and the non-Noetherian dimension methods for root-adjointing rings (Asgharzadeh [2]) are standard and not claimed. (8) The additive second-Tor formula is a dimension-shift consequence of the report's tensor calculation, not an independent discovery; the flat face resolution,  $\text{fd}_{A_d} D = d$ , the lattice syzygies and the tensor torsion are the report's and are credited. (9) The  $D$ -module decomposition of  $\text{Tor}_1$  is not an  $A$ -module splitting. (10) The vanishing of  $\text{Tor}_2$  is not Tor-independence ( $\text{Tor}_1 \neq 0$ ). (11) The disjointness test creates no field extensions and does not decide linear disjointness without access to the arithmetic. (12) Reconstruction needs the boundary embedding and the distinguished 1; an abstract  $\text{Tor}_2$  group, or an abstract derived equivalence, is not claimed to recover an order. (13) The DGA model is not claimed cofibrant or  $E_\infty$ -formal, and it determines no higher operations. (14) The minimal polynomial of an arbitrary element of degree at least three cannot be read from a two-generator  $\text{Tor}_2$ . (15)  $\text{Tor}_{>0}(D, E) = 0$  for every module  $E$  killed by the tail, despite  $\text{pd } D = d + 1$ ; the nonvanishing uses the target  $A_d$ . (16) On the finite-support core the augmentation to the coefficient field exists, so "every set-sized target factors through  $\text{ct}$ " is false there; the shadow theorem concerns the specified constant-term modules. (17) The resolutions control the specified face, lattice and quotient modules, not all ideals; the length-two flat resolution and the Tor calculations are asserted for  $T_{\mathbb{Q}}$  and the class ring, not for  $A_d$  with  $d \geq 2$ , whose tail is not flat.

**22.** (1) AI-assisted and unrefereed, with complete written arguments; no Lean build, no Wolfram computation and no independent referee review. (2) No named longstanding conjecture is claimed settled; the finite-certificate and no-universal-set package is a proposed contribution, its priority is not certified, and its novelty statement is a qualified assessment of selected material (and is incomplete, [Section 1.4](#)). (3) The bare collapse of integer-valued rational functions of one variable has the D-ring antecedent (Gunji–McQuillan, through Liu’s Proposition 1.12, the 1976 paper not read independently; the domain must be infinite), and the Newton basis and the binomial-ring viewpoint are classical (Elliott [17], Cahen–Chabert [8], the monograph not re-read in full); neither is claimed. (4) Normal forms, Hahn operations and  $\text{Frac}(\mathbf{Oz}) = \mathbf{No}$  (L’Innocente–Mantova, Proposition 2.4.5) and the repository’s splitting, constant-term and fraction background are inherited; no unreviewed geometric assertion of the repository is imported. (5) The certificates are relative to exact field operations and exact membership in  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$ , not algorithms for arbitrary descriptions of surreal numbers; equality, gcd, support bounds and membership are not presumed decidable, and choosing  $b$  above an arbitrary support is a construction, not an effective operation. (6) Neither the rational bound  $(d+1)^m + m(2d^2 + 2d + 1)^{m-1}$  nor the Gaussian grid is claimed optimal, and the anisotropic refinement is available only after the presentation has been inspected. (7) The lower bound  $(d+1)^m$  concerns fixed point-value tests chosen independently of the coefficients, not algorithms that inspect the coefficients. (8) Point counts count exact evaluations, not bit operations or floating-point samples. (9) The no-set-test theorem needs the full proper class and fails for set-sized rings; the fixed-workspace extension needs an available fresh exponent, and its failure for Archimedean value groups is a failure of the method, not a proof that every finite test fails. (10) Arbitrary functions, entire Hahn series, surreal exponentials, infinite-dimensional operators and Diophantine varieties are not classified (entire series preserving the omnific part of a fixed workspace are classified in [55]; batch-32 status note after [Corollary 16.67](#)), nor are algebraic or semialgebraic maps ( $|X|$  preserves  $\mathbf{Oz}$  and is only piecewise polynomial; 25 has since classified the globally  $C^\infty$  semialgebraic ones, [Theorem 16.102](#), and shown that finitely smooth ones need not be finitely piecewise polynomial, [Theorem 16.112](#); batch 34). (11) The birational corollary assumes a rational inverse in both directions; no bijective polynomial map is claimed to have one, no Jacobian-conjecture claim is made, the coordinates need not have coefficients in  $\Lambda$ , and the automorphism group is not classified. (12) Image ideals are not assumed principal, and no greatest-common-divisor conclusion is drawn. (13) The tensor isomorphisms are read through finite sums of pure tensors or in a containing universe; they do not assert that  $\mathbf{Oz}$  is a set. (14) Gaussian arithmetic is not replaced by a Gaussian binomial basis; the ordinary Gaussian universal-set problem (Byszewski–Frączyk–Szumowicz [7], of which only the abstract and the bibliographic record were inspected; Volkov–Petrov [41]) is neither solved nor presented as newly discovered. (15)  $\omega^\gamma$  is the normal-form monomial, not an analytic exponential; no algebraic closedness of  $\mathbf{No}[i]$ , differentiation or topology is used; ct is not a global standard part and is multiplicative only on nonnegative supports. (16) The 1,776 finite assertions (exact algebra and sampled lexicographic inequalities) verify no arbitrary support, infinite Hahn summability, cut above an arbitrary set, class-sized quantifier, proper-class alternative or originality; the lexicographic model illustrates the quantification over all exponents and does not replace it. (17) Only selected repository files were read, some responses were truncated, the repository was neither checked out nor built, and an empty search was not taken as evidence of absence. (18) Independent scrutiny may find simplifications, broader antecedents or errors.

**23.** (1) The answers concern the cyclic exponent groups  $\mathbb{Z}a$  only, for the real pair and its Gaussian counterpart; they classify neither arbitrary ordered exponent groups nor support restrictions, and the proof does not classify the groups with a discrete convex component. (2) The Hahn field over  $\mathbb{Q}a$

is open; the Puiseux result concerns bounded denominators only, is not an identification with that Hahn field, and does not say that the limit fibre is isomorphic to a reduced finite-stage fibre, whose image in it may grow. (3) The Boolean isomorphism types for  $k = \mathbb{R}$  and  $\mathbb{C}$  are not determined: the independent family does not make the algebra free, and neither completeness, nor infinite joins, nor a product representation of the ring, nor a free presentation follows from the count of  $2^{2^{\aleph_0}}$  branches. (4) The weak global and global dimensions of the arithmetic fibre are open; the periodic resolution is exact only over the coefficient fibre, and no change-of-rings statement is made for the arithmetic fibre or for **Oz**. (5) The residue criterion is a finite algebraic criterion for represented algebraic Laurent elements, not a decision procedure for arbitrary real coefficients or arbitrary surreal normal forms; substituting  $T = 0$  in a singular equation need not separate branches. (6) The integer part of the Laurent field does not extend to the normalization; no single refinement removes all nilpotents; the ramification filtration is not a valuation. (7) The finite checks (18,040 assertions in 12 categories, most of them clipped-order compositions and threshold checks) count assertions, not theorems; they establish neither integral closure, nor the existence of finite places or prime extensions, nor arbitrary-support summability, nor the normal-form embedding, nor cardinalities, nor exactness of an infinite resolution. (8) AI-assisted and unrefereed; no Lean or other proof-assistant check, no repository build and no independent review; a compiled PDF is a document check. (9) Priority is not certified: the literature comparison was targeted, and the repository comparison covered selected sections of this report, not all reports or the history; nothing claims to settle a named factorization conjecture, uniqueness of surreal factorization, a global surcomplex analytic calculus or the general theory of surreal supports. (10) Integral dependence, normalization of curves, valuation domains, quadratic square classes and periodic resolutions are classical; the residue criterion is a proved pullback-style tool, not a claim to have invented pullback normalization; the period-two mechanism has antecedents in matrix factorizations; the Newton–Puisseux theorem, the lifting of idempotents across nil ideals and the uniqueness of the countable atomless Boolean algebra are classical and not claimed. (11) No continuum hypothesis and no regularity of  $2^{\aleph_0}$  are used; all objects are sets, with ordinary choice and no class Zorn lemma, class module category or measurable cardinal; the repository’s descent theorem is compared, not used, and no unrefereed repository theorem is an input. (12) The questions concern precisely specified rings and do not assert that the cyclic calculation settles an unrestricted Hahn problem; divisibility of the exponent group alone is not a proof of radicality. (13) The formalization plan claims no implemented declaration and no repository build.

**24.** (1) AI-assisted, unrefereed research draft with written proofs; no Lean or other proof-assistant check and no Wolfram computation; historical priority is not certified; the generic equivalence of global choice with the class maximal-ideal principle may be familiar folklore, and the monomial and support-prime statements may have antecedents in the generalized-series or ordered-monoid literature. (2) [Theorem 14.30](#) gives no lower bound for **Oz** itself: it does not show that a nonarithmetic maximal ideal of **Oz**, or one above every  $1 + \omega^a$ , implies global choice; its converse uses arbitrary specially built class rings, and the maximal-ideal theorem for set rings yields only choice for sets. (3) The class recursion is a class definition, not an effective algorithm; the finite-equation test quantifies over all ring elements and may be undecidable, and no algorithm for ideal membership or decision procedure for membership in a chosen maximal ideal is given. (4) No canonical or choice-free maximal ideal is constructed; uniformity is relative to a chosen global well-order, not parameter-free canonicity. (5) The continuum branching is a lower bound within one fibre, not a spectrum: not every maximal ideal of the fibre is indexed by a branch, and  $\mathcal{Q}_a^k$  is not claimed to be a ring of functions on Cantor space. (6) No cardinality is asserted for the collection of all class ideals and no class-sized Zariski

space is formed; families of class ideals are single class relations on parameters and ring elements. (7) Containing an algebraically closed class subfield does not make a residue field algebraically closed; the residue fields are not claimed isomorphic to  $\mathbf{No}[i]$ , and their embeddings in  $\mathbf{No}[i]$  need not be surjective. (8) The maps to  $\mathbf{No}[i]$  need not preserve order, strong summability, arbitrary infinite sums, exponentiation or any analytic structure; no analytic extension of complex functions to all surcomplex arguments is claimed, and the algebraic closedness of  $\mathbf{No}[i]$  is background, not reproved. (9) “Residually surcomplex” means a separating family of homomorphisms: not residual finiteness, not a representation in a set-indexed product, not a single injective homomorphism with maximal kernel; no product indexed by a proper class of class fields is used. (10) [Theorem 14.57](#) needs both term-closure and strong sum-closure; it is not a classification of all prime ideals and does not say that every omnific prime is a monomial prime. (11) No factorization conjecture is used or addressed (refinement, unique factorization of generalized series); the existence and cutoff theorems classify neither all prime ideals, nor all maximal ideals of a binomial fibre, nor all residue-field extensions of the embedded Hahn core. (12) The finite checks (133,770 assertions in six groups: greedy ideals in  $\mathbb{Z}/n\mathbb{Z}$  and in products of  $\mathbb{F}_2$ , finite geometric remainders, dyadic algebra over  $\mathbb{F}_{257}$ , finite branch filters, lexicographic scale examples) certify no class recursion, infinite Hahn sum, arbitrary surreal support, global choice or priority. (13) The field cores, the dyadic product algebras and the Cantor-type subalgebra are this report’s (07’s), reproved and credited, not claimed; the greedy Krull method, the compatibility mechanism of Hodges, Ern  and Entin (checked through Entin’s corrected version, not against every line of Ern ’s paper or Hodges’s), the characteristic-zero Hahn closure (Mac Lane through Knight–Lange, Mac Lane’s paper not fully inspected), the rescaling automorphism, the universality and back-and-forth method and  $\text{Jac}(A) = 0$  are credited or called elementary; Conway and Gonshor are cited as standard, not reread. (14) Only the comparison with set-sized targets uses the unrefereed universal theorem; the constructions and separation proofs do not. (15) The rescaling [Proposition 14.43](#) shows that the cutoff is an invariant of the embedded fibre at a chosen scale, not that the abstract fibres are nonisomorphic, and the ordinal family changes the scale, so it gives no larger same-fibre count. (16) The repository comparison covered the pinned report and selected guides, not every report or unintegrated companion, and absence from the excerpts was not taken as absence; the literature search was targeted; the delivered Wikipedia link served for orientation only. (17) The eleven hypotheses 24’s audit calls critical are kept: GB has no choice and the unrestricted principle is not the restricted omnific one; ring elements are sets and cosets are coded by least representatives or Scott codes; a stage stores a set of generators or a set partial embedding, never a proper-class ideal; ideal membership and rejection witnesses are finite sums; every Hahn series has a set support and the geometric cancellation rests on summability, not convergence;  $b > na$  must hold for *every* ordinary  $n$ ; closure over a class exponent group is reduced polynomial by polynomial to set groups; a field core is not algebraic closedness; the support-prime classification assumes both closure properties; the maps to  $\mathbf{No}[i]$  need not preserve order, summability, infinite sums, exponentiation or analytic structure; uniform families are relations, and no class of all class ideals and no proper-class-indexed product is formed. (18) Its audit’s record of what was not performed is kept: the proofs had internal review only, not independent peer review; the finite checks ran under Python 3.13.5; no Lean, proof-assistant or Wolfram verification was performed; the PDF was built with pdfLaTeX and inspected visually. (19) The eight plausible but incorrect arguments of 24’s Section 11.2 are printed at the end of [Section 14.9](#). (20) Its questions are not claimed resolved.

**25.** (1) AI-assisted and unrefereed, with complete written arguments; no Lean formalization or build, no Wolfram computation and no independent proof review; novelty is proposed relative to

the inspected sources, priority is not certified, and no named longstanding conjecture (factorization, birthdays, surreal exponentiation) is claimed settled. (2) The repository was read selectively through a connector whose long responses were sometimes truncated, not checked out or built; newly delivered and unintegrated manuscripts were not exhaustively audited, an empty search was not taken as evidence of absence, and it is not asserted that no equivalent statement occurs in them or in the wider literature. (3) The finite-difference strategy, real polynomial rigidity, finite-order detection of smoothness and the nonflatness of tame functions are classical (Jones–Thomas–Wilkie, Bhardwaj–McCulloch–Ramachandran–Woo, Miller, Bochnak–Coste–Roy) and not claimed; the Bhardwaj et al. preprint is cited in its inspected version v1, without transferring its numerical growth constants; Miller’s paper is a bibliographic and contextual antecedent, with limited full-text access, whose specific argument is reproved; the Bochnak–Coste–Roy monograph was not reread in full. (4) The bound  $D^2$  is sufficient, not claimed optimal, and concerns scalar graph descriptions; a degree bound for a full vector graph must first be converted by output elimination. (5) The graph degree is that of a quantifier-free description; a bound for formulas with unresolved quantifiers is a different notion. (6) The mean-value and Rolle theorems are used only for semialgebraic functions at finite order; nothing is asserted about unconstrained surreal-valued calculus. (7) The numerical-polynomial normal form is exact algebra, not an algorithm on symbolic descriptions: equality, support membership and exact coefficient operations are not assumed decidable; the Newton basis and the decompositions are inherited from this report. (8)  $ct$  is multiplicative only on nonnegative supports, not on **No**. (9) The bounded-window theorem asserts no regularity at the boundary of the box and supplies no holomorphic cutoff. (10) The shears are maps of the lattice as a set, not ring automorphisms; the Jacobian statement is a triangular calculation, not a claim about the Jacobian conjecture; the surcomplex shears are not holomorphic in general. (11) The surcomplex results are real-algebraic; no global surreal exponential or sine and no topology in which all real analytic series converge is presupposed. (12) The Gaussian split is not a Gaussian binomial-basis theorem; ordinary binomial polynomials need not be  $\mathbb{Z}[i]$ -valued. (13) The rank-one obstruction concerns the uniform monomial multiplier only, not the classification of semialgebraic preservers in rank one; nonmonomial multipliers and restricted input domains need separate analysis. (14)  $K_{cvx}(\mathbb{Z}a)$  is used pointwise, through set-sized divisible hulls; no union of supports over a proper class, no class Zorn lemma, no set of all class ideals and no class-indexed product is used, and “bounded” in the two-end quotient is order-theoretic, with no compactness. (15) Nothing is classified for arbitrary class functions, all finite-smoothness preservers, arbitrary omnific ideals, all polynomial automorphisms in several variables or holomorphic functions in an unrestricted surreal calculus; no physical prediction is made. (16) The no-set-test theorem excludes coefficient-independent tests only, not coefficient-adapted certificates or algorithms; it is a proper-class quantifier phenomenon, not a running-time statement. (17) The finite checks (1,578 exact assertions in 18 groups) verify finite identities, finite binomial jets, finite Newton grids and sampled lexicographic comparisons, not infinite supports, class quantifiers, general semialgebraic calculus, the complete theorems or novelty; the PDF build is a document check, and the PowerShell build script was not executed. (18) Some ingredients are elementary consequences of classical real algebraic geometry; independent review may find stronger antecedents, a shorter formulation or errors. (19) The integer-part formula of **Oz** needs the correction for a negative infinitesimal remainder, and no argument drops it.

**26.** (1) AI-assisted and proof-bearing, not refereed and not checked in Lean or by a machine; the arbitrary-cone formula and the detector construction are proposed as original, but the searches did not establish priority, and no named general conjecture about **No** is claimed solved. (2)  $\text{pd}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}) = h + 1$  is not claimed for  $h \geq 2$ ; only the interval  $\max\{2, h\} \leq \text{pd} \leq h + 1$  is proved, and



21's exact orthant value is credited, not used (for simplicial cones see the merge's isomorphism with  $A_d$ ). (3) No all-module weak or global dimension of  $A_{\mathcal{C}}$ ,  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  is asserted. (4) No extension to unrestricted infinite Hahn supports is asserted; every element has finite support. (5) The face-lattice reconstruction needs the marked monomial grading; an unmarked recognition theorem is not claimed. (6) "Core" does not mean integer part of the fraction field, and a partial exponent cone must not be replaced by the positive cone of the ambient order. (7) The numbers  $\beta_{p,G}$  count summands of a reduced flat resolution, not minimal free ranks, and the reduced resolution is not unique. (8) The geometric monomial class excludes arithmetic ideals such as  $pA_{\mathcal{C}}$  and restrictions on admitted coefficients. (9) Miller's real-polyhedral theory [73] is not identified with these resolutions, and Geist–Miller's all-module theorems [72] are neither reproved nor claimed; cellular and lcm-lattice methods [70, 71] are classical. (10) Theorem 13.85 concerns modules acting through  $\text{ct}$  only, not all modules over a core or the class ring; the scalar-quotient calculation and the common fraction field are elementary comparisons, not new theorems. (11) The nonflatness concerns the displayed two-ray relation for  $d \geq 2$ ; it does not deny the embeddings, the fraction field or the retraction, and asserts nothing about embeddings of one-dimensional cores. (12) The homological-epimorphism theorem makes no recollement or compact-generation claim. (13) The finite checks (52,080 exact assertions, among them 188 face quotients, 1,329 further antichains, 56 cancelled pairs and 2,457 cap checks, with a fixed seed and no exhaustive search) do not verify filtered colimits, exactness in every rational degree, the dual-basis argument, the topological contraction for every polytope or normal forms, and are not an independent proof or referee report. (14) The repository was compared selectively (this report's lines 1–210 and 3300–3520 at the pin, its README, the root and catalogue READMEs and an automorphism-report README), not all reports or companion manuscripts, and an empty connector search was not taken as absence. (15) All homological statements concern set-sized rings and modules; the full omnific ring appears only as the target of an embedding and of a finite relation.

**27.** (1) An author-side, internally reviewed, unrefereed draft with no Lean verification; historical priority is provisional, and no Riemann hypothesis, omnific factorization conjecture or published automorphism problem is claimed solved. (2) No henselization or spectrum of the proper-class ring  $\mathbf{Oz}$  is asserted; all rings, spectra and henselizations are set-sized. (3)  $\text{Idem}(\mathcal{G})$  and  $\mathbf{X}$  are not classified: atomlessness, isolated points and, for  $\kappa = \aleph_0$ , the Cantor space are open, and "maximal" is a cardinal statement only. (4)  $\mathcal{L}^h$  and  $\mathcal{L}^h[i]$  are local, not products of branches; nontrivial idempotents exist only in the generic fibre. (5) The imaginary unit lives only in  $\mathcal{G}$ ;  $\mathcal{L}^h$  has no square root of  $-1$  and residue field  $\mathbb{Q}$ . (6) The support projection is not a ring map, and the specializations are maps of finite algebras, not evaluations of Laurent series at zero. (7) The rings are not Noetherian and no Krull dimension is determined; no infinitesimal smoothness criterion applies. (8) The Gaussian decomposition does not assert  $\mathbf{X} \not\cong \mathbf{X} \sqcup \mathbf{X}$ . (9) The valuation count imports de Felipe–Teissier [78]; nothing is claimed about the components themselves or their topology compared with  $\mathbf{X}$ . (10) The finite étale equivalence concerns  $\mathcal{L}^h$ , not  $B_{\mathbb{E}}$  or  $\mathbf{Oz}$ , and the rank classification over  $\mathcal{G}$  is of isomorphism classes, not morphisms. (11) Only  $K_0(\mathcal{G})$  and finite projectives are computed; no higher  $K$ -theory or derived formula is asserted. (12) No birthday bound is asserted; seeds are contained in  $\mathbb{E}$ , but the operations of a prior workspace are not preserved without further closure. (13) The finite checks (5,760 exact SymPy assertions, 5,586 of them Boolean atoms; the count is not a measure of proof strength) prove no support, localization-denominator, prime-existence, transfinite, cardinality or priority statement. (14) Henselian algebra, Boolean powers and the Stacks Project results are standard, de Felipe–Teissier's theorem is imported, and the splitter mechanism is the repository's (20), re-derived. (15) Independent checking of the whole proof, in particular



of the bridge from supports to localization, is warranted; the exact Stone space, atomlessness, enlargement of the hull, mixed-characteristic analogues and class-level geometry are not settled. (16) The repository changed during 27's inspection, so its guide reads are not a complete same-commit audit; its search for "hensel" was not exhaustive (and missed existing occurrences, [Section 15.11](#)); its targeted literature searches found no primary source for the construction, which does not certify novelty. (17) No regularity of  $\kappa$ , continuum hypothesis, GCH or large cardinal is used. (18) Its twelve questions are proposed directions and may already be answered in literature 27 did not locate.

**The merge.** (1) The results marked [merge] are consequences of the sources' lemmas; they are no more refereed than the sources and are not formalized. (2) No general law is asserted for the generator counts. (3) For full-support Hahn rings over a prescribed set group, exact thresholds are computed only in [Examples 10.4](#) and [10.5](#); the general questions of 04 and 10 stay open. (4) The support-threshold theorem is stated for  $k \in \{\mathbb{R}, \mathbb{C}\}$  as in 04 and 11, not for every set field. (5) No Gaussian analogue of 10's congruence-preservation criterion is given. (6) Nothing is claimed about the homological dimensions of the class ring **Oz**; the weak global dimensions of the cores remain undetermined ( $\text{pd}_{A_d} D$  for  $d \geq 2$ , undetermined before the fifth batch, is  $d + 1$  by 21's [Corollary 13.47](#)). (7) The finite-support cores of 12 and 14 are outside [Theorem 5.1](#): they have set-sized images detecting the tail. (8) [Proposition 13.12\(v\)](#) needs a countably generated tail; among the five constructions it settles only  $A_\kappa^{08}$  with  $\text{cf } \kappa = \aleph_0$ . (9) The extensions of 17's results to  $\kappa = \aleph_0$  and of its Tor formula to the cores say nothing about the class ring's homological dimensions, and  $R_{\aleph_0}^{11}$ , like the cores, is outside [Theorem 5.1](#). (10) The comparison of Gaussian tests concerns the sizes of two explicit certificates, not the minimal size of a test, and [Remark 15.52](#) by itself decides nothing about  $\text{Idem}(\mathbb{B}_{\mathbb{R}})$  (its candidates are decided in [Proposition 15.68](#)). (11) [Corollary 15.59](#) is a descent, not a membership test for  $\mathcal{N}_\kappa$ , and [Proposition 15.68](#) decides the idempotents  $(1 + \bar{u}_H)/2$  only for  $H = \iota_b(h)$  with  $h \in TK_G[T]$ , in particular for positive  $H \in \Pi$  supported in one cyclic group; other  $H$  are not decided. (12) [Remark 15.75](#) concerns the splitting results only; the extension of branch maps and 19's detection still use global choice. (13) The extension of 21's lattice theorem to every unital  $D$  ([Theorem 13.48](#)) and [Remark 13.49](#) give no projective dimension for the five constructions or for  $R_\kappa^{11}$  with uncountable  $\kappa$ ; the remark shows only that (H2) cannot be dropped from [Proposition 13.12\(v\)](#). (14) The comparison of 22's and 16's rational certificates ([Remark 16.71](#)) concerns the sizes of two explicit certificates, not the least size of a test. (15) [Remark 16.88](#) says nothing for  $D = k$ , and the remark after [Proposition 16.86](#) decides nothing about finite certificates in a workspace without fresh exponents. (16) The comparisons of [Remark 15.119](#) and [Example 15.87](#) concern the real pair; [Remark 15.102](#) concerns the bounded-denominator Puiseux ring only and determines neither the isomorphism type nor the order density of its Boolean algebra; [Remark 15.114](#) rests on applying the proof of [Proposition 15.76](#) verbatim to the coefficient field  $\overline{\mathbb{Q}} \cap \mathbb{R}$ , and nothing is claimed for noncyclic groups or for the Hahn field over  $\mathbb{Q}a$ . (17) The eighth-batch merge results need 24's foundation where they use maximal ideals: the extension of point evaluations in [Remark 14.47](#) uses global choice, is not canonical and does not land in  $\mathcal{E}_a$ ; [Remark 14.28](#) gives class ultrafilters in GBC only; and the copy of  $\mathbf{No}[i]$  in [Corollary 14.44](#) makes the residue fields neither algebraically closed nor isomorphic to  $\mathbf{No}[i]$ . (18) [Remark 16.111](#) concerns monomial multipliers and the real pair; [Remark 16.120](#) shows only that global smoothness cannot be weakened to  $C^k$  in [Corollary 16.118](#), not what the exceptional sets of finitely smooth maps are; and the batch-34 status of [Question 18.20](#) re-scopes 22's comment, it does not supply a piecewise classification. (19) The isomorphism of a rational simplicial cone core with  $A_d$  transfers 21's exact projective dimensions to simplicial face quotients only; for nonsimplicial quotients with  $h \geq 2$  the endpoint stays open, and [Remark 13.86](#)

gives a lower bound for the weak global dimension, not its value. (20) The identification of  $27$ 's  $\sqrt{\Pi\mathcal{N}}$  with  $\Pi\mathcal{N}$  uses  $19$ 's [Theorem 15.29](#), which  $27$  neither uses nor needs, and the branch counts compared in [Remark 15.143](#) concern three different rings; nothing is inferred about  $\mathbf{Oz}$  itself.

### 18.3 Open questions and their status

The sources pose the following questions, each as a question suggested by the proofs rather than as an established open problem. Several coincide and are merged.

**Question 18.1** (Least detecting targets over a fixed exponent group; 03, 04, 07, 08, 09, 10, 11, 13, 15). For a set-sized ordered group  $\Gamma$ , a family of admissible supports and a specified element (or the whole positive-support ideal), what is the least cardinality of a ring target detecting it, and which intrinsic support conditions characterize it?

*Status.* Open in general. The field bound ([Proposition 10.3](#)) is a lower bound that is exact for the five constructions and for the two examples of [Section 10.2](#); there it answers 04's "optimal small targets" question for its own example, and for 10's question "when is the bound attained" it shows that the bound need not be attained (in the rational example, and for  $A_{\aleph_1}^{03}$  if CH fails). 08's criterion and 11's coinitality theorem give sufficient conditions (07's "sharp set-sized analogues" and 09's "quantitative problem" are the same question). 13's question (how large a target must be to detect a prescribed series, and whether its divisibility bound is optimal) and 15's "optimal local cardinal bounds" are this question as well: their bounds are the monomial count of [Proposition 10.1\(ii\)](#), which the field bound dominates; the threshold theorem ([Theorem 10.10](#)) answers them for the five constructions, and [Example 10.5](#) shows that the monomial bound need not be attained.

**Question 18.2** (Cardinal arithmetic without the matching hypothesis; 06). For  $\kappa$  uncountable regular with  $\kappa^{<\kappa} > \kappa$ , what is the least size of a target detecting a nonzero element of  $\Pi_\kappa^{06}$ , and is there an integer part of size exactly  $\kappa$  with the same lower bound?

*Status.* Answered. The least size is  $\kappa^{<\kappa}$  in ZFC ([Theorem 11.11](#), a merge result from 06's collision lemma); 08 constructs, for every infinite  $\kappa$  in ZFC, an integer part of size exactly  $\kappa$  with threshold  $\kappa$  ([Theorem 11.16](#)), without claiming its fraction field.

**Question 18.3** (Higher projective dimension; 06). What is  $\text{pd}_A \mathbb{Z}$  for the set-sized integer parts, and how does it depend on the cofinality of the principal-ideal chain?

*Status.* Partly answered.  $\text{fd}_A D = 1$  and  $\text{pd}_A D \geq 2$  are known for all five constructions. If the tail is countably generated, that is if the principal-ideal chain has countable cofinality, then  $\text{pd}_A D = 2$  ([Proposition 13.12\(v\)](#), a merge result from 12's telescope); among the five constructions this is the case exactly for  $A_\kappa^{08}$  with  $\text{cf } \kappa = \aleph_0$ . For chains of uncountable cofinality (03, 06, 09, 11, and 08 with  $\text{cf } \kappa > \aleph_0$ ) the question stays open. For comparison, the ordered-cone cores have  $\text{pd } D = 2$  and the coordinate-cone cores  $\text{pd}_{A_d} D = d + 1$  (21, [Corollary 13.47](#)), although their tails are countably generated: they fail (H2) ([Remark 13.49](#)); 17's countable lexicographic model has  $\text{pd } D = 2$  and  $\text{pd } L(M) = 2$  ([Remark 13.32](#)), and 17 asks the same question for its uncountable models ([Question 18.15](#)).

**Question 18.4** (Nonarithmetic primes and their residue fields; 06, 07). Which prime or maximal class ideals of  $\mathbf{Oz}$  have nonzero constant-term image and zero intersection with  $\mathbb{Z}$ , and which class fields occur as  $\text{Frac}(\mathbf{Oz}/P)$  for primes  $P \not\subseteq \Pi$ ? Can the embedded field  $\mathcal{F}_a$  distinguish such primes?

*Status.* Open. Necessary conditions: [Corollary 14.10](#), [Proposition 14.12](#), and [Remark 14.13](#). *Status (batch 34).* Partly answered, with global choice. The existence clause is answered by 24: every maximal class ideal containing  $1 + \omega^a$ , or any nonconstant element with unit constant term, is a nonarithmetic prime with  $P \cap \mathbb{Z} = 0$  and  $P \not\subseteq \Pi$ , and such ideals exist in GBC ([Theorem 14.34](#), [Corollary 14.35](#), and [Proposition 14.33](#)); they separate the points of  $\mathbf{Oz}$  ([Theorem 14.50](#)), and there are at least  $2^{\aleph_0}$  of them above each  $1 + \omega^a$  and a family indexed by  $\mathbf{On} \times \mathbb{Z}_2$  across scales ([Theorems 14.46](#) and [14.48](#)). On the second clause: every maximal residue field of characteristic 0 embeds in  $\mathbf{No}[i]$  ([Theorem 14.53](#)) and, above  $1 + \omega^a$ , contains a copy of  $\mathbf{No}[i]$  ([Corollary 14.44](#)); which fields occur is not determined ([Questions 18.30](#) and [18.31](#)). On the third: the embedded  $\mathcal{F}_a$  meets every maximal ideal above  $1 + \omega^a$  trivially, a field mapping injectively to every nonzero quotient, so by itself it does not distinguish them; the maximal ideals of [Theorem 14.46](#) are separated by the dyadic idempotents over  $\mathcal{E}_a$ , and the monomial cutoff [Theorem 14.37](#) separates the scales [merge]. The classification of nonarithmetic primes stays open, as does existence without global choice ([Questions 18.28](#) and [18.29](#)). The primes that are term-closed and strongly sum-closed are classified, and none of characteristic 0 is maximal ([Theorem 14.57](#)).

**Question 18.5** (The ambient binomial quotient; 07). Which normal-form invariants describe the elements and ideals of  $\mathcal{Q}_a$  beyond the Cantor algebra; which idempotents lie outside it; how are cyclotomic decompositions at different scales related?

*Status.* Open. *Status (batch 34).* Partly addressed by 24. Every monomial  $\omega^b$ ,  $b \geq 0$ , is zero or a unit in  $\mathcal{Q}_a$ , zero exactly when  $a \ll b$  ([Theorem 14.37](#)); the whole tail above  $\text{cvx}(\mathbb{Z}a)$  can be discarded,  $\mathcal{Q}_a$  being the binomial quotient of the elements supported in  $\text{cvx}(\mathbb{Z}a)$  ([Proposition 14.40](#)); the binomial quotients at separated scales are comaximal, and those at different scales are abstractly isomorphic by rescaling ([Corollary 14.42](#) and [Proposition 14.43](#)), which relates the cyclotomic decompositions at different scales; and every branch of the Cantor algebra lies in a maximal ideal ([Theorem 14.46](#)). Idempotents outside the Cantor algebra, the nilradical and the prime ideals of  $\mathcal{Q}_a$  remain open. 24 asks the same question in sharper form (its Question 12.6): determine the nilradical and the Jacobson radical of  $\mathcal{Q}_a^k$ ; is  $(1 + \omega^a)$  radical; how much of the idempotent algebra is generated by finite rational-exponent blocks; and what information beyond the monomial cutoff and the dyadic branch data determines a maximal ideal? 24's comments: the reduction to  $\text{cvx}(\mathbb{Z}a)$  should be used first, but it still leaves a proper-class infinitesimal exponent core and general infinite supports, and finite polynomial squarefreeness does not by itself give radicality in the full non-Noetherian Hahn support ring.

**Question 18.6** (Residue fields of the set-sized models; 08, 09). Which characteristic-zero fields of the threshold cardinality occur as maximal residue fields of  $A_\kappa^{08}$  and of  $A_{\lambda,D}^{09}$ , and how many maximal ideals are there?

*Status.* Open; existence, cardinality and at least  $\lambda$  nonarithmetic maximal ideals (for 09) are proved.

**Question 18.7** (Extra structure on the workspace; 08). Can prescribed residue-field behaviour be combined with closure under a surreal derivation or an exponential, or with an initial embedding, without destroying the finite-coordinate invariant or the cardinality?

*Status.* Open.

**Question 18.8** (Intermediate rings; 04, 11, 15). Which rings between the finite-support ring and  $D + \Pi_k$  (for instance, given by allowed families of support order types) still have set-sized reflection  $D$ ? Is enough closure under subordinate division also necessary?

*Status.* Open; [Theorem 9.1](#) settles the case of cardinal support bounds. 09's question on support bounds below a regular cardinal is addressed for the groups of 06 and 11 ([Sections 11.2](#) and [11.5](#)), where the support sizes of inverses and quotients are checked.

**Question 18.9** (Beyond ring observations; 04). Which weaker multiplicative compatibility conditions still force factorization through the residue, and which structured families of set-sized local models can retain a specified family of scales?

*Status.* Open.

**Question 18.10** (Invisible class domains and automorphisms; 10, 11, 15). What are the ideals, class-valued representations and factorization theory of proper-class quotient domains invisible to all set-sized rings? How much can an automorphism of  $\mathbf{Oz}$  do inside  $\Pi$  while fixing  $\text{ct}$ ?

*Status.* Partly answered (batch 31). The first sentence is open. The second is answered in the report on omnific-preserving automorphisms [48] (`opa:as:cor:invisible`, from its automatic-strongness theorem `opa:as:thm:main` and its factorization `opa:as:thm:factor`, proved by that report's sources 12 and 13 and independently by its source 14). Every ring automorphism  $\sigma$  of  $\mathbf{Oz}$  satisfies  $\text{ct} \circ \sigma = \text{ct}$ , and extends uniquely to a strong automorphism of  $\mathbf{No}$  fixing  $\mathbb{R}$  pointwise, of the form  $M_{\chi, \tau} \circ u$ . Here  $\tau$  is an order automorphism of the exponents,  $\chi$  a positive character and  $u$  a strong leading-term-fixing automorphism preserving  $\mathbf{Oz}$ , classified by that report's convex-support criterion `opa:thm:criterion`. On  $\Pi$  it therefore acts first by  $u$ , then by an exponent automorphism and a positive rescaling of monomials; every such composition occurs.

**Question 18.11** (Normalization; 03). Give a membership test for  $\mathcal{N}$  on mixed supports; describe the integral closure over a fixed finitely generated exponent subgroup; decide the descent  $\mathcal{N} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}$ ; find the algebra generation number of  $\mathcal{N}_{\kappa}$  between  $\text{cf } \kappa$  and  $\kappa^{\aleph_0}$ ; find a class-sized replacement for the valuative criterion of integrality; describe the infinite set-sized images of  $\mathcal{N}$ .

*Status.* Partly answered. For the last clause, 19 classifies the set-sized *domain* images of  $\mathcal{N}$  and  $\mathcal{N}_{\mathbb{C}}$  (they are  $\mathbb{Z}$  and  $\mathbb{F}_p$ , [Theorem 15.46](#)) and shows that every set-sized image factors through  $\mathcal{N}/\Pi\mathcal{N} \cong \mathbb{Z}[\text{Idem}]$  ([Corollary 15.45](#) and [Theorem 15.38](#)); images with zero divisors are not classified. 20 shows that  $\text{Idem}(\mathcal{N}/\Pi\mathcal{N})$  is atomless and a proper class and that no set-indexed family of set-sized images is jointly faithful ([Theorems 15.73](#) and [15.79](#)); its isomorphism type is open ([Question 18.18](#)). The descent clause is answered affirmatively:  $\mathcal{N} \cap F_{\kappa}^{03} = \mathcal{N}_{\kappa}$  ([Corollary 15.59](#), a merge result from 20's support projection). The clause on a fixed exponent subgroup is partly answered: for every set-sized subgroup  $G$ ,  $\mathcal{N} \cap K_G$  is the integral closure  $\mathcal{N}(G)$  of the omnific integers supported in  $G$ , so integrality witnesses at larger scales never help, and for  $G = \mathbb{Z}a$  the rational one-scale slice is determined exactly ([Theorems 15.57](#) and [15.60](#)); a description of  $\mathcal{N}(G)$  in general is open. *Status* (batch 32). For  $G = \mathbb{Z}a$  the clause is answered by 23:  $\mathcal{N}(\mathbb{Z}a)$  is the set of elements of  $K_{\mathbb{Z}a}$  integral over  $\mathbb{R}[\omega^a]$  whose residues at the finitely many places over  $\omega^a = 0$  of each finite subextension are algebraic integers, and its  $\mathbb{R}$ -linear span is the integral closure of  $\mathbb{R}[\omega^a]$  ([Theorems 15.91](#) and [15.93](#); the same holds for the Gaussian pair with  $\mathbb{C}$  in place of  $\mathbb{R}$ ). This is a criterion, not an algorithm ([Question 18.17\(c\)](#)), and for noncyclic finitely generated  $G$  a description stays open. The membership test, the generation number and the valuative criterion stay open; 19 asks the same membership question, for supports in one finitely generated exponent subgroup first, allowing integrality witnesses at larger scales, and 20 asks for effective support-local certificates ([Question 18.17\(c\)](#)). 04's question on Diophantine fibers belongs to the sibling report [47], and so does 13's question on fibers of polynomials with nonlinear factors.

**Question 18.12** (Homological dimensions of the cores; 12, 14, 21, merge). For  $d \geq 2$ , is  $\text{pd}_{A_d} D$  equal to  $d$  or to  $d + 1$ ? When  $K \neq \text{Frac } D$ , what are the weak global dimensions of  $A_d$  (at least  $d + 1$ ) and of  $A_0 = A_1$  (at least 2)? How do cores over non-simplicial cones, their arbitrary ideals and their homological localizations behave?

*Status.* Partly answered. The first clause is answered by 21:  $\text{pd}_{A_d} D = d + 1$  for every  $d \geq 1$  (Corollary 13.47); in rank one this is the merge value 2 of Proposition 13.12, and over every ordered-cone core  $\text{pd } D = 2$ . The weak global dimensions stay open, when  $K \neq \text{Frac } D$ , with lower bounds  $d + 1$  and 2 (Section 13.5); for  $D = K$  the rank-one weak global dimension is 1, as shown after Theorem 13.48. 21 adds  $\text{gl. dim } A_d \geq d + 2$  when  $K \neq \text{Frac } D$  (Theorem 13.48) and asks whether the bounds  $\text{gl. dim } A_d \geq d + 2$  and  $\text{w. gl. dim } A_d \geq d + 1$  are sharp, which is merged here. Cores over non-simplicial cones, arbitrary ideals and homological localizations stay open.

*Status (batch 34).* The non-simplicial clause is answered by 26 for flat dimension, for the quotients by radical coefficient-full monomial ideals: over the core  $A_{\mathcal{C}}$  of any pointed full-dimensional rational polyhedral cone,  $\text{fd}(A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}})$  is the top degree of the reduced homology of finite join complexes, with explicit cap-module witnesses (Theorem 13.70); the quotient keeping a face  $F$  has flat dimension the largest dimension of a face disjoint from  $F$  (Theorem 13.75), so  $\text{fd } D = d$  for every cone, the cone is simplicial exactly when all facet quotients have flat dimension one (Theorem 13.79), and cube cones have facet quotients of flat dimension  $d - 1$  (Theorem 13.81). Projective dimensions are bracketed,  $\max\{2, h\} \leq \text{pd} \leq h + 1$  (Corollary 13.71), and for simplicial cones the upper endpoint holds for proper-face quotients, a simplicial core being  $A_d$  (21's Corollary 13.47; merge). 26 asks the same two questions (its Questions 13.1, 13.2), merged here: for a radical geometric quotient with  $h = \text{fd} \geq 2$ , when is  $\text{pd} = h + 1$  and can it equal  $h$ ; is the upper endpoint universal for rational cones and arbitrary  $D \subseteq K$ ; and what are  $\text{w. gl. dim } A_{\mathcal{C}}$  and  $\text{gl. dim } A_{\mathcal{C}}$  for nonsimplicial cones, with explicit hypotheses on  $D$  and  $K$ ? 26's comments: a plausible step is to combine the cap detector with the non-splitting of a countable telescope to build a nonzero  $\text{Ext}^{h+1}$  class, controlling compatibility across the face differential, which flat detection alone does not do; its theorem gives  $\text{w. gl. dim } A_{\mathcal{C}} \geq d$ , not an all-module upper bound; Geist–Miller's orthant theorems [72] address a stronger all-module problem in their setting; and extending a total Koszul or diagonal resolution to a nonsimplicial arithmetic core differs from resolving only the face quotients. The lower bound is already  $d + 1$  when  $K \neq \text{Frac } D$ , by lattice modules (Remark 13.86, a merge remark). Arbitrary ideals, homological localizations and the exact weak global and global dimensions stay open.

**Question 18.13** (Beyond one coefficient lattice; 14, 21). Replace the single coefficient lattice by a finite filtration of lattices indexed by support levels and determine which extension data survive multiplication; characterize all finitely generated rank-one modules with double dual  $B$ ; determine how coefficient rings that are not principal ideal domains change the tensor kernel.

*Status.* Open. 21 asks the first part again, for the derived tensor products of finitely generated omnific ideals whose generators lie at several independent boundary levels, and notes that its common-scale DGA (Theorem 13.50) does not cover them; its question is merged here.

**Question 18.14** (Fresh scales; 16). (a) Can the factorial-residue Gaussian certificate be replaced by an optimal or near-optimal grid, and how does a regular basis of  $\text{Int}(\mathbb{Z}[i])$  interact with the omnific splitting? (b) Under what hypotheses are the formal inverse branches of a polynomial map in several variables at independently fresh coordinates classified by finitely many forbidden support blocks (triangular automorphisms exclude a direct analogue of one-variable surjectivity rigidity)? (c) For a fixed set-sized Hahn subfield and its nonnegative-support ring, which coefficient subgroups admit enough dominant extensions inside the field to recover the image gap and the rational certificates?



*Status.* Partly answered. (a) is partly addressed by 18: its Gaussian simplex test is smaller than the factorial-residue certificate for  $d \geq 3$ , and every ordinary testing set for  $\text{Int}(\mathbb{Z}[i])$ , for instance the Volkov–Petrov sets of size  $O(d)$  in one variable, also tests  $\mathbf{Oz}[i]$  (Proposition 16.64); the optimal size is not determined. 22 reduces the constant-test part of (a) exactly to the ordinary problem: a finite subset of  $\mathbb{Z}[i]^m$  tests preservation of  $\mathbf{Oz}[i]$  in a coordinate degree space if and only if it tests  $\text{Int}(\mathbb{Z}[i]^m)$  there (Theorem 16.82); its question on sharp Gaussian constant tests (how far known ordinary Gaussian universal sets improve the surcomplex certificate once combined with the remote denominator stage, distinguishing coordinate from total degree in several variables) is merged here, and 22 claims no optimality for its grid  $C_{d+1}^m$ . For (c), 18’s discrete-gap arguments, unlike the fresh-scale ones, do hold in fixed Hahn workspaces (Corollary 16.67); and 22 answers the part on rational certificates for its single-scale certificates: a coefficient subgroup  $G$  of the value group  $\Gamma$  has a fresh exponent in  $\Gamma$  exactly when the convex subgroup it generates is proper, and then the Laurent blocks, the detector and the certificates hold in the workspace (Proposition 16.86). The image gap in a fixed workspace, the corresponding condition for 16’s multi-exponent probes (Theorem 16.49), and the existence of other finite certificates when that convex subgroup is all of  $\Gamma$  stay open, as does (b).

**Question 18.15** (Relations of lattice ideals; 17, 21). (a) Classify the kernels of omnific matrices with several monomial scales, and determine which coefficient-lattice data beyond  $q$  and  $\ell$  they need. (b) Determine the projective dimensions of  $\Pi_\kappa^{11}$  and  $J_g(M)$  over  $R_\kappa^{11}$ , and an upper bound for its weak global dimension. (c) How much of a finitely generated coefficient algebra do the product lattices, Hom modules and mixed multiplication kernels together reconstruct?

*Status.* Open, except (b) for  $\kappa = \aleph_0$ :  $\text{pd } \Pi_{\aleph_0}^{11} = 1$  and  $\text{pd } J_g(M) = 2$  for  $r \geq 2$  (Remark 13.32, a merge result). For uncountable  $\kappa$ , (b) is Question 18.3 for  $R_{\Gamma_\kappa^{11}, < \kappa}^{11}$ ; (a) is also 14’s question (Question 18.13). (c) is partly addressed by 21: with the distinguished 1 and the boundary embedding, the self multiplication kernel  $\mathcal{T}(\mathfrak{o}, \mathfrak{o})$  recovers the multiplication of an order (Theorem 13.56), and 17’s Theorem 13.39 recovers the order from  $\text{End}_A(J_\gamma(\mathfrak{o}))$ ; 21 asks which parts of these framed data can be recovered intrinsically from a category of omnific quotient modules with its tensor structure, which is merged into (c).

**Question 18.16** (Ordinary-output loci; 18). (a) Which hypotheses on  $f \in \mathcal{N}_n(\Lambda)$  exclude positive-dimensional fibres and make its ordinary-output locus a finite union of translated ordinary lattices, explaining both the linewise rigidity and the example  $XY$ ? (b) For a polynomial of degree  $d$  without a rich target, which patterns of the numbers  $\#\mathcal{L}_{f,b} \in \{0, \dots, d\}$  occur as  $b$  varies? (c) Can Gaussian test points in  $\mathbf{Oz}[i]$  that are not ordinary reduce the least number of test points below the ordinary minimum? (d) Can the cluster theorem be formalized for an abstract discretely ordered ring and specialized through the normal-form bridge?

*Status.* Open. The real analogue of (c) is settled by 16: for  $\mathbf{Oz}^n$  no set of omnific points smaller than  $|L|$  tests a lower-set space  $V_L$  (Theorem 16.46); for  $\mathbf{Oz}[i]$  the lower bound  $d + 1$  from the dimension is all that is known. 22’s lifting theorem (Theorem 16.82) concerns test points in  $\mathbb{Z}[i]^m$  only and does not bear on (c).

**Question 18.17** (The arithmetic fibre of the normalization; 19). (a) Determine  $\text{Idem}(\mathbf{B}_{\mathbb{R}})$ : is it trivial, does it have atoms, can independent families of nontrivial idempotents be constructed, and is it a set? (b) For which fixed exponent groups and support classes is the extended ideal radical in the corresponding real integral closure, and when does real-rooted lifting give a Boolean-power fibre although set-family divisibility fails? (c) Given finitely many integral elements with monic



witnesses, can a common Boolean partition and its algebraic-integer coefficients be extracted by a concrete symbolic procedure, including the equality tests in the quotient?

*Status.* Partly answered. (a) is answered by 20:  $\text{Idem}(\mathbb{B}_{\mathbb{R}})$  is nontrivial and atomless, it is a proper class with no set-sized order-dense subset, the explicit family of [Theorem 15.72](#) is independent and indexed by all ordinals, and free families of every set size exist over every set-sized subring ([Theorems 15.66](#) and [15.71](#) to [15.73](#)); the same holds for  $\mathbb{B}_{\mathbb{C}}$  ([Corollary 15.77](#)). The earlier reduction of (a) to instances of the membership clause of [Question 18.11](#) ([Remark 15.52](#)) is superseded: 20 needs no membership test, and the candidates of that remark are nontrivial ([Proposition 15.68](#), a merge result). What remains of (a) is the isomorphism type ([Question 18.18](#)). (b) is partly answered: for every nonzero set-sized ordered group the reduced fibre  $\mathcal{C}(G)$  is a Boolean power of  $\overline{\mathbb{Z}}$ , and one dominant coordinate adds free Boolean families ([Proposition 15.76](#) and [Theorem 15.70](#)); whether  $\Pi(G)\mathcal{N}(G)$  is radical, and whether  $\text{Idem}(\mathcal{C}(G))$  is atomless, in a fixed workspace was left open by 20, which asks the same; for cyclic groups it is now decided (batch 32, below). (c) is open; 20 asks it for algebraic Laurent elements given by equations and branch data, with complexity bounds, and settles the one-scale rational case ([Theorem 15.60](#)); 23 gives a criterion for cyclic groups (below). [Theorem 15.42](#) holds throughout. 19's membership question is merged into [Question 18.11](#).

*Status (batch 32).* (b) is decided for every nonzero cyclic group  $G = \mathbb{Z}a$  by 23, in opposite directions:  $\Pi(\mathbb{Z}a)\mathcal{N}(\mathbb{Z}a)$  is not radical, its fibre having nilpotents of every index ([Theorem 15.86](#)), and  $\text{Idem}(\mathcal{C}(\mathbb{Z}a))$  is atomless of cardinality and order density  $2^{\aleph_0}$  ([Corollary 15.110](#) and [Theorem 15.111](#)), also for the Gaussian pair. For the bounded-denominator Puiseux support class over  $\mathbb{Q}a$  the extended ideal is radical ([Theorem 15.100](#)), and there real-rooted lifting gives a Boolean-power fibre although set-family divisibility fails ([Remark 15.102](#), a merge observation). For other groups and support classes (b) stays open, and the Hahn field over  $\mathbb{Q}a$  is [Question 18.21](#). 23 asks the same question (its [Question 10.2](#)): for a prescribed set-sized  $G$ , which properties of  $G$  and of the admitted supports determine radicality of  $\Pi(G)\mathcal{N}(G)$ , and can the one-variable ramification filtration be replaced by a finite-rank or cut-valued filtration when  $G$  has several Archimedean layers? 23's comments: the cyclic negative and the Puiseux positive results are two test cases that any criterion must distinguish; a next model is a lexicographic sum of two cyclic groups, where the coefficient block is a field with its own integer part and the residue-basis step of [Theorem 15.93](#) needs a replacement; the proof does not classify groups with a discrete convex component. (c) is addressed for cyclic groups by 23's residue criterion ([Theorem 15.91](#) and [Remark 15.92](#)), which is a criterion and not an algorithm. 23 asks (its [Question 10.7](#)), for coefficients in a number field or a computable presentation of  $\overline{\mathbb{Q}}$ , for an algorithm that takes a finite algebraic function field and a chosen Laurent branch at infinity, decides membership in  $\mathcal{N}(\mathbb{Z}a)$ , computes  $\nu_T(f)$ , and produces a monic certificate and the least refinement killing a nilpotent residue. 23's comments: the outputs are determined (normalize at the finite places, test the residues, compute the finitely many orders), and what remains is a representation and complexity problem; singular plane equations can hide branches, so normalization cannot in general be replaced by factoring one specialized polynomial; one would bound the certificate degree  $dr \deg Q$  of [Remark 15.92](#) in the input degree and height and decide when smaller certificates exist.

**Question 18.18** (The class Boolean algebra of the fibre; 20). Which further Boolean invariants determine  $\text{Idem}(\mathbb{B}_{\mathbb{R}})$ ? Can its automorphisms, or its possible infinite joins, be characterized directly in terms of support-local arithmetic?

*Status.* Open. Known: it is atomless, a proper class with no set-sized order-dense subset, and relatively free over every set-sized subring in every set size ([Theorems 15.71](#) and [15.73](#)); this does not make it free, complete or saturated.

**Question 18.19** (Finite certificates at new scales; 22). (a) What is the smallest coefficient-uniform remote test set for rational functions with given degree bounds and a prescribed fresh  $t$ ; can the bivariate count  $5d^2 + 6d + 3$  of [Theorem 16.70](#) be improved with an explicit coefficient-independent choice of the ordinary coordinates? (b) Can total-degree bounds, Newton polytopes or prescribed monomial supports replace the rectangular degree space and give substantially smaller tests? (c) For which useful notations of surreal coefficients can a certified group bound and a symbolic fresh scale be produced effectively? (d) Given sign-sequence descriptions of a support bound, what ordinal birthday bounds hold for the witness  $t$  and for a bad input produced by the resultant method? (e) Can sufficient fresh-scale conditions for a bounded-degree family be expressed without naming normal-form supports, for instance in the pair **(No, Oz)**, possibly with predicates for convex subgroups? (f) Can the support-block and finite-certificate proofs be added as a small, independently auditable extension of the repository’s Hahn layer?

*Status.* Open. 22’s comments: in (a) the proof ignores the structure of the resultant and uses one bound in every direction, and an optimality claim must say whether the test set may depend on the coefficients or only on their support group and degrees; 16’s probe certificate uses fewer values with  $m + 1$  fresh exponents ([Remark 16.71](#)). In (b) the dimension lower bound holds for any finite-dimensional polynomial space with a rich input set, achieving it with integer-valued nodes is an arithmetic issue, and on the rational side sparse resultants must still exclude cancellation and degree loss; lower-set spaces, including total degree, are already tested optimally ([Theorem 16.46](#)), and 16’s probes give total-degree rational certificates ([Corollary 16.50](#)). In (c) a first implementation would use finite normal forms with exponents in a computable ordered group and the fresh scale as an extra lexicographic generator, a second infinite supports with certified bounding data; coefficient equality, support membership, support bounding and ring membership should be separate parts of the input contract, none following from the other three (compare 16’s non-claim (4) and [Question 18.17\(c\)](#)). In (d) the construction of the fresh exponent as  $\omega$  times a bound proves existence only; bounds should come from the representation used, with every multiplication and monomial operation accounted for. In (e) a merely transcendental test point is not enough, since its order relative to the coefficient scales is essential, and the certificate is not quantifier elimination for the pair. In (f) the milestones are the order embedding  $(n, g) \mapsto nb + g$  on lexicographic blocks, reverse well-ordering and local finiteness of the Laurent image, multiplicativity, the nonnegative-support detector, the resultant bound and the grid lemma, and the explicit deceptive function, with the proper-class conclusions proved through an explicit small-set bounding interface; a finite-rank toy model would not establish the full claims ([Section 19.2](#)).

**Question 18.20** (Arithmetic maps beyond affine polynomial space; 22). (a) Which single-valued algebraic or semialgebraic maps preserve **Oz**, and do their branches admit analogous certificates? (b) Which affine varieties admit a counterpart of the constant-grid and remote-direction certificate for their omnific points? (c) How are the polynomial automorphisms whose two directions preserve **Oz<sup>m</sup>** organized by their ordinary specializations and purely infinite parts? (d) How far do the finite certificates and the absence of set-sized tests extend to rings  $D \oplus \Pi_k$  for other infinite subrings  $D$  of  $k$ , for instance rings of algebraic integers?

*Status.* Open, with (d) partly addressed. 22’s comments: in (a) a global polynomial conclusion is false ( $|X|$  preserves **Oz** and is only piecewise polynomial), and the target is a piecewise classification with explicit branch hypotheses and finite complexity bounds, controlling algebraic branches and their negative-support tails. In (b) the affine proof uses the Zariski density of an ordinary grid and free movement in one coordinate, neither automatic on a variety; a first theorem could fix an explicit dominant parametrization and track its exceptional locus, and varieties with very few

ordinary integral points need a different mechanism. In (c) [Corollary 16.84](#) removes denominators but does not classify the group, which the triangular binomial shears of [Example 16.85](#) show to be larger than the group with coefficients in  $\mathbf{Oz}$ ; a start is the subgroup generated by affine maps and triangular binomial shears, together with the kernel of specialization, and nothing is implied about all automorphisms in higher dimension. This is not the question on automorphisms of the ring  $\mathbf{Oz}$  in [Question 18.10](#). In (d) the detector is insensitive to  $D$ , the polynomial step needs integer-valued polynomial theory over  $D$ , and the deceiver needs a constant outside  $D$  with a suitable cleared denominator; 16 already proves rational collapse, the transfer  $\mathcal{N}_r(\mathcal{A}_{D,k}) = \text{Int}(D^r) \oplus \Pi_k[\mathbf{X}]$ , the probe certificates and the one-variable removal of exceptions for every unital  $D \subseteq k$  ([Theorems 16.42](#), [16.49](#) and [16.52](#) and [Proposition 16.45](#)), and the absence of set-sized tests holds for every  $D \neq k$  ([Remark 16.88](#), a merge result); finite constant test sets for  $\text{Int}(D^m)$  and the lifting theorem for general  $D$  remain open.

*Status (batch 34).* The first clause of (a) is answered by 25 for globally smooth maps, in every dimension: a globally  $C^\infty$  semialgebraic map  $\mathbf{No}^r \rightarrow \mathbf{No}^{r'}$  preserves  $\mathbf{Oz}^r$  if and only if its coordinates are integer-valued polynomials, with explicit binomial expansions, and global  $C^{\mathbb{D}^2}$  regularity suffices at scalar graph degree  $\mathbb{D}$  ([Theorem 16.102](#) and [Corollary 16.103](#)); the same holds over every real closed field with a cofinal discretely ordered subring, and for  $\mathbf{Oz}[i]$  in real coordinates ([Theorem 16.115](#)). A semialgebraic preserver of  $\mathbf{Oz}$  whose graph lies on one irreducible plane curve is a polynomial without any smoothness hypothesis ([Corollary 16.99](#)). Every semialgebraic preserver is polynomial on its two tails, the quotient by the bounded-region preservers being  $\mathcal{N}_1(B)^2$  ([Theorem 16.95](#) and [Corollary 16.96](#)); but for every ordinary  $k$  a semialgebraic  $C^k$  preserver agrees on  $\mathbf{Oz}$  with no finite family of polynomials ([Theorem 16.112](#)). So 22's target of a piecewise classification with finitely many polynomial pieces is attainable only on the tails, or with algebraic branches as pieces; the bounded-region preservers are [Question 18.37](#). The certificate clause of (a) is answered negatively for samples chosen independently of the coefficients, even for a globally smooth branch of an irreducible quadratic graph ([Theorem 16.117](#)), while a globally smooth semialgebraic nonpreserver fails on a proper class of inputs ([Corollaries 16.118](#) and [16.119](#)). 25 asks for coefficient-adapted certificates (its [Question 12.7](#)), merged into (a): given a quantifier-free algebraic graph description and exact access to its surreal coefficients, can preservation be certified or refuted by finitely many coefficient-adapted lattice points and finitely many support conditions? 25's comments: its no-set theorem leaves this adaptive question open, and an answer should distinguish symbolic decidability from the existence of a finite mathematical certificate. 25 also asks (its [Question 12.9](#)), merged into (b), to extend the smooth classification from affine space to semialgebraic manifolds or algebraic varieties with a specified omnific integral model, and which geometric conditions replace coordinate cofinality and tensor interpolation; 25's comments: the affine proof does not transplant, since a variety may have few omnific points and no cofinal coordinate lines, and the omnific Diophantine geometry of the repository [\[47\]](#) should be a separate input with its hypotheses audited. The rest of (b), and (c), (d), stay as above. The dynamics report [\[79\]](#) poses a dynamical relative of (a): its reciprocal rational maps have exactly prescribed infinitely large finite periodic points, all in  $\mathbf{Oz}[i]$ , and a numerator–denominator presentation with purely infinite Gaussian omnific coefficients, but are deliberately not shown to map  $\mathbf{Oz}[i]$  into itself (`dyn:rd:thm:omnific`); it asks which polynomial or rational maps preserve  $\mathbf{Oz}$  or  $\mathbf{Oz}[i]$  on a natural invariant domain while realizing specified infinite cycles (`dyn:rd:q:omnific`). Neither is an answer to a clause here.

**Question 18.21** (The full rational Hahn workspace; 23). Let  $K_{\mathbb{Q}a} = k((X^{\mathbb{Q}a}))$ , with arbitrary reverse well-ordered supports, let  $\mathcal{A}(\mathbb{Q}a)$  be its omnific or Gaussian omnific integer part and  $\Pi(\mathbb{Q}a)$  its positive-support ideal. Is  $\Pi(\mathbb{Q}a) \text{IC}_{K_{\mathbb{Q}a}}(\mathcal{A}(\mathbb{Q}a))$  radical?

*Status.* Open. 23's comments: [Theorem 15.100](#) answers the bounded-denominator subcase affirmatively, and the obstruction to repeating its proof is exact: an element of  $K_{\mathbb{Q}a}$ , and the coefficients of its integrality equation, need not lie in a finite cyclic refinement. A first intermediate case would allow supports generated by finitely many rationally independent real exponents, with a restriction preventing positive supports from accumulating at zero. Divisibility of the exponent group alone is not a proof of radicality; the supports of the coefficients must be controlled as well. For the real pair  $K_{\mathbb{Q}a}$  is real closed, so [Theorem 15.51](#) would apply if its axiom (ii) held there; the support  $\{a/n : n \geq 1\}$  blocks the monomial proof of (ii) (the paragraph after that theorem), and whether (ii) holds is not decided [merge].

**Question 18.22** (The uncountable Boolean isomorphism type; 23). Determine the isomorphism types of  $\text{Idem}(\mathbf{B}(\mathbb{Z}a))$  and  $\text{Idem}(\mathbf{B}^{\text{co}}(\mathbb{Z}a))$  for  $k = \mathbb{R}$  and  $k = \mathbb{C}$ . Are there invariants, beyond atomlessness, cardinality, order density and relative splitting, that distinguish them from familiar Boolean algebras?

*Status.* Open. 23's comments: the countable answer of [Theorem 15.118](#) is a baseline, and one would like to describe the Stone space directly from compatible places of finite function fields over  $T = 0$ . The explicit family gives  $2^{2^{\aleph_0}}$  points on the coefficient side ([Corollary 15.112](#)), but neither a product representation of the ring nor a free presentation of the algebra follows from this count; countable joins, saturation and automorphisms should be asked separately, and relative splitting is weaker than all of them together. This is the fixed-cyclic counterpart of [Question 18.18](#), which concerns the proper-class algebra  $\text{Idem}(\mathbf{B}_{\mathbb{R}})$ ; by [Remark 15.113](#) the idempotents of  $\mathbf{B}(\mathbb{Z}a)$  and  $\mathbf{C}(\mathbb{Z}a)$  are the same.

**Question 18.23** (Arithmetic residue gluing; 23). Describe concretely the embedding  $\mathbf{B}(\mathbb{Z}a) \hookrightarrow \mathbf{B}^{\text{co}}(\mathbb{Z}a)$  and the extension  $0 \rightarrow \mathbf{J}/T\mathcal{B} \rightarrow \mathbf{B}(\mathbb{Z}a) \rightarrow \mathbf{C}(\mathbb{Z}a) \rightarrow 0$ . How much of  $D$  can be recovered intrinsically from this extension and its multiplication, without naming the inclusion  $D \subseteq k$ ?

*Status.* Open. 23's comments: the residue criterion says which elements of  $\mathcal{B}$  lie in  $\mathcal{N}(\mathbb{Z}a)$ , and the task is to organize these conditions uniformly over all finite fields and all compatible branches. The strict inclusion of [Proposition 15.117](#) is an intrinsic arithmetic signal: multiplication by a nilpotent admits coefficients that cannot occur in the arithmetic fibre. A Boolean-power or locally constant description must keep this nonreduced information, not only the reduced data, for which see [Proposition 15.76](#) and [Remark 15.114](#).

**Question 18.24** (Homological dimensions of the arithmetic fibre; 23). What are the weak global and global dimensions of  $\mathbf{B}(\mathbb{Z}a)$ ? Can the kernels  $\bar{z}_n^{n-j} \mathbf{B}^{\text{co}}(\mathbb{Z}a)$  be resolved over  $\mathbf{B}(\mathbb{Z}a)$  so as to compute a nonzero sequence of higher Tor groups?

*Status.* Open. 23's comments: the infinite dimensions of  $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$  ([Theorem 15.116](#)) cannot be pulled back through the inclusion, since the alternating maps cease to be a resolution over  $\mathbf{B}(\mathbb{Z}a)$  ([Proposition 15.117](#)); a direct approach would start from the  $\mathbf{B}(\mathbb{Z}a)$ -module structure of  $\mathbf{B}^{\text{co}}(\mathbb{Z}a)$ , the modules  $\bar{z}_n^r \mathbf{B}^{\text{co}}(\mathbb{Z}a)$  and their relation modules. This is not the question on projective dimensions of the constant module over the finite-support cores ([Question 18.12](#)); those rings, modules and maps are different.

**Question 18.25** (The completed coefficient ring; 23). Describe  $\widehat{\mathcal{B}} = \varprojlim_r \mathcal{B}/T^r \mathcal{B}$  intrinsically. Is there a useful representation by compatible completed local valuation rings, and which restricted-product or branch-compatibility conditions are needed?

*Status.* Open. 23's comments: by [Corollary 15.107](#) the arithmetic completion reduces to this problem together with the known first fibre. With  $2^{2^{80}}$  branches over zero, a naive identification with an unrestricted product of local completions needs proof and should not be assumed, and completion in the integer powers of  $T$  should be distinguished from completion for the filtration indexed by all positive rationals.

**Question 18.26** (Refinement beyond monomial root maps; 23). For finite algebraic covers replacing  $T = U^m$ , which branchwise ramification data determine the kernel on arithmetic fibres? Is there an exact analogue of [\(15.30\)](#) when different branches have different ramification indices?

*Status.* Open. 23's comments: the uniform multiplier  $m$  is what makes the threshold a single scalar, and a nonuniform cover needs the full tuple of branch orders, not only their minimum. Finite normalization and lying over give a concrete formulation, but compatibility with the arithmetic integer parts must be checked: a map of function fields alone need not induce the desired map of fibres.

**Question 18.27** (A formal development; 23). Formalize the cyclic nonradicality theorem, the residue criterion and the refinement law in Lean, with an abstract algebraic-function-field layer separate from the embedding into the constructed surreal field.

*Status.* Open. 23's plan: first the short argument for  $\mathbf{z}_n$ , with finite normalization and one valuation normalized by  $v(T) = 1$ ; then the coefficient-span lemma and the exact ideal test; the Boolean splitting needs further finite-extension and ramification infrastructure, and the periodic resolution follows once the annihilator identity is available. No declaration has been implemented, and no repository build was run for 23 ([Section 19.2](#)).

**Question 18.28** (Choice strength for the omnific ring; 24). Over GB with choice for sets, does every  $(1 + \omega^a) \subseteq \mathbf{Oz}$  have a maximal class extension? Does the existence of such extensions for all  $a > 0$  imply global choice? What changes when a single value, such as  $a = 1$ , is fixed?

*Status.* Open. 24's comments: [Theorem 14.30](#) cannot answer this restriction, since its converse builds class polynomial rings whose variables encode all finite partial choice functions; a first step would be to see whether that compatibility encoding can be interpreted in a fixed binomial quotient without assuming the global choice class.

**Question 18.29** (Definable rather than globally selected kernels; 24). Can a nonarithmetic maximal ideal of  $\mathbf{Oz}$  be defined in the pure set-theoretic language from a set parameter, in natural models of set theory? Can one specify a branch of the dyadic algebra for which a definable maximal extension exists?

*Status.* Open. 24's comments: its uniformity is relative to a chosen global well-order, not parameter-free canonicity; absoluteness and definability should be studied separately from existence in an expanded class model, and a construction stable under a useful class of universe extensions would be much stronger than the choice-dependent recursion.

**Question 18.30** (Are the binomial residue fields algebraically closed?; 24). For a maximal ideal  $M \supseteq (1 + \omega^a)$ , must  $\mathbf{Oz}/M$  be algebraically closed? If not, which algebraic extensions occur over it? Ask the same for  $\mathbf{Oz}[i]$ .

*Status.* Open. 24's comments: by [Corollary 14.52](#) a positive answer would identify these residue fields abstractly with  $\mathbf{No}[i]$  in GBC; the embedded algebraically closed field ([Corollary 14.44](#)) supplies



roots only for polynomials with coefficients in it; a concrete test is to understand integral extensions of the binomial quotient and which of their elements descend into its maximal fields, and a first target is a root-existence theorem for monic polynomials with coefficients in a finite extension of the embedded dyadic and Hahn core. 24 calls this the most consequential next step.

**Question 18.31** (Size over the Hahn core; 24). How large can a family algebraically independent over the embedded  $\mathcal{F}_a(i)$  be in a binomial residue field? Can the residue field equal that core, or a prescribed set-generated extension of it? Which branch choices force additional transcendence?

*Status.* Open. 24's comments: the explicit core is useful because it is a fixed base for algebraic dependence, not only a cardinal bound; the dyadic roots are algebraic over it, so the continuum of branch choices is not by itself a transcendence-degree argument.

**Question 18.32** (Branching beyond the dyadic subalgebra; 24). At one fixed scale  $a$ , is there a family of distinct maximal class ideals indexed by every set cardinal? Can larger independent idempotent families be built in  $\mathcal{Q}_a$ , and which compatible finite-algebra specializations extend with prescribed residue-field properties?

*Status.* Open. The known lower bound at a fixed scale is  $2^{\aleph_0}$  ([Theorem 14.46](#)). 24's comments: the ordinal family changes the embedded scale and rescaling identifies the abstract fibres, so it cannot be counted again; new independent equations inside one fibre are needed. For comparison, 20's fresh-scale splitting ([Theorems 15.66](#) and [15.71](#)) builds free Boolean families of every set size in the normalization fibre  $\mathcal{N}/\Pi\mathcal{N}$ , a different ring [merge].

**Question 18.33** (Removing one support-closure hypothesis; 24). Classify the term-closed prime ideals of **Oz** without assuming strong sum-closure, and the strongly sum-closed ones without assuming term-closure. Does one closure property force the other under a natural additional algebraic or definability condition?

*Status.* Open. 24's comments: the proof of [Theorem 14.57](#) uses the two properties at different points, one to extract a monomial support restriction and the other to rebuild a series from its terms; a counterexample or a redundancy theorem would locate the reach of coefficientwise methods, and neither implication should be assumed.

**Question 18.34** (Formal verification at the right foundational level; 24). Can the finite-algebra and support-identity package be formalized over the repository's constructed surreal field, with the class-choice theorem formalized separately in an explicit model of GB?

*Status.* Open. 24's plan: first the degree and unit lemmas, the geometric multiplier, the monomial inverse certificates, the dyadic idempotents and the convex truncation kernels; maximal ideals of set rings in a larger Lean universe can then check the algebraic consequences of maximality, but do not verify the metatheorem over two-sorted GB, since a Lean type is set-sized relative to a larger universe and the strength of global choice must not be hidden in a universe change; the last step is an explicit syntactic or semantic formalization of the class recursion and the choice equivalence ([Section 19.2](#)).

**Question 18.35** (The optimal differentiability threshold; 25). For each graph degree  $D$ , determine the least  $k(D)$  such that every globally  $C^{k(D)}$  semialgebraic preserver of a cofinal discretely ordered ring is polynomial. Is a linear bound in  $D$  possible, or can polynomial contact force quadratic growth?



*Status.* Open; for  $\mathbf{Oz}$ ,  $1 + \lfloor (D - 2)/4 \rfloor \leq s(D) \leq D^2$  and  $s(1) = 1$  (Corollary 16.113). 25's comments: the proof of Theorem 16.100 bounds the contact order of  $Q(X, Y) = 0$  with  $Y = P(X)$  by  $D \deg P$ , and improving that estimate under the arithmetic tail constraint is a concrete route; the lower bound is linear in  $D$  but proves no optimality of  $D^2$ .

**Question 18.36** (Bounds for full vector graphs; 25). Given a degree- $D$  description of the full graph of a map  $E^r \rightarrow E^{r'}$ , find an explicit regularity threshold in terms of  $D, r, r'$  without first using an unspecified quantifier-elimination bound for the scalar coordinate graphs. Can output elimination keep useful dimension-free bounds?

*Status.* Open. 25's comments: Theorem 16.102 gives the dimension-free bound  $D^2$  for scalar graphs and for vector maps whose coordinate graphs each have that degree; eliminating the other outputs from one full graph can increase degree, and quantifying that increase is a distinct problem.

**Question 18.37** (Bounded-region structure; 25). Classify the ideal  $\text{SA}_b(\mathbf{Oz})$  of bounded-region semialgebraic lattice preservers. Which of its elements arise by multiplying a semialgebraic function over  $K_H$  by a multiplier supported in  $H^\uparrow$ , followed by a finite cutoff?

*Status.* Open. 25's comments: the exact quotient Corollary 16.96 reduces the classification of all semialgebraic preservers to this ideal and a known numerical-polynomial ring; absorption gives a large explicit subclass, not a converse. 25 names this the focused next problem, with regularity and exponent rank tracked together.

**Question 18.38** (Multipliers for one algebraic family; 25). For a fixed algebraic branch  $\phi$ , determine all  $c \in \mathbf{No}$  with  $c\phi(x) \in \mathbf{Oz}$  on a prescribed omnific interval. Can cancellation in a nonmonomial  $c$  allow substantially smaller supports than absorption of all of  $K_H$  requires?

*Status.* Open. 25's comments: Theorem 16.105 answers the uniform field-absorption question exactly and Theorem 16.108 the monomial square-root case; the family-specific problem between them is not settled.

**Question 18.39** (Rank and local flexibility; 25). For a fixed divisible ordered exponent group  $\Gamma$ , when does its normal-form integer part admit a semialgebraic preserver of a prescribed finite smoothness order that is not finitely piecewise polynomial?

*Status.* Open. 25's comments: a pair  $0 < a \ll b$  in  $\Gamma$  suffices (Proposition 16.121); its failure in rank one rules out one mechanism (Proposition 16.122), and necessity for all constructions is not proved.

**Question 18.40** (Beyond semialgebraicity; 25). Which polynomially bounded o-minimal expansions of real closed Hahn fields admit the same cofinal-discrete-ring tail theorem? Under what hypotheses does global infinite differentiability still continue the tail polynomial through the whole domain?

*Status.* Open. 25's comments: the finite-difference part needs polynomial growth, derivative control and eventual monotonicity, and the global step the appropriate nonflatness theorem; exponential structures need different growth conclusions, and the classical literature [66, 67] is a guide, not an automatic transfer.

**Question 18.41** (Groups of finite-smoothness lattice diffeomorphisms; 25). Study the group of semialgebraic  $C^k$  diffeomorphisms of  $\mathbf{No}^r$  preserving  $\mathbf{Oz}^r$ . How much of it is generated by numerical-polynomial automorphisms, affine arithmetic maps and the bounded-window shears of Corollary 16.114?

*Status.* Open. 25’s comments: the globally smooth subgroup is polynomial by [Theorem 16.102](#), which does not classify its automorphisms; the finite-smoothness group contains bounded-window elements of Jacobian determinant one, and its relation to the smooth subgroup is unexplored. The polynomial subgroup is the subject of [Question 18.20\(c\)](#).

**Question 18.42** (Complex regularity versus real cutoffs; 25). Find minimal regularity assumptions under which a real-semialgebraic lattice preserver of  $\mathbf{Oz}[i]$  satisfying the Cauchy–Riemann equations is polynomial. Can the  $C^\infty$  hypothesis of [Corollary 16.116](#) be reduced uniformly at fixed graph complexity?

*Status.* Open. 25’s comments: the shear counterexamples are not holomorphic, so they do not settle it, and a proof should stay within an explicit surreal differential framework rather than import complex analytic continuation.

**Question 18.43** (Arithmetic invisibility inside the purely infinite ideal; 25). Classify the semialgebraic maps whose lattice values all lie in  $\Pi$ , modulo bounded-region modifications. More generally, prescribe the ordinary constant-term map on  $\mathbf{Oz}^r$  and determine the allowed bounded-window deformations.

*Status.* Open. 25’s comments: for smooth maps the numerical-polynomial decomposition and coefficientwise constant terms answer it ([Corollary 16.103](#)); finite regularity brings in the radical examples, whose geometry is invisible to every output constant term.

**Question 18.44** (Formal verification; 25). Formalize the cofinal tail-extrapolation lemma, semi-algebraic finite-difference rigidity and convex-support absorption, and combine them with the repository’s actual omnific ring. Which part of real closed semialgebraic calculus can be reused without new axioms?

*Status.* Open. 25’s comments: the core algebraic identities are finite and good early targets; the delicate obligations are semialgebraic Rolle and monotonicity over arbitrary real closed fields, the class-to-set support reduction, and the exact support of a rational-power Hahn expansion ([Section 19.2](#)).

**Question 18.45** (Flat extensions of cones; 26). For pointed cones  $\mathcal{C} \subseteq \mathcal{C}'$  in the same rational space, characterize flatness of  $A_{\mathcal{C}} \rightarrow A_{\mathcal{C}'}$ . Which new exponent comparisons force the two-ray obstruction of [Theorem 13.88](#)?

*Status.* Open. 26’s comments: a necessary condition is that no newly admitted quotient of distinct old ray monomials creates the forbidden unit-constant relation; whether a finite set of such comparisons suffices, or higher syzygies add obstructions when all two-ray tests pass, is to be decided.

**Question 18.46** (Recognition without named monomials; 26). Does the abstract ring  $A_{\mathcal{C}}$ , with  $D$  named but no grading, determine the face lattice of  $\mathcal{C}$ ? When must an isomorphism preserve the geometric radical ideals?

*Status.* Open. 26’s comments: [Corollary 13.78](#) solves only the marked version; an unmarked theorem needs an intrinsic characterization of the coefficient-full geometric ideals, which connects these invariants to the repository’s embedding and automorphism program.

**Question 18.47** (Infinite-support extensions; 26). Which Hahn-type completions of  $A_{\mathcal{C}}$  keep the principal exhaustion of each  $I_F$ , the detector formula and the finite flat resolutions?

*Status.* Open. 26’s comments: the issue is whether one positive shift can be subtracted from all exponents of a series; finite supports allow a minimum over finitely many inequalities, while an infinite support can approach a boundary face with no uniform inward margin, so a theorem must impose a support condition excluding this or replace principal exhaustion.

**Question 18.48** (Exponent-density hypotheses; 26). Replace  $\mathbb{Q}^d$  by a subgroup  $\Gamma \subseteq \mathbb{R}^d$ . Which conditions on  $\Gamma \cap F$  for each face are sufficient, or necessary, for flat face ideals and for (13.22)?

*Status.* Open. 26’s comments: the proof uses countable sequences in relative interiors tending to zero and enough subtraction stability, not every rational exponent; comparison with the dense-subgroup orthant case of Geist–Miller [72, Section 4] should separate the existence of a flat resolution from the telescope bound on projective dimension.

**Question 18.49** (Beyond polyhedral cones; 26). Can an infinite face poset give a usable analogue of the join-homology formula? How do the number of faces, the cofinality of inward shifts and homological dimension interact for nonpolyhedral cones?

*Status.* Open. 26’s comments: finite-label cancellation is no longer automatic, and finite supports do not force finitely many labels globally; locally polyhedral cones and increasing unions of rational cones are controlled first cases.

**Question 18.50** (Arithmetic rings beyond principal ideal domains; 26). For a general domain  $D \subseteq K$ , determine the flat dimensions over  $A_{\mathcal{C}}$  of the modules acting through  $D$ .

*Status.* Open. 26’s comments: Theorem 13.85 uses the torsion theory and flatness of torsion-free modules over a principal ideal domain; over a non-Prüfer domain further homology of  $D$  survives, and a theorem should separate the cone contribution from the  $D$ -module contribution without simply adding dimensions where torsion removes the tail.

**Question 18.51** (Efficient reduced resolutions; 26). Can the top join homology and a detector certificate be computed without all  $2^n$  subsets of generators? Can the reduced differentials be chosen canonically or functorially?

*Status.* Open. 26’s comments: its program favours auditability over efficiency; for face-prime quotients the disjoint-face formula already replaces subset enumeration by a maximum over the face poset, and general antichains may admit discrete-Morse or interval reductions compatible with Lemma 13.67, the lcm-lattice techniques being classical [71].

**Question 18.52** (Characteristic-sensitive examples; 26). Which finite face antichains have join complexes whose homology depends on the characteristic of  $K$ , and how small can such a cone and antichain be?

*Status.* Open. 26’s comments: Corollary 13.82 gives a route through torsion in squarefree monomial topology; the omnific applications are in characteristic zero, and a minimal example would show which parts of the arbitrary-field theorem carry arithmetic information.

**Question 18.53** (Derived structure beyond self-Tor; 26). Describe the derived restriction along  $A_{\mathcal{C}} \rightarrow A_{\mathcal{C}}/\mathfrak{r}_{\mathcal{F}}$ , including the subcategory invisible to tensoring with the quotient and any compact-generation properties.

*Status.* Open. 26’s comments: the homological epimorphism is proved (Theorem 13.72); a recollement or compactness statement needs more hypotheses and proofs, and the contrast between vanishing self-Tor and nonzero detector Tor is a starting point, not a substitute for that analysis.

**Question 18.54** (A staged formalization; 26). Formalize the disjoint-face formula, separating finite combinatorics, filtered monomial ideals and the embedding into the surreal field.

*Status.* Open. 26's plan: encode finite rational cones and joins; prove principal exhaustion; prove the simplex and cap calculations; formalize finite scalar cancellation; then import the normal-form embedding. The equational nonflatness theorem is a small independent target, the finite relation criterion being in Mathlib (Mathlib.RingTheory.Flat.EquationalCriterion); 26 supplies no Lean code (Section 19.2).

**Question 18.55** (The exact Boolean algebra; 27). For the hull of Proposition 15.123, determine  $\text{Idem}(\mathcal{G})$  up to isomorphism. Is it atomless? Can  $\mathbf{X}$  have isolated points? For  $\kappa = \aleph_0$ , can the construction be refined so that  $\mathbf{X}$  is the Cantor space?

*Status.* Open. 27's comments: a free subalgebra of rank  $\kappa$  and  $|\mathbf{X}| = 2^\kappa$  decide none of these; a first step is to classify which further pointed étale neighbourhoods produce idempotents outside the algebra generated by the  $\hat{e}_\alpha$ . Compare Questions 18.18 and 18.22 for the Boolean algebras of 20's and 23's fibres.

**Question 18.56** (Dependence on the hull; 27). How far do the isomorphism type of  $\mathcal{G}$ , or the homeomorphism type of  $\mathbf{X}$ , depend on the choices of clearing and division in the hull? Is there a natural equivalence class of such spaces attached to a seed field?

*Status.* Open. 27's comments: the theorem is existential; it keeps the cardinal and the seed but proves no canonical minimal hull or uniqueness of its local algebra.

**Question 18.57** (Enlarging an integer part; 27). For induced omnific integer parts  $B \subseteq B'$  with compatible localizations  $\mathfrak{L} \rightarrow \mathfrak{L}'$ , when is the map of generic henselian fibres injective, and when do branches merge or appear?

*Status.* Open. 27's comments: the splitter atoms survive every global normalization witness, which protects the certified family but not idempotents created by unrelated étale neighbourhoods.

**Question 18.58** (Normalization versus henselization; 27). Relate the reduced normalization fibre  $\text{IC}_{\mathbb{E}}(\mathfrak{L})/\sqrt{n \text{IC}_{\mathbb{E}}(\mathfrak{L})}$  to the Stone space  $\mathbf{X}$ . Can the finite-pattern argument give an explicit correspondence between normalization choices and henselian minimal primes?

*Status.* Open. 27's comments: a solution should separate residue-field extensions from geometric splitting and account for the forced passage from  $\mathbb{E}$  to  $\mathbb{E}[i]$ ; a map on the whole normalization fibre is stronger than the finite atom survival proved. The full-class fibre  $\mathcal{N}/\Pi\mathcal{N} \cong \mathbb{Z}[\text{Idem}]$  (Theorem 15.38) is the class analogue of that normalization fibre [merge].

**Question 18.59** (Specialization in the local spectrum; 27). Determine the prime chains between the  $2^\kappa$  minimal primes and the maximal ideal of  $\mathfrak{L}^h$ . Which Krull dimensions occur among hulls of cardinality  $\kappa$ , and which specialization patterns can be prescribed?

*Status.* Open. 27's comments: the generic fibre is zero-dimensional, which says nothing about the dimension of the local ring; branching and dimension should be studied separately.

**Question 18.60** (The valuation components; 27). What are the value groups, residue fields and internal topologies of the connected components of  $\text{RZ}(\mathfrak{L})$ ? Can valuations realizing all explicit Boolean assignments be built directly from the scale data?

*Status.* Open. 27's comments: [Corollary 15.142](#) counts components through an external theorem; a direct valuative realization would explain how order-infinite elements acquire positive value at these non-order-convex centres.

**Question 18.61** (Mixed-characteristic constant-term primes; 27). Replace  $\Pi_{\mathbb{E}}$  by  $\text{ct}^{-1}(p\mathbb{Z}) \cap B_{\mathbb{E}}$  for an ordinary prime  $p$ . What remains of maximal branching, forced complexification and the strict Galois product, and what changes at  $p = 2$ ?

*Status.* Open. 27's comments: the quadratic formulas use 2 and 16 as units after rationalization, and at a mixed-characteristic centre the residue extension and the completion differ, so the proofs do not transfer. By [Theorem 6.2](#) the corresponding ideal of  $\mathbf{Oz}$  is  $p\mathbf{Oz} = p\mathbb{Z} + \Pi$ , which already contains  $\Pi$  [merge].

**Question 18.62** (Finite étale algebras before localization; 27). Classify finite étale algebras over  $B_{\mathbb{E}}$  and compare their restrictions to the rational constant-term localization and the ordinary prime localizations. Is there a useful arithmetic gluing description?

*Status.* Open. 27's comments:  $\text{Fin}\acute{\text{Et}}(\mathfrak{L}^h) \simeq \text{Fin}\acute{\text{Et}}(\mathbb{Q})$  is henselian local algebra, not a classification over  $B_{\mathbb{E}}$  ([Proposition 15.141](#)).

**Question 18.63** (Birthday-bounded realizations; 27). For which ordinals  $\lambda$  can the hull be built inside the surreals born before  $\lambda$ ? How do closure requirements, regularity or large-cardinal hypotheses improve bounds on  $\lambda$ ?

*Status.* Open. 27's comments: one seed can have a very long sign expansion, and a fresh support bound may leave a chosen cutoff; controlled birthday realizations would connect the local algebra to the foundational reports without confusing cardinality with birthday.

**Question 18.64** (Compatible class-level geometry; 27). Can the directed system of set-sized omnific augmentation rings and their henselizations be organized into a class-sized geometric object without treating the class of all primes as a set? Which invariants are stable?

*Status.* Open. 27's comments: an answer must state its foundations and transition maps; 27 avoids class spectra and class ultrafilters. 24's maximal class ideals ([Theorem 14.24](#)) show one foundation, global choice, in which class ideals of  $\mathbf{Oz}$  can be built [merge].

**Question 18.65** (Higher invariants and descent; 27). How far do finite-partition methods extend to higher  $K$ -theory, derived categories and cohomological invariants of  $\mathcal{G}$  and  $\mathfrak{L}^h$ ? Which invariants of  $\mathfrak{L}^h$  become locally constant after passage to  $\mathcal{G}$ ?

*Status.* Open. 27's comments: filtered colimits and product formulas are natural starting points; only  $K_0(\mathcal{G})$  and the finite projectives are proved.

**Question 18.66** (Formal verification of the branch bridge; 27). Can the monic certificates, the support projection and the survival of finite branches be formalized in Lean and connected to a library construction of henselization and Boolean spectra?

*Status.* Open. 27's plan: first [Lemma 15.126](#), with its quantification over every denominator (a proof of the quadratic identity alone would miss the difficulty); then, in order, the finite quadratic identities, monic certificates, support projection, survival under localization, the hull, and henselization, spectra and cardinality, which need general commutative-algebra and topology infrastructure beyond the surreal normal-form interface ([Section 19.2](#)).

## 19 Verification record and formalization routes

### 19.1 Finite checks

Each manuscript ships a program that checks finite identities with exact arithmetic. They are kept byte-identical in `code/` with their recorded outputs in `data/`; all were rerun for the eight merges on copies (Python 3.14.4, SymPy 1.14.0) and reproduce the recorded outputs, up to line endings, the recorded Python version, for 18 the recorded time stamp and for 23 its hash of the unshipped manuscript.

| Source | What is checked                                                                                                                                                                                                                              | Checks   | Needs  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------|
| 03     | finite telescopes in a symbolic lattice, the unit series coefficients, lexicographic comparisons                                                                                                                                             | 2,362    | stdlib |
| 04     | 1,950 finite telescopes in $\mathbb{Q} \oplus_{\text{lex}} \mathbb{Q}$ with remainders and positive supports, 600 constant-term and 900 augmentation products, 3 support failures and 3 distinguishing examples                              | 5,406    | stdlib |
| 06     | geometric remainders and supports, constant terms, boundary cases, Euler derivation, real complex-structure blocks                                                                                                                           | 17,586   | stdlib |
| 07     | weighted geometric identities, binomial remainders, first idempotents, dyadic refinement maps                                                                                                                                                | 621      | SymPy  |
| 08     | telescopes, constant terms, Hahn Leibniz identities, binomial powers, root corrections                                                                                                                                                       | 1,277    | stdlib |
| 09     | telescopes, truncated comaximality, coefficient maps, finite-support evaluation, Gaussian derivation, gcd identities                                                                                                                         | 597      | stdlib |
| 10     | divided differences, Newton reconstruction, digit indicators and lifts, telescopes, rational differences                                                                                                                                     | 171,395  | SymPy  |
| 11     | finite division identities, positive quotient supports, augmentation and conjugation identities, Gaussian derivation                                                                                                                         | 1,858    | stdlib |
| 12     | homology of the Boolean tensor complex for $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$ on a five-point grid in ranks 1–4, with $d \circ d = 0$ ( <a href="#">Theorem 13.7</a> )                                                                | 780      | SymPy  |
| 13     | 51 finite geometric remainders and their positivity, a cubic norm determinant, a Laurent norm identity, a rank-deficient point, a constant-term product                                                                                      | 6 groups | stdlib |
| 14     | ranks for $\alpha^d = 2$ , homogenized Rees equations, quadratic torsion generators, $P^2 = 2\mathbb{Z}[\sqrt{10}]$ and the mod-5 obstruction, action associativity and tensor balancing                                                     | 2,189    | SymPy  |
| 15     | finite telescopes and divisor truncations, constant-term products, finite-support evaluation, Gaussian matrix evaluation                                                                                                                     | 820      | stdlib |
| 16     | Newton reconstruction, lower-set interpolation matrices, Gaussian counterexample, 150 centered defects, 22 formal inverse branches                                                                                                           | 1,354    | SymPy  |
| 17     | coefficient multiplication matrices and kernels (quadratic, cubic, full orders, mixed, independent coefficients), the Pell matrix, fractional-linear identities, descent ranks $q, \ell$                                                     | 347      | SymPy  |
| 18     | Newton expansions on finite-support $\mathbb{Q}[H]$ , $\mathbb{Q}(i)[H]$ models, centering formulas, the composition obstruction, rational differences, Lagrange leading coefficients, nilpotent matrices, Gaussian examples (20 categories) | 499      | SymPy  |



| Source | What is checked                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Checks  | Needs  |
|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|--------|
| 19     | integral idempotent and doubling identities, spectral projectors, the Catalan expansion to degree 13, 216 endpoint signs of the lifting (24 polynomials, degrees 1–8)                                                                                                                                                                                                                                                                                                                      | 253     | SymPy  |
| 20     | quadratic identities and branch specializations, simple-zero witnesses for 31 radicand products and all 16 four-branch assignments, initial Laurent coefficients to degree 15, Boolean atoms in 1–6 variables and the symmetric difference, 875 support-projection examples and one failure of multiplicativity                                                                                                                                                                            | 6,674   | SymPy  |
| 21     | Koszul squares and transition identities in dimensions 1–6, finite telescope identities and support barriers; integral multiplication-kernel bases, ranks and discriminants of $\mathbb{Z}[\sqrt{2}]$ , $\mathbb{Z}[\sqrt{3}]$ , $\mathbb{Z}[\sqrt[3]{2}]$ , $\mathbb{Z}[\varphi]$ , and the product rank for $\mathbb{Q}(\sqrt{2})$ , $\mathbb{Q}(\sqrt{3})$                                                                                                                              | 1,608   | stdlib |
| 22     | Newton reconstruction in 1–3 variables, integral binomial values, grid determinants and integral inverses, resultant coordinate-degree bounds, good specializations (degrees, coprimality, nonzero bad-locus polynomial), first negative Laurent terms, the geometric remainder, Gaussian examples, fixed-divisor ideals, triangular inverses, sampled lexicographic block signs (22 categories, seed 23092026)                                                                            | 1,776   | SymPy  |
| 23     | binomial power identities of $z_n$ for $n \leq 20$ to order 18 and the displayed coefficients of $z_2$ , refinement thresholds for $n \leq 100$ , $m \leq 50$ and the table for $z_{12}$ , monic refinement certificates, the idempotent identity of $y_c$ for $c = 1, 4, \dots, 1600$ , finite Boolean word and partition models in $r \leq 8$ variables, truncated-polynomial annihilators and alternating compositions for $n \leq 40$ , clipped refinement composition (12 categories) | 18,040  | stdlib |
| 24     | greedy maximal-ideal selection in $\mathbb{Z}/n\mathbb{Z}$ with rejection witnesses and in finite products of $\mathbb{F}_2$ , finite geometric remainders, roots and primitive idempotents of $T^{2^n} + 1$ over $\mathbb{F}_{257}$ with the binary refinement, finite branch filters, finite-rank lexicographic examples of the zero-or-unit cutoff (6 groups, seed 20260923)                                                                                                            | 133,770 | stdlib |
| 25     | extrapolation with arbitrary rational steps, high differences, Newton reconstruction on finite grids, binomial square jets, nonzero binomial coefficients, endpoint exponents and odd root multiplicities, the quadratic witness's exponents and coefficient, rank-two and rank-one exponent comparisons (18 groups, seed 20260923)                                                                                                                                                        | 1,578   | stdlib |
| 26     | face lattices of orthants, cones over polygons, a cube and an octahedron; diagonal scalar homology of the simplex resolution against reduced homology of the join complexes; actual equal-label cancellations; full multiplicity functions of 188 face quotients and 1,329 further antichains; 2,457 rational cap checks                                                                                                                                                                   | 52,080  | stdlib |
| 27     | the splitter equation and squarefree discriminant, both specializations, initial Laurent coefficients, the generic idempotent, forced complexification, 5,586 Boolean atoms in 1–6 variables, finite support projection, Gaussian product maps, a finite Galois splitting matrix (8 groups)                                                                                                                                                                                                | 5,760   | SymPy  |

These checks catch sign, index and coefficient errors in finite special cases. None of them verifies an infinite support, a class-size argument, a cardinal computation, the existence of a maximal ideal, an Ext computation, a primality statement, or novelty; those rest on the written proofs (all sources). The merge results were not machine-checked. For the second merge an independent exact computation, not shipped, rechecked [Theorem 13.7](#) in 1,626 cases (multidegree and subset  $F$ , including every proper  $F$ , over  $\mathbb{Q} \subset \mathbb{Q}(\sqrt{2})$  in ranks up to 3) and the formulas of [Corollary 13.23](#) for  $d \leq 8$ ,  $n \leq 12$ ; like the shipped checks it verifies finite cases only. For the third merge, independent exact computations, not shipped, rechecked the Pell matrix, the multiplication kernels and self-Tor endpoints for  $r \leq 5$ , the irreducibility of  $X^r - (1 + i)$  over  $\mathbb{Q}(i)$  for  $r \leq 6$  and five  $\mathrm{GL}_2(\mathbb{Z})$  cases (17); the composition obstruction of [Theorem 16.63](#) in 27 random cases and the formula for  $\eta_f$  (18); the idempotent identities, the Catalan expansion and 36 random cases of the lifting bound (19); and the sizes of the Gaussian certificates compared in [Proposition 16.64](#) for  $n \leq 3$ ,  $d \leq 5$ . For the fourth merge an independent exact computation, not shipped, rechecked in 18 checks the certificate (15.24), the values of  $\sigma_{\pm}$ , the many-branch identity of [Theorem 15.70](#), the simple zeros used for the square classes, and the identities behind [Proposition 15.68](#). For the fifth merge an independent exact computation, not shipped, rechecked in 78 checks the converse formula of [Theorem 13.46](#) in 12 cases, the exponent identities behind it, the Koszul transition identity, the ranks of  $\mathrm{Tor}_2$  and  $\mathrm{Ind}_2$  and the discriminants of the four orders, and the product rank for  $\mathbb{Q}(\sqrt{2}), \mathbb{Q}(\sqrt{3})$ ; like 21's checks it proves no infinite Ext nonvanishing. For the sixth merge an independent exact computation, not shipped, rechecked in 101 checks the resultant and the partial fractions of [Example 16.87](#), the counts after [Theorem 16.70](#) and the comparison of [Remark 16.71](#) for  $m \leq 4$ ,  $d \leq 5$ , the bound (16.26) in twelve random trivariate cases, the inverse shears of [Example 16.85](#), the two Gaussian denominators and the coefficients of [Example 16.74](#); it proves no statement about infinite supports. For the seventh merge 23's program was rerun on a copy: all 18,040 assertions pass in the same 12 categories, and the rewritten record differs from the shipped one only in the Python version and in its hash of the manuscript, which is `null` because the manuscript is not shipped beside the program (and, on Windows, in line endings). An independent exact computation, not shipped, rechecked in 2,642 checks (most of them the exponent inequality of axiom (ii) used in [Remark 15.102](#)) the relations (15.28) for  $n \leq 7$  and (15.33), the refinement certificates of [Theorem 15.89](#), the Laurent coefficients and the full-class factorization of [Example 15.87](#), the residues of the Boolean words of [Theorem 15.111](#) modulo  $T$  for three parameters with the squarefreeness of their radicands, and the cancellation identity behind [Theorem 15.115](#); like 23's program it verifies no integral closure, place or cardinality. For the eighth merge 24's and 25's programs (standard library only) were rerun on copies with an explicit output path (Python 3.14.4): 24 passed all 133,770 assertions in the same six groups and 25 all 1,578 in the same 18 groups, and the rewritten records differ from `data/24-class-residues-verification.json` and `data/25-semialgebraic-smoothness-verification.json` only in the recorded Python version (3.14.4 for 3.13.5) and, on Windows, in line endings; 24's shipped `.txt` record is byte-identical to its JSON record. Neither program proves a class recursion, an infinite Hahn identity, a semialgebraic calculus theorem or a proper-class statement. An independent exact computation (SymPy), not shipped, rechecked in 300 checks the identification of 24's first idempotents with 07's modulo  $\xi_2^4 + 1$  ([Remark 14.45](#)), the squarefreeness and absence of real roots of  $T^{2^n} + 1$  for  $n \leq 5$ , finite geometric remainders behind (14.26), the Boolean-ring criterion of [Remark 14.28](#) for pairs in the algebra of subsets of a four-element set, the coefficients  $\frac{1}{2}$  and  $-\frac{1}{8}$  of (16.42) for  $b = 1, \dots, 4$ , the constant of the  $k$ th derivative in the proof of [Theorem 16.112](#) for  $k \leq 6$  with the degree count of [Corollary 16.113](#), and the vanishing orders behind [Remark 16.120](#); it verifies no class recursion, infinite support or semialgebraic calculus. 26's and 27's programs were also rerun on copies (Python 3.14.4, SymPy 1.14.0): 26 passed all 52,080 assertions with the same counts (188, 1,329, 56, 2,457) and

rewrote a record, beside itself, identical to `data/26-face-topology-verification_results.json` except for the elapsed time; 27 passed all 5,760 assertions in the same eight groups, and its record, written only with `-output`, equals `data/27-henselian-branching-verification.json` up to line endings on Windows. Neither verifies a filtered colimit, a localization denominator, a prime or a cardinality. An independent exact computation (SymPy), not shipped, rechecked in 2,726 checks the multiplicities (13.21) and the flat dimension  $\mu_{\mathcal{C}}(F)$  for every proper nonzero face of the cones over a square and a cube, by computing the reduced homology of every join complex  $K_{<G}$  directly, the square-cone multiplicities of the examples, the splitter identity and the one-radical factorization of Remark 15.122 with its six Laurent coefficients, the identity  $j^2 = -1$  of Proposition 15.128 and the inverse of the Gaussian decomposition; it verifies no colimit, detector module or henselization.

## 19.2 Formalization routes

The source-era assertion that the repository has no omnific module is now obsolete. The ledger `docs/FORMALIZATION.md` maps the actual omnific and Gaussian rings, their retractions, degree and unit laws, common-divisor support arguments and the quadratic definition of the purely infinite ideals through the sibling report's labels. This shared material does not prove the full-class quotient theorem or the cardinal threshold results here. The following routes remain proposals for those statements. First formalize the scaled-field collision lemma and the separation lemma, which need no surreals (06, 04, 07). Then the countable telescope in a set-sized Hahn workspace, with its two support inequalities, proved coefficientwise and not by convergence of partial sums (04, 06, 07, 08, 10). The set-sized criteria (Theorems 10.7 and 10.8) avoid proper-class syntax and are good first targets (08). The full-class theorem needs an explicit universe interface in which the target and the collision index are small and the exponent class supplies smaller scales below every small support; a universe-polymorphic theorem about an arbitrary field does not supply them (04, 08, 10, 11). For the constructions, the finite-coordinate invariant of 08 must be preserved through field generation, real closure and truncation, and the module bound at a cardinal must use cyclic quotients or the direct collision, not an endomorphism ring (08, 06). The homological part comes last, over set-sized rings only (06). For 07's quotient, the support characterization of  $\mathbf{Oz}[\omega^{-a}]$  and the exact binomial kernel should be kept separate from the finite Chinese-remainder algebra. The repository's small-normal-form modules (`SmallNormalForm.lean`, `SmallNormalFormTruncation.lean`, `SmallNormalFormEvaluation.lean`, `SignSequenceValuationBalls.lean`, see [51]) are possible prerequisites (04); they are not a certificate for any result here. The later manuscripts add three proposals. 12 would define its cores as subrings of a monoid algebra cut out by the constant-coefficient predicate, with the directed-union equality, exactness of the finite-support simplex complex and the degreewise classification of Theorem 13.7 as the main obligations, and the surreal embedding last. 14 would start from an abstract domain  $A = D \oplus P$  with  $P$  a  $k$ -linear ideal in which every element is a product of two, a hypothesis the finite-support model satisfies. 16 separates a finite part (simple-root recursion, centering, the leading-term argument, finite differences), a Hahn part (the block embedding) and an actual-surreal part (a common set-sized exponent workspace and a fresh exponent). The third batch adds three more. 17 separates a finite layer (saturated lattices, constant kernels, scalar multipliers and coefficient product matrices over a principal ideal domain), a support layer (the small-family gap and the union of principal ideals) and a homological layer over the set-sized models only. 18 would formalize the cluster theorem for an abstract discretely ordered ring and specialize it through the normal-form bridge. 19 would formalize the determinant contraction over any ring with a retraction, radicality under power divisibility, saturation, the root-forcing estimate, the spectral projectors and the doubling, and keep the class representation theorem and

its ultrafilter foundation separate. 20 proposes its set-sized rank extension (Theorem 15.70) as a first target: Hahn coefficient projection, the finite determinant certificate for radical membership, iterated Hahn fields and the presentation and specialization of a multiquadratic algebra, all over set-sized ordered groups; the full-surreal theorem needs in addition the normal-form bridge, a bound above a set of exponents and universe management, and the class-ultrafilter representations should stay a separate foundational module. The fifth batch adds one. 21 would formalize its dimension theorem on the finite-support monoid algebra before any surreal embedding: the face ideals, the coefficientwise simplex contraction, the Koszul transitions, the mapping cone and the last Hom differential, with the nonvanishing lemma that a finitely supported element cannot have prescribed nonzero coefficients at infinitely many distinct exponents; its second part would start from an abstract ring  $D \oplus \Pi$  with  $k \oplus \Pi$ , a common positive monomial factor and a flat tail, whose instantiation at the class rings is a separate obligation. The sixth batch adds one. 22 proposes a small extension of the repository's Hahn layer: the order embedding  $(n, g) \mapsto nb + g$  on lexicographically ordered blocks, reverse well-ordering and local finiteness of the Laurent image, preservation of multiplication, the nonnegative-support detector, the resultant degree bound with the grid lemma, and the explicit deceptive function, with the proper-class conclusions proved through an explicit small-set bounding interface; a finite-rank toy model would not establish the full claims (Question 18.19(f)). The seventh batch adds one. 23 would first prove its  $\mathbf{z}_n$  argument from finite normalization and one valuation, then the coefficient-span lemma and the exact ideal test, with an abstract algebraic-function-field layer separate from the embedding into the surreal field; the Boolean splitting needs further ramification infrastructure, and the periodic resolution follows from the annihilator identity (Question 18.27). The eighth batch adds two. 24 would first verify the degree and unit lemmas, the geometric multiplier, the monomial inverse certificates, the dyadic idempotents and the convex truncation kernels over the constructed surreal field, and formalize the class recursion and the choice equivalence separately, in an explicit model of GB, without hiding global choice in a change of universe (Question 18.34). 25 would start from the cofinal tail extrapolation, semialgebraic finite-difference rigidity and convex-support absorption, with semialgebraic Rolle and monotonicity over arbitrary real closed fields, the class-to-set support reduction and the exact support of a rational-power Hahn expansion as the delicate obligations (Question 18.44). 26 would encode finite rational cones and joins, principal exhaustion, the simplex and cap calculations and scalar cancellation before the surreal embedding, with the equational nonflatness theorem as a small separate target (Question 18.54); 27 would start with the survival of every finite branch under localization (Question 18.66). None was executed.

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